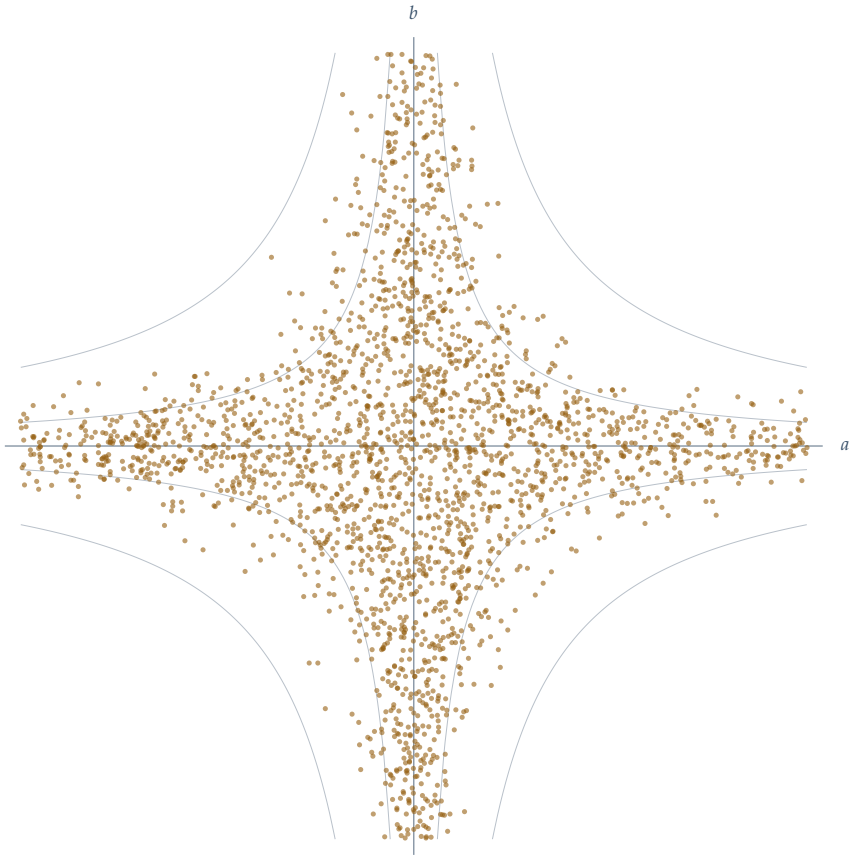
THE MISSING TEXTBOOKS

Singular Learning Theory for the Rest of Us

From $y = abx$ to Neural Networks

Written by Claude

Conceived and edited by Xiaoyu Li



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Preface

This book began with a list of textbooks that should exist but do not. One entry observed that neural networks are singular statistical models: many different weights give the same function, the Fisher information is degenerate, and the asymptotic theory behind the Laplace approximation, the Bayesian information criterion, and Akaike's criterion does not apply to them. The entry noted that the correct theory exists. Sumio Watanabe built it from 1999 on, and it replaces the familiar half the number of parameters by a single invariant of the loss, the learning coefficient. It has recently been used to measure how the complexity of a network changes during training. But Watanabe's monograph of 2009 asks its readers to know resolution of singularities before they begin, and the theory has a reputation for being hard to learn. The entry proposed a textbook for the rest of us: one family of toy models, starting from the model $y = abx$ with two parameters, on which every concept is computed by hand; Hironaka's theorem on resolution of singularities used as a black box, stated precisely and never proved; and a last part that meets the empirical results of deep learning.

The resulting book has fifteen chapters in four parts. Part I sets the classical theory against the running toy models and computes the free energy of the simplest of them completely, which turns out to contain most of the theory in miniature. Part II is about the learning coefficient as a geometric quantity: the rate at which the volume of nearly true parameters shrinks, computed by blowing up the parameter space or by reading a Newton polygon, with a catalogue of the values that are known. Part III is about Watanabe's theorems on the free energy, the generalization error, the criteria WAIC and WBIC, and the posterior's response to temperature: it proves the central result of each completely for a class of models where the computation can be carried through, states the general theorem with its source, and shows how a posterior moves between singularities as the data grow. Part IV asks what all this says about trained networks: the local learning coefficient, its changes during training, and what singular learning theory does not explain. Each chapter states what the regular theory says about its subject, computes the singular answer by hand on a toy model, proves its central theorem completely, in general or for such a class of models, measures the result in a short script, records it on a ledger card, and closes by saying where the theory stops.

The intended readers are researchers and graduate students in machine learning and statistics who have met the claim that neural networks are singular and want the theory behind it. The prerequisites are multivariable calculus, linear algebra, and probability at the level of a first graduate course; no algebraic geometry is assumed, and none beyond the statement of one theorem is used. Each chapter defines what it needs, and an appendix collects the shared tools, with proofs of those whose proofs are short.

Two cautions belong at the front. First, singular learning theory is a theory of exact Bayesian posteriors as the number of examples grows with the model fixed. Deep learning trains one network by stochastic gradient

descent, often with more parameters than examples. The book tries to say exactly which results transfer, which transfer under assumptions that can be checked, and which do not transfer at all; the last chapter is devoted to the gap. Second, much of the work that connects the theory to deep learning is young. Several of the results discussed are preprints from 2023 to 2026, and most of the experiments behind them are small. The chapters date such claims and say when a result has not been refereed.

A NOTE ON AUTHORSHIP. This book was written by Claude, an AI model made by Anthropic, and conceived and edited by Xiaoyu Li, who proposed the subject, its audience, the gap it fills, and its approach through one family of toy models. Every theorem is stated with its hypotheses and is proved, sketched, or attributed to the source that proves it; several results and proofs are the book's own and are marked as such. Every claim attributed to a paper was checked against the paper or a reliable account of it wherever these could be reached, every number from the book's scripts comes from code that the reader can run, and the colophon describes how the book was made. Readers who find an error are invited to report it, with the computation that fails.

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Notation

The chapters share the notation below. Where a chapter uses one of these letters for something else, such as S for a set or v for a variance, it defines the local meaning where it introduces it, and the notes at the end of each chapter translate the notation of its main sources into this one.

Data, models and losses

$X^n = (X_1, \dots, X_n)$	the data, independent with distribution q, the truth
(x_i, y_i)	an input and an output in a regression $y = f(w, x) + \varepsilon$, with $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ and $\sigma = 1$ unless stated
$p(x \mid w)$, $w \in W \subset \mathbb{R}^d$	a model and its parameter; W is compact unless stated
$\varphi(w)$	the prior density, uniform on $[-1, 1]^d$ unless stated
a, b; $a_1, \dots, a_L$; A, B	the parameters of the running models (two scalars; a chain; two matrices)
$L_n(w)$, $L(w)$	the empirical loss $-\frac{1}{n}\sum_i \log p(X_i \mid w)$ and its expectation
S, S_n	the entropy of q and its empirical version $-\frac{1}{n}\sum_i \log q(X_i)$
w_0, W_0	a true (optimal) parameter, and the set of all of them
$K(w)$, $K_n(w)$	$L(w) - L(w_0) \geq 0$, the Kullback–Leibler divergence $\mathrm{KL}(q \,\|\, p(\cdot \mid w))$ in the realizable case, and $L_n(w) - L_n(w_0)$

Posteriors and free energies

β	an inverse temperature; the tempered posterior has density proportional to $\varphi(w)e^{-n\beta L_n(w)}$
$\mathbb{E}_w\big[f\big]$, $\mathbb{E}_w^\beta\big[f\big]$	the posterior expectation of $f(w)$ at $\beta = 1$ and at β
$p^*(x)$	the Bayes predictive density $\mathbb{E}_w\big[p(x \mid w)\big]$
F_n, $F_n(\beta)$	the free energy $-\log \int \varphi(w)e^{-nL_n(w)}\,\mathrm{d}w$ (the negative log marginal likelihood), and its tempered version
$F_n - nS_n$	the normalized free energy; in the realizable case, where $L_n(w_0) = S_n$, it is $-\log \int \varphi(w)e^{-nK_n(w)}\,\mathrm{d}w$
$\bar{F}(n)$	the population free energy $-\log \int \varphi(w)e^{-nK(w)}\,\mathrm{d}w$

The learning coefficient

$\zeta(z)$	the zeta function $\int K(w)^z \varphi(w)\,\mathrm{d}w$
λ, m	the learning coefficient (the real log canonical threshold of K with respect to φ) and its multiplicity: $-\lambda$ is the largest pole of ζ and m its order
$V(\varepsilon)$, $v(t)$	the volume $\int_{K(w)<\varepsilon} \varphi(w)\,\mathrm{d}w$ and the state density $V'(t)$
$\lambda(w^*)$, $\hat{\lambda}(w^*)$	the local learning coefficient at a point w^* and an estimate of it
ν	the singular fluctuation

Generalization and criteria

G_n, T_n	the Bayes generalization loss $-\mathbb{E}_X[\log p^*(X)]$ and the Bayes training loss $-\frac{1}{n}\sum_i \log p^*(X_i)$; $G_n - S$ is the generalization error in the realizable case, and $G_n - L(w_0)$ in general
V_n	the functional variance $\sum_i \mathrm{Var}_w[\log p(X_i \mid w)]$
WAIC, AIC	$T_n + V_n/n$ and $L_n(\hat{w}) + \frac{d}{n}$: estimates of the generalization loss, on the per-sample scale of L_n (statistical software reports $2n$ times these)
WBIC, BIC	$\mathbb{E}_w^\beta[nL_n(w)]$ at $\beta = 1/\log n$, and $nL_n(\hat{w}) + \frac{d}{2}\log n$: approximations of the free energy, on its scale

Probability and asymptotics

Probabilities and expectations $\mathbb{P}$ and $\mathbb{E}$ without a subscript are over the data (and fresh test points); a subscript names other randomness. We write $\mathcal{N}(\mu, \sigma^2)$ for the normal distribution and Unif for the uniform one. The symbols O_p and o_p refer to $n \to \infty$ with the randomness of the data; $a_n \sim b_n$ means $a_n/b_n \to 1$, and $a \asymp b$ means that each is bounded by a constant multiple of the other. Logarithms are natural, and $\mathbb{N} = \{0, 1, 2, \dots\}$.

Prologue

Two parameters

Take two numbers a and b, multiply them, and use the product as the slope of a line through the origin:

$$y = abx + \varepsilon, \qquad x \sim \mathcal{N}(0,1), \quad \varepsilon \sim \mathcal{N}(0,1).$$

This is the smallest model in this book, and the one we return to in every chapter. Fit it, in the Bayesian way, to data that are pure noise: n pairs (x_i, y_i) drawn with $y_i = \varepsilon_i$, so that the truth is the slope 0. Put a uniform prior on the square $[-1,1]^2$, and look at the posterior distribution of (a,b).

A statistician trained on regular models expects to see a small Gaussian blob. The model has two parameters; there is a maximum-likelihood estimate $(\hat{a},\hat{b})$; the posterior should concentrate around it at the rate $n^{-1/2}$, with covariance $(nI)^{-1}$, where I is the Fisher information; and the logarithm of the marginal likelihood should be the maximized log-likelihood minus $\frac{d}{2}\log n = \log n$, up to a bounded term. That is what the Bayesian information criterion of Schwarz (1978) says, and it is what every textbook proof of the Laplace approximation delivers.

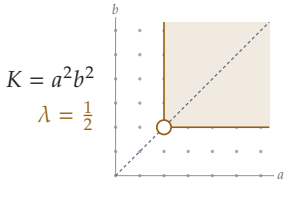
None of it happens. Figure P.1 shows draws from the posterior for three sample sizes. They do not form a blob. They crowd onto the cross $ab = 0$, the set of all parameters that give the true slope, and they spread along both arms. As n grows, the arms get thinner much faster than they get shorter: the posterior does pile up at the crossing point in the end, but only at the rate $1/\log n$, and at $n = 10^6$ nearly two thirds of its mass, 64% on average over data sets, still lies where $\max(|a|,|b|) > 0.1$. There is no single maximum-likelihood point to aim at, either: the likelihood depends on (a,b) only through the product $c = ab$, and every point of the square on the hyperbola $ab = \hat{c}$ maximizes it.

The marginal likelihood misbehaves too. Write F_n for its negative logarithm, the *free energy*, and nS_n for the negative log-likelihood of the truth, so that $F_n - nS_n$ is what the data charge the model for not knowing the truth in advance. Figure P.2 plots the average of $F_n - nS_n$ over two hundred data sets against n, for our model and for the regular model $y = ax_1 + bx_2$, which also has two parameters, also has the uniform prior on $[-1,1]^2$, and also has the truth 0. The regular model pays exactly what Schwarz predicts: one unit of $\log n$ for its two parameters, plus a constant. Our model pays about half as much, and a little less again: its free energy grows like

$$F_n - nS_n = \tfrac{1}{2}\log n - \log\log n + O_p(1). \tag{0.1}$$

The measured slope against $\log n$ over the range of the figure is 0.39, below $\frac{1}{2}$ because of the term $-\log\log n$, and the bronze curve, computed from the exact formula of Chapter 3, passes through all nine measured intervals.

The two numbers in (0.1), the coefficient $\frac{1}{2}$ of $\log n$ and the coefficient 1 of $-\log\log n$, are the subject of this book. The first is the *learning coefficient*



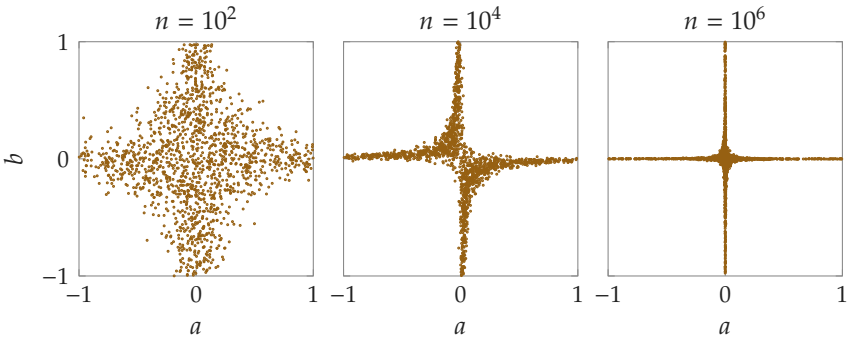


Figure P.1. Draws from the posterior of $y = abx$ when the truth is the slope 0, with the uniform prior on $[-1, 1]^2$: 1500 exact draws for one data set at each sample size (data seeds 102, 104, 106; draw seeds 202, 204, 206). The draws first fix the product $c = ab$ from its one-dimensional posterior, then a given c; Chapter 2 derives the method. The posterior collapses onto the cross $ab = 0$ quickly and toward its center only very slowly (Theorem 2.5).

λ of the model at its truth, a number that algebraic geometers know as the real log canonical threshold of the loss; the second is $m - 1$, where m is the learning coefficient's *multiplicity*. For a regular model with d parameters, $\lambda = \frac{d}{2}$ and $m = 1$, and the classical formulas are the special case. For $y = abx$ at the truth 0, $\lambda = \frac{1}{2}$ and $m = 2$: two parameters that learn like one, with a bonus.

Why two parameters learn like one

The explanation is geometric, and it fits in a paragraph. The data can only see the product ab, so the parameters (a, b) and $(ta, b/t)$ give the same distribution for every $t \neq 0$: the model folds its parameter space onto the interval of slopes. Learning the truth means locating the slope ab to within about $n^{-1/2}$, and the posterior puts its mass on the parameters that do so, which are those of the thin region $\{|ab| \lesssim n^{-1/2}\}$ around the cross. What matters for the free energy is the prior mass of that region. A regular model with two parameters must locate both coordinates, and the region it keeps is a disc of radius $n^{-1/2}$, of area about n^{-1}; the free energy pays the logarithm of the inverse, $\log n$. Our model keeps a cross-shaped region whose area is about $n^{-1/2} \log n$: the arms are $n^{-1/2}$ thick far from the center and fatter near it. The free energy pays $\frac{1}{2} \log n - \log \log n$. In general, the volume $V(\varepsilon)$ of the parameters whose loss is within ε of the truth's shrinks like $\varepsilon^{\lambda} (\log \frac{1}{\varepsilon})^{m-1}$ as $\varepsilon \to 0$, and the free energy pays $\lambda \log n - (m - 1) \log \log n$. Chapters 3 and 4 make this precise.

The same geometry controls everything else a statistician asks of a model. The expected Kullback–Leibler divergence from the truth to the Bayes predictive distribution, the generalization error, is λ/n to first order, not $\frac{d}{2n}$ (Chapter 9). The penalties of the information criteria change accordingly (Chapters 10 and 11). And the posterior does not settle near one point: when several regions of the parameter space nearly fit the data, it moves

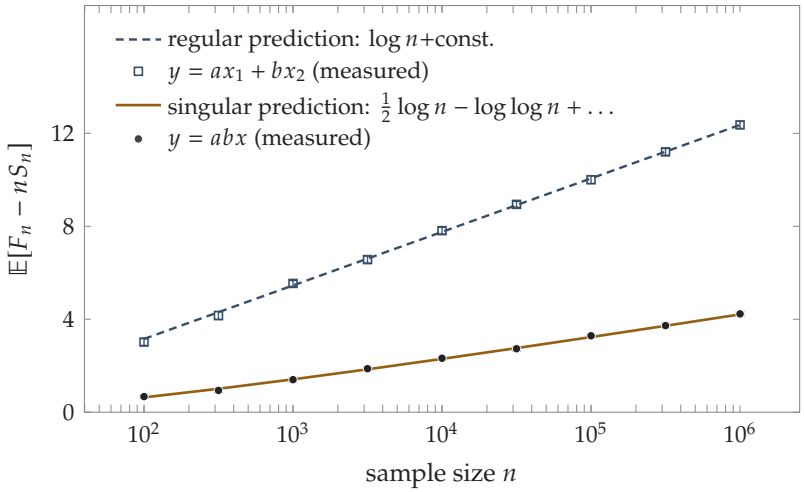


Figure P.2. The normalized free energy $F_n - nS_n$ of two models with two parameters each, averaged over 200 data sets at each of nine sample sizes $n = 10^{k/2}$, $k = 4, \ldots, 12$, with Student-t 95% intervals; the truth is 0 and the prior uniform on $[-1, 1]^2$ for both (data seeds $10^4 k + r$ and $2 \cdot 10^4 k + r$, $r < 200$). Squares: the regular model $y = ax_1 + bx_2$ (Gaussian integral). Dots: the model $y = abx$ (exact quadrature in $c = ab$). Dashed: Laplace's prediction of the mean, $\log n - 1 - \log(2\pi) + \log 4$. Solid: the exact formula of Computation 3.1 with S_{xx} replaced by n, averaged over $Z_n \sim \mathcal{N}(0, 1)$; it keeps the slowly varying factor and tends to $\frac{1}{2}\log n - \log\log n - \frac{1}{2} + \log(4/\sqrt{2\pi})$, the limit of Theorem 3.4 for this model (Corollary 3.5). Script: `figures/scripts/00-prologue-free.py`.

between them as n grows, trading accuracy against learning coefficient (Chapter 12).

Why this matters for neural networks

A model with a cross for its set of true parameters can look like a curiosity designed to break a theorem. It is not; it is the smallest piece of a neural network. Take one hidden unit with input weight b and output weight a. With a tanh activation its contribution is $a\tanh(bx)$, which is abx to first order near $b = 0$; with a ReLU it is $a\,\mathrm{relu}(bx)$, which is exactly $ab\,\mathrm{relu}(x)$ when $b > 0$. Either way the function depends on the two weights through their product near the point where the unit is switched off, and the unit's parameters fold just as ours do. Networks have many more folds: permuting the hidden units of a layer, flipping the signs of a tanh unit's weights, rescaling the two sides of a ReLU, and switching units off all produce different weights with the same function (Dinh et al. 2017; Fukumizu and Amari 2000; Sussmann 1992). In the language of statistics, neural networks are *singular models*: their parameters are not identifiable, and their Fisher information is degenerate at the parameters that matter (Watanabe 2009; Wei et al. 2023). The formulas of the regular theory, and the intuitions built on them, such as counting parameters to measure complexity, do not apply.

The theory that does apply was built by Sumio Watanabe from 1999 on

name	model	λ	m	$\frac{d}{2}$
the regular plane	$y = ax_1 + bx_2$	1	1	1
the product model	$y = abx$	$\frac{1}{2}$	2	1
the cusp model	$y = ab^2x$	$\frac{1}{4}$	1	1
the quartic model	$y = ax_1 + b^2x_2$	$\frac{3}{4}$	1	1
the chain	$y = a_1a_2\cdots a_Lx$	$\frac{1}{2}$	L	$\frac{L}{2}$
the tanh neuron	$y = a\tanh(bx)$	$\frac{1}{2}$	2	1
the ReLU neuron	$y = a\,\mathrm{relu}(bx)$	$\frac{1}{2}$	2	1
the mixture	$(1-a)\mathcal{N}(0,1) + a\mathcal{N}(b,1)$	$\frac{1}{2}$	2	1

Table P.1. Some of the running models, at the truth $w_0 = 0$ (for the mixture, at the truth $\mathcal{N}(0,1)$), with independent standard normal inputs, unit noise, and the uniform prior on $[-1,1]^d$; for the tanh neuron, $x \sim \mathrm{Unif}[-1,1]$ and the prior on $[-2,2]^2$, and for the mixture, $a \in [0,1]$ and $b \in [-2,2]$. The learning coefficient λ and its multiplicity m are proved in Chapters 2 to 6; $\frac{d}{2}$ is what the regular theory would charge.

(Watanabe 1999, 2001a, 2009, 2018). It computes the learning coefficient from the geometry of the loss, with the help of a deep theorem of algebraic geometry, Hironaka's resolution of singularities (Hironaka 1964a,b), and it proves that this one number replaces $\frac{d}{2}$ in the free energy, where the multiplicity adds the term in $\log\log n$, and in the generalization error. Layered neural networks were among its examples from its first journal papers (Watanabe 2001a,b). Since about 2020 the theory has been applied to modern deep learning (Wei et al. 2023): the learning coefficient can be estimated locally at a trained network from samples of a localized posterior (Lau et al. 2025), and its changes during training have been used to locate stages of development in small transformers (Hoogland et al. 2025). How much of this carries over to networks trained by stochastic gradient descent, often with more parameters than examples, is a real question, and the last part of this book treats it as one.

What this book does

Watanabe's theory has a reputation for being hard to learn, and the reputation is earned: its general theorems rest on resolution of singularities and on empirical processes on the resolved parameter space, his monograph of 2009 opens with both, and the connections to deep learning are scattered across papers that assume the theory. This book takes a different road. It starts from $y = abx$ and a handful of relatives, Table P.1, on which every concept of the theory can be computed by hand: the free energy, the posterior, the learning coefficient and its multiplicity, the generalization error, the criteria WAIC and WBIC. It adds machinery only when a computation demands it. The one piece of machinery it does not build is resolution of singularities: Hironaka's theorem is stated precisely in the appendix and used as a black box, and every computation of a learning coefficient in the book is done by explicit changes of coordinates that the reader can check.

The book has four parts. *Part I* sets the classical theory against the running

models (Chapter 1), shows how they violate its assumptions (Chapter 2), and computes the free energy of the models whose loss depends on the parameters through one coefficient, which turns out to contain most of the theory in miniature (Chapter 3). *Part II* is geometry. The learning coefficient is the rate at which the volume of nearly true parameters shrinks (Chapter 4); it is computed by blowing up the parameter space until the loss becomes a monomial (Chapter 5), or, for many losses, read off a Newton polygon in one step (Chapter 6); and its values are known for a few families of models in use (Chapter 7). *Part III* is statistics: Watanabe's theorems on the free energy (Chapter 8) and the generalization error (Chapter 9), the criteria WAIC (Chapter 10) and WBIC (Chapter 11), and the way the posterior chooses between singularities (Chapter 12). *Part IV* turns to trained networks: the local learning coefficient and how to estimate it (Chapter 13), its changes during training (Chapter 14), and the gap between Bayesian posteriors and stochastic gradient descent (Chapter 15).

Every chapter has the same anatomy. *The question* says what the chapter is about, and a slate box says what the regular theory answers. *By hand* computes the singular answer on one or more of the running models, in numbered steps the reader can check. *The theorem* proves the central result completely; when the general proof needs machinery the book does not build, the chapter proves the theorem under a stated simplifying hypothesis, states the general theorem separately with its source, and says exactly what the general proof adds. *Measuring it* runs a short script, available with the book, and compares what it measures with both theories: the singular prediction is drawn in bronze, the regular one in slate. *The ledger* records the result on a card, and the epilogue collects the fifteen cards. *Where the theory stops* names what was assumed, where the result fails, and what is open. Exercises and notes close the chapter; the notes translate the notation of the main sources into the book's.

How to read. The chapters are written to be read in order, but each defines what it uses and points to where its tools come from. A short course for a reader who wants the main line and the connection to deep learning is Chapters 1 to 4, then Chapters 8, 9, 13 and 15. A reader who wants only the geometry can read Part II after Chapter 3. The prerequisites are multivariable calculus, linear algebra, probability at the level of a first graduate course (laws of large numbers, the central limit theorem, and convergence in law), maximum likelihood and Bayes' rule, and the basics of training neural networks. No algebraic geometry is assumed.

What this book does not do. It does not prove Hironaka's theorem, and it does not reproduce the empirical-process theory that Watanabe needs for his theorems in full generality; it states both precisely and proves the theorems completely where they can be proved by elementary means. It treats models with finitely many parameters as the number of examples grows, which is the setting of the theory; it says at each step what that leaves out of deep learning. And it does not claim that singular learning theory explains why neural networks generalize. It explains why parameter counts are the wrong measure of a singular model's complexity in Bayesian

learning from many examples, and what replaces them; how far that goes toward explaining deep learning is the subject of Chapter 15.

Notes

The Bayesian information criterion is due to Schwarz (1978), who derived it for exponential families; Haughton (1988) extended it to curved families and proved its consistency. Akaike's criterion, its frequentist counterpart, is due to Akaike (1974). The free energy formula for singular models, with the learning coefficient in place of $\frac{d}{2}$, is Watanabe's: his review (Watanabe 2024) traces to Watanabe (1999) the finding that the learning coefficient determines the accuracy of Bayesian estimation; Watanabe (2001a) proved the formula for the averaged free energy; his monograph (Watanabe 2009) gives it for the free energy as a random variable; and the two books (Watanabe 2009, 2018) develop its consequences. Resolution of singularities in characteristic zero is the theorem of Hironaka (1964a,b). The symmetries of networks with tanh units were classified by Sussmann (1992); the effect of the resulting singularities on learning dynamics was studied by Fukumizu and Amari (2000). The argument that the singularity of neural networks is a feature for Bayesian learning, not a defect, is made in Wei et al. (2023), and the local learning coefficient as a measure of a trained network's complexity is introduced in Lau et al. (2025).

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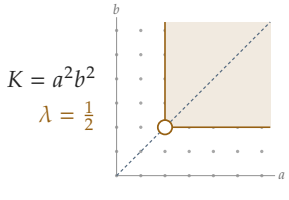
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Part I

Where Regular Theory Ends

Classical statistics charges a model $\frac{d}{2}\log n$ for its d parameters, and it is right when the parameters are identifiable and the Fisher information is invertible. Neural networks satisfy neither condition. Two parameters are enough to see what goes wrong and to compute, by hand, what replaces the classical answer.



1 Regular Models

What the Laplace approximation promises

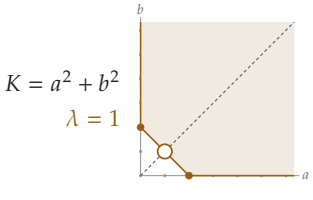
THE MODEL — $y = ax_1 + bx_2$

IN THIS CHAPTER. What does a parameter cost? When the true parameter is a single interior point at which the Hessian of the loss is invertible, the classical answer is $\frac{1}{2}\log n$ in the free energy and $\frac{1}{2n}$ in the generalization error. We compute both by hand for the regular plane $y = ax_1 + bx_2$, whose posterior, marginal likelihood and predictive distribution are exactly Gaussian; with a flat prior its generalization error is exactly $1/(n-1)$. We prove the Laplace approximation of the free energy, $F_n = nL_n(\hat{w}) + \frac{d}{2}\log\frac{n}{2\pi} + \frac{1}{2}\log\det J - \log\varphi(w_0) + o_p(1)$, and with it BIC, and we prove the generalization error $\frac{d}{2n}$ for Gaussian linear models. Quadrature on three regular models, one of them a normal mixture whose likelihood is not Gaussian, confirms every prediction, down to the $1/n$ error of the Laplace approximation. Everything rests on identifiability, an invertible Hessian and an interior truth, and neural networks violate the first two.

1.1 The question: what does a parameter cost?

Suppose we have trained two networks on the same data, a small one and a large one, and must decide which to trust. A Bayesian compares their marginal likelihoods; a frequentist estimates their test losses. Both comparisons have classical large-sample answers, and both answers count parameters. MacKay (1992) built a Bayesian framework for backpropagation networks on the Laplace approximation of the marginal likelihood, in which the evidence penalizes overflexible models and an effective number of well-determined parameters measures complexity; in 2021 the approximation was offered again, with a software library, for the posteriors of deep networks (Daxberger et al. 2021). The frequentist counterpart is Akaike's criterion, which penalizes the maximized log-likelihood in proportion to the number of parameters (Akaike 1974). This chapter asks what these classical answers promise, under which hypotheses, and why. Every later chapter measures against them.

THE SETTING. The data $X^n = (X_1, \ldots, X_n)$ are independent with distribution q, the truth. A model is a family of densities $p(x \mid w)$ indexed by $w \in W \subset \mathbb{R}^d$, with a prior density φ on W. The empirical and the population loss are $L_n(w) = -\frac{1}{n}\sum_i \log p(X_i \mid w)$ and $L(w) = -\mathbb{E}_q[\log p(X \mid w)]$, a true parameter w_0 minimizes L, and $K(w) = L(w) - L(w_0) \geq 0$. The *free energy* is



the negative logarithm of the *marginal likelihood*,

$$F_n = -\log \int_W \varphi(w) e^{-nL_n(w)}\, \mathrm{d}w;$$

the posterior odds of two models are their prior odds times $e^{F_n^{(2)} - F_n^{(1)}}$, so a Bayesian prefers the smaller free energy.[1] The truth is *realizable* if $q = p(\cdot \mid w_0)$. Then $nL_n(w_0) = nS_n$, and the normalized free energy $F_n - nS_n = -\log \int \varphi e^{-nK_n}$, with $K_n = L_n - L_n(w_0)$, is what the data charge the model for not knowing w_0. The posterior predicts with the *predictive density* $p^*(x) = \mathbb{E}_w\big[p(x \mid w)\big]$, whose *Bayes generalization loss* is $G_n = -\mathbb{E}_X[\log p^*(X)]$ for a fresh $X \sim q$; in the realizable case the generalization error $G_n - S = \mathrm{KL}(q \,\|\, p^*)$ is what we lose per test point by not knowing w_0. In a regression model the observation is a pair $X = (x, y)$ and the densities are conditional on the input.[2]

[1] The name is from statistical physics: with $nL_n(w)$ as the energy of the configuration w, F_n is the free energy at temperature 1. Chapter 11 varies the temperature.

[2] The input density, the same for every w, cancels from every quantity in this book except S and S_n, which we therefore define with the conditional density $q(y \mid x)$.

Definition 1.1 (Regular and singular truths). The truth q is *regular* for the model if L has a single minimizer w_0 on W, the point w_0 lies in the interior of W, and the Hessian $J = \nabla^2 L(w_0)$ is positive definite. Otherwise it is *singular*.

Regularity is a property of the truth relative to the model, a convention the book takes from Watanabe (2013). In the realizable case J is the Hessian of the Kullback–Leibler divergence K, and it equals the *Fisher information* at w_0 whenever $\int p(x \mid w)\, \mathrm{d}x = 1$ may be differentiated twice under the integral sign. The running model of this chapter is the *regular plane* $y = ax_1 + bx_2 + \varepsilon$, with $x \sim \mathcal{N}(0, I_2)$, $\varepsilon \sim \mathcal{N}(0,1)$, the prior uniform on $[-1,1]^2$ and the truth 0. Its loss is $L(w) = L(w_0) + \frac{1}{2}|w - w_0|^2$, so $J = I_2$ at every truth.

THE REGULAR ANSWER · the price of a parameter. If the truth is regular and realizable, each of the d parameters costs $\frac{1}{2}\log n$ in the free energy (Schwarz 1978),

$$F_n = nS_n + \frac{d}{2}\log n + O_p(1),$$

and $\frac{1}{2n}$ in the expected generalization error, while the training loss underestimates the test loss by $\frac{d}{n}$ (Akaike 1974):

$$\mathbb{E}[G_n] - S = \frac{d}{2n} + o\Big(\frac{1}{n}\Big).$$

For the product model $y = abx$ of the prologue at its truth $ab = 0$, neither formula is right. Its loss $K(a,b) = \frac{1}{2}a^2b^2$ vanishes on the whole cross $\{ab = 0\}$, and its Hessian at the origin is zero. The free energy grows like $\frac{1}{2}\log n - \log\log n$, not $\log n$ (Figure P.2; Chapter 3), and the generalization error is $\frac{1}{2n}$ to first order, not $\frac{1}{n}$ (Chapter 9). The regular answer's hypotheses fail, and its formula overcharges. To see what replaces it we must know exactly what it promises, under which hypotheses, and where in its proof each hypothesis is used. That is this chapter's question; its answers are the slate dashed lines that every later figure draws for comparison.

1.2 By hand: the regular plane

Every quantity of the regular plane is a Gaussian integral, because its energy is exactly quadratic in the parameter. We work in d dimensions: the model is $y = w^\top x + \varepsilon$ with $w \in \mathbb{R}^d$, $x \sim \mathcal{N}(0, I_d)$ and $\varepsilon \sim \mathcal{N}(0,1)$; $d = 2$ is the regular plane and $d = 1$ the regular line $y = ax + \varepsilon$. The data enter through $S_{xx} = \sum_i x_i x_i^\top$ and $S_{xy} = \sum_i x_i y_i$, and the least-squares estimate $\hat{w} = S_{xx}^{-1} S_{xy}$ maximizes the likelihood.[3] The tool is the Gaussian integral $\int_{\mathbb{R}^d} e^{-\frac{1}{2}(w-m)^\top P (w-m)}\, dw = (2\pi)^{d/2} \det(P)^{-1/2}$ (Lemma A.2).

[3] The law of S_{xx} is the Wishart distribution with n degrees of freedom and scale I_d; Lemma 1.2 collects the three facts about it that we need.

COMPUTATION 1.1 · THE REGULAR PLANE WITH A GAUSSIAN PRIOR

1 *The energy is a quadratic.* Expanding $nL_n(w) = \frac{1}{2}\sum_i (y_i - w^\top x_i)^2 + \frac{n}{2}\log(2\pi)$ around $\hat{w}$, the linear term vanishes:

$$nL_n(w) = nL_n(\hat{w}) + \tfrac{1}{2}(w - \hat{w})^\top S_{xx} (w - \hat{w}).$$

2 *The posterior is Gaussian.* Take the prior $\mathcal{N}(0, \tau^2 I_d)$. Completing the square after adding its exponent $|w|^2/2\tau^2$ shows that the posterior is $\mathcal{N}(m_n, P_n^{-1})$, with precision $P_n = S_{xx} + \tau^{-2} I$ and mean $m_n = P_n^{-1} S_{xy}$, the ridge-regression estimate with penalty τ^{-2}. The constant left over is $\frac{1}{2}\hat{w}^\top (\tau^2 I + S_{xx}^{-1})^{-1} \hat{w}$, by the identity $S_{xx} - S_{xx} P_n^{-1} S_{xx} = (\tau^2 I + S_{xx}^{-1})^{-1}$.

3 *The free energy is explicit.* The prior's constant is $(2\pi\tau^2)^{-d/2}$, the Gaussian integral gives $(2\pi)^{d/2} \det(P_n)^{-1/2}$, and $\tau^2 P_n = I + \tau^2 S_{xx}$, so

$$F_n = nL_n(\hat{w}) + \tfrac{1}{2}\log\det\!\left(I + \tau^2 S_{xx}\right) + \tfrac{1}{2}\hat{w}^\top \left(\tau^2 I + S_{xx}^{-1}\right)^{-1} \hat{w}. \quad (1.1)$$

4 *Each parameter costs $\frac{1}{2}\log n$.* By the law of large numbers $S_{xx}/n \to I$ and $\hat{w} \to w_0$ almost surely, so the second term of (1.1) is $\frac{d}{2}\log n + d\log\tau + o(1)$ and the third tends to $|w_0|^2/2\tau^2$. The first is exact: with the noise vector $\boldsymbol{\varepsilon}$ and the projection Q onto the span of the input columns, $nL_n(\hat{w}) - nL_n(w_0) = -\frac{1}{2}\boldsymbol{\varepsilon}^\top Q \boldsymbol{\varepsilon}$, a $-\frac{1}{2}\chi^2_d$ variable for every $n \geq d$. Hence

$$F_n - nS_n = \frac{d}{2}\log n + d\log\tau + \frac{|w_0|^2}{2\tau^2} - \frac{1}{2}\chi^2_d + o_p(1).$$

5 *The predictive is Gaussian.* At a new input, $y = w^\top x + \varepsilon$ with $w \sim \mathcal{N}(m_n, P_n^{-1})$, so $p^*(y \mid x) = \mathcal{N}\!\left(y;\, m_n^\top x,\, 1 + s_n(x)\right)$ with $s_n(x) = x^\top P_n^{-1} x$: the uncertainty about w widens the predictive.

6 *The generalization error.* The divergence from $\mathcal{N}(\mu_1, 1)$ to $\mathcal{N}(\mu_2, 1+s)$ is $\frac{1}{2}\log(1+s) + \frac{1+(\mu_1-\mu_2)^2}{2(1+s)} - \frac{1}{2}$. With $\mu_1 = w_0^\top x$, $\mu_2 = m_n^\top x$ and $s = s_n(x)$, averaging over the test input gives

$$G_n - S = \mathbb{E}_x\left[\frac{1}{2}\log(1+s) - \frac{s}{2(1+s)} + \frac{\left(x^\top(m_n - w_0)\right)^2}{2(1+s)}\right]. \quad (1.2)$$

7 *Each parameter costs* $\frac{1}{2n}$. Since $s_n(x) \approx |x|^2/n$, the first two terms are $O(n^{-2})$; the vector $m_n - w_0$ has covariance about I/n, so the last term has mean about $\mathbb{E}|x|^2/2n = \frac{d}{2n}$. Theorem 1.7 makes this exact.

A quadratic energy turns every integral into a Gaussian one, and the only work is completing squares. Three facts about Gaussian inputs make the expectations over the data exact as well.

Lemma 1.2 (Inverse Wishart facts). *Let $x_1, \dots, x_n$ be independent $\mathcal{N}(0, I_d)$ vectors, $n \geq d$, and $S_{xx} = \sum_i x_i x_i^\top$. Then S_{xx} is invertible with probability one, and:*

(i) for every unit vector u, $1/(u^\top S_{xx}^{-1} u)$ has the χ^2_{n-d+1} law;

(ii) if $n \geq d+2$, then $\mathbb{E}\,\mathrm{tr}(S_{xx}^{-1}) = d/(n-d-1)$;

(iii) if $x \sim \mathcal{N}(0, I_d)$ is independent of S_{xx}, then $x^\top S_{xx}^{-1} x$ has the law of U/V, with $U \sim \chi^2_d$ and $V \sim \chi^2_{n-d+1}$ independent.

Proof. The plan is to reduce everything to one diagonal entry of S_{xx}^{-1}. Let $\mathbf{X}$ be the $n \times d$ matrix with rows $x_i^\top$, so that $S_{xx} = \mathbf{X}^\top \mathbf{X}$. Conditionally on the other columns of $\mathbf{X}$, which span a subspace of dimension at most $d-1 < n$, column j is a standard Gaussian vector of $\mathbb{R}^n$ and lies in that subspace with probability zero; so the columns are linearly independent and S_{xx} is invertible almost surely.

Part (i). For an orthogonal O, $\mathbf{X}O$ has the law of $\mathbf{X}$, and $(\mathbf{X}O)^\top \mathbf{X}O = O^\top S_{xx} O$; choosing $Oe_1 = u$ shows that $u^\top S_{xx}^{-1} u$ has the law of $(S_{xx}^{-1})_{11}$. Write $\mathbf{X} = [\mathbf{x}^{(1)}, \mathbf{X}']$ and let Q' project onto the orthogonal complement of the column space of $\mathbf{X}'$, of dimension $n-d+1$ almost surely ($Q' = I$ if $d = 1$). Block inversion gives $(S_{xx}^{-1})_{11} = 1/|Q'\mathbf{x}^{(1)}|^2$, and conditionally on $\mathbf{X}'$ the squared length of the projection of the standard Gaussian vector $\mathbf{x}^{(1)}$ onto a fixed subspace of dimension $n-d+1$ is χ^2_{n-d+1}, whatever $\mathbf{X}'$ is.

Part (ii). For $V \sim \chi^2_k$ with $k \geq 3$, the Gamma integral (Definition A.4) gives $\mathbb{E}[1/V] = 2^{k/2-1}\Gamma(\frac{k}{2}-1)/(2^{k/2}\Gamma(\frac{k}{2})) = 1/(k-2)$. By (i) with $u = e_j$, each diagonal entry of S_{xx}^{-1} is such a $1/V$ with $k = n-d+1 \geq 3$; add d of them.

Part (iii). Write $x = |x|\,u$. The length and the direction of a standard Gaussian vector are independent, $|x|^2 \sim \chi^2_d$, and both are independent of S_{xx}. By (i), conditionally on $(u, |x|)$ the variable $V = 1/(u^\top S_{xx}^{-1} u)$ is χ^2_{n-d+1} whatever u is; so V is independent of $(u, |x|)$, and $x^\top S_{xx}^{-1} x = |x|^2/V$. ∎

COMPUTATION 1.2 · THE EXACT GENERALIZATION ERROR OF THE REGULAR PLANE

1 *The flat prior.* As $\tau \to \infty$ in Computation 1.1, the posterior becomes $\mathcal{N}(\hat{w}, S_{xx}^{-1})$ and the predictive $\mathcal{N}(\hat{w}^\top x, 1 + s(x))$ with $s(x) = x^\top S_{xx}^{-1} x$. (With the uniform prior on the square, the posterior is this Gaussian restricted to the square, which by Computation 1.3 removes an exponentially small mass.)

2 *Average over the noise.* Given the inputs, $\hat{w} - w_0 = S_{xx}^{-1} \sum_i x_i \varepsilon_i$ has covariance S_{xx}^{-1}, so the conditional mean of $(x^\top(\hat{w} - w_0))^2$ is $s(x)$, which cancels the second term of (1.2):

$$\mathbb{E}\big[G_n - S \;\big|\; x_1, \dots, x_n\big] = \tfrac{1}{2}\mathbb{E}_x \log\big(1 + x^\top S_{xx}^{-1} x\big).$$

3 *Average over the inputs.* By Lemma 1.2(iii), $\log(1 + x^\top S_{xx}^{-1} x)$ has the law of $\log(U + V) - \log V$ with $U + V \sim \chi^2_{n+1}$. Since $\mathbb{E} \log \chi^2_k = \psi(k/2) + \log 2$, with $\psi = \Gamma'/\Gamma$,

$$\mathbb{E}[G_n] - S = \frac{1}{2}\Big(\psi\big(\tfrac{n+1}{2}\big) - \psi\big(\tfrac{n-d+1}{2}\big)\Big). \tag{1.3}$$

4 *The regular plane.* For $d = 2$, the recurrence $\psi(z+1) = \psi(z) + 1/z$ at $z = \frac{n-1}{2}$ gives $\mathbb{E}[G_n] - S = \frac{1}{n-1}$ exactly, for every $n \geq 2$ and every truth w_0. For the regular line, (1.3) is $\frac{1}{2n} + \frac{1}{4n^2} + O(n^{-3})$. The plug-in predictive $\mathcal{N}(\hat{w}^\top x, 1)$ does worse: its error $K(\hat{w}) = \frac{1}{2}|\hat{w} - w_0|^2$ has mean $\frac{d}{2(n-d-1)}$ for $n \geq d + 2$ by Lemma 1.2(ii), which is $\frac{1}{n-3}$ for the plane.

COMPUTATION 1.3 · THE UNIFORM PRIOR: LAPLACE'S ERROR IS A BOX PROBABILITY

1 *The free energy.* With the uniform prior on $[-1, 1]^d$, of density 2^{-d}, step 1 of Computation 1.1 gives, exactly,

$$F_n = nL_n(\hat{w}) - \frac{d}{2}\log(2\pi) + \frac{1}{2}\log\det S_{xx} + d\log 2 - \log \Pi_n, \tag{1.4}$$

where Π_n is the probability that a $\mathcal{N}(\hat{w}, S_{xx}^{-1})$ vector lies in the box.

2 *The comparison.* The Laplace approximation of Theorem 1.4 below, with the observed information $J_n = S_{xx}/n$ and $\varphi(\hat{w}) = 2^{-d}$, is the same expression without $-\log \Pi_n$. Its error is exactly $-\log \Pi_n$, which lies between $1 - \Pi_n$ and $(1 - \Pi_n)/\Pi_n$.

3 *The boundary is far.* Coordinate j of the Gaussian leaves $[-1, 1]$ with probability at most $2\bar{\Phi}((1 - |\hat{w}_j|)/\sigma_j)$, where $\sigma_j^2 = (S_{xx}^{-1})_{jj} \approx \frac{1}{n}$ and $\bar{\Phi}$ is the standard normal tail. At the truth 0, $\hat{w} = O_p(n^{-1/2})$, so the boundary is $\sqrt{n} + O_p(1)$ posterior standard deviations away and $1 - \Pi_n = e^{-n/2 + O_p(\sqrt{n})}$: for the plane at $n = 100$ its median over data sets is $3 \cdot 10^{-18}$. When the box does cut the posterior the error is not small: its mean is 0.15 at $n = 10$, and 0.36 at $n = 100$ for the truth $(0.9, 0)$ (Exercise 1.11).

WHERE $\frac{d}{2}$ COMES FROM. The coefficient of $\log n$ is a scaling exponent of volume. At the truth 0, $K(w) = \frac{1}{2}|w|^2$, and the prior mass of the parameters within ε of the truth's loss is the area of a disc of radius $\sqrt{2\varepsilon}$ times the density $\frac{1}{4}$: $V(\varepsilon) = \frac{\pi}{2}\varepsilon$ for $\varepsilon \leq \frac{1}{2}$. By Fubini's theorem

$\int \varphi e^{-nK} = \int_0^\infty n e^{-nt} V(t)\,\mathrm{d}t = \frac{\pi}{2n} + O(e^{-n/2})$, so the population free energy is $\bar{F}(n) = -\log \int \varphi e^{-nK} = \log n - \log \frac{\pi}{2} + O(e^{-n/2})$. Each direction in which K grows quadratically contributes a factor $\varepsilon^{1/2}$ to the volume and $\frac{1}{2}\log n$ to the free energy. This exponent is the *learning coefficient* λ, which algebraic geometers call the real log canonical threshold of K (the same number; Chapter 4): $\lambda = 1 = \frac{d}{2}$ for the plane, with multiplicity $m = 1$. For the product model, $K = \frac{1}{2}a^2b^2$ gives $V(\varepsilon) = \sqrt{2\varepsilon}\,(1 + \frac{1}{2}\log\frac{1}{2\varepsilon})$: the exponent of a single parameter, and a logarithm.

1.3 The theorem: Laplace's approximation of the free energy

The regular plane is typical: for every regular truth the energy is quadratic where it matters, and the free energy is a Gaussian integral up to a vanishing error. We write $\ell(x,w) = \log p(x \mid w)$, $|\cdot|$ for the Euclidean norm, and $\|\cdot\|$ for the operator norm.

Assumption 1.3 (Regularity). The data $X_1, X_2, \ldots$ are independent with law q, and:

(R1) $W \subset \mathbb{R}^d$ is compact, φ is a probability density on W, and for q-almost every x the map $w \mapsto \ell(x,w)$ is continuous on W.

(R2) L is finite and continuous on W and has a unique minimizer w_0, which lies in the interior of W.

(R3) $\sup_{w\in W}|L_n(w) - L(w)| \to 0$ in probability.

(R4) For some $r_0 > 0$ the ball $U = \{w : |w - w_0| < r_0\}$ lies in W, and for q-almost every x the map $w \mapsto \ell(x,w)$ is three times continuously differentiable on U, with all partial derivatives of orders one to three bounded on U by $M(x)$, where $\mathbb{E}_q[M(X)] < \infty$.

(R5) The Hessian $J = \nabla^2 L(w_0)$ is positive definite.

(R6) φ is continuous at w_0 and $\varphi(w_0) > 0$.

(R7) (For Corollary 1.6(ii) only.) $\mathbb{E}_q|\nabla\ell(X,w_0)|^2 < \infty$.

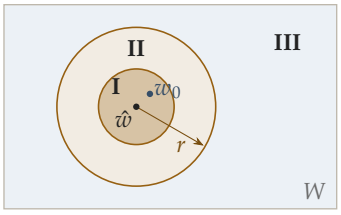
Conditions (R2) and (R5) say that the truth is regular, and (R3), a *uniform law of large numbers*, holds whenever $w \mapsto \ell(x,w)$ is continuous and $\mathbb{E}_q \sup_W|\ell(X,w)| < \infty$ (Exercise 1.1). Under (R4), dominated convergence allows us to differentiate under the expectation on U, so $\nabla L(w_0) = -\mathbb{E}\nabla\ell(X,w_0) = 0$ and $J = -\mathbb{E}\nabla^2\ell(X,w_0)$.[4] A minimizer $\hat{w}$ of L_n over W exists by (R1); when there are several, every statement holds for each.[5]

Theorem 1.4 (The Laplace approximation of the free energy). *Under (R1)–(R6), every minimizer $\hat{w}$ of L_n over W satisfies*

$$F_n = nL_n(\hat{w}) + \frac{d}{2}\log\frac{n}{2\pi} + \frac{1}{2}\log\det J - \log\varphi(w_0) + o_p(1), \qquad (1.5)$$

[4] Statisticians call J the expected and $J_n = \nabla^2 L_n(\hat{w})$ the observed information. In the realizable case the information identity $\mathbb{E}[\nabla\ell\,\nabla\ell^{\mathsf{T}}] = -\mathbb{E}[\nabla^2\ell]$ makes J the Fisher information.

[5] We ignore measurability: L_n is continuous, so the infima over compact sets that define our events are measurable.



I $|w-\hat{w}| < \delta_n$: a Gaussian integral
II $\delta_n \le |w-\hat{w}| < r$: a Gaussian tail
III $|w-\hat{w}| \ge r$: at most $e^{-n\eta}$

Figure 1.1. The three regions of the proof of Theorem 1.4, with $\delta_n = n^{-2/5}$ and a fixed radius r.

and the same holds with J replaced by the observed information $J_n = \nabla^2 L_n(\hat{w})$ *and $\varphi(w_0)$ by $\varphi(\hat{w})$.*

The proof cuts W into the three regions of Figure 1.1, and it needs to know how L_n behaves near its minimum.[6]

Lemma 1.5 (The empirical loss near its minimum). *Assume (R1)–(R5), let $\hat{w}$ minimize L_n over W, and let $\mu > 0$ be the smallest eigenvalue of J. There is $r_1 > 0$ such that for every $r \in (0, r_1]$ there are $\eta > 0$ and events Ω_n with $\mathbb{P}(\Omega_n) \to 1$ on which:*

(i) $|\hat{w} - w_0| < r/2$ *and* $\nabla L_n(\hat{w}) = 0$;

(ii) $L_n(w) - L_n(\hat{w}) \ge \eta$ *for every* $w \in W$ *with* $|w-\hat{w}| \ge r$;

(iii) the eigenvalues of $J_n = \nabla^2 L_n(\hat{w})$ *are at least* $\mu/2$, *and for* $|w-\hat{w}| < r$ *we have* $L_n(w) - L_n(\hat{w}) \ge \frac{\mu}{4}|w-\hat{w}|^2$ *and*

$$\left| L_n(w) - L_n(\hat{w}) - \tfrac{1}{2}(w-\hat{w})^{\top} J_n (w-\hat{w}) \right| \le C|w-\hat{w}|^3,$$

with the constant $C = \frac{1}{6} d^{3/2}(\mathbb{E}M + 1)$.

Moreover $\hat{w} \to w_0$ and $J_n \to J$ in probability, and $\sup_{|w-w_0|\le r_n} \|\nabla^2 L_n(w) - J\| \to 0$ *in probability whenever $r_n \to 0$ in probability.*

Proof. The plan is to get (i) and (ii) from uniform convergence and identifiability, and (iii) from Taylor's theorem with the envelope M. Let $\bar{M}_n = \frac{1}{n}\sum_i M(X_i)$, which tends to $\mathbb{E}M$ almost surely. On the convex set U the third derivatives of L_n are bounded by $\bar{M}_n$, so each entry of $\nabla^2 L_n$ changes by at most $\sqrt{d}\,\bar{M}_n|w'-w|$ between points w, w' of U; as the operator norm is at most d times the largest entry,

$$\|\nabla^2 L_n(w') - \nabla^2 L_n(w)\| \le d^{3/2}\bar{M}_n |w'-w| \qquad (w, w' \in U). \tag{1.6}$$

Taylor's theorem with the Lagrange remainder bounds the remainder of order three of L_n at w by $\frac{1}{6}\bar{M}_n(\sum_j |w'_j - w_j|)^3 \le \frac{1}{6} d^{3/2}\bar{M}_n |w'-w|^3$. By the strong law, whose hypothesis is (R4), $\nabla^2 L_n(w_0) = -\frac{1}{n}\sum_i \nabla^2 \ell(X_i, w_0) \to J$ almost surely.

Choose $r_1 \le r_0/2$ with $2d^{3/2}(\mathbb{E}M+1)r_1 \le \mu/4$, and let $r \le r_1$. On the event $\{\bar{M}_n \le \mathbb{E}M + 1\} \cap \{\|\nabla^2 L_n(w_0) - J\| \le \mu/4\}$, (1.6) gives $\|\nabla^2 L_n(w) - J\| \le \mu/2$ on $B(w_0, 2r)$, so every eigenvalue of $\nabla^2 L_n$ there is at least $\mu/2$. The set

[6] The inner radius $\delta_n = n^{-a}$ needs $n\delta_n^2 \to \infty$, so that the inner ball holds the whole Gaussian, and $n\delta_n^3 \to 0$, so that the cubic term is negligible on it: every a strictly between $\frac{1}{3}$ and $\frac{1}{2}$ works, and we take $a = \frac{2}{5}$.

$W_r = \{w \in W : |w - w_0| \geq r/2\}$ is compact and, by (R2), L has on it a minimum larger than $L(w_0)$, which we write $L(w_0) + 3\eta$. Let Ω_n be the intersection of the event above with $\{\sup_W |L_n - L| < \eta\}$; then $\mathbb{P}(\Omega_n) \to 1$ by (R3). On Ω_n, for every $w \in W_r$,

$$L_n(w) > L(w) - \eta \geq L(w_0) + 2\eta > L_n(w_0) + \eta \geq L_n(\hat{w}) + \eta.$$

So $\hat{w} \notin W_r$, that is, $|\hat{w} - w_0| < r/2$; then $\hat{w}$ is interior to U and $\nabla L_n(\hat{w}) = 0$, which is (i). If $|w - \hat{w}| \geq r$, then $w \in W_r$, and the display gives (ii). For (iii), let $|w - \hat{w}| < r$, so that the segment from $\hat{w}$ to w lies in $B(w_0, 2r)$, and let $\tilde{H} = \int_0^1 (1-t)\nabla^2 L_n(\hat{w} + t(w - \hat{w}))\,\mathrm{d}t$, whose eigenvalues are at least $\frac{1}{2} \cdot \frac{\mu}{2}$. Taylor's theorem with integral remainder and $\nabla L_n(\hat{w}) = 0$ give

$$L_n(w) - L_n(\hat{w}) = (w - \hat{w})^\top \tilde{H} (w - \hat{w}) \geq \frac{\mu}{4} |w - \hat{w}|^2,$$

and the bound on the remainder of order three at $\hat{w}$, with $\bar{M}_n \leq \mathbb{E}M + 1$, gives the second inequality. Finally, (i) for every $r \leq r_1$ gives $\hat{w} \to w_0$; on $|w - w_0| \leq r_n < r_0$, (1.6) gives $\|\nabla^2 L_n(w) - J\| \leq \|\nabla^2 L_n(w_0) - J\| + d^{3/2}\bar{M}_n r_n \to 0$, and $r_n = |\hat{w} - w_0|$ gives $J_n \to J$. ∎

Proof of Theorem 1.4. The plan is to cut W into the three regions of Figure 1.1 and to show that the integral

$$Z_n = \int_W \varphi(w) e^{-n(L_n(w) - L_n(\hat{w}))}\,\mathrm{d}w$$

satisfies $n^{d/2} Z_n \to (2\pi)^{d/2} \varphi(w_0) \det(J)^{-1/2}$ in probability. Since $F_n = nL_n(\hat{w}) - \log Z_n$ and the limit is positive, taking logarithms then gives (1.5).

By (R6) there is $r_\varphi > 0$ with $\varphi \leq 2\varphi(w_0)$ on $B(w_0, r_\varphi)$. Fix $r \leq \min(r_1, r_\varphi/2)$ in Lemma 1.5, with its η, C and Ω_n, and let $\delta_n = n^{-2/5}$, so that $n\delta_n^2 = n^{1/5} \to \infty$ and $n\delta_n^3 = n^{-1/5} \to 0$. Let $\xi \sim \mathcal{N}(0, I_d)$. We work on Ω_n, with n so large that $\delta_n < r$.

Region III, $|w - \hat{w}| \geq r$. By Lemma 1.5(ii) the integrand is at most $\varphi(w) e^{-n\eta}$, so this part of Z_n is at most $e^{-n\eta} = o(n^{-d/2})$.

Region II, $\delta_n \leq |w - \hat{w}| < r$. Here $|w - w_0| < \frac{3}{2} r < r_\varphi$, and Lemma 1.5(iii) bounds the integrand by $2\varphi(w_0) e^{-n\mu|w - \hat{w}|^2/4}$, which is a Gaussian density of covariance $\frac{2}{n\mu} I$ times $2\varphi(w_0)(4\pi/n\mu)^{d/2}$. So this part of Z_n is at most $2\varphi(w_0)(4\pi/n\mu)^{d/2} \mathbb{P}(|\xi| \geq \delta_n \sqrt{n\mu/2})$, which is $o(n^{-d/2})$ because $\delta_n \sqrt{n} \to \infty$.

Region I, $|w - \hat{w}| < \delta_n$. By Lemma 1.5(iii), $n(L_n(w) - L_n(\hat{w}))$ differs from $\frac{n}{2}(w - \hat{w})^\top J_n (w - \hat{w})$ by at most $\rho_n = Cn\delta_n^3 \to 0$. Let ω_n be the largest value of $|\varphi(v) - \varphi(w_0)|$ for $|v - w_0| \leq \delta_n + |\hat{w} - w_0|$; it tends to zero in probability by (R6), since $\hat{w} \to w_0$. So this part of Z_n lies between $(\varphi(w_0) \mp \omega_n) e^{\mp \rho_n} \gamma_n$, where by the Gaussian integral

$$\gamma_n = \int_{|u| < \delta_n} e^{-nu^\top J_n u/2}\,\mathrm{d}u = \left(\frac{2\pi}{n}\right)^{d/2} \frac{1 - \mathbb{P}\big(|J_n^{-1/2}\xi| \geq \delta_n \sqrt{n}\big)}{\det(J_n)^{1/2}}.$$

The eigenvalues of J_n are at least $\mu/2$, so $|J_n^{-1/2}\xi| \leq |\xi|\sqrt{2/\mu}$ and the probability tends to zero; with $J_n \to J$, $n^{d/2}\gamma_n \to (2\pi)^{d/2} \det(J)^{-1/2}$ in probability.

Adding the three parts gives the plan's limit. For the second form of the theorem, $\log\det J_n \to \log\det J$ and $\log\varphi(\hat{w}) \to \log\varphi(w_0)$ in probability, by Lemma 1.5 and (R6). ∎

Every hypothesis was used once: compactness (R1) gave $\hat{w}$; uniform convergence and identifiability (R2, R3) disposed of region III; smoothness with an integrable envelope (R4) made the energy quadratic; the invertible Hessian (R5) made the quadratic a Gaussian; the interior truth kept the ball $B(w_0, 2r)$, and with it the Gaussian, inside W; and the prior (R6) produced $\varphi(w_0)$. The theorem is Laplace's method for a fixed function (Theorem A.1), made random; the lemma controls how L_n changes with n. If $\varphi(w_0) = 0$, the right side of (1.5) is infinite and the free energy grows faster than $\frac{d}{2}\log n$; if w_0 is on the boundary, region I is cut and a random term survives (Exercise 1.8). The factor $\varphi(\hat{w})(2\pi/n)^{d/2}\det(J_n)^{-1/2}$, the prior density at the fit times the volume of the posterior, multiplies the best-fit likelihood $e^{-nL_n(\hat{w})}$ in the evidence; MacKay calls it the *Occam factor* (MacKay 2003, Chapter 28). For a regular truth it shrinks like $n^{-1/2}$ per parameter.[7]

[7] For one parameter with a uniform prior of width σ_w, MacKay's Occam factor is $\sigma_{w|D}/\sigma_w$, the posterior width over the prior width.

Corollary 1.6 (BIC and the regular answer). *Assume (R1)–(R6).*

(i) With $\mathrm{BIC} = nL_n(\hat{w}) + \frac{d}{2}\log n$, $F_n - \mathrm{BIC} \to \frac{1}{2}\log\det J - \log\varphi(w_0) - \frac{d}{2}\log(2\pi)$ *in probability.*

(ii) If also (R7) holds, let $\mathcal{I} = \mathbb{E}_q[\nabla\ell(X, w_0)\nabla\ell(X, w_0)^\top]$ *and* $Z \sim \mathcal{N}(0, \mathcal{I})$. *Then* $\sqrt{n}(\hat{w} - w_0) \to \mathcal{N}(0, J^{-1}\mathcal{I}J^{-1})$ *and* $n(L_n(w_0) - L_n(\hat{w})) \to \frac{1}{2}Z^\top J^{-1}Z$ *in law, and, also in law,*

$$F_n - nL_n(w_0) - \tfrac{d}{2}\log\tfrac{n}{2\pi} - \tfrac{1}{2}\log\det J + \log\varphi(w_0) \to -\tfrac{1}{2}Z^\top J^{-1}Z.$$

(iii) If moreover the truth is realizable and $\mathcal{I} = J$, *then* $Z^\top J^{-1}Z \sim \chi^2_d$, $nL_n(w_0) = nS_n$, *and* $F_n - nS_n = \frac{d}{2}\log n + O_p(1)$.

Proof. Part (i) is Theorem 1.4 rearranged. For (ii) the plan is to linearize ∇L_n around w_0 and then expand L_n around $\hat{w}$. By (R4) the score has mean $\mathbb{E}\nabla\ell(X, w_0) = -\nabla L(w_0) = 0$, so by (R7) and the central limit theorem $\sqrt{n}\,\nabla L_n(w_0) \to \mathcal{N}(0, \mathcal{I})$ in law. On Ω_n, $\nabla L_n(\hat{w}) = 0$, and the fundamental theorem of calculus gives $0 = \nabla L_n(w_0) + \bar{H}_n(\hat{w} - w_0)$ with $\bar{H}_n = \int_0^1 \nabla^2 L_n(w_0 + t(\hat{w} - w_0))\,\mathrm{d}t \to J$ in probability (Lemma 1.5). By Slutsky's lemma (Definition A.13), $\sqrt{n}(\hat{w} - w_0) = -\bar{H}_n^{-1}\sqrt{n}\,\nabla L_n(w_0)$ converges in law to $J^{-1}Z$. Taylor's theorem around $\hat{w}$ gives $L_n(w_0) - L_n(\hat{w}) = (w_0 - \hat{w})^\top\tilde{H}_n(w_0 - \hat{w})$ with $\tilde{H}_n = \int_0^1 (1-t)\nabla^2 L_n(\hat{w} + t(w_0 - \hat{w}))\,\mathrm{d}t \to \frac{1}{2}J$, so $n(L_n(w_0) - L_n(\hat{w}))$ converges in law to $\frac{1}{2}(J^{-1}Z)^\top J(J^{-1}Z) = \frac{1}{2}Z^\top J^{-1}Z$; subtracting it from (1.5) gives the last statement. For (iii), $J^{-1/2}Z \sim \mathcal{N}(0, I_d)$ when $\mathcal{I} = J$. ∎

The limit in (ii) is random: $F_n - nS_n - \frac{d}{2}\log n$ converges in law, not in probability, and its fluctuation is the likelihood ratio of the fitted model against the truth.[8] Every free energy theorem of the book has this form, with $\frac{d}{2}$ replaced by the learning coefficient (Theorems 3.4 and 8.6). Taking

[8] In the realizable case with $\mathcal{I} = J$, part (ii) contains Wilks's theorem for a simple hypothesis: twice the log-likelihood ratio, $2n(L_n(w_0) - L_n(\hat{w}))$, tends to χ^2_d.

expectations in (iii), and granting the uniform integrability that convergence in law does not supply,

$$\mathbb{E}[F_n - nS_n] = \frac{d}{2}\log\frac{n}{2\pi e} + \frac{1}{2}\log\det J - \log\varphi(w_0) + o(1), \qquad (1.7)$$

which is $\log n - \log\frac{\pi e}{2}$ for the regular plane, the slate dashed line of Figure P.2.

THE POSTERIOR AND THE CHOICE BETWEEN MODELS. The same three regions show that the posterior law of $\sqrt{n}(w - \hat{w})$ converges in total variation to $\mathcal{N}(0, J^{-1})$, in probability (Exercise 1.2). The classical statement asks less. In the *Bernstein–von Mises theorem* of van der Vaart (1998, Theorem 10.1), the data are independent with law P_{θ_0}, θ_0 an interior point of $\Theta \subset \mathbb{R}^d$, and the model is dominated and differentiable in quadratic mean at θ_0 with nonsingular Fisher information I_{θ_0}; for every $\varepsilon > 0$ there are tests ϕ_n with $P^n_{\theta_0}\phi_n \to 0$ and $\sup_{|\theta-\theta_0|\geq\varepsilon} P^n_\theta(1-\phi_n) \to 0$; and the prior has a Lebesgue density continuous in a neighborhood of θ_0 and positive at θ_0. Then the posterior law of $\sqrt{n}(\theta - \theta_0)$ is within $o_p(1)$ in total variation of $\mathcal{N}(\Delta_n, I^{-1}_{\theta_0})$, with $\Delta_n = I^{-1}_{\theta_0} n^{-1/2}\sum_i \dot{\ell}_{\theta_0}(X_i)$ the normalized score. The hypothesis that fails in a singular model is the nonsingular information: wherever the true parameters form a curve the Fisher information is singular (Proposition 2.1), and the posterior of the product model spreads uniformly over scales (Theorem 2.5).

Schwarz (1978) derived his criterion as the leading terms of the asymptotic expansion of the Bayes solution to the problem of choosing among models of different dimensions, terms that do not depend on the prior.[9] His setting, as Haughton (1988) describes it, was independent observations from an exponential family with models that are affine subspaces of the natural parameter space. Haughton extended the expansion, with its next term, to models that are smooth curved submanifolds, and proved that the criterion made of the three leading terms selects the true model with probability tending to one. Under our hypotheses Corollary 1.6 gives the consistency of Bayesian model choice and of BIC: a model with a larger population loss pays order n more free energy, and of two models that contain the truth the one with $d_2 - d_1$ more parameters pays $\frac{d_2-d_1}{2}\log n + O_p(1)$ more (Exercise 1.6). Akaike's criterion, on the book's per-sample scale AIC $= L_n(\hat{w}) + \frac{d}{n}$, is not consistent: between nested true models $2n(\mathrm{AIC}^{(1)} - \mathrm{AIC}^{(2)})$ is the likelihood-ratio statistic minus $2(d_2 - d_1)$, the statistic tends to $\chi^2_{d_2-d_1}$ in law (Wilks 1938), and AIC picks the larger model with probability tending to $\mathbb{P}(\chi^2_{d_2-d_1} > 2(d_2 - d_1))$, 0.157 for one extra parameter. AIC is built for prediction, the subject of the next section.

[9] Schwarz's paper fills four pages of the *Annals of Statistics*.

1.4 The theorem: the generalization error is $\frac{d}{2n}$

Theorem 1.7 (The generalization error of a regular model). *(i) Let $x \sim \mathcal{N}(0, I_d)$ and $y = w_0^\top x + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0,1)$ independent of x and $w_0 \in \mathbb{R}^d$, and let the model be $y = w^\top x + \mathcal{N}(0,1)$ with the prior*

$\mathcal{N}(0, \tau^2 I_d)$, $0 < \tau < \infty$. *Then* $n(\mathbb{E}[G_n] - S) \to \frac{d}{2}$.

(ii) Let the truth be realizable by a model that satisfies the conditions (A.1)–(A.3) of Watanabe (2010): W is compact and defined by analytic inequalities, the prior is a smooth positive function times a nonnegative analytic one, and $w \mapsto \log(q(x)/p(x \mid w))$ extends to an analytic map from a complex neighborhood of W into $L^s(q)$, $s \geq 6$, with moment conditions on its supremum. If the truth is regular and $\varphi(w_0) > 0$, then $\mathbb{E}[G_n] - S = \frac{d}{2n} + o(\frac{1}{n})$.

Proof of (i). The plan is to compute the expectation over the noise exactly, as in Computation 1.2, and to pass to the limit over the inputs with Lemma 1.2. Write $\kappa = \tau^{-2}$, $P_n = S_{xx} + \kappa I$, $s(x) = x^\top P_n^{-1} x$, so that (1.2) holds with $m_n = P_n^{-1} S_{xy}$.

The noise. Since $P_n^{-1} S_{xx} = I - \kappa P_n^{-1}$, we have $m_n - w_0 = P_n^{-1} \sum_i x_i \varepsilon_i - \kappa P_n^{-1} w_0$, whose second-moment matrix given the inputs is $P_n^{-1} - \kappa P_n^{-2} + \kappa^2 P_n^{-1} w_0 w_0^\top P_n^{-1}$. In (1.2) the term $s(x)$ then cancels as in Computation 1.2, and

$$\mathbb{E}\left[G_n - S \;\middle|\; x_1, \dots, x_n\right] = \tfrac{1}{2}\mathbb{E}_x \log\bigl(1 + s(x)\bigr) + \mathbb{E}_x\left[\frac{\kappa^2 (x^\top P_n^{-1} w_0)^2 - \kappa |P_n^{-1} x|^2}{2(1 + s(x))}\right].$$

Bounds. For $s \geq 0$, $s - \frac{s^2}{2} \leq \log(1+s) \leq s$. For a positive semidefinite B, $\mathbb{E}_x[x^\top B x] = \operatorname{tr} B$ and $\mathbb{E}_x[(x^\top B x)^2] = (\operatorname{tr} B)^2 + 2\operatorname{tr}(B^2) \leq 3(\operatorname{tr} B)^2$ (diagonalize B and use $\mathbb{E} Z^4 = 3$, Lemma A.2). With $B = P_n^{-1}$, $(x^\top P_n^{-1} w_0)^2 \leq |P_n^{-1} x|^2 |w_0|^2$ and $\mathbb{E}_x |P_n^{-1} x|^2 = \operatorname{tr}(P_n^{-2}) \leq (\operatorname{tr} P_n^{-1})^2$, these give

$$\left|\mathbb{E}\left[G_n - S \;\middle|\; x_1, \dots, x_n\right] - \tfrac{1}{2} \operatorname{tr} P_n^{-1}\right| \leq c\, (\operatorname{tr} P_n^{-1})^2, \qquad c = \tfrac{3}{4} + \tfrac{1}{2}(\kappa^2 |w_0|^2 + \kappa).$$

The limit. It remains to show that $n\mathbb{E}[\operatorname{tr} P_n^{-1}] \to d$ and $n\mathbb{E}[(\operatorname{tr} P_n^{-1})^2] \to 0$. Almost surely $S_{xx}/n \to I$, so $n \operatorname{tr} P_n^{-1} \to d$ and $n(\operatorname{tr} P_n^{-1})^2 \to 0$. Since $P_n \geq S_{xx}$ and $P_n \geq \kappa I$, both are dominated: $0 \leq n \operatorname{tr} P_n^{-1} \leq g_n$ and $0 \leq n(\operatorname{tr} P_n^{-1})^2 \leq \frac{d}{\kappa} g_n$ with $g_n = n \operatorname{tr} S_{xx}^{-1}$, where $g_n \to d$ almost surely and $\mathbb{E} g_n = \frac{nd}{n-d-1} \to d$ by Lemma 1.2(ii). If $0 \leq f_n \leq g_n$, $f_n \to f$ and $g_n \to g$ almost surely, and $\mathbb{E} g_n \to \mathbb{E} g < \infty$, then $\mathbb{E} f_n \to \mathbb{E} f$, by Fatou's lemma applied to f_n and to $g_n - f_n$. Applied twice, this gives the two limits. ∎

Proof sketch of (ii). Under (A.1)–(A.3), Watanabe (2010) proves that $n\mathbb{E}[G_n - S]$ converges (Theorem 1) and that the limit is the learning coefficient λ for the posterior at inverse temperature 1 (Corollary 2); for a regular truth with w_0 interior and $\varphi(w_0) > 0$, $\lambda = \frac{d}{2}$ (Watanabe 2013, Lemma 3). This is Theorem 9.9 with $\lambda = \frac{d}{2}$. The mechanism, as a heuristic: let $\dot\ell = \nabla \ell(\cdot, w_0)$, $\ddot\ell = \nabla^2 \ell(\cdot, w_0)$, and let the posterior mean and covariance of $u = w - w_0$ be $\mu_n = O_p(n^{-1/2})$ and $\Sigma_n = O_p(n^{-1})$. Expanding $e^{\ell(x,w) - \ell(x,w_0)}$ to second order and averaging over the posterior, $p^*/q = 1 + r$ with $r = \mu_n^\top \dot\ell + \frac{1}{2}\operatorname{tr}[(\ddot\ell + \dot\ell \dot\ell^\top)(\Sigma_n + \mu_n \mu_n^\top)] + O_p(n^{-3/2})$. Then $G_n - S = -\mathbb{E}_q \log(1 + r) = -\mathbb{E}_q r + \frac{1}{2}\mathbb{E}_q r^2 + O_p(n^{-3/2})$, where $\mathbb{E}_q r = 0$ exactly, as p^* and q are densities, and $\mathbb{E}_q r^2 = \mu_n^\top J \mu_n + O_p(n^{-3/2})$, as $\mathbb{E}_q[\dot\ell \dot\ell^\top] = J$. The posterior spread has dropped out: $G_n - S = \frac{1}{2}\mu_n^\top J \mu_n + O_p(n^{-3/2})$, and

since μ_n differs from $\hat{w} - w_0$ by $O_p(\frac{1}{n})$, Corollary 1.6 gives $n(G_n - S) \to \frac{1}{2}\chi^2_d$ in law, of mean $\frac{d}{2}$. Omitted is the control of the remainders and of the posterior's tails in expectation, which Watanabe obtains from his general theory (Watanabe 2009, Chapter 6); Watanabe (2018, Chapter 4) treats the regular case. ■

AKAIKE'S PENALTY. For the least-squares plug-in, step 4 of Computation 1.1 gives $L_n(\hat{w}) - S_n = -\frac{1}{2n}\chi^2_d$, so with Computation 1.2 the training loss underestimates the test loss by $\frac{d}{2(n-d-1)} + \frac{d}{2n} = \frac{d}{n} + O(n^{-2})$. In every regular realizable model the expansions of Corollary 1.6 give $n(L(\hat{w}) - S)$ and $n(S_n - L_n(\hat{w}))$ the same limit $\frac{1}{2}\chi^2_d$, so the gap is $\frac{d}{n}$ to first order, granting uniform integrability. Correcting for it is the criterion AIC $= L_n(\hat{w}) + \frac{d}{n}$, which Akaike (1998) introduced in 1973 as an estimate of the expected log-likelihood of the fitted model.[10] The Bayes predictive has the same gap between its generalization and training losses (Watanabe 2010), and WAIC estimates it without counting parameters (Chapter 10).

[10] Software reports AIC on the deviance scale, $-2\log(\text{maximum likelihood}) + 2d$, which is $2n$ times the book's per-sample value.

THE FREE ENERGY AND THE GENERALIZATION ERROR ARE ONE FACT. The predictive density p^* of the first n observations satisfies $F_{n+1} - F_n = -\log p^*(X_{n+1})$, the ratio of the two marginal likelihoods; taking expectations, $\mathbb{E}[G_n] = \mathbb{E}[F_{n+1}] - \mathbb{E}[F_n]$, and summing, $\mathbb{E}[F_n - nS_n] = \sum_{k<n}(\mathbb{E}[G_k] - S)$. So $\frac{d}{2n}$ is the increment of $\frac{d}{2}\log n$, although an expansion with an $o(1)$ error cannot be differenced to prove it, which is why Theorem 1.7 needs its own proof. Clarke and Barron (1990) proved the cumulative statement: the relative entropy between the distribution of n observations and the Bayes mixture, which is $\mathbb{E}[F_n - nS_n]$, is $\frac{d}{2}\log n + c + o(1)$, with a constant c that they identify and that depends only on the prior density and the Fisher information at the truth; (1.7) has exactly this form.

1.5 Measuring it

Two scripts measure the two promises on three regular models. For the regular line and the regular plane, at the truth 0 with the uniform prior on $[-1, 1]^d$, every quantity of Section 1.2 is exact, (R1)–(R7) hold because the log-likelihood is quadratic in w, and the measurement checks those computations and the limits of Corollary 1.6 and Theorem 1.7. The third model tests the theorem where the Laplace approximation is not exact: the normal mixture $(1-a)\mathcal{N}(0,1) + a\mathcal{N}(b,1)$ of Chapter 10, at the truth $(\frac{1}{2}, 2)$, with the prior uniform on $[0,1] \times [-4,4]$ (a wider range of b than the running one, to keep the posterior inside the box). At this truth the mixture is regular: distinct Gaussian translates are linearly independent, so only $(\frac{1}{2}, 2)$ gives the truth; quadrature gives $J = \left(\begin{smallmatrix} 2.20 & 0.45 \\ 0.45 & 0.32 \end{smallmatrix}\right)$, with eigenvalues 0.22 and 2.30; and near $(\frac{1}{2}, 2)$ the derivatives of the log-likelihood up to order three are bounded by a multiple of $(1 + |x|)^3$. At the truth $\mathcal{N}(0,1)$ of Chapter 10 the same model is singular, since every parameter with $a = 0$ or $b = 0$ is then true: regularity belongs to the truth.

THE FREE ENERGY. The script `01-regular-free.py` computes F_n by quadrature, for 200 data sets at nine sizes from 10^2 to 10^6 for the linear models and

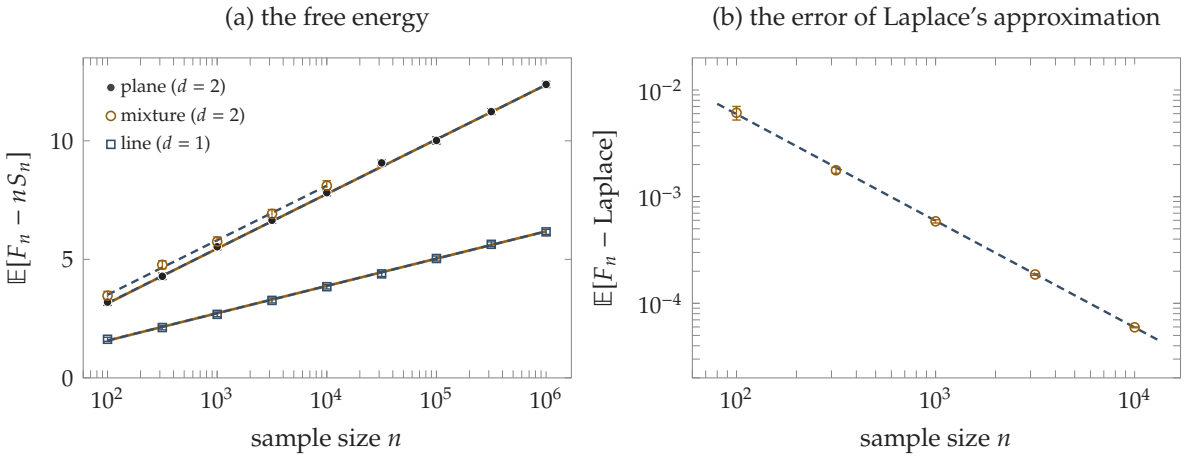


Figure 1.2. The free energy of three regular models, by quadrature. (a) The mean of $F_n - nS_n$ for the regular line (squares) and the regular plane (dots), 200 data sets per point, and for the mixture $(1-a)\mathcal{N}(0,1) + a\mathcal{N}(b,1)$ at its regular truth $(\frac{1}{2}, 2)$ (circles), 100 data sets per point, with Student-t 95% intervals. Dashed: the regular prediction (1.7), which is $\frac{1}{2}\log\frac{n}{2\pi e} + \log 2$, $\log\frac{n}{2\pi e} + \log 4$ and $\log\frac{n}{2\pi e} + \frac{1}{2}\log\det J + \log 8$; the singular theory predicts the same lines, since $\lambda = \frac{d}{2}$ and $m = 1$. Solid, under the dashed lines: the exact means of the linear models, from the Wishart expectation of $\log\det S_{xx}$, within 0.016 of the dashed lines. (b) The mean of F_n minus its Laplace approximation with the observed information, for the mixture, with 95% intervals; dashed: the second-order prediction $0.594/n$. Seeds are fixed in `figures/scripts/01-regular-free.py`.

for 100 data sets at five sizes up to 10^4 for the mixture (on a 20×20 grid in coordinates whitened by the observed information, which agrees with finer grids to $3 \cdot 10^{-7}$). In Figure 1.2(a) the least-squares slopes of the means against $\log n$ are 0.499, 0.998 and 0.992, against $\frac{d}{2} = \frac{1}{2}$, 1 and 1, and 22 of the 23 intervals contain the prediction (1.7); the miss (the plane at $n = 10^{4.5}$, by 2.6 standard errors) is what one should expect of one interval in twenty. For the linear models, F_n minus the Laplace approximation is $-\log\Pi_n$, at most $3 \cdot 10^{-9}$ in these runs. For the mixture the error is real (Figure 1.2(b)): n times its mean is 0.61, 0.56, 0.59, 0.59 and 0.596 at $n = 10^2, 10^{2.5}, \ldots, 10^4$, against 0.594 from the next term of Laplace's method (Exercise 1.5). The theorem's $o_p(1)$ is an $O(1/n)$ here, with the predicted coefficient.

THE GENERALIZATION ERROR. The script `01-regular-gen.py` averages $n(G_n - S)$ over 2000 data sets per size for the linear models and over 200 for the mixture. For the linear models it draws (S_{xx}, S_{xy}) from their exact joint law (Wishart, and Gaussian given S_{xx}; at $n = 100$ this agrees with data drawn point by point, 1.009 ± 0.043 against 1.048 ± 0.044, exact 1.010) and evaluates (1.2) by Gauss–Hermite quadrature, neglecting the posterior mass outside the box, at most $2 \cdot 10^{-8}$. For the mixture it integrates $q\log(q/p^*)$ with the predictive p^* of the whitened grid. In Figure 1.3 all 18 intervals of the linear models contain the exact values of Computation 1.2; four of the five intervals of the mixture contain 1, and the fifth, 0.87 ± 0.13 at $n = 10^{3.5}$, misses it by 0.006. Each script runs in one to two minutes.

Nothing in these figures departs from the regular theory, and nothing should: they are the reference from which the bronze curves of later chapters

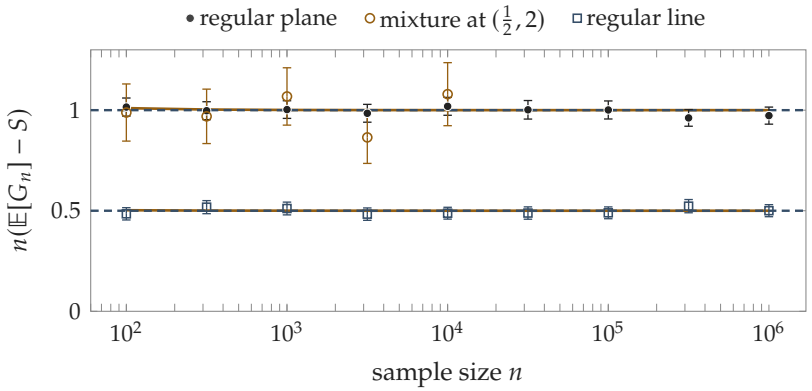


Figure 1.3. The Bayes generalization error times n for the regular line (squares) and plane (dots), 2000 data sets per point, and for the mixture at its regular truth (circles), 200 data sets per point, with Student-t 95% intervals. Dashed: $\frac{d}{2}$ (Theorem 1.7), which is also the singular theory's λ. Solid: the exact values (1.3). Seeds are fixed in `figures/scripts/01-regular-gen.py`.

depart. In Chapter 3 the product model, with the same two parameters as the regular plane, pays $\frac{1}{2}\log n - \log\log n$, and in Chapter 9 the cubic cusp has $n(\mathbb{E}[G_n] - S) \to \frac{1}{6}$.

1.6 The ledger

For a regular truth the classical theory keeps its promises, with explicit constants and complete proofs. The free energy is the Laplace approximation up to $o_p(1)$, and each parameter costs $\frac{1}{2}\log n$ (Theorem 1.4 and Corollary 1.6); the posterior is asymptotically normal; Bayesian model choice and BIC are consistent; and each parameter costs $\frac{1}{2n}$ in the expected generalization error (Theorem 1.7), exactly $\frac{1}{n-1}$ in all for the regular plane with a flat prior. In the language of the rest of the book a regular truth has learning coefficient $\lambda = \frac{d}{2}$ and multiplicity $m = 1$, and every formula of the singular theory reduces to the regular one.

LEDGER · $y = ax_1 + bx_2$

QUANTITY	the free energy $F_n - nS_n$ and the Bayes generalization error $\mathbb{E}[G_n] - S$
REGULAR THEORY	$\frac{d}{2}\log n + O_p(1)$ and $\frac{d}{2n} + o(\frac{1}{n})$: half a unit of $\log n$, and $\frac{1}{2n}$, per parameter (Theorems 1.4 and 1.7)
SINGULAR THEORY	the same, with $\lambda = \frac{d}{2}$ and $m = 1$: a regular truth is the special case
RUNNING EXAMPLE	the regular plane: $\lambda = 1$; $\mathbb{E}[F_n - nS_n] = \log n - \log\frac{\pi e}{2} + o(1)$; $\mathbb{E}[G_n] - S = \frac{1}{n-1}$ exactly with a flat prior
MEASURED	slopes 0.499, 0.998, 0.992 of $\mathbb{E}[F_n - nS_n]$ against $\log n$ (line, plane, mixture at a regular truth); $n(\mathbb{E}[G_n] - S)$ at $\frac{d}{2}$; Laplace's error $0.594/n$ on the mixture, as predicted

RESTS ON	a unique true parameter, interior to W, with a positive definite Hessian; smoothness with integrable envelopes; uniform convergence of L_n; a prior positive at the truth; d fixed as $n \to \infty$

1.7 Where the theory stops

WHAT WAS ASSUMED. The theorem needs a single true parameter. Networks have many: permuting the hidden units of a layer, or flipping the signs of a tanh unit's weights, gives finitely many copies of each true parameter, which only lowers the free energy by the logarithm of their number (Exercise 1.3); but rescaling the two sides of a ReLU unit, or switching a unit off, gives curves and higher-dimensional sets of true parameters (Chapter 2), on which the Hessian is singular (Proposition 2.1). The interior truth fails when the truth sits at the edge of the parameter space, such as a variance component equal to zero or a weight at the end of its range; the posterior is then a cut Gaussian and a random term survives in the free energy (Exercise 1.8). ReLU networks are not smooth everywhere, and they are not analytic, as part (ii) of Theorem 1.7 requires. The prior must be positive at the truth; a prior that vanishes there changes the exponent of n (Chapter 5). The posterior is the exact Bayesian posterior, not a training run. And d is fixed while $n \to \infty$, whereas networks are often trained with more parameters than examples.

WHERE IT BREAKS. The classical theory fails in models much simpler than networks, and in networks themselves. For a normal mixture the likelihood-ratio statistic for one component against two does not have the χ^2 limit of Wilks (1938): with the means restricted to $[-M, M]$ its limit is governed by the supremum of a Gaussian process over $[-M, M]$, which tends to infinity with M (Brazzale and Mameli 2024; Hartigan 1985). With the truth on the boundary the limits are typically mixtures of χ^2 laws (Self and Liang 1987). For multilayer networks whose truth is a smaller network the likelihood ratio has a larger order, at least $\log n$ with two redundant hidden units (Fukumizu 2003). On the Bayesian side, Watanabe (2001) proved that for nonidentifiable models such as layered networks the free energy is $\lambda \log n - (m-1) \log \log n$ plus a constant, with 2λ at most the number of parameters, so BIC can only overcharge; Drton and Plummer (2017) show that BIC's penalty does not follow the large-sample behavior of the marginal likelihood of singular models, and Amari et al. (2006) survey how AIC, BIC and MDL fail for multilayer perceptrons and mixtures. The approximation also breaks when the dimension grows: it is not valid when d is comparable with n, and Shun and McCullagh (1995) give an example with d of order $n^{1/2}$ in which its relative error is of order one. In deep learning the Laplace approximation is evaluated at a trained network, whose Hessian is degenerate wherever the network is singular; Wei et al. (2023) argue that it is inappropriate for singular models, and Lotfi et al. (2022) report that the marginal likelihood itself can be negatively correlated with generalization when it is used to compare architectures.

OPEN PROBLEMS. Within its hypotheses the regular theory answers this chapter's questions completely, and its refinements to order $1/n$ (Exercise 1.5) have long been standard. The open problems begin where the hypotheses fail, and they are the subject of this book. What replaces $\frac{d}{2}$ when the Hessian is singular has an answer, the learning coefficient, but computing it for networks of realistic size is open (Chapter 7). Whether a Laplace-type formula built from local information at a trained network can approximate the free energy of a singular model to order $\log n$ is open in general; the singular BIC of Drton and Plummer (2017) and the local learning coefficient of Chapter 13 are partial answers (Exercise 1.13).

Exercises

1.1 CONSOLIDATION. (A uniform law of large numbers.) Let W be compact, $w \mapsto \ell(x, w)$ continuous for every x, and $\mathbb{E}\sup_W |\ell(X, w)| < \infty$. Show that L is continuous and $\sup_W |L_n - L| \to 0$ almost surely. *(Hint: for $w \in W$ let u_ρ and l_ρ be the supremum and infimum of $\ell(x, \cdot)$ over the ball of radius ρ around w; then $\mathbb{E}[u_\rho - l_\rho] \to 0$ as $\rho \to 0$; cover W by finitely many such balls and apply the strong law to their u's and l's.)*

1.2 CONSOLIDATION. (Posterior normality.) Under (R1)–(R6), show that the total variation distance between the posterior law of $\sqrt{n}(w - \hat{w})$ and $\mathcal{N}(0, J^{-1})$ tends to zero in probability. *(Hint: on region I of Figure 1.1, the ratio of the posterior density to the density of $\mathcal{N}(\hat{w}, (nJ_n)^{-1})$ is a product of three factors that tend to 1 uniformly; both laws put vanishing mass outside region I; and Scheffé's lemma compares $\mathcal{N}(0, J_n^{-1})$ with $\mathcal{N}(0, J^{-1})$.)*

1.3 CONSOLIDATION. (Symmetric copies.) Let $g_1, \dots, g_K$ be maps of W onto itself that preserve volume and leave $p(x \mid w)$ and φ unchanged, let W be the union of the sets $g_k(W_1)$, which overlap only in null sets, and let (R1)–(R6) hold on W_1 for the prior $K\varphi$, with true parameter w_1. Show that the free energy is the Laplace approximation (1.5) at w_1 minus $\log K$, up to $o_p(1)$. For the tanh neuron $y = a\tanh(bx)$ at a truth with $a_0 b_0 \neq 0$, the symmetry $(a, b) \mapsto (-a, -b)$ lowers the free energy by $\log 2$. Why does this not contradict Theorem 1.4?

1.4 CONSOLIDATION. (The volume of a nondegenerate minimum.) Under (R1), (R2) and (R4)–(R6), show that as $\varepsilon \to 0$

$$V(\varepsilon) = \int_{K<\varepsilon} \varphi \;\sim\; \frac{\varphi(w_0)(2\pi\varepsilon)^{d/2}}{\Gamma(\frac{d}{2}+1)\sqrt{\det J}},$$

and check it against the regular plane. Deduce with Lemma 4.3 that $\bar{F}(n) = \frac{d}{2}\log\frac{n}{2\pi} + \frac{1}{2}\log\det J - \log\varphi(w_0) + o(1)$, so that a regular truth has $\lambda = \frac{d}{2}$ and $m = 1$. *(Hint: near w_0, K lies between $\frac{1}{2}(1 \mp \eta)(w - w_0)^\top J(w - w_0)$.)*

1.5 CONSOLIDATION. ⋆ (The error term of Laplace's method.) (a) Let f have a unique minimum at an interior point w^* of an interval, with $f''(w^*) > 0$, and let g be smooth. With f_k, g_k the derivatives at w^*,

show that $\int e^{-n(f-f(w^*))}g = \sqrt{2\pi/(nf_2)}\left[g_0+\frac{1}{n}\left(\frac{g_2}{2f_2}-\frac{g_1f_3}{2f_2^2}-\frac{g_0f_4}{8f_2^2}+\frac{5g_0f_3^2}{24f_2^3}\right)+O(n^{-2})\right]$. (b) For n tosses of a coin with k heads, the uniform prior and $\hat\theta = k/n$ in a fixed compact subset of $(0,1)$, compute F_n from the Beta integral and show, from (a) and from Stirling's series, that it exceeds the approximation of Theorem 1.4 by $\frac{1}{n}\left(\frac{13}{12}-\frac{1}{12\hat\theta(1-\hat\theta)}\right)+O(n^{-2})$. (c) In d dimensions with a flat prior, show that the excess is $-c/n+O(n^{-2})$ with $c = -\frac{1}{8}f_{ijkl}H^{ij}H^{kl}+\frac{1}{8}f_{ijk}f_{lmn}H^{ij}H^{kl}H^{mn}+\frac{1}{12}f_{ijk}f_{lmn}H^{il}H^{jm}H^{kn}$, summed over repeated indices, H^{ij} the inverse Hessian. This is the prediction $0.594/n$ of Figure 1.2(b). *(Hint: expand $e^{-n(\text{cubic}+\text{quartic})}$ to second order and use the Gaussian moments of orders four and six.)*

1.6 BY HAND. (BIC by hand.) (a) Under the hypotheses of Corollary 1.6(ii) for two models, prove that the posterior probability of the model with the smaller population loss, or of the smaller of two models that contain the truth, tends to one. (b) Compare the regular line $y = ax_1$ with the regular plane $y = ax_1 + bx_2$, uniform priors, on data from $y = a_0x_1+\varepsilon$, $|a_0|<1$. Neglecting the box probabilities in (1.4), show that $F_n^{\text{plane}} - F_n^{\text{line}} = \frac{1}{2}\log V - \frac{1}{2}\chi - \frac{1}{2}\log(2\pi)+\log 2$, where the likelihood-ratio statistic $\chi \sim \chi^2_1$ and $V = S_{22}-S_{12}^2/S_{11} \sim \chi^2_{n-1}$ are independent. Deduce that the posterior prefers the plane with probability $\mathbb{P}(\chi^2_1 > \log(2V/\pi))$, about 0.042 at $n=100$ and 0.003 at $n=10^4$, whereas AIC prefers it with probability 0.157 at every n.

1.7 BY HAND. (The ridge prior and the evidence.) In Computation 1.1 let $S_{xx} = nI_d$. Show that F_n is smallest at $\hat\tau^2 = |\hat w|^2/d - 1/n$ when this is positive and at $\tau = 0$ otherwise (MacKay's evidence maximization). At the truth 0, show that the evidence switches the parameters off with probability $\mathbb{P}(\chi^2_d \le d)$, which is $1-e^{-1}\approx 0.63$ for $d=2$; at a truth $w_0\neq 0$, $\hat\tau^2 \to |w_0|^2/d$.

1.8 BY HAND. (A truth on the boundary.) Fit the regular line with the prior uniform on $[0,1]$ to data from $a_0 = 0$. With $A = \sum_i x_i^2$, the unconstrained $\hat a = \sum_i x_iy_i/A$ and $Z = \hat a\sqrt{A} \sim \mathcal{N}(0,1)$, show that up to exponentially small terms $F_n - nS_n = \frac{1}{2}\log\frac{A}{2\pi} - \frac{1}{2}Z^2 - \log\Phi(Z)$ and that the posterior of $\sqrt{A}a$ is $\mathcal{N}(Z,1)$ cut to $[0,\infty)$. Using that $\Phi(Z)$ is uniform, show that $\mathbb{E}[F_n - nS_n] = \frac{1}{2}\log\frac{n}{2\pi}+\frac{1}{2}+O(\frac{1}{n})$, one unit more than for an interior truth with the same prior density. Which hypothesis fails?

1.9 VARIATION. (Unknown noise variance.) Give the regular plane an unknown noise standard deviation σ, with the prior uniform on $[-1,1]^2\times[\frac{1}{2},2]$ and the truth $(0,0,1)$. Show that Assumption 1.3 holds with $d=3$ and $J = \operatorname{diag}(1,1,2)$, and that (1.7) becomes $\frac{3}{2}\log\frac{n}{2\pi e}+\frac{1}{2}\log 2+\log 6$.

1.10 VARIATION. (A misspecified regular model.) Fit the regular line, with unit noise variance in the model, to data $y = \sigma_0\varepsilon$, $\sigma_0\neq 1$. Show that $J=1$ and $\mathcal{I}=\sigma_0^2$, so that Corollary 1.6(ii) gives the fluctuation $-\frac{1}{2}\sigma_0^2\chi^2_1$.

Show that the plug-in predictive has $\mathbb{E}[L(\hat{a})] - L(w_0) = \frac{\sigma_0^2}{2(n-2)} \approx \frac{\operatorname{tr}(IJ^{-1})}{2n}$, while the Bayes predictive with the flat prior has $\mathbb{E}[G_n] - L(w_0) = \frac{1}{2}(\psi(\frac{n+1}{2}) - \psi(\frac{n}{2}))$, exactly as in the realizable case: Watanabe's λ/n does not need realizability (Chapter 9).

1.11 MEASUREMENT. (The Laplace error at small n.) Compute $-\log \Pi_n$ of Computation 1.3 for the regular plane at $n = 3, 5, 10, 20, 30$, with the data generator of the script `01-regular-free.py`. Observe means of about 1.26, 0.60, 0.15, 0.013 and 0.0014 (the last is carried by rare data sets; its median is $3 \cdot 10^{-5}$), and explain the decay with the bound of Computation 1.3. Repeat for the truth $(0.9, 0)$, and explain why the error is still about 0.36 at $n = 100$.

1.12 MEASUREMENT. (A truth near the singular set.) In the mixture part of the script, set $b_0 = 1$: the smallest eigenvalue of J drops from 0.22 to 0.073, and the coefficient of Exercise 1.5 becomes $-c = -4.5$ (at $b_0 = \frac{1}{2}$: 0.014 and -112). Measure $n\mathbb{E}[F_n - \text{Laplace}]$ for n from 10^3 to $10^{4.5}$ and watch it approach -4.5. Below $n = 10^3$ the log-likelihood is sometimes far from quadratic. Relate the sample size at which the regular theory becomes accurate to the smallest eigenvalue of J; Chapter 12 lets the truth approach the singularity as n grows.

1.13 RESEARCH. (The Laplace approximation of a network.) Practical Laplace approximations for networks (Daxberger et al. 2021; MacKay 1992) evaluate a curvature matrix at a trained network and add a prior precision to it, which makes the degenerate directions count as regular ones. For the tanh neuron at the truth 0 with a Gaussian prior, compute what such an approximation gives for $F_n - nS_n$ as a function of n and the prior precision, and compare with $\frac{1}{2}\log n - \log\log n$ (Chapter 3). Open: a formula of this kind that is accurate to order $\log n$ at a degenerate point.

Notes and References

The approximation carries the name of Laplace, whose 1774 memoir was the first widely read presentation of inverse probability; Stigler (1986) introduces it and gives an English translation. Tierney and Kadane (1986) made it a tool for posterior moments and marginal densities, and Kass and Raftery (1995) review its use for Bayes factors, describing BIC as a rough approximation to the log Bayes factor; for nested hypotheses and a normal prior that carries the information of one observation, the error of BIC improves from $O_p(1)$ to $O_p(n^{-1/2})$ (Kass and Wasserman 1995). Theorem 1.4 and Corollary 1.6 present the standard argument, with an envelope condition that makes every step checkable. The exact error $1/(n-1)$ of the regular plane and its digamma form (1.3) are the book's computation; we have not found them in the literature.

Akaike introduced his criterion in 1973 (Akaike 1998) and named it AIC in Akaike (1974). Our statement of van der Vaart's Theorem 10.1 follows the preprint of Requadt et al. (2025, Theorem 1). Under misspecification the

posterior is still approximately normal, centered near the minimizer of the Kullback–Leibler divergence, but its credible sets are not confidence sets (Kleijn and van der Vaart 2012). In the minimum description length tradition the same $\frac{d}{2}\log n$ is the parametric complexity of a regular model: Rissanen (1996) sharpened the code length with the Fisher information, and Grünwald (2004, eq. (2.21)) writes the complexity as $\frac{d}{2}\log\frac{n}{2\pi}+\log\int\sqrt{\det I(w)}\,\mathrm{d}w+o(1)$. Watanabe treats the regular case as a special case of his general theory: the regular values of his four errors are listed in Watanabe (2010), and Watanabe (2018, Chapter 4) is devoted to regular posteriors.

TRANSLATING THE LITERATURE. Schwarz (1978) and Haughton (1988) write $S(Y,n,j)$ for the logarithm of the posterior probability of model j, up to a constant that does not depend on the model: our $-F_n^{(j)}$ plus the log prior probability of the model. Their leading terms $n\sup_{\psi\in m_j}(Y\psi-b(\psi))-\frac{1}{2}k_j\log n$ are our $-\mathrm{BIC}$, with k_j our d. Akaike's AIC $=-2\log(\text{maximum likelihood})+2k$ is $2n$ times ours. Van der Vaart (1998) writes θ, P_θ, $\dot\ell_\theta$ and I_θ for our w, $p(\cdot\mid w)$, $\nabla\ell(\cdot,w)$ and, in the realizable case, J. Watanabe (2013) writes F for our F_n and $J(w)$ for the Hessian of L, and says that q is regular for $p(x\mid w)$; Watanabe (2009) calls a model regular if it is identifiable with positive definite Fisher information at every parameter, and counts the regular models among the singular ones. Watanabe (2010) writes B_g for our G_n-S. The D_n of Clarke and Barron (1990) is our $\mathbb{E}[F_n-nS_n]$, and MacKay's evidence is e^{-F_n}.

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2 Singular Models

When parameters fold

THE MODEL — $y = a\tanh(bx)$

IN THIS CHAPTER. Different weights of a neural network can compute the same function, and where a unit is switched off they do so along whole curves and planes. We compute the true sets and Fisher information of the running models, and the posterior of $y = abx$ at its singular truth exactly, through the posterior of the product ab. We prove that the Fisher information is singular wherever the true set contains a curve, and that under the posterior $\log|a|/\log n$ converges to the uniform distribution on $[-\frac{1}{2}, 0]$: the posterior spreads evenly over all scales from $n^{-1/2}$ to 1, and no smooth coordinates make it asymptotically normal. Exact draws up to $n = 10^{16}$ confirm the law and its slow convergence. For networks the true sets are known only in special cases, and how the posterior spreads over them is open.

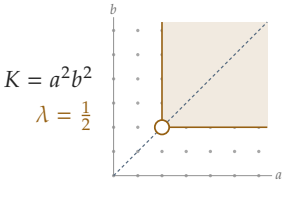
2.1 The question: two weights, one function

Train a network with one hidden layer of h units with the tanh activation,

$$f_w(x) = d + \sum_{j=1}^{h} a_j \tanh(b_j x + c_j), \tag{2.1}$$

twice, from two random initializations, and compare the trained weights. They differ, and yet the two networks may compute exactly the same function. Exchanging two hidden units changes w and not f_w; so does negating the three weights (a_j, b_j, c_j) of one unit, because tanh is odd. For a unit with the rectified linear activation, multiplying its incoming weight and bias by $\alpha > 0$ and dividing its outgoing weight by α changes nothing either. These *generic symmetries* act on the whole parameter space: exchanges and negations give each parameter of a tanh network finitely many twins, and rescalings give each parameter of a ReLU network curves of them.

Other moves exist only at special parameters. If $a_j = 0$, then the jth unit is switched off, and its incoming weight and bias can take every value: a whole plane of parameters computes one function. If $b_j = 0$, then the unit outputs the constant $a_j \tanh(c_j)$, which the output bias can absorb; if $(b_j, c_j) = (b_k, c_k)$, then the two units act as one with outgoing weight $a_j + a_k$. Such parameters are called *reducible*. They form a set of measure zero, but it is exactly the set on which a network with more units than it needs represents a simpler function.



The smallest instance is one unit without biases, fitted to pure noise. The *tanh neuron* of this book is the model

$$y = a\tanh(bx) + \varepsilon, \qquad x \sim \mathrm{Unif}[-1,1], \quad \varepsilon \sim \mathcal{N}(0,1), \tag{2.2}$$

with the uniform prior on $W = [-2,2]^2$ and the truth $y = \varepsilon$, so that the true regression function is $f \equiv 0$.[1] Every parameter with $a = 0$ or $b = 0$ computes the zero function, so the parameters that realize the truth form a cross. Near its center $a\tanh(bx) = abx + O(ab^3)$: the tanh neuron is the product model $y = abx$ of the prologue, slightly bent, and the product model is what we compute with.

In the notation of Chapter 1, a model $p(x \mid w)$, $w \in W \subset \mathbb{R}^d$, fitted to data from a truth q has the average loss $L(w) = -\mathbb{E}_q[\log p(X \mid w)]$, minimized at a true parameter w_0; $K(w) = L(w) - L(w_0) \geq 0$, and $W_0 = \{w \in W : K(w) = 0\}$ is the *set of true parameters*. The *Fisher information* at w is the covariance of the score,

$$I(w) = \mathbb{E}_{p(\cdot\mid w)}\left[\nabla_w \log p(X \mid w)\, \nabla_w \log p(X \mid w)^{\top}\right],$$

and in the realizable case $I(w) = \nabla^2 K(w)$ at every true parameter w when two derivatives may be taken under the integral sign (Computation 2.2). Following Watanabe (2010, Definition 2), the pair of truth and model is *regular* if W_0 is a single point w_0 and $\nabla^2 L(w_0)$ is positive definite, with w_0 in the interior of W (a requirement this book adds), and *singular* otherwise; a singular model is a model with singular truths of interest.[2]

THE REGULAR ANSWER · the shape of the posterior. Suppose that the model is differentiable in quadratic mean at an interior true parameter w_0 with nonsingular Fisher information $I(w_0)$, that for every $\varepsilon > 0$ some sequence of tests separates w_0 from $\{w : \|w - w_0\| \geq \varepsilon\}$ uniformly, and that the prior has a density that is continuous and positive at w_0. Then, by the Bernstein–von Mises theorem (van der Vaart 1998, Theorem 10.1), the posterior law of $\sqrt{n}(w - w_0)$ is close in total variation to $\mathcal{N}(\Delta_n, I(w_0)^{-1})$, in probability, with $\Delta_n = I(w_0)^{-1} n^{-1/2} \sum_i \nabla_w \log p(X_i \mid w_0)$. The posterior concentrates at one point, at the rate $n^{-1/2}$, in the shape of a Gaussian with covariance $(nI(w_0))^{-1}$.

For the tanh neuron at $f \equiv 0$ the hypotheses fail twice: the true set is a cross, and at its center the Fisher information is the zero matrix (Computation 2.2). The prologue's Figure P.1 showed what happens instead for the product model: the posterior draws crowd onto the cross and spread along both arms. This chapter asks what the set of true parameters of a singular model looks like, what the Fisher information does on it, and where the posterior goes when there is no single point to go to.

[1] The inputs of this model are uniform on $[-1,1]$, not standard normal as for the other running models: with Gaussian inputs its loss is not analytic in b at $b = 0$ (Exercise 2.1), and Watanabe's general theorems assume analyticity.

[2] Watanabe's 2009 book uses the words differently: there a *model* is regular if it is identifiable with a positive definite Fisher information at every parameter, and the singular models include the regular ones. This book follows his usage from 2010 on.

2.2 By hand: crosses, fibers, and the exact posterior

We compute the true sets of the running models, the Fisher information of the product model, and the posterior of the product model, exactly. For a regression $y = f(w,x) + \varepsilon$ with standard normal noise,

$$K(w) = \tfrac{1}{2}\mathbb{E}_x\big[(f(w,x) - f(w_0,x))^2\big], \tag{2.3}$$

because $-\log p(y \mid x,w) = \tfrac{1}{2}(y - f(w,x))^2 + \tfrac{1}{2}\log(2\pi)$ and the residual $y - f(w_0,x) = \varepsilon$ has mean zero and is independent of x.

COMPUTATION 2.1 · THE TRUE SETS OF THE CAST

1 *The product model* $y = abx$, with $x \sim \mathcal{N}(0,1)$ and the uniform prior on $[-1,1]^2$. For the truth $y = c_0 x + \varepsilon$, (2.3) gives $K = \tfrac{1}{2}(ab - c_0)^2$. At the singular truth $c_0 = 0$, $K = \tfrac{1}{2}a^2b^2$ and W_0 is the cross $\{a = 0\} \cup \{b = 0\}$. At a truth with $0 < |c_0| < 1$, W_0 is the part of the hyperbola $ab = c_0$ in the square, two arcs through its interior.

2 *The tanh neuron* (2.2). Since $\tanh^2 = 1 - \cosh^{-2}$,

$$K(a,b) = \frac{a^2}{2}\cdot\frac{1}{2}\int_{-1}^{1}\big(1 - \cosh^{-2}(bx)\big)\,dx = \frac{a^2}{2}\Big(1 - \frac{\tanh b}{b}\Big),$$

with the value 0 at $b = 0$, and W_0 is the cross. The series $1 - \frac{\tanh b}{b} = \frac{b^2}{3} - \frac{2b^4}{15} + \frac{17b^6}{315} - \cdots$ converges for $|b| < \frac{\pi}{2}$, the distance to the poles $\pm\frac{i\pi}{2}$ of tanh; and since $\tanh(b)/b$ is analytic on the strip $|\operatorname{Im} b| < \frac{\pi}{2}$, K is analytic on a neighborhood of W. The ratio $r(b) = 6K/(a^2b^2) = 3\mathbb{E}[x^2(\tanh(bx)/(bx))^2]$ decreases in $|b|$, because $\tanh(u)/u$ is positive and decreases on $(0,\infty)$ (its derivative has the sign of $u - \frac{1}{2}\sinh 2u < 0$); so $r(2) = \frac{3}{4}(1 - \frac{1}{2}\tanh 2) = 0.388 \le r(b) \le r(0) = 1$, and $0.388\cdot\frac{1}{6}a^2b^2 \le K \le \frac{1}{6}a^2b^2$ on W.

3 *The ReLU neuron* $y = a\,\mathrm{relu}(bx)$, with $x \sim \mathcal{N}(0,1)$ and the uniform prior on $[-1,1]^2$. Since $\mathrm{relu}(bx)^2 = b^2x^2\mathbf{1}\{bx > 0\}$ and $\mathbb{E}[x^2\mathbf{1}\{x > 0\}] = \frac{1}{2}$, $K = \frac{1}{4}a^2b^2$ exactly, and W_0 is the cross. The model itself is not analytic: for $x \neq 0$, $b \mapsto \mathrm{relu}(bx)$ has a kink at $b = 0$.

4 *The mixture* $p(x \mid a,b) = (1-a)\phi(x) + a\phi(x-b)$, with ϕ the standard normal density, $(a,b) \in [0,1]\times[-2,2]$, the uniform prior, and the truth ϕ. Write $p = \phi\,(1 + ag)$ with $g(x) = e^{bx - b^2/2} - 1$, so that $\mathbb{E}g = 0$ and $\mathbb{E}g^2 = e^{b^2} - 1$ under the truth. With $\psi(t) = t - \log(1+t) \ge 0$, $K = \mathbb{E}[-\log(1+ag)] = \mathbb{E}[\psi(ag)]$, and Taylor's theorem for ψ gives $K = \frac{a^2}{2}(e^{b^2} - 1) + O(a^3)$ uniformly in $|b| \le 2$. So $W_0 = \{a = 0\} \cup \{b = 0\}$, a cross whose arm $\{a = 0\}$ lies on the boundary of W, and $0.149\,a^2b^2 \le K \le 13.4\,a^2b^2$ on W (Exercise 2.2). But K is not analytic in a at $a = 0$ when $b \neq 0$: its kth Taylor coefficient in a is $(-1)^k\mathbb{E}[g^k]/k$, which grows like $e^{k(k-1)b^2/2}$.

5 *Rank-one regression* $y = b\,a^\top x + \varepsilon$, with $a \in \mathbb{R}^N$, $b \in \mathbb{R}^M$, $x \sim \mathcal{N}(0, I_N)$ and the truth 0: $K = \tfrac{1}{2}\|a\|^2\|b\|^2$, and $W_0 = \{a = 0\} \cup \{b =$

model, prior box	$K(a,b)$	c_1, c_2	analytic
$y = abx$, $[-1,1]^2$	$\frac{1}{2}a^2b^2$	$\frac{1}{2}$, $\frac{1}{2}$	yes
$y = a\tanh(bx)$, $[-2,2]^2$	$\frac{a^2}{2}\left(1-\frac{\tanh b}{b}\right)$	0.0647, $\frac{1}{6}$	yes
$y = a\,\mathrm{relu}(bx)$, $[-1,1]^2$	$\frac{1}{4}a^2b^2$	$\frac{1}{4}$, $\frac{1}{4}$	K yes, model no
mixture, $[0,1]\times[-2,2]$	$\mathbb{E}[\psi(ag)]$	0.149, 13.4	no, in a

Table 2.1. Four models whose set of true parameters is a cross, at the truth "nothing to learn" (Computation 2.1); inputs $x \sim \mathcal{N}(0,1)$, except $x \sim \mathrm{Unif}[-1,1]$ for the tanh neuron. On the whole box $c_1a^2b^2 \le K \le c_2a^2b^2$, and all four have $\lambda = \frac{1}{2}$ and $m = 2$, against $\frac{d}{2} = 1$. The lower constants of the tanh neuron and the mixture are $\frac{1}{6}r(2)$ and $\frac{1}{2}\min_{|b|\le 2}\mathbb{E}[g^2/(1+g_+)]/b^2$ (Exercise 2.2), computed numerically.

0} is the union of two linear subspaces, of dimensions M and N, that meet at the origin.

In every case the singular truth is "nothing to learn", and the true set is where a unit, or a mixture component, is switched off: its outgoing weight vanishes, or its incoming weight does (Table 2.1). Where K lies between two multiples of a^2b^2 on the whole box, the prior volume $V(\varepsilon)$ of $\{K < \varepsilon\}$ is squeezed between those of $\{a^2b^2 < \varepsilon/c_2\}$ and $\{a^2b^2 < \varepsilon/c_1\}$, both of order $\varepsilon^{1/2}\log\frac{1}{\varepsilon}$ (Chapter 4); so the tanh neuron, the ReLU neuron and the mixture share the product model's learning coefficient $\lambda = \frac{1}{2}$ and multiplicity $m = 2$, by the comparison rule of Proposition 5.4. (The learning coefficient is the real log canonical threshold of K; Chapter 4 shows they are the same number.) The comparison does not transfer analyticity: the ReLU neuron and the mixture are not analytic models, so Watanabe's general theorems do not apply to them verbatim, and mixtures need separate treatment, as in the study of WAIC and WBIC for mixture models by Watanabe (2021). The tanh neuron with uniform inputs is analytic, which is why the book uses it.

Truths away from the cross behave differently. At a truth $0 < |c_0| < 1$ of the product model the true set is a smooth curve: still singular, with $\lambda = \frac{1}{2}$ but $m = 1$.[3] At a truth (a_0, b_0) with $a_0b_0 \neq 0$ the tanh neuron is regular up to its sign symmetry: the Taylor coefficients ab and $-\frac{1}{3}ab^3$ of $a\tanh(bx)$ at $x = 0$ determine b^2 and ab, so $W_0 = \{\pm(a_0, b_0)\}$, where the Fisher information is positive definite.[4]

COMPUTATION 2.2 · THE FISHER INFORMATION

1 For a regression with standard normal noise the score is $(y - f(w,x))\nabla_w f(w,x)$, and under $p(\cdot \mid w)$ the residual $y - f(w,x)$ is standard normal and independent of x; so $I(w) = \mathbb{E}_x\left[\nabla_w f\,\nabla_w f^\top\right]$.

2 Differentiating (2.3) twice under the integral sign, $\nabla^2 K(w) = \mathbb{E}_x[\nabla_w f\,\nabla_w f^\top] + \mathbb{E}_x[(f(w,x) - f(w_0,x))\nabla_w^2 f(w,x)]$, and the second term vanishes at a true parameter. So $\nabla^2 K(w) = I(w)$ on W_0, for every regression model of Computation 2.1 except the ReLU neuron at $b = 0$, where f is not differentiable in b.

[3] With $c_0 = 1$ the hyperbola meets $[-1,1]^2$ only at the corners $\pm(1,1)$, where $K \approx \frac{1}{2}(u+v)^2$ in the distances $u, v \ge 0$ to the corner, and the learning coefficient becomes 1. The book's generic truths cross the interior of the box.

[4] But nearly singular: at $(1,1)$ its eigenvalues are 0.0021 and 0.36, so the posterior standard deviation along the weak direction is about $(0.0021\,n)^{-1/2}$, which is 0.69 at $n = 10^3$, a sixth of the width of the box.

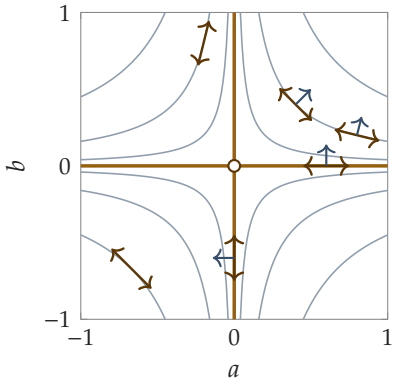


Figure 2.1. The fibers of the product model in $[-1,1]^2$: the cross $ab = 0$ (bronze) and the hyperbolas $ab = \pm 0.04, \pm 0.16, \pm 0.45$ (slate). At each marked point the bronze double arrow is the null direction of the Fisher information $I(a,b)$, tangent to the fiber; the slate arrows point along the gradient of ab, the direction of the nonzero eigenvalue. At the origin, the open circle, $I = 0$.

3 For the product model $\nabla_w f = (bx, ax)$, so, with $\mathbb{E}x^2 = 1$,

$$I(a,b) = \begin{pmatrix} b^2 & ab \\ ab & a^2 \end{pmatrix} = \begin{pmatrix} b \\ a \end{pmatrix} \begin{pmatrix} b & a \end{pmatrix}.$$

4 Its determinant is $a^2b^2 - (ab)^2 = 0$ everywhere. Away from the origin the eigenvalue a^2+b^2 belongs to (b,a), the gradient of $c = ab$, and the eigenvalue 0 to $(a,-b)$, along which $\mathrm{d}c = b\,\mathrm{d}a + a\,\mathrm{d}b$ vanishes: the tangent of the hyperbola $\{ab = \text{const}\}$ through (a,b). At the origin $I = 0$.

5 For the tanh neuron $\nabla_w f = (\tanh(bx),\ ax\cosh^{-2}(bx))$, so $I(a,0) = \mathrm{diag}(0, \frac{1}{3}a^2)$, $I(0,b) = \mathrm{diag}(1 - \frac{\tanh b}{b}, 0)$ and $I(0,0) = 0$: the null direction is the direction of the arm.

The Fisher information of the product model is singular everywhere, at the truth and away from it, for one reason: through every parameter passes a curve, a hyperbola $\{ab = c\}$ or an arm of the cross, along which the distribution of the data does not change, and its tangent is a null direction (Figure 2.1). We call the set of parameters that give one distribution a *fiber* of the model; the fibers of the product model are the hyperbolas $\{ab = c\}$, $c \neq 0$, and the cross. The information about c itself is $\mathbb{E}x^2 = 1$ per observation, and $I(a,b) = \nabla c\, \nabla c^\top$ is that information pulled back to the parameters.[5]

The posterior of the product model can be computed exactly at every n, and the computation shows how the fibers organize it. Take the singular truth $c_0 = 0$, so that the y_i are pure noise, and write $S_{xx} = \sum_i x_i^2$, $S_{xy} = \sum_i x_i y_i$, $\hat{c} = S_{xy}/S_{xx}$ and $s = S_{xx}^{-1/2}$; write Φ for the standard normal distribution function and Π_n for the posterior distribution of $w = (a,b)$ at $\beta = 1$.

[5] Jeffreys' prior, proportional to $\sqrt{\det I(w)}$, is therefore identically zero for the product model and cannot be normalized.

COMPUTATION 2.3 · THE POSTERIOR OF $y = abx$, EXACTLY

1 *The data see only* $c = ab$. Expanding the square, $nL_n(a,b) = \frac{(ab-\hat{c})^2}{2s^2} + \frac{1}{2}\left(\sum_i y_i^2 - S_{xy}^2/S_{xx}\right) + \frac{n}{2}\log(2\pi)$, and only the first term depends on (a,b). With the prior constant $\frac{1}{4}$ cancelled, the

posterior density on $W = [-1,1]^2$ is proportional to $e^{-(ab-\hat{c})^2/(2s^2)}$.

2 *Coordinates along the fibers.* On $\{a \neq 0\}$ use (a, c) with $c = ab$: then $b = c/a$, $\mathrm{d}a\,\mathrm{d}b = \mathrm{d}a\,\mathrm{d}c/|a|$, and $|b| \le 1$ becomes $|c| \le |a|$. The posterior of (a, c) has density proportional to $e^{-(c-\hat{c})^2/(2s^2)}/|a|$ on $\{|c| \le |a| \le 1\}$; the line $a = 0$ has measure zero.

3 *The coefficient.* Since $\int_{|c|\le|a|\le 1} \mathrm{d}a/|a| = 2\log\frac{1}{|c|}$, the posterior of c has density proportional to $e^{-(c-\hat{c})^2/(2s^2)}\log\frac{1}{|c|}$ on $[-1,1]$: a Gaussian of width $s \approx n^{-1/2}$ times $\frac{1}{2}\log\frac{1}{|c|}$, the prior density of c (Chapter 3).

4 *The fiber.* Given $c \neq 0$, the conditional density of a is proportional to $1/|a|$ on $\{|c| \le |a| \le 1\}$: the sign of a is a fair coin, $\log|a|$ is uniform on $[\log|c|, 0]$, and $b = c/a$; the data do not enter.

5 *The marginal of a.* Integrating c over $[-|a|, |a|]$ in step 2, with $Y = \hat{c} + s\xi$ for $\xi \sim \mathcal{N}(0,1)$,

$$\Pi_n(\mathrm{d}a) \propto \frac{1}{|a|}\left[\Phi\Big(\frac{|a|-\hat{c}}{s}\Big) - \Phi\Big(\frac{-|a|-\hat{c}}{s}\Big)\right]\mathrm{d}a = \frac{\mathbb{P}(|Y| \le |a|)}{|a|}\,\mathrm{d}a.$$

6 *The scale.* In $u = \log|a|$, both signs together, the posterior density is proportional to $\mathbb{P}(|Y| \le e^u)$ on $u \le 0$: about $2e^u\Phi'(\hat{c}/s)/s$ when $e^u \ll s$, and about 1 when $e^u \gg s + |\hat{c}|$, so flat in $\log|a|$ from $|a| \approx n^{-1/2}$ to $|a| = 1$. Since $\int_{-\infty}^{u} \mathbb{P}(|Y| \le e^v)\,\mathrm{d}v = \mathbb{E}[(u - \log|Y|)_+]$, with $t_+ = \max(t, 0)$,

$$\Pi_n\big(\log|a| \le u\big) = \frac{\mathbb{E}[(u-\log|Y|)_+]}{\mathbb{E}[(-\log|Y|)_+]} \qquad (u \le 0).$$

What one had to notice is that the likelihood is constant on the fibers $\{ab = c\}$, so that the posterior factors into a one-dimensional posterior of c and the prior's own distribution on each fiber; and that the uniform prior, conditioned on a hyperbola, is uniform in $\log|a|$ (not in arc length; Exercise 2.5). The data locate c to within $O(n^{-1/2})$ of 0 and say nothing about where along the fiber (a, b) lies, and the fiber over such a c reaches from $|a| \approx n^{-1/2}$ to $|a| = 1$: the posterior spreads $\log|a|$ evenly over about $\frac{1}{2}\log n$. Steps 3 and 4 also give an exact sampler: draw c from its posterior by inverting its distribution function on a grid, then $\log|a|$ uniformly from $[\log|c|, 0]$, a random sign, and $b = c/a$.[6]

In logarithmic coordinates (Figure 2.2) a hyperbola $|ab| = |c|$ becomes the segment $\log_{10}|a| + \log_{10}|b| = \log_{10}|c|$ between the axes, and the draws spread evenly along it. For the data set at $n = 10^{12}$, $\log_{10}|\hat{c}| = -5.92$, 96% of the 1500 draws lie within one decade of the line $\log_{10}|a| + \log_{10}|b| = -6$, and the median of $\log_{10}|a|$ is -3.01, halfway along it. As n grows the segment lengthens like $\log n$ while its thickness stays bounded.

[6] This is the sampler of the prologue's Figure P.1, `sk.sample_product_posterior` in the book's script kit.

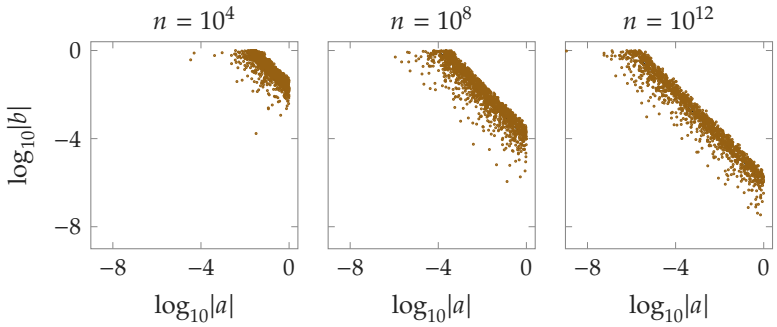


Figure 2.2. Exact posterior draws of $y = abx$ at the singular truth in logarithmic coordinates: 1500 draws for one data set at each sample size (statistics seeds 204, 208, 212; draw seeds 304, 308, 312), by the sampler of Computation 2.3. Dashed: the segment $\log_{10}|a| + \log_{10}|b| = -\frac{1}{2}\log_{10} n$. The cross of Figure P.1 becomes a segment whose length grows like $\log n$. Script: `figures/scripts/02-singular-scales.py`.

2.3 The theorem: folds make the Fisher information singular

Computation 2.2 found a null direction of the Fisher information along the fiber through each point. Nothing about the product model was needed for that, only a curve of true parameters and two derivatives.

Proposition 2.1 (Folds make the Fisher information singular). *Let w_0 be an interior point of W with $K(w_0) = 0$, and suppose that K is twice continuously differentiable on a neighborhood of w_0.*

(i) If $\gamma : (-\delta, \delta) \to W_0$ is a C^1 curve with $\gamma(0) = w_0$ and $\gamma'(0) = v \neq 0$, then $\nabla^2 K(w_0)\, v = 0$.

(ii) If W_0 contains a k-dimensional C^1 submanifold M through w_0, then the tangent space of M at w_0 lies in the kernel of $\nabla^2 K(w_0)$, and $\operatorname{rank} \nabla^2 K(w_0) \leq d - k$.

(iii) In the realizable case, if $I(w_0) = \nabla^2 K(w_0)$, which holds when two derivatives may be taken under the integral sign (Computation 2.2), then $I(w_0)v = 0$ in (i), and the pair is singular.

Proof. For (i), let U be a neighborhood of w_0 inside the interior of W on which K is twice continuously differentiable. For all small $|t|$, $\gamma(t) \in U$ and $K(\gamma(t)) = 0 = \min_W K$, because $K \geq 0$; so $\gamma(t)$ is an interior minimum of K, and $\nabla K(\gamma(t)) = 0$. The map $t \mapsto \nabla K(\gamma(t))$ is therefore identically zero near $t = 0$, and by the chain rule, ∇K being C^1 on U and γ being C^1, its derivative at 0 is $\nabla^2 K(w_0)\gamma'(0) = \nabla^2 K(w_0)v = 0$. For (ii), every tangent vector v of M at w_0 is the velocity $\gamma'(0)$ of a C^1 curve γ in $M \subseteq W_0$ through w_0, so $\nabla^2 K(w_0)v = 0$ by (i); the kernel contains a k-dimensional space, and the rank is at most $d - k$. For (iii), $I(w_0)v = \nabla^2 K(w_0)v = 0$ with $v \neq 0$, and the pair is singular already because W_0 is not a point. ∎

The converse fails: the quartic model $y = ax_1 + b^2 x_2$ of Chapter 5 has

a single true parameter and $\nabla^2 K(0) = \operatorname{diag}(1, 0)$. A fold is one way to be singular, a degenerate minimum another, and the learning coefficient of Chapter 4 measures both. On a k-dimensional piece of W_0 at most $r = d - k$ directions carry information; when the rank is exactly r, the learning coefficient near such a point is $\frac{r}{2}$ rather than $\frac{d}{2}$ (Chapter 13), and where pieces cross, like the arms at the origin of the product model, the rank drops further and the learning coefficient lies between $\frac{r}{2}$ and $\frac{r}{2} + \frac{d-r}{4}$ for the rank r there (Theorem 13.3).

FOLDS IN NETWORKS. For networks with one hidden layer of tanh units the folds are known completely: Farrugia-Roberts (2023) characterizes, for every parameter, the parameters that compute its function. We state the classical theorem behind this as that paper restates it, for the networks (2.1) with input $x \in \mathbb{R}$, whose parameter $w \in \mathbb{R}^{3h+1}$ lists the triples (a_j, b_j, c_j) and the output bias d. Call w *reducible* if $a_j = 0$ for some j, or $b_j = 0$ for some j, or $(b_j, c_j) = \pm(b_k, c_k)$ for some $j \neq k$, and *irreducible* otherwise.[7]

Theorem 2.2 (Sussmann 1992). *(i) A parameter $w \in \mathbb{R}^{3h+1}$ is reducible if and only if $f_w = f_{w'}$ for some parameter w' of a network with fewer than h hidden units.*

(ii) If w is irreducible, then the parameters $w' \in \mathbb{R}^{3h+1}$ with $f_{w'} = f_w$ are exactly those obtained from w by permuting the hidden units and negating the triples (a_j, b_j, c_j) of some of them.

[7] We have read the theorem in this restatement, not in Sussmann's paper. His title speaks of minimal nets; by part (i), these are the irreducible ones.

These $2^h h!$ moves give $2^h h!$ distinct parameters when w is irreducible (a negation fixing a unit forces $a_j = 0$, a permutation moving one forces $(b_j, c_j) = \pm(b_k, c_k)$), so at an irreducible truth W_0 is finite; Chen et al. (1993) describe the group of these moves for multilayer tanh networks, a direct product of Weyl groups. At a reducible truth, by part (i) the truth of every network with more units than it needs, the picture changes.

EXAMPLE 2.3 (Two units, one needed). Let $h = 2$ and let the truth be $f^*(x) = d^* + a^* \tanh(b^* x + c^*)$ with $a^* \neq 0 \neq b^*$, fitted as a regression with standard normal noise and inputs uniform on $[-1, 1]$; then $K(w) = 0$ means $f_w = f^*$ on $[-1, 1]$, hence on $\mathbb{R}$, since both are analytic in x. By Theorem 2.2(i) such a $w \in \mathbb{R}^7$ is reducible, and in each case of reducibility it computes the function of a one-unit parameter, to which part (ii) applies. If $a_2 = 0$: $(a_1, b_1, c_1) = \pm(a^*, b^*, c^*)$, $d = d^*$, and (b_2, c_2) is free, two planes. If $b_2 = 0$: $(a_1, b_1, c_1) = \pm(a^*, b^*, c^*)$ and $d + a_2 \tanh c_2 = d^*$ with (a_2, c_2) free, two curved surfaces. If $(b_1, c_1) = \sigma(b_2, c_2)$ with $\sigma = \pm 1$: $(b_1, c_1) = \epsilon(b^*, c^*)$, $d = d^*$ and $a_1 + \sigma a_2 = \epsilon a^*$ for a sign ϵ, four lines. With the units exchanged in the first two cases, W_0 is the union of eight two-dimensional pieces and four lines in $\mathbb{R}^7$, which meet along lines and at points; by Proposition 2.1 the Fisher information has rank at most 5 on the two-dimensional pieces, at most 6 on the lines, and less where pieces cross.

For ReLU networks the positive rescaling of each unit is already a continuous symmetry, so every true set contains curves; Dinh et al. (2017) used it to change the sharpness of a minimum arbitrarily without changing

the function, Bona-Pellissier et al. (2023) give conditions under which a deep ReLU network is determined by its function up to permutation and rescaling, and Grigsby et al. (2023) show that some ReLU networks have further, hidden symmetries.

The folds shape gradient descent too. Fukumizu and Amari (2000) showed that a critical point of a network with $h-1$ hidden units gives lines of critical points of the network with h units, which can be local minima or saddles and can cause plateaus of training; Amari et al. (2006), Wei et al. (2008) and Cousseau et al. (2008) analyze learning near such singularities. On the product model, gradient flow from $(\varepsilon, \varepsilon)$ needs a time of about $\frac{1}{2c_0}\log\frac{c_0}{\varepsilon^2}$ to leave the origin toward the truth $c_0 > 0$ (Exercise 2.7); Chapter 15 returns to the dynamics.

2.4 The theorem: the posterior spreads over scales

We now turn Computation 2.3 into a theorem with a complete proof. Besides S_{xx}, S_{xy}, $\hat{c}$ and s, put

$$Z_n = \frac{S_{xy}}{\sqrt{S_{xx}}} = \frac{\hat{c}}{s}, \qquad \ell_n = \tfrac{1}{2}\log S_{xx} = -\log s,$$

and write $t_+ = \max(t, 0)$ and $t_- = \max(-t, 0)$. Under the singular truth, given $x_1, \dots, x_n$, the statistic $S_{xy} = \sum_i x_i \varepsilon_i$ is normal with mean 0 and variance S_{xx}, so $Z_n \sim \mathcal{N}(0,1)$ exactly, for every n; and $S_{xx}/n \to 1$ almost surely by the strong law of large numbers. The proof needs one fact about the logarithm of a normal variable.

Lemma 2.4 (The logarithm of a Gaussian). *Let $z \in \mathbb{R}$, let $\xi \sim \mathcal{N}(0,1)$, and let $X = \log|z+\xi|$. Then $\mathbb{E}X_- \le \sqrt{2/\pi}$ and $\mathbb{E}X_+ \le \log(1+|z|) + \sqrt{2/\pi}$. Consequently, with $\kappa(z) = \log(1+|z|) + \sqrt{2/\pi}$, for every real L*

$$\left|\mathbb{E}[(L-X)_+] - L_+\right| \le \kappa(z).$$

Proof. The density of $z+\xi$ is at most $1/\sqrt{2\pi}$, and $X_- > 0$ only where $|z+\xi| < 1$, so $\mathbb{E}X_- \le \frac{1}{\sqrt{2\pi}}\int_{-1}^{1}\log\frac{1}{|y|}\,dy = \sqrt{2/\pi}$. Since $1+|z|+|\xi| \le (1+|z|)(1+|\xi|)$ and $\log(1+|\xi|) \le |\xi|$, $X_+ \le \log(1+|z|+|\xi|) \le \log(1+|z|) + |\xi|$, and $\mathbb{E}|\xi| = \sqrt{2/\pi}$. For the consequence, let first $L \ge 0$. Pointwise $(L-X)_+ \ge L - X$, and $(L-X)_+ \le L + X_-$ (if $X \ge 0$, then the left side is at most L; if $X < 0$, then it equals $L + X_-$); taking expectations, $L - \mathbb{E}X_+ \le \mathbb{E}(L-X)_+ \le L + \mathbb{E}X_-$, and both $\mathbb{E}X_+$ and $\mathbb{E}X_-$ are at most $\kappa(z)$. If $L < 0$, then $0 \le (L-X)_+ \le X_-$ pointwise, so $0 \le \mathbb{E}(L-X)_+ \le \kappa(z)$, while $L_+ = 0$. ∎

Theorem 2.5 (The posterior spreads uniformly over scales). *Let $X_i = (x_i, y_i)$, $i = 1, \dots, n$, be independent, with $x_i \sim \mathcal{N}(0,1)$ and $y_i \sim \mathcal{N}(0,1)$ independent of x_i: the product model $y = abx + \varepsilon$ at its singular truth. Let the prior be uniform on $W = [-1,1]^2$, let Π_n be the posterior at $\beta = 1$, and let $\xi \sim \mathcal{N}(0,1)$ be independent of the data.*

(i) For every data set with $S_{xx} > 0$ and every $u \le 0$,

$$\Pi_n\big(\log|a| \le u\big) = \frac{\mathbb{E}_\xi\big[(u - \log|\hat{c} + s\xi|)_+\big]}{\mathbb{E}_\xi\big[(-\log|\hat{c} + s\xi|)_+\big]}. \tag{2.4}$$

(ii) With $\kappa_n = \log(1 + |Z_n|) + \sqrt{2/\pi}$, on every data set with $\ell_n > \kappa_n$,

$$\sup_{u \le 0}\Big|\Pi_n\big(\log|a| \le u\big) - \Big(1 + \frac{u}{\ell_n}\Big)_+\Big| \le \frac{2\kappa_n}{\ell_n - \kappa_n}. \tag{2.5}$$

(iii) As $n \to \infty$,

$$\sup_{\tau \le 0}\Big|\Pi_n\big(\log|a| \le \tau \log n\big) - (1 + 2\tau)_+\Big| = O_p\Big(\frac{1}{\log n}\Big).$$

In particular $\Pi_n(\log|a| \le \tau \log n) \to 1 + 2\tau$ in probability for every $\tau \in [-\frac{1}{2}, 0]$: the posterior law of $\log|a|/\log n$ converges weakly, in probability, to $\mathrm{Unif}(-\frac{1}{2}, 0)$.

(iv) The same holds with b in place of a.

Proof. The plan is to read (i) off Computation 2.3, to compare its numerator and denominator with $u + \ell_n$ and ℓ_n by Lemma 2.4, and to let $n \to \infty$ in the resulting bound. Fix a data set with $S_{xx} > 0$, an event of probability one.

For (i), steps 1, 2 and 5 of Computation 2.3 are exact: the posterior density on W is proportional to $e^{-(ab-\hat{c})^2/(2s^2)}$, and for $a \neq 0$ the substitution $c = ab$ gives $\int_{-1}^{1} e^{-(ab-\hat{c})^2/(2s^2)}\,\mathrm{d}b = \sqrt{2\pi}\,s\,\mathbb{P}(|Y| \le |a|)/|a|$ with $Y = \hat{c} + s\xi$. For $u \le 0$, substitute $|a| = e^v$ on each half-line and exchange integral and expectation by Tonelli's theorem, which applies because the integrand $\mathbf{1}\{\log|Y| \le v\}$ is nonnegative ($Y \neq 0$ almost surely):

$$\begin{aligned}\int_{|a| \le e^u} \frac{\mathbb{P}(|Y| \le |a|)}{|a|}\,\mathrm{d}a &= 2\int_{-\infty}^{u} \mathbb{P}\big(\log|Y| \le v\big)\,\mathrm{d}v \\ &= 2\,\mathbb{E}\Big[\int_{-\infty}^{u} \mathbf{1}\{\log|Y| \le v\}\,\mathrm{d}v\Big] = 2\,\mathbb{E}\big[(u - \log|Y|)_+\big].\end{aligned}$$

Dividing the value at u by the value at $u = 0$ gives (2.4). The denominator is positive because $\mathbb{P}(|Y| < 1) > 0$, and finite by the next paragraph.

For (ii), $\hat{c} = sZ_n$ gives $Y = s(Z_n + \xi)$; with $X = \log|Z_n + \xi|$, $u - \log|Y| = u + \ell_n - X$. Write $N = \mathbb{E}_\xi(u + \ell_n - X)_+$ and $D = \mathbb{E}_\xi(\ell_n - X)_+$ for the numerator and the denominator of (2.4), and $A = (u + \ell_n)_+$. By Lemma 2.4 with $z = Z_n$, $|N - A| \le \kappa_n$ and $|D - \ell_n| \le \kappa_n$; and $0 \le A \le \ell_n$, because $u \le 0 < \kappa_n < \ell_n$. Then $D \ge \ell_n - \kappa_n > 0$, and splitting the difference of the ratios into a numerator error and a denominator error,

$$\Big|\frac{N}{D} - \frac{A}{\ell_n}\Big| = \frac{|\ell_n(N - A) - A(D - \ell_n)|}{\ell_n D} \le \frac{\ell_n\kappa_n + \ell_n\kappa_n}{\ell_n(\ell_n - \kappa_n)} = \frac{2\kappa_n}{\ell_n - \kappa_n}.$$

Since $A/\ell_n = (1 + u/\ell_n)_+$, this is (2.5).

For (iii), put $u = \tau \log n$ and $\alpha_n = \log n/\ell_n$. On $\{\ell_n > \kappa_n\}$, by (2.5) and the triangle inequality, the supremum in (iii) is at most $2\kappa_n/(\ell_n - \kappa_n) + \sup_{\tau \le 0} |(1+\alpha_n\tau)_+ - (1+2\tau)_+|$. For $0 < \alpha < \beta$ the function $\tau \mapsto (1+\alpha\tau)_+ - (1+\beta\tau)_+$ on $(-\infty, 0]$ equals $(\beta-\alpha)|\tau| \le 1 - \alpha/\beta$ on $[-1/\beta, 0]$, equals $1 + \alpha\tau \le 1 - \alpha/\beta$ on $[-1/\alpha, -1/\beta]$, and vanishes below $-1/\alpha$; so the second supremum is at most $|\alpha_n - 2|/\max(\alpha_n, 2) \le \frac{1}{2}|\alpha_n - 2| = |\log(n/S_{xx})|/\log S_{xx}$. Now $S_{xx}/n \to 1$ almost surely, so $\ell_n/\log n \to \frac{1}{2}$ and $\log(n/S_{xx}) \to 0$, and κ_n has the same law for every n, because $Z_n \sim \mathcal{N}(0,1)$. Given $\varepsilon > 0$, choose M with $\mathbb{P}(\kappa_n > M) < \varepsilon$. For all large n, with probability at least $1 - 2\varepsilon$, $\kappa_n \le M$, $\ell_n \ge \frac{1}{3}\log n$ and $|\log(n/S_{xx})| \le 1$; if also $\log n \ge 6M$, then $\ell_n - \kappa_n \ge \frac{1}{6}\log n$ and the supremum is at most $(12M+2)/\log n$. This is the $O_p(1/\log n)$ bound. The distribution function of $\mathrm{Unif}(-\frac{1}{2}, 0)$ is $(1+2\tau)_+$ for $\tau \le 0$ and 1 for $\tau > 0$, where the posterior one is 1 as well, since $|a| \le 1$; uniform convergence to a continuous distribution function gives the weak convergence.

For (iv), the map $(a, b) \mapsto (b, a)$ preserves W, Lebesgue measure and ab, hence the posterior, so $\log|b|$ has the posterior law of $\log|a|$. ■

Under a regular posterior a parameter lives at one scale, $n^{-1/2}$; here $\log|a|$ is spread evenly over $\frac{1}{2}\log n$ units of scale. The posterior does concentrate at the origin, but only logarithmically: by (2.4), $\Pi_n(|a| > \delta)$ is about $2\log(1/\delta)/(\log n + 0.58)$, which is 0.32 for $\delta = 0.1$ at $n = 10^6$ and still 0.02 at $n = 10^{100}$.[8]

[8] So the arms of the prologue's Figure P.1 do shorten as n grows, but only in posterior mass and only at the rate $1/\log n$: at $n = 10^6$ nearly two thirds of the draws, 64% on average over data sets, have $\max(|a|, |b|) > 0.1$.

Remark 2.6 (The finite-n law). Since $(L - X)_+ = L - X + (X - L)_+$, the numerator of (2.4) is $u + \ell_n - \mu(Z_n) + \mathbb{E}_\xi(X - u - \ell_n)_+$ with $\mu(z) = \mathbb{E}\log|z + \xi|$, and the last term is negligible once $u + \ell_n$ is large; so the posterior p-quantile of $\log|a|$ is about $-(1-p)(\ell_n - \mu(Z_n))$ (we do not bound the error). As $\mathbb{E}\ell_n = \frac{1}{2}\log n + O(n^{-1})$ and $\mathbb{E}\mu(Z_n) = \mathbb{E}\log|\mathcal{N}(0,2)| = -\frac{\gamma}{2}$, with $\gamma = 0.5772$ Euler's constant, the average p-quantile of $\log|a|/\log n$ is about

$$-\frac{1-p}{2}\Big(1 + \frac{\gamma}{\log n}\Big); \tag{2.6}$$

for the quartiles at $n = 10^9$ this is -0.385, -0.257, -0.128, against the limits -0.375, -0.25, -0.125. The convergence is genuinely as slow as $1/\log n$.

Remark 2.7 (The scales and the $\log\log n$). The denominator of (2.4) is the factor $\log n$ of the marginal likelihood. By step 1 of Computation 2.3, $nK_n(a, b) = (ab - \hat{c})^2/(2s^2) - Z_n^2/2$, and the proof of (i) gives, exactly,

$$\int_W \tfrac{1}{4} e^{-nK_n(w)}\,\mathrm{d}w = \frac{\sqrt{2\pi}}{2}\, s\, e^{Z_n^2/2}\, \mathbb{E}_\xi\big[(-\log|\hat{c} + s\xi|)_+\big].$$

The expectation is $\ell_n + O_p(1)$ by Lemma 2.4, so $F_n - nS_n = \frac{1}{2}\log n - \log\log n + O_p(1)$, as Theorem 3.4 proves for every one-coefficient model; the $\log\log n$ is the logarithm of the number of scales that the posterior occupies.

The first consequence answers the regular answer box: no change of coordinates, no centering and no rate rescues the Gaussian.

Corollary 2.8 (No Gaussian limit in smooth coordinates). *In the setting of Theorem 2.5, let Ψ be a C^1 diffeomorphism from an open set containing W onto an open subset of $\mathbb{R}^2$, let $\hat\theta_n \in \mathbb{R}^2$ and $\sigma_n > 0$ be statistics, and let Q be a probability measure on $\mathbb{R}^2$ with a density bounded by a constant B. Then the total variation distance between the posterior law of $(\Psi(w) - \hat\theta_n)/\sigma_n$ and Q does not converge to 0 in probability. In particular the conclusion of the Bernstein–von Mises theorem fails in every system of C^1 coordinates, with every centering and every rate.*

Proof. The plan is to show that Q, moved and rescaled, puts most of its mass on a bounded range of scales of the distance to $\Psi(0)$, while by Theorem 2.5 the posterior puts vanishing mass on every such range. Write $\theta_0 = \Psi(0)$, $\rho(w) = \|\Psi(w) - \theta_0\|$, $\|w\|_\infty = \max(|a|, |b|)$ and $T_n = (\Psi(w) - \hat\theta_n)/\sigma_n$.

Step 1: ρ is comparable with $\|w\|_\infty$. The derivative of Ψ is bounded on the compact convex set W, so $\rho \le c_2\|w\|_\infty$ by the mean value inequality. The ratio $\rho(w)/\|w\|_\infty$ is continuous and positive on $W \setminus \{0\}$, since Ψ is injective, and its lower limit at 0 is at least the smallest singular value $\sigma_* > 0$ of the invertible matrix $D\Psi(0)$; so it exceeds $\sigma_*/2$ near 0 and has a positive minimum on the compact rest of W, and $\rho \ge c_1\|w\|_\infty$ for some $c_1 > 0$.

Step 2: bounded ranges of scales have vanishing posterior mass. Fix $\ell > 0$ and $J = [v, v+\ell]$, and let $J' = [v - \log c_2,\ v + \ell - \log c_1]$, of length $\ell' = \ell + \log(c_2/c_1)$. If $\log\rho(w) \in J$, then $\log\|w\|_\infty \in J'$, and $\|w\|_\infty$ is $|a|$ or $|b|$; by part (iv), $\Pi_n(\log\rho \in J) \le 2\,\Pi_n(\log|a| \in J')$. By (2.5), the posterior distribution function of $\log|a|$, which is continuous, is within $\varepsilon_n = 2\kappa_n/(\ell_n - \kappa_n)$ of the $(1/\ell_n)$-Lipschitz function $\min\{1, (1 + u/\ell_n)_+\}$, for $u \le 0$ and also for $u > 0$, where both equal 1. Hence on $\{\ell_n > \kappa_n\}$

$$\sup_{v\in\mathbb{R}} \Pi_n\big(\log\rho \in [v, v+\ell]\big) \le \frac{2\ell'}{\ell_n} + 4\varepsilon_n,$$

which tends to 0 in probability, as in the proof of (iii).

Step 3: a limit Q would put mass on a bounded range of scales. Suppose, to the contrary, that $\delta_n = \|\Pi_n(T_n \in \cdot) - Q\|_{\mathrm{TV}} \to 0$ in probability. Choose $M \ge 1$ with $Q(\{\|t\| \le M\}) \ge \frac{7}{8}$ and $\eta \in (0, 1]$ with $\pi\eta^2 B \le \frac{1}{8}$, so that every disc of radius η has Q-mass at most $\frac{1}{8}$; put $\ell = \log(3M/\eta) \ge \log 3$ and $D_n = \|\hat\theta_n - \theta_0\|$. If $D_n \ge 2M\sigma_n$, then $\|T_n\| \le M$ implies $\|\Psi(w) - \hat\theta_n\| \le \frac{1}{2}D_n$, so $\frac{1}{2}D_n \le \rho \le \frac{3}{2}D_n$, a range of length $\log 3$ in $\log\rho$, with posterior probability at least $\frac{7}{8} - \delta_n$. If $D_n < 2M\sigma_n$, then $\|T_n\| \le M$ implies $\rho < 3M\sigma_n$, while $\{\rho < \eta\sigma_n\}$ is the event that T_n lies in the disc of radius η about $(\theta_0 - \hat\theta_n)/\sigma_n$, of posterior probability at most $\frac{1}{8} + \delta_n$; so $\eta\sigma_n \le \rho < 3M\sigma_n$, a range of length ℓ, with posterior probability at least $\frac{7}{8} - \delta_n - \frac{1}{8} - \delta_n$. In both cases

$$\sup_{v\in\mathbb{R}} \Pi_n\big(\log\rho \in [v, v+\ell]\big) \ge \tfrac{3}{4} - 2\delta_n,$$

which is at least $\frac{1}{2}$ with probability tending to one, contradicting step 2. ∎

The coefficient, by contrast, is learned in the regular way: the posterior law of $(ab - \hat c)/s$ converges in total variation to $\mathcal{N}(0, 1)$ (Exercise 2.3). Only

the parametrization is singular, as for a network whose weights fold while its output does not. The second consequence concerns point estimates.

Corollary 2.9 (MAP, posterior mean, and the scale of $|a|$). *In the setting of Theorem 2.5, on every data set with $0 < |\hat{c}| < 1$:*

(i) the posterior density is maximal exactly on $\{(a,b) \in W : ab = \hat{c}\}$, two arcs of a hyperbola on which $|a|$ takes every value in $[|\hat{c}|, 1]$; so the maximum a posteriori estimate, which for the uniform prior is the maximum-likelihood estimate in W, is a curve;

(ii) the posterior mean of w is $(0,0)$;

(iii) the posterior mean of $|a|$ is $\mathbb{E}_\xi[(1-|\hat{c}+s\xi|)_+]/\mathbb{E}_\xi[(-\log|\hat{c}+s\xi|)_+]$, and $\log n \cdot \mathbb{E}_w[|a|] \to 2$ in probability, while the posterior median of $|a|$ is $n^{-1/4+o_p(1)}$.

Proof. The plan is to read each claim off the posterior density and the integrals of Theorem 2.5. For (i), the posterior density is proportional to $e^{-(ab-\hat{c})^2/(2s^2)} \le 1$, with equality exactly when $ab = \hat{c}$, and the points of W with $ab = \hat{c}$ are $(a, \hat{c}/a)$ with $|\hat{c}| \le |a| \le 1$. For (ii), the map $(a,b) \mapsto (-a,-b)$ preserves W, Lebesgue measure and ab, hence the posterior, so $\mathbb{E}_w[a] = \mathbb{E}_w[-a] = 0$, and likewise $\mathbb{E}_w[b] = 0$. For (iii), the inner integral of the proof of Theorem 2.5(i) and Tonelli's theorem give $\int_W |a|\, e^{-(ab-\hat{c})^2/(2s^2)}\,\mathrm{d}w = \sqrt{2\pi}\, s \int_{-1}^{1} \mathbb{P}(|Y| \le |a|)\,\mathrm{d}a = 2\sqrt{2\pi}\, s\, \mathbb{E}(1-|Y|)_+$, and the normalizing integral is $2\sqrt{2\pi}\, s\, \mathbb{E}(-\log|Y|)_+$. The numerator lies between $1 - \mathbb{E}|Y| \ge 1 - |\hat{c}| - s$ and 1, so it tends to 1 almost surely; the denominator is $\ell_n + O_p(1)$ by Lemma 2.4, and $\ell_n/\log n \to \frac{1}{2}$. The median follows from Theorem 2.5(iii) with $\tau = -\frac{1}{4}$. ∎

In the regular theory these summaries agree to $o_p(n^{-1/2})$. Here the maximizers form a curve from scale $n^{-1/2}$ to scale 1, the mean is a point near no maximizer, and the typical size of $|a|$ is $n^{-1/4}$ while its mean is $2/\log n$.

2.5 Measuring it

Theorem 2.5 is exact at every n through (2.4), so a measurement can test both the limit and the finite-n law. The script `02-singular-scales.py` (in `figures/scripts`; about 35 seconds) simulates the product model at its singular truth at eight sample sizes from 10^2 to 10^{16}: the posterior depends on the data only through (S_{xx}, S_{xy}), and under the truth $S_{xx} \sim \chi^2_n$ and $S_{xy} \mid S_{xx} \sim \mathcal{N}(0, S_{xx})$ exactly, so it draws these two statistics. For each size it simulates 200 data sets, draws 2000 exact posterior samples from each with the sampler of Computation 2.3, and records the posterior quartiles of $\log|a|/\log n$; a measured point is their mean over data sets, with a Student-t 95% interval, and the prediction is the same mean computed from (2.4) by Gauss–Hermite quadrature over $Z_n \sim \mathcal{N}(0,1)$, with ℓ_n at its mean. The regular comparison is the regular plane $y = ax_1 + bx_2$ with the same prior and truth, whose posterior of a is $\mathcal{N}(\hat{a}, v)$ (the box truncates it negligibly), with $\hat{a}/\sqrt{v} \sim \mathcal{N}(0,1)$ and $1/v \sim \chi^2_{n-1}$. The sampler passes a check against

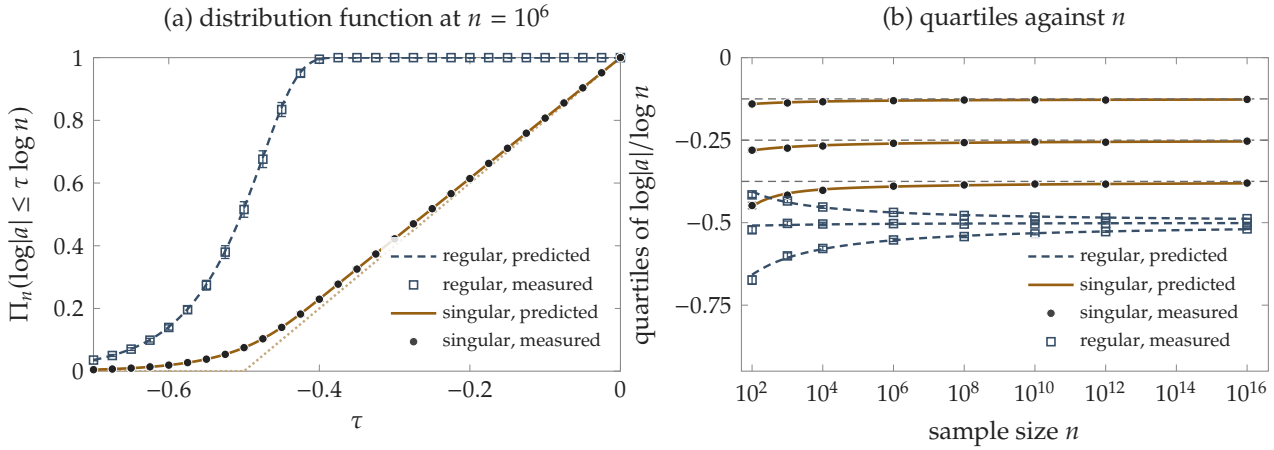


Figure 2.3. The posterior of $\log|a|/\log n$ for the product model $y = abx$ at its singular truth (dots; bronze curves) and for the regular plane $y = ax_1 + bx_2$ (squares; slate curves), with the uniform prior on $[-1, 1]^2$. Each point is the mean over 200 data sets, simulated through (S_{xx}, S_{xy}), of a posterior quantity estimated from 2000 exact draws, with Student-t 95% intervals. (a) The distribution function at $n = 10^6$. Bronze solid: the exact prediction (2.4), averaged over data sets; bronze dotted: the limit Unif$(-\frac{1}{2}, 0)$ of Theorem 2.5; slate dashed: the regular prediction $2\Phi(n^{\tau+1/2}/\sqrt{2}) - 1$. (b) The three quartiles against n, from 10^2 to 10^{16}. Bronze: the exact predictions, approaching the limits $-\frac{3}{8}$, $-\frac{1}{4}$, $-\frac{1}{8}$ (thin dashed); slate dashed: the regular predictions, collapsing onto $-\frac{1}{2}$. Seeds: data 2026 (a), $5000 + \log_{10} n$ (b); draws $10^4 + r$ (a), $10^5 \log_{10} n + r$ (b) for data set r. Script: `figures/scripts/02-singular-scales.py`.

the exact answer: on each data set the quartiles of its draws differ from the exact quartiles of the same posterior by 0.0040 to 0.0053 on average, the size of the Monte Carlo error alone (standard error about 0.005).

In Figure 2.3(a), at $n = 10^6$, the 29 measured points of the product model lie within 0.0008 of the exact prediction, and every interval contains it; for the regular plane 28 of 29 intervals contain the regular prediction. The regular posterior puts $\log|a|/\log n$ between -0.68 and -0.43 with probability 0.9, and its spread in $\log|a|$ does not grow with n; the singular posterior spreads it over all of $[-\frac{1}{2}, 0]$, close to the uniform law, with a gap of at most 0.075 that is the finite-n effect of Remark 2.6.

In Figure 2.3(b) the singular quartiles approach $-\frac{3}{8}$, $-\frac{1}{4}$, $-\frac{1}{8}$ as slowly as (2.6) says: at $n = 10^{16}$ they are -0.3809, -0.2539, -0.1270 predicted and -0.3800, -0.2530, -0.1265 measured, with half-widths 0.0014, 0.0012, 0.0008. The regular quartiles collapse onto $-\frac{1}{2}$, from $(-0.674, -0.522, -0.416)$ at $n = 10^2$ to $(-0.519, -0.501, -0.488)$ at 10^{16}. Of the 24 singular intervals 22 contain the prediction, the two misses (at $n = 10^3$ and 10^{12}) lying at the edge of their intervals; of the 24 regular ones 21 do, the three misses being the three quartiles at $n = 10^2$, from the same 200 data sets, 1.3 half-widths away. At $n = 10^2$ the two models are hard to tell apart by their quartiles; what separates them is that the singular spread of $\log|a|$ grows like $\frac{1}{4}\log n$ and the regular one does not.

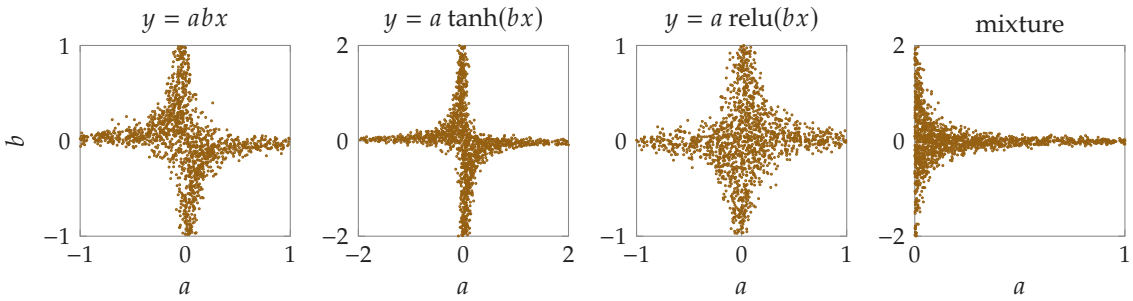


Figure 2.4. Posterior draws for the four models of Table 2.1 at the truth "nothing to learn", one data set of size $n = 10^3$ each (data seed 1000, draw seeds 2000 to 2003): 1500 draws from the posterior computed by midpoint quadrature on a grid of 600×600 cells (300×1200 for the mixture, whose weight a lies in $[0, 1]$ and location b in $[-2, 2]$). Script: `figures/scripts/02-singular-zoo.py`.

Figure 2.4 shows that the picture is not special to the product model (its script runs in about 40 seconds). For the product model the grid quartiles of $\log|a|/\log n$ are -0.403, -0.267, -0.133, against -0.406, -0.266, -0.133 from (2.4) for the same data, which checks the grid. The ReLU neuron gives almost the same numbers, -0.403, -0.264, -0.132; Exercise 2.4 shows why. The tanh neuron gives -0.399, -0.240, -0.073 and the mixture -0.500, -0.309, -0.151: their boxes and their constants c_1, c_2 differ from the product model's, and an $O(1)$ change in $\log|a|$ moves $\log|a|/\log n$ by $O(1/\log n)$, about 0.1 at $n = 10^3$. Whether their scale laws have the same limit we have not proved (Exercise 2.11).

2.6 The ledger

The regular picture, a Gaussian at one point, needs a single true parameter with an invertible Fisher information, and a fold destroys both: wherever the true set contains a curve, its tangent is a null vector of the Fisher information (Proposition 2.1). The posterior of the product model is the prior's own distribution along the fibers $\{ab = c\}$, mixed over the fibers the data allow; it spreads $\log|a|$ uniformly over $\frac{1}{2}\log n$ units of scale (Theorem 2.5), so that no smooth coordinates make it Gaussian (Corollary 2.8), and the logarithm of the normalizer of that uniform law is the $-\log\log n$ of Chapter 3.

LEDGER · $y = a\tanh(bx)$	
QUANTITY	the set of true parameters W_0, the Fisher information on it, and the shape of the posterior
REGULAR THEORY	W_0 a point; Fisher information invertible; the posterior close to a Gaussian of width $n^{-1/2}$ (Bernstein–von Mises)
SINGULAR THEORY	W_0 contains curves, on which the Fisher information is singular (Proposition 2.1); $\log\|a\|/\log n \to \mathrm{Unif}(-\frac{1}{2}, 0)$ under the posterior (Theorem 2.5); no Gaussian limit

	in C^1 coordinates (Corollary 2.8)
RUNNING EXAMPLE	$y = abx$: W_0 the cross, $I(a,b) = (b,a)^\top(b,a)$; posterior quartiles of $\log\lvert a\rvert/\log n \to -\frac{3}{8}, -\frac{1}{4}, -\frac{1}{8}$; tanh neuron: the same cross, $K \asymp a^2b^2$
MEASURED	quartiles -0.3800, -0.2530, -0.1265 (half-widths at most 0.0014) at $n = 10^{16}$, predicted -0.3809, -0.2539, -0.1270; regular plane -0.519, -0.501, -0.488
RESTS ON	realizability at the singular truth; Gaussian noise of known variance; the uniform prior on $[-1,1]^2$, positive on both arms; the exact posterior

2.7 Where the theory stops

WHAT WAS ASSUMED. A truth exactly at the singular point: at a truth $0 < |c_0| < 1$ the spread stops at the scale $|c_0|$, and the posterior converges to a fixed law on the hyperbola (Exercise 2.5). Gaussian noise of known variance, which makes the likelihood exactly Gaussian in c and Z_n exactly standard normal; the inputs enter only through S_{xx}, so every input law with $\log S_{xx}/\log n \to 1$ in probability gives the same limit. A uniform prior, positive on both arms of the cross. The exact posterior at $\beta = 1$, although the proof goes through at every fixed inverse temperature β, with s replaced by $(\beta S_{xx})^{-1/2}$ and Z_n by $\sqrt{\beta}\, Z_n$, and the limit survives at $\beta = 1/\log n$, where the rate in (iii) becomes $\log\log n/\log n$. Not analyticity: the ReLU neuron obeys the same law (Exercise 2.4); but Watanabe's general theorems assume it, which is why the tanh neuron takes uniform inputs and the mixture is not covered verbatim. And patience: at the rate $1/\log n$, the finite-n law (2.4), not its limit, describes the posterior at all practical sample sizes.

WHERE IT BREAKS. Priors that vanish on the singular set change the picture. Jeffreys' prior is identically zero for the product model; where the Fisher information degenerates only on the singular set it vanishes there, and Watanabe (2024) states that with Jeffreys' prior the learning coefficient is at least $\frac{d}{2}$, which removes the advantage of the singularity. A prior proportional to $|a|^h$, $h > 0$, which vanishes on one arm only, keeps $\lambda = \frac{1}{2}$ but lowers m to 1 and moves the posterior onto the other arm, with no spreading over scales (Exercise 2.9). Maximum likelihood fares worse: the maximum-likelihood set of the product model is a curve (Corollary 2.9), and the classical likelihood-ratio theory fails, with a statistic that is not bounded in probability for one against two normal components with unrestricted means (Hartigan 1985, as later accounts state it) and of order at least $\log n$ for perceptrons with two or more redundant hidden units (Fukumizu 2003, from the abstract). In networks the true sets are unions of pieces of different dimensions (Example 2.3), and no theorem we know describes how the posterior divides its mass among the pieces and over scales; Theorem 2.5 covers the product model and the ReLU neuron only. And for feedforward ReLU networks Wei et al. (2023) check, in the arXiv version we read, that the Fisher information is degenerate at every nontrivial weight vector: there the regular picture fails everywhere.

OPEN PROBLEMS. First, the scale law beyond one coefficient: for the tanh neuron, the mixture (Exercise 2.11), and networks at reducible truths, find the limit law of the posterior along W_0, its division among the pieces and its distribution over scales along each. Watanabe's theory describes the posterior through a resolution of singularities, in the variable $nK(w)$ (Watanabe 2009, 2024); as of October 2026 we have not found the distribution over scales along W_0 stated, even for the product model. Second, the geometry of W_0 for deep networks. Farrugia-Roberts (2023) characterizes the equivalence classes of reducible single-hidden-layer tanh networks and outlines extensions to modern architectures as future work; for ReLU networks Grigsby et al. (2025) show that the functional dimension varies across parameter space and give disconnected fibers, and the experiments of Grigsby et al. (2023) indicate that the probability that a network at initialization has no hidden symmetries decreases toward 0 with depth and increases toward 1 with width and input dimension, a trend for which we know of no proof. Third, the crossings of W_0 in networks of moderate size, where the learning coefficient drops, have not been catalogued as far as we know; Chapters 7 and 13 collect what is known.

Exercises

2.1 CONSOLIDATION. ★ (The tanh neuron.) (a) Show that $\mathbb{E}\tanh^2(bx) = 1 - \frac{\tanh b}{b}$ for $x \sim \mathrm{Unif}[-1,1]$, and expand it to order b^4. (b) Show that $u - \frac{u^3}{3} \le \tanh u \le u$ for $u \ge 0$, and deduce $\frac{b^2}{3} - \frac{2b^4}{15} \le \mathbb{E}\tanh^2(bx) \le \frac{b^2}{3}$. (c) With $x \sim \mathcal{N}(0,1)$, show that $b^2 - 2b^4 \le \mathbb{E}\tanh^2(bx) \le b^2$, but that the Taylor series of $\mathbb{E}\tanh^2(bx)$ at $b = 0$ diverges for every $b \ne 0$. *(Hint: in (c), the coefficient of b^{2k} is that of u^{2k} in $\tanh^2 u$, which decays only geometrically because of the poles at $\pm\frac{i\pi}{2}$, times $\mathbb{E}x^{2k} = (2k-1)!!$.)*

2.2 CONSOLIDATION. ★ (The mixture.) (a) Show that $\mathbb{E}g^2 = e^{b^2} - 1$, $\mathbb{E}g^3 = e^{3b^2} - 3e^{b^2} + 2$, and $K = \frac{a^2}{2}\mathbb{E}g^2 - \frac{a^3}{3}\mathbb{E}g^3 + O(a^4)$ uniformly in $|b| \le 2$. (b) Using $\psi(t) \ge t^2/(2(1+t_+))$, $\psi(t) \le \frac{t^2}{2}$ for $t \ge 0$ and $\psi(t) \le t^2/(2(1+t))$ for $-1 < t < 0$, show that $\frac{a^2}{2}\mathbb{E}[g^2/(1+g_+)] \le K \le a^2(e^{b^2}-1)$. (c) Show that the kth Taylor coefficient of $a \mapsto K(a,b)$ at 0 is $(-1)^k\mathbb{E}[g^k]/k$, at least $2^{-k-1}e^{k(k-1)b^2/2}/k$ for large even k, so that the series diverges for every $a > 0$ when $b \ne 0$. *(Hint: for (b), on $\{g < 0\}$, $1 + ag \ge 1 + g = e^{bx - b^2/2}$.)*

2.3 CONSOLIDATION. (The coefficient is regular.) Show that under the posterior $t = (ab - \hat{c})/s$ has a density proportional to $\phi(t)\,(\ell_n - \log|Z_n + t|)$ on $\{|Z_n + t| \le e^{\ell_n}\}$, and deduce from Lemma 2.4 that its total variation distance from $\mathcal{N}(0,1)$ is at most $3\kappa_n/(2(\ell_n - \kappa_n))$ when $\ell_n > \kappa_n$. Conclude that the posterior law of $(\log|a|, \log|b|)/\log n$ converges weakly, in probability, to the uniform law on the segment from $(-\frac{1}{2}, 0)$ to $(0, -\frac{1}{2})$ (Figure 2.2).

2.4 CONSOLIDATION. (The ReLU neuron obeys the same law.) Show that on $\{b > 0\}$ the energy of the ReLU neuron depends on (a,b) only through ab, with features $\mathrm{relu}(x_i)$, and on $\{b < 0\}$ only through $-ab$, with

features relu$(-x_i)$; deduce that Theorem 2.5(iii) holds for it. Where does the proof use analyticity?

2.5 BY HAND. (A generic truth.) Let the truth be $y = c_0 x + \varepsilon$ with $0 < |c_0| < 1$. (a) Using steps 2 to 4 of Computation 2.3, show that the posterior converges weakly, in probability, to the law on W_0 under which the sign of a is uniform, $\log|a|$ is uniform on $[\log|c_0|, 0]$, and $b = c_0/a$. (b) Show that its density with respect to arc length on W_0 is $|a|/(2\log\frac{1}{|c_0|}\sqrt{a^4 + c_0^2})$, the prior density divided by $\|\nabla(ab)\|$ and normalized. (c) Explain why neither the Bernstein–von Mises theorem nor the scale law holds.

2.6 BY HAND. ★ (Two units, one needed, without biases.) Let $f_w(x) = a_1\tanh(b_1 x) + a_2\tanh(b_2 x)$, $x \sim \mathrm{Unif}[-1,1]$, with the truth $a^*\tanh(b^* x)$, $a^* b^* \neq 0$. (a) Show with Theorem 2.2 that W_0 is the union of twelve lines in $\mathbb{R}^4$, and list them. (b) Compute the rank and the kernel of $I(w)$ at $w = (a^*, b^*, 0, \beta)$ for $\beta \notin \{0, \pm b^*\}$, for $\beta = b^*$ and for $\beta = 0$, and match the kernels with the lines through these points. (c) Near $(a^*, b^*, 0, 0)$, expand $f_w - f^*$ to first order in $a_1 - a^*$ and $b_1 - b^*$ and to lowest order in (a_2, b_2), and find a product model inside the network. *(Hint:* $\tanh(\beta x)$, $\tanh(b^* x)$ *and* $x\cosh^{-2}(b^* x)$ *are linearly independent when* $|\beta| \notin \{0, |b^*|\}$*: compare their complex poles, starting nearest to the origin.)*

2.7 BY HAND. (A plateau at the fold.) Run gradient flow on $K(a,b) = \frac{1}{2}(ab - c_0)^2$, $c_0 > 0$, from $(\varepsilon, \varepsilon)$ with $0 < \varepsilon^2 < c_0$. Show that $a = b$ for all $t \geq 0$, that $v = a^2$ solves $\dot v = 2v(c_0 - v)$, and that v reaches $\frac{c_0}{2}$ at the time $\frac{1}{2c_0}\log(\frac{c_0}{\varepsilon^2} - 1)$. How does the plateau depend on the distance of the start from the singular point?

2.8 VARIATION. ★ (A Gaussian prior.) Replace the uniform prior by the standard normal density on $\mathbb{R}^2$. Show that given $c = ab$ the density of $u = \log|a|$ is proportional to $\exp(-\frac{1}{2}(e^{2u} + c^2 e^{-2u}))$, and deduce that the scale law of Theorem 2.5 still holds. *(Hint: on* $[\log|c|, 0]$ *the exponent lies in* $[-1, 0]$*; outside, it falls off doubly exponentially.)*

2.9 VARIATION. (A prior that vanishes on one arm.) Let the prior be $\frac{h+1}{4}|a|^h$ on $[-1,1]^2$ with $h > 0$. (a) Show that the zeta function of $K = \frac{1}{2}a^2 b^2$ has poles at $-\frac{1}{2}$ and $-\frac{h+1}{2}$, so $\lambda = \frac{1}{2}$ and $m = 1$. (b) Show that given c the density of $u = \log|a|$ is proportional to e^{hu} on $[\log|c|, 0]$, and deduce that $\Pi_n(\log|a| \leq \tau\log n) \to 0$ in probability for every $\tau < 0$. Where does the posterior go?

2.10 MEASUREMENT. (The chain.) For $y = a_1 a_2 a_3 x$ at the truth 0, show that given $c = a_1 a_2 a_3$ the vector $(\log|a_i|)_{i\leq 3}$ is uniform on the simplex $\{u_i \leq 0,\ \sum_i u_i = \log|c|\}$, and that $\log|a_1|/\log n$ converges in law under the posterior to $-\frac{1}{2}B$, $B \sim \mathrm{Beta}(1,2)$, with distribution function $(1+2\tau)^2$ on $[-\frac{1}{2}, 0]$ and quartiles -0.250, -0.146, -0.067. Adapt the sampler of `02-singular-scales.py` (the density of c now carries $(\log\frac{1}{|c|})^2$), measure the quartiles at $n = 10^4, 10^8$ and 10^{16}, and compare.

2.11 RESEARCH. (Scale laws without a coefficient.) For the tanh neuron the energy does not depend on (a, b) through one coefficient, and the proof of Theorem 2.5 does not apply. Prove or refute that $\log|a|/\log n$ converges to $\mathrm{Unif}(-\frac{1}{2}, 0)$ under its posterior, and do the same for the mixture. One route compares nK_n with a product model's energy in the coefficient $ab\sqrt{r(b)/3}$ by a sandwich as in Chapter 8. We know of no published treatment.

Notes and References

In the early 1990s the non-identifiability of networks was studied as a question about functions (Chen et al. 1993; Sussmann 1992), and the statistical side followed (Sections 2.3 and 2.7). Watanabe's analysis took up exactly these models early: Watanabe (2001) proved that the stochastic complexity of nonidentifiable learning machines has the form $\lambda \log n - (m-1)\log\log n + O(1)$, with $2\lambda \le d$ for multilayer networks, and Watanabe (2007) argued that almost all learning machines in use are singular, with neither the maximum-likelihood estimator nor the posterior converging to a normal distribution (for both we rely on the abstracts). The theory is developed in Watanabe (2009, 2018) and reviewed in Watanabe (2024); Wei et al. (2023) addressed it to deep-learning researchers.

Theorem 2.5, its corollaries and the two remarks are the book's own; we have not found the uniform law over scales in print, though its ingredients are elementary and its normalizer is the factor $(\log n)^{m-1}$ of Watanabe's free energy formula.

TRANSLATING THE LITERATURE. Watanabe (2009) and, following him, Wei et al. (2023) call a *model* regular when it is identifiable with a positive definite Fisher information, and strictly singular otherwise; our regular and singular describe a truth relative to a model, as in Watanabe (2010) and later; Wei et al. also integrate the Fisher information against the true distribution, which agrees with ours at true parameters. Watanabe (2024) writes $\theta \in \Theta$ for our $w \in W$ and $\pi(\theta)$ for our prior $\varphi(w)$. Farrugia-Roberts (2023) writes (a_i, b_i, c_i, d) as we do in (2.1), and $F[w]$ for the functional equivalence class of w, our W_0 (within W) when w is a true parameter. Van der Vaart (1998) writes θ_0, I_{θ_0} and Δ_{n,θ_0} for our w_0, $I(w_0)$ and Δ_n.

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3 The Free Energy by Hand

The whole theory in one coefficient

THE MODEL — $y = abx,\ y = ab^2x$

IN THIS CHAPTER. What does Bayesian model comparison charge a model whose parameters the data see only through their product? When the likelihood depends on the parameters through one scalar coefficient $c = g(w)$, the free energy is a Gaussian window of width about $n^{-1/2}$ integrated against the prior density of c. We compute it by hand for $y = abx$, $y = ab^2x$ and their relatives, and prove that a density $|c - c_0|^{2\lambda-1}(\log \frac{1}{|c-c_0|})^{m-1}$ at the truth gives $F_n - nS_n = \lambda \log n - (m-1)\log\log n + O_p(1)$, with an explicit limit law, convergence of means, and a correction of order $1/\log n$ when $m \geq 2$. Exact quadrature up to $n = 10^8$ confirms every term, the slow one included. The theorem needs a scalar coefficient, Gaussian noise and a fixed truth; Watanabe's theorem removes the first assumption at the price of resolution of singularities.

3.1 The question: what does a product cost?

A hidden unit with input weight b and output weight a contributes $a\tanh(bx) \approx abx$ near the point where it is switched off, and $a\,\mathrm{relu}(bx) = ab\,\mathrm{relu}(x)$ when $b > 0$ (Chapter 2): there the data see the two weights only through their product. The smallest model with this feature is the *product model*

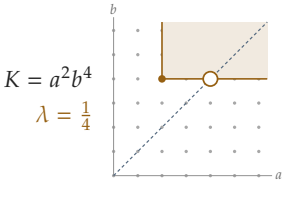
$$y = abx + \varepsilon, \qquad x \sim \mathcal{N}(0,1), \quad \varepsilon \sim \mathcal{N}(0,1), \quad (a,b) \in [-1,1]^2,$$

with the uniform prior, fitted to data that are pure noise, so that the true slope is $c_0 = 0$. A Bayesian compares models by their *marginal likelihoods*, the probabilities that they assigned to the data before seeing them. The negative logarithm of the marginal likelihood is the *free energy*[1]

$$F_n = -\log \int_W \varphi(w) \prod_{i=1}^{n} p(y_i \mid x_i, w)\, \mathrm{d}w = -\log \int_W \varphi(w) e^{-nL_n(w)}\, \mathrm{d}w,$$

and $F_n - nS_n = -\log \int_W \varphi(w) e^{-nK_n(w)}\, \mathrm{d}w$, where nS_n is the negative log-likelihood of the truth, is what the data charge the model for not knowing the truth in advance. The prologue measured it for the product model (Figure P.2): it grows like $\frac{1}{2}\log n - \log\log n$. This chapter derives that formula with every constant, for every regression $y = g(w)x + \varepsilon$, whose likelihood depends on its parameters through the one number $g(w)$.

[1] The name comes from statistical physics: e^{-F_n} is the normalizing constant, or partition function, of the posterior, a Gibbs measure with energy nL_n at temperature 1.



THE REGULAR ANSWER · the free energy. If the model is regular with d parameters, then $F_n = nL_n(\hat{w}) + \frac{d}{2}\log n + O_p(1)$, where $\hat{w}$ is the maximum-likelihood estimate, and since $nL_n(\hat{w}) - nS_n = O_p(1)$, also $F_n - nS_n = \frac{d}{2}\log n + O_p(1)$ (Schwarz 1978); Theorem 1.4 gives the bounded term up to $o_p(1)$. For $y = abx$, with $d = 2$, the data should charge $\log n$.

The regular answer, Schwarz's *Bayesian information criterion*, comes from Laplace's method (Chapter 1): each of the d directions in which the posterior has width $n^{-1/2}$ contributes $\frac{1}{2}\log n$. For $y = abx$ the method has nothing to expand around. The likelihood is maximized on the whole hyperbola $ab = \hat{c}$, where the Hessian of nL_n, the matrix $S_{xx}(b,a)^\top(b,a)$ with $S_{xx} = \sum_i x_i^2$, has rank one, so that Laplace's approximation, which contains half the logarithm of its determinant, returns $-\infty$. The Fisher information $\left(\begin{smallmatrix} b^2 & ab \\ ab & a^2 \end{smallmatrix}\right)$ is singular everywhere and zero at the center of the cross $ab = 0$ of true parameters (Computation 2.2 and Proposition 2.1), the posterior spreads along the cross (Theorem 2.5), and the prologue measured a slope of 0.39 against $\log n$, not 1.

The data see one number, the slope $c = ab$, so whatever replaces $\frac{d}{2}$ must depend on how the prior spreads that number. The question of this chapter is: *what is F_n for $y = abx$, exactly, and for which models does the same computation work?*

3.2 By hand: one coefficient, one integral

Write $S_{xx} = \sum_i x_i^2$, $S_{xy} = \sum_i x_i y_i$ and $S_{yy} = \sum_i y_i^2$. With unit noise variance the energy of the product model is $nL_n(a,b) = \frac{n}{2}\log(2\pi) + \frac{1}{2}\sum_i (y_i - abx_i)^2$, and the truth is $c_0 = 0$, so that $y_i = \varepsilon_i$.

COMPUTATION 3.1 · THE FREE ENERGY OF $y = abx$

1 *The energy depends on (a,b) only through $c = ab$.* Expanding the square,

$$nL_n(a,b) = \tfrac{n}{2}\log(2\pi) + \tfrac{1}{2}S_{yy} - cS_{xy} + \tfrac{1}{2}c^2 S_{xx}.$$

At the truth $c = 0$ this is $nS_n = \frac{n}{2}\log(2\pi) + \frac{1}{2}S_{yy}$. Completing the square with $\hat{c} = S_{xy}/S_{xx}$ and $Z_n = S_{xy}/\sqrt{S_{xx}}$ gives

$$nK_n(a,b) = \frac{S_{xx}}{2}(ab - \hat{c})^2 - \frac{Z_n^2}{2}.$$

2 *The data enter through two numbers.* Given the inputs, $S_{xy} = \sum_i x_i \varepsilon_i$ is normal with mean 0 and variance S_{xx}. So Z_n is standard normal, exactly and for every n, and independent of S_{xx}, which has the chi-square distribution with n degrees of freedom. As a function of c, the likelihood is a Gaussian window of width $S_{xx}^{-1/2} \approx n^{-1/2}$ centered Z_n widths away from the truth.

3 *The density of $c = ab$.* For $|a|$ and $|b|$ independent and uniform on $[0,1]$, and $0 < s \le 1$,

$$\mathbb{P}(|ab| \le s) = \mathbb{P}(|a| \le s) + \int_s^1 \frac{s}{a}\,\mathrm{d}a = s + s\log\frac{1}{s}.$$

Differentiating, and splitting the mass evenly between the two signs, c has the density $\rho(c) = \frac{1}{2}\log\frac{1}{|c|}$ on $[-1,1]$.

4 *One integral.* Integrating e^{-nK_n} against the prior is integrating the window against ρ:

$$F_n - nS_n = -\frac{Z_n^2}{2} - \log\int_{-1}^{1} e^{-S_{xx}(c-\hat{c})^2/2}\,\tfrac{1}{2}\log\tfrac{1}{|c|}\,\mathrm{d}c.$$

Substitute $c = t/\sqrt{S_{xx}}$, so that $\sqrt{S_{xx}}(c-\hat{c}) = t - Z_n$, $\mathrm{d}c = \mathrm{d}t/\sqrt{S_{xx}}$ and $\log\frac{1}{|c|} = \frac{1}{2}\log S_{xx} - \log|t|$:

$$F_n - nS_n = \tfrac{1}{2}\log S_{xx} - \frac{Z_n^2}{2} - \log\int_{|t|<\sqrt{S_{xx}}} e^{-(t-Z_n)^2/2}\,\tfrac{1}{2}\big(\tfrac{1}{2}\log S_{xx} - \log|t|\big)\,\mathrm{d}t.$$

5 *The integral is elementary.* Dropping the restriction $|t| < \sqrt{S_{xx}}$ changes the integral by at most a constant times $e^{-S_{xx}/8}\log S_{xx}$ when $|Z_n| \le \frac{1}{2}\sqrt{S_{xx}}$. With $U \sim \mathcal{N}(0,1)$ and $h(z) = \mathbb{E}\log|z+U|$, the two Gaussian integrals are $\sqrt{2\pi}$ and $\sqrt{2\pi}\,h(Z_n)$, so that, up to that exponentially small error,

$$F_n - nS_n = \tfrac{1}{2}\log S_{xx} - \log\big(\tfrac{1}{2}\log S_{xx} - h(Z_n)\big) - \frac{Z_n^2}{2} + \tfrac{1}{2}\log\tfrac{2}{\pi}.$$

6 *Read off the asymptotics.* Since $S_{xx} = n(1 + O_p(n^{-1/2}))$ and $h(Z_n) = O_p(1)$, the second term is $-\log\log n + \log 2 + O_p(1/\log n)$, and

$$F_n - nS_n = \tfrac{1}{2}\log n - \log\log n + \tfrac{1}{2}\log\tfrac{8}{\pi} - \frac{Z_n^2}{2} + O_p\Big(\frac{1}{\log n}\Big).$$

The exactness of step 2 is special to Gaussian noise.[2] At $n = 10^4$, with $S_{xx} = n$ and $Z_n = 0$, step 5 gives $F_n - nS_n = 4.6052 - \log 5.2404 - 0.2258 = 2.7230$ nats, because $h(0) = \mathbb{E}\log|U| = -\frac{1}{2}(\gamma + \log 2) = -0.6352$, with $\gamma = 0.5772$ Euler's constant.[3] Step 6 gives 2.8522; the difference is the $O_p(1/\log n)$ term. The regular plane $y = ax_1 + bx_2$, with the same prior and at the same point, pays $\log n + \log\frac{2}{\pi} = 8.7588$ nats: the product model, with as many parameters, is cheaper by six nats, a factor of about 420 in marginal likelihood.

WHAT ONE HAD TO NOTICE. Three things. The likelihood depends on (a,b) only through $c = ab$, so the integral over the square is an integral against

[2] With other noise of unit variance the Gaussian model no longer contains the truth, and nS_n must be replaced by $nL_n(w_0)$, the negative log-likelihood of the model at $c = 0$. The other steps hold unchanged, but Z_n is standard normal only in the limit, by the central limit theorem.

[3] Differentiate $\mathbb{E}|U|^s = 2^{s/2}\Gamma(\frac{s+1}{2})/\sqrt{\pi}$ at $s = 0$, and use $\Gamma'(\frac{1}{2})/\Gamma(\frac{1}{2}) = -\gamma - 2\log 2$.

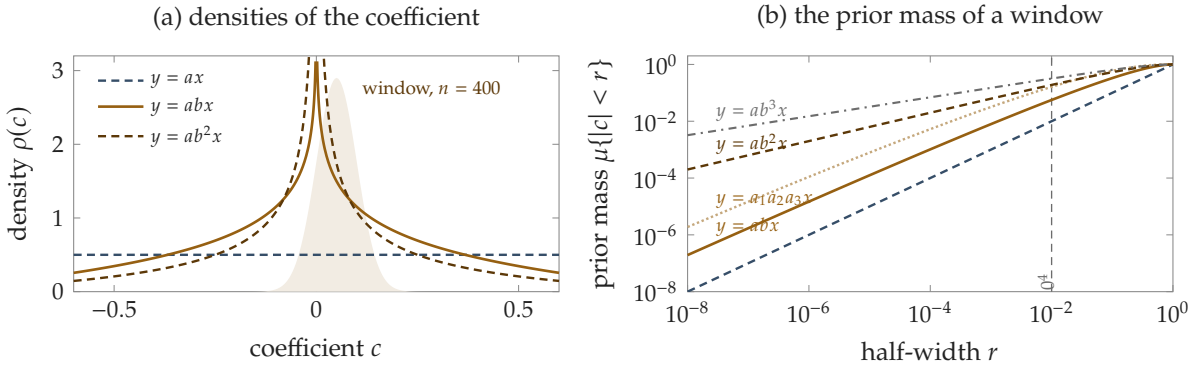


Figure 3.1. The prior seen through the coefficient, for the uniform prior on $[-1,1]^d$. (a) The density ρ of $c = g(w)$ for the regular line $y = ax$ (flat), the product model $y = abx$ (a logarithmic spike) and the cusp model $y = ab^2x$ (a power spike), with a likelihood window $e^{-n(c-\hat{c})^2/2}$ for $n = 400$ and $\hat{c} = 0.05$ (shaded, arbitrary vertical scale). (b) The prior mass $\mu\{|c| < r\}$ of a window of half-width r, from the closed forms of Computation 3.2; its slope tends to 2λ: 1 for the first three models (up to logarithms), $\frac{1}{2}$ and $\frac{1}{3}$ for the cusps. The free energy is about minus the logarithm of the mass at $r = n^{-1/2}$ (dashed: $n = 10^4$). A Monte Carlo check with 10^7 prior draws (seed 31) agrees with all twenty values tested, at $r = 10^{-1}, \dots, 10^{-4}$, within its 95% Clopper–Pearson intervals. Script: `figures/scripts/03-coefficient-mass.py`.

the law of c under the prior. The data reduce to S_{xx} and Z_n, and Z_n is exactly standard normal. And the cross of true parameters has become a logarithmic singularity of ρ at $c = 0$, which a window of width $n^{-1/2}$ sees as the factor $\frac{1}{2}\log n$: that factor is the $-\log\log n$. The same facts hold for every model $y = g(w)x + \varepsilon$ with a scalar coefficient; only the density changes (Computation 3.2, Figure 3.1).

COMPUTATION 3.2 · DENSITIES OF COEFFICIENTS

1 *The chain* $c = a_1 a_2 \cdots a_L$. For a_i uniform on $[-1,1]$, $\mathbb{P}(-\log|a_i| > u) = \mathbb{P}(|a_i| < e^{-u}) = e^{-u}$, so $-\log|a_i|$ is exponential with mean 1, and $-\log|c|$, a sum of L independent such variables, has the Gamma density $u^{L-1}e^{-u}/(L-1)!$. Changing variables to $s = e^{-u}$, and splitting the mass between the signs,

$$\rho_L(c) = \frac{1}{2(L-1)!}\left(\log\frac{1}{|c|}\right)^{L-1} \qquad (|c| \le 1).$$

For $L = 1$ this is the uniform density $\frac{1}{2}$ of the regular line; for $L = 2$ it is Computation 3.1.

2 *The cusp* $c = ab^2$. For $0 < s \le 1$, $\mathbb{P}(|a|b^2 \le s) = \mathbb{P}(b^2 \le s) + \int_{\sqrt{s}}^1 \frac{s}{b^2}\,\mathrm{d}b = 2\sqrt{s} - s$, so $\rho(c) = \frac{1}{2}(|c|^{-1/2} - 1)$.

3 *The family* $c = ab^k$, $k \ge 2$. In the same way $\mathbb{P}(|a||b|^k \le s) = s^{1/k} + s\int_{s^{1/k}}^1 b^{-k}\,\mathrm{d}b = \frac{ks^{1/k}-s}{k-1}$, so $\rho(c) = \frac{|c|^{1/k-1}-1}{2(k-1)}$; for the cubic cusp $y = ab^3x$, $\rho(c) = \frac{1}{4}(|c|^{-2/3} - 1)$.

4 *Read off.* In each case $\rho(c) = |c|^{2\lambda-1}(\log \frac{1}{|c|})^{m-1}\ell(c)$ near $c = 0$, with $\ell(c) \to \ell_0 > 0$ and (λ, m, ℓ_0) equal to $(\frac{1}{2}, L, \frac{1}{2(L-1)!})$ for the chain, $(\frac{1}{4}, 1, \frac{1}{2})$ for the cusp, and $(\frac{1}{2k}, 1, \frac{1}{2(k-1)})$ for ab^k.

THE CUSP BY HAND. With the cusp's density, step 4 of Computation 3.1 becomes, after the same substitution and up to an exponentially small error,

$$F_n - nS_n = \tfrac{1}{4}\log S_{xx} - \frac{Z_n^2}{2} - \log\Big(\tfrac{1}{2}\int_{\mathbb{R}} e^{-(t-Z_n)^2/2}|t|^{-1/2}\,\mathrm{d}t - \tfrac{1}{2}\sqrt{2\pi}\,S_{xx}^{-1/4}\Big).$$

The subtracted term is $O(n^{-1/4})$, so $F_n - nS_n = \frac{1}{4}\log n + O_p(1)$, with no $\log\log n$ because the density has no logarithm. At $Z_n = 0$ the integral is $2^{1/4}\Gamma(\frac{1}{4}) = 4.3116$, and at $n = 10^4$ the free energy is 1.53 nats (1.59 with the $O(n^{-1/4})$ term): cheaper still.

THE RULE, AND WHY IT HOLDS. In both computations the exponent of $|c - c_0|$ in the density gave the coefficient of $\log n$, and the power of the logarithm gave that of $\log\log n$: a density $|c - c_0|^{2\lambda-1}(\log \frac{1}{|c-c_0|})^{m-1}$ at the truth gives $F_n - nS_n \approx \lambda\log n - (m-1)\log\log n$. Heuristically, the window keeps the coefficients within about $n^{-1/2}$ of the truth and discards the rest, so $e^{-(F_n - nS_n)}$ is, up to a bounded factor, the prior mass of the window (Figure 3.1(b)):

$$\mu\{|c - c_0| < n^{-1/2}\} \asymp \int_0^{n^{-1/2}} \delta^{2\lambda-1}\Big(\log\frac{1}{\delta}\Big)^{m-1}\,\mathrm{d}\delta \asymp n^{-\lambda}\Big(\frac{1}{2}\log n\Big)^{m-1}.$$

Its negative logarithm is $\lambda\log n - (m-1)\log\log n + O(1)$, and Section 3.3 turns the heuristic into a proof. This is the singular version of the Bayesian *Occam factor* of MacKay (2003, ch. 28): not a ratio of a posterior width to a prior width, but the prior mass of a window around a singular point.[4] The exponent λ is the *learning coefficient* and m its *multiplicity*; as numbers attached to K and the prior they are the *real log canonical threshold* and its multiplicity, the same two numbers under another name (Proposition 3.7 connects the definitions).

[4] For the product model the window keeps the region $|ab| \lesssim n^{-1/2}$ around the cross, of area about $n^{-1/2}\log n$: the arms are $n^{-1/2}$ thick far from the center and fatter near it, which is where the logarithm comes from.

3.3 The theorem: from the density to the free energy

The computations used three facts: the likelihood factors through a scalar coefficient, the data enter through a window whose position is standard normal, and the prior density of the coefficient has a power-and-logarithm singularity at the truth. We now prove the rule under exactly these hypotheses, for a *one-coefficient model*.

Assumption 3.1 (A one-coefficient regression model).

(i) The inputs $x_1, x_2, \ldots$ are independent real random variables with a common distribution, and $\mathbb{E}[x^2] = 1$.

(ii) The outputs are $y_i = c_0 x_i + \varepsilon_i$ for a fixed $c_0 \in \mathbb{R}$, with $\varepsilon_1, \varepsilon_2, \ldots$ independent $\mathcal{N}(0,1)$ variables, independent of the inputs.

(iii) The model is $p(y \mid x, w) = (2\pi)^{-1/2}\exp(-\tfrac{1}{2}(y - g(w)x)^2)$ with a measurable $g\colon W \to \mathbb{R}$ and a prior probability density φ on $W \subseteq \mathbb{R}^d$. The law of the coefficient $c = g(w)$ when $w \sim \varphi$ is denoted μ.

(iv) There are $\eta \in (0,1)$, $\lambda > 0$, an integer $m \geq 1$, and a bounded function ℓ on $0 < |c - c_0| < \eta$ with one-sided limits $\ell_+ = \lim_{c \downarrow c_0} \ell(c)$ and $\ell_- = \lim_{c \uparrow c_0} \ell(c)$ and $\ell_+ + \ell_- > 0$, such that on the interval $|c - c_0| < \eta$ the law μ has the density

$$\rho(c) = |c - c_0|^{2\lambda - 1}\Big(\log \frac{1}{|c - c_0|}\Big)^{m-1}\ell(c) \qquad (0 < |c - c_0| < \eta).$$

The two-sided case is ℓ continuous and positive at c_0; the case $\ell_- = 0$ covers a coefficient that cannot fall below the truth, such as $c = a^2$ at $c_0 = 0$ (Exercise 3.7). Nothing is assumed about μ away from c_0, and neither compactness of W nor analyticity of g, which Watanabe's general theorem assumes, is needed.[5] The inputs need only a second moment; the noise must be Gaussian of known variance, which makes the energy exactly quadratic in c. Write

$$S_{xx} = \sum_{i=1}^{n} x_i^2, \qquad T_n = \sum_{i=1}^{n} x_i \varepsilon_i, \qquad \hat{c} = c_0 + \frac{T_n}{S_{xx}}, \qquad Z_n = \frac{T_n}{\sqrt{S_{xx}}}$$

on the event $\{S_{xx} > 0\}$, and $Z_n = 0$ on $\{S_{xx} = 0\}$; $\hat{c} = S_{xy}/S_{xx}$ is the least-squares slope.

[5] A prior with an atom at c_0, such as a spike-and-slab prior that switches the coefficient off with probability $p > 0$, violates part 4. Then the integral in (3.2) below is at least $pe^{-Z_n^2/2}$, and $F_n - nS_n \leq \log \frac{1}{p}$ stays bounded.

Lemma 3.2 (Exact representation). *Assume parts 1–3 of Assumption 3.1. On the event $\{S_{xx} > 0\}$, for every $w \in W$,*

$$nK_n(w) = \frac{S_{xx}}{2}\big(g(w) - \hat{c}\big)^2 - \frac{Z_n^2}{2}, \tag{3.1}$$

and

$$F_n - nS_n = -\frac{Z_n^2}{2} - \log \int_{\mathbb{R}} e^{-S_{xx}(c - \hat{c})^2/2}\,\mu(\mathrm{d}c). \tag{3.2}$$

On $\{S_{xx} = 0\}$, $F_n - nS_n = 0$. Conditionally on $x_1, \dots, x_n$, the variable Z_n is standard normal on $\{S_{xx} > 0\}$; if $\mathbb{P}(x = 0) = 0$, then $Z_n \sim \mathcal{N}(0,1)$ is independent of S_{xx}.

Proof. The plan is to expand the squares, complete the square in the coefficient, and push the prior forward to the coefficient. Write $c = g(w)$. Since $y_i - cx_i = \varepsilon_i - (c - c_0)x_i$, expanding the squares in $nL_n(w) = \frac{n}{2}\log(2\pi) + \frac{1}{2}\sum_i (y_i - cx_i)^2$ and in $nS_n = \frac{n}{2}\log(2\pi) + \frac{1}{2}\sum_i \varepsilon_i^2$ gives $nK_n(w) = \frac{1}{2}S_{xx}(c - c_0)^2 - (c - c_0)T_n$. On $\{S_{xx} > 0\}$, completing the square gives (3.1), because $\hat{c} - c_0 = T_n/S_{xx}$ and $T_n^2/S_{xx} = Z_n^2$; on $\{S_{xx} = 0\}$ every x_i vanishes, so $K_n \equiv 0$ and $F_n - nS_n = 0$. The integrand of $F_n - nS_n = -\log \int_W \varphi e^{-nK_n}$ depends on w only through $g(w)$, and $\int_W f(g(w))\varphi(w)\,\mathrm{d}w = \int_{\mathbb{R}} f(c)\,\mu(\mathrm{d}c)$ for every measurable $f \geq 0$, by the definition of μ; with $f(c) = \exp(-S_{xx}(c -$

$\hat{c})^2/2 + Z_n^2/2)$ this is (3.2). Finally, given the inputs, T_n is a fixed linear combination of the independent $\mathcal{N}(0,1)$ variables ε_i, so it is $\mathcal{N}(0, S_{xx})$, and Z_n is $\mathcal{N}(0,1)$ on $\{S_{xx} > 0\}$. If $\mathbb{P}(x = 0) = 0$, then $S_{xx} > 0$ almost surely, and since this conditional law does not depend on the inputs, Z_n is independent of them. ∎

The lemma reduces the free energy to one dimension, at every n. The next lemma is the analysis: what a *Gaussian window* of width $s^{-1/2}$ sees when it is placed z widths from the singular point of ρ.

Lemma 3.3 (A Gaussian window on a singular density). *Let ρ, η, λ, m, ℓ and $\ell_\pm$ be as in part 4 of Assumption 3.1, and write $\ell_{\operatorname{sgn} t}$ for ℓ_+ when $t > 0$ and for ℓ_- when $t < 0$. For $s > 1$ and $z \in \mathbb{R}$ put*

$$J_s(z) = s^{\lambda}\left(\tfrac{1}{2}\log s\right)^{1-m} \int_{|c-c_0|<\eta} e^{-s(c-c_0-z/\sqrt{s})^2/2}\,\rho(c)\,\mathrm{d}c,$$

$$J(z) = \int_{\mathbb{R}} e^{-(t-z)^2/2}\,|t|^{2\lambda-1}\ell_{\operatorname{sgn} t}\,\mathrm{d}t.$$

Then J is continuous and positive, and $\sup_{|z|\le M}|J_s(z) - J(z)| \to 0$ as $s \to \infty$, for every $M > 0$. Moreover, there is a constant C such that $J_s(z) \le C(1+|z|)^{(2\lambda-1)_+}$ for all $s \ge e^2$ and $z \in \mathbb{R}$.

Proof. The plan is to rescale the coefficient by $\sqrt{s}$, which turns the window into a fixed Gaussian and the density into a function that converges pointwise, and then to use dominated convergence.

Rescaling. Substitute $c = c_0 + t/\sqrt{s}$. Then $|c-c_0|^{2\lambda-1} = s^{1/2-\lambda}|t|^{2\lambda-1}$, $\mathrm{d}c = s^{-1/2}\,\mathrm{d}t$, and $\log\frac{1}{|c-c_0|} = \frac{1}{2}\log s\,(1+u_s(t))$ with $u_s(t) = \log\frac{1}{|t|}\big/\frac{1}{2}\log s$. The powers of s cancel, and $J_s(z) = \int_{\mathbb{R}} e^{-(t-z)^2/2}|t|^{2\lambda-1} f_s(t)\,\mathrm{d}t$ with

$$f_s(t) = \mathbf{1}\{0 < |t| < \eta\sqrt{s}\}\,\big(1+u_s(t)\big)^{m-1}\,\ell\big(c_0 + t/\sqrt{s}\big).$$

A bound on f_s. Let $\bar{\ell} = \sup\ell$, and let $s \ge e^2$, so that $\frac{1}{2}\log s \ge 1$. If $0 < |t| < \eta\sqrt{s}$, then $\log\frac{1}{|t|} > -\frac{1}{2}\log s$, because $\eta < 1$, so $1 + u_s(t) > 0$; moreover $1 + u_s(t) \le 1$ when $|t| \ge 1$, and $1 + u_s(t) \le 1 + \log\frac{1}{|t|}$ when $|t| < 1$. With $\log_+ v = \max(\log v, 0)$, this gives

$$0 \le f_s(t) \le \bar{\ell}\,b(t), \qquad b(t) = \big(1 + \log_+\tfrac{1}{|t|}\big)^{m-1}.$$

Convergence. Fix $t \ne 0$. As $s \to \infty$, eventually $|t| < \eta\sqrt{s}$, $u_s(t) \to 0$, and $c_0 + t/\sqrt{s}$ tends to c_0 from the side of $\operatorname{sgn} t$; so $f_s(t) \to \ell_{\operatorname{sgn} t}$. For $|z| \le M$ and every t, $(t-z)^2 \ge (|t|-M)_+^2$, so

$$\sup_{|z|\le M}|J_s(z) - J(z)| \le \int_{\mathbb{R}} e^{-(|t|-M)_+^2/2}|t|^{2\lambda-1}|f_s(t) - \ell_{\operatorname{sgn} t}|\,\mathrm{d}t.$$

The integrand tends to 0 for every $t \ne 0$. It is at most $2\bar{\ell}$ times the function $e^{-(|t|-M)_+^2/2}|t|^{2\lambda-1}b(t)$, which is integrable because $2\lambda - 1 > -1$, because b grows only like a power of $\log\frac{1}{|t|}$ near 0, and because of the Gaussian

factor. Dominated convergence gives the uniform convergence; the same domination shows that J is continuous, and $J > 0$ because $\ell_\pm \geq 0$ and $\ell_+ + \ell_- > 0$.

The bound. By the bound on f_s, $J_s(z) \leq \bar{\ell} \int_{\mathbb{R}} e^{-(t-z)^2/2}|t|^{2\lambda-1}b(t)\,\mathrm{d}t$. The part over $|t| \leq 1$ is at most $\int_{|t|\leq 1}|t|^{2\lambda-1}b(t)\,\mathrm{d}t < \infty$. On $|t| > 1$, $b = 1$; if $2\lambda - 1 \leq 0$ that part is at most $\sqrt{2\pi}$, and if $p = 2\lambda - 1 > 0$, then $|t|^p \leq 2^p(|t-z|^p + |z|^p)$ makes it at most $2^p\left(\int e^{-v^2/2}|v|^p\,\mathrm{d}v + \sqrt{2\pi}|z|^p\right)$. Both cases give the bound $C(1+|z|)^{(2\lambda-1)_+}$. ∎

Theorem 3.4 (The free energy of a one-coefficient model). *Under Assumption 3.1, let $R_n = F_n - nS_n - \lambda \log n + (m-1)\log\log n$, and for $z \in \mathbb{R}$ let*

$$\chi(z) = (m-1)\log 2 - \log \int_{\mathbb{R}} e^{-t^2/2+zt}\,|t|^{2\lambda-1}\ell_{\operatorname{sgn} t}\,\mathrm{d}t. \tag{3.3}$$

Let Z_n be as in Lemma 3.2, which is standard normal given the inputs on an event whose probability tends to one. Then:

(i) $R_n - \chi(Z_n) \to 0$ in probability. Consequently

$$F_n - nS_n = \lambda \log n - (m-1)\log\log n + O_p(1),$$

and R_n converges in law to $\chi(Z)$, where $Z \sim \mathcal{N}(0,1)$.

(ii) $\mathbb{E}[R_n] \to \mathbb{E}[\chi(Z)]$, and $\mathbb{E}|\chi(Z)| < \infty$.

Proof. The plan is to split the integral in (3.2) into the window $|c - c_0| < \eta$, which Lemma 3.3 controls, and the rest, which the Gaussian factor makes negligible; part 2 then follows from bounds that make (R_n) uniformly integrable. Write $S = S_{xx}$.

Step 1: decomposition. On $\{S > 1\}$ write the integral in (3.2) as $A_n + B_n$, with A_n the part over $|c - c_0| < \eta$. Since $\hat{c} = c_0 + Z_n/\sqrt{S}$, the definition of J_s gives $A_n = S^{-\lambda}(\frac{1}{2}\log S)^{m-1}J_S(Z_n)$, and the identity $e^{-t^2/2+zt} = e^{z^2/2}e^{-(t-z)^2/2}$ turns (3.3) into $\chi(z) = (m-1)\log 2 - \frac{z^2}{2} - \log J(z)$. Inserting A_n into (3.2) and using $\log(\frac{1}{2}\log S) - \log\log n = \log\frac{\log S}{\log n} - \log 2$, we get on $\{S > 1\}$

$$R_n = \chi(Z_n) + \lambda \log\frac{S}{n} - (m-1)\log\frac{\log S}{\log n} - \log\frac{J_S(Z_n)}{J(Z_n)} - \log\Big(1 + \frac{B_n}{A_n}\Big). \tag{3.4}$$

Step 2: the corrections vanish. By the strong law of large numbers and part 1 of Assumption 3.1, $S/n \to 1$ almost surely, so $\mathbb{P}(S > 1) \to 1$ and the second and third terms of (3.4) tend to 0. Fix $\epsilon > 0$ and choose M with $\mathbb{P}(|Z_n| > M) \leq \epsilon$ for all n, which is possible because Z_n is standard normal or 0 given the inputs. With $j_M = \min_{|z|\leq M} J(z) > 0$, on $\{|Z_n| \leq M\}$ we have $|J_S(Z_n)/J(Z_n) - 1| \leq j_M^{-1}\sup_{|z|\leq M}|J_S(z) - J(z)|$, which tends to 0 almost surely by Lemma 3.3. For the last term, if $|\hat{c} - c_0| \leq \eta/2$ then $|c - \hat{c}| \geq \eta/2$ whenever $|c - c_0| \geq \eta$, so $B_n \leq e^{-S\eta^2/8}$, because μ is a probability measure; and $A_n \geq S^{-\lambda}J_S(Z_n)$ when $S \geq e^2$. For large n the event $\{n/2 \leq S \leq 2n,\ |Z_n| \leq M,\ J_S(Z_n) \geq j_M/2\}$ has probability at least $1 - 2\epsilon$, and on it

$|\hat c - c_0| = |Z_n|/\sqrt{S} \le \eta/2$ and $B_n/A_n \le 2j_M^{-1}(2n)^\lambda e^{-n\eta^2/16} \to 0$. As ϵ was arbitrary, $R_n - \chi(Z_n) \to 0$ in probability. Since Z_n is standard normal given the inputs on $\{S > 0\}$ and $\mathbb{P}(S > 0) \to 1$, $\chi(Z_n)$ converges in law to $\chi(Z)$, and so does R_n, by Slutsky's lemma (Definition A.13); a sequence that converges in law is bounded in probability. This proves part 1.

Step 3: an upper bound. Part 2 follows once (R_n) is uniformly integrable. Suppose $\ell_+ > 0$ (the case $\ell_- > 0$ is the mirror image), and shrink η so that $\ell \ge \ell_+/2$ on $(c_0, c_0 + \eta)$. Let $n > \eta^{-2}$ and $r = n^{-1/2}$. For $c_0 < c < c_0 + r$, the inequality $(c - \hat c)^2 \le 2(c - c_0)^2 + 2(\hat c - c_0)^2$ gives $S(c - \hat c)^2/2 \le S/n + Z_n^2$, and since $\log \frac{1}{\delta} \ge \log \frac{1}{r}$ for $\delta \le r$,

$$\mu\{c_0 < c < c_0 + r\} \ge \frac{\ell_+}{2}\Big(\log\frac{1}{r}\Big)^{m-1}\int_0^r \delta^{2\lambda-1}\,\mathrm{d}\delta = \frac{\ell_+}{4\lambda}\,n^{-\lambda}\big(\tfrac12\log n\big)^{m-1}.$$

So the integral in (3.2) is at least $e^{-S/n - Z_n^2}$ times this mass, and on $\{S > 0\}$

$$R_n \le \frac{Z_n^2}{2} + \frac{S}{n} + C_1, \qquad C_1 = (m-1)\log 2 + \log\frac{4\lambda}{\ell_+}; \tag{3.5}$$

on $\{S = 0\}$, $R_n = -\lambda\log n + (m-1)\log\log n$, which is negative for large n.

Step 4: a lower bound. The integral in (3.2) is at most $\mu(\mathbb{R}) = 1$, so $R_n \ge -Z_n^2/2 - \lambda\log n$ always. On $G_n = \{S \ge n/2,\ |Z_n| \le \eta\sqrt{S}/2\}$ we do better: there $B_n \le e^{-n\eta^2/16}$, as in Step 2, and $A_n \le C(1 + |Z_n|)^p S^{-\lambda}(\frac12\log S)^{m-1}$ with $p = (2\lambda - 1)_+$, by Lemma 3.3. Since $s \mapsto s^{-\lambda}(\log s)^{m-1}$ decreases for $s \ge e^{(m-1)/\lambda}$, for large n and $S \ge n/2$ we have $S^{-\lambda}(\frac12\log S)^{m-1} \le 2^\lambda n^{-\lambda}(\frac12\log n)^{m-1}$, and also $e^{-n\eta^2/16} \le n^{-\lambda}(\frac12\log n)^{m-1}$. Hence, for large n, on G_n,

$$R_n \ge -\frac{Z_n^2}{2} - p\log(1 + |Z_n|) - C_2, \qquad C_2 = \log(2^\lambda C + 1) - (m-1)\log 2. \tag{3.6}$$

Step 5: uniform integrability. By (3.5), (3.6) and Step 4, for large n,

$$|R_n| \le Y_n + \lambda\log n\,\mathbf{1}_{G_n^c}, \qquad Y_n = \frac{Z_n^2}{2} + \frac{S}{n} + p\log(1 + |Z_n|) + |C_1| + |C_2|.$$

The family (Y_n) is uniformly integrable, because $\mathbb{P}(|Z_n| > u) \le \mathbb{P}(|Z| > u)$ for every $u > 0$ and $S/n \to 1$ in mean by the law of large numbers in L^1. The second term has mean $\lambda\log n\,\mathbb{P}(G_n^c) \to 0$, because $\mathbb{P}(G_n^c)$ decays exponentially: given the inputs on $\{S \ge n/2\}$, $\mathbb{P}(|Z_n| > \eta\sqrt{S}/2) \le 2e^{-\eta^2 n/16}$, and $\mathbb{P}(S < n/2) \le e^{\theta n/2}(\mathbb{E}e^{-\theta x^2})^n \le e^{-\theta n/4}$ for a small $\theta > 0$ with $\mathbb{E}[e^{-\theta x^2}] \le 1 - \frac34\theta$, which exists because $(1 - e^{-\theta x^2})/\theta$ increases to x^2 as $\theta \downarrow 0$ (monotone convergence). Nonnegative variables whose means tend to 0 form a uniformly integrable sequence, and so does a sum of two such sequences. Hence (R_n) is uniformly integrable, convergence in law gives $\mathbb{E}[R_n] \to \mathbb{E}[\chi(Z)]$, and $\mathbb{E}|\chi(Z)| \le \sup_n \mathbb{E}|R_n| < \infty$. ∎

The limit laws of the running models are values of Kummer's function ${}_1F_1(a;b;v) = \sum_{k\ge0}\frac{(a)_k}{(b)_k}\frac{v^k}{k!}$, where $(a)_k = a(a+1)\cdots(a+k-1)$.

model	coefficient	$\rho(c), 0<\lvert c\rvert<1$	$\ell(c)$	λ	m	$\ell_\pm$	$\chi(0)$	$\mathbb{E}\chi(Z)$	$\frac{d}{2}$
$y=ax$	a	$\frac12$	$\frac12$	$\frac12$	1	$\frac12$	−0.226	−0.726	$\frac12$
$y=abx$	ab	$\frac12\log\frac{1}{\lvert c\rvert}$	$\frac12$	$\frac12$	2	$\frac12$	0.467	−0.033	1
$y=a_1a_2a_3x$	$a_1a_2a_3$	$\frac14\log^2\frac{1}{\lvert c\rvert}$	$\frac14$	$\frac12$	3	$\frac14$	1.854	1.354	$\frac32$
$y=ab^2x$	ab^2	$\frac12(\lvert c\rvert^{-1/2}-1)$	$\frac12(1-\lvert c\rvert^{1/2})$	$\frac14$	1	$\frac12$	−0.768	−1.073	1
$y=ab^3x$	ab^3	$\frac14(\lvert c\rvert^{-2/3}-1)$	$\frac14(1-\lvert c\rvert^{2/3})$	$\frac16$	1	$\frac14$	−0.446	−0.671	1

Table 3.1. The running one-coefficient models at the truth 0, with the uniform prior on $[-1,1]^d$: the density of the coefficient, the factor ℓ of Assumption 3.1, the learning coefficient and multiplicity, and the limit law of R_n (Corollary 3.5). The value $\chi(0)$ is also the constant of the population free energy $-\log\int e^{-nK}\varphi = \lambda\log n-(m-1)\log\log n+\chi(0)+o(1)$. The means $\mathbb{E}\chi(Z)$ of the cusps are computed by 80-node Gauss–Hermite quadrature in `03-coefficient-free.py`; the others are $\chi(0)-\frac12$.

Corollary 3.5 (The running models). *Let the inputs and the noise be as in parts 1–2 of Assumption 3.1, let the prior be uniform on $[-1,1]^d$, and let $c_0=0$. For the five models of Table 3.1, Assumption 3.1 holds with the listed λ, m and $\ell_+=\ell_-$, and the limit law of Theorem 3.4 is*

$$\chi(z)=(m-1)\log 2-\log\Big(\ell_+\,2^\lambda\,\Gamma(\lambda)\;{}_1F_1\big(\lambda;\tfrac12;\tfrac{z^2}{2}\big)\Big).$$

For $\lambda=\frac12$ this is $(m-1)\log 2-\log(\ell_+\sqrt{2\pi})-\frac{z^2}{2}$. In particular, for the product model,

$$F_n-nS_n=\tfrac12\log n-\log\log n+\tfrac12\log\tfrac{8}{\pi}-\tfrac12 Z_n^2+o_p(1),$$
$$\mathbb{E}[F_n-nS_n]=\tfrac12\log n-\log\log n+\tfrac12\log\tfrac{8}{\pi}-\tfrac12+o(1).$$

Proof. The plan is to read the hypotheses off the densities computed by hand and to evaluate (3.3) as a power series. By Computations 3.1 and 3.2, in each case ℓ is bounded on $0<|c|<1$ and tends to the listed value as $c\to0$, so Assumption 3.1 holds with every $\eta<1$. For $\ell_+=\ell_-$ the integral in (3.3) is $2\ell_+\int_0^\infty t^{2\lambda-1}e^{-t^2/2}\cosh(zt)\,dt$. Expanding $\cosh(zt)$, integrating term by term (monotone convergence), and using $\int_0^\infty t^{a-1}e^{-t^2/2}\,dt=2^{a/2-1}\Gamma(\frac a2)$ (Definition A.4, after $\tau=t^2/2$), $\Gamma(\lambda+k)=\Gamma(\lambda)(\lambda)_k$ and $(2k)!=4^kk!(\frac12)_k$ gives

$$\ell_+2^\lambda\sum_{k\ge0}\Gamma(\lambda+k)\frac{2^kz^{2k}}{(2k)!}=\ell_+2^\lambda\Gamma(\lambda)\sum_{k\ge0}\frac{(\lambda)_k}{(\frac12)_k}\frac{(z^2/2)^k}{k!}.$$

For $\lambda=\frac12$ the series is $e^{z^2/2}$; for the product model $(m-1)\log2-\log(\frac12\sqrt{2\pi})=\frac12\log\frac{8}{\pi}$, and part 2 of Theorem 3.4 with $\mathbb{E}[Z^2/2]=\frac12$ gives the mean. ∎

The bronze curve of the prologue's Figure P.2 is this corollary at finite n: the formula of step 4 of Computation 3.1 with S_{xx} replaced by n, averaged over Z_n; its limit $\frac12\log n-\log\log n-\frac12+\log\frac{4}{\sqrt{2\pi}}$ is the corollary's, since

$\log\frac{4}{\sqrt{2\pi}} = \frac{1}{2}\log\frac{8}{\pi}$. For the chain of every length L the same proof gives $\chi(z) = \frac{1}{2}\log\frac{2^{2L-1}((L-1)!)^2}{\pi} - \frac{z^2}{2}$.

3.4 The theorem: the limit law and its slow approach

The limit law is a function of the position of the window in units of its width; its $-\frac{z^2}{2}$ part is the energy $nK_n = -Z_n^2/2$ of the least-squares fit. It is a limit in law only: R_n does not converge in probability, because Z_n and Z_{4n} are correlated but not equal (Exercise 3.4). And when $m \geq 2$ it is approached slowly, because the logarithm in ρ brings a correction of order $1/\log n$.

Proposition 3.6 (The approach to the limit). *Assume Assumption 3.1, $\mathbb{E}[x^4] < \infty$, and $|\ell(c) - \ell_{\operatorname{sgn}(c-c_0)}| \leq C|c-c_0|^\alpha$ for $0 < |c-c_0| < \eta$, with some $C, \alpha > 0$. Let $\kappa(z)$ be the mean of $\log|t|$ under the probability density on $\mathbb{R}$ proportional to $e^{-(t-z)^2/2}|t|^{2\lambda-1}\ell_{\operatorname{sgn} t}$. Then*

$$R_n = \chi(Z_n) + \frac{2(m-1)}{\log n}\,\kappa(Z_n) + O_p\Big(\frac{1}{(\log n)^2}\Big). \tag{3.7}$$

If $\lambda = \frac{1}{2}$ and $\ell_+ = \ell_-$, then $\kappa(z) = \mathbb{E}\log|z+U|$ with $U \sim \mathcal{N}(0,1)$, and $\mathbb{E}[\kappa(Z)] = -\gamma/2$ for $Z \sim \mathcal{N}(0,1)$.

Proof. The plan is to expand the factor $(1+u_s)^{m-1}$ in the proof of Lemma 3.3 to first order in $1/\log s$, uniformly over $|z| \leq M$, and to return to (3.4). Write $H = \frac{1}{2}\log s \geq 1$ and $u = \log\frac{1}{|t|}$, so that $(1+u_s(t))^{m-1} = 1+(m-1)u/H+r$ with $|r| \leq \sum_{k\geq 2}\binom{m-1}{k}|u|^k H^{-k} \leq (1+|u|)^{m-1}/H^2$, and write $\ell(c_0 + t/\sqrt{s}) = \ell_{\operatorname{sgn} t} + e(t)$ with $|e(t)| \leq C|t|^\alpha s^{-\alpha/2}$. With $J_1(z) = \int_{\mathbb{R}} \log|t|\; e^{-(t-z)^2/2}|t|^{2\lambda-1}\ell_{\operatorname{sgn} t}\,\mathrm{d}t$, inserting both expansions into the formula for J_s gives, uniformly over $|z| \leq M$,

$$J_s(z) = J(z) - \frac{m-1}{H}\,J_1(z) + O\Big(\frac{1}{H^2} + s^{-\alpha/2}\Big):$$

the error terms from r, from e, and from the region $|t| \geq \eta\sqrt{s}$ that f_s excludes are bounded by H^{-2}, by $s^{-\alpha/2}$ and by $e^{-(\eta\sqrt{s}-M)^2/4}$, each times an integral $\int e^{-(|t|-M)_+^2/4}|t|^{2\lambda-1}(1+|\log|t||)^{m-1}(1+|t|^\alpha)\,\mathrm{d}t < \infty$. Since $J \geq j_M > 0$ and $\kappa = J_1/J$ is bounded on $|z| \leq M$, taking logarithms gives $-\log(J_s(z)/J(z)) = (m-1)\kappa(z)/H + O(H^{-2} + s^{-\alpha/2})$ uniformly there.

In (3.4), Chebyshev's inequality and $\mathbb{E}[x^4] < \infty$ give $S/n = 1 + O_p(n^{-1/2})$, so the second term is $O_p(n^{-1/2})$, the third is $O_p(n^{-1/2}/\log n)$, and $2/\log S = 2/\log n + O_p(n^{-1/2}(\log n)^{-2})$; Step 2 of the proof of Theorem 3.4 showed that the last term is smaller than every power of n with probability as close to one as we like. Since $\mathbb{P}(|Z_n| > M)$ is small uniformly in n for large M, the expansion at $z = Z_n$ gives (3.7). If $\lambda = \frac{1}{2}$ and $\ell_+ = \ell_-$, the density defining κ is that of $\mathcal{N}(z,1)$, so $\kappa(z) = \mathbb{E}\log|z+U|$; for independent Z and U, $Z+U$ has the law of $\sqrt{2}\,V$ with $V \sim \mathcal{N}(0,1)$, and $\mathbb{E}[\kappa(Z)] = \frac{1}{2}\log 2 - \frac{1}{2}(\gamma + \log 2) = -\frac{\gamma}{2}$. ∎

For the product model and every chain the first-order term has mean $-(m-1)\gamma/\log n$: -0.063 at $n = 10^4$ and -0.031 at $n = 10^8$ for $y = abx$, twice

that for the chain of length three, whose mean remainder, by the finite-n prediction of Section 3.5, comes within 0.05 of its limit only for $n > 10^{11}$. We proved the expansion in probability only; for the means it would need the uniform integrability of $(\log n)^2(R_n - \chi(Z_n))$, which we do not prove, and Section 3.5 checks it numerically.

THE COEFFICIENT DENSITY IS THE STATE DENSITY. The exponents of ρ are the learning coefficient and multiplicity in the sense of Part II, where they are defined through the volume of nearly true parameters and the poles of the *zeta function* $\zeta(z) = \int K(w)^z \varphi(w)\,\mathrm{d}w$ (Chapter 4).[6] For one coefficient the definitions can be compared directly.

[6] Physicists would call ρ, or the density of $K(w)$ under the prior, a density of states; Watanabe calls the latter the state density.

Proposition 3.7 (Density, volume and zeta function). *Assume parts 3 and 4 of Assumption 3.1, and let $K(w) = \frac{1}{2}(g(w) - c_0)^2$, the population counterpart of K_n when $\mathbb{E}[x^2] = 1$. Then:*

(i) the volume function $V(\varepsilon) = \int_{K(w)<\varepsilon} \varphi(w)\,\mathrm{d}w$ *equals $\mu\{|c - c_0| < \sqrt{2\varepsilon}\}$, and $V(\varepsilon) \sim \lambda^{-1} 2^{\lambda-m}(\ell_+ + \ell_-)\,\varepsilon^\lambda (\log \frac{1}{\varepsilon})^{m-1}$ as $\varepsilon \to 0$;*

(ii) the zeta function is finite for $-\lambda < z \le 0$ and infinite at $z = -\lambda$, and $(z+\lambda)^m \zeta(z) \to 2^{\lambda-m}(m-1)!\,(\ell_+ + \ell_-)$ as $z \downarrow -\lambda$.

PROOF. The plan is to reduce both statements to one integral near c_0. The definition of μ gives the formula for V, and $\zeta(z) = 2^{-z}\int_{\mathbb{R}} |c - c_0|^{2z}\mu(\mathrm{d}c)$. With $\tilde\ell(\delta) = \ell(c_0 + \delta) + \ell(c_0 - \delta)$, put $I(\sigma, r) = \int_0^r \delta^{2\sigma-1}(\log\frac{1}{\delta})^{m-1}\tilde\ell(\delta)\,\mathrm{d}\delta$ for $\sigma > 0$ and $0 < r \le \eta$. For part 1, $V(\varepsilon) = I(\lambda, r)$ with $r = \sqrt{2\varepsilon}$, and for $r \le e^{-1}$ the substitution $\delta = rv$ gives

$$I(\lambda, r) = r^{2\lambda}\Big(\log\frac{1}{r}\Big)^{m-1}\int_0^1 v^{2\lambda-1}\Big(1 + \frac{\log(1/v)}{\log(1/r)}\Big)^{m-1}\tilde\ell(rv)\,\mathrm{d}v.$$

As $r \to 0$ the integral tends to $(\ell_+ + \ell_-)/(2\lambda)$ by dominated convergence, with the bound $2\bar\ell\, v^{2\lambda-1}(1 + \log\frac{1}{v})^{m-1}$, and $r^{2\lambda}(\log\frac{1}{r})^{m-1} \sim (2\varepsilon)^\lambda 2^{1-m}(\log\frac{1}{\varepsilon})^{m-1}$. For part 2, $\zeta(z)$ is $2^{-z}I(z+\lambda, \eta)$ plus a part from $|c - c_0| \ge \eta$ that is at most $(\eta^2/2)^z$ for $z \le 0$, and $I(\sigma, \eta)$ is finite exactly when $\sigma > 0$, because $\tilde\ell \ge \frac{1}{2}(\ell_+ + \ell_-) > 0$ near 0. The substitutions $\delta = e^{-v}$ and $y = 2\sigma v$ give

$$(2\sigma)^m I(\sigma, \eta) = \int_{2\sigma\log(1/\eta)}^{\infty} e^{-y} y^{m-1}\,\tilde\ell\big(e^{-y/(2\sigma)}\big)\,\mathrm{d}y \to (m-1)!\,(\ell_+ + \ell_-)$$

as $\sigma \downarrow 0$, by dominated convergence. With $\sigma = z + \lambda$, multiplying by $2^{-z} \to 2^\lambda$ and dividing by 2^m gives the limit; the part from $|c - c_0| \ge \eta$ disappears after multiplication by σ^m. ∎

So λ is the exponent of the volume and the abscissa of convergence of ζ, and m the power of $\log\frac{1}{\varepsilon}$ in the volume and the order of the blow-up of ζ at $-\lambda$. For the running models ζ is meromorphic with its largest pole at $-\lambda$, of order m: $2^{-z}(2z+1)^{-L}$ for the chain (Exercise 3.1), $2^{-z}(4z+1)^{-1}(2z+1)^{-1}$ for the cusp, $2^{-z}(6z+1)^{-1}(2z+1)^{-1}$ for the cubic cusp. For analytic K in general the meromorphic continuation goes back to Bernshtein and Gel'fand (1969) and Atiyah (1970), and Theorem 4.6 relates the volume and

the zeta function. In one coefficient, ρ determines Watanabe's *state density*: $v(t) = V'(t) = \big(\rho(c_0 + \sqrt{2t}) + \rho(c_0 - \sqrt{2t})\big)/\sqrt{2t}$.

3.4.1 What Watanabe's theorem adds

For a general model the free energy is a d-dimensional integral, and Watanabe's theorem says that the conclusion of Theorem 3.4 survives: if the model is realizable and satisfies his analytic conditions (Assumption A.10), and λ and m are the learning coefficient and multiplicity of K, then $F_n = nS_n + \lambda \log n - (m-1)\log\log n + R_n$ with R_n converging in law. This is Theorem 2 of Watanabe (2013), who credits the expectation version to Watanabe (2001) and the version as a random variable to his book (Watanabe 2009), without a theorem number; Chapter 8 states it with its hypotheses (Theorem 8.6) and proves it for models whose mean is linear in a vector of coefficients (Theorem 8.4). Each step of our proof has a counterpart. The reduction to one coefficient becomes resolution of singularities (Hironaka 1964), a black box (Theorem A.8): in local coordinates $K = u^{2k}$ and $\varphi(w)\,\mathrm{d}w = b(u)|u^h|\,\mathrm{d}u$, and h and k determine λ and m. The window lemma becomes the asymptotics of the state density near 0. The normal variable Z_n becomes an empirical process $\xi_n(u)$ with $nK_n = nu^{2k} - \sqrt{n}\,u^k\xi_n(u)$, whose convergence in law to a Gaussian process (Theorem A.11) needs the relatively finite variance condition. Watanabe's review (Watanabe 2024) writes the limit as $\chi(\xi) = -\log\sum \iint t^{\lambda-1}e^{-t+\sqrt{t}\,\xi(u)}\,\mathrm{d}t\;\mathrm{d}u^*$, summed over local coordinates, with $\mathrm{d}u^*$ carried by the part of the resolved space where the coordinates that attain λ vanish. For one coefficient that part is c_0 approached from either side, $\mathrm{d}u^*$ has mass $2^{\lambda-m}\ell_\pm$ on the two sides, $\xi = \pm\sqrt{2}\,Z$ there, and $t \to t^2/2$ turns his formula into (3.3) (Exercise 3.2). The general proof adds Hironaka's theorem, uniform control of an empirical process, and a multiplicity that counts coordinates rather than powers of a logarithm. It does not add a rate: we know of no general counterpart of Proposition 3.6.

3.5 Measuring it

The script `03-coefficient-free.py` computes $F_n - nS_n$ for the five models of Table 3.1 at 13 sample sizes from 10^2 to 10^8, with 200 data sets per model and sample size; at the truth $c_0 = 0$ the data are pure noise. For $n \le 10^5$ it simulates the data sets in full, and for larger n it draws (S_{xx}, Z_n) from their exact joint law (Lemma 3.2). It evaluates (3.2) by adaptive quadrature with the kit's `coefficient_free_energy`, to a relative error of about 10^{-10};[7] on simulated data at $n \le 10^4$ its values agree to 10^{-11} with the kit's `free_energy_chain` and with step 5 of Computation 3.1 (`03-coefficient-checks.py`). Two predictions need no simulation: the limit of Theorem 3.4, and the *finite-n prediction*, which evaluates (3.2) at $S_{xx} = n$ and averages over Z_n by Gauss–Hermite quadrature, keeping the slowly varying factor that the limit drops.

Figure 3.2 shows three things. First, BIC is wrong by a growing amount: at $n = 10^8$ it charges 17.9 nats for $y = abx$ and 27.1 for the chain, against

[7] The routine integrates over $\hat{c} \pm 40$ window widths, splits the interval at the singular point $c = 0$, where ρ is infinite but integrable, and calls adaptive Gauss–Kronrod quadrature on each piece.

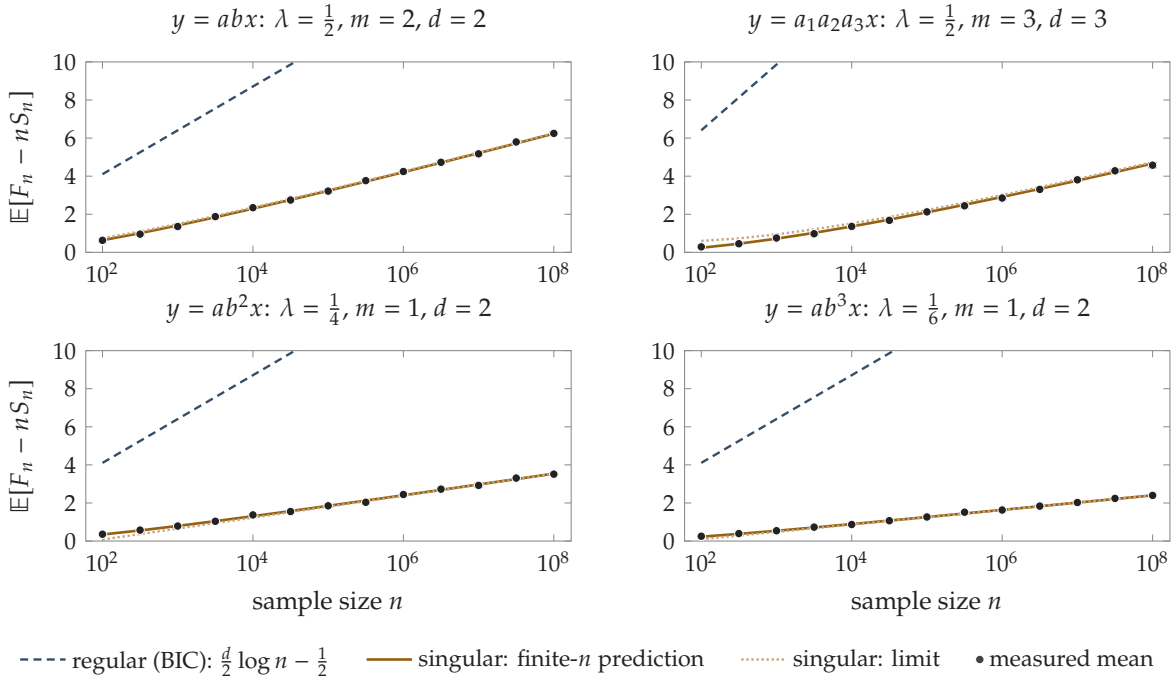


Figure 3.2. The normalized free energy of four one-coefficient models at the truth 0, with inputs and noise $\mathcal{N}(0,1)$ and the uniform prior on $[-1,1]^d$. Dots: the mean of $F_n - nS_n$ over 200 data sets at each of 13 sample sizes, by exact quadrature, with Student-t 95% intervals (seeds $10^6(j+1) + 1000k + r$ for model j, sample size k, data set r). Solid bronze: the finite-n prediction, the exact representation (3.2) at $S_{xx} = n$ averaged over Z_n. Dotted bronze: the limit $\lambda \log n - (m-1)\log\log n + \mathbb{E}\chi(Z)$ of Theorem 3.4. Dashed slate: the mean of Schwarz's BIC, $\frac{d}{2}\log n - \frac{1}{2}$, which leaves the panels at $n \approx 4 \times 10^4$ ($d = 2$) and $n \approx 10^3$ ($d = 3$). Script: `figures/scripts/03-coefficient-free.py`.

measured means of 6.25 and 4.58. Second, the finite-n prediction runs through the points: 58 of the 65 means lie within their 95% intervals of it.[8] Third, the slopes are biased as the $\log\log n$ term predicts: straight lines in $\log n$ fitted to the means over $10^2 \le n \le 10^8$ have slopes 0.500, 0.413, 0.324, 0.233 and 0.159 for the regular line, the product model, the chain and the two cusps, against $\lambda = \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{4}, \frac{1}{6}$; for the product model and the chain the deficit is the local slope $\lambda - (m-1)/\log n$ at $n = 10^5$, 0.413 and 0.326, and for the cusps it comes from the corrections of order $n^{-1/4}$ and $n^{-1/3}$.

Figure 3.3 shows what Figure 3.2 cannot: the distance $R_n - \chi(Z_n)$, which Theorem 3.4 says tends to 0 in probability. Subtracting $\chi(Z_n)$ data set by data set removes the noisy part of R_n, and since $\mathbb{E}\chi(Z_n) = \mathbb{E}\chi(Z)$, the mean is also the gap $\mathbb{E}[R_n] - \mathbb{E}\chi(Z)$. For the cusps it falls fast, from 0.262 to 0.0075 and from 0.125 to 0.0012, at the rates $n^{-1/4}$ and $n^{-1/3}$ at which their factors ℓ approach their limits. For the product model and the chain it falls slowly, from −0.108 to −0.030 and from −0.399 to −0.065, and tracks the finite-n prediction (60 of 65 means within their intervals). The first-order term of Proposition 3.6 describes the product model from $n \approx 10^4$ on (−0.063

[8] The Student-t interval undercovers here, because $F_n - nS_n$ contains the skewed term $-Z_n^2/2$: in 40,000 simulated sets of 200 draws of $-Z^2/2$ its coverage was 94.3%, and 7 or more misses out of 65 then have probability 0.075.

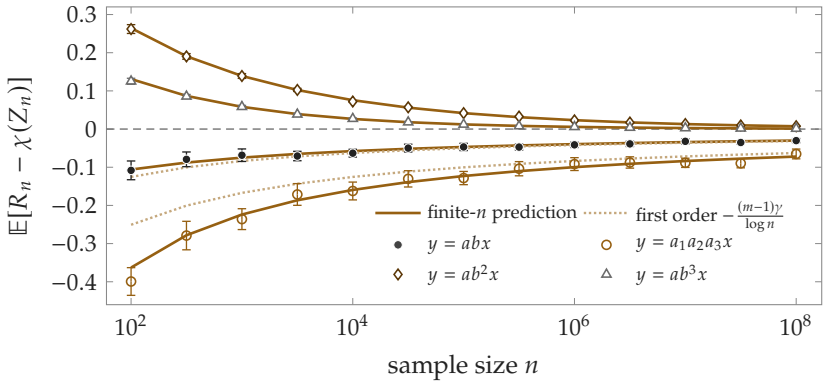


Figure 3.3. The distance to the limit law, data set by data set: the mean of $R_n - \chi(Z_n)$ over the 200 data sets of Figure 3.2, with $Z_n = S_{xy}/\sqrt{S_{xx}}$ from each data set, and Student-t 95% intervals. Theorem 3.4 says that $R_n - \chi(Z_n) \to 0$ in probability. Solid bronze: the finite-n prediction; dotted bronze: the first-order term $-(m-1)\gamma/\log n$ of Proposition 3.6, for the models with $m \geq 2$. The regular line, not drawn, stays within 0.007 of 0. Script: `figures/scripts/03-coefficient-free.py`.

predicted, -0.063 ± 0.010 measured) and three quarters of the chain's gap (-0.125 against -0.162 ± 0.023 at $n = 10^4$); the rest is the $(\log n)^{-2}$ term. No measurement contradicts the theorem.

3.6 The ledger

For a model whose likelihood depends on its parameters through one scalar coefficient, the free energy is a one-dimensional integral (Lemma 3.2), and its asymptotics are read off the density of the coefficient at the truth: the exponent gives λ, the power of the logarithm $m - 1$ (Theorem 3.4). The remainder converges in law and in mean to an explicit function of one normal variable, and is within $O_p(1/\log n)$ of it when $m \geq 2$ (Proposition 3.6). The measurements confirm all three statements, and show that the product model's $-\log\log n$ biases its slope over every range of n that can be simulated.

LEDGER · $y = abx$, $y = ab^2x$	
QUANTITY	The normalized free energy $F_n - nS_n$ of a model whose likelihood depends on w through one coefficient $c = g(w)$.
REGULAR THEORY	$\frac{d}{2}\log n + O_p(1)$ (BIC): $\log n$ for $y = abx$ and $y = ab^2x$, $\frac{3}{2}\log n$ for $y = a_1a_2a_3x$.
SINGULAR THEORY	$\lambda \log n - (m-1)\log\log n + O_p(1)$ when the coefficient's prior density behaves like $\lvert c - c_0\rvert^{2\lambda-1}(\log \frac{1}{\lvert c-c_0\rvert})^{m-1}$ at the truth; the remainder tends to $\chi(Z)$ in law and in mean (Theorem 3.4), with a $1/\log n$ correction when $m \geq 2$ (Proposition 3.6).

RUNNING EXAMPLE	$y = abx$: $\lambda = \frac{1}{2}$, $m = 2$, $\chi(Z) = \frac{1}{2}\log\frac{8}{\pi} - \frac{Z^2}{2}$; $y = ab^2x$: $\lambda = \frac{1}{4}$, $m = 1$.
MEASURED	Exact quadrature, 200 data sets per n, $10^2 \le n \le 10^8$: slopes 0.41, 0.32, 0.23, 0.16 against $\lambda = \frac{1}{2}, \frac{1}{2}, \frac{1}{4}, \frac{1}{6}$; $\mathbb{E}[R_n - \chi(Z_n)] = -0.030$ for $y = abx$ and 0.0075 for $y = ab^2x$ at $n = 10^8$.
RESTS ON	One scalar coefficient; Gaussian noise of known variance; a fixed realizable truth; the exact posterior at $\beta = 1$; a prior law of c with the stated form at c_0; $\mathbb{E}[x^2] = 1$ (and $\mathbb{E}[x^4] < \infty$ for the rate).

3.7 Where the theory stops

WHAT WAS ASSUMED. Four things, and a trained network violates each. First, the likelihood depends on the parameters through one scalar: exact for $y = g(w)x$, false for the tanh neuron, whose mean is abx only to first order, for the quartic model $y = ax_1 + b^2x_2$, which has two coefficients, and for every network with two hidden units. (The ReLU neuron is a borderline case: $a\,\mathrm{relu}(bx)$ is $ab\,\mathrm{relu}(x)$ for $b > 0$ and $-ab\,\mathrm{relu}(-x)$ for $b < 0$, two one-coefficient models glued along $b = 0$.) Second, Gaussian noise of known variance, which makes the energy exactly quadratic in c and Z_n exactly normal; an unknown variance adds $\frac{1}{2}\log n$ (Exercise 3.8). Third, a fixed truth: the limit holds for each c_0 but not uniformly in c_0, and a truth at distance $n^{-1/2}$ from the singular point behaves like the singular point (Exercise 3.9; Chapter 12). Fourth, the exact posterior at temperature 1 under a prior whose coefficient law has the stated form at c_0. The prior is part of the answer: a Gaussian prior changes the constant (Exercise 3.6), and the prior $\varphi(a, b) = \frac{1}{2}|a|$, which vanishes on the arm $a = 0$ of the cross, gives $c = ab$ the density $1 - |c|$, without a logarithm, so that $m = 1$.

WHERE IT BREAKS. The bounded term is not small, and it does not settle down at the sample sizes one can afford: in Figure 3.3 the product model's remainder is still 0.030 from its limit at $n = 10^8$, and the chain's 0.065. Drton and Plummer (2017), whose singular BIC uses λ and m, remark that the bounded remainder is generally very difficult to estimate and that they knew of no higher-order approximations in statistically relevant singular settings. The lower-order terms matter most when learning coefficients are similar. Fitted to the same data at the truth 0, the product model beats the regular line $y = ax$ by the log *Bayes factor* $\log(\frac{1}{2}\log S_{xx} - h(Z_n))$ (Exercise 3.11): on average 1.58 nats at $n = 10^4$, 2.25 at $n = 10^8$, and 3 only at $n \approx 10^{17}$. Inside one model whose prior mixes the law of the cusp's coefficient with that of the chain of length five, the posterior odds favor the chain's region, whose learning coefficient is twice as large, on average until $n \approx 10^{13}$ (Exercise 3.12). This is the situation of the loss–complexity trade-off that Carroll et al. (2025) use to read the development of small transformers, in a preprint that assumes the lower-order terms of competing local free energies to cancel, noting that otherwise their difference enters the posterior odds, especially at small n. Skalse (2023) argues in a blog

post that the limit $n \to \infty$ is not where the generalization of deep networks needs explaining; our numbers make a narrower point: even for $y = abx$ the terms below $\lambda \log n$ still move at every practical n.

OPEN PROBLEMS. For one-coefficient models the free energy is completely described, exactly by Lemma 3.2, in the limit by Theorem 3.4, and to first order in $1/\log n$ by Proposition 3.6. The open questions begin with two coefficients (Chapter 8); two are precise. First, compute the coefficient of $1/\log n$ in $\mathbb{E}[F_n - nS_n]$ for a model with several coefficients and $m \geq 2$, such as rank-one regression with $M = N \geq 2$ inputs and outputs; we know of no such computation. Second, Carroll et al. (2025) leave to future work the theory of the lower-order terms of local free energies and of their role in the posterior's choice between regions; for most networks even the multiplicity of a local free energy is unknown, and an estimator of m and of the constant from posterior samples, with error bars, would be a first step (Exercise 3.12).

Exercises

3.1 CONSOLIDATION. (Products of uniforms.) (a) Show that $\mathbb{E}|a|^{2z} = (2z+1)^{-1}$ for a uniform on $[-1,1]$ and $z > -\frac{1}{2}$, and deduce that the zeta function of the chain is $2^{-z}(2z+1)^{-L}$. (b) Derive ρ_L of Computation 3.2 by induction on L. (c) Show that $\mu\{|c| < r\} = r\sum_{k<L} \frac{1}{k!}(\log\frac{1}{r})^k$ for $0 < r \leq 1$, and check Proposition 3.7 for the chain.

3.2 CONSOLIDATION. (The limit laws.) (a) Derive the values of $\chi(0)$ and $\mathbb{E}\chi(Z)$ in Table 3.1 for the regular line, the product model and the chain. (b) Show that $\int_0^\infty t^{2\lambda-1}e^{-t^2/2+zt}\,\mathrm{d}t = 2^{\lambda-1}\int_0^\infty \tau^{\lambda-1}e^{-\tau+\sqrt{2\tau}\,z}\,\mathrm{d}\tau$, and check the translation of Watanabe's $\chi(\xi)$ in Section 3.4.1.

3.3 CONSOLIDATION. (The population free energy, and Gibbs.) (a) For the product model show that $\bar{F}(n) = -\log\int e^{-nK}\varphi$ equals $\frac{1}{2}\log n - \log(\log n + \gamma + \log 2) + \frac{1}{2}\log\frac{8}{\pi}$ up to an exponentially small error. *(Hint:* $\int_0^\infty e^{-t^2/2}\log t\,\mathrm{d}t = -\frac{1}{4}\sqrt{2\pi}(\gamma+\log 2)$.*)* (b) Show that $F_n - nS_n = \min_\nu\{\mathbb{E}_\nu[nK_n] + \mathrm{KL}(\nu\,\|\,\varphi)\}$ over probability densities ν on W, and deduce $\mathbb{E}[F_n - nS_n] \leq \bar{F}(n)$ for every model. (c) For the product model show that $\bar{F}(n) - \mathbb{E}[F_n - nS_n] \to \frac{1}{2}$.

3.4 CONSOLIDATION. (No convergence in probability; other scales.) Let $x \sim \mathcal{N}(0,1)$. (a) Show that (Z_n, Z_{4n}) converges in law to a normal pair with correlation $\frac{1}{2}$, and deduce that R_n does not converge in probability for the product model. *(Hint: $T_{4n} - T_n$ is independent of T_n.)* (b) If $\mathbb{E}[x^2] = \sigma_x^2$, show that $R_n \to \chi(Z) + \lambda\log\sigma_x^2$ in law. (c) At inverse temperature β, show that $F_n(\beta) - n\beta S_n - \lambda\log n + (m-1)\log\log n \to \chi(\sqrt{\beta}Z) + \lambda\log\beta$ in law.

3.5 BY HAND. (The family $y = a^j b^k x$.) (a) For $1 \leq j < k$, not both even, show that $c = a^j b^k$ has the density $\frac{|c|^{1/k-1} - |c|^{1/j-1}}{2(k-j)}$, so that $\lambda = \frac{1}{2k}$, $m = 1$ and $\ell_\pm = \frac{1}{2(k-j)}$; check $\chi(0) = -0.446$ for the cubic cusp. What

changes when j and k are both even? (b) For $j = k$ odd, show that $\rho(c) = \frac{1}{2k^2}|c|^{1/k-1}\log\frac{1}{|c|}$, so that $\lambda = \frac{1}{2k}$ and $m = 2$.

3.6 BY HAND. (A Gaussian prior.) Let a, b be independent $\mathcal{N}(0,1)$ under the prior. (a) Show that $c = ab$ has the density $\frac{1}{\pi}K_0(|c|)$, with $K_0(v) = \int_0^\infty e^{-v\cosh u}\,\mathrm{d}u$. (b) Using $K_0(v) = \log\frac{2}{v} - \gamma + o(1)$, show that $\lambda = \frac{1}{2}$, $m = 2$, $\ell_\pm = \frac{1}{\pi}$ and $\chi(z) = \frac{1}{2}\log(2\pi) - \frac{z^2}{2}$, which exceeds the uniform prior's limit by $\log\frac{\pi}{2} = 0.45$ nats. Explain the sign.

3.7 BY HAND. (A one-sided coefficient.) Let $y = a^2x + \varepsilon$, with a uniform on $[-1,1]$ and $c_0 = 0$. (a) Show that $\rho(c) = \frac{1}{2}c^{-1/2}$ on $(0,1]$, so that $\lambda = \frac{1}{4}$, $m = 1$, $\ell_+ = \frac{1}{2}$ and $\ell_- = 0$. (b) Show that $\chi(z) = -\log(\frac{1}{2}\int_0^\infty e^{-t^2/2+zt}t^{-1/2}\,\mathrm{d}t)$, with $\chi(0) = -0.075$ and $\mathbb{E}\chi(Z) = -0.231$. (c) Why is the cusp's $\chi(0)$ smaller by exactly $\log 2$, and why does $\chi(z)$ grow as $z \to -\infty$?

3.8 VARIATION. ★ (Unknown noise variance.) Add a noise scale to the model, $y = g(w)x + \sigma\varepsilon$, with a prior density for σ that is continuous and positive at $\sigma_0 = 1$, an interior point of its compact support. Show that $F_n - nS_n = (\lambda + \frac{1}{2})\log n - (m-1)\log\log n + O_p(1)$. *(Hint: integrate σ out by Laplace's method (Theorem A.1): up to bounded terms and terms free of c, this gives $\frac{n}{2}\log Q(c) + \frac{1}{2}\log n$ with $Q(c) = \sum_i(y_i - cx_i)^2$, and $\frac{n}{2}\log\frac{Q(c)}{Q(\hat{c})}$ is nearly the window of Lemma 3.2. For σ uniform on $[\frac{1}{2}, 2]$ and $\sum_i \varepsilon_i^2 = n$ the extra cost is $\frac{1}{2}\log\frac{n}{\pi} + \log\frac{3}{2} + o(1)$.)*

3.9 VARIATION. (A truth near the singular point.) For the product model with a truth $|c_0| < \frac{1}{2}$: (a) show that, up to an exponentially small error, $F_n - nS_n = \frac{1}{2}\log S_{xx} - \log\big(\frac{1}{2}\log S_{xx} - h(\sqrt{S_{xx}}\,c_0 + Z_n)\big) - \frac{Z_n^2}{2} + \frac{1}{2}\log\frac{2}{\pi}$. (b) Deduce that $c_0 = \tau n^{-1/2}$ with τ fixed gives the expansion of $c_0 = 0$, while a fixed $c_0 \neq 0$ gives $\frac{1}{2}\log n - \log\log\frac{1}{|c_0|} + \frac{1}{2}\log\frac{2}{\pi} - \frac{Z_n^2}{2} + o_p(1)$, with $\lambda = \frac{1}{2}$ and $m = 1$. (c) For $c_0 = 0.1$ the slope of $\bar{F}(n)$ against $\log n$ is 0.45, 0.53, 0.50 at $n = 10^2, 10^3, 10^4$; explain the crossover near $n \approx c_0^{-2}$, and show that $\bar{F}(n)$ exceeds its value at $c_0 = 0$ (0.82 more at $n = 10^4$) by an amount growing like $\log\log n$.

3.10 MEASUREMENT. (Fitting λ and m.) Fit $A\log n - B\log\log n + C$ to the means computed by `03-coefficient-free.py`, weighted by their intervals: for the product model $A = 0.49 \pm 0.01$ and $B = 0.82 \pm 0.15$, for the chain $B = 1.76 \pm 0.12$ (standard errors), against $m - 1 = 1$ and 2. With m fixed, the slope of the means plus $(m-1)\log\log n$ is 0.509 and 0.515. Explain why λ is well determined and m is not, and which term of Proposition 3.6 biases B downward.

3.11 MEASUREMENT. (A Bayes factor between equal learning coefficients.) Fit $y = ax$ and $y = abx$ to the same data at the truth 0. (a) Show that $F_n(y{=}ax) - F_n(y{=}abx) = \log(\frac{1}{2}\log S_{xx} - h(Z_n))$ up to an exponentially small error. (b) Compute its mean: 1.58 nats at $n = 10^4$, 2.25 at $n = 10^8$, and 3 only at $n \approx 10^{17}$. (c) What does this say about choosing between

models that differ only in their multiplicity?

3.12 RESEARCH. (Lower-order terms in the posterior's choice of region.) Let the prior be the equal mixture of two disjoint parameter spaces, one with $c = ab^2$ (the cusp) and one with $c = a_1 \cdots a_5$ (the chain, $\lambda = \frac{1}{2}$, $m = 5$), each uniform. (a) Show that the posterior odds of the chain's region are $\exp(F_A - F_B)$, with F_A and F_B the free energies of the cusp and of the chain, and that the cusp's region wins as $n \to \infty$. (b) Compute with (3.2) that $\mathbb{E}[F_A - F_B]$ is 0.89, 0.19 and -0.37 at $n = 10^4$, 10^{12} and 10^{14}. (c) Open: estimate the multiplicity and the constant of a network's local free energies from posterior samples, with error bars, and decide when they change the posterior's choice (Carroll et al. 2025).

Notes and References

The free energy formula with the learning coefficient in place of $\frac{d}{2}$ is Watanabe's: his review (Watanabe 2024) traces it to Watanabe (1999); Watanabe (2001) proved that the averaged free energy is $\lambda \log n - (m - 1) \log \log n$ plus a constant; the book (Watanabe 2009) gives the version as a random variable, which Watanabe (2013) restates as Theorem 2; and Watanabe (2018) develops its consequences. The one-coefficient theory of this chapter is a special case, with the book's own proofs. The product model is the smallest reduced-rank regression (Aoyagi and Watanabe 2005), and the chain the narrowest deep linear network (Aoyagi 2024).

Densities of products of independent variables are classical: Craig (1936) studied the product of two normal variables, and Epstein (1948) used the Mellin transform, which does for products what the Fourier transform does for sums. Up to the factor 2^{-z} and a change of variable, the zeta function of Proposition 3.7 is the Mellin transform of ρ, which is why the chain's zeta function is the Lth power of a uniform variable's (Exercise 3.1). The window heuristic is Bayesian Occam's razor: MacKay (2003, ch. 28) writes the evidence as the best-fit likelihood times an Occam factor, the ratio of the posterior to the prior accessible volume. Drton and Plummer (2017) turned the free energy formula into a model-selection criterion that uses both λ and m; Drton et al. (2009) and Lin (2011) treat Bayesian integrals of algebraic models, and Wei et al. (2023) argue for deep-learning readers that singularity is a feature of Bayesian learning, not a defect.

TRANSLATING THE LITERATURE. Watanabe (2013) writes F for our F_n, the Bayes free energy, and expands it around $nL_n(w_0)$, our nS_n in the realizable case; his R_n is ours. Watanabe (2001) calls the free energy the stochastic complexity, a name the 2009 book keeps beside the evidence (its §6.2 is "Evidence and stochastic complexity"). The review (Watanabe 2024) writes θ, Θ, $\pi(\theta)$ for our w, W, $\varphi(w)$; its t, $\mathrm{d}u^*$ and ξ translate as in Section 3.4.1. Drton and Plummer (2017) write $\lambda_i(\pi_0)$ and $m_i(\pi_0)$ for our λ and m of a model $\mathcal{M}_i$ at the truth π_0; their proxy $P(Y_n \mid \hat{\pi}_i, \mathcal{M}_i)(\log n)^{m_i(\pi_0)-1}/n^{\lambda_i(\pi_0)}$ for the marginal likelihood is our expansion of e^{-F_n} with the remainder dropped and e^{-nS_n} replaced by the maximized likelihood.

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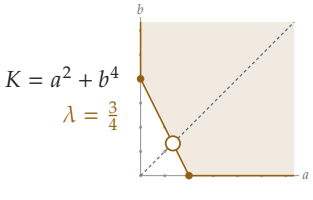
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Part II

The Learning Coefficient

The number that replaces $\frac{d}{2}$ is the learning coefficient λ: the rate at which the volume of nearly true parameters shrinks. It is computed by changing coordinates until the loss becomes a monomial, which is what resolution of singularities guarantees and what Newton polygons do in one step.



4 Volume

The learning coefficient as a scaling exponent

THE MODEL — $y = a^2x_1 + b^2x_2$

IN THIS CHAPTER. How much prior mass do the parameters hold that are almost as good as the truth, and how fast does it shrink? For a regular model it falls like $\varepsilon^{d/2}$; for the quartic cup $y = a^2x_1 + b^2x_2$, whose only true parameter has a zero Hessian, we compute $V(\varepsilon) = 1.311\,\varepsilon^{1/2}$ exactly, and the volumes of the other running models by scaling and by logarithmic coordinates. We prove that $V(\varepsilon) \sim C\varepsilon^{\lambda}(\log\frac{1}{\varepsilon})^{m-1}$ gives the population free energy $\lambda \log n - (m-1)\log\log n + O(1)$, that a normal-crossing function has $\lambda = \min_j \frac{h_j+1}{2k_j}$, and that $-\lambda$ is the largest pole of the zeta function and m its order, in both directions. Monte Carlo with up to 10^8 prior draws recovers every exponent to within 0.001 once the logarithm is modeled, and shows the price of volumes: about $100/V(\varepsilon)$ draws for 10% accuracy, which no model with many parameters can pay.

4.1 The question: how much room near the truth?

A trained network sits in a region of weight space where the loss is nearly minimal, and deep learning has long asked how large such regions are. Hochreiter and Schmidhuber (1997) searched for *flat minima*, large connected regions in which the error stays approximately constant; later measures of flatness mostly use the eigenvalues of the Hessian, which Dinh et al. (2017) showed to depend on the parametrization.[1] A Bayesian asks the question with a prior. If $K(w) = L(w) - L(w_0)$ is the excess loss of a parameter over the truth and φ the prior, the parameters that are almost as good as the truth hold the prior mass

$$V(\varepsilon) = \int_{\{K(w)<\varepsilon\}} \varphi(w)\,\mathrm{d}w, \qquad \varepsilon > 0, \tag{4.1}$$

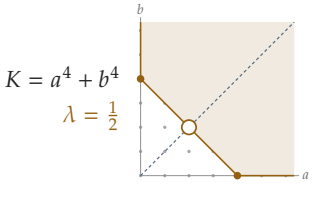
the *volume function* of K with respect to φ.[2]

Chapter 3 met this function in one dimension: when the likelihood depends on w only through a scalar coefficient c, $V(\varepsilon)$ is the prior mass of the window $|c - c_0| < \sqrt{2\varepsilon}$, and the free energy is, up to a term bounded in probability, minus the logarithm of that mass at $\varepsilon = 1/n$ (Theorem 3.4 and Proposition 3.7). The model of this chapter, the *quartic cup* $y = a^2x_1 + b^2x_2$ plus standard Gaussian noise, with $x \sim \mathcal{N}(0, I_2)$ and the uniform prior on $[-1,1]^2$, fitted to pure noise, has two coefficients and the loss

$$K(a,b) = \tfrac{1}{2}\mathbb{E}_x\left[(a^2x_1 + b^2x_2)^2\right] = \tfrac{1}{2}\left(a^4 + b^4\right).$$

[1] A prior mass does not: if $w = g(u)$ is a smooth bijection and the prior is carried along as a density, the set $\{K < \varepsilon\}$ and its prior mass are the same in both coordinates.

[2] For the uniform prior on a box, $V(\varepsilon)$ is the Lebesgue volume of $\{K < \varepsilon\}$ divided by the volume of the box. Watanabe, and Lau et al. (2025), speak of volumes, and so do we.



Its single true parameter $(0,0)$ has a zero Hessian, so the pair is singular. The free energy is no longer an integral in one variable, but the population free energy $-\log \int e^{-nK}\varphi$ depends on K only through its law under the prior, which V encodes: the volume function replaces the coefficient density.

THE REGULAR ANSWER · the volume near a nondegenerate minimum. Let K be continuous on a compact W with a single zero w_0, an interior point near which K is three times continuously differentiable with a positive definite Hessian J, and let φ be continuous and positive at w_0. Then $\{K < \varepsilon\}$ is, for small ε, nearly the ellipsoid $\frac{1}{2}(w-w_0)^\top J(w-w_0) < \varepsilon$, and

$$V(\varepsilon) \sim \varphi(w_0)\,\frac{\pi^{d/2}\,(2\varepsilon)^{d/2}}{\Gamma(\frac{d}{2}+1)\sqrt{\det J}} \qquad (\varepsilon \to 0)\,:$$

half a power of ε per parameter. For the regular plane $y = ax_1 + bx_2$ and the uniform prior on $[-1,1]^2$, $V(\varepsilon) = \frac{\pi}{2}\varepsilon$ exactly for $\varepsilon \le \frac{1}{2}$.

The constant is the volume of an ellipsoid (Definition A.4); its logarithm contains the curvature term $-\frac{1}{2}\log\det J$ of Laplace's formula (Theorem 1.4). For the quartic cup the formula fails at its first step: $J = 0$, and the formula returns an infinite volume, although $V \le 1$. For the product model $y = abx$ the true parameters form the cross $ab = 0$, on whose arms J has rank one and at whose center it vanishes. The question of this chapter is: *how fast does $V(\varepsilon)$ shrink for a singular model, what does the rate control, and how is it related to the zeta function, through which the book defines the learning coefficient?*

4.2 By hand: four volumes

COMPUTATION 4.1 · THE VOLUME OF THE QUARTIC CUP

1 *The set is a scaled copy of one set.* Since $K(ta,tb) = t^4K(a,b)$, the set $\{K < \varepsilon\}$ is $\{u^4+v^4<1\}$ scaled by $(2\varepsilon)^{1/4}$ in both directions, and for $\varepsilon \le \frac{1}{2}$ it lies in the square.

2 *That set has an explicit area.* By symmetry and the substitution $s = v^4$, the area of $\{u^4+v^4<1\}$ is

$$A_4 = 4\int_0^1 (1-v^4)^{1/4}\,\mathrm{d}v = \int_0^1 s^{-3/4}(1-s)^{1/4}\,\mathrm{d}s = B\big(\tfrac{1}{4},\tfrac{5}{4}\big),$$

which is $\Gamma(\frac{1}{4})^2/(2\sqrt{\pi}) = 3.7081$ (Definition A.4).

3 *The volume is an exact power.* Scaling by $(2\varepsilon)^{1/4}$ multiplies areas by $(2\varepsilon)^{1/2}$, and the prior density is $\frac{1}{4}$, so for $0 < \varepsilon \le \frac{1}{2}$

$$V(\varepsilon) = \tfrac{1}{4}A_4\,(2\varepsilon)^{1/2} = C\varepsilon^{1/2}, \qquad C = \frac{\Gamma(\frac{1}{4})^2}{4\sqrt{2\pi}} = 1.3110.$$

Dividing ε by ten divides the volume by $\sqrt{10}$, the regular plane's by ten.

4 *The state density and the zeta function.* The law of K under the prior has the density $v(t) = V'(t) = \frac{C}{2} t^{-1/2}$ on $(0, \frac{1}{2})$, and the zeta function $\zeta(z) = \int K^z \varphi$ integrates t^z against this law:

$$\zeta(z) = \frac{C}{2} \int_0^{1/2} t^{z-1/2} \, \mathrm{d}t + \int_{\{K \geq 1/2\}} K^z \varphi = \frac{C \, 2^{-z-1/2}}{2z+1} + \int_{\{K \geq 1/2\}} K^z \varphi.$$

Since $\frac{1}{2} \leq K \leq 1$ on $\{K \geq \frac{1}{2}\}$, the last integral is an entire function of z. So ζ has one pole, simple, at $-\frac{1}{2}$, and $(z + \frac{1}{2})\zeta(z) \to \frac{C}{2} = 0.6555$.

5 *The population free energy.* The partition function factorizes, $Z(n) = \int e^{-nK} \varphi = \big(\int_0^1 e^{-na^4/2} \, \mathrm{d}a\big)^2$; extending the integral to $(0, \infty)$ changes it by a relative error below $e^{-n/2}$ for $n \geq 3$, and $\int_0^\infty e^{-na^4/2} \, \mathrm{d}a = \Gamma(\frac{5}{4})(2/n)^{1/4}$. So, up to an exponentially small error,

$$\bar{F}(n) = -\log Z(n) = \tfrac{1}{2} \log n - \log\big(\sqrt{2}\,\Gamma(\tfrac{5}{4})^2\big) = \tfrac{1}{2} \log n - 0.1500.$$

At $n = 10^4$ the cup pays 4.455 nats; the regular plane, with $\bar{F}(n) = \log n - \log \frac{\pi}{2}$, pays 8.759.

6 *The same volume, in the coefficients.* In the coefficients $c = (a^2, b^2)$ the loss is $\frac{1}{2}|c|^2$, the loss of the regular plane; what differs is the prior. Each $c_j = a_j^2$ has the density $\frac{1}{2} c_j^{-1/2}$ on $(0, 1]$, so c has the density $(4\sqrt{c_1 c_2})^{-1}$, infinite at the truth. Over the quarter disc $|c| < \sqrt{2}\varepsilon$, in polar coordinates, it integrates to $\frac{1}{4}\sqrt{2}\varepsilon \int_0^{\pi/2} (\cos\theta \sin\theta)^{-1/2} \, \mathrm{d}\theta = \frac{1}{4}\sqrt{2}\varepsilon \, A_4$, step 3 again.

WHAT ONE HAD TO NOTICE. Two things. Homogeneity makes every sublevel set a scaled copy of one set, so the volume is an exact power whose exponent is the number of coordinates divided by the degree: $\frac{2}{4}$ here, $\frac{2}{2} = \frac{d}{2}$ for the regular plane. And in its coefficients the quartic cup is the regular plane on a quadrant: the fold $a \mapsto a^2$ piles the prior up at the truth, and a parametrization that folds onto the truth pays less.[3] Figure 4.1 shows the sets: the regular plane's disc shrinks like $\varepsilon^{1/2}$ in each direction, the cup like $\varepsilon^{1/4}$, the quartic model $K = \frac{1}{2}(a^2 + b^4)$ like $\varepsilon^{1/2}$ in a and $\varepsilon^{1/4}$ in b, and the product model's cross keeps arms of full length that only thin.

EVERY SUM OF POWERS. Scaling works for every sum of powers of separate coordinates. Let $K = \frac{1}{2} \sum_{j=1}^d w_j^{2k_j}$ with positive integers k_j, under the uniform prior on $[-1, 1]^d$, and let $F(u) = \sum_j u_j^{2k_j}$ and $\lambda = \sum_j \frac{1}{2k_j}$. Scaling gives $\mathrm{vol}\{F < t\} = t^\lambda \, \mathrm{vol}\{F < 1\}$, and computing $\int_{\mathbb{R}^d} e^{-F}$ both as a product of one-dimensional integrals and by slicing along the values of F gives $\mathrm{vol}\{F < 1\}\,\Gamma(\lambda + 1) = \prod_j 2\Gamma(1 + \frac{1}{2k_j})$. Since $\{K < \varepsilon\} = \{F < 2\varepsilon\}$ lies in the

[3] The constant has a classical name: $\int_0^{\pi/2} (1 - \frac{1}{2}\sin^2\theta)^{-1/2} \, \mathrm{d}\theta = \Gamma(\frac{1}{4})^2/(4\sqrt{\pi}) = 1.8541$ is the complete elliptic integral of the first kind at modulus $1/\sqrt{2}$, and $C = \frac{\sqrt{2}}{2} \times 1.8541$.

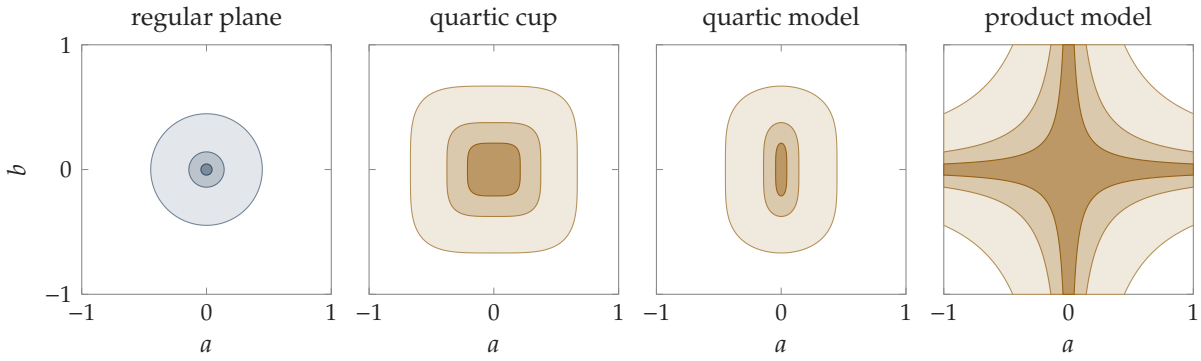


Figure 4.1. The sets $\{K < \varepsilon\}$ in the prior box $[-1,1]^2$ for $\varepsilon = 10^{-1}, 10^{-2}, 10^{-3}$ (light to dark), for $K = \frac{1}{2}(a^2+b^2)$, $\frac{1}{2}(a^4+b^4)$, $\frac{1}{2}(a^2+b^4)$ and $\frac{1}{2}a^2b^2$. Their prior masses are 0.157, 0.0157, 0.0016; 0.415, 0.131, 0.041; 0.261, 0.046, 0.008; and 0.807, 0.418, 0.184. Script: `figures/scripts/04-volume-shapes.py`.

box for $\varepsilon \le \frac{1}{2}$,

$$V(\varepsilon) = \frac{\prod_{j=1}^{d} \Gamma\!\left(1+\frac{1}{2k_j}\right)}{\Gamma(1+\lambda)}\,(2\varepsilon)^{\lambda}, \qquad \lambda = \sum_{j=1}^{d} \frac{1}{2k_j} \qquad \left(0 < \varepsilon \le \frac{1}{2}\right). \tag{4.2}$$

The exponents add: each coordinate contributes the reciprocal of its degree. The quartic model has $\lambda = \frac{1}{2}+\frac{1}{4} = \frac{3}{4}$, the sextic model $\frac{1}{2}(a^2+b^6)$ of Chapter 8 has $\frac{2}{3}$, and the mixed cup $\frac{1}{2}(a^4+b^6)$ of Chapter 6 has $\frac{5}{12}$. Products need a second trick, the one with which Chapter 3 found the density of a product of uniform variables.

COMPUTATION 4.2 · PRODUCTS IN LOGARITHMIC COORDINATES

1. *Logarithmic coordinates turn a uniform prior into exponentials.* If a is uniform on $[-1,1]$, then $T_a = \log\frac{1}{|a|}$ has $\mathbb{P}(T_a > t) = \mathbb{P}(|a| < e^{-t}) = e^{-t}$ for $t \ge 0$: it is exponential with mean 1. Under the uniform prior on $[-1,1]^2$, T_a and T_b are independent.

2. *A product's sublevel set becomes a half-plane.* For the product model, with $s = \sqrt{2\varepsilon}$ and $\ell = \log\frac{1}{s}$, $\{\frac{1}{2}a^2b^2 < \varepsilon\} = \{T_a + T_b > \ell\}$. The sum of two independent standard exponentials has the tail $e^{-\ell}(1+\ell)$, so for $\varepsilon \le \frac{1}{2}$

$$V(\varepsilon) = s\left(1+\log\frac{1}{s}\right) \sim \frac{1}{\sqrt{2}}\,\varepsilon^{1/2}\log\frac{1}{\varepsilon} \qquad (\varepsilon \to 0)\,:$$

   $\lambda = \frac{1}{2}$ and $m = 2$. The logarithm comes from the tie between two exponentials with the same rate.

3. *Unequal rates: the slower one wins.* For the cusp model $y = ab^2x$, $K = \frac{1}{2}a^2b^4$ and $\{K < \varepsilon\} = \{T_a + 2T_b > \ell\}$, where $2T_b$ is exponential with mean 2. Conditioning on T_a,

$$\mathbb{P}(T_a + 2T_b > \ell) = e^{-\ell} + \int_0^{\ell} e^{-t}e^{-(\ell-t)/2}\,\mathrm{d}t = 2e^{-\ell/2} - e^{-\ell},$$

model	K	$V(\varepsilon)$ for $\varepsilon \le \frac{1}{2}$	λ	m	C	$\frac{d}{2}$
regular plane	$\frac{1}{2}(a^2+b^2)$	$\frac{\pi}{2}\,\varepsilon$	1	1	1.5708	1
quartic cup	$\frac{1}{2}(a^4+b^4)$	$1.3110\,\varepsilon^{1/2}$	$\frac{1}{2}$	1	1.3110	1
quartic model	$\frac{1}{2}(a^2+b^4)$	$1.4699\,\varepsilon^{3/4}$	$\frac{3}{4}$	1	1.4699	1
sextic model	$\frac{1}{2}(a^2+b^6)$	$1.4457\,\varepsilon^{2/3}$	$\frac{2}{3}$	1	1.4457	1
mixed cup	$\frac{1}{2}(a^4+b^6)$	$1.2662\,\varepsilon^{5/12}$	$\frac{5}{12}$	1	1.2662	1
product model	$\frac{1}{2}a^2b^2$	$s(1+\ell)$	$\frac{1}{2}$	2	0.7071	1
cusp model	$\frac{1}{2}a^2b^4$	$2s^{1/2}-s$	$\frac{1}{4}$	1	2.3784	1
cubic cusp	$\frac{1}{2}a^2b^6$	$\frac{1}{2}(3s^{1/3}-s)$	$\frac{1}{6}$	1	1.6837	1
chain, $L=3$	$\frac{1}{2}a_1^2a_2^2a_3^2$	$s(1+\ell+\frac{1}{2}\ell^2)$	$\frac{1}{2}$	3	0.1768	$\frac{3}{2}$

Table 4.1. Volumes of the running models under the uniform prior on $[-1,1]^d$, exact for $\varepsilon \le \frac{1}{2}$, with $s=\sqrt{2\varepsilon}$ and $\ell=\log\frac{1}{s}$; C is the constant in $V(\varepsilon)\sim C\varepsilon^{\lambda}(\log\frac{1}{\varepsilon})^{m-1}$. The first five are sums of powers, (4.2); the last four are monomials (Computation 4.2, and Proposition 4.4 for C). Every volume is checked by quadrature in `figures/scripts/04-volume-checks.py`.

so $V(\varepsilon)=2(2\varepsilon)^{1/4}-(2\varepsilon)^{1/2}$: $\lambda=\frac{1}{4}$, $m=1$, and the faster exponential leaves a correction of relative size $\varepsilon^{1/4}$.

4 *The rule.* With $T_j=\log\frac{1}{|w_j|}$, a prior factor $|w_j|^{h_j}$ makes T_j exponential with rate h_j+1, and $\prod_j w_j^{2k_j}<\varepsilon$ when $\sum_j 2k_jT_j>\log\frac{1}{\varepsilon}$, a sum of exponentials with rates $\frac{h_j+1}{2k_j}$. Its tail decays at the smallest rate, and each tie adds a power of $\log\frac{1}{\varepsilon}$: this is Proposition 4.4.

5 *State densities and zeta functions.* For the product model $v(t)=\log\frac{1}{2t}\big/(2\sqrt{2t})$ on $(0,\frac{1}{2})$, and $\zeta(z)=2^{-z}\big(\frac{1}{2}\int_{-1}^{1}|a|^{2z}\,da\big)^2=2^{-z}(2z+1)^{-2}$: a double pole at $-\frac{1}{2}$, with $(z+\frac{1}{2})^2\zeta(z)\to\frac{\sqrt{2}}{4}=0.3536$. For the cusp, $\zeta(z)=2^{-z}(2z+1)^{-1}(4z+1)^{-1}$: a simple pole at $-\frac{1}{4}$, and one at $-\frac{1}{2}$ that gives the correction of step 3.

Table 4.1 collects the two rules that the rest of Part II elaborates: for a sum of powers of separate coordinates the exponents add; for a product the smallest exponent wins, and ties add powers of the logarithm. The quartic cup and the product model have the same learning coefficient $\frac{1}{2}$ for different reasons: the cup because each coordinate contributes $\frac{1}{4}$, the product because a small product needs only one small factor. Every analytic loss can be brought, piece by piece, to a monomial times a positive function by a change of coordinates (Hironaka's theorem, Theorem A.8); Chapter 5 does this by hand.

4.3 The theorem: from the volume to the free energy

Definition 4.1 (Volume, state density, zeta function, partition function). Let $W \subseteq \mathbb{R}^d$ be measurable, φ a probability density on W, and $K\colon W \to [0,\infty)$ measurable. The volume function is $V(\varepsilon) = \varphi\{w : K(w) < \varepsilon\}$ for $\varepsilon > 0$. If $\varphi\{K = 0\} = 0$ and the law of $K(w)$ under $w \sim \varphi$ has a density v on $(0,\infty)$, then v is the *state density*, and $V(\varepsilon) = \int_0^\varepsilon v(t)\,\mathrm{d}t$. The *zeta function* is

$$\zeta(z) = \int_{\{K>0\}} K(w)^z \varphi(w)\,\mathrm{d}w$$

at the complex z for which the integral converges absolutely. The *partition function* is $Z(s) = \int_W e^{-sK(w)}\varphi(w)\,\mathrm{d}w$ and the *population free energy* is $\bar{F}(s) = -\log Z(s)$, for $s > 0$.

All four depend only on the law of K under the prior: V is its distribution function, ζ its Mellin transform (Definition A.5), and Z its Laplace transform.

Lemma 4.2 (Slicing by the value of the loss). *In the setting of Definition 4.1, let $g\colon (0,\infty) \to [0,\infty)$ be measurable and $G(t) = \int_t^\infty g(u)\,\mathrm{d}u$ for $t \geq 0$. Then*

$$\int_W G\big(K(w)\big)\varphi(w)\,\mathrm{d}w = \int_0^\infty g(t)\,V(t)\,\mathrm{d}t.$$

Both sides may be infinite. In particular $Z(s) = s\int_0^\infty e^{-st}V(t)\,\mathrm{d}t$ for $s > 0$, and $\int_W K^{-\sigma}\varphi = \sigma\int_0^\infty t^{-\sigma-1}V(t)\,\mathrm{d}t$ for $\sigma > 0$, with $K^{-\sigma} = \infty$ where $K = 0$.

Proof. For each w, $G(K(w)) = \int_0^\infty g(t)\mathbf{1}\{K(w) < t\}\,\mathrm{d}t$. Integrate against φ and exchange the integrals, which Tonelli's theorem allows because the integrand is nonnegative; the inner integral $\int_W \mathbf{1}\{K(w) < t\}\varphi(w)\,\mathrm{d}w$ is $V(t)$. For the special cases take $g(t) = se^{-st}$, with $G(t) = e^{-st}$, and $g(t) = \sigma t^{-\sigma-1}$, with $G(t) = t^{-\sigma}$ for $t > 0$ and $G(0) = \infty$. ∎

Lemma 4.3 (From the volume to the free energy). *In the setting of Definition 4.1, suppose that*

$$V(\varepsilon) \sim C\varepsilon^\lambda\Big(\log\frac{1}{\varepsilon}\Big)^{m-1} \qquad (\varepsilon \downarrow 0)$$

for constants $C > 0$ and $\lambda > 0$ and an integer $m \geq 1$. Then, as $s \to \infty$ through the real numbers,

$$\int_W e^{-sK(w)}\varphi(w)\,\mathrm{d}w \sim C\,\Gamma(\lambda+1)\,s^{-\lambda}(\log s)^{m-1}, \tag{4.3}$$

and therefore $\bar{F}(s) = \lambda\log s - (m-1)\log\log s - \log\big(C\,\Gamma(\lambda+1)\big) + o(1)$.

Proof. The plan is to write the partition function as an integral of V against an exponential kernel, to rescale so that V is evaluated where the hypothesis applies, and to pass to the limit by dominated convergence. By Lemma 4.2 and the substitution $t = x/s$,

$$\frac{Z(s)}{s^{-\lambda}(\log s)^{m-1}} = \int_0^\infty e^{-x} R_s(x)\,\mathrm{d}x, \qquad R_s(x) = \frac{V(x/s)}{s^{-\lambda}(\log s)^{m-1}}.$$

Pointwise limit. Fix $x > 0$. As $s \to \infty$, $x/s \to 0$ and $\log \frac{s}{x}/\log s \to 1$, so the hypothesis gives $R_s(x) \to C x^\lambda$.

Domination. Choose $t_0 \in (0,1)$ such that $V(t) \le 2Ct^\lambda(\log \frac{1}{t})^{m-1}$ for $0 < t < t_0$, and let $s \ge e$. If $x < st_0$, then $R_s(x) \le 2Cx^\lambda\big(\log \frac{s}{x}/\log s\big)^{m-1}$, and $\log \frac{s}{x}/\log s = 1 + \log \frac{1}{x}/\log s$ is at most 1 when $x \ge 1$ and at most $1 + \log \frac{1}{x}$ when $x < 1$, because $\log s \ge 1$. If $x \ge st_0$, then $V \le 1$ and $(\log s)^{m-1} \ge 1$ give $R_s(x) \le s^\lambda \le (x/t_0)^\lambda$. So for every $x > 0$

$$R_s(x) \le 2Cx^\lambda\Big(1 + \log^+ \frac{1}{x}\Big)^{m-1} + \Big(\frac{x}{t_0}\Big)^\lambda,$$

with $\log^+ u = \max(\log u, 0)$, and e^{-x} times the right side is integrable on $(0,\infty)$ because $\lambda > 0$. Dominated convergence gives $\int_0^\infty e^{-x} R_s(x)\,\mathrm{d}x \to C\int_0^\infty x^\lambda e^{-x}\,\mathrm{d}x = C\,\Gamma(\lambda+1)$, which is (4.3); taking logarithms gives the statement about $\bar F$. ∎

The lemma is step 5 of Computation 4.1 in general, where $\sqrt{2}\,\Gamma(\frac{5}{4})^2 = 1.1619 = C\,\Gamma(\frac{3}{2})$. Chapter 8 applies it at $s = (1\pm\eta)^2 n$, that is, to the multiples $(1\pm\eta)^2 K$ that sandwich the random loss up to additive terms (Theorem 8.4). Its converse is Karamata's Tauberian theorem (Theorem A.6), so V and Z determine each other's leading terms.[4] When $m \ge 2$ the limit is approached slowly: for the product model $Z(n) = \frac{1}{2}\sqrt{\pi/(2n)}\,(\log n + \gamma + \log 2)$ up to an exponentially small error, with γ Euler's constant, so its ratio to the lemma's prediction is 1.069 at $n = 10^8$ and reaches 1.01 only at $n \approx 10^{55}$.

The next result computes the volume of every monomial times a positive factor, under a prior of the same kind; by Hironaka's theorem it is the local building block of every learning coefficient in the book.

[4] Theorems that pass from a function to its transform are called Abelian, and those that pass back, which need an extra condition such as monotonicity, Tauberian; Bingham et al. (1987) is the standard reference.

Proposition 4.4 (Volumes of normal-crossing functions). *Let $B = \prod_{j=1}^d [-r_j, r_j]$ with $r_j > 0$. Let $k_1, \dots, k_d$ and $h_1, \dots, h_d$ be nonnegative integers, not all k_j zero, and let a and b be continuous functions on B with $a > 0$ and $b \ge 0$. On B put*

$$K(w) = a(w)\prod_{j=1}^d w_j^{2k_j}, \qquad \varphi(w) = b(w)\prod_{j=1}^d |w_j|^{h_j},$$

and $V(\varepsilon) = \int_{B\cap\{K<\varepsilon\}} \varphi$. For j with $k_j > 0$ let $\lambda_j = \frac{h_j+1}{2k_j}$; let $\lambda = \min_j \lambda_j$, let J be the set of indices with $\lambda_j = \lambda$, and let m be its number of elements. Write $w' = (w_j)_{j\notin J}$, $B' = \prod_{j\notin J}[-r_j, r_j]$, and $(0_J, w')$ for the point whose

coordinates in J are 0. Then $V(\varepsilon) = (C + o(1))\,\varepsilon^{\lambda}(\log\frac{1}{\varepsilon})^{m-1}$ as $\varepsilon \downarrow 0$, with

$$C = \frac{1}{(m-1)!\,\lambda\prod_{j\in J} k_j}\int_{B'} \frac{b(0_J, w')}{a(0_J, w')^{\lambda}} \prod_{j\notin J} |w_j|^{h_j - 2k_j\lambda}\,\mathrm{d}w', \tag{4.4}$$

a finite integral, read as $b(0)a(0)^{-\lambda}$ when J contains every index. If $b(0) > 0$, then $C > 0$ and $V(\varepsilon) \sim C\varepsilon^{\lambda}(\log\frac{1}{\varepsilon})^{m-1}$.

Proof. The plan is to reduce to the positive orthant of the unit cube, to compute the volume exactly in the coordinates of J by the logarithmic coordinates of Computation 4.2, to integrate the other coordinates by dominated convergence with a and b frozen at $w_J = 0$, and to control the error of freezing by continuity. Write $L = \log\frac{1}{\varepsilon}$ and $\pi(u') = \prod_{j\notin J} u_j^{h_j - 2k_j\lambda}$.

Step 1: reductions. The coordinate hyperplanes are null sets, so V is the sum of the volumes of the 2^d orthants. A reflection $w \mapsto (\sigma_j w_j)_j$ maps an orthant onto $\prod_j [0, r_j]$ and changes only a and b, into $a(\sigma\,\cdot)$ and $b(\sigma\,\cdot)$. The substitution $w_j = r_j u_j$ maps $\prod_j [0, r_j]$ onto $[0,1]^d$ and keeps the form of K and $\varphi\,\mathrm{d}w$, with a and b multiplied by $\prod_j r_j^{2k_j}$ and $\prod_j r_j^{h_j+1}$; the formula for C does not change, because $h_j + 1 = 2k_j\lambda$ for $j \in J$ and $r_j^{h_j+1-2k_j\lambda} u_j^{h_j-2k_j\lambda}\,\mathrm{d}u_j = w_j^{h_j-2k_j\lambda}\,\mathrm{d}w_j$ for $j \notin J$. It therefore suffices to prove, for all continuous $a > 0$ and $b \geq 0$ on $[0,1]^d$, that $V(\varepsilon) = (C_+ + o(1))\varepsilon^{\lambda}L^{m-1}$ with

$$C_+ = c_0 \int_{[0,1]^{J^c}} \frac{b(0_J, u')}{a(0_J, u')^{\lambda}}\,\pi(u')\,\mathrm{d}u', \qquad c_0 = \frac{1}{2^m(m-1)!\,\lambda\prod_{j\in J} k_j}. \tag{4.5}$$

Summing over the orthants then gives (4.4): at $u_J = 0$ the signs of the coordinates in J do not change the integrand, which gives the factor 2^m, and the other signs reassemble the integral over B'. The integral is finite because $h_j - 2k_j\lambda > -1$ for $j \notin J$ (if $k_j > 0$ this says $\lambda < \lambda_j$; if $k_j = 0$ it says $h_j > -1$), a is bounded below and b above.

Step 2: the coordinates in J, exactly. For $\eta > 0$ let $G(\eta) = \int_{[0,1]^J} \mathbf{1}\{\prod_{j\in J} u_j^{2k_j} < \eta\} \prod_{j\in J} u_j^{h_j}\,\mathrm{d}u_J$. For $j \in J$ substitute $u_j = e^{-t_j/(h_j+1)}$, $t_j \geq 0$: then $u_j^{h_j}\,\mathrm{d}u_j$ becomes $(h_j+1)^{-1}e^{-t_j}\,\mathrm{d}t_j$, and $u_j^{2k_j} = e^{-t_j/\lambda}$ because $\frac{2k_j}{h_j+1} = \frac{1}{\lambda}$. The condition reads $\sum_{j\in J} t_j > \lambda\log\frac{1}{\eta}$; under the densities e^{-t_j} the sum has the Gamma density $t^{m-1}e^{-t}/(m-1)!$, whose tail at $x \geq 0$ is $e^{-x}\sum_{i<m} x^i/i!$ (both equal 1 at $x = 0$ and have the derivative $-x^{m-1}e^{-x}/(m-1)!$). With $c_J = \prod_{j\in J}(h_j+1)^{-1}$,

$$G(\eta) = c_J\,\eta^{\lambda}\sum_{i=0}^{m-1}\frac{(\lambda\log\frac{1}{\eta})^i}{i!} \quad (0 < \eta \leq 1), \qquad G(\eta) = c_J \quad (\eta \geq 1). \tag{4.6}$$

Step 3: frozen coefficients. Let $F \geq 0$ be bounded and measurable and α measurable with values in $[\alpha_-, \alpha_+] \subset (0, \infty)$, both functions of $u' = (u_j)_{j\notin J}$ only, and let $M(u') = \prod_{j\notin J} u_j^{2k_j}$. By Fubini's theorem the volume of

$\{\alpha M \prod_{j\in J} u_j^{2k_j} < \varepsilon\}$ under $F(u')\prod_j u_j^{h_j}\,\mathrm{d}u$ is

$$I(\varepsilon) = \int_{[0,1]^{J^c}} F(u') \prod_{j\notin J} u_j^{h_j}\, G\Big(\frac{\varepsilon}{\alpha(u')M(u')}\Big)\,\mathrm{d}u'$$

(no integral when J contains every index). We claim that $I(\varepsilon)/(\varepsilon^\lambda L^{m-1}) \to c_0 \int F\alpha^{-\lambda}\pi\,\mathrm{d}u'$; note that $c_J\lambda^{m-1}/(m-1)! = c_0$. Fix u' with $M(u') > 0$, which excludes a null set, and let $\eta = \varepsilon/(\alpha M)$. As $\varepsilon \to 0$, $\eta \to 0$, and by (4.6), since $\log\frac{1}{\eta} = L + \log(\alpha M)$,

$$\frac{G(\eta)}{\varepsilon^\lambda L^{m-1}} = \frac{c_J}{(\alpha M)^\lambda} \sum_{i<m} \frac{\lambda^i}{i!} \frac{(\log\frac{1}{\eta})^i}{L^{m-1}} \to \frac{c_0}{(\alpha M)^\lambda}.$$

For domination let $L \ge \max(1, \log\alpha_+)$. If $\eta \le 1$, then $0 \le \log\frac{1}{\eta} \le 2L$, because $M \le 1$, so $G(\eta) \le c_J e^{2\lambda}\eta^\lambda L^{m-1}$; if $\eta > 1$, then $G(\eta) = c_J \le c_J\eta^\lambda$. In both cases $G(\eta)/(\varepsilon^\lambda L^{m-1}) \le c_J e^{2\lambda}\alpha_-^{-\lambda} M(u')^{-\lambda}$, and $F\prod_{j\notin J} u_j^{h_j} M^{-\lambda} = F\pi$ is integrable by Step 1. Dominated convergence proves the claim, which for $F = \alpha = 1$ is the proposition with $a = b = 1$.

Step 4: unfreezing. Write $u^\circ = (0_J, u')$, $a_- = \min a$ and $b_+ = \max b$, and fix $\eta' \in (0, \frac{1}{2})$. By uniform continuity there is $\delta \in (0,1)$ such that $|b(u) - b(u^\circ)| \le \eta'$ and $(1-\eta')a(u^\circ) \le a(u) \le (1+\eta')a(u^\circ)$ whenever $\max_{j\in J} u_j < \delta$. Split the cube into $N = \{\max_{j\in J} u_j < \delta\}$ and its complement.

The complement contributes $o(\varepsilon^\lambda L^{m-1})$. If $u_{j_1} \ge \delta$ for some $j_1 \in J$, then $K < \varepsilon$ forces $\prod_{j\ne j_1} u_j^{2k_j} < \varepsilon/(a_-\delta^{2k_{j_1}})$, so the contribution is at most $\sum_{j_1\in J} b_+ V_{j_1}(\varepsilon/(a_-\delta^{2k_{j_1}}))$, with V_{j_1} the volume of $\prod_{j\ne j_1} u_j^{2k_j}$ on $[0,1]^{d-1}$ under $\prod_{j\ne j_1} u_j^{h_j}\,\mathrm{d}u$. If no k_j with $j \ne j_1$ is positive, $V_{j_1}(\eta) = 0$ for $\eta \le 1$. Otherwise Steps 2 and 3 with $F = \alpha = 1$ apply to that monomial, whose smallest ratio is λ, attained $m-1$ times, when $m \ge 2$, and some $\lambda' > \lambda$ when $m = 1$; so the contribution is $O(\varepsilon^\lambda L^{m-2})$ or $O(\varepsilon^{\lambda'} L^d)$, both $o(\varepsilon^\lambda L^{m-1})$.

On N we have $b \le b(u^\circ) + \eta'$ and $K \ge (1-\eta')a(u^\circ)\prod_j u_j^{2k_j}$, so the volume of $N \cap \{K < \varepsilon\}$ is at most the frozen integral $I_+(\varepsilon)$ with $F = b(u^\circ) + \eta'$ and $\alpha = (1-\eta')a(u^\circ)$. We also have $b \ge (b(u^\circ)-\eta')_+$ and $K \le (1+\eta')a(u^\circ)\prod_j u_j^{2k_j}$ on N, so that volume is at least the frozen integral $I_-(\varepsilon)$ with $F = (b(u^\circ)-\eta')_+$ and $\alpha = (1+\eta')a(u^\circ)$, minus the part of $I_-(\varepsilon)$ outside N, which is $o(\varepsilon^\lambda L^{m-1})$ as above. Step 3 applied to I_+ and I_- gives

$$\limsup_{\varepsilon\to 0} \frac{V(\varepsilon)}{\varepsilon^\lambda L^{m-1}} \le \frac{c_0}{(1-\eta')^\lambda} \int \frac{b(u^\circ)+\eta'}{a(u^\circ)^\lambda}\,\pi\,\mathrm{d}u',$$

$$\liminf_{\varepsilon\to 0} \frac{V(\varepsilon)}{\varepsilon^\lambda L^{m-1}} \ge \frac{c_0}{(1+\eta')^\lambda} \int \frac{(b(u^\circ)-\eta')_+}{a(u^\circ)^\lambda}\,\pi\,\mathrm{d}u'.$$

As $\eta' \downarrow 0$ both bounds tend to C_+ by dominated convergence. Finally, if $b(0) > 0$, then $b(0_J, w') > 0$ for w' near 0, and the integral in (4.4) is positive. ∎

The proof works verbatim for real $k_j \geq 0$ and $h_j > -1$, with $|w_j|^{2k_j}$ for $w_j^{2k_j}$. The constants of the four monomials in Table 4.1 are the proposition with $a = \frac{1}{2}$ and $b = 2^{-d}$. A check with non-constant factors, $K = (1+\frac{1}{2}u_1+u_2^2)\,u_1^2u_2^4$ and $\varphi = (2+\sin u_1+u_2)|u_1|$ on $[-1,1]^2$, gives $V(\varepsilon)/\varepsilon^{1/4} = 5.29322$ at $\varepsilon = 10^{-8}$ and 5.29364 at 10^{-16} by quadrature, against $C = 5.29364$. On $[0,1]^d$ with $a = b = 1$, λC_+ is the total mass of the measure $\mathrm{d}u^*$ of Watanabe (2013, Lemma 4). The volume also gives rules of Chapter 5 (Proposition 5.4) in one line: if $c_1K_1 \leq K_2 \leq c_2K_1$, then $V_1(\varepsilon/c_2) \leq V_2(\varepsilon) \leq V_1(\varepsilon/c_1)$, so comparable losses share λ and m (Exercise 4.3).

4.4 The theorem: the volume and the zeta function

The book defines the learning coefficient through the zeta function. The volume function and the zeta function are two transforms of one law, so they carry the same information; this section says how.

Definition 4.5 (Learning coefficient and multiplicity). In the setting of Definition 4.1, let K be bounded and $\varphi\{K = 0\} = 0$. Let $\lambda > 0$, and suppose that the integral defining ζ converges for $\Re z > -\lambda$ and that ζ extends to a meromorphic function on an open set containing the closed half-plane $\{\Re z \geq -\lambda\}$, whose only pole there is $-\lambda$, of order m. Then λ is the *learning coefficient* of K with respect to φ, and m is its *multiplicity*; λ is also called the *real log canonical threshold*.

For analytic losses (part (iv) below), which include every running model except the mixture, $-\lambda$ is simply the largest pole. A pole of order m means that near $-\lambda$, $\zeta(z) = (z+\lambda)^{-m}P(z) + h(z)$ with a polynomial P, a convergent power series h, and $P(-\lambda) \neq 0$, the *leading coefficient*.

Theorem 4.6 (The volume and the zeta function). *In the setting of Definition 4.1, let K be bounded, $\varphi\{K = 0\} = 0$, and $\lambda > 0$.*

(i) (Convergence.) *The real x with $\int_{\{K>0\}} K^x\varphi < \infty$ form an interval containing $[0,\infty)$; let $-\sigma_c \leq 0$ be its left end point. On $\Re z > -\sigma_c$ the integral defining ζ converges absolutely and ζ is holomorphic, and for $-\sigma_c < \Re z < 0$*

$$\zeta(z) = -z\int_0^\infty t^{z-1}\,V(t)\,\mathrm{d}t. \tag{4.7}$$

(ii) (From the volume to the zeta function.) *If $V(\varepsilon) \sim C\,\varepsilon^{\lambda}(\log\frac{1}{\varepsilon})^{m-1}$ with $C > 0$ and an integer $m \geq 1$, then $\sigma_c = \lambda$, the integral diverges at $z = -\lambda$, and*

$$(z+\lambda)^m\zeta(z) \to \lambda\,C\,(m-1)! \qquad (z \downarrow -\lambda \text{ through real } z). \tag{4.8}$$

Consequently, if ζ extends meromorphically to a neighborhood of $-\lambda$, then $-\lambda$ is a pole of order exactly m with leading coefficient $\lambda C(m-1)!$, and no pole lies in $\Re z > -\lambda$.

(iii) (From the zeta function to the volume.) *Conversely, if ζ satisfies the hypothesis of Definition 4.5, with leading coefficient A at $-\lambda$, then $A > 0$ and*

$$V(\varepsilon) \sim \frac{A}{\lambda\,(m-1)!}\,\varepsilon^{\lambda}\Big(\log\frac{1}{\varepsilon}\Big)^{m-1} \qquad (\varepsilon \downarrow 0).$$

(iv) (Analytic losses; statement.) *Let $W = \{w : \pi_1(w) \geq 0, \dots, \pi_k(w) \geq 0\}$ be compact with nonempty interior, with analytic π_j; let $\varphi = \varphi_1\varphi_2$ with $\varphi_1 \geq 0$ analytic and $\varphi_2 > 0$ infinitely differentiable; and let $K \geq 0$ be analytic on a neighborhood of W with $\varphi\{K = 0\} = 0$. Then ζ extends to a meromorphic function on $\mathbb{C}$ whose poles are negative rational numbers. If it has a pole, then its largest pole $-\lambda$ and that pole's order m satisfy the hypothesis of Definition 4.5, and $V(\varepsilon) \sim C\varepsilon^{\lambda}(\log\frac{1}{\varepsilon})^{m-1}$ with $C > 0$.*

So the learning coefficient is the exponent of the volume, the multiplicity is one more than the power of its logarithm, and the leading coefficient is $\lambda C(m-1)!$: 0.6555 for the quartic cup and 0.3536 for the product model, as Computations 4.1 and 4.2 found. Parts (i) and (ii) are elementary; part (iii) rests on a Tauberian theorem of Delange (1954). Part (iv) is stated: the continuation is the theorem of Bernshtein and Gel'fand (1969) and Atiyah (1970), which Watanabe states for the zeta functions of statistical models (Watanabe 2010, p. 3584), and given a resolution of singularities Chapter 5 derives the rest from Proposition 4.4 (Theorem 5.3).

Proof of (i) and (ii). The plan is to read convergence off Lemma 4.2, to obtain holomorphy from power series, and to prove (4.8) by an Abelian argument in logarithmic coordinates. Let $T = \max(1, \sup K)$.

(i). If $x < y$ are real and $\int K^x\varphi < \infty$, then $K^y \leq K^x + T^{|y|}$ on $\{K > 0\}$, because $K^y \leq K^x$ where $K \leq 1$ and $K^y \leq T^{|y|}$ where $K > 1$; so these x form an interval, which contains $[0, \infty)$ because $K^x \leq T^x$ for $x \geq 0$. Let $\Re z_0 > -\sigma_c$, and choose $r > 0$ with $\Re z_0 - r > -\sigma_c$. For $|z - z_0| < r$ and $K(w) > 0$,

$$K^z = K^{z_0}e^{(z-z_0)\log K} = \sum_{k\geq 0}\frac{(z-z_0)^k}{k!}\,K^{z_0}(\log K)^k,$$

and the series of absolute values is $K^{\Re z_0}e^{|z-z_0||\log K|} \leq K^{\Re z_0 - r} + K^{\Re z_0 + r}$, which is integrable. So the series may be integrated term by term, and near z_0 the function ζ is a convergent power series in $z - z_0$: it is holomorphic. For (4.7) let $\Re z < 0$. For $K(w) > 0$, $K(w)^z = -z\int_{K(w)}^{\infty} t^{z-1}\,\mathrm{d}t$; the integral of $|z|\,t^{\Re z - 1}\mathbf{1}\{t > K(w)\}\varphi(w)$ over t and w is $\frac{|z|}{|\Re z|}\int K^{\Re z}\varphi < \infty$, so Fubini's theorem allows the exchange, and $\varphi\{0 < K < t\} = V(t)$ because $\varphi\{K = 0\} = 0$.

(ii), convergence. For $\sigma > 0$, Lemma 4.2 gives $\int K^{-\sigma}\varphi = \sigma\int_0^{\infty} t^{-\sigma-1}V(t)\,\mathrm{d}t$, and the part over $[1, \infty)$ is at most 1. Near $t = 0$, $V(t)$ lies between $\frac{C}{2}t^{\lambda}(\log\frac{1}{t})^{m-1}$ and $2Ct^{\lambda}(\log\frac{1}{t})^{m-1}$, so the part over $(0, 1]$ is finite exactly when $\sigma < \lambda$. Hence $\sigma_c = \lambda$, and the integral diverges at $-\lambda$.

(ii), the limit. Let $0 < \sigma < \lambda$ and $\delta = \lambda - \sigma$. The substitution $t = e^{-u}$ gives $\zeta(-\sigma) = \sigma \int_0^\infty e^{-\delta u} f(u)\,\mathrm{d}u + \theta_\sigma$ with $f(u) = e^{\lambda u} V(e^{-u})$ and $0 \le \theta_\sigma \le 1$. By hypothesis $f(u) \sim C u^{m-1}$ as $u \to \infty$, and $0 \le f(u) \le e^{\lambda u}$. Given $\eta' > 0$, choose $U \ge 1$ with $|f(u) - C u^{m-1}| \le \eta' u^{m-1}$ for $u \ge U$. Since $\int_0^\infty e^{-\delta u} u^{m-1}\,\mathrm{d}u = \Gamma(m)\delta^{-m}$,

$$\left| \delta^m \int_U^\infty e^{-\delta u} f(u)\,\mathrm{d}u - C \int_{\delta U}^\infty e^{-y} y^{m-1}\,\mathrm{d}y \right| \le \eta' \Gamma(m),$$

while the integral over $[0, U]$, multiplied by δ^m, lies between 0 and $\delta^m U e^{\lambda U}$. As $\delta \downarrow 0$ the integral over $y \ge \delta U$ tends to $\Gamma(m)$ and $\delta^m U e^{\lambda U}$ to 0; since η' was arbitrary, $\delta^m \int_0^\infty e^{-\delta u} f(u)\,\mathrm{d}u \to C\,\Gamma(m)$. Multiplying by $\sigma \to \lambda$, and noting $\delta^m \theta_\sigma \to 0$, gives (4.8).

(ii), the consequence. If ζ extends meromorphically near $-\lambda$, its expansion there is $\sum_{k \ge -p} c_k (z+\lambda)^k$ with $c_{-p} \ne 0$, and $(z+\lambda)^m \zeta(z)$ behaves like $c_{-p}(z+\lambda)^{m-p}$, which has a finite nonzero limit only if $p = m$. By (i) no pole lies in $\Re z > -\lambda$. ∎

Proof of (iii). The plan is to apply Delange's Tauberian theorem to a monotone function built from the law of $\log \frac{1}{K}$, and to return to V by an integration by parts. Replacing K by K/T, $T = \max(1, \sup K)$, multiplies $\zeta(z)$ by T^{-z} and A by T^λ, and replaces $V(\varepsilon)$ by $V(T\varepsilon)$; the claim is unchanged, so we may assume $K \le 1$. Let μ be the law of $U = \log \frac{1}{K(w)}$ under $w \sim \varphi$, a law on $[0, \infty)$; then $\zeta(z) = \int e^{-zu} \mu(\mathrm{d}u)$ for $\Re z > -\lambda$, and $V(e^{-t}) = \mu((t, \infty))$.

A monotone function and its transform. Fix $\theta > 0$ and, for $t \ge 0$, let

$$\alpha(t) = \int_{[0,t]} e^{(\lambda+\theta)u} \mu(\mathrm{d}u),$$

a nondecreasing nonnegative function. For real $s > \theta$, Tonelli's theorem gives

$$\int_0^\infty e^{-st} \alpha(t)\,\mathrm{d}t = \int e^{(\lambda+\theta)u} \Big(\int_u^\infty e^{-st}\,\mathrm{d}t \Big) \mu(\mathrm{d}u) = \frac{\zeta(s-\lambda-\theta)}{s},$$

which is finite because $s - \lambda - \theta > -\lambda$. So the Laplace transform f of α converges for $\Re s > \theta$ and equals $\zeta(s-\lambda-\theta)/s$ there. By hypothesis f extends holomorphically to every point of the line $\Re s = \theta$ other than θ, and near θ, $f(s) = (s-\theta)^{-m} g(s) + h(s)$ with $g(s) = P(s-\lambda-\theta)/s$, h holomorphic at θ, and $g(\theta) = A/\theta > 0$, because $A \ne 0$ and $(z+\lambda)^m \zeta(z) > 0$ for real $z > -\lambda$. These are the hypotheses of Delange (1954, Théorème III), with $a = \theta > 0$ and $\omega = m$, not an integer ≤ 0; its conclusion is $\alpha(t) \sim B e^{\theta t} t^{m-1}$ as $t \to \infty$, with $B = \frac{A}{\theta\,(m-1)!}$.

Back to the volume. Since $e^{-(\lambda+\theta)u} = (\lambda+\theta) \int_u^\infty e^{-(\lambda+\theta)y}\,\mathrm{d}y$, Tonelli's theorem and $\int_{(t,y]} \mathrm{d}\alpha = \alpha(y) - \alpha(t)$ give

$$V(e^{-t}) = \int_{(t,\infty)} e^{-(\lambda+\theta)u}\,\mathrm{d}\alpha(u) = (\lambda+\theta) \int_t^\infty e^{-(\lambda+\theta)y} \alpha(y)\,\mathrm{d}y - e^{-(\lambda+\theta)t} \alpha(t).$$

As $t \to \infty$, the first term is $\frac{\lambda+\theta}{\lambda} B e^{-\lambda t} t^{m-1}(1+o(1))$, because $\int_t^\infty e^{-\lambda y} y^{m-1}\,\mathrm{d}y \sim e^{-\lambda t} t^{m-1}/\lambda$, and the second is $B e^{-\lambda t} t^{m-1}(1+o(1))$. Their difference is $B(\frac{\theta}{\lambda} + o(1)) e^{-\lambda t} t^{m-1}$, which with $\varepsilon = e^{-t}$ is the claim. ■

The proof never bounded ζ along vertical lines; positivity of the law of K did that work, through Delange's theorem.[5] The more familiar route, Mellin inversion, shows where the logarithms come from but needs such bounds.

[5] The hypothesis that the integral converges for $\Re z > -\lambda$ can be dropped. After scaling to $K \le 1$: if ζ were holomorphic at the left end point $-\sigma_c > -\lambda$ of its half-plane of convergence, its Taylor series at a nearby real point x, with coefficients $\int K^x (\log K)^k \varphi/k!$, would converge a little to the left of $-\sigma_c$ with terms of one sign there, and Tonelli's theorem would make the integral converge there too.

WHY A POLE OF ORDER m GIVES THE $(m-1)$ST POWER OF THE LOGARITHM. By (4.7), $-\zeta(z)/z$ is the Mellin transform of V, so $V(t) = \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} t^{-z}\,\frac{-\zeta(z)}{z}\,\mathrm{d}z$ for $-\lambda < c < 0$. Move the line of integration left, past $-\lambda$. Since $t^{-z} = t^\lambda \sum_k (\log\frac{1}{t})^k (z+\lambda)^k/k!$, the residue at a pole $A(z+\lambda)^{-m} + \dots$ is t^λ times a polynomial of degree $m-1$ in $\log\frac{1}{t}$ with leading term $\frac{A}{\lambda}(\log\frac{1}{t})^{m-1}/(m-1)!$, and the new line contributes $O(t^{\lambda+\delta})$. Watanabe derives the expansion of the state density of a normal-crossing function this way, by the inverse Mellin transform (Watanabe 2024, §4.1); Watanabe (2013, Lemma 4) quotes the result as Theorem 4.6 of Watanabe (2009). ◇

WHERE THE ZETA FUNCTION COMES FROM. In his address to the 1954 International Congress of Mathematicians in Amsterdam, I. M. Gel'fand asked whether, for a polynomial P and a test function ϕ, the integral $\int_{\{P>0\}} P^z \phi\,\mathrm{d}x$ is a meromorphic function of z with poles in finitely many arithmetic progressions, and asked for the residues (Gel'fand 1957, §3, pp. 262–263). It was answered with Hironaka's resolution of singularities (Hironaka 1964) by I. N. Bernstein and S. I. Gel'fand (Bernshtein and Gel'fand 1969),[6] and independently by Atiyah (1970), who used the result to divide distributions by analytic functions: after a resolution the integral is a finite sum of integrals of monomials times smooth functions, whose poles are negative rationals. Bernshtein (1972) proved the continuation for polynomials without resolution, from an identity $b(z)P^z = D(z)P^{z+1}$ with a polynomial b and a differential operator D polynomial in z; b is now called the *Bernstein–Sato polynomial*, and Kashiwara (1976) proved that its roots are negative rationals. Watanabe (1999, 2001) brought the zeta function into statistics.

[6] The English translations of his papers spell the name Bernshtein, as the citations here do.

4.5 Measuring it

The script `04-volume-mc.py` estimates the volumes of the regular plane and of the quartic cup, product, quartic and sextic models, under the uniform prior on $[-1,1]^2$, by plain Monte Carlo: it draws N points from the prior with the kit's `volume_fraction`, counts those with $K < \varepsilon$ at the fifteen levels $\varepsilon = 10^{-1}, 10^{-1.5}, \dots, 10^{-8}$, and reports each fraction with its exact Clopper–Pearson 95% interval (Section A.8). The count is binomial with mean $NV(\varepsilon)$, so the relative standard error is about $(NV)^{-1/2}$ and 10% accuracy needs $N \approx 100/V(\varepsilon)$: at $\varepsilon = 10^{-8}$, 7.2×10^4 draws for the product model, 7.6×10^5 for the cup, 1.5×10^7 for the sextic model, 6.8×10^7 for the quartic model and 6.4×10^9 for the regular plane. The script uses 10^7, 10^7, 3×10^7, 10^8 and 10^8 draws (seeds 401–405), at least 153 inside every level

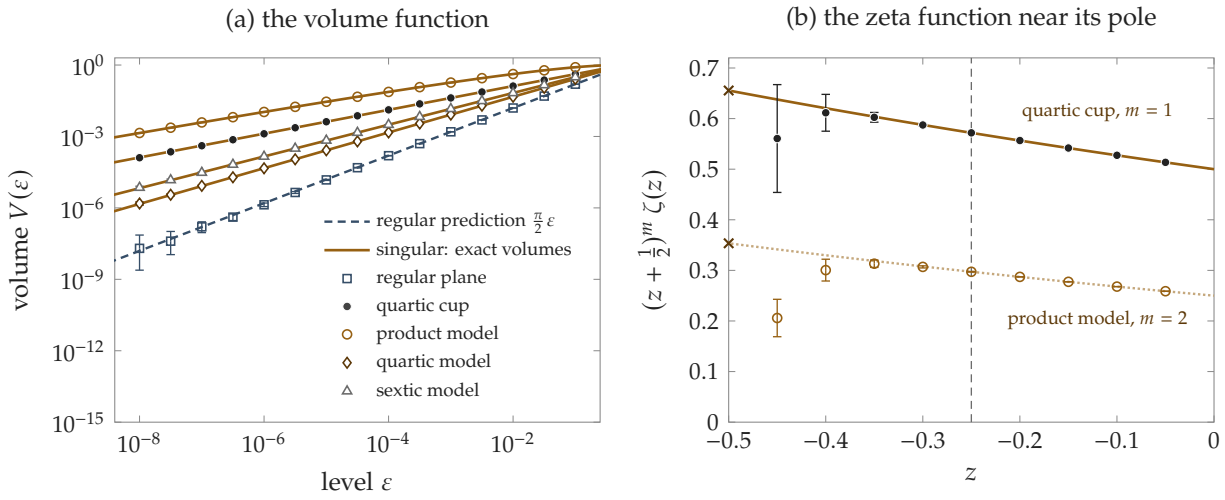


Figure 4.2. Monte Carlo against exact values under the uniform prior on $[-1,1]^2$. (a) The fraction of N prior draws with $K < \varepsilon$ at fifteen levels, with exact Clopper–Pearson 95% intervals (mostly smaller than the marks), for the regular plane ($N = 10^8$, seed 401), the quartic cup (10^7, 402), the product model (10^7, 403), the quartic model (10^8, 404) and the sextic model (3×10^7, 405). Solid bronze: exact volumes (Table 4.1). Dashed slate: the regular prediction, the $\varepsilon^{d/2}$ law, which the regular plane follows exactly. (b) $(z+\frac{1}{2})^m\zeta(z)$ for the quartic cup ($m=1$) and the product model ($m=2$). Bronze lines: exact values (closed form; polar quadrature for the cup). Crosses: the limits $\lambda C(m-1)!$ of Theorem 4.6(ii). Marks (filled for the cup, hollow for the product): Monte Carlo means of K^z over 10^7 prior draws (seeds 411, 412) with nominal 95% intervals, mean ±1.96 standard errors; on and left of the dashed line $z=-\frac{1}{4}$ the variance of K^z is infinite and the intervals mean nothing. Scripts: `figures/scripts/04-volume-mc.py`, `04-volume-zeta.py`.

for the singular models; it and `04-volume-zeta.py` each run in under a minute.

Figure 4.2(a) shows the measured volumes against the exact ones. The exact value lies inside the 95% interval at 74 of the 75 pairs of model and level; the miss is the regular plane at $\varepsilon = 10^{-5.5}$, with 444 draws against an expected 497 (nested counts share their draws, so the next two levels are low as well). The chord slopes between neighboring levels, with exact 95% intervals from the binomial law of the inner count given the outer one, cover the exact chord slopes at 55 of the 56 places for the singular models; the product model's local slope $\frac{1}{2} - 1/(2+\log\frac{1}{2\varepsilon})$ is 0.331 at $\varepsilon = 10^{-2}$, 0.405 at 10^{-4} and 0.449 at 10^{-8}.

Weighted least squares in $\log V$, over the levels $10^{-8} \le \varepsilon \le 10^{-2}$ that hold at least 100 draws ($\varepsilon \ge 10^{-6}$ for the regular plane), with 95% intervals from 2000 parametric-bootstrap replicates of the multinomial counts in the shells between levels (seed 499), turns the volumes into exponents. A straight line in log–log coordinates gives 1.0006 [0.9987, 1.0025] for the regular plane, 0.5001 [0.4990, 0.5011] for the cup, 0.7494 [0.7486, 0.7503] for the quartic model and 0.6665 [0.6653, 0.6676] for the sextic model. For the product model it gives 0.3905 [0.3901, 0.3909], the average local

slope, which understates λ by more than a fifth. The leading factor $(\log \frac{1}{\varepsilon})^{m-1}$ alone overcorrects, to 0.520 $[0.519, 0.520]$, because the exact volume $\varepsilon^{1/2}(c_0 + c_1 \log \frac{1}{\varepsilon})$ has $c_0/c_1 = 1.31$, not small against $\log \frac{1}{\varepsilon}$ on this range. The exact form (4.6), ε^{λ} times a polynomial of degree $m-1$ in $\log \frac{1}{\varepsilon}$, gives 0.5001 $[0.4988, 0.5013]$, with $c_1 = 0.708$ against the exact $\frac{1}{\sqrt{2}} = 0.707$.

Figure 4.2(b) measures the other side of the theorem. The exact curves reach the predicted limits: at $z = -0.4999$ they are at 0.65548 and 0.35353, against 0.65551 and 0.35355. Plain Monte Carlo of $\zeta(z) = \mathbb{E}[K^z]$ agrees with the exact values to within 0.1% for $z \geq -0.25$ and falls below them further left: at $z = -0.45$ it returns 82 ± 15 for the product model against 136.6, and 11.2 ± 2.1 for the cup against 12.75. By parts (i) and (ii) the variance of K^z, $\zeta(2z) - \zeta(z)^2$, is infinite for $z \leq -\frac{\lambda}{2}$: there the few draws closest to the true set carry most of the integral and are usually missing, so the estimate is typically too small and its nominal interval too narrow. The pole that defines λ is invisible to the most direct estimator of ζ.

4.6 The ledger

The prior mass within ε of the truth shrinks like $\varepsilon^{\lambda}(\log \frac{1}{\varepsilon})^{m-1}$, and the two exponents are the learning coefficient and its multiplicity: the exponents of the population free energy (Lemma 4.3), the largest pole of the zeta function and its order (Theorem 4.6), and, for normal crossings, explicit functions of the exponents of the loss and of the prior (Proposition 4.4). The measurements confirm every exponent and show two costs: seeing λ by Monte Carlo costs about $\varepsilon^{-\lambda}$ draws, and a multiplicity $m \geq 2$ biases every slope by about $(m-1)/\log \frac{1}{\varepsilon}$ unless the logarithm is modeled.

LEDGER · $y = a^2x_1 + b^2x_2$

QUANTITY	The volume function $V(\varepsilon) = \varphi\{K < \varepsilon\}$, the prior mass of nearly true parameters, and its scaling exponent.
REGULAR THEORY	$V(\varepsilon) \sim C\varepsilon^{d/2}$: an ellipsoid, ε^1 for two parameters; $\bar{F}(n) = \frac{d}{2}\log n + O(1)$.
SINGULAR THEORY	$V(\varepsilon) \sim C\varepsilon^{\lambda}(\log \frac{1}{\varepsilon})^{m-1}$ with $-\lambda$ the largest pole of ζ, of order m (Theorem 4.6); $\bar{F}(n) = \lambda \log n - (m-1)\log\log n + O(1)$ (Lemma 4.3); $\lambda = \min_j \frac{h_j+1}{2k_j}$ for normal crossings (Proposition 4.4).
RUNNING EXAMPLE	Quartic cup: $V(\varepsilon) = 1.311\,\varepsilon^{1/2}$ exactly for $\varepsilon \leq \frac{1}{2}$, so $\lambda = \frac{1}{2}$, $m = 1$, against $\frac{d}{2} = 1$; $\bar{F}(n) = \frac{1}{2}\log n - 0.150$. Product model: $V \sim 0.707\,\varepsilon^{1/2}\log \frac{1}{\varepsilon}$, $\lambda = \frac{1}{2}$, $m = 2$.
MEASURED	Monte Carlo, 10^7 to 10^8 prior draws: exponents 1.0006, 0.5001, 0.5001 (with the logarithm; 0.39 without), 0.7494, 0.6665 against 1, $\frac{1}{2}$, $\frac{1}{2}$, $\frac{3}{4}$, $\frac{2}{3}$; ζ by Monte Carlo fails for $z < -\lambda/2$.
RESTS ON	A known population loss $K \geq 0$; a volume of the power-logarithm form (guaranteed for analytic K through resolution); a prior with $\varphi\{K = 0\} = 0$, positive near the

true set; bounded K and Delange's Tauberian theorem for the converse.

4.7 Where the theory stops

WHAT WAS ASSUMED. First, that we know K. The volume belongs to the population loss, which in practice is unknown; a practitioner has K_n, whose sublevel sets fluctuate on the scale $n^{-1/2}$, much larger than the scale $1/n$ at which the free energy reads the volume. For models linear in their coefficients this moves the free energy only by a term bounded in probability (Theorem 8.4); for other models it is part of Watanabe's theorem. Second, the power-logarithm form of V: a theorem for analytic K; for losses comparable with an analytic one, such as the tanh neuron with Gaussian inputs or the mixture, comparison gives $V(\varepsilon) \asymp \varepsilon^{\lambda}(\log \frac{1}{\varepsilon})^{m-1}$, hence λ and m as exponents, but neither C nor the existence of the limit; for a ReLU network, whose loss is not analytic, nothing in this chapter guarantees it. Third, a prior that is positive near the true parameters and puts no mass on them. An atom at the truth makes $V(0+) > 0$ and the free energy bounded; a prior that vanishes on the true set raises λ. Jeffreys' prior vanishes wherever the Fisher information is degenerate and turns the quartic cup into the regular plane, with $\lambda = 1$ (Exercise 4.9), in line with Watanabe's statement that Jeffreys' prior gives $\lambda \geq \frac{d}{2}$ (Watanabe 2024). Fourth, the shape of the parameter set: for the product model with the truth $ab = 1$, the true set meets $[-1,1]^2$ only at two corners, $V(\varepsilon) \sim \frac{1}{2}\varepsilon$, and $\lambda = 1$, not the $\frac{1}{2}$ of a truth whose curve crosses the interior.

WHERE IT BREAKS. Volumes cannot be measured where they matter. Plain Monte Carlo needs about $100\,\varepsilon^{-\lambda}/C$ draws for 10% accuracy, 1.3×10^{13} already at $\varepsilon = 10^{-2}$ for a regular model with ten parameters, and a network's learning coefficient is a sum of contributions from many directions. The learning coefficients of networks are therefore estimated through the mean energy of a tempered posterior (Chapters 11 and 13), which the second asymptotic in part (i) of Theorem A.6 ties to the same λ, and not through volumes; the zeta function is no easier (Figure 4.2(b)). Even exact volumes show their exponents slowly when $m \geq 2$, and this qualifies the statement that singular points dominate. Skalse (2023) argues, in a blog post, that singular points need not dominate without analyticity and infinite data, with the example $\min(|x|, |y|)$. Its volume is explicit: for $K = \frac{1}{2}\min(a^2, b^2)$ on $[-1,1]^2$, $V(\varepsilon) = 2\sqrt{2\varepsilon} - 2\varepsilon$, so $\lambda = \frac{1}{2}$ and $m = 1$, and the square $|a|, |b| < 0.1$ holds a share of $V(\varepsilon)$ that tends to 0.1, its share of the arms: the center never dominates. For the analytic product model, with the same λ, that share tends to one, but slowly: by Computation 4.2 it is $1 - 2\log 10/(1 + \log \frac{1}{\sqrt{2}\varepsilon})$, which is 0.53 at $\varepsilon = 10^{-8}$ (0.54 in a Monte Carlo check with 10^7 draws) and 0.96 only at $\varepsilon = 10^{-100}$.

OPEN PROBLEMS. Two. First, estimate volumes in high dimension: for analytic K on a box in $\mathbb{R}^d$, give an estimator of $V(\varepsilon)$, or of λ through V, with controlled relative error at a cost polynomial in d and $\log \frac{1}{\varepsilon}$. Nested

sampling (Skilling 2006) estimates this kind of function, the prior mass above a sequence of likelihood levels, by drawing from the prior restricted to shrinking sets; near a singular true set those sets are thin and branched, and we found no analysis of the method for singular losses. Second, estimate the multiplicity: give an estimator of m from N prior draws whose error probability decays at a stated rate in N, for normal-crossing K; we do not know of one. The local estimators of Chapter 13 face the same question at a trained network, where m shifts the estimate of λ by about $(m-1)/\log n$.

Exercises

4.1 CONSOLIDATION. (Sums of powers.) (a) Derive (4.2) in detail and check the constants 1.4699, 1.4457 and 1.2662 of Table 4.1. (b) Show that $V(\varepsilon) = \frac{\pi}{6}(2\varepsilon)^{3/2}$ for $\frac{1}{2}(a^2+b^2+c^2)$ on $[-1,1]^3$, and $V(\varepsilon) = 1.1107\,\varepsilon^{1/2}$ for $\frac{1}{2}(a^2+b^2)^2$ on $[-1,1]^2$; explain why the last constant lies between $1.3110/\sqrt{2}$ and 1.3110. (c) For the mixed cup $\frac{1}{2}(a^4+b^6)$, show that ζ has one pole, simple, at $-\frac{5}{12}$, with $(z+\frac{5}{12})\zeta(z) \to 0.5276$, and that $\bar{F}(n) = \frac{5}{12}\log n - 0.1155 + o(1)$; compare λ with the Newton diagram through $(4,0)$ and $(0,6)$ (Theorem 6.4).

4.2 CONSOLIDATION. (The Abelian lemma at work.) (a) For the product model, compute the ratio of $Z(n)$ to the prediction of Lemma 4.3 at $n = 10^4$ and 10^8 (1.138 and 1.069). (b) Let $K = \frac{1}{2}(c^2 + a^2b^2)$ on $[-1,1]^3$. Show that $Z(n)$ factorizes, that $Z(n) \sim \frac{\pi}{4} n^{-1} \log n$, and conclude with Karamata's Tauberian theorem (Theorem A.6) that $V(\varepsilon) \sim \frac{\pi}{4}\varepsilon \log \frac{1}{\varepsilon}$: $\lambda = 1$, $m = 2$.

4.3 CONSOLIDATION. (Rules from the volume.) (a) Prove $V_1(\varepsilon/c_2) \le V_2(\varepsilon) \le V_1(\varepsilon/c_1)$ if $c_1 K_1 \le K_2 \le c_2 K_1$. (b) For the tanh neuron with $x \sim \mathrm{Unif}[-1,1]$ and the uniform prior on $[-2,2]^2$, $K = \frac{a^2}{2}\mathbb{E}[\tanh^2(bx)]$; use that $\tanh(u)/u$ decreases on $(0,\infty)$ to show $0.2323\,\frac{a^2b^2}{6} \le K \le \frac{a^2b^2}{6}$, and deduce that $V(\varepsilon)$ is within constant factors of $\varepsilon^{1/2}\log\frac{1}{\varepsilon}$. (c) Show that if φ is continuous and positive at an interior true parameter near which K is of class C^2, then $V(\varepsilon) \ge c\,\varepsilon^{d/2}$ for small ε, so $\lambda \le \frac{d}{2}$.

4.4 CONSOLIDATION. (Monomials.) Let $a = b = 1$ in Proposition 4.4, on $[-1,1]^d$. Show that $\zeta(z) = \prod_j 2/(2k_j z + h_j + 1)$ and $(z+\lambda)^m \zeta(z) \to \lambda C(m-1)!$. Show also that $V(\varepsilon)/\prod_j \frac{2}{h_j+1}$ is the probability that $\sum_{j:k_j>0} E_j > \log\frac{1}{\varepsilon}$ for independent exponential E_j of rates λ_j.

4.5 BY HAND. (A prior that vanishes on an arm.) Let $K = \frac{1}{2}a^2b^4$ on $[-1,1]^2$. (a) With the prior $\frac{1}{2}|a|$, show that $\zeta(z) = 2^{-z}(z+1)^{-1}(4z+1)^{-1}$ and $V(\varepsilon) = \frac{4}{3}(2\varepsilon)^{1/4} - \frac{1}{3}(2\varepsilon)$, and check $C = \frac{4}{3}\,2^{1/4}$ against Proposition 4.4. (b) With the prior $\frac{1}{2}|b|$, show that $\zeta(z) = 2^{-z}(2z+1)^{-2}$ and that V is exactly the product model's volume; explain this by the change of variables $c = b|b|$. A prior that vanishes on the arm $b = 0$, where the cusp's $\frac{1}{4}$ lives, raises λ to $\frac{1}{2}$ and creates a logarithm.

4.6 BY HAND. (Three variables.) (a) For $K = \frac{1}{2}a^2b^4c^6$ on $[-1,1]^3$, find $\lambda = \frac{1}{6}$, $m = 1$ and $C = \frac{9}{2}2^{1/6}$ from Proposition 4.4, and show from the tail of $T_a + 2T_b + 3T_c$ that $V(\varepsilon) = \frac{9}{2}s^{1/3} - 4s^{1/2} + \frac{1}{2}s$ with $s = \sqrt{2\varepsilon}$. (b) For $K = \frac{1}{2}a^2b^4c^4$ find $\lambda = \frac{1}{4}$, $m = 2$ and $C = 2^{-3/4}$, and compute $V(\varepsilon)/(\varepsilon^{1/4}\log\frac{1}{\varepsilon})$ at $\varepsilon = 10^{-12}$ by quadrature (0.580, against $C = 0.595$). Why is the convergence so slow?

4.7 BY HAND. (State densities.) (a) For the product model, check that $\int_0^{1/2} t^z v(t)\,dt = 2^{-z}(2z+1)^{-2}$. (b) For the chain of length L, show that $\sqrt{2t}\,(L-1)!\,v(t) = (\frac{1}{2}\log\frac{1}{2t})^{L-1}$ on $(0,\frac{1}{2})$. (c) For a one-coefficient model with coefficient density ρ (Chapter 3) and $K = \frac{1}{2}(c-c_0)^2$, show that $\sqrt{2t}\,v(t) = \rho(c_0+\sqrt{2t}) + \rho(c_0-\sqrt{2t})$.

4.8 VARIATION. (A Gaussian prior.) Let a, b be independent $\mathcal{N}(0,1)$ under the prior. Show that the quartic cup has $V(\varepsilon) \sim 0.8346\,\varepsilon^{1/2}$ and $\bar{F}(n) = \frac{1}{2}\log n + 0.302 + o(1)$, and, using the density $\frac{1}{\pi}K_0(|c|)$ of $c = ab$ from Chapter 3, a Bessel function with $K_0(v) = \log\frac{2}{v} - \gamma + o(1)$, that the product model has $V(\varepsilon) \sim \frac{\sqrt{2}}{\pi}\varepsilon^{1/2}\log\frac{1}{\varepsilon}$. Which results of this chapter need a bounded parameter set?

4.9 VARIATION. (Jeffreys' prior.) For the quartic cup, show that the Fisher information at (a,b) is $\mathrm{diag}(4a^2, 4b^2)$, so that Jeffreys' prior on $[-1,1]^2$ is $|ab|$; that under it $c = (a^2, b^2)$ is uniform on $[0,1]^2$; and that $V(\varepsilon) = \frac{\pi}{2}\varepsilon$ for $\varepsilon \le \frac{1}{2}$, the volume of the regular plane, so $\lambda = 1 = \frac{d}{2}$. What does this say about the claim that singular models are cheap?

4.10 MEASUREMENT. (The cost of Monte Carlo.) (a) With 10^8 draws in `04-volume-mc.py`, adapted to $K = \frac{1}{2}(a^2+b^2+c^2)$ on $[-1,1]^3$, down to which ε do the counts stay above 100? (About 7.7×10^{-5}.) (b) For $K = \frac{1}{2}|w|^2$ on $[-1,1]^d$, compute the draws needed for 10% accuracy at $\varepsilon = 10^{-2}$ when $d = 10$ and $d = 50$ (1.3×10^{13} and 1.9×10^{72}).

4.11 MEASUREMENT. (The zeta function near its pole.) (a) Show that $\mathrm{Var}(K^z) = \zeta(2z) - \zeta(z)^2$ is finite exactly for $z > -\lambda/2$. (b) Run `04-volume-zeta.py` with ten other seeds, record the estimates of $\zeta(-0.45)$ for the product model, compare their median and mean with 136.6, and explain why an unbiased estimator is usually far too small.

4.12 RESEARCH. (Volumes in high dimension.) Design an estimator of λ for an analytic loss on $[-1,1]^d$ from the volumes of nested sublevel sets, by nested sampling (Skilling 2006) or by Markov chains restricted to $\{K < \varepsilon_k\}$, and analyze its cost on the chain $\frac{1}{2}(a_1\cdots a_L)^2$ and on rank-one regression (Chapter 7), where λ and m are known. The open part is a guarantee at a cost polynomial in d for singular K, whose sublevel sets are thin neighborhoods of branched sets.

4.13 RESEARCH. (Estimating the multiplicity.) For normal-crossing K, test $m = 1$ against $m = 2$ from the empirical volumes of N prior draws at levels $\varepsilon_1 > \cdots > \varepsilon_k$, using the exact forms (4.6), and find how N must

grow to keep both error probabilities below 0.05 as the levels shrink. Open: the same for general analytic K, whose polynomial in $\log \frac{1}{\varepsilon}$ has unknown coefficients and whose lower poles add corrections.

Notes and References

That λ is the exponent of a volume runs through Watanabe's work: the review (Watanabe 2024) states that λ is the limit of $\log V(\varepsilon)/\log \varepsilon$, and Wei et al. (2023), who argue from the volume that 2λ is the meaningful count of a singular model's effective parameters, cite Theorem 7.1 of Watanabe (2009) for this characterization (arXiv version). Lau et al. (2025) define the local learning coefficient through such volumes near a trained parameter (their Definition 1), as Chapter 13 does.

Watanabe (2013, Lemma 4) gives the leading term of the state density of a normal-crossing function, and Arnold et al. (1988, pp. 215–262) develop the corresponding expansions of oscillatory integrals; sublevel sets and Newton polyhedra return in Chapter 6. Théorème III of Delange (1954), a generalization of Ikehara's Tauberian theorem, needs only a nondecreasing function and the behavior of its Laplace transform on the boundary line of convergence. The proofs of Lemmas 4.2 and 4.3, Proposition 4.4 and parts (i)–(iii) of Theorem 4.6 are our presentations of standard arguments; the reduction of part (iii) to Delange's theorem is ours, and we have not seen it in the singular-learning literature.

Translating the literature. The review (Watanabe 2024) writes θ, Θ and $\pi(\theta)$ for our w, W and $\varphi(w)$, and $v(t) = \int \delta(t - K(\theta))\pi(\theta)\,\mathrm{d}\theta$ for the state density, our $v = V'$. Lau et al. (2025) write $V(\varepsilon)$ for the Lebesgue volume of $\{w \in B(w^*) : L(w) - L(w^*) < \varepsilon\}$, without a prior, and $c\,\varepsilon^{\lambda(w^*)}(-\log \varepsilon)^{m(w^*)-1}$ for its asymptotics; our V with the uniform prior on the ball $B(w^*)$ is theirs divided by its volume. Saito (2007) defines the real log canonical threshold of f as the supremum of the c for which $|f|^{-c}$ is locally integrable; for $f = K$ and a prior bounded above and away from zero this is the σ_c of Theorem 4.6(i), which part (ii) identifies with λ, so $\lambda = \mathrm{rlct}(K)$, and $\lambda = \frac{1}{2}\,\mathrm{rlct}(f)$ when $K = f^2$. Lin (2011) defines the real log canonical threshold of K through $\int |K|^{-z}\varphi$, whose poles are at $z = \lambda$ rather than $-\lambda$, and that of an ideal $\langle f_1, \dots, f_k \rangle$ through $(\sum_i f_i^2)^{1/2}$, which gives 2λ when K is comparable with $\sum_i f_i^2$. In the language of Bingham et al. (1987), our hypothesis says that V is regularly varying at 0 with index λ.

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5 Blow-Ups

Resolution of singularities by hand

THE MODEL — $y = ax_1 + b^2x_2$

IN THIS CHAPTER. The quartic model's loss $\frac{1}{2}(a^2 + b^4)$ has a Hessian of rank one at the truth: counting parameters predicts $\lambda = 1$, counting nonzero Hessian eigenvalues $\frac{1}{2}$, and the truth is $\frac{3}{4}$. We compute it by hand with two blow-ups, changes of coordinates that replace the singular point by a curve of directions until the loss is a monomial; the cusp $(a^2 - b^3)^2$ needs three and gives $\frac{5}{12}$. We prove that every resolution computes the volume and the zeta function, so that λ and m are read off its exponents, and the rules of the trade, among them $\lambda = \frac{d-s}{2}$ near a smooth s-dimensional set of true parameters across which the loss grows quadratically. Measured volumes agree with the charts to six digits and approach the predicted constants, slowly for the cusp. Hironaka's theorem guarantees a resolution for every analytic loss, but finding one is another matter: for most networks none is known.

5.1 The question: what does a flat direction cost?

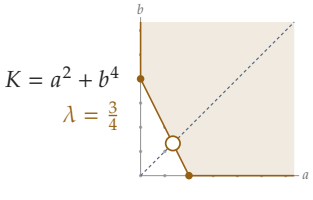
The Hessian of a trained network typically has a bulk of eigenvalues near zero and a few large ones (Sagun et al. 2016),[1] and a tempting reading is that the effective number of parameters is the rank of the Hessian. The model of this chapter tests that reading with two parameters. Fit $y = ax_1 + b^2x_2 + \varepsilon$, with $x \sim \mathcal{N}(0, I_2)$ and $\varepsilon \sim \mathcal{N}(0, 1)$, to pure noise, with the uniform prior on $[-1, 1]^2$. The loss of this *quartic model* relative to the truth $a = b = 0$ is

$$K(a, b) = \tfrac{1}{2}\mathbb{E}_x\left[(ax_1 + b^2x_2)^2\right] = \tfrac{1}{2}\left(a^2 + b^4\right),$$

whose Hessian at the truth is $\operatorname{diag}(1, 0)$: along b the loss is flat to second order. Counting parameters gives $\lambda = \frac{d}{2} = 1$, and counting nonzero Hessian eigenvalues gives $\frac{1}{2}$. Neither is right: the free energy of the quartic model grows like $\frac{3}{4}\log n$, with measured slope 0.749 (Chapter 8). The flat direction is not free, because the loss grows like b^4 along it; it costs a quarter where a regular direction costs a half.

Chapter 4 explained where such numbers come from when K is a *monomial*: if $K = \prod_j w_j^{2k_j}$ near 0 and the prior behaves like $\prod_j |w_j|^{h_j}$, each coordinate with $k_j > 0$ offers the candidate $(h_j + 1)/(2k_j)$, the learning coefficient is the smallest candidate, and its multiplicity is the number of candidates that tie (Proposition 4.4). The product model's a^2b^2 has candidates $\frac{1}{2}$ and $\frac{1}{2}$, so $\lambda = \frac{1}{2}$ and $m = 2$. The quartic $a^2 + b^4$ is not a monomial, and neither is the

[1] Sagun et al. (2016) is a preprint; Chapter 13 returns to Hessian-based measures of complexity.



loss of a network. The regular theory handles one kind of non-monomial loss.

THE REGULAR ANSWER · the Morse lemma. If $K \geq 0$ is analytic, $K(w_0) = 0$ at an interior point w_0, and $\nabla^2 K(w_0)$ is positive definite, then there are analytic coordinates u centered at w_0 in which $K = u_1^2 + \cdots + u_d^2$ (Theorem A.9 with $r = d$). Then $\{K < \varepsilon\}$ is a ball of radius $\sqrt{\varepsilon}$, $V(\varepsilon) \sim c\,\varepsilon^{d/2}$, and $\lambda = \frac{d}{2}$, $m = 1$ for a prior positive at w_0 (Watanabe 2013, Lemma 3). Every parameter costs one half.

The quartic model's Hessian is degenerate, and cleverer coordinates cannot help. A function is *normal crossing* at a point if, in some analytic coordinates centered there, it is a monomial times an analytic function that does not vanish at the point; near the point it then vanishes exactly on a union of coordinate hyperplanes. The function $a^2 + b^4$ vanishes only at the origin, and a change of coordinates that is invertible near the origin cannot turn a point into a union of lines (nor can it for the regular $\sum_i u_i^2$, $d \geq 2$). The way out is a change of variables that replaces the singular point by a whole curve and is invertible everywhere else. Integrals do not notice it, because the curve is a null set; the loss does, because it can become a monomial. Such changes are *blow-ups*. The question of this chapter is how to find them for a given loss, and why the exponents they produce are the learning coefficient and its multiplicity.

5.2 By hand: blowing up a point

THE BLOW-UP OF THE PLANE. Every point $w \neq 0$ of the plane lies on exactly one line through the origin, and the origin lies on all of them. The *blow-up* of the origin separates the lines there: its points are the pairs (w, ℓ) of a line ℓ through the origin and a point $w \in \ell$, and the map $g(w, \ell) = w$ forgets the line. Away from $w = 0$ the line is determined by w, so g is one-to-one there; over the origin sits a whole curve of points $(0, \ell)$, one for each direction, the *exceptional divisor* E.[2] Two charts describe it, chart I by the slope v of the line $b = va$ and the abscissa u, chart II by $a = ub$ and the ordinate v:

$$\text{I: } (a,b) = (u, uv),\ \det g' = u; \qquad \text{II: } (a,b) = (uv, v),\ \det g' = v. \tag{5.1}$$

[2] "Divisor" is the geometers' word for a hypersurface, here a curve. The directions through the origin form the real projective line, a circle, and near E the blow-up is a Möbius band.

The exceptional divisor is the axis $\{u = 0\}$ in chart I, whose point $(0, v)$ is the direction of slope v, and the axis $\{v = 0\}$ in chart II. A line $b = ca$ becomes the line $\{v = c\}$ of chart I, so lines that met at the origin no longer meet (Figure 5.1); where both charts apply, (u, v) in chart I is $(1/v, uv)$ in chart II. The map g is analytic, proper (preimages of compact sets are compact), and an analytic isomorphism off the null sets E and $\{0\}$, so integrals can be computed on the charts. The part $\{|b| \leq |a| \leq 1\}$ of the square is the image of $[-1,1]^2$ in chart I, and the part $\{|a| \leq |b| \leq 1\}$ that in chart II, so for every integrable F

$$\int_{[-1,1]^2} F(a,b)\,\mathrm{d}a\,\mathrm{d}b = \int_{[-1,1]^2} \big(F(u,uv)|u| + F(uv,v)|v|\big)\,\mathrm{d}u\,\mathrm{d}v. \tag{5.2}$$

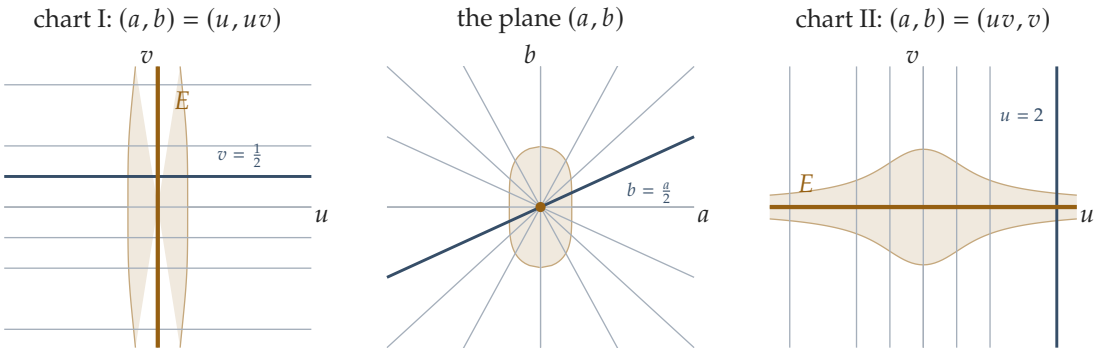


Figure 5.1. The blow-up of the origin of the plane. Middle: eight lines through the origin and the set $\{a^2 + b^4 < 0.05\}$ (shaded). Left and right: the two charts (5.1), which show all of the lines but the vertical one (chart I) and the horizontal one (chart II). Each line through the origin becomes a line crossing the exceptional divisor E (bronze) once, at its direction: $b = a/2$ (dark slate) is $\{v = \frac{1}{2}\}$ in chart I and $\{u = 2\}$ in chart II, and the origin has become all of E. In chart I the shaded set is a strip of nearly constant width around E, because $a^2 + b^4 = u^2 \times$ (a unit) there; in chart II it still pinches at the center, where $a^2 + b^4 = v^2(u^2 + v^2)$ is not normal crossing (Computation 5.1).

In $\mathbb{R}^d$ the blow-up of the origin has d charts, $w_j = u_j$ and $w_i = u_j u_i$ for $i \neq j$, with Jacobian $|u_j|^{d-1}$.

NORMAL CROSSING AND THE RECIPE. A change of variables $w = g(u)$ turns the loss into $K \circ g$ and the prior measure $\varphi(w)\,\mathrm{d}w$ into the weight $\varphi(g(u))|\det g'(u)|\,\mathrm{d}u$. The volume and the zeta function of Chapter 4 depend only on this pair, and they are monomial computations when both members are monomials in the same coordinates.

Definition 5.1 (Normal crossing). Let $\kappa \geq 0$ be analytic and $\omega \geq 0$ measurable near a point P of a d-dimensional analytic manifold. The pair (κ, ω) is *normal crossing at* P if there are analytic coordinates u centered at P in which, near P,

$$\kappa = u_1^{2k_1} \cdots u_d^{2k_d}\, a(u), \qquad \omega = |u_1|^{h_1} \cdots |u_d|^{h_d}\, b(u),$$

with nonnegative integers k_j, h_j and positive infinitely differentiable functions a, b. The *candidates* at P are the numbers $(h_j + 1)/(2k_j)$ for the j with $k_j > 0$.

The exponents of κ are even because $\kappa \geq 0$, and a positive factor such as a or b, called a *unit*, changes no exponent. The recipe, which Section 5.3 justifies, has four steps.

(i) Pull the loss and the prior back to the charts of a blow-up, multiplying the prior by $|\det g'|$.

(ii) Check normal crossing at every point of the exceptional divisor in every chart, not only at the centers.

(iii) Where the check fails, blow up that point, and repeat.

(iv) When every point is normal crossing, λ is the smallest candidate, and m the largest number of coordinates attaining it at one point.

Theorem A.8 guarantees that the process can be completed for every analytic loss; in general it needs blow-ups along smooth submanifolds,[3] but for two variables point blow-ups suffice. We drop the factor $\frac{1}{2}$ of the quartic loss, which changes neither λ nor m, since the volume function of cK is $V(\varepsilon/c)$.

[3] In $\mathbb{R}^d$ the blow-up along $\{w_1 = \cdots = w_k = 0\}$ has the charts $w_1 = u_1$, $w_i = u_1 u_i$ for $2 \le i \le k$, the other coordinates unchanged, with Jacobian $|u_1|^{k-1}$.

COMPUTATION 5.1 · THE QUARTIC LOSS $a^2 + b^4$ BY TWO BLOW-UPS

1 Near the origin $K = a^2 + b^4$ vanishes only at 0, and its Hessian there is $\operatorname{diag}(2, 0)$: neither the Morse lemma nor normal crossing. The prior is $\frac{1}{4}$ on $[-1, 1]^2$.

2 Blow up the origin. In chart I, $K = u^2(1 + u^2v^4)$ with Jacobian $|u|$. The factor $1 + u^2v^4$ is a unit, so the pair is normal crossing with $k = (1, 0)$, $h = (1, 0)$, and the candidate along $E_1 = \{u = 0\}$ is $\frac{1+1}{2} = 1$.

3 In chart II, $K = v^2(u^2 + v^2)$ with Jacobian $|v|$. At the points $(u_0, 0)$ with $u_0 \neq 0$ the factor $u^2 + v^2$ is a unit and the candidate is again 1. At the center, the direction of the b-axis, along which K grows only like b^4, the factor vanishes: not normal crossing.

4 Blow up the center of chart II. In chart II.1, $(u, v) = (s, st)$, so $(a, b) = (s^2t, st)$, $K = s^4t^2(1 + t^2)$, and the Jacobian is $|v| \cdot |s| = s^2|t|$: candidates $\frac{2+1}{4} = \frac{3}{4}$ for s and $\frac{1+1}{2} = 1$ for t. In chart II.2, $(u, v) = (st, t)$, so $(a, b) = (st^2, t)$, $K = t^4(1 + s^2)$, and the Jacobian is t^2: candidate $\frac{3}{4}$.

5 The factors $1 + t^2$ and $1 + s^2$ are units everywhere, so every point of charts I, II.1 and II.2, which cover the twice blown-up plane, is normal crossing. The candidates are 1 on E_1 and $\frac{3}{4}$ on the new curve E_2, which is $\{s = 0\}$ in II.1 and $\{t = 0\}$ in II.2; where E_1 and E_2 cross they differ. Hence $\lambda = \frac{3}{4}$ and $m = 1$, against $\frac{d}{2} = 1$ and half the Hessian rank, $\frac{1}{2}$.

6 Check by scaling: $a = \varepsilon^{1/2}x$, $b = \varepsilon^{1/4}y$ maps $\{a^2 + b^4 < \varepsilon\}$ onto $\{x^2 + y^4 < 1\}$, of area $B(\frac{1}{4}, \frac{3}{2}) = 3.4961$, and for $\varepsilon \le 1$ the set lies in the square, so $V(\varepsilon) = 0.87402\,\varepsilon^{3/4}$ exactly.

ONE CHART AND THE CONSTANT. Chart II.2 alone, $(a, b) = (st^2, t)$, maps $\{t \neq 0\}$ one-to-one onto $\{b \neq 0\}$, the plane without a null set. It is not proper, since $s = a/b^2$ is unbounded near the a-axis, but the change of variables formula needs no properness. If $t^4(1 + s^2) < \varepsilon \le 1$, then $|a| \le \sqrt{\varepsilon}$ and $|b| \le \varepsilon^{1/4}$, so the square imposes nothing, and

$$\begin{aligned} V(\varepsilon) &= \frac{1}{4}\int_{\mathbb{R}}\int_{\mathbb{R}} \mathbf{1}\{t^4(1+s^2) < \varepsilon\}\, t^2 \,\mathrm{d}t\,\mathrm{d}s \\ &= \frac{1}{4}\int_{\mathbb{R}} \frac{2}{3}\Big(\frac{\varepsilon}{1+s^2}\Big)^{3/4} \mathrm{d}s = \frac{B(\frac{1}{2}, \frac{1}{4})}{6}\,\varepsilon^{3/4}, \end{aligned} \tag{5.3}$$

which is the same $0.87402\,\varepsilon^{3/4}$. The parameter s labels the parabola $a = sb^2$ along which a point approaches the origin, and $s \in \mathbb{R} \cup \{\infty\}$ runs over the exceptional curve E_2: the constant is an integral over E_2, whose density $(1+s^2)^{-3/4}$ says how much prior mass approaches the truth along each parabola. This chart is one chart of a *weighted blow-up*, the one that Chapter 6 reads off the Newton polygon. For the model's loss $\frac{1}{2}(a^2+b^4)$ the volume is $0.87402\,(2\varepsilon)^{3/4} = 1.4699\,\varepsilon^{3/4}$, and the Abelian lemma (Lemma 4.3) gives $\bar{F}(n) = \frac{3}{4}\log n - \log(1.4699\,\Gamma(\frac{7}{4})) + o(1) = \frac{3}{4}\log n - 0.301 + o(1)$, as in Chapter 8.

THREE MORE BY HAND. The regular a^2+b^2 becomes $u^2(1+v^2)$ in chart I, with Jacobian $|u|$: one blow-up, candidate $1 = \frac{d}{2}$. Squaring halves the candidate: $(a^2+b^2)^2$ becomes $u^4(1+v^2)^2$, so $\lambda = \frac{1}{2}$, and indeed $V(\varepsilon) = \frac{\pi}{4}\varepsilon^{1/2}$. The product model at a generic truth, $(ab-c)^2$ with $0 < c < 1$, needs no blow-up: near a point of the hyperbola $ab = c$, where $a \neq 0$, the functions $(u, v) = (ab - c, a)$ are analytic coordinates with Jacobian determinant $-a$, in which $K = u^2$; so $\lambda = \frac{1}{2}$, $m = 1$, a case of Proposition 5.5 below.

WHAT ONE HAD TO NOTICE. Each direction of approach to the origin sees a simpler function than the loss, and a blow-up makes the directions into coordinates. Along most directions the quartic loss grows like a^2; along the b-axis it grows like b^4, and that direction, blown up again, carries the learning coefficient. The candidate $\frac{3}{4}$ combines the order 4 of the loss along E_2 with the order 2 of the Jacobian, which records how much area the two blow-ups squeeze near E_2; the Hessian sees only the first blow-up.

5.3 The theorem: what a resolution computes

Resolutions are not unique, but they all give the same answer, because they compute the volume and the zeta function, which do not refer to coordinates. The *learning coefficient* and the *real log canonical threshold* of K with respect to φ are one number, with $-\lambda$ the largest pole of $\zeta(z) = \int K^z \varphi$ (Chapter 4): the first name is Watanabe's, for its statistical role, the second the geometers'. The prior $\varphi \geq 0$ is bounded and need not be normalized. The engine is the monomial computation of Chapter 4 with a weight that may vanish, since a partition of unity will cut the weight into pieces.

Lemma 5.2 (Monomials with a continuous weight). *Let $k, h \in \mathbb{N}^d$ with $k \neq 0$, and let $c\colon [0,1]^d \to [0,\infty)$ be continuous. Put $r_j = (h_j+1)/(2k_j)$ when $k_j > 0$ and $r_j = \infty$ when $k_j = 0$, $\lambda = \min_j r_j$, $J = \{j : r_j = \lambda\}$ and $m = |J|$. Write $u = (u_J, u')$, with u' the coordinates outside J, and $u^h = \prod_j u_j^{h_j}$. Then the integral over the face $\{u_J = 0\}$*

$$C_c = \frac{1}{(m-1)!\,\lambda \prod_{j\in J} 2k_j} \int_{[0,1]^{d-m}} c(0_J, u') \prod_{j \notin J} u_j^{h_j - 2k_j\lambda}\, \mathrm{d}u' \qquad (5.4)$$

converges, and as $\varepsilon \to 0$

$$\int_{[0,1]^d} \mathbf{1}\{u^{2k} < \varepsilon\}\, u^h\, c(u)\, \mathrm{d}u = \big(C_c + o(1)\big)\, \varepsilon^{\lambda} \Big(\log \frac{1}{\varepsilon}\Big)^{m-1}.$$

If c is positive at some point of the face, then $C_c > 0$.

It is Proposition 4.4 restated for one orthant of the cube, with K exactly a monomial and a weight that may vanish, which is the form the proof of the theorem uses.

PROOF. Extend c to $[-1,1]^d$ by $c(u) = c(|u_1|, \dots, |u_d|)$, a continuous function. The integrand $\mathbf{1}\{u^{2k} < \varepsilon\}|u^h|c(u)$ is even in every coordinate, so its integral over $[-1,1]^d$ is 2^d times the integral in the lemma. Proposition 4.4, with $B = [-1,1]^d$, the factor $a \equiv 1$ and the weight $b = c \geq 0$, gives the integral over $[-1,1]^d$ as $(C + o(1))\varepsilon^\lambda(\log \frac{1}{\varepsilon})^{m-1}$ with a finite constant C, which is $2^d C_c$: its face integral runs over $[-1,1]^{d-m}$, which gives the factor 2^{d-m}, and $\prod_{j\in J} k_j = 2^{-m}\prod_{j\in J} 2k_j$. If $c(0_J, u'_0) > 0$, then $c(0_J, \cdot) > 0$ near u'_0 in the face, where the integrand of C_c is positive almost everywhere, so $C_c > 0$. ∎

The theorem allows the prior to be cut off by the boundary of the parameter set; in the charts this is recorded by *coordinate orthants*, sets $\{u : \sigma_j u_j \geq 0 \text{ for all } j\}$ with signs $\sigma_j \in \{1, -1\}$.

Theorem 5.3 (The learning coefficient on a resolution). *Let $O \subset \mathbb{R}^d$ be open, $K\colon O \to [0,\infty)$ real analytic, and $\varphi\colon O \to [0,\infty)$ bounded and measurable, zero outside a compact set $W \subset O$; let $W_0 = \{w \in W : K(w) = 0\}$. Let M be a d-dimensional real analytic manifold and $g\colon M \to O$ a proper real analytic map that, for some closed null set $N \subset O$ with null preimage, is an analytic isomorphism from $M \setminus g^{-1}(N)$ onto $O \setminus N$. Suppose that every point $P \in g^{-1}(W_0)$ is the center of a* coordinate cube*: analytic coordinates u near P, with $u(P) = 0$ and with $[-1,1]^d$ inside their range, in which, for $u \in [-1,1]^d$,*

$$K(g(u)) = u^{2k}a(u), \qquad \varphi(g(u))\,|\det g'(u)| = |u^h|\,b(u)\,\mathbf{1}_R(u), \tag{5.5}$$

with $k, h \in \mathbb{N}^d$, a and b positive and infinitely differentiable, and R a union of closed coordinate orthants, possibly empty. Let λ_P be the smallest candidate $(h_j+1)/(2k_j)$ of the cube and m_P the number of indices j attaining it, with $\lambda_P = \infty$ if R is empty, and suppose that $\lambda_P < \infty$ for some P. Then the smallest λ_P is attained; call it λ, and let m be the largest m_P with $\lambda_P = \lambda$.

(i) (Pulling back.) For every measurable $F\colon [0,\infty) \to [0,\infty]$, $\int_W F(K(w))\varphi(w)\,\mathrm{d}w = \int_M F(K(g(u)))\varphi(g(u))|\det g'(u)|\,\mathrm{d}u$; in particular V and ζ can be computed on M.

(ii) (Volume.) $V(\varepsilon) \sim C\varepsilon^\lambda(\log\frac{1}{\varepsilon})^{m-1}$ as $\varepsilon \to 0$, for a constant $C > 0$.

(iii) (Zeta function.) ζ extends to a meromorphic function on $\mathbb{C}$ whose poles are negative rational numbers; the largest is $-\lambda$, of order m.

(iv) (Local values.) For $w \in W_0$, let $\lambda(w)$ be the smallest λ_P over $P \in g^{-1}(w)$, and $m(w)$ the largest m_P among those P with $\lambda_P = \lambda(w)$. If $\lambda(w) < \infty$, both are attained, and for all small $r > 0$, $V_{w,r}(\varepsilon) = \int_{\{K<\varepsilon\}\cap B(w,r)} \varphi \asymp \varepsilon^{\lambda(w)}(\log\frac{1}{\varepsilon})^{m(w)-1}$.

In particular, λ and m depend neither on the resolution nor on the cubes.

The integral over M is computed chart by chart, $|\det g'(u)|\,du$ being the pull-back of Lebesgue measure. In Computation 5.1, M is the twice blown-up plane, N the origin, and the cubes are small squares around the points of $E_1 \cup E_2$.

Proof. The plan is to change variables, cover $g^{-1}(W_0)$ by finitely many cubes in which $K \circ g$ is exactly a monomial, cut the integrals into pieces with a partition of unity, apply Lemma 5.2 to each piece, and recover the zeta function from the volume.

Step 1: pulling back. On $M \setminus g^{-1}(N)$ the map g is a diffeomorphism onto $O \setminus N$, so by the change of variables formula in charts, $\int_{O\setminus N} G(w)\,dw = \int_{M\setminus g^{-1}(N)} G(g(u))|\det g'(u)|\,du$ for every measurable $G \geq 0$. Both N and $g^{-1}(N)$ are null; take $G = F(K)\varphi$.

Step 2: exact monomials and finitely many cubes. Let $P \in g^{-1}(W_0)$, with cube coordinates u and, say, $k_1 > 0$. The functions $x_1 = u_1 a(u)^{1/(2k_1)}$ and $x_j = u_j$ for $j \geq 2$ have a diagonal Jacobian matrix with positive entries at $u = 0$, so they are coordinates near P, by the inverse function theorem. In them $K \circ g = x^{2k}$; x_1 has the sign of u_1, so R is unchanged; and the weight is $|x^h|\,\tilde{b}(x)\,\mathbf{1}_R(x)$ with $\tilde{b} = b\,a^{-h_1/(2k_1)}|\det(\partial u/\partial x)|$ positive and smooth. So on a small cube $Q_P = \{|x_j| \leq \delta_P \text{ for all } j\}$ the pair satisfies (5.5) with $a \equiv 1$ and the same k, h, R. Let Q'_P be the open cube of half the size. The set $g^{-1}(W_0)$ is compact, because g is proper, so finitely many $Q'_{P_1}, \dots, Q'_{P_N}$ cover it, and on the compact set $g^{-1}(W) \setminus \bigcup_i Q'_{P_i}$ the function $K \circ g$ is at least some $\delta_0 > 0$. Choose smooth $\psi_i \geq 0$ on M, supported in Q_{P_i}, with $\sum_i \psi_i \leq 1$ everywhere and $\sum_i \psi_i = 1$ on the compact set $\{u \in g^{-1}(W) : K(g(u)) \leq \delta_0/2\}$: bump functions equal to 1 on the half-cubes, divided by their sum and multiplied by a smooth cutoff that equals 1 on that set and vanishes outside the union of the half-cubes, which contains the set.

Step 3: the volume. Let $\omega = (\varphi \circ g)|\det g'|$. For $\varepsilon < \delta_0/2$, (i) gives $V(\varepsilon) = \sum_i V_i(\varepsilon)$ with $V_i(\varepsilon) = \int_{Q_{P_i}} \mathbf{1}\{x^{2k} < \varepsilon\}\,|x^h|\,\tilde{b}\,\psi_i\,\mathbf{1}_R\,dx$ in the coordinates and exponents of the i-th cube. Split Q_{P_i} into its closed orthants, keep those in R, and reflect and rescale each onto $[0,1]^d$, which multiplies x^{2k} and the weight by constants. By Lemma 5.2, $V_i(\varepsilon) = (C_i + o(1))\varepsilon^{\lambda_i}(\log\frac{1}{\varepsilon})^{m_i-1}$ with $C_i \geq 0$ and $(\lambda_i, m_i) = (\lambda_{P_i}, m_{P_i})$. Let λ_* be the smallest λ_i and m_* the largest m_i with $\lambda_i = \lambda_*$; then $V(\varepsilon) = (C_* + o(1))\varepsilon^{\lambda_*}(\log\frac{1}{\varepsilon})^{m_*-1}$, with $C_* \geq 0$ the sum of the C_i over the i with $(\lambda_i, m_i) = (\lambda_*, m_*)$. Conversely, let $P \in g^{-1}(W_0)$ have $R \neq \emptyset$. The integrand of (i) is nonnegative, so $V(\varepsilon)$ is at least its integral over the cube of P, which by Lemma 5.2, applied to one orthant of R with the weight $\tilde{b} > 0$, is at least $c_P\varepsilon^{\lambda_P}(\log\frac{1}{\varepsilon})^{m_P-1}$ with $c_P > 0$. Comparing the bounds as $\varepsilon \to 0$, every such P has $\lambda_P > \lambda_*$, or $\lambda_P = \lambda_*$ and $m_P \leq m_*$. Since (λ_*, m_*) is attained at one of the P_i, the minimum λ is attained, $(\lambda, m) = (\lambda_*, m_*)$, and $C_* \geq c_{P_i} > 0$. This proves (ii).

Step 4: the zeta function. For $\Re z > 0$, $\zeta(z) = \sum_i \int (x^{2k})^z |x^h|\tilde{b}\,\psi_i \mathbf{1}_R\,dx + E(z)$, where $E(z) = \int (1 - \sum_i \psi_i)(K \circ g)^z \omega$. On the support of that integrand $K \circ g$ lies between $\delta_0/2$ and $\max K$, so E is entire. After the same splitting into orthants, each other piece is a finite sum of integrals $\int_{[0,1]^d} \prod_j y_j^{2k_j z + h_j}\,c(y)\,dy$ with c infinitely differentiable on $[0,1]^d$, which are meromorphic on $\mathbb{C}$ with

poles among the numbers $-(h_j+1+\ell)/(2k_j)$, $\ell \in \mathbb{N}$, by Definition A.5(ii) applied in one coordinate at a time. So ζ is meromorphic on $\mathbb{C}$, its poles are negative rational numbers, and by Step 3 none lies to the right of $-\lambda$. To see that $-\lambda$ is a pole of order exactly m, write $K^z = |z| \int_K^\infty t^{z-1}\,\mathrm{d}t$ for real $z<0$; by Tonelli's theorem, for $-\lambda < z < 0$,

$$\zeta(z) = |z| \int_0^\infty t^{z-1} V(t)\,\mathrm{d}t.$$

The part over $t \geq \delta$ is at most $V(\infty)\delta^z$, bounded near $z=-\lambda$. On $t<\delta$, (ii) gives $V(t) = (C+\eta(t))t^\lambda(\log\frac{1}{t})^{m-1}$ with $\eta(t)\to 0$, and the substitution $t=e^{-s}$ gives $\int_0^\delta t^{\sigma-1}(\log\frac{1}{t})^{m-1}\,\mathrm{d}t = \Gamma(m)\sigma^{-m}+O(1)$ as $\sigma = z+\lambda \downarrow 0$. Letting $\delta\to 0$, $(z+\lambda)^m\zeta(z)\to \lambda\,\Gamma(m)\,C>0$ as $z\downarrow -\lambda$, so the pole at $-\lambda$ has order exactly m.

Step 5: local values. Let $w \in W_0$ with $\lambda(w)<\infty$. The fiber $g^{-1}(w)$ is compact; cover it by finitely many half-cubes Q'_P, $P\in g^{-1}(w)$, with union U. Since g is proper, $g^{-1}(\bar{B}(w,r))\subset U$ for small r; otherwise points outside U with images tending to w would accumulate at a point of $g^{-1}(w)\setminus U$, which is empty. The argument of Step 3, with functions ψ_i for these finitely many cubes whose sum is 1 on $g^{-1}(\bar{B}(w,r))$, bounds $V_{w,r}(\varepsilon)$ above by a multiple of $\varepsilon^{\lambda'}(\log\frac{1}{\varepsilon})^{m'-1}$ for the best pair (λ',m') of the cubes; and for every $P\in g^{-1}(w)$ a small cube around P maps into $B(w,r)$, so $V_{w,r}(\varepsilon)\geq c\,\varepsilon^{\lambda_P}(\log\frac{1}{\varepsilon})^{m_P-1}$. As in Step 3, $(\lambda',m')=(\lambda(w),m(w))$, both attained. Finally, (ii) and (iii) determine λ and m through V and ζ, which do not refer to g. ∎

WHERE RESOLUTIONS COME FROM. Hironaka's theorem supplies them.[4] In the form of Theorem A.8, every analytic f with $f(w^*)=0$ has, near w^*, a proper analytic map g that is an isomorphism off the zero set and makes $f\circ g$ and $\det g'$ normal crossing in the same coordinates. Applied to K, or to $K\varphi_1$ when the prior is an analytic factor $\varphi_1\geq 0$ times a smooth positive one, it gives the hypotheses of Theorem 5.3 near each interior true parameter; resolving the defining functions of W together with K makes the region a union of orthants in each chart. Watanabe (2013, Theorem 1) states the result over a neighborhood of the true parameters as his *standard representation*, $K(g(u)) = u^{2k}$ and $\varphi(g(u))|g'(u)| = b(u)|u^h|$ with $b>0$, which is (5.5) with $a\equiv 1$; he states that its proof is given in Theorem 6.1 of his 2009 book. With it, every analytic K with such a prior has a learning coefficient, λ is a positive rational number, and $1\leq m\leq d$. The black box is the existence of g.

[4] Hironaka's two papers fill 217 pages of the *Annals of Mathematics* (Hironaka 1964a,b).

WHEN ONE BLOW-UP IS NOT ENOUGH. A loss that vanishes on a curve with a singular point needs more blow-ups. The *cusp* $a^2=b^3$ is the simplest singular curve, and its loss $(a^2-b^3)^2$ is the square of a function that changes sign, like the loss of a realizable regression.

COMPUTATION 5.2 · THE CUSP $(a^2-b^3)^2$ BY THREE BLOW-UPS

1 Write $K=f^2$ with $f=a^2-b^3$, and blow up the origin. In chart I, $f=u^2(1-uv^3)$ and $K=u^4\times$(a unit) near $E_1=\{u=0\}$, with

Jacobian $|u|$: candidate $\frac{1+1}{4} = \frac{1}{2}$. In chart II, $f = v^2(u^2 - v)$, with Jacobian $|v|$; away from the center the candidate along E_1 is again $\frac{1}{2}$. At the center, the curve $v = u^2$, the cusp seen in chart II and called its *strict transform*, is tangent to E_1: not normal crossing.

2 Blow up the center of chart II. In chart II.2, $(u, v) = (st, t)$, $f = t^3(s^2t - 1)$ and the Jacobian is t^2, so $K = t^6 \times$ (a unit) with candidate $\frac{2+1}{6} = \frac{1}{2}$ along $E_2 = \{t = 0\}$. In chart II.1, $(u, v) = (s, st)$, $f = s^3t^2(s - t)$, the Jacobian is $s^2|t|$, and $K = s^6t^4(s - t)^2$: three curves pass through the center, $E_2 = \{s = 0\}$, $E_1 = \{t = 0\}$ and the strict transform $\{s = t\}$, and three curves through a point cannot all be coordinate axes.

3 Blow up the center of chart II.1. In the chart $(s, t) = (p, pq)$, $K = p^{12}q^4(q - 1)^2$ with Jacobian $p^4|q|$; in the chart $(s, t) = (pq, q)$, $K = p^6q^{12}(p - 1)^2$ with Jacobian p^2q^4. The new curve E_3, $\{p = 0\}$ and $\{q = 0\}$ respectively, has candidate $\frac{4+1}{12} = \frac{5}{12}$. It meets E_1 at $q = 0$, the strict transform at $q = 1$ and E_2 at $p = 0$ in the second chart, each time transversally; the other curves have candidates $\frac{1}{2}$, $\frac{0+1}{2}$ and $\frac{1}{2}$.

4 Every point is now normal crossing, and $\lambda = \frac{5}{12}$, $m = 1$: the singular point lowers λ below the value $\frac{1}{2}$ of the smooth points of the curve.

5 The constant. In the first chart, $p^{12}q^4(q - 1)^2 < \varepsilon$ for $|p| < (\varepsilon/(q^4(q - 1)^2))^{1/12}$, and over these p the weight $\frac{1}{4}p^4|q|$ integrates to $\frac{1}{10}|q|\,(\varepsilon/(q^4(q - 1)^2))^{5/12}$. The second chart covers the rest of E_3 through $q = 1/p$, and

$$V(\varepsilon) \sim \frac{\varepsilon^{5/12}}{10} \int_{\mathbb{R}} |q|^{-2/3}|1 - q|^{-5/6}\,\mathrm{d}q$$
$$= \frac{B(\frac{1}{3}, \frac{1}{6}) + B(\frac{1}{2}, \frac{1}{6}) + B(\frac{1}{3}, \frac{1}{2})}{10}\,\varepsilon^{5/12} = 1.9906\,\varepsilon^{5/12}.$$

As for the quartic, the constant is an integral over the exceptional curve that carries λ, with the exponents $1 - 4 \cdot \frac{5}{12} = -\frac{2}{3}$ and $0 - 2 \cdot \frac{5}{12} = -\frac{5}{6}$ of (5.4) where other curves cross it. Scaling confirms it: $a = s^{1/2}x$, $b = s^{1/3}y$ with $s = \sqrt{\varepsilon}$ turns $\{|a^2 - b^3| < s\}$ into $\{|x^2 - y^3| < 1\}$, of area $7.9622 = 4 \times 1.9906$.

REAL, NOT COMPLEX. Algebraic geometry usually studies the *log canonical threshold* over the complex numbers, and it can differ from the learning coefficient. The loss $(a^2 + b^2 + c^2)^2$ on $[-1, 1]^3$ becomes $u^4(1 + v^2 + w^2)^2$ in the chart $(a, b, c) = (u, uv, uw)$, with Jacobian u^2, so $\lambda = \frac{3}{4}$, and indeed $V(\varepsilon) = \frac{1}{8} \cdot \frac{4\pi}{3}\varepsilon^{3/4}$. Over the complex numbers $1 + v^2 + w^2$ vanishes on a smooth surface, which gives the candidate $\frac{0+1}{2} = \frac{1}{2}$; it has no real points, and the volume never sees it. The real threshold is at least the complex one (Saito 2007, Corollary 2), so a complex computation only bounds λ from below.

5.4 The theorem: rules of the trade

Most learning coefficients in the literature come not from full resolutions but from rules that compare, add and multiply losses with known coefficients. The rules are statements about volumes and hold beyond analytic losses. Say that (K, φ) has *learning coefficient* λ *and multiplicity* m *in the volume sense* if there are constants $0 < c \le C$ with

$$c\,\varepsilon^{\lambda}\Big(\log\frac{1}{\varepsilon}\Big)^{m-1} \le V(\varepsilon) \le C\,\varepsilon^{\lambda}\Big(\log\frac{1}{\varepsilon}\Big)^{m-1} \qquad \text{for all small } \varepsilon > 0.$$

The pair (λ, m) is then unique. For an analytic K with a resolution, Theorem 5.3 gives these bounds with the pole data of ζ; for non-analytic losses the volume sense is the one that makes sense: the tanh neuron with Gaussian inputs, whose K is not analytic, has $K \asymp a^2 b^2$ on its prior box and so, by rule (ii) below, the product model's pair. Write $(\lambda, m) \preceq (\lambda', m')$ if $\lambda < \lambda'$, or $\lambda = \lambda'$ and $m \ge m'$: the smaller pair has the larger volume.

Proposition 5.4 (Rules of the trade). *Let $\varphi, \varphi_1, \varphi_2$ be bounded priors vanishing outside a compact set W, and let the losses below be nonnegative on W.*

(i) (Invariance.) If Φ is an analytic isomorphism from an open set O' onto an open set $O \supset W$, then $(K \circ \Phi, (\varphi \circ \Phi)|\det \Phi'|)$ and (K, φ) have the same volume and zeta functions, hence the same (λ, m).

(ii) (Comparison.) If (K_1, φ) and (K_2, φ) have (λ_1, m_1) and (λ_2, m_2) and $K_1 \le c\,K_2$, then $(\lambda_1, m_1) \preceq (\lambda_2, m_2)$. If (K_1, φ) has (λ, m) and $c_1 K_1 \le K_2 \le c_2 K_1$ with $c_1, c_2 > 0$, then (K_2, φ) has (λ, m). If (K, φ_1) has (λ, m) and $c_1\varphi_1 \le \varphi_2 \le c_2\varphi_1$, then (K, φ_2) has (λ, m).

(iii) (Sums.) Let $w = (w', w'') \in \mathbb{R}^{d'} \times \mathbb{R}^{d''}$, $K(w) = K'(w') + K''(w'')$ and $\varphi(w) = \varphi'(w')\varphi''(w'')$. If (K', φ') and (K'', φ'') have (λ', m') and (λ'', m''), then (K, φ) has $(\lambda' + \lambda'',\ m' + m'' - 1)$.

(iv) (Products.) With the same notation and $K = K'K''$, suppose the zeta functions of (K', φ') and (K'', φ'') are meromorphic on $\mathbb{C}$ with real poles, largest poles $-\lambda'$ and $-\lambda''$, of orders m' and m'', and that their defining integrals converge for real $z > -\lambda'$ and $z > -\lambda''$ respectively. Then the zeta function of (K, φ) is meromorphic on $\mathbb{C}$ with real poles, with largest pole $-\min(\lambda', \lambda'')$ of order m' if $\lambda' < \lambda''$, m'' if $\lambda'' < \lambda'$, and $m' + m''$ if $\lambda' = \lambda''$.

(v) ($\lambda \le \frac{d}{2}$.) If w_0 is an interior point of W with $K(w_0) = 0$, K is twice continuously differentiable near w_0, and $\varphi \ge c > 0$ near w_0, then $V(\varepsilon) \ge c'\,\varepsilon^{d/2}$ for small ε; so $\lambda \le \frac{d}{2}$ whenever (K, φ) has a learning coefficient.

(vi) (Global is the minimum of local.) Let K be continuous on W, $W_0 = \{w \in W : K(w) = 0\}$ nonempty, and suppose every $w \in W_0$ has a radius $r_w > 0$ such that $(K, \varphi \mathbf{1}_{B(w, r_w)})$ has a pair $(\lambda(w), m(w))$. Then the smallest $\lambda(w)$ is attained; call it λ; and (K, φ) has (λ, m) with m the largest $m(w)$ for which $\lambda(w) = \lambda$.

By Theorem 5.3(iv), the hypothesis of (vi) holds for every analytic K with a resolution, with $\lambda(w)$ and $m(w)$ the *local learning coefficient* and multiplicity at w, which Chapter 13 estimates at trained networks.

PROOF. The plan is to bound volume functions above and below by one another; only (iv) uses the zeta function. Throughout, $L = \log \frac{1}{\varepsilon}$.

(i) By the change of variables formula, both pairs give the same $\int F(K)\varphi$ for every measurable $F \geq 0$, in particular for $F = \mathbf{1}_{[0,\varepsilon)}$ and $F(t) = t^z$.

(ii) If $K_1 \leq cK_2$, then $\{K_2 < \varepsilon\} \subset \{K_1 < c\varepsilon\}$, so for small ε

$$c_2' \varepsilon^{\lambda_2} L^{m_2-1} \leq V_2(\varepsilon) \leq V_1(c\varepsilon) \leq C'' \varepsilon^{\lambda_1} L^{m_1-1},$$

with constants from the two definitions; as $\varepsilon \to 0$ this forces $\lambda_1 < \lambda_2$, or $\lambda_1 = \lambda_2$ and $m_1 \geq m_2$. If $c_1 K_1 \leq K_2 \leq c_2 K_1$, the same inclusions give $V_1(\varepsilon/c_2) \leq V_2(\varepsilon) \leq V_1(\varepsilon/c_1)$, both of the order $\varepsilon^{\lambda} L^{m-1}$. For the priors, $c_1 V_{\varphi_1} \leq V_{\varphi_2} \leq c_2 V_{\varphi_1}$.

(iii) Since $K', K'' \geq 0$, the set $\{K < \varepsilon\}$ contains $\{K' < \varepsilon/2\} \times \{K'' < \varepsilon/2\}$ and is contained in $\{K' < \varepsilon\} \times \{K'' < \varepsilon\}$. By Fubini's theorem $V'(\frac{\varepsilon}{2}) V''(\frac{\varepsilon}{2}) \leq V(\varepsilon) \leq V'(\varepsilon) V''(\varepsilon)$, and both bounds are of the order $\varepsilon^{\lambda'+\lambda''} L^{(m'-1)+(m''-1)}$.

(iv) By Tonelli's theorem $\zeta(z) = \zeta'(z)\zeta''(z)$ for $\Re z > 0$, and the product continues meromorphically with real poles. For real $z > -\lambda''$ the integral defining ζ'' converges, so $\zeta''(z) = \int K''^z \varphi'' > 0$; for analytic losses with a resolution this convergence is Step 4 of the proof of Theorem 5.3. If $\lambda' < \lambda''$, then ζ'' is holomorphic and positive at $-\lambda'$, so the product has a pole of order m' there and none to its right. If $\lambda' = \lambda''$, the leading Laurent terms multiply and the orders add.

(v) Since $K \geq 0$ is minimal at the interior point w_0, $\nabla K(w_0) = 0$, and Taylor's formula gives $K(w) \leq M|w - w_0|^2$ on a ball $B(w_0, \rho)$ where $\varphi \geq c$, with M half the largest norm of the Hessian there. So $\{K < \varepsilon\}$ contains the ball of radius $\sqrt{\varepsilon/M}$ about w_0 once $\varepsilon < M\rho^2$, and $V(\varepsilon) \geq c\, v_d (\varepsilon/M)^{d/2}$, with v_d the volume of the unit ball. If also $V(\varepsilon) \leq C\varepsilon^{\lambda} L^{m-1}$, then $\lambda \leq \frac{d}{2}$.

(vi) The set W_0 is compact. Cover it by finitely many balls $B(w_i, r_{w_i})$; outside their union $K \geq \delta > 0$ on W. Let (λ_*, m_*) be the smallest of the pairs $(\lambda(w_i), m(w_i))$ for $\leq$, and V_w the volume of $(K, \varphi \mathbf{1}_{B(w,r_w)})$. For $\varepsilon < \delta$, $V(\varepsilon) \leq \sum_i V_{w_i}(\varepsilon) \leq C\varepsilon^{\lambda_*} L^{m_*-1}$, and $V(\varepsilon) \geq V_{w_i}(\varepsilon) \geq c\, \varepsilon^{\lambda_*} L^{m_*-1}$ for the i attaining (λ_*, m_*); so (K, φ) has (λ_*, m_*). For every $w \in W_0$, $V(\varepsilon) \geq V_w(\varepsilon)$, which as in (ii) gives $(\lambda_*, m_*) \leq (\lambda(w), m(w))$; hence λ_* is the smallest $\lambda(w)$ and m_* the largest $m(w)$ among the points attaining it. ∎

WATANABE'S EXAMPLE. Watanabe (2013) resolves $K = (ab + c)^2 + a^2 b^4$, "the Kullback–Leibler distance of a neural network model" from his 2009 book, with four charts, such as $(a, b, c) = (a_3, b_3, a_3 b_3 (b_3 c_3 - 1))$, where $K = a_3^2 b_3^4 (c_3^2 + 1)$ with Jacobian $|a_3 b_3^2|$ and candidates 1 and $\frac{3}{4}$; he finds $\lambda = \frac{3}{4}$, $m = 1$. The rules give this in two lines. The shear $c' = ab + c$ has Jacobian 1 and makes $K = c'^2 + a^2 b^4$, a sum in separate variables; on $\{K < \varepsilon\}$ with $\varepsilon \leq 0.1$, $|ab| = |a|^{1/2} (|a| b^2)^{1/2} \leq \varepsilon^{1/4}$, so $|c| < 1$, and the cube in (a, b, c') has the same small volumes as the cube in (a, b, c). The loss c'^2 has $(\frac{1}{2}, 1)$, the monomial $a^2 b^4$ has $(\frac{1}{4}, 1)$, and by (iii) $\lambda = \frac{3}{4}$, $m = 1$. For $|a| \geq \sqrt{\varepsilon}$ the substitution $b = |a|^{-1/2} y$ gives $\{c'^2 + a^2 b^4 < \varepsilon\}$ the area $|a|^{-1/2} B(\frac{1}{4}, \frac{3}{2}) \varepsilon^{3/4}$, so

with the prior $\frac{1}{8}$, $V(\varepsilon) \sim \frac{1}{2}B(\frac{1}{4}, \frac{3}{2})\varepsilon^{3/4} = 1.748\,\varepsilon^{3/4}$. The local value is $\frac{3}{4}$ all along the a-axis and 1 along the b-axis away from the origin.

The last rule concerns the commonest shape of a set of true parameters: a smooth manifold along which the loss grows quadratically in every transverse direction. Following Wei et al. (2023), we call such a set *minimally singular*; a function with such a minimum set is called Morse–Bott, after Bott (1954). Chapter 14 applies the rule to the toy model of superposition, as Chen et al. (2023) do.

Proposition 5.5 (Minimally singular sets). *Let $K \geq 0$ be twice continuously differentiable near $w^* \in \mathbb{R}^d$, with $K(w^*) = 0$. Suppose that near w^* the zero set $\{K = 0\}$ is a C^2 submanifold S of dimension s, $0 \leq s < d$, and that $\nabla^2 K(w^*)$ has rank $d - s$; equivalently, it is positive definite on the orthogonal complement of the tangent space $T_{w^*}S$, on which it vanishes. Let φ be bounded, and bounded below by a positive constant near w^*. Then for every small $r > 0$,*

$$V_{w^*,r}(\varepsilon) = \int_{\{K<\varepsilon\}\cap B(w^*,r)} \varphi \asymp \varepsilon^{(d-s)/2} :$$

the local learning coefficient at w^* is $\frac{d-s}{2}$ and its multiplicity is 1.

The case $s = 0$ is the regular answer. The Hessian vanishes on $T_{w^*}S$ because $\nabla K \equiv 0$ on S: differentiating along a curve γ in S gives $\nabla^2 K\,\gamma' = 0$; so the hypothesis is that the rank is as large as it can be. Theorem 13.3 bounds $\lambda(w^*)$ between $\frac{r}{2}$ and $\frac{r}{2} + \frac{d-r}{4}$ for a Hessian of rank r: minimally singular sets attain the lower bound, and the quartic model ($r = 1$, $d = 2$) the upper one.

Proof. The plan is to straighten S and compare K with the squared distance to it. Because S is a C^2 submanifold, a C^2 diffeomorphism Φ from a neighborhood of $0 \in \mathbb{R}^{d-s} \times \mathbb{R}^s$ onto a neighborhood of w^*, with $\Phi(0) = w^*$, maps $\{x = 0\}$ onto S. Then $\tilde{K} = K \circ \Phi$ is C^2 and nonnegative, with $\tilde{K}(0, y) = 0$ and, since each $(0, y)$ is a minimum, $\nabla_x \tilde{K}(0, y) = 0$. Taylor's formula in x gives

$$\tilde{K}(x, y) = x^\top H(x, y)\, x, \qquad H(x, y) = \int_0^1 (1 - \tau)\, \nabla_x^2 \tilde{K}(\tau x, y)\, \mathrm{d}\tau,$$

with H continuous and $H(0, 0) = \frac{1}{2}\nabla_x^2 \tilde{K}(0, 0)$. At the critical point w^* the Hessian transforms as a quadratic form, $\nabla^2 \tilde{K}(0) = \Phi'(0)^\top \nabla^2 K(w^*) \Phi'(0)$, of rank $d - s$. Its y-block vanishes because $\tilde{K}(0, y) \equiv 0$, and its mixed block because $\nabla_x \tilde{K}(0, y) \equiv 0$; so the x-block, positive semidefinite because $\tilde{K} \geq 0$, has rank $d - s$ and is positive definite. By continuity there are $\rho > 0$ and $0 < c_1 \leq c_2$ with $c_1|x|^2 \leq \tilde{K}(x, y) \leq c_2|x|^2$ on the box $Q_\rho = \{|x|_\infty \leq \rho,\ |y|_\infty \leq \rho\}$, where $\tilde{\varphi} = (\varphi \circ \Phi)|\det \Phi'|$ is bounded above and below. For $\rho' \leq \rho$ and small ε, the set $\{x : c|x|^2 < \varepsilon\}$ is a ball inside the cube of side $2\rho'$, of volume $v_{d-s}(\varepsilon/c)^{(d-s)/2}$, so the volume of $(\tilde{K}, \tilde{\varphi})$ on $Q_{\rho'}$ is $\asymp \varepsilon^{(d-s)/2}$. Finally, for small r there is $\rho' < \rho$ with $\Phi(Q_{\rho'}) \subset B(w^*, r) \subset \Phi(Q_\rho)$, and by the change of variables formula $V_{w^*,r}$ lies between the volumes on $Q_{\rho'}$ and Q_ρ. ∎

THE RULES ON THE CAST. On the product model's cross away from the origin the Hessian of a^2b^2 has rank 1 and the local pair is $(\frac{1}{2}, 1)$; at the origin it is the monomial value $(\frac{1}{2}, 2)$, the global pair by (vi). For rank-one regression at a point with $a \neq 0$, $b = 0$, the zero set $\{b = 0\}$ has dimension N and the Hessian is $\|a\|^2 I_M$ in b, so the local value is $\frac{M}{2}$, against the global $\frac{\min(M,N)}{2}$ (Chapters 7 and 13).

5.5 Measuring it

The script `05-blowups-check.py` checks the theorem on the quartic $a^2 + b^4$, the cusp $(a^2 - b^3)^2$ and Watanabe's $(ab + c)^2 + a^2b^4$, with the uniform prior on $[-1, 1]^d$, $\lambda = \frac{3}{4}, \frac{5}{12}, \frac{3}{4}$, $m = 1$ and the constants $C = 0.87402, 1.99056, 1.74804$ computed above. It computes the volume at 27 values of ε from 10^{-1} to 10^{-14} in three ways: by quadrature of the exact lengths of the sections of $\{K < \varepsilon\}$ in the original coordinates, which uses nothing about the resolution; as the sum over the leaf charts of Computations 5.1 and 5.2, each a unit square by (5.2) applied at every blow-up, with the Jacobian as weight; and by counting which of 10^7 prior draws (seeds 5005 to 5007) have $K < \varepsilon$, with Clopper–Pearson 95% intervals, wherever at least 20 draws fall in the set. It runs in under a minute.

The chart sums agree with the original volumes to a relative $1.7 \cdot 10^{-7}$ (quartic) and $7.8 \cdot 10^{-7}$ (cusp) at every ε, and to 10^{-12} at most values of ε.[5] The Monte Carlo intervals contain the exact volume at 12 of 13, 26 of 27 and 14 of 14 values of ε (correlated, through shared draws). In Figure 5.2(a) the slopes are the learning coefficients, not $\frac{d}{2}$. In panel (b) the quartic's ratio to the prediction is 1, because $V(\varepsilon) = C\varepsilon^{3/4}$ exactly for $\varepsilon \leq 1$; Watanabe's is 0.858 at $\varepsilon = 10^{-2}$ and 0.9999 at 10^{-14}; the cusp's is 0.656, 0.841, 0.926 and 0.966 at $\varepsilon = 10^{-2}, 10^{-6}, 10^{-10}, 10^{-14}$. The cusp converges slowly because its next pole is close: the curves with candidate $\frac{1}{2}$ give a correction of order $\varepsilon^{1/2}$, relative order $\varepsilon^{1/12}$, and the script finds $V(\varepsilon) = C\varepsilon^{5/12} - 1.000\,\varepsilon^{1/2}$ to four digits from 10^{-6} to 10^{-14}. The deficit is the thin band around the curve $a^2 = b^3$ that the square cuts off where the curve leaves it: in the scaled coordinates below Computation 5.2 the band beyond $y = s^{-1/3}$ has area $4s^{1/6}$ to leading order, which removes $\frac{1}{4} \cdot 4\varepsilon^{1/2}$ from V. Least-squares slopes of $\log V$ over $[10^{-14}, 10^{-6}]$ are 0.7500, 0.4095 and 0.7494; local slopes between 10^{-13} and 10^{-14} are 0.7500, 0.4134 and 0.7500.

The script also estimates $\zeta(z) = \mathbb{E}[K(w)^z]$ under the prior by Monte Carlo in the original coordinates, at $z > -\lambda/4$, where K^z has a finite fourth moment,[6] and on the charts by quadrature in a form that shows the pole: for the quartic, charts II.1 and II.2 contribute $\frac{1}{4z+3}$ times $\int_0^1 t^{2z+1}(1 + t^2)^z \, dt$ and $\int_0^1 (1 + s^2)^z \, ds$. Table 5.1 shows the two agreeing within the intervals, and $(z + \lambda)\zeta(z)$ approaching the residue λC of Step 4 of the proof; the residues computed on the charts, 0.65551439 and 0.82939962, equal λC to eight digits.

A resolution also says what the posterior looks like near a singularity. The script `05-blowups-posterior.py` computes the Gibbs measure $\pi_n \propto e^{-nK}\varphi$ of the quartic model's population loss $K = \frac{1}{2}(a^2 + b^4)$, the posterior with

[5] Near the corners $(\pm 1, 1)$, where the cusp leaves the square, the sections in charts I and II.2 change their number of intervals; adaptive quadrature that is not told so loses the sixth digit.

[6] For $z \leq -\lambda/2$ the variance of K^z is infinite, and the sample mean, dominated by rare huge terms, is unreliable at every practical sample size.

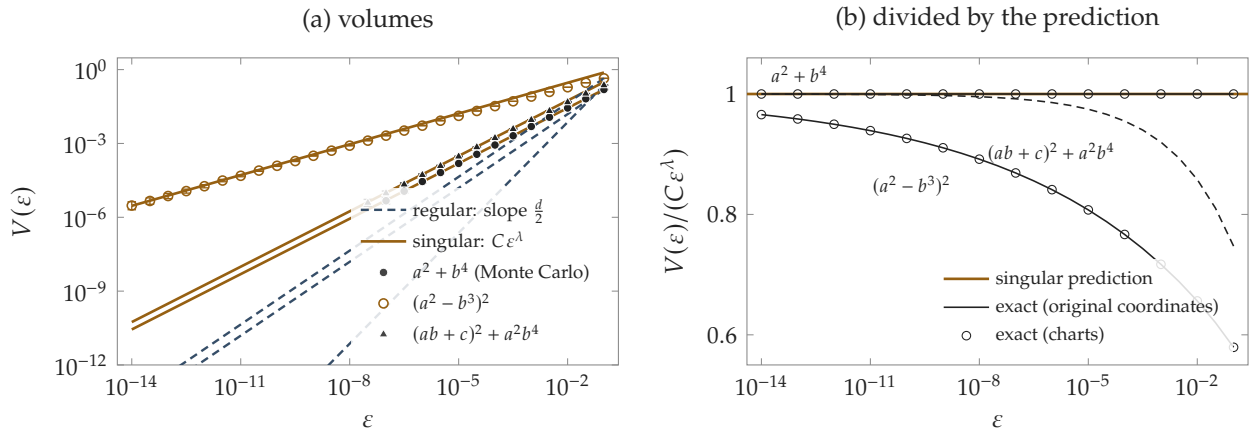


Figure 5.2. The volume $V(\varepsilon)$ of three losses with the uniform prior on $[-1,1]^d$. (a) Monte Carlo estimates from 10^7 prior draws (seeds 5005 to 5007) with Clopper–Pearson 95% intervals, where at least 20 draws fall in the set; bronze: the prediction $C\varepsilon^\lambda$ of Theorem 5.3 with $\lambda = \frac{3}{4}, \frac{5}{12}, \frac{3}{4}$ and the constants of Computations 5.1 and 5.2 and Watanabe's example; slate dashed: slope $\frac{d}{2}$ through the value at $\varepsilon = 10^{-1}$. (b) The exact volumes divided by $C\varepsilon^\lambda$: lines from sections in the original coordinates (dashed for Watanabe's example), open circles from the sums over the leaf charts, which agree with the lines to 10^{-6}. Script: `figures/scripts/05-blowups-check.py`.

the fluctuation of the data removed, whose normalizer is the partition function of Chapter 4. In the coordinates $s = a/b^2$ and $E = nK$ of chart II.2, the computation behind (5.3) gives π_n the density $\propto E^{\lambda-1}e^{-E}(1+s^2)^{-3/4}$ while $E < n/2$: the energy nK is Gamma$(\frac{3}{4}, 1)$ at every n, independent of the point s of the exceptional curve where the parameter approaches the truth; a regular model with two parameters would have Gamma$(1,1)$. The script does not use this: it computes π_n by midpoint quadrature on a 1500×1500 grid in the original coordinates, on a box holding all but e^{-60} of the mass, for $n = 10^2, 10^4, 10^6$. In Figure 5.3 the three distribution functions of nK coincide with that of Gamma$(\frac{3}{4})$ to within 0.0008, and the mean of nK is 0.7500 at all three n, against the regular 1. Among 2000 draws at $n = 10^4$ the Spearman correlation of $|\theta|$, $\theta = \arctan s$, with nK is 0.006, and the quartiles of θ are -1.187, -0.004, 1.125, against ∓ 1.144 and 0 for the density $\propto |\cos\theta|^{-1/2}$. With data the posterior acquires a random tilt in these coordinates (Theorem A.11 and Chapter 8); that $\mathbb{E}^\beta_w[nK]$ is about λ/β is Proposition 11.4.

5.6 The ledger

A learning coefficient is computed by changing coordinates until the loss and the prior are monomials. Blow-ups do it and change no integral, so λ and m are read off the exponents on the charts of a resolution, every resolution gives the same answer, and the constants of the volumes are integrals over exceptional curves. Most coefficients in practice come from rules: comparison, sums, products, minima of local values, and $\frac{d-s}{2}$ on

a^2+b^4, $\lambda=\frac{3}{4}$				$(a^2-b^3)^2$, $\lambda=\frac{5}{12}$			
z	Monte Carlo	charts	$(z+\lambda)\zeta$	z	Monte Carlo	charts	$(z+\lambda)\zeta$
-0.05	1.06103 ± 0.00005	1.06107	0.7428	-0.025	1.07211 ± 0.00005	1.07214	0.4199
-0.10	1.13168 ± 0.00011	1.13177	0.7357	-0.05	1.15564 ± 0.00011	1.15571	0.4238
-0.15	1.21432 ± 0.00020	1.21450	0.7287	-0.10	1.36889 ± 0.00032	1.36913	0.4336
-0.50		2.73545	0.6839	-0.30		4.58629	0.5351
-0.70		13.21920	0.6610	-0.40		44.72631	0.7454
$-\frac{3}{4}$		residue	0.6555	$-\frac{5}{12}$		residue	0.8294

Table 5.1. The zeta function of two losses with the uniform prior on $[-1,1]^2$. Monte Carlo: mean of K^z over 10^7 prior draws (seed 5050) with a Student 95% interval, for $z > -\lambda/4$ only. Charts: quadrature on the leaf charts. Last columns: $(z+\lambda)\zeta(z)$ from the charts, tending to the residue λC at $-\lambda$; slowly for the cusp, whose next pole is at $-\frac{1}{2}$. Script: `figures/scripts/05-blowups-check.py`.

minimally singular sets.

LEDGER · $y = ax_1 + b^2x_2$

QUANTITY	the learning coefficient λ and its multiplicity m, computed on a resolution
REGULAR THEORY	$\frac{d}{2}$ and $m = 1$: the Morse lemma makes a nondegenerate minimum a sum of squares, which one blow-up resolves
SINGULAR THEORY	$\lambda = \min(h_j+1)/(2k_j)$ over the charts of a resolution with $K \circ g = u^{2k}$ and $\varphi\lvert\det g'\rvert = b\lvert u^h\rvert$, m the most ties at one point (Theorem 5.3); the rules of Propositions 5.4 and 5.5
RUNNING EXAMPLE	the quartic model $K = \frac{1}{2}(a^2+b^4)$: two blow-ups, $\lambda = \frac{3}{4}$, $m = 1$, $V(\varepsilon) = 1.4699\,\varepsilon^{3/4}$, against $\frac{d}{2} = 1$ and half the Hessian rank, $\frac{1}{2}$; the cusp $(a^2-b^3)^2$: three blow-ups, $\frac{5}{12}$
MEASURED	chart sums equal original volumes to 10^{-6}; local slopes 0.7500, 0.4134, 0.7500 at $\varepsilon = 10^{-14}$; under the Gibbs measure $\propto e^{-nK}\varphi$ the energy nK has mean 0.7500 and law Gamma($\frac{3}{4}$) to 0.0008
RESTS ON	analyticity (Hironaka's theorem); a resolution in which the prior and the boundary of W are normal crossing too; a prior bounded above and below near the true parameters; knowing K

5.7 Where the theory stops

WHAT WAS ASSUMED. Theorem 5.3 assumes an analytic loss and a resolution in hand in which the prior, cut off by the boundary of the parameter set, is normal crossing too. The boundary matters: the product model at the truth $ab = 1$ on $[-1,1]^2$ has true parameters only at the corners $\pm(1,1)$, where $(ab-1)^2 = (s+t-st)^2$, with $s = 1-a \geq 0$ and $t = 1-b \geq 0$, vanishes at one point of a quadrant; the blow-up $s = x$, $t = xy$ gives $x^2(1+y-xy)^2$

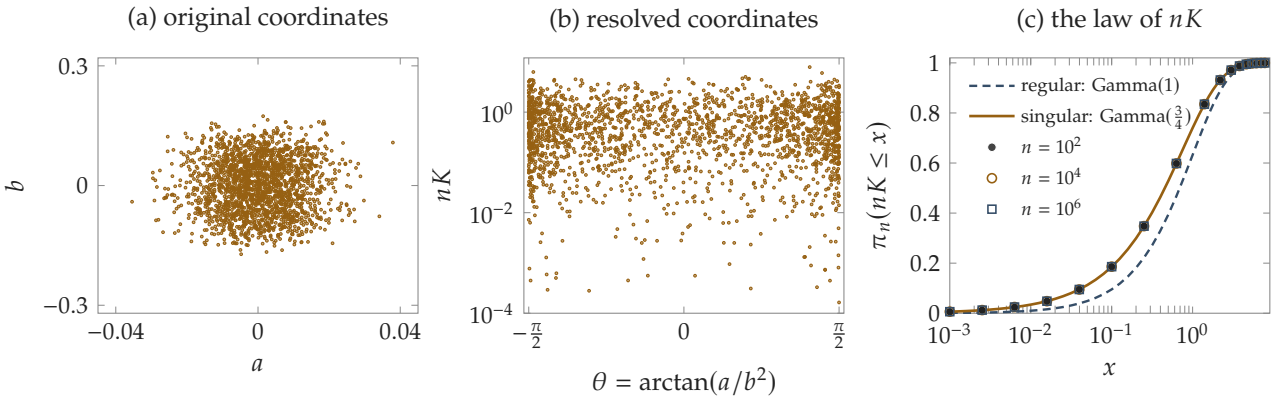


Figure 5.3. The Gibbs measure $\pi_n \propto e^{-nK}\varphi$ of the quartic model's population loss $K = \frac{1}{2}(a^2 + b^4)$, uniform prior on $[-1, 1]^2$, by midpoint quadrature on a 1500×1500 grid in the original coordinates. (a) 2000 draws at $n = 10^4$ (seed 5004). (b) The same draws in the coordinates of chart II.2: the point θ of the exceptional curve and the energy nK, which are independent. (c) The distribution function of nK at three sample sizes, against Gamma($\frac{3}{4}$) (bronze) and the regular Gamma(1) (slate dashed); the measured values differ from Gamma($\frac{3}{4}$) by at most 0.0008. Script: `figures/scripts/05-blowups-posterior.py`.

with Jacobian x, so $\lambda = 1$, not the $\frac{1}{2}$ of a hyperbola crossing the square (Exercise 5.10). The rules assume priors bounded above and below near the true parameters; a prior that vanishes there raises λ (Exercise 5.9), and Jeffreys' prior gives $\lambda \ge \frac{d}{2}$ (Watanabe 2024). And all of it assumes that the population loss K is known; in practice it is not, and Chapters 11 and 13 estimate λ from data.

WHERE IT BREAKS. Analyticity is not a technicality. The loss $K(a) = e^{-1/a^2}$ is infinitely differentiable and vanishes only at 0,[7] but $\{K < \varepsilon\} = \{|a| < (\log \frac{1}{\varepsilon})^{-1/2}\}$: with the uniform prior on $[-1, 1]$, $V(\varepsilon) = (\log \frac{1}{\varepsilon})^{-1/2}$, no power of ε, a zeta function that diverges for every $z < 0$, and a population free energy that grows like $\frac{1}{2} \log\log n$. No resolution exists for it; Watanabe (2010) notes the corresponding failure for a prior factor $e^{-1/\|w\|^2}$. ReLU networks are analytic only piecewise; the ReLU neuron escapes because its $K = \frac{1}{4}a^2b^2$ happens to be a polynomial. Second, Hironaka's theorem is an existence theorem. Constructive proofs exist (Bierstone and Milman 1997; Villamayor 1989), but the learning coefficients known exactly come from resolutions found by hand for particular families (Chapter 7), such as reduced-rank regression (Aoyagi and Watanabe 2005) and deep linear networks (Aoyagi 2024). Third, complex computations answer a different question: for $(a^2 + b^2 + c^2)^2$ the complex threshold is $\frac{1}{2}$ and the learning coefficient $\frac{3}{4}$, the value $\frac{n}{d}$ that Saito (2007, Proposition 3.3) proves for a zero of order d in n variables whose leading form has no other real zero. Finally, the asymptotics can be far away: the cusp's volume is 3.4% below its leading term at $\varepsilon = 10^{-14}$, and a slope fitted over the range $[10^{-6}, 10^{-1}]$ that Monte Carlo reaches easily is 0.385, against $\lambda = 0.417$.

[7] All its derivatives vanish at 0, so its Taylor series there is 0: the standard smooth function that is not analytic.

OPEN PROBLEMS. Computing the learning coefficients of the networks people train is open beyond the families of Chapter 7: exact values are known for one tanh layer with a single input, but not for one tanh layer with several inputs and outputs and more than four hidden units, nor for networks with two nonlinear layers. We know of no practical algorithm for the real log canonical threshold in many variables. The complex one is minus the largest root of the Bernstein–Sato polynomial (Kollár 1997), which computer algebra finds in small examples, but it only bounds λ from below. And a theory of learning coefficients for losses that are smooth or piecewise analytic but not analytic, which would cover ReLU networks without luck, would need a replacement for Hironaka's theorem; Chapter 13 measures such networks anyway.

Exercises

5.1 CONSOLIDATION. (The blow-up as a manifold.) Show that the charts (5.1) agree on their overlap through $(u, v) \mapsto (1/v, uv)$, that g is proper, and that E is a circle. Derive (5.2).

5.2 CONSOLIDATION. (The constant of the quartic.) Show that the residue of ζ at $-\frac{3}{4}$ for $a^2 + b^4$ is $\frac{1}{8}\big(\int_{-1}^{1}(1+s^2)^{-3/4}\,\mathrm{d}s + \int_{-1}^{1}|t|^{-1/2}(1+t^2)^{-3/4}\,\mathrm{d}t\big)$, that $t = 1/s$ turns the second integral into $\int_{|s|>1}(1+s^2)^{-3/4}\,\mathrm{d}s$, and hence that $C = \frac{1}{6}B(\frac{1}{2}, \frac{1}{4}) = \frac{1}{4}B(\frac{1}{4}, \frac{3}{2})$.

5.3 CONSOLIDATION. (Units and powers.) Show that $a^2b^2 + a^4b^4$ has $(\lambda, m) = (\frac{1}{2}, 2)$, that $a^2b^2c^2$ has $(\frac{1}{2}, 3)$ with no blow-up (check it with the product rule), and with one blow-up that $(a^2 + b^2)^k$ has $\lambda = \frac{1}{k}$ and $(a^2 + b^2 + c^2)^k$ has $\lambda = \frac{3}{2k}$.

5.4 CONSOLIDATION. (The chain, point by point.) Use Proposition 5.5 and the monomial rule to find the local pair at every true parameter of the chain $(a_1a_2a_3)^2$ on $[-1, 1]^3$, boundary points included, and deduce the global pair $(\frac{1}{2}, 3)$ from rule (vi).

5.5 BY HAND. (A loss with two branches.) Show that $K = a^2b^2 + a^4 = a^2(a^2 + b^2)$ becomes $u^2v^4(1 + u^2)$ in chart II, so that $\lambda = \frac{1}{2}$ and $m = 2$, the product model's pair, although K is not a monomial. Check that $V(\varepsilon) \sim \frac{1}{4}\varepsilon^{1/2}\log\frac{1}{\varepsilon}$ with the uniform prior.

5.6 BY HAND. (Tangent parabolas.) Resolve $K = (a^2 - b^4)^2$, which vanishes on the parabolas $a = \pm b^2$, by two point blow-ups, and show that $\lambda = \frac{3}{8}$, $m = 1$. *(Hint: in the chart $(a, b) = (st^2, t)$, $K = t^8(s-1)^2(s+1)^2$ with Jacobian t^2.)*

5.7 BY HAND. ★ (A higher cusp.) Show that $K = (a^2 - b^5)^2$ needs four point blow-ups, that on the last exceptional curve $K \sim p^{20}$ with Jacobian $|p|^6$, and that $\lambda = \frac{7}{20}$, $m = 1$. *(Hint: the first blow-up gives $f = v^2(u^2 - v^3)$ in chart II, a cusp again, with an extra v^2 in f and v in the Jacobian. Check by scaling $a = s^{1/2}x$, $b = s^{1/5}y$.)*

5.8 BY HAND. ★ (A network-shaped loss.) Let $K = (ab - cd)^2$ near $0 \in \mathbb{R}^4$, the loss of a sum of two products. Blow up the origin; in the chart $a = u$, $b = uv_2$, $c = uv_3$, $d = uv_4$ show that $K = u^4(v_2 - v_3v_4)^2$ with Jacobian $|u|^3$, that $v_2 - v_3v_4$ can serve as a coordinate, and that $\lambda = \frac{1}{2}$, $m = 1$: the singular point does not lower the value at the smooth points of the zero set. What does the Newton polyhedron of K predict (Chapter 6), and why does it fail here? *(Hint: the candidates are* $\frac{3+1}{4}$ *and* $\frac{1}{2}$*.)*

5.9 VARIATION. (A prior that vanishes.) Give $a^2 + b^4$ the prior proportional to $|b|^h$, $h \geq 0$, on $[-1, 1]^2$. Redo Computation 5.1 with the prior in the weight and show that $\lambda = \frac{3+h}{4}$, which exceeds $\frac{d}{2} = 1$ for $h > 1$; with the prior proportional to $|a|^h$, show that $\lambda = \frac{3+2h}{4}$. Why does this not contradict rule (v)?

5.10 VARIATION. (A corner.) Show that for $K = (ab - 1)^2$ with the uniform prior on $[-1, 1]^2$, $V(\varepsilon) \sim \frac{1}{4}\varepsilon$, so $\lambda = 1$, while for $(ab - c)^2$ with $0 < c < 1$, $V(\varepsilon) \sim \log(\frac{1}{c})\, \varepsilon^{1/2}$. Explain the difference with the coordinates $s = 1 - a$, $t = 1 - b$ and one blow-up.

5.11 MEASUREMENT. (Two poles.) Fit $V(\varepsilon) = A\varepsilon^{\lambda} + B\varepsilon^{1/2}$, with A, B, λ free, to the cusp's exact volumes on $[10^{-14}, 10^{-6}]$. How close does λ come to $\frac{5}{12}$? Repeat with the Monte Carlo volumes and say what limits the accuracy.

5.12 MEASUREMENT. (The posterior with data.) Change the posterior script to sample the posterior $\propto e^{-nK_n}\varphi$ of the quartic model for one data set at $n = 10^4$, and plot the draws in the coordinates (θ, nK). Is nK still independent of θ? Compare the mean of nK over 100 data sets with $\frac{3}{4}$, and relate the difference to the Gibbs generalization error of Chapter 9.

5.13 RESEARCH. (Resolution by machine.) Implement recursive point blow-ups for polynomials $K(a, b) \geq 0$ with $K(0, 0) = 0$, tracking strict transforms and Jacobians, and test it on the chapter's examples and on $(a^2 - b^{2k+1})^2$, $k \leq 5$. How does the number of charts grow? Find a loss for which the Newton-polygon shortcut of Chapter 6 fails and the blow-ups do not.

Notes and References

Resolution of singularities in characteristic zero is the theorem of Hironaka (1964a,b). Bernshtein and Gel'fand (1969) and Atiyah (1970) used it to continue $\int |f|^z \varphi$ meromorphically; Atiyah's real analytic local form is the version that Lin (2011, Theorem 3.3) attributes to Atiyah and says Watanabe (2009) uses as his Theorem 2.3. In learning theory, Watanabe (2001) gives an algorithm for λ by resolution, and Watanabe's standard representation (Watanabe 2013, Theorem 1) is the form in which the later theory uses Hironaka's theorem; his review (Watanabe 2024) restates it. Constructive resolution goes back to Villamayor (1989) and Bierstone and Milman (1997).

In algebraic statistics, Drton et al. (2009, Chapter 5) and Sullivant (2018, Chapter 17) treat Bayesian integrals of algebraic models, and Lin (2011, 2017) develops the rules of Proposition 5.4 for ideals, among them comparison, sums and products, and the Jacobian-rank bound that Chapter 13 uses (Propositions 3.4 and 3.7 and Corollary 4.15 of the paper, in the numbering of its arXiv version); Watanabe (2024, Section 3) lists the first three without proof. Saito (2007) proves that the real threshold is at least the complex one and that it can be larger; Kollár (1997) surveys the complex theory.

Theorem 5.3 is the book's own presentation of standard arguments: the monomial volumes of Proposition 4.4, a partition of unity on the resolved space, and the Mellin continuation of Definition A.5. The cusp's three blow-ups are its classical minimal embedded resolution; the constants as integrals over exceptional curves and the sum-rule shortcut for Watanabe's example, whose loss he attributes to Example 1.6 of the 2009 book, are ours. Proposition 5.5 is proved here by comparison with the squared distance to the critical set; nondegenerate critical manifolds are Bott (1954)'s, and the proposition is the case of equality in the Hessian lower bound of Chapter 13.

TRANSLATING THE LITERATURE. Watanabe (2013) writes the standard representation as $K(g(u)) = u^{2k}$ and $\varphi(g(u))|g'(u)| = b(u)|u^h|$, our (5.5) with $a \equiv 1$ and $|\det g'|$ for his $|g'|$, and calls λ the real log canonical threshold and m its multiplicity. Lin (2011, 2017) writes $\mathrm{RLCT}_\Omega(f;\varphi)$ for the pair (threshold, multiplicity) of a function f with amplitude φ on a region Ω: for $f = K$ our (λ, m), and for an ideal $\langle f_1, \dots, f_r\rangle$ the pair $(2\lambda, m)$ when $K \asymp \sum_i f_i^2$. Saito (2007) writes $\mathrm{rlct}(f)$ for the supremum of the c with $|f|^{-c}$ locally integrable: λ for $f = K$, 2λ for $K = f^2$. Algebraic geometers write $\mathrm{lct}(f)$ for the complex analogue; Kollár (1997) writes it $c_0(f)$ and computes it on a resolution as $\min_j (a_j + 1)/b_j$, with b_j the order of f along the divisor E_j and a_j its discrepancy: for $f = K$, our $2k_j$ and h_j. Watanabe (2024) writes θ, Θ, $\pi(\theta)$ for our w, W, $\varphi(w)$.

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6 Newton Polygons

The toric shortcut

THE MODEL — $y = a^2x_1 + b^3x_2$

IN THIS CHAPTER. Can the learning coefficient be read off the exponents of the loss? For the model $y = a^2x_1 + b^3x_2$, whose loss is $K = \frac{1}{2}(a^4 + b^6)$, logarithmic coordinates turn the set of nearly true parameters into a polyhedron and its volume exponent into the value of a linear program, $\frac{5}{12}$, whose dual finds the point $(t, t) = (\frac{12}{5}, \frac{12}{5})$ where the diagonal meets the Newton polygon of K. We prove that $\lambda = 1/t$ for every sum of even monomials with positive coefficients in two variables, with multiplicity $m = 2$ exactly when the diagonal meets a vertex, compute the zeta function on the cones of the polygon's normal fan, and state with sources the theorem for nondegenerate functions of d variables. Exact quadrature for thirty random sums reproduces the polygon's λ to 3×10^{-7} and its m to 10^{-3}, while Monte Carlo slopes at practical ε fall short by about 0.09 when $m = 2$. The polygon depends on the coordinates: the losses of the product model at a generic truth and of a linear network with two hidden units are degenerate, and for them $1/t$ is only an upper bound.

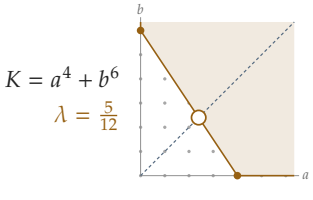
6.1 The question: reading λ off the exponents

The learning coefficients computed so far came from changes of coordinates. Chapter 5 blew up the origin twice to find $\lambda = \frac{3}{4}$ for $a^2 + b^4$, and three times for the cusp $(a^2 - b^3)^2$, where $\lambda = \frac{5}{12}$; each computation ended with the loss a monomial times a positive function in every chart, and with λ read off the monomial's exponents (Theorem 5.3). Do the exponents of the loss itself, before a change of coordinates, already contain the answer? The loss of a linear network with Gaussian inputs is a polynomial in the weights, and so are the losses of most toy models of this book: their exponents are what one sees first.

The model of this chapter is the *mixed cup* $y = a^2x_1 + b^3x_2 + \varepsilon$, with independent standard normal inputs x_1, x_2, unit noise, the uniform prior on $[-1, 1]^2$, and the truth $(a, b) = (0, 0)$, so that y is pure noise. Its loss is

$$K(a, b) = \tfrac{1}{2}\mathbb{E}_x\left[(a^2x_1 + b^3x_2)^2\right] = \tfrac{1}{2}\left(a^4 + b^6\right),$$

whose Hessian at the truth is zero: the data see a only through a^2 and b only through b^3.



THE REGULAR ANSWER · exponents. If K has a nondegenerate minimum at w_0, then in suitable analytic coordinates centered at w_0, $K = \sum_{i=1}^{d} w_i^2$ (the Morse lemma, the case $r = d$ of Theorem A.9). Every exponent that matters is 2, and $\lambda = \frac{d}{2}$ with $m = 1$ (Chapter 1).

For the mixed cup the regular answer says nothing: there are no squares, and counting parameters gives $\frac{d}{2} = 1$. Since K is a sum of a function of a and a function of b, the rule for sums of functions of separate variables (Proposition 5.4) gives $\lambda = \frac{1}{4} + \frac{1}{6} = \frac{5}{12}$ and $m = 1$: each coordinate contributes the reciprocal of its exponent. That rule stops at sums of separate monomials. The losses $a^6 + a^2b^2 + b^6$ and $a^4b^2 + a^2b^4$, the emblems of Parts III and IV, mix the variables.[1]

The learning coefficient of a function is what algebraic geometers call its *real log canonical threshold*; the two names denote one number. The question of the chapter is: *given the exponents of the monomials of K, what are λ and m, and when is the answer read off the exponents the true one?* The answer was found in the 1970s for the decay of oscillatory integrals,[2] and it is drawn at the foot of the chapter's first page: mark the exponents of K, take the convex hull of the quadrants above them, and find where the diagonal enters it. For the mixed cup it enters at $(t, t) = (\frac{12}{5}, \frac{12}{5})$, and $1/t = \frac{5}{12}$.

[1] Each opening page of this book carries the Newton diagram of its emblem, and each part page that of its part; this chapter explains the drawing.

[2] An oscillatory integral $\int e^{i\tau S(w)}\phi(w)\,dw$ decays like a power of τ, possibly times a power of $\log\tau$, as $\tau \to \infty$. For a phase $S \geq 0$ it obeys the same Newton-polyhedron estimates as the volume of $\{S < \varepsilon\}$ (Greenblatt 2010, Theorem 1.6).

6.2 By hand: Newton diagrams of the cast

Write a polynomial as $K = \sum_{v\in\mathcal{E}} c_v a^{v_1} b^{v_2}$, with $\mathcal{E}$ the set of exponents v whose coefficients c_v are not zero. The *Newton polygon* $\Gamma(K)$ is the convex hull of the quadrants $v + \mathbb{R}^2_{\geq 0}$, $v \in \mathcal{E}$. Its boundary comes down a vertical ray, runs through finitely many vertices and edges, and leaves along a horizontal ray. The diagonal enters $\Gamma(K)$ at a point (t, t), and t is the *Newton distance*. The rule this chapter proves and tests is

$$\lambda = \frac{1}{t}, \qquad m = \begin{cases} 2 & \text{if } (t,t) \text{ is a vertex of } \Gamma(K), \\ 1 & \text{otherwise.} \end{cases} \tag{6.1}$$

Figure 6.1 draws three Newton diagrams, and Table 6.1 applies the rule to the cast of the book. It reproduces every value computed in Chapters 3 to 5, and it fails on three of the last four rows.

COMPUTATION 6.1 · THE MIXED CUP IN LOGARITHMIC COORDINATES

1. The prior is uniform on $[-1, 1]^2$ and $K = \frac{1}{2}(a^4 + b^6)$ is even in each variable, so the volume of nearly true parameters is $V(\varepsilon) = \text{area}\{(a, b) \in (0, 1]^2 : a^4 + b^6 < 2\varepsilon\}$.
2. Substitute $a = e^{-u_1}$, $b = e^{-u_2}$ with $u \in [0, \infty)^2$, so that $da\,db = e^{-u_1-u_2}\,du$, $a^4 = e^{-4u_1}$ and $b^6 = e^{-6u_2}$. Write $L = \log\frac{1}{\varepsilon}$.
3. A sum of two positive numbers lies between the larger and twice the larger. So $a^4 + b^6 < 2\varepsilon$ forces $4u_1 > L - \log 2$ and $6u_2 > L - \log 2$, and it is implied by $4u_1 > L$ and $6u_2 > L$. The set is squeezed

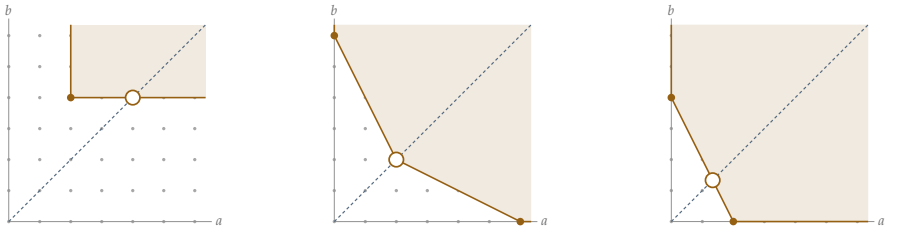


Figure 6.1. Newton diagrams. Gray dots mark the possible exponents (i, j) of $a^i b^j$, $0 \le i, j \le 6$; the shaded region is the Newton polygon, its bronze boundary passes through the vertices, the dashed slate line is the diagonal, and the ringed point is (t, t). For a^2b^4 the diagonal meets the horizontal ray at $(4, 4)$; for $a^6 + a^2b^2 + b^6$ it meets the vertex $(2, 2)$, and $m = 2$. The polygon of $(a - b^2)^2 = a^2 - 2ab^2 + b^4$ is that of $a^2 + b^4$, but its learning coefficient is $\frac{1}{2}$, not $1/t = \frac{3}{4}$.

between two translated quadrants, whose weights $\int e^{-u_1-u_2}\,du$ are $e^{-L/4-L/6} = \varepsilon^{5/12}$ and $2^{5/12}\varepsilon^{5/12}$: for $\varepsilon \le \frac{1}{2}$,

$$\varepsilon^{5/12} \le V(\varepsilon) \le 2^{5/12}\varepsilon^{5/12} = 1.3348\,\varepsilon^{5/12}.$$

4 The exponent is the value of a linear program: in the rescaled coordinates u/L the quadrant is $P = \{4u_1 \ge 1,\ 6u_2 \ge 1\}$, the weight $e^{-L(u_1+u_2)}$ is largest where $u_1 + u_2$ is smallest, and $\min_P(u_1 + u_2) = \frac{1}{4} + \frac{1}{6} = \frac{5}{12}$, at the corner $(\frac{1}{4}, \frac{1}{6})$.

5 The dual program, to maximize $\theta_1 + \theta_2$ over $\theta \ge 0$ subject to $\theta_1(4, 0) + \theta_2(0, 6) \le (1, 1)$, has the optimum $\theta = (\frac{1}{4}, \frac{1}{6})$ with the same value. Dividing $\theta_1(4, 0) + \theta_2(0, 6) = (1, 1)$ by $\theta_1 + \theta_2$ exhibits $(\frac{12}{5}, \frac{12}{5})$ as a convex combination of the exponents: the dual finds the point where the diagonal meets the Newton polygon, and $\lambda = 1/t$.

6 Exactly, scaling a by $(2\varepsilon)^{1/4}$ and b by $(2\varepsilon)^{1/6}$ gives $V(\varepsilon) = C_6(2\varepsilon)^{5/12} = 1.2662\,\varepsilon^{5/12}$ for $\varepsilon \le \frac{1}{2}$, with $C_6 = \int_0^1 (1 - v^6)^{1/4}\,dv = \Gamma(\frac{1}{6})\Gamma(\frac{5}{4})/(6\Gamma(\frac{17}{12})) = 0.9486$. The partition function factorizes, and $\bar{F}(n) = \frac{5}{12}\log n - \log\big(2^{5/12}\Gamma(\frac{5}{4})\Gamma(\frac{7}{6})\big) = \frac{5}{12}\log n - 0.1155$ up to exponentially small terms, as Lemma 4.3 predicts. At $n = 10^4$ the mixed cup pays 3.722; the regular plane pays $\log n - \log\frac{\pi}{2} = 8.759$.

What one had to notice is that *logarithmic coordinates* turn monomials into linear functions, and a sum of monomials with positive coefficients into something within a constant factor of its largest term.[3] The set of nearly true parameters becomes a polyhedron, its volume an exponential integral over the polyhedron, and the rate of decay of that integral the value of a *linear program*, whose dual lives in the plane of exponents (Figure 6.2).

[3] This comparison of the sum with its largest term is the only place where the proof of Theorem 6.4 looks at the coefficients. Its lower half fails as soon as a coefficient is negative, and nondegeneracy is the condition that repairs it.

model	K	vertices of $\Gamma(K)$	Newton $(\frac{1}{t}, m)$	true (λ, m)
regular plane	$\frac{1}{2}(a^2+b^2)$	$(0,2),(2,0)$	$(1,1)$	$(1,1)$
product model	$\frac{1}{2}a^2b^2$	$(2,2)$	$(\frac{1}{2},2)$	$(\frac{1}{2},2)$
cusp model	$\frac{1}{2}a^2b^4$	$(2,4)$	$(\frac{1}{4},1)$	$(\frac{1}{4},1)$
quartic model	$\frac{1}{2}(a^2+b^4)$	$(0,4),(2,0)$	$(\frac{3}{4},1)$	$(\frac{3}{4},1)$
quartic cup	$\frac{1}{2}(a^4+b^4)$	$(0,4),(4,0)$	$(\frac{1}{2},1)$	$(\frac{1}{2},1)$
mixed cup	$\frac{1}{2}(a^4+b^6)$	$(0,6),(4,0)$	$(\frac{5}{12},1)$	$(\frac{5}{12},1)$
emblem of Part III	$a^6+a^2b^2+b^6$	$(0,6),(2,2),(6,0)$	$(\frac{1}{2},2)$	$(\frac{1}{2},2)$
emblem of Part IV	$a^4b^2+a^2b^4$	$(2,4),(4,2)$	$(\frac{1}{3},1)$	$(\frac{1}{3},1)$
	$(a-b)^2$	$(0,2),(2,0)$	$(1,1)$	$(\frac{1}{2},1)$
	$(a-b^2)^2$	$(0,4),(2,0)$	$(\frac{3}{4},1)$	$(\frac{1}{2},1)$
	$(a^2-b^2)^2$	$(0,4),(4,0)$	$(\frac{1}{2},1)$	$(\frac{1}{2},2)$
cusp curve	$(a^2-b^3)^2$	$(0,6),(4,0)$	$(\frac{5}{12},1)$	$(\frac{5}{12},1)$

Table 6.1. The rule (6.1) on the cast. Upper block: sums of even monomials with positive coefficients, where Theorem 6.4 proves the rule; the true values come from Chapters 3 to 5 and, for the two emblems, from the theorem. Lower block: functions with cancellations; the true values come from Computation 6.3 and Theorem 6.9. Script: `figures/scripts/06-newton-byhand.py`.

COMPUTATION 6.2 · A VERTEX ON THE DIAGONAL: $a^6+a^2b^2+b^6$

1 The point $(2,2)$ lies below the segment from $(0,6)$ to $(6,0)$, so all three exponents are vertices, the diagonal enters the polygon at the vertex $(2,2)$, $t=2$, and the rule predicts $\lambda=\frac{1}{2}$ and $m=2$.

2 Each monomial is below K and K is below three times the largest, so with $R(\delta)=\{(a,b)\in(0,1]^2 : a^6<\delta,\ a^2b^2<\delta,\ b^6<\delta\}$ we have $R(\varepsilon/3)\subseteq\{K<\varepsilon\}\subseteq R(\varepsilon)$.

3 For $a<\delta^{1/3}$ the binding constraint on b is $b<\delta^{1/6}$; for $\delta^{1/3}\le a<\delta^{1/6}$ it is $b<\delta^{1/2}/a$. Hence

$$\operatorname{area} R(\delta)=\delta^{1/3}\delta^{1/6}+\int_{\delta^{1/3}}^{\delta^{1/6}}\frac{\delta^{1/2}}{a}\,\mathrm{d}a=\delta^{1/2}\bigl(1+\tfrac{1}{6}\log\tfrac{1}{\delta}\bigr),$$

and $(\frac{\varepsilon}{3})^{1/2}(1+\frac{1}{6}\log\frac{3}{\varepsilon})\le V(\varepsilon)\le\varepsilon^{1/2}(1+\frac{1}{6}\log\frac{1}{\varepsilon})$: $\lambda=\frac{1}{2}$ and $m=2$, as predicted.

4 The logarithm counts scales. Every scale $a\in(\varepsilon^{1/3},\varepsilon^{1/6})$ contributes equally, because there a^2b^2 dominates and the region is a hyperbolic strip; in Figure 6.2 this range is the thick edge of P, of length $\frac{1}{6}$ in u_1. Quadrature (Section 6.5) gives $V(10^{-8})=3.866\times10^{-4}$, between the bounds 2.456×10^{-4} and 4.070×10^{-4}, and $V(\varepsilon)/(\varepsilon^{1/2}\log\frac{1}{\varepsilon})$ equal to 0.210 at $\varepsilon=10^{-8}$ and 0.174 at 10^{-50}, approaching $\frac{1}{6}$. For the product model a^2b^2 the constant is $\frac{1}{2}$: its strip runs over all scales from $\varepsilon^{1/2}$ to 1.

The rule looks only at exponents, so it cannot see cancellations between coefficients.

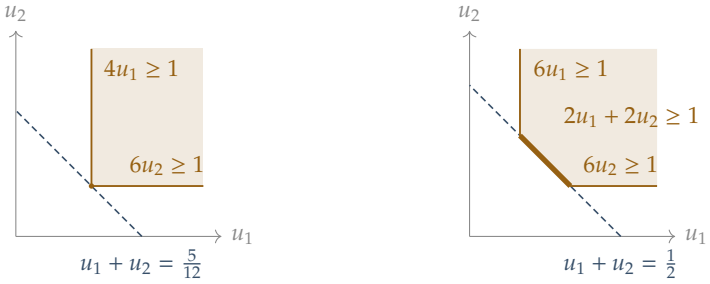


Figure 6.2. The dual picture. In the coordinates $a = e^{-Lu_1}$, $b = e^{-Lu_2}$ the set where every monomial of K is below $\varepsilon = e^{-L}$ is the shaded polyhedron P, with one bronze constraint for each vertex of the Newton polygon. Over P the prior weight $e^{-L(u_1+u_2)}$ is largest on the dashed line $u_1 + u_2 = 1/t$, which touches P at a corner for the mixed cup and along the thick edge for $a^6 + a^2b^2 + b^6$; there the volume collects equal contributions from a range of scales of length proportional to L, and $m = 2$.

COMPUTATION 6.3 · WHEN THE SHORTCUT FAILS

1. In $(a - b)^2 = a^2 - 2ab + b^2$ the exponent $(1, 1)$ lies on the segment between $(2, 0)$ and $(0, 2)$, so the polygon is that of $a^2 + b^2$: $t = 1$, and the rule predicts $\lambda = 1$, the value of a regular model with two parameters.
2. The truth is $\lambda = \frac{1}{2}$: the change $u = a - b$, $w = b$ has Jacobian 1 and turns K into u^2 (invariance under analytic isomorphisms, Proposition 5.4). On the prior square, exactly, $V(\varepsilon) = \sqrt{\varepsilon} - \frac{\varepsilon}{4}$.
3. In logarithmic coordinates step 3 of Computation 6.1 breaks: on the diagonal $a = b > 0$ all three monomials have size a^2 and $K = 0$. The face polynomial of the edge, $(a - b)^2$, vanishes at points with $a \neq 0$ and $b \neq 0$.
4. The same happens to $(a - b^2)^2$, whose polygon is that of the quartic model while $u = a - b^2$ turns it into u^2, so $\lambda = \frac{1}{2} < \frac{3}{4}$; and to $(a^2 - b^2)^2 = (a - b)^2(a + b)^2$, which is u^2w^2 in the coordinates $u = a - b$, $w = a + b$, so that $\lambda = \frac{1}{2} = 1/t$ but $m = 2$. The cusp curve $(a^2 - b^3)^2 = a^4 - 2a^2b^3 + b^6$ has the polygon of the mixed cup and an edge polynomial that vanishes on the curve $a^2 = b^3$, yet three blow-ups give $\lambda = \frac{5}{12} = 1/t$ and $m = 1$ (Chapter 5). Cancellation is a danger, not a verdict; Theorem 6.9 says when it matters.

6.3 The theorem: Newton's polygon in the plane

The computations suggest a proof: compare K with its largest monomial, pass to logarithmic coordinates, and solve a linear program. This section carries it out for the functions on which no cancellation can occur, computes their zeta function on the cones of the normal fan, and extracts the half of the rule that survives cancellations.

6.3.1 Newton polygons and linear programs

Definition 6.1 (Newton polygon). Let $K(a,b) = \sum_{v\in\mathcal{E}} c_v a^{v_1} b^{v_2}$ be a polynomial, or a power series convergent near 0, whose exponent set $\mathcal{E} = \{v : c_v \neq 0\} \subset \mathbb{N}^2$ does not contain $(0,0)$. Its *Newton polygon* is $\Gamma = \operatorname{conv}(\mathcal{E}) + \mathbb{R}^2_{\geq 0}$ and its *Newton distance* is $t = \min\{\tau \geq 0 : (\tau,\tau) \in \Gamma\}$. For a nonzero $\eta \in \mathbb{R}^2_{\geq 0}$, the function $v \mapsto \langle v, \eta\rangle$ attains its minimum over Γ, which is $\mu(\eta) = \min_{v\in\mathcal{E}}\langle v,\eta\rangle$, on a *face* of Γ (a vertex, a compact edge, or one of the two boundary rays), and η is an inner normal of that face. The *diagonal face* is the face that contains (t,t) in its relative interior. For a compact face F the *face polynomial* is $K_F = \sum_{v\in\mathcal{E}\cap F} c_v a^{v_1} b^{v_2}$. A *positive even sum* is a polynomial K as above with every $c_v > 0$ and every exponent even.

The minimum over Γ equals the minimum over $\mathcal{E}$ because a point of Γ is a convex combination of exponents plus a vector $w \geq 0$, and $\langle w,\eta\rangle \geq 0$; and $t \geq \frac{1}{2}$, because every point of Γ has coordinate sum at least 1. A positive even sum is nonnegative, vanishes at 0, and is a function of $|a|$ and $|b|$.

Order the vertices of Γ as $V^1,\dots,V^r$ with increasing first coordinates. Let $\eta_0 = (1,0)$ and $\eta_r = (0,1)$, the inner normals of the two rays, and for $1 \leq k < r$ let η_k be the primitive integer vector perpendicular to the edge $[V^k, V^{k+1}]$, whose coordinates are positive. These are the rays of the *normal fan*; C_k denotes the cone spanned by η_{k-1} and η_k, $|u|_1 = u_1 + u_2$ the coordinate sum of $u \in \mathbb{R}^2_{\geq 0}$, and $\mu_k = \mu(\eta_k)$.

Lemma 6.2 (The normal fan and the linear program). *Let Γ and t be as in Definition 6.1.*

(i) The cones $C_1,\dots,C_r$ cover $\mathbb{R}^2_{\geq 0}$ and meet only along rays; the vertex V^k minimizes $\langle\cdot,u\rangle$ over Γ exactly when $u \in C_k$, so $\mu(u) = \langle V^k, u\rangle$ there.

(ii) (Weak duality.) $\mu(u) \leq t|u|_1$ for every $u \in \mathbb{R}^2_{\geq 0}$.

(iii) (Strong duality.) Equality $\mu(u) = t|u|_1$ holds exactly on the normal cone $\mathcal{N}$ of Γ at (t,t): $\mathcal{N} = C_k$ if $(t,t) = V^k$, and $\mathcal{N}$ is the ray through η_k if (t,t) lies in the relative interior of the face with inner normal η_k. On the polyhedron $P = \{u \in \mathbb{R}^2_{\geq 0} : \langle v,u\rangle \geq 1 \text{ for every } v \in \mathcal{E}\}$ the minimum of $|u|_1$ is $1/t$, attained exactly on $F^* = \{u \in \mathcal{N} : |u|_1 = 1/t\}$: the segment from η_{k-1}/μ_{k-1} to η_k/μ_k in the vertex case, the point η_k/μ_k otherwise.

(iv) For every ray with $\mu_j > 0$, $|\eta_j|_1/\mu_j \geq 1/t$, with equality exactly for the rays in $\mathcal{N}$.

Proof. The plan is to read (i) off the convexity of the boundary and (ii)–(iv) off the fact that (t,t) lies on it. (i) The edge from V^k to V^{k+1} is perpendicular to η_k and goes right and down, so $\langle V^{k+1} - V^k, u\rangle = g_k(u_1\eta_{k,2} - u_2\eta_{k,1})$ with $g_k > 0$; it is positive, zero or negative as the angle of u is smaller than, equal to or larger than that of η_k. By convexity the angles of $\eta_0,\dots,\eta_r$ increase

strictly from 0 to $\frac{\pi}{2}$. Hence the cones cover the quadrant and meet only along rays, and for $u \in C_k$ the values $\langle V^j, u\rangle$ do not increase up to $j = k$ and do not decrease after it, so V^k minimizes over the vertices, hence over $\Gamma = \operatorname{conv}\{V^j\} + \mathbb{R}^2_{\geq 0}$; for $u \notin C_k$ a neighboring vertex does strictly better. (ii) Write $(t,t) = \sum_v \theta_v v + w$ with θ a probability vector on $\mathcal{E}$ and $w \geq 0$; for $u \geq 0$, $t|u|_1 = \langle (t,t), u\rangle \geq \sum_v \theta_v \langle v, u\rangle \geq \mu(u)$. (iii) Equality in (ii) says that (t,t) minimizes $\langle \cdot, u\rangle$ over Γ, that is, $u \in \mathcal{N}$. If $(t,t) = V^k$, then $\mathcal{N} = C_k$ by (i). If (t,t) lies in the relative interior of the face with normal η_k and direction e, then $(t,t) \pm se \in \Gamma$ for small $s > 0$, so every $u \in \mathcal{N}$ is orthogonal to e and is a multiple of η_k; and $\eta_k \in \mathcal{N}$, because (t,t) lies on the face where $\langle \cdot, \eta_k\rangle$ is smallest. For $u \in P$, (ii) gives $1 \leq \mu(u) \leq t|u|_1$, with equality exactly when $u \in \mathcal{N}$ and $|u|_1 = 1/t$, and such u lie in P. The rays of $\mathcal{N}$ meet the line $|u|_1 = 1/t$ at $\eta/(t|\eta|_1) = \eta/\mu(\eta)$. (iv) Apply (ii) and (iii) to $u = \eta_j$. ∎

Minimizing $|u|_1$ over P is a linear program whose dual, maximizing $\sum_v \theta_v$ over $\theta \geq 0$ with $\sum_v \theta_v v \leq (1,1)$, has the value $1/t$; parts (ii) and (iii) are weak and strong duality.

6.3.2 The formula

The proof bounds the volume function from both sides; the next lemma, a two-sided form of the Abelian lemma of Chapter 4, turns such bounds into the learning coefficient and its multiplicity.

Lemma 6.3 (Two-sided volume bounds). *Let $K \geq 0$ be bounded and measurable on W, let φ be a prior density on W, and suppose that for some $\lambda > 0$, integer $m \geq 1$, constants $0 < c \leq C$ and $\varepsilon_0 \in (0,1)$,*

$$c\, g(\varepsilon) \leq V(\varepsilon) \leq C\, g(\varepsilon) \quad \text{for } 0 < \varepsilon < \varepsilon_0, \qquad g(\varepsilon) = \varepsilon^{\lambda} \big(\log \tfrac{1}{\varepsilon}\big)^{m-1}. \tag{6.2}$$

Then $\bar{F}(n) = \lambda \log n - (m-1)\log\log n + O(1)$. Moreover $\zeta(z) = \int_W K^z \varphi$ is holomorphic on $\Re z > -\lambda$, and $\zeta(x)(x+\lambda)^m$ stays between two positive constants for real $x \in (-\lambda, -\lambda + \delta)$, for some $\delta > 0$. Consequently, if ζ extends meromorphically to a neighborhood of $-\lambda$, then $-\lambda$ is a pole of order exactly m, and no pole has a larger real part.

Proof. The plan is to write both transforms as integrals of V and compare them with the same integrals of g. As in the proof of Theorem A.6, $Z(n) = \int_W e^{-nK}\varphi = n\int_0^\infty e^{-ns} V(s)\,\mathrm{d}s$. Choose $\varepsilon_1 \leq \varepsilon_0$ with $\varepsilon_1 < e^{-(m-1)/\lambda}$, so that g increases on $(0, \varepsilon_1]$, and let ν be the measure on $[0, \varepsilon_1]$ with $\nu([0,s)) = g(s)$. Part (i) of Theorem A.6 gives $J(n) = \int e^{-ns}\nu(\mathrm{d}s) = n\int_0^{\varepsilon_1} e^{-ns} g(s)\,\mathrm{d}s + e^{-n\varepsilon_1} g(\varepsilon_1) \sim \Gamma(\lambda+1) n^{-\lambda} (\log n)^{m-1}$. Splitting $Z(n)$ at ε_1 and using (6.2) below ε_1 and $V \leq 1$ above it, $c(J(n) - e^{-n\varepsilon_1} g(\varepsilon_1)) \leq Z(n) \leq CJ(n) + e^{-n\varepsilon_1}$, and taking logarithms gives the expansion of $\bar{F}(n) = -\log Z(n)$.

For real $x \in (-\lambda, 0)$, Tonelli's theorem and $K^x = \int_K^\infty (-x) s^{x-1}\,\mathrm{d}s$ give $\zeta(x) = (-x)\int_0^\infty s^{x-1} V(s)\,\mathrm{d}s$. The part over $s \geq \varepsilon_0$ is at most ε_0^x. Below ε_0 we use (6.2); with $\kappa = x + \lambda$ and $\ell_0 = \log \frac{1}{\varepsilon_0}$, the substitution $s = \varepsilon_0 e^{-\rho}$ turns

$\int_0^{\varepsilon_0} s^{x-1} g(s)\,\mathrm{d}s$ into $\varepsilon_0^\kappa \int_0^\infty e^{-\kappa\rho}(\rho+\ell_0)^{m-1}\,\mathrm{d}\rho$, and expanding the binomial,

$$\int_0^{\varepsilon_0} s^{x-1} g(s)\,\mathrm{d}s = \varepsilon_0^\kappa \sum_{j=0}^{m-1} \binom{m-1}{j} \frac{j!\,\ell_0^{m-1-j}}{\kappa^{j+1}} = \frac{(m-1)!}{\kappa^m}\big(1+O(\kappa)\big).$$

Since $-x \to \lambda > 0$, $\zeta(x)\kappa^m$ stays between positive constants near $\kappa = 0$, and $\zeta(x) < \infty$ for $x > -\lambda$. As $|K^z| = K^{\Re z}$, the integral defining ζ converges absolutely and locally uniformly on $\Re z > -\lambda$, where ζ is therefore holomorphic. If ζ is meromorphic near $-\lambda$ with Laurent series $\sum_{j\ge -p} a_j(z+\lambda)^j$, $a_{-p} \neq 0$, then $\zeta(x)(x+\lambda)^p \to a_{-p}$, and the two-sided bound forces $p = m$. ∎

Theorem 6.4 (Newton's polygon formula in the plane). *Let K be a positive even sum with Newton distance t, and let $m = 2$ if (t,t) is a vertex of its Newton polygon and $m = 1$ otherwise. Let the prior φ be a density on $W = [-1,1]^2$ that is bounded, and bounded below by a positive constant near 0. Then:*

(i) there are constants $0 < c \le C$ and $\varepsilon_0 > 0$ with $c\,\varepsilon^{1/t}(\log\frac{1}{\varepsilon})^{m-1} \le V(\varepsilon) \le C\,\varepsilon^{1/t}(\log\frac{1}{\varepsilon})^{m-1}$ for $0 < \varepsilon < \varepsilon_0$;

(ii) $\bar F(n) = \frac{1}{t}\log n - (m-1)\log\log n + O(1)$ as $n \to \infty$;

(iii) ζ is holomorphic on $\Re z > -\frac{1}{t}$, and $\zeta(x)(x+\frac{1}{t})^m$ stays between two positive constants as $x \downarrow -\frac{1}{t}$; for the uniform prior, ζ is meromorphic on a half-plane containing $\{\Re z \ge -\frac{1}{t}\}$ and $-\frac{1}{t}$ is its largest pole, of order m (Proposition 6.5).

The learning coefficient of K on W is therefore $\lambda = 1/t$, with multiplicity m.

The true set of a positive even sum is the origin, an axis, or both axes; the theorem is global on W, and the origin determines λ. Its hypotheses on the prior hold for every continuous prior that is positive at the origin.

Proof. The plan is that of Computation 6.1: reduce to the uniform prior on the positive quadrant, compare K with its largest monomial, and compute the volume of the largest monomial's sublevel set in logarithmic coordinates with Lemma 6.2. Parts (ii) and (iii) then follow from (i) by Lemma 6.3, except the meromorphy, which is Proposition 6.5. Write $L = \log\frac{1}{\varepsilon}$ and, for $H \ge 0$, $A_H(\varepsilon) = \mathrm{area}\{(a,b) \in (0,1]^2 : H(a,b) < \varepsilon\}$.

Step 1: reduction. Let $\varphi \le \Phi$ on W and $\varphi \ge \phi_0 > 0$ on $[-r_0, r_0]^2$, $0 < r_0 \le 1$. By evenness, $V(\varepsilon) \le 4\Phi A_K(\varepsilon)$, and $V(\varepsilon) \ge 4\phi_0 r_0^2 A_{\tilde K}(\varepsilon)$ with $\tilde K(a,b) = K(r_0 a, r_0 b)$, a positive even sum with the same exponents. So it suffices to show, for every positive even sum, that $A_K(\varepsilon)$ lies between constant multiples of $\varepsilon^{1/t}L^{m-1}$ for small ε.

Step 2: the largest monomial. Let $M(a,b) = \max_{v\in\mathcal{E}} a^{v_1}b^{v_2}$ on $(0,1]^2$, and let c_+ and c_- be the largest and the smallest coefficient. Then $c_- M \le K \le |\mathcal{E}|c_+ M$, and

$$A_M\big(\varepsilon/(|\mathcal{E}|c_+)\big) \le A_K(\varepsilon) \le A_M(\varepsilon/c_-). \tag{6.3}$$

Step 3: logarithmic coordinates. The map $u \mapsto (e^{-u_1}, e^{-u_2})$ is a diffeomorphism of the open quadrant onto $(0,1)^2$ with Jacobian $e^{-|u|_1}$, and

$M(e^{-u_1}, e^{-u_2}) = e^{-\mu(u)}$. Since μ is positively homogeneous and $\{\mu = 1\}$ is a null set,

$$A_M(e^{-L}) = \int_{\{\mu(u)>L\}} e^{-|u|_1}\,\mathrm{d}u = L^2 \int_P e^{-L|u|_1}\,\mathrm{d}u = L^2 \int_{1/t}^{\infty} e^{-Lh}\psi(h)\,\mathrm{d}h.$$

The last step uses the coordinates (u_1, h), $h = |u|_1$, of Jacobian 1, and $\psi(h)$ is the length of the interval $\{u_1 : (u_1, h - u_1) \in P\}$, an interval because P is convex, and empty for $h < 1/t$ by Lemma 6.2(iii). Let $\psi_0 = \psi(1/t)$, the length of the projection of F^* to the first axis: $\psi_0 > 0$ if (t, t) is a vertex and $\psi_0 = 0$ otherwise.

Step 4: the shape of P near F^*. We show that for some $B \geq 1$ and every $h \geq 1/t$,

$$\psi_0 + \left(h - \tfrac{1}{t}\right) \leq \psi(h) \leq B\left(\psi_0 + h - \tfrac{1}{t}\right). \tag{6.4}$$

Lower bound: P is closed upwards, because the exponents are ≥ 0. If $u^{\pm}$ are the endpoints of F^* with $u_1^- \leq u_1^+$, the points $u^- + (0, h - \frac{1}{t})$ and $u^+ + (h - \frac{1}{t}, 0)$ lie in the convex slice $P \cap \{|u|_1 = h\}$, and the segment between them projects to an interval of length $\psi_0 + h - \frac{1}{t}$. Upper bound in the vertex case: the slice lies in $\{u \geq 0 : |u|_1 = h\}$, so $\psi(h) \leq h \leq \max(1, \frac{1}{t\psi_0})(\psi_0 + h - \frac{1}{t})$. Otherwise $F^* = \{u^*\}$ and (t, t) lies in the relative interior of a face Δ on which $\langle \cdot, u^* \rangle$ attains its minimum 1. Take $w^{\pm} = (t, t) \pm \delta e \in \Delta$, with e a direction of Δ and $\delta > 0$; they lie on the line $\{\langle \cdot, u^* \rangle = 1\}$, which avoids 0, so they are linearly independent. For $u \in P$, $\langle w^{\pm}, u \rangle \geq \mu(u) \geq 1 = \langle w^{\pm}, u^* \rangle$, so $\kappa_{\pm} = \langle w^{\pm}, u - u^* \rangle \geq 0$ and $|u|_1 - \frac{1}{t} = \frac{1}{2t}\langle w^+ + w^-, u - u^* \rangle = \frac{1}{2t}(\kappa_+ + \kappa_-)$. Expanding $u - u^*$ in the basis dual to (w^+, w^-) gives $|u_1 - u_1^*| \leq C_\Delta(\kappa_+ + \kappa_-) = 2tC_\Delta(|u|_1 - \frac{1}{t})$, so $\psi(h) \leq 4tC_\Delta(h - \frac{1}{t})$.

Step 5: conclusion. Integrating (6.4), $\int_{1/t}^{\infty} e^{-Lh}(\psi_0 + h - \frac{1}{t})\,\mathrm{d}h = e^{-L/t}(\psi_0 L^{-1} + L^{-2})$, so for every $L > 0$

$$e^{-L/t}(\psi_0 L + 1) \leq A_M(e^{-L}) \leq B\, e^{-L/t}(\psi_0 L + 1). \tag{6.5}$$

By (6.3), $A_K(e^{-L})$ lies between the values of (6.5) at $L + \log(|\mathcal{E}|c_+)$ and at $L + \log c_-$; a bounded shift of L changes $e^{-L/t}$ and, for large L, $\psi_0 L + 1$ by bounded factors. So $A_K(\varepsilon)$ lies between constant multiples of $\varepsilon^{1/t}(\psi_0 L + 1)$, which is of the order of $\varepsilon^{1/t} L^{m-1}$ by Step 3, and Step 1 proves (i). ∎

For the product model, where $M = K = a^2b^2$ and $\psi(h) = h$, the lower bound in (6.5) is the exact volume $\varepsilon^{1/2}(1 + \frac{1}{2}\log\frac{1}{\varepsilon})$. The multiplicity is one plus the dimension of the face of P on which the linear program attains its optimum.

6.3.3 The zeta function on the normal fan

On the cone C_k the largest monomial is $a^{V_1^k} b^{V_2^k}$, and a linear change of variables in logarithmic coordinates that maps the quadrant onto C_k turns it into a monomial in new variables, and every other monomial into that monomial times powers of the new variables.

Proposition 6.5 (The zeta function on the normal fan). *Let K be a positive even sum, with the notation of Lemma 6.2, and let the prior be uniform on $[-1,1]^2$. Let $D_k = |\det(\eta_{k-1},\eta_k)|$, $A_k(z) = \mu_{k-1}z + |\eta_{k-1}|_1$, $B_k(z) = \mu_k z + |\eta_k|_1$, and*

$$Q_k(\xi) = \sum_{v\in\mathcal{E}} c_v\, \xi_1^{\langle v,\eta_{k-1}\rangle-\mu_{k-1}}\xi_2^{\langle v,\eta_k\rangle-\mu_k},$$

a polynomial with nonnegative integer exponents and constant term c_{V^k}. Then for $\Re z > 0$

$$\zeta(z) = \sum_{k=1}^{r} D_k \int_0^1\!\!\int_0^1 \xi_1^{A_k(z)-1}\xi_2^{B_k(z)-1} Q_k(\xi)^z \,\mathrm{d}\xi_1\,\mathrm{d}\xi_2, \tag{6.6}$$

and ζ extends meromorphically to the half-plane $\Re z > -\min_j(|\eta_j|_1+1)/\mu_j$ (minimum over j with $\mu_j>0$), which contains $\{\Re z \ge -\frac{1}{t}\}$. There its poles lie among the numbers $-|\eta_j|_1/\mu_j$, and the largest is $-\frac{1}{t}$, of order m. If $(t,t) = V^k$, then $(z+\frac{1}{t})^2\zeta(z) \to D_k c_{V^k}^{-1/t}/(\mu_{k-1}\mu_k) = \frac{\psi_0}{t}c_{V^k}^{-1/t}$ as $z\to -\frac{1}{t}$; otherwise the pole is simple with a positive residue.

Proof. The plan is to change variables on each cone, to continue each integral by splitting off the values of the integrand on two edges of the unit square, and to locate the largest pole with Lemma 6.2. By evenness, $\zeta(z) = \int_{(0,1]^2} K^z \,\mathrm{d}a\,\mathrm{d}b$. The map $\Psi_k(\xi) = (\xi_1^{\eta_{k-1,1}}\xi_2^{\eta_{k,1}}, \xi_1^{\eta_{k-1,2}}\xi_2^{\eta_{k,2}})$, the composite of $\xi\mapsto -\log\xi$, the linear map $s\mapsto s_1\eta_{k-1}+s_2\eta_k$ and $u\mapsto e^{-u}$, maps $(0,1)^2$ diffeomorphically onto the points of $(0,1)^2$ whose logarithmic coordinates lie in the interior of C_k, with Jacobian $D_k\xi_1^{|\eta_{k-1}|_1-1}\xi_2^{|\eta_k|_1-1}$; by Lemma 6.2(i) these images cover $(0,1)^2$ up to a null set. Since $a^{v_1}b^{v_2}\circ\Psi_k = \xi_1^{\langle v,\eta_{k-1}\rangle}\xi_2^{\langle v,\eta_k\rangle}$, $K\circ\Psi_k = \xi_1^{\mu_{k-1}}\xi_2^{\mu_k}Q_k$; the exponents of Q_k are nonnegative, and both vanish only for V^k, the one exponent minimizing both $\langle\cdot,\eta_{k-1}\rangle$ and $\langle\cdot,\eta_k\rangle$. This proves (6.6), and $c_{V^k}\le Q_k\le\sum_v c_v$ on $[0,1]^2$.

Fix k, write $A = A_k$, $B = B_k$, and let $\chi = Q_k^z = \exp(z\log Q_k)$, infinitely differentiable on $[0,1]^2$ and entire in z. Split χ into four parts, the last of which collects what remains:

$$\chi(\xi) = \chi(0,0) + \big[\chi(\xi_1,0)-\chi(0,0)\big] + \big[\chi(0,\xi_2)-\chi(0,0)\big] + \chi_{12}(\xi).$$

By the mean value theorem the brackets are at most $C\xi_1$ and $C\xi_2$, and $\chi_{12} = \int_0^{\xi_1}\int_0^{\xi_2}\partial_1\partial_2\chi$ is at most $C\xi_1\xi_2$, uniformly for z in compact sets. Integrating term by term for $\Re A, \Re B > 0$,

$$\int_{[0,1]^2}\xi_1^{A-1}\xi_2^{B-1}\chi = \frac{\chi(0,0)}{AB} + \frac{I_1}{B} + \frac{I_2}{A} + I_{12},$$

with $I_1 = \int_0^1 \xi_1^{A-1}[\chi(\xi_1,0)-\chi(0,0)]\,\mathrm{d}\xi_1$, I_2 the analogous integral in ξ_2, and $I_{12} = \int \xi_1^{A-1}\xi_2^{B-1}\chi_{12}$, all three holomorphic as long as $\Re A > -1$ and $\Re B > -1$. This continues each term to that region, with poles only where A or B vanishes, that is, at numbers $-|\eta_j|_1/\mu_j \le -\frac{1}{t}$, those equal to $-\frac{1}{t}$ coming

from rays in $\mathcal{N}$ (Lemma 6.2(iv)); and the region contains $\{\Re z \geq -\frac{1}{t}\}$ because $(|\eta_j|_1 + 1)/\mu_j > \frac{1}{t}$.

If $(t,t) = V^k$, both rays of C_k lie in $\mathcal{N}$, so that $A_k(z) = \mu_{k-1}(z + \frac{1}{t})$ and $B_k(z) = \mu_k(z + \frac{1}{t})$. Then the term $D_k\chi(0,0)/(A_kB_k)$ has a double pole, with the coefficient $D_k c_{V^k}^{-1/t}/(\mu_{k-1}\mu_k) > 0$, and every other term has at most a simple pole. Since $\mu_j = t|\eta_j|_1$ for these two rays, the length ψ_0 of the projection of the segment F^* is $|\eta_{k-1,1}/\mu_{k-1} - \eta_{k,1}/\mu_k| = tD_k/(\mu_{k-1}\mu_k)$. Otherwise one ray η_k lies in $\mathcal{N}$, shared by at most two cones. In a cone whose first ray is η_k, $B(-\frac{1}{t}) > 0$ and the coefficient of $1/A$ is $\chi(0,0)/B + I_2 = \int_0^1 \xi_2^{B-1}\chi(0,\xi_2)\,\mathrm{d}\xi_2$, so the residue there is $\frac{D}{\mu(\eta_k)}\int_0^1 \xi_2^{B(-1/t)-1}Q(0,\xi_2)^{-1/t}\,\mathrm{d}\xi_2 > 0$; a cone whose second ray is η_k is treated symmetrically. The residues add, and the pole is simple. ∎

For the mixed cup the rays are $(1,0)$, $(3,2)$ and $(0,1)$, and (6.6) reads $\zeta(z) = 2^{-z}\left[2\int_0^1(1+\xi^4)^z\,\mathrm{d}\xi + 3\int_0^1(1+\xi^6)^z\,\mathrm{d}\xi\right]/(12z+5)$: one simple pole, at $-\frac{5}{12}$, with residue 0.5276, which is $\frac{5}{12}$ times the constant 1.2662 of Computation 6.1, as the correspondence of Theorem 4.6 requires. For $a^6 + a^2b^2 + b^6$ the double pole has the coefficient $\frac{1}{12}$, which corresponds to $V(\varepsilon) \sim \frac{1}{6}\varepsilon^{1/2}\log\frac{1}{\varepsilon}$, the constant approached in Computation 6.2. The computation works for every prior that is the restriction to W of a smooth function with $\varphi(0) > 0$, with χ multiplied by the sum of the prior over the four quadrants; iterating the integration by parts as after Definition A.5 continues ζ to all of $\mathbb{C}$.

The maps Ψ_k are the charts of a *toric modification*, one monomial map for each cone of the normal fan.[4] When every cone is *smooth*, $D_k = 1$, the charts glue to a manifold on which K is a monomial times a positive function: a resolution of singularities in the sense of Theorem A.8, found without a blow-up chosen by hand and without the existence theorem of Hironaka (1964).[5] For (6.6) smoothness is not needed, because the change of variables happens inside the open quadrant.

[4] Toric varieties are named after the torus $(\mathbb{C}\setminus\{0\})^d$, which they contain as a dense open set; here the real torus $(\mathbb{R}\setminus\{0\})^d$, where no coordinate vanishes, plays its part.

[5] For $a^2 + b^4$ the fan has the rays $(1,0)$, $(2,1)$ and $(0,1)$; the cone spanned by $(2,1)$ and $(0,1)$ has $D = 2$, and adding the ray $(1,1)$ makes every cone smooth. The three charts of the refined fan are those of the two point blow-ups of Chapter 5.

6.3.4 The half that survives cancellation

The lower bound in (6.5) used only the exponents, and it survives cancellations, because a power series is never much larger than the largest of its vertex monomials.

Corollary 6.6 (The Newton bound). *Let $K \geq 0$ be real analytic near $0 \in \mathbb{R}^2$, $K(0) = 0$, $K \not\equiv 0$; let t be the Newton distance of its Taylor series and m_Γ the multiplicity that (6.1) assigns to its polygon; and let the prior be bounded below by a positive constant near 0. Then $V(\varepsilon) \geq c\,\varepsilon^{1/t}(\log\frac{1}{\varepsilon})^{m_\Gamma - 1}$ for small ε. Hence $\lambda(0) \leq 1/t$, and $m(0) \geq m_\Gamma$ if $\lambda(0) = 1/t$.*

Proof. The plan is to bound K by a multiple of its largest vertex monomial and to apply the lower half of the proof of Theorem 6.4. Let $\mathcal{V}$ be the vertex set of Γ, $M(w) = \max_{v\in\mathcal{V}}|a|^{v_1}|b|^{v_2}$, $\bar{v} = \max_{v\in\mathcal{V}}|v|_1$, and choose $r \in (0, \frac{1}{2}]$ such that the Taylor series $\sum_\alpha c_\alpha w^\alpha$ converges absolutely on $[-2r, 2r]^2$ and the prior is at least $\phi_0 > 0$ on $[-r,r]^2$. Each exponent α lies in Γ, so

$\alpha \geq \sum_i \theta_i V^i$ coordinatewise for a probability vector θ on $\mathcal{V}$. For $|a|, |b| \leq 2r$ the entries of $w/(2r)$ are at most 1 in absolute value; dropping the factors of $\alpha - \sum_i \theta_i V^i$ and bounding a weighted geometric mean by its largest factor,

$$|w^\alpha| = (2r)^{|\alpha|_1} \left|\left(\frac{w}{2r}\right)^\alpha\right| \leq (2r)^{|\alpha|_1} \prod_i \left|\left(\frac{w}{2r}\right)^{V^i}\right|^{\theta_i} \leq (2r)^{|\alpha|_1 - \bar{v}} M(w).$$

Summing, $K \leq C_K M$ on $[-2r, 2r]^2$, with $C_K = (2r)^{-\bar{v}} \sum_\alpha |c_\alpha| (2r)^{|\alpha|_1}$. Substituting $w = rw'$ and using $M(rw') \leq M(w')$, $V(\varepsilon) \geq 4\phi_0 r^2 A_M(\varepsilon/C_K)$, and Steps 3–5 of the proof of Theorem 6.4, with only the lower bound in (6.4), give $A_M(e^{-L}) \geq e^{-L/t}(\psi_0 L + 1)$; they used neither signs nor parities. For analytic K the local volume satisfies $V(\varepsilon) \sim C' \varepsilon^{\lambda(0)} (\log \frac{1}{\varepsilon})^{m(0)-1}$ (Theorem 4.6; the meromorphy of ζ behind it goes back to Atiyah (1970) and Bernshtein and Gel'fand (1969)), and comparing rates gives the rest. ∎

6.4 The theorem: nondegenerate and degenerate functions

The proof of Theorem 6.4 used positivity once, in the lower bound $K \geq c_- M$. What it needs is that cancellations among the monomials of a face never make K much smaller than its largest monomial.

Definition 6.7 (Nondegenerate functions). Let $K \geq 0$ be real analytic near $0 \in \mathbb{R}^d$ with $K(0) = 0$, and let $\Gamma(K) = \operatorname{conv}(\mathcal{E}) + \mathbb{R}^d_{\geq 0}$ be the Newton polyhedron of its Taylor series. The function K is *nondegenerate* if for every compact face F of $\Gamma(K)$ the face polynomial $K_F = \sum_{v \in \mathcal{E} \cap F} c_v w^v$ has no zero in the torus $(\mathbb{R} \setminus \{0\})^d$.

Positive even sums are nondegenerate. Varchenko's definition asks instead that ∇K_F have no zero on the torus; for $K \geq 0$ the two agree. Indeed $K(\tau^{\eta_1} w_1, \ldots, \tau^{\eta_d} w_d) = \tau^{\mu(\eta)}(K_F(w) + o(1))$ as $\tau \downarrow 0$ for η normal to F alone, so $K_F \geq 0$ and its zeros are critical points; conversely such an η is positive, so $\mu(\eta) > 0$, and Euler's identity $\sum_j \eta_j w_j \partial_j K_F = \mu(\eta) K_F$ makes every critical point of K_F a zero. The Newton distance and the diagonal face are defined as in the plane.

Theorem 6.8 (Newton polyhedra in d variables). *Let $K \geq 0$ be real analytic near $0 \in \mathbb{R}^d$, $K(0) = 0$, $K \not\equiv 0$; let t be its Newton distance and k the dimension of its diagonal face; and let the prior be smooth with $\varphi(0) > 0$ and supported near 0, so that V is the local volume.*

(i) In every analytic coordinate system centered at 0, $V(\varepsilon) \geq c\, \varepsilon^{1/t} (\log \frac{1}{\varepsilon})^{d-k-1}$ for small ε; so $\lambda(0) \leq 1/t$, and $m(0) \geq d - k$ if $\lambda(0) = 1/t$ (Greenblatt 2010, Theorem 1.2(a)).

(ii) If K is nondegenerate, $V(\varepsilon)$ lies between constant multiples of $\varepsilon^{1/t} (\log \frac{1}{\varepsilon})^{d-k-1}$, and $\lambda(0) = 1/t$, $m(0) = d - k$ (Greenblatt 2010, Theorem 1.2(a, b)). For ideals: if $K \asymp \sum_i f_i^2$ and $\langle f_1, \ldots, f_s \rangle$ is monomial or, more generally, sos-nondegenerate, then $(2\lambda(0), m(0))$ is given by the Newton polyhedron of the ideal (Lin 2017, Theorem 1.3).

In two variables $d-k$ is 2 for a vertex and 1 for an edge or a ray, which is (6.1), and (i) is Corollary 6.6. Greenblatt's estimates concern $\int_{\{|S|<\varepsilon\}}\phi$ with a smooth cutoff $\phi\geq 0$, $\phi(0)>0$, the local volume function of $S=K$. Varchenko's oscillatory-integral theorem also assumes $t>1$; for a function with a sign neither the sublevel estimates nor the oscillatory ones need it (Greenblatt 2010, Theorem 1.6(a)), so for losses it plays no role.[6]

[6] For a function that changes sign it matters: $a^2+b^2-c^2$ satisfies Varchenko's condition and has $t=\frac{2}{3}$, but the volume of $\{|a^2+b^2-c^2|<\varepsilon\}$ near 0 shrinks like ε, not like $\varepsilon^{3/2}$ (Greenblatt 2010, Section 2).

Proof sketch. The plan is Proposition 6.5 in every dimension, with nondegeneracy in place of positivity. (1) Refine the normal fan of $\Gamma(K)$ to a fan of simplicial cones whose generators are bases of $\mathbb{Z}^d$. (2) A d-dimensional cone with generators $\eta^1,\dots,\eta^d$ gives the chart $w_j=\pm\prod_i \xi_i^{\eta^i_j}$; the charts glue to an analytic manifold with a proper map onto a neighborhood of 0 that is an isomorphism off the coordinate hyperplanes. (3) In each chart $K=\xi_1^{\mu(\eta^1)}\cdots\xi_d^{\mu(\eta^d)}Q(\xi)$, and the Jacobian is $\prod_i|\xi_i|^{|\eta^i|_1-1}$. On $\{\xi_i=0:i\in I\}$, Q is, up to a monomial, the face polynomial of the face dual to the cone spanned by $\{\eta^i:i\in I\}$, and the points over the origin are those for which that face is compact; nondegeneracy keeps Q away from 0 there (Lin 2017, Theorem 4.8), so every chart is in normal-crossing form. (4) Theorem 5.3 gives λ as the minimum of $|\eta|_1/\mu(\eta)$ over generators and m as the largest number of minimizing generators in one cone. (5) The linear program of Lemma 6.2 in d variables (Exercise 6.4) shows that the minimum is $1/t$, attained exactly in the normal cone of the diagonal face, a cone of dimension $d-k$; its refinement supplies $d-k$ minimizing generators in one cone, and no cone has more, because they are linearly independent. Omitted: smooth refinements, properness, and Q over noncompact faces; see Lin (2017, Section 4), who credits the desingularization to Varchenko (1976) and Arnold et al. (1988, Section 8.3), and Watanabe (2024, Section 3) for the statistical setting. ∎

Degenerate functions are where the coordinates matter. In two variables Greenblatt characterized exactly when the Newton distance is still right; the order of a zero of a polynomial is the smallest order of a partial derivative that does not vanish there.

Theorem 6.9 (Degenerate functions of two variables; Greenblatt). *Let $K\geq 0$ be real analytic near $0\in\mathbb{R}^2$, $K(0)=0$, $K\not\equiv 0$, with Newton distance t and diagonal face Δ, and the prior as in Theorem 6.8.*

(i) $\lambda(0)=1/t$, unless Δ is a compact edge whose face polynomial K_Δ has a zero in $(\mathbb{R}\setminus\{0\})^2$ of order larger than t; then $\lambda(0)<1/t$.

(ii) If $\lambda(0)=1/t$, then $m(0)=2-\dim\Delta$, unless Δ is a compact edge and K_Δ has a zero in $(\mathbb{R}\setminus\{0\})^2$ of order exactly t; then $m(0)=2$.

This is Theorem 1.5 of Greenblatt (2010), whose multiplicity is our $m-1$, and it settles the lower block of Table 6.1. The edge polynomials $(a-b)^2$, $(a-b^2)^2$, and $(a+b)^2$ of $(a+b)^2+b^4$ (whose λ is $\frac{3}{4}$, by the substitution $u=a+b$) vanish to order 2 on curves in the torus, and 2 exceeds $t=1$, $\frac{4}{3}$ and 1. The edge polynomial of $(a^2-b^2)^2$ vanishes to order $2=t$, so $\lambda=\frac{1}{2}$

and $m = 2$. That of the cusp curve vanishes to order $2 < t = \frac{12}{5}$, so the rule is right although K is degenerate. Alternatively, $f = a^2 - b^3$ has only zeros of order 1 on its edge, and 1 does not exceed its Newton distance $\frac{6}{5}$; by Greenblatt's Theorem 1.2, $\operatorname{area}\{|f| < \delta\} \asymp \delta^{5/6}$, and $\{K < \varepsilon\} = \{|f| < \varepsilon^{1/2}\}$ gives $\lambda = \frac{5}{12}$.

A degenerate polygon can often be repaired by a change of coordinates, as in Computation 6.3. In two variables it always can: every real analytic function has *adapted coordinates*, in which the reciprocal of the Newton distance gives the exact decay rate of the associated oscillatory integral.[7] The other repair keeps the coordinates and changes the generators. Lin (2017, Proposition 3.6) proves that *if two sets* $\{f_1, \dots, f_r\}$ *and* $\{g_1, \dots, g_s\}$ *of functions generate the same ideal of the ring of functions real analytic on a compact semianalytic set* Ω*, then* $\sum_i f_i^2$ *and* $\sum_j g_j^2$ *have the same learning coefficient and multiplicity on* Ω (his real log canonical threshold of an ideal is 2λ in our units). The proof is one line: each $g_j = \sum_i h_{ji} f_i$ with h_{ji} bounded on Ω, so $\sum_j g_j^2 \le C \sum_i f_i^2$ by the Cauchy–Schwarz inequality, symmetrically, and comparable functions have the same λ and m (Proposition 5.4). It makes λ a property of the ideal, not of the generators one writes down, and Lin builds Newton polyhedra of ideals on it. Theorem 6.4 is the two-variable case of his formula for monomial ideals, since a positive even sum is comparable with the sum of the squares of the monomials $a^{v_1/2} b^{v_2/2}$, $v \in \mathcal{E}$.

[7] The notion is Arnold's. Varchenko gave sufficient conditions for adaptedness and constructed adapted coordinates; Greenblatt (2010, Section 1) credits him with their existence for every real analytic phase, while Ikromov and Müller (2011) describe his result as covering functions without multiple components and prove existence for all real analytic functions and all smooth functions of finite type.

6.5 Measuring it

Two scripts test the rule on functions whose volume can be computed independently, with the uniform prior on $[-1, 1]^2$, so that $V(\varepsilon)$ is the area of $\{K < \varepsilon\}$ in $(0, 1]^2$.

`06-newton-random.py` draws thirty random positive even sums, with two to five distinct exponents from $\{0, 2, 4, 6\}^2 \setminus \{(0, 0)\}$ and coefficients log-uniform on $[0.2, 5]$ (generator seed 6); eight of them have a vertex on the diagonal. The script computes t and m in rational arithmetic and measures $V(\varepsilon)$ in two ways. The first is quadrature: since K increases in b, $V(\varepsilon) = \int_0^1 \beta_\varepsilon(a)\,\mathrm{d}a$ with $\beta_\varepsilon(a) = \sup\{b \le 1 : K(a, b) < \varepsilon\}$; in the coordinate $x = -\log a$ the root is found by Newton's method and the integral by adaptive Gauss–Kronrod quadrature with relative tolerance 10^{-12}. Logarithmic coordinates reach $\varepsilon = e^{-4000}$ without underflow,[8] and the routine reproduces the exact volumes of a^2b^2, $a^4 + b^6$ and $a^2 + b^4$ to 10^{-14}.[9] The exponent and the multiplicity are fitted by least squares of $\log V$ on $-L$, $\log L$, 1 and $1/L$ at $L = \log \frac{1}{\varepsilon} = 250, 500, \dots, 4000$. The second is what a practitioner would do: 2×10^6 Monte Carlo points (seed $100 + k$ for the k-th function) and the slope of $\log V$ between $\varepsilon = 10^{-2}$ and 10^{-4}, with an exact 95% interval from the Clopper–Pearson interval for the fraction of points below 10^{-2} that are also below 10^{-4}. The functions with cancellations have exact volume formulas, evaluated with 40-digit arithmetic and fitted in the same way at $L = 200$ to 1000.

The quadrature fits (Figure 6.3) agree with the polygon's prediction to within 2.5×10^{-7} in λ, and the fitted multiplicities are within 0.001 of the

[8] That is $\varepsilon \approx 10^{-1737}$. No experiment sees such an ε, but the theorem is a statement about $\varepsilon \to 0$, and the quadrature can follow it there.

[9] The routine uses the polygon only to rescale the integrand and to place panel boundaries; a blind variant that uses neither changes no $\log V$ by more than 5×10^{-13}.

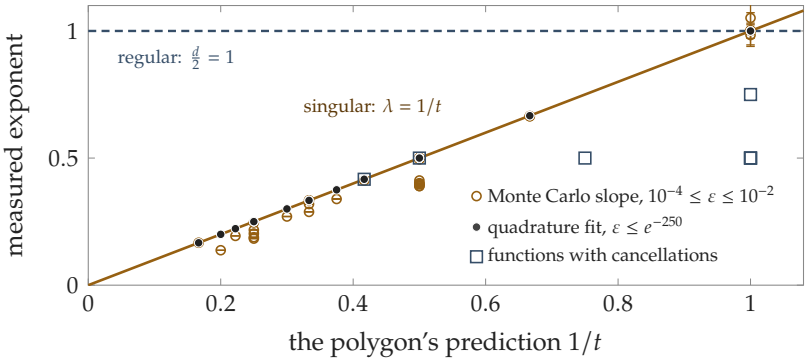


Figure 6.3. The polygon's prediction against measured volume exponents. Dots: thirty random positive even sums, exponent fitted to exact quadrature of $V(\varepsilon)$ at $\log \frac{1}{\varepsilon} = 250$ to 4000; they lie on the bronze line to within 3×10^{-7}. Circles: the same functions, slope of $\log V$ between $\varepsilon = 10^{-2}$ and 10^{-4} from 2×10^6 Monte Carlo points each (seeds 100 to 129), with exact 95% intervals, mostly smaller than the marks; the eight with $m = 2$ lie about 0.09 below the line. Squares: functions with cancellations, exponents fitted to exact volumes: from the left the cusp curve and $(a^2 - b^2)^2$ on the line, $(a - b^2)^2$ at $(\frac{3}{4}, \frac{1}{2})$, $(a + b)^2 + b^4$ at $(1, \frac{3}{4})$, and at $(1, \frac{1}{2})$ the functions $(a - b)^2$, $\frac{1}{2}(a_1 b_1 + a_2 b_2)^2$ and the product model at the truth $ab = \frac{1}{4}$ in local coordinates. Script: `figures/scripts/06-newton-random.py`.

polygon's, eight of them 2; a variant of the quadrature that does not use the polygon gives the same values, so this agreement was not arranged. The Monte Carlo slopes deviate from $1/t$ by -0.013 on average for the twenty-two functions with $m = 1$ (range -0.063 to $+0.051$) and by -0.092 for the eight with $m = 2$ (range -0.110 to -0.061); only eleven of the thirty 95% intervals contain $1/t$. The intervals are correct about the Monte Carlo error and silent about the bias of a slope taken before the asymptotic regime begins.

`06-newton-slopes.py` shows where the bias comes from (Figure 6.4). For the mixed cup the volume is an exact power; the Monte Carlo slopes over five decades are 0.417, 0.418, 0.416, 0.413 and 0.417, and four of the five intervals contain $\frac{5}{12}$. For $a^6 + a^2 b^2 + b^6$ the factor $\log \frac{1}{\varepsilon}$ lowers every slope, asymptotically by $1/\log \frac{1}{\varepsilon}$ and at practical ε by less (Exercise 6.12): the exact local slope is 0.386 at $\varepsilon = 10^{-1.5}$, 0.443 at $10^{-5.5}$ and 0.490 at 10^{-40}. For $a^4 b^6 + a^6 b^4$, with $\lambda = \frac{1}{5}$ and $m = 1$, the second pole of ζ, at $-\frac{1}{4}$, is close, the correction to V is a factor $1 - O(\varepsilon^{1/20})$, and the exact slope is 0.141 at $10^{-3.5}$ and 0.196 only at 10^{-20}. Both biases are predicted by the theory, by the multiplicity and by the cones of Proposition 6.5, and both are invisible to a confidence interval; they also lower the Monte Carlo slopes of $(a^2 - b^2)^2$, to 0.414, and of the cusp curve, to 0.382.

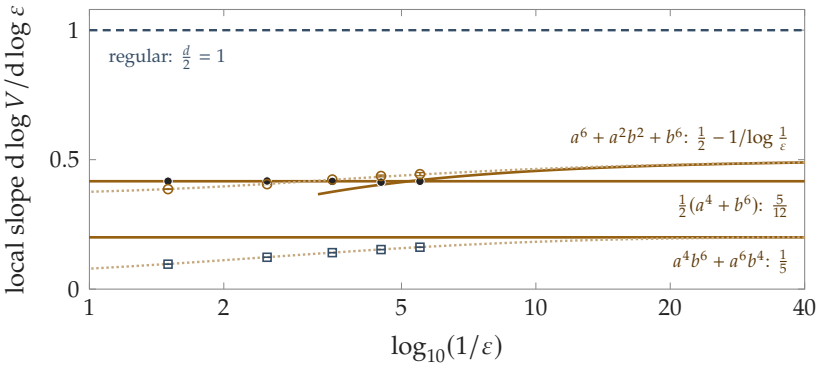


Figure 6.4. What a slope measures: local slopes of $\log V$ against $\log \varepsilon$ for three positive even sums. Solid bronze: the slopes of the leading term $C\varepsilon^{\lambda}(\log \frac{1}{\varepsilon})^{m-1}$ of V (Proposition 6.5), λ for $m = 1$ and $\lambda - 1/\log \frac{1}{\varepsilon}$ for $m = 2$ (drawn from $\varepsilon = 10^{-3}$). Dotted bronze: exact local slopes by quadrature (central differences in $\log \varepsilon$); for the mixed cup they coincide with $\frac{5}{12}$. Marks (dots for $\frac{1}{2}(a^4 + b^6)$, circles for $a^6 + a^2b^2 + b^6$, squares for $a^4b^6 + a^6b^4$): Monte Carlo slopes over the decades from $[10^{-2}, 10^{-1}]$ to $[10^{-6}, 10^{-5}]$, 4×10^6 uniform points per function (seeds 61 to 63), with exact 95% intervals. Script: `figures/scripts/06-newton-slopes.py`.

6.6 The ledger

For sums of even monomials with positive coefficients in two variables, the learning coefficient is read off the exponents, with a complete proof: $\lambda = 1/t$, and $m = 2$ exactly when the diagonal enters the Newton polygon at a vertex. In every dimension $1/t$ bounds λ from above, with equality for nondegenerate functions (stated and sketched).

LEDGER · $y = a^2x_1 + b^3x_2$	
QUANTITY	The learning coefficient read off the exponents of K.
REGULAR THEORY	Every exponent that matters is 2: $\lambda = \frac{d}{2}$, $m = 1$.
SINGULAR THEORY	$\lambda = 1/t$, $m = 2$ if the diagonal meets a vertex, else 1 (Theorem 6.4); $\lambda \le 1/t$ in every coordinate system (Corollary 6.6 and Theorem 6.8).
RUNNING EXAMPLE	$y = a^2x_1 + b^3x_2$, $K = \frac{1}{2}(a^4 + b^6)$: $t = \frac{12}{5}$, $\lambda = \frac{5}{12}$, $m = 1$, against $\frac{d}{2} = 1$.
MEASURED	Thirty random sums: quadrature exponents within 3×10^{-7} of $1/t$, multiplicities within 10^{-3}; Monte Carlo slopes at $\varepsilon \ge 10^{-4}$ low by 0.09 when $m = 2$.
RESTS ON	Nondegeneracy in the chosen coordinates (positive coefficients in the proof); analyticity at the point; a prior positive there.

6.7 Where the theory stops

WHAT WAS ASSUMED. The complete proof assumes positive coefficients and even exponents, a form of nondegeneracy that can be checked by looking. The general theorem assumes nondegeneracy, a property of K in a coordinate system, not of the model: $(a-b)^2$ is degenerate in the coordinates (a, b) and nondegenerate in $(a-b, b)$. It assumes analyticity at the point, since the polygon is drawn from a Taylor series; the loss of a ReLU network is only piecewise analytic, and on a kink there is no Taylor series. It is local: the learning coefficient of a model is the minimum of the local values over its true set (Proposition 5.4), so in general a polygon is needed at every true point. And the prior must be positive at the point: a prior vanishing like $|a|^{h_1}|b|^{h_2}$ replaces the diagonal by the ray through (h_1+1, h_2+1) (Exercise 6.9).

WHERE IT BREAKS. The losses of neural networks are usually degenerate in their natural coordinates, for a structural reason: a nondegenerate $K \geq 0$ vanishes near 0 only on coordinate hyperplanes (Exercise 6.10 for two variables), and the true sets of networks are in general not of that form. At a true point (a_0, b_0) of the product model with generic truth $ab = c_0 \neq 0$, in the coordinates $\alpha = a - a_0$, $\beta = b - b_0$, the loss is $\frac{1}{2}(b_0\alpha + a_0\beta + \alpha\beta)^2$, whose polygon has the vertices $(2, 0)$ and $(0, 2)$: the rule predicts the regular value 1, while $\lambda = \frac{1}{2}$; the edge polynomial vanishes to order $2 > t = 1$, as Theorem 6.9 requires. The linear network $y = (a_1b_1 + a_2b_2)x$, with two hidden units, has $K = \frac{1}{2}(a_1b_1 + a_2b_2)^2$; in four variables the diagonal meets the Newton polyhedron at the exponent $(1, 1, 1, 1)$ of the cross term, inside the edge from $(2, 2, 0, 0)$ to $(0, 0, 2, 2)$, and the rule predicts $\lambda = 1$ with $m = 3$. The truth is $\lambda = \frac{1}{2}$, $m = 1$: K depends on $c = a_1b_1 + a_2b_2$, whose density under the uniform prior is continuous at 0 with the value $\frac{1}{4}\int_{-1}^{1} \log^2|x|\,\mathrm{d}x = 1$, so Theorem 3.4 applies. With one hidden unit the network is the product model and the rule is exact; the second unit makes the true set a quadric cone.[10] Reduced-rank regression behaves the same way: at the zero truth with $M = N = H = 2$ the polyhedron of $\frac{1}{2}\|BA\|_F^2$ predicts $\lambda = 2$, while the formula of Aoyagi and Watanabe (2005) gives $\frac{3}{2}$ (Exercise 6.13). Even where the rule is exact, slopes at practical ε are biased (Figure 6.4), and the same correction, $-(m-1)\log\log n$, affects the estimators of Chapters 11 and 13.

[10] The Aoyagi–Watanabe formula of Chapter 7 with one input, one output, two hidden units and a truth of rank 0 gives the same $\lambda = \frac{1}{2}$ and $m = 1$.

OPEN PROBLEMS. In two variables adapted coordinates always exist (Ikromov and Müller 2011). In three or more variables, Greenblatt asks in which situations a coordinate change exists after which the Newton polyhedron gives sharp estimates, and expects the answer to need higher-dimensional resolution of singularities (Greenblatt 2010, Section 1). For singular learning theory the question is concrete: for which architectures and truths is there a short, explicit sequence of blow-ups after which the loss is nondegenerate in every chart, so that the rest is a linear program? We know of no general answer, even for two-layer networks with analytic activations. A second question is computational: deciding nondegeneracy means deciding, face by face, whether a polynomial system has a real solution in the torus.

Exercises

6.1 CONSOLIDATION. Show that adding to a positive even sum a positive even monomial with exponent in its Newton polygon changes neither λ nor m, and that $(a+b)^2$ and $(a^2-b^2)^2$ violate the rule. For a^6+b^6 and $a^6+a^2b^4+b^6$ check with Proposition 6.5 that $V(\varepsilon)\sim C\varepsilon^{1/3}$ with $C=0.9638$ and 0.9133.

6.2 CONSOLIDATION. Let $K\geq 0$ be analytic near $0\in\mathbb{R}^d$ with $K(0)=0$ and $K\not\equiv 0$. Show that every exponent of its Taylor series has coordinate sum at least 2, so that $t\geq\frac{2}{d}$, and conclude from Theorem 6.8(i) that $\lambda(0)\leq\frac{d}{2}$. When does the rule (6.1) give $\frac{d}{2}$?

6.3 CONSOLIDATION. Derive the regular answer from Theorem 6.8: if $K=\sum_{i=1}^{d}w_i^2+O(|w|^3)$, its Newton polyhedron has the vertices $2e_i$, the Newton distance is $\frac{2}{d}$, the diagonal face is the simplex spanned by the vertices, and K is nondegenerate.

6.4 CONSOLIDATION. ⋆ Prove Lemma 6.2(ii)–(iii) and the bounds of Theorem 6.4(i) for positive even sums of d variables, with $m=d-k$ for a diagonal face of dimension k. *(Hint: the face of P where $|u|_1$ is minimal has dimension $d-k-1$; slice P by the hyperplanes $|u|_1=h$ and bound the slices' volumes near $h=\frac{1}{t}$ between constant multiples of $(h-\frac{1}{t})^k$, as in (6.4).)*

6.5 BY HAND. Apply the rule to the cusp models a^2b^4 and a^2b^6 and, with $m=d-k$, to the chain $a^2b^2c^2$, say which face the diagonal meets, and recover the values $(\frac{1}{4},1)$, $(\frac{1}{6},1)$ and $(\frac{1}{2},3)$ of Chapter 3.

6.6 BY HAND. In three variables find $1/t$ and the dimension k of the diagonal face for $a^2b^2+b^2c^2+c^2a^2$, $a^2b^2+c^2$ and $a^2+b^2+c^2$. *(Hint: $(1/t,3-k)$ is $(\frac{3}{4},1)$, $(1,2)$ and $(\frac{3}{2},1)$; check the second with the sum rule of Proposition 5.4.)*

6.7 BY HAND. Show from (6.6) that under the uniform prior the zeta function of a^2+b^4 is $\big[\int_0^1(1+\xi^2)^z\,d\xi+2\int_0^1(1+\xi^4)^z\,d\xi\big]/(4z+3)$, and check numerically that $\frac{1}{3}$ of the bracket at $z=-\frac{3}{4}$ is the constant 0.8740 of the exact volume $V(\varepsilon)=0.8740\,\varepsilon^{3/4}$.

6.8 BY HAND. For which real c is $K_c=a^4+ca^2b^2+b^4$ nonnegative, and for which nondegenerate? Show that $V(\varepsilon)=C_c\varepsilon^{1/2}$ for small ε, with $C_c=\frac{1}{8}\int_0^{2\pi}k_c(\theta)^{-1/2}\,d\theta$ and $k_c(\theta)=K_c(\cos\theta,\sin\theta)$, as long as $c>-2$, and explain why m jumps to 2 at $c=-2$. *(Hint: $C_1=0.8429$, $C_0=0.9270$, $C_{-1}=1.0783$.)*

6.9 VARIATION. Let the prior be proportional to $|a|^{h_1}|b|^{h_2}$ near 0. Redo the proof of Theorem 6.4 with the weight $e^{-\langle h+1,u\rangle}$, and show that $\lambda=1/t_h$, where $t_h(h_1+1,h_2+1)$ is the point at which the ray through (h_1+1,h_2+1) enters the Newton polygon. For the product model with $h=(1,0)$ show that $\lambda=\frac{1}{2}$ and $m=1$.

6.10 VARIATION. Show that a nondegenerate $K \geq 0$ of two variables vanishes near 0 only on the coordinate axes. *(Hint: in a chart, the face polynomials are the restrictions of Q to the edges $\xi_1 = 0$ and $\xi_2 = 0$.)* Deduce that the product model at a generic truth is degenerate in the coordinates $(a - a_0, b - b_0)$, and find coordinates in which it is not.

6.11 MEASUREMENT. Sample $u \in \mathbb{R}^2_{\geq 0}$ with density proportional to $e^{-|u|_1}$ on $\{|u|_1 \geq h_0\}$, $h_0 = (L + \log c_-)/t$, and show with Lemma 6.2(ii) that $(h_0 + 1)e^{-h_0}$ times the fraction of samples with $K < \varepsilon$ is unbiased for $V(\varepsilon)$. Use it to estimate the exponents of the thirty functions of `06-newton-random.py` at $\varepsilon = e^{-100}$ and e^{-200}. Which half of Theorem 6.4 does the sampler take for granted?

6.12 MEASUREMENT. Modify `06-newton-slopes.py` to correct the Monte Carlo slopes of $a^6 + a^2b^2 + b^6$ for the factor $\log \frac{1}{\varepsilon}$, using $m = 2$. How much bias remains at $\varepsilon = 10^{-3}$, and where does it come from? *(Hint: $V(\varepsilon) = \varepsilon^{1/2}(c_1 \log \frac{1}{\varepsilon} + c_0)(1 + o(1))$; estimate c_0/c_1 from the quadrature.)*

6.13 RESEARCH. Show that the Newton distance of the reduced-rank regression loss $\frac{1}{2}\|BA\|_F^2$ at the zero truth, $B \in \mathbb{R}^{M\times H}$, $A \in \mathbb{R}^{H\times N}$, is $t = 2/(H \min(M, N))$. For $M = N = H = 2$ compare $1/t = 2$ with the value $\frac{3}{2}$ of Theorem 7.5; for $H = 1$ explain why the rule is exact (Theorem 7.2). Then look for a blow-up after which the loss for $M = N = H = 2$ is nondegenerate in every chart; no general procedure of this kind is known.

Notes and References

Newton polyhedra entered singularity theory in the 1970s: Kouchnirenko (1976) computed Milnor numbers from them, and Varchenko (1976) the decay of oscillatory integrals for phases nondegenerate with respect to their Newton polyhedra; the second volume of Arnold et al. (1988) contains Varchenko's theorems (Lin cites its Section 8.3). Greenblatt (2010) proved the sublevel-set estimates of Theorems 6.8 and 6.9, starting, as we did, from a decomposition of a neighborhood of the origin according to which monomials dominate.[11] In complex geometry the log canonical threshold of a monomial ideal is the reciprocal of the Newton distance (Howald 2001).

In singular learning theory, Newton diagrams were used by Yamazaki and Watanabe (2004) for mixtures of binomial distributions and by Yamazaki et al. (2010) for the Bayesian generalization error, and Watanabe's review notes that for nondegenerate Newton diagrams the resolution is found by a toric modification (Watanabe 2024, Section 3). Lin (2011, 2017) developed the theory for statistics: real log canonical thresholds of ideals, the τ-distance for monomial priors, sos-nondegeneracy, and toric resolution. Drton et al. (2009, Chapter 5) and Sullivant (2018, Chapter 17) treat Bayesian integrals of algebraic models in textbook form, and Watanabe (2009) develops resolution of singularities for learning theory at book length.

The proof of Theorem 6.4 by logarithmic coordinates and linear programming, Lemma 6.3, and the closed form (6.6) are the book's own presentation;

[11] Greenblatt cuts the neighborhood into slivers on which the vertex monomials of one face dominate all others (his Section 2); in logarithmic coordinates the slivers become neighborhoods of the cones of the normal fan.

the result is the two-variable case of Lin's theorem for monomial ideals (Lin 2017, Theorem 1.3 and Proposition 4.12), which he proves by toric resolution. The example $(a+b)^2+b^4$ is Lin's Remark 4.13. Theorem numbers of Greenblatt (2010) and Lin (2017) refer to their arXiv versions v3, which we read.

TRANSLATING THE LITERATURE. Greenblatt (2010) writes S for our K, n for the number of variables d, d for the Newton distance t, $N(S)$ for the Newton polyhedron, $C(S)$ for the diagonal face, k for its dimension, and $I_{|S|,\phi}(\varepsilon) = \int_{\{|S|<\varepsilon\}} \phi$ for the local volume function $V(\varepsilon)$; his growth index is our λ and his multiplicity our $m-1$. Lin (2017) writes ω for w, $\mathcal{P}(f)$ for the Newton polyhedron, l for t and θ for m; his $\mathrm{RLCT}_\Omega(f;\varphi)$ of a function $f \geq 0$ is our (λ, m), but for an ideal $\langle f_1, \dots, f_r\rangle$ it is $(2\lambda, m)$ for $K = \sum f_i^2$, because his zeta function is $\int (\sum f_i^2)^{-z/2}|\varphi|$; his amplitude ω^τ is our prior factor $|w|^\tau$, and his τ-distance our t_h with $h = \tau$ (Exercise 6.9). Watanabe's "Newton diagram" is our polygon, his "toric modification" our charts Ψ_k.

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7 A Catalogue

Learning coefficients of models in use

THE MODEL — $y = BAx$

IN THIS CHAPTER. A linear layer of width H from N inputs to M outputs, $y = BAx$, has $H(M+N)$ parameters, and the regular theory charges half a unit of $\log n$ for each. Its smallest case, rank-one regression $y = b\,a^{\top}x$, learns a zero truth with the learning coefficient $\min(M,N)/2$: we compute this by hand in polar coordinates and prove it for box priors, the zeta function being a product of a factor for $|a|$ and one for $|b|$, whose poles collide, giving multiplicity 2, exactly when $M = N$. A reduction proved here brings every truth of reduced-rank regression to a zero truth of a smaller problem; with it we check the Aoyagi–Watanabe formula, stated with its case distinctions, and collect what is known for deep linear networks, tanh networks, mixtures, Boltzmann machines and Bayesian networks, each with its status. Two-temperature estimates from Metropolis samples, validated where the tempered means are exactly known, give 1.40, 2.50, 2.50 and 3.44 at $n = 10^5$ for four reduced-rank cases with $\lambda = \frac{3}{2}, \frac{5}{2}, \frac{5}{2}, \frac{7}{2}$ and $\frac{d}{2} = 3, 3, 6, 6$, the two with $m = 2$ short by about the predicted logarithmic correction. Exact values are known for few architectures, and for deep nonlinear networks only bounds.

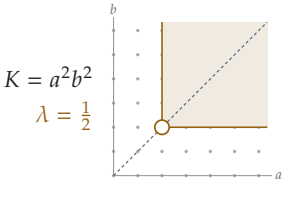
7.1 The question: what a layer costs

A low-rank adapter fine-tunes a weight matrix of a pretrained network by training only a correction BA, with $B \in \mathbb{R}^{M\times H}$, $A \in \mathbb{R}^{H\times N}$ and a small rank H (Hu et al. 2021, a preprint).[1] The same factorization is the bottleneck of a linear autoencoder and, in statistics since Anderson (1951), *reduced-rank regression*: multivariate regression with a coefficient matrix of rank at most H. In this book's notation the model is

$$y = BAx + \varepsilon, \qquad x \sim \mathcal{N}(0, I_N), \quad \varepsilon \sim \mathcal{N}(0, I_M), \tag{7.1}$$

with M outputs, N inputs and $d = H(M+N)$ parameters. The parametrization is far from one-to-one: (gA, Bg^{-1}) gives the same product as (A, B) for every invertible $H \times H$ matrix g, and the products fill the set of matrices of rank at most H, of dimension $H(M+N-H)$ when $H \le \min(M,N)$. Choosing H from data, or asking how much the data pay for a trained adapter, needs the learning coefficient of (7.1) at a truth C_0 of some rank $r \le H$.

[1] Hu et al. (2021) initialize A at random and B at zero, so that training starts from $BA = 0$, at a point of the branch $\{B = 0\}$ of the set of parameters whose product vanishes.



THE REGULAR ANSWER · half a unit of $\log n$ per parameter. If the truth is regular for a model with d parameters, then $\lambda = \frac{d}{2}$ and $m = 1$: $F_n - nS_n = \frac{d}{2}\log n + O_p(1)$ (Theorem 1.4), and the Bayes generalization error is $\frac{d}{2n} + o(\frac{1}{n})$ (Theorem 1.7). For a model that is not identifiable, common practice replaces d by the dimension of its set of distributions; for the rank of a reduced-rank regression that is $H(M+N-H)$ (Drton and Plummer 2017).

Neither count is right at the truths that matter. Take $H = 1$, *rank-one regression* $y = b\,a^\top x + \varepsilon$ with $a \in \mathbb{R}^N$ and $b \in \mathbb{R}^M$, so that $d = M + N$ and the rank-one matrices form a set of dimension $M + N - 1$. At the truth $C_0 = 0$, $K(a,b) = \frac{1}{2}|a|^2|b|^2$ (Computation 2.1); the true set $\{a = 0\} \cup \{b = 0\}$ is two linear spaces, of dimensions M and N, crossing at the origin, where the Hessian of K vanishes. We shall see that $\lambda = \min(M,N)/2$: for $M = N = 3$ it is $\frac{3}{2}$, against $\frac{d}{2} = 3$ and $\frac{1}{2}(M+N-1) = \frac{5}{2}$, and with $M = N$ the free energy gains a $-\log\log n$ besides (Theorem 8.6). At a truth of rank one the dimension count is right. The learning coefficient at a truth is the minimum of the local ones over the true set, attained at its most singular points (Proposition 5.4), and parameter counts do not see those points; this is why BIC with the dimension $H(M+N-H)$ tends to choose ranks that are too small.[2]

This chapter asks: *What are the learning coefficients and multiplicities of the models in use — reduced-rank regression, deep linear networks, networks with a hidden layer of tanh units, mixtures, Boltzmann machines, Bayesian networks — at the truths of interest? Which values are proved and which only bounded, and how close do estimates from posterior samples come to them?*

[2] Drton and Plummer (2017, Example 2.2) give this overpenalization of models that contain the truth without being minimal as the theoretical explanation of the tendency that simulation studies had found.

7.2 By hand: rank one, depth, and a tanh neuron

The loss of rank-one regression depends on (a,b) only through the lengths $|a|$ and $|b|$. A prior that respects this, the uniform distribution on the product $\mathbb{B}^N \times \mathbb{B}^M$ of the unit balls, makes every quantity exact; Section 7.3 replaces it by the book's box prior.

COMPUTATION 7.1 · RANK-ONE REGRESSION ON A PRODUCT OF BALLS

1 *Polar coordinates.* Under the uniform prior on $\mathbb{B}^N \times \mathbb{B}^M$ the vectors a and b are independent, and so are $r = |a|$ and $s = |b|$. A ball of radius $t \le 1$ has t^N times the volume of the unit ball, so $\mathbb{P}(r \le t) = t^N$ and r has the density Nt^{N-1} on $[0,1]$; likewise s has the density Mt^{M-1}. The loss is $K = \frac{1}{2}r^2s^2$; the directions a/r and b/s do not enter it.

2 *The zeta function.* For $\Re z > 0$, $\mathbb{E}r^{2z} = \int_0^1 t^{2z} N t^{N-1}\,\mathrm{d}t = \frac{N}{2z+N}$, likewise for s, and independence gives

$$\zeta(z) = \mathbb{E}\left[(\tfrac{1}{2}r^2s^2)^z\right] = 2^{-z}\,\mathbb{E}r^{2z}\,\mathbb{E}s^{2z} = \frac{2^{-z}NM}{(2z+N)(2z+M)},$$

a meromorphic function on $\mathbb{C}$ with poles at $-\frac{N}{2}$ and $-\frac{M}{2}$ and nowhere else.

3 *Read off.* If $N < M$, the largest pole is $-\frac{N}{2}$, and it is simple: $\lambda = \frac{N}{2}$, $m = 1$; symmetrically if $M < N$. If $M = N$, the two poles fall on each other, and $-\frac{N}{2}$ is a double pole: $\lambda = \frac{N}{2}$, $m = 2$. In every case $\lambda = \min(M, N)/2$.

4 *The volume.* Put $\delta = \sqrt{2\varepsilon} \le 1$, so that $K < \varepsilon$ exactly when $rs < \delta$. Given s, the probability that $r < \delta/s$ is 1 for $s \le \delta$ and $(\delta/s)^N$ for $s > \delta$, so

$$V(\varepsilon) = \delta^M + \int_\delta^1 M s^{M-1} \Big(\frac{\delta}{s}\Big)^N \, \mathrm{d}s = \begin{cases} \delta^N\big(1 + N \log \frac{1}{\delta}\big), & M = N, \\ \dfrac{M\delta^N - N\delta^M}{M - N}, & M \neq N. \end{cases}$$

For $N < M$, $V(\varepsilon) \sim \frac{M}{M-N}(2\varepsilon)^{N/2}$, a pure power; for $M = N$, $V(\varepsilon) \sim \frac{N}{2}(2\varepsilon)^{N/2} \log \frac{1}{\varepsilon}$, a power times one logarithm, the signature of $m = 2$ (Chapter 4).

5 *Numbers.* For $M = N = 3$ and $\varepsilon = 10^{-4}$, $\delta = 0.0141$ and $V(\varepsilon) = \delta^3(1 + 3\log\frac{1}{\delta}) = 3.90 \times 10^{-5}$. A regular model with six parameters keeps a prior fraction of order $\varepsilon^3 = 10^{-12}$ within ε of its truth, and one with five, the dimension of the rank-one matrices, of order 10^{-10}.

At a truth of rank one, $C_0 = b_0 a_0^\top \neq 0$, the model is as tame as a singular model can be. Its true set $\{(a_0/t, t b_0) : t \neq 0\}$ is a smooth curve, and at each of its points (a, b) the differential $(\alpha, \beta) \mapsto \beta a^\top + b\,\alpha^\top$ of $(a, b) \mapsto ba^\top$ has rank $M + N - 1$, vanishing exactly on the multiples of the tangent $(-a, b)$.[3] The Hessian of $K = \frac{1}{2}\|ba^\top - C_0\|_F^2$ there is the Gram matrix of this differential, with the tangent line as kernel, and the rule for minimally singular sets (Proposition 5.5) gives $\lambda = \frac{1}{2}(M + N - 1)$ and $m = 1$, which is $\frac{5}{2}$ for $M = N = 3$, provided the curve meets the interior of the prior's support.

[3] The tangent is the infinitesimal form of the rescaling $(a, b) \mapsto (a/t, tb)$, the case $H = 1$ of the symmetry $(A, B) \mapsto (gA, Bg^{-1})$; the symmetry is the only degeneracy at such a point.

COMPUTATION 7.2 · DEPTH, WIDTH, AND A NONLINEARITY

1 *The chain* $y = a_1 a_2 \cdots a_L x$ of Chapter 3, with the uniform prior on $[-1, 1]^L$ and the truth 0, has $K = \frac{1}{2}a_1^2 \cdots a_L^2$. The coordinates are independent and $\mathbb{E}|a_j|^{2z} = \int_0^1 t^{2z}\,\mathrm{d}t = \frac{1}{2z+1}$, so

$$\zeta(z) = 2^{-z} \prod_{j=1}^{L} \mathbb{E}|a_j|^{2z} = \frac{2^{-z}}{(2z+1)^L},$$

with one pole, of order L, at $-\frac{1}{2}$: $\lambda = \frac{1}{2}$ and $m = L$, against $\frac{d}{2} = \frac{L}{2}$. Each factor is a ball of dimension one, and the L poles coincide. In rank-one regression the two factors are balls of dimensions N and M: the smaller dimension sets λ, and the poles coincide only when $M = N$. Depth stacks equal poles; width moves them apart.

2 *The tanh neuron* $y = a\tanh(bx)$, with $x \sim \mathrm{Unif}[-1,1]$, the uniform prior on $[-2,2]^2$ and the truth 0, has $K = \frac{a^2}{2}(1 - \frac{\tanh b}{b})$, and Computation 2.1 showed that $0.388 \cdot \frac{1}{6}a^2b^2 \le K \le \frac{1}{6}a^2b^2$ on the box. Under this prior $\mathbb{E}|a|^{2z} = \frac{1}{2}\int_0^2 t^{2z}\,\mathrm{d}t = \frac{4^z}{2z+1}$, so $\frac{1}{6}a^2b^2$ has $\zeta(z) = 6^{-z}\left(\frac{4^z}{2z+1}\right)^2$, with a double pole at $-\frac{1}{2}$. By comparison (Proposition 5.4), $\lambda = \frac{1}{2}$ and $m = 2$: the nonlinearity changes the constants, not the exponents.

The neuron doubled is not the product model doubled. For two units, $f(x) = a_1\tanh(b_1x) + a_2\tanh(b_2x)$ with $d = 4$, Aoyagi's formula in Section 7.4 gives $\lambda = \frac{2}{3}$ and $m = 1$ at the truth 0, which Exercise 7.7 derives by hand: the second unit adds $\frac{1}{6}$, where the regular count adds 1.

What one had to notice. In all three computations the prior is a product and the loss is, up to constant factors, a product of functions of the factors, so the zeta function is the product of the factors' zeta functions.[4] A factor that depends on the length of a vector uniform on a ball in $\mathbb{R}^k$ contributes a simple pole at $-\frac{k}{2}$; the largest pole of the product sets λ, and poles that coincide add their orders. Products of losses take the minimum of the learning coefficients, sums of losses in separate variables add them (Proposition 5.4). Reduced-rank regression with $H \ge 2$ is neither, and Section 7.4 reduces it to them where it can.

[4] Chapter 3 met the same structure through the density of the product $c = a_1 \cdots a_L$: the zeta function is the Mellin transform of the law of K, and Mellin transforms turn products of independent variables into products of transforms.

7.3 The theorem: rank-one regression

For $k \ge 1$ let $\omega_k = \pi^{k/2}/\Gamma(\frac{k}{2}+1)$ be the volume of the unit ball of $\mathbb{R}^k$ ($\omega_1 = 2$, $\omega_2 = \pi$, $\omega_3 = \frac{4}{3}\pi$), so that its sphere has area $k\omega_k$. For a box $A \subset \mathbb{R}^k$, a product of compact intervals, with the origin in its interior, write $|A|$ for its volume and $\rho_A > 0$ for the largest radius of a ball centered at the origin and contained in A.

Lemma 7.1 (The length of a uniform point of a box). *Let $A \subset \mathbb{R}^k$ be a box with the origin in its interior and a uniform on A. Then $\mathbb{P}(|a| \le t) = \omega_k t^k/|A|$ for $0 \le t \le \rho_A$. The function $\zeta_A(z) = \mathbb{E}|a|^{2z}$, defined for $\Re z > -\frac{k}{2}$, extends to a meromorphic function on $\mathbb{C}$ whose only pole is a simple pole at $-\frac{k}{2}$ with residue $k\omega_k/(2|A|)$; for real $z > -\frac{k}{2}$, $\zeta_A(z)$ is the finite and positive expectation $\mathbb{E}|a|^{2z}$.*

Proof. The plan is to split A into a ball, where polar coordinates give the pole, and the rest, which gives an entire function. For $t \le \rho_A$ the ball of radius t lies in A, which gives the distribution function. Let D be the ball of radius ρ_A. In polar coordinates, for $\Re z > -\frac{k}{2}$,

$$\int_D |a|^{2z}\,\mathrm{d}a = k\omega_k \int_0^{\rho_A} t^{2z+k-1}\,\mathrm{d}t = \frac{k\omega_k\,\rho_A^{2z+k}}{2z+k}.$$

On $A \setminus D$ the length $|a|$ lies between ρ_A and the largest length R_A of a point of A, so for each such a the function $z \mapsto |a|^{2z} = e^{2z\log|a|}$ is entire and bounded

by $\max(\rho_A^{2\Re z}, R_A^{2\Re z})$, uniformly on compact sets of z; differentiation under the integral sign shows that $E_A(z) = |A|^{-1}\int_{A\setminus D}|a|^{2z}\,\mathrm{d}a$ is entire. Hence $\zeta_A(z) = k\omega_k\rho_A^{2z+k}/(|A|(2z+k)) + E_A(z)$ for $\Re z > -\frac{k}{2}$, and the right side is meromorphic on $\mathbb{C}$ with one pole, simple, at $-\frac{k}{2}$, of residue $k\omega_k\rho_A^0/(2|A|) = k\omega_k/(2|A|)$. For real $z > -\frac{k}{2}$ the defining integral converges and its integrand is positive. ∎

Theorem 7.2 (Rank-one regression at the zero truth). *Let $M, N \geq 1$ and $y = b\,a^\top x + \varepsilon$ with $a \in \mathbb{R}^N$, $b \in \mathbb{R}^M$, $x \sim \mathcal{N}(0, I_N)$ and $\varepsilon \sim \mathcal{N}(0, I_M)$ independent of x. Let the truth be $y = \varepsilon$, and the prior uniform on $W = A\times B$, where $A \subset \mathbb{R}^N$ and $B \subset \mathbb{R}^M$ are boxes with the origin in their interiors. Then $K(a,b) = \frac{1}{2}|a|^2|b|^2$, and:*

(i) The zeta function extends to a meromorphic function on $\mathbb{C}$ with poles in $\{-\frac{N}{2}, -\frac{M}{2}\}$. Its largest pole is $-\frac{1}{2}\min(M,N)$, of order 2 if $M = N$ and 1 if $M \neq N$. So $\lambda = \frac{1}{2}\min(M,N)$, with $m = 2$ if $M = N$ and $m = 1$ if $M \neq N$.

(ii) As $\varepsilon \to 0$, $V(\varepsilon) \sim C\varepsilon^\lambda(\log\frac{1}{\varepsilon})^{m-1}$, with $C = 2^{N/2}\frac{\omega_N}{|A|}\mathbb{E}|b|^{-N}$ if $N < M$, where b is uniform on B and the expectation is finite; with $C = 2^{M/2}\frac{\omega_M}{|B|}\mathbb{E}|a|^{-M}$ if $M < N$, where a is uniform on A; and with $C = 2^{N/2-1}N\omega_N^2/(|A||B|)$ if $M = N$.

Proof. The plan is the computation of Computation 7.1, with Lemma 7.1 in place of the exact densities on balls. Since ε is independent of x and has mean zero, $K(a,b) = \frac{1}{2}\mathbb{E}|b\,a^\top x|^2 = \frac{1}{2}|b|^2\mathbb{E}[(a^\top x)^2] = \frac{1}{2}|a|^2|b|^2$.

(i) Under the uniform prior on $A\times B$, a and b are independent and uniform on A and on B, so for $\Re z > 0$

$$\zeta(z) = \mathbb{E}\big[(\tfrac{1}{2}|a|^2|b|^2)^z\big] = 2^{-z}\,\zeta_A(z)\,\zeta_B(z),$$

and by Lemma 7.1 the right side is meromorphic on $\mathbb{C}$, with poles at $-\frac{N}{2}$ and $-\frac{M}{2}$ only. Suppose first that $N < M$. Near $-\frac{N}{2}$ the factor $2^{-z}\zeta_B(z)$ is holomorphic, because $-\frac{N}{2} > -\frac{M}{2}$, and its value there is $2^{N/2}\mathbb{E}|b|^{-N} > 0$ by the last claim of the lemma; the factor ζ_A has a simple pole there with residue $N\omega_N/(2|A|) \neq 0$. So $-\frac{N}{2}$ is a simple pole of ζ, and no pole lies to its right: $\lambda = \frac{N}{2}$ and $m = 1$. The case $M < N$ is the same with a and b exchanged. If $M = N$, both ζ_A and ζ_B have simple poles with nonzero residues at $-\frac{N}{2}$, and $2^{-z} \neq 0$, so ζ has a pole of order exactly 2 there, its only pole: $\lambda = \frac{N}{2}$ and $m = 2$.

(ii) Put $\delta = \sqrt{2\varepsilon}$, so that $K < \varepsilon$ exactly when $|a||b| < \delta$, and let $G(t) = \mathbb{P}(|a| < t)$, which is $\omega_N t^N/|A|$ for $t \leq \rho_A$. Conditioning on b and splitting according to whether $|b| \geq \delta/\rho_A$,

$$V(\varepsilon) = \mathbb{E}\big[G(\delta/|b|)\big] = \frac{\omega_N\delta^N}{|A|}\,\mathbb{E}\big[|b|^{-N};\ |b| \geq \delta/\rho_A\big] + \mathbb{E}\big[G(\delta/|b|);\ |b| < \delta/\rho_A\big].$$

Once $\delta \leq \rho_A\rho_B$, the second term is at most $\mathbb{P}(|b| < \delta/\rho_A) = \omega_M(\delta/\rho_A)^M/|B|$. If $N < M$, then $\mathbb{E}|b|^{-N}$ is finite by the lemma, the expectation in the first

term increases to it as $\delta \downarrow 0$ by monotone convergence, and the second term is $O(\delta^M) = o(\delta^N)$; so $V(\varepsilon) \sim \frac{\omega_N}{|A|}\mathbb{E}|b|^{-N}\,\delta^N$, which is the claim since $\delta^N = 2^{N/2}\varepsilon^{N/2}$. The case $M < N$ follows by exchanging a and b. If $M = N$, then $|b|$ has the density $N\omega_N t^{N-1}/|B|$ on $[0, \rho_B]$, so for $\delta \le \rho_A\rho_B$

$$\begin{aligned}\mathbb{E}\big[|b|^{-N};\ |b| \ge \delta/\rho_A\big] &= \frac{N\omega_N}{|B|}\int_{\delta/\rho_A}^{\rho_B}\frac{\mathrm{d}t}{t} + \mathbb{E}\big[|b|^{-N};\ |b| > \rho_B\big]\\ &= \frac{N\omega_N}{|B|}\log\frac{1}{\delta} + O(1),\end{aligned}$$

while the second term of V is $O(\delta^N)$. Hence $V(\varepsilon) = \frac{N\omega_N^2}{|A||B|}\delta^N\log\frac{1}{\delta} + O(\delta^N)$, and since $\log\frac{1}{\delta} = \frac{1}{2}\log\frac{1}{\varepsilon} - \frac{1}{2}\log 2$, this is $C\varepsilon^{N/2}\log\frac{1}{\varepsilon}\,(1 + o(1))$ with the stated C. ∎

For $M = N = 1$ the theorem is the product model, with $\lambda = \frac{1}{2}$ and $m = 2$ (Chapter 3); on $[-1,1]^2$ it gives $C = 2^{-1/2}$, the constant of the exact volume $\sqrt{2\varepsilon}\,(1 + \frac{1}{2}\log\frac{1}{2\varepsilon})$ of Chapter 4. For $M = 1$ and $N = 2$, the model $y = b\,(a_1x_1 + a_2x_2)$, it gives $\lambda = \frac{1}{2}$ with $m = 1$: one more input parameter removes the logarithm. The box enters only through C. For $M = N = 3$ and $A = B = [-1,1]^3$, $C = \pi^2\sqrt{2}/12 = 1.163$, and the computation of (ii), carried to its end (Exercise 7.2), gives for every $\varepsilon \le \frac{1}{2}$

$$V(\varepsilon) = (2\varepsilon)^{3/2}\Big(\frac{\pi^2}{24}\log\frac{1}{2\varepsilon} + c_2\Big), \qquad c_2 = \frac{\pi}{3}J_1 + \frac{\pi^2}{36} = 0.5905, \tag{7.2}$$

with $J_1 = \mathbb{E}[|a|^{-3};\ |a| > 1] = 0.3021$ for a uniform on the cube. Monte Carlo with 2×10^7 prior draws gives $V(10^{-2}) = 6.23 \times 10^{-3}$ (Clopper–Pearson interval $[6.19, 6.26] \times 10^{-3}$) against 6.22×10^{-3} from (7.2); the leading term alone gives 5.36×10^{-3}. Priors bounded above and below by positive constants near the origin, and inputs with a positive definite second-moment matrix Σ, for which $K = \frac{1}{2}|b|^2 a^\top\Sigma a$, leave λ and m unchanged, by comparison (Proposition 5.4), though not the constants; Exercise 7.10 treats a Gaussian prior, whose support is not compact.

The theorem gives the global coefficient, the minimum of the local ones over the true set. At a point $(a^*, 0)$ with $a^* \ne 0$ the factor $|a|^2$ stays away from zero, K is comparable to $|b|^2$, and the local value is $\frac{M}{2}$ with $m = 1$; at $(0, b^*)$ with $b^* \ne 0$ it is $\frac{N}{2}$ (Chapter 13 computes both). So when $N < M$ the minimum $\frac{N}{2}$ is attained along the whole branch $\{a = 0\}$, and when $M = N$ the two branches tie and their crossing point carries the second logarithm.

7.4 The theorem: reduced rank, and the catalogue

For $H \ge 2$ the product BA is a sum of H rank-one terms that share the input and output spaces, and the zeta function no longer factors. A reduction, proved here, removes the rank of the truth, and the formula of Aoyagi and Watanabe (2005) covers every case. Aoyagi's papers compute the real log canonical threshold of K with respect to the prior, which is the learning coefficient under another name (Chapter 4). For the model (7.1) with an input second-moment matrix $\Sigma > 0$ and a truth C_0 of rank r,

$K(A,B) = \frac{1}{2}\operatorname{tr}\big((BA - C_0)\,\Sigma\,(BA - C_0)^\top\big)$, and a true parameter (A,B) is *deepest* if $\operatorname{rank} A = \operatorname{rank} B = r$, the least possible. For w in the true set write $\lambda(w)$ and $m(w)$ for the learning coefficient and multiplicity of K restricted to a small neighborhood of w, with a prior continuous and positive at w; by comparison they depend on neither (Proposition 5.4).

Proposition 7.3 (Reduction to a zero truth). *Let $1 \le r \le \min(M,N,H)$, and let (λ_0, m_0) be the local learning coefficient and multiplicity at the origin of the zero-truth loss $K'(A',B') = \frac{1}{2}\|B'A'\|_F^2$, $B' \in \mathbb{R}^{(M-r)\times(H-r)}$, $A' \in \mathbb{R}^{(H-r)\times(N-r)}$, taken to be $(0,1)$ if one of $M-r$, $N-r$, $H-r$ is zero. Then every true parameter w satisfies $\lambda(w) \ge \frac{1}{2}r(M+N-r) + \lambda_0$, with equality and $m(w) = m_0$ at every deepest w, and with $m(w) \le m_0$ wherever equality holds. Consequently, for a prior that is bounded on a compact W and continuous and positive at some deepest true parameter interior to W,*

$$\lambda = \tfrac{1}{2}r(M+N-r) + \lambda_0, \qquad m = m_0.$$

Proof. The plan is to bring a true parameter to a normal form by symmetries, to change coordinates so that K becomes, up to constant factors, a sum of squares of $r(M+N-r)$ coordinates plus the zero-truth loss in the remaining blocks, and to compare true parameters by homogeneity. All norms are Frobenius norms.

Step 1: normal form. Let $w = (A,B)$ with $BA = C_0$, and choose invertible $P \in \mathbb{R}^{M\times M}$ and $Q \in \mathbb{R}^{N\times N}$ with $PC_0Q = E_r = \left(\begin{smallmatrix} I_r & 0 \\ 0 & 0 \end{smallmatrix}\right)$. Split $AQ = (A_1', A_2')$ after its first r columns, and let B_1' be the first r rows of PB. From $(PB)(AQ) = E_r$ we get $B_1'A_1' = I_r$, so A_1' has rank r and $\ker B_1'$, of dimension $H-r$, meets the column space of A_1' only in 0. With $g^{-1} = (A_1', X)$, the columns of X a basis of $\ker B_1'$, we have $g \in GL(H)$, $gA_1' = \left(\begin{smallmatrix} I_r \\ 0 \end{smallmatrix}\right)$ and $B_1'g^{-1} = (I_r, 0)$. Comparing the blocks of $(PBg^{-1})(gAQ) = E_r$ then gives

$$gAQ = \begin{pmatrix} I_r & 0 \\ 0 & \alpha \end{pmatrix}, \qquad PBg^{-1} = \begin{pmatrix} I_r & 0 \\ 0 & \beta \end{pmatrix}, \qquad \beta\alpha = 0,$$

with $\alpha \in \mathbb{R}^{(H-r)\times(N-r)}$ and $\beta \in \mathbb{R}^{(M-r)\times(H-r)}$, and w is deepest exactly when $\alpha = 0$ and $\beta = 0$. The map $\Phi(A,B) = (gAQ, PBg^{-1})$ is a linear automorphism of the parameter space, and $K \circ \Phi^{-1}(A,B) = \frac{1}{2}\operatorname{tr}\big(D\Sigma D^\top\big)$ with $D = P^{-1}(BA - E_r)Q^{-1}$ lies between two positive multiples of $\frac{1}{2}\|BA - E_r\|^2$, both being positive definite quadratic forms in $BA - E_r$. By invariance and comparison (Proposition 5.4), $(\lambda(w), m(w))$ is the local value of $\frac{1}{2}\|BA - E_r\|^2$ at $\Phi(w)$. From now on $C_0 = E_r$, $\Sigma = I$, and w is in normal form.

Step 2: coordinates. Write $A = \left(\begin{smallmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{smallmatrix}\right)$ in the block sizes of the normal form, so that at w, $A_{11} = B_{11} = I_r$, the blocks $A_{12}, A_{21}, B_{12}, B_{21}$ vanish, $A_{22} = \alpha$ and $B_{22} = \beta$. Near w, where A_{11} and B_{11} are invertible, the blocks of $BA - E_r$ are $P' = B_{11}A_{11} + B_{12}A_{21} - I_r$, $Q' = B_{11}A_{12} + B_{12}A_{22}$, $R' = B_{21}A_{11} + B_{22}A_{21}$ and $S = B_{21}A_{12} + B_{22}A_{22}$. Replacing (A_{11}, A_{12}, B_{21}) by (P', Q', R') and keeping the other blocks is an analytic isomorphism near w, with inverse $A_{11} = B_{11}^{-1}(P' + I_r - B_{12}A_{21})$, $A_{12} = B_{11}^{-1}(Q' - B_{12}A_{22})$ and $B_{21} = (R' - B_{22}A_{21})A_{11}^{-1}$. To express S in the new coordinates, let

$Y = (B_{11}A_{11})^{-1} = (I - B_{12}A_{21} + P')^{-1}$ and $Y_0 = (I - B_{12}A_{21})^{-1}$, so that $Y - Y_0 = -YP'Y_0$, and collect in L the terms that contain R', Q' or $Y - Y_0$. Substituting the inverse,

$$S = (R' - B_{22}A_{21})\,Y\,(Q' - B_{12}A_{22}) + B_{22}A_{22} = B_{22}\big(I + A_{21}Y_0B_{12}\big)A_{22} + L.$$

Each entry of L is a sum of products of an entry of P', Q' or R' with functions analytic and bounded near w, so $\|L\| \le c(\|P'\| + \|Q'\| + \|R'\|)$ near w for some constant c. Since $A_{21}(I - B_{12}A_{21}) = (I - A_{21}B_{12})A_{21}$, we have $I + A_{21}Y_0B_{12} = (I - A_{21}B_{12})^{-1} =: G$, which is I at w and invertible near it.

Step 3: comparison. Put $T = \|P'\|^2 + \|Q'\|^2 + \|R'\|^2$ and $S' = B_{22}GA_{22}$. Then $K = \frac{1}{2}(T + \|S\|^2)$, and since $\|S\|^2 \le 2\|S'\|^2 + 2\|L\|^2$, $\|S'\|^2 \le 2\|S\|^2 + 2\|L\|^2$ and $\|L\|^2 \le 3c^2T$, near w the loss K lies between two positive multiples of

$$\tilde{K} = \tfrac{1}{2}T + \tfrac{1}{2}\|B'_{22}A_{22}\|^2, \qquad B'_{22} = B_{22}G.$$

Replacing B_{22} by B'_{22}, with A_{21} and B_{12} fixed, is one more analytic isomorphism, and it fixes w because $G = I$ there.

Step 4: the local value at w. In the coordinates P', Q', R', B'_{22}, A_{22}, B_{11}, B_{12} and A_{21}, the function $\tilde{K}$ is the sum of a positive definite quadratic form in the $r^2 + r(N-r) + (M-r)r = r(M+N-r)$ entries of P', Q', R', whose learning coefficient is $\frac{1}{2}r(M+N-r)$ with $m = 1$, and the zero-truth loss K' in (B'_{22}, A_{22}), near the point (β, α); the coordinates B_{11}, B_{12}, A_{21} do not enter. The prior, carried to these coordinates, is continuous and positive at the image of w, hence bounded above and below on a small box around it by multiples of the uniform prior, which is a product over the groups. Integrating out the free coordinates and applying the rule for sums of functions of separate variables (Proposition 5.4) gives $\lambda(w) = \frac{1}{2}r(M+N-r) + \lambda'(\beta,\alpha)$ and $m(w) = m'(\beta,\alpha)$, with (λ', m') the local values of K'. At a deepest w, $(\beta,\alpha) = (0,0)$ and $(\lambda', m') = (\lambda_0, m_0)$. If one of $M-r$, $N-r$, $H-r$ is zero, K' vanishes identically, leaving its variables free, and contributes $(0,1)$.

Step 5: homogeneity. For $t > 0$ the dilation $(\beta,\alpha) \mapsto (t\beta, t\alpha)$ is an analytic isomorphism with $K'(t\beta,t\alpha) = t^4K'(\beta,\alpha)$, so the local values of K' at (β,α) and at $(t\beta,t\alpha)$ coincide. Let U_0 be a box around the origin small enough to realize (λ_0, m_0), and choose t so small that a box U around $(t\beta,t\alpha)$, small enough to realize its local values, lies in U_0. With the same uniform density, the volume of $\{K' < \varepsilon\} \cap U$ is at most that of $\{K' < \varepsilon\} \cap U_0$, so by the volume asymptotics of Chapter 4 either $\lambda'(\beta,\alpha) > \lambda_0$, or the two are equal and $m'(\beta,\alpha) \le m_0$. With Step 4 this proves the inequality and its cases of equality.

Step 6: the global value. The global λ is the minimum over the true set of the local values with the actual prior, and m the largest multiplicity among the points that attain it (Proposition 5.4). With a bounded prior, the volume function near a true parameter w is at most a constant times the one with a prior positive at w, so the local value with the actual prior is at least $\lambda(w)$, with multiplicity at most $m(w)$ if the two agree; at the interior deepest parameter where the prior is positive, they coincide. So $\lambda = \frac{1}{2}r(M+N-r) + \lambda_0$ and $m = m_0$. ∎

Corollary 7.4. *Under the hypotheses of Proposition 7.3 on the prior, reduced-rank regression with true rank $r = 1$ has (i) $\lambda = \frac{1}{2}(M+N-1)$ and $m = 1$ if $H = 1$; (ii) $\lambda = \frac{1}{2}(M+N-1)+\frac{1}{2}\min(M-1, N-1)$ if $H = 2$, with $m = 2$ if $M = N \geq 2$ and $m = 1$ otherwise. In particular, for $(M, N, H) = (2, 2, 1)$, $(2, 2, 2)$, $(3, 3, 1)$ and $(3, 3, 2)$, $\lambda = \frac{3}{2}, 2, \frac{5}{2}, \frac{7}{2}$ with $m = 1, 2, 1, 2$.*

Proof. For $H = r = 1$ the zero-truth problem has $H - r = 0$, so $(\lambda_0, m_0) = (0, 1)$. For $H = 2$ and $r = 1$ it is rank-one regression with $M-1$ outputs and $N-1$ inputs at the zero truth, and Theorem 7.2, applied on arbitrarily small boxes around the origin, gives its local value there, $\frac{1}{2}\min(M-1, N-1)$ with multiplicity 2 if $M = N$ and 1 otherwise; if $\min(M, N) = 1$, a dimension is zero and the value is $(0, 1)$. ∎

Theorem 7.5 (Aoyagi and Watanabe 2005). *In reduced-rank regression (7.1), with $B \in \mathbb{R}^{M\times H}$, $A \in \mathbb{R}^{H\times N}$, a truth of rank $r \leq \min(M, N, H)$ and a prior as in Proposition 7.3, let $\iota = 0$ if $M + N + H + r$ is even and $\iota = 1$ if it is odd. The learning coefficient and its multiplicity are as follows.*

(i) If $N + r \leq M + H$, $M + r \leq N + H$ and $H + r \leq M + N$, then

$$\lambda = \frac{2(H+r)(M+N) - (M-N)^2 - (H+r)^2 + \iota}{8}, \qquad m = 1 + \iota.$$

(ii) If $M + H < N + r$, then $\lambda = \frac{1}{2}(HM - Hr + Nr)$ and $m = 1$.

(iii) If $N + H < M + r$, then $\lambda = \frac{1}{2}(HN - Hr + Mr)$ and $m = 1$.

(iv) If $M + N < H + r$, then $\lambda = \frac{1}{2}MN$ and $m = 1$.

The cases are exclusive and exhaustive: if (i) fails, one of its inequalities fails, and two of (ii)–(iv) together would force $H < r$, $M < r$ or $N < r$. We have not seen the original paper. We quote the four cases from the restatement of Hayashi and Watanabe (2017, eq. (6)), which omits the multiplicities, and the multiplicities from Watanabe (2013, Table 3) and Drton and Plummer (2017, Example 2.2), whose values the parity rule reproduces; the hypotheses, as the restatements give them, are Gaussian noise with identity covariance, inputs with a positive definite second-moment matrix, and a prior positive at the truth. Five checks, computed by `07-catalogue-formulas.py`:

- For $H = 1$ the formula gives $\frac{1}{2}\min(M, N)$ at $r = 0$, with $m = 2$ exactly when $M = N$ (Theorem 7.2), and $\frac{1}{2}(M + N - 1)$ with $m = 1$ at $r = 1$ (Corollary 7.4).
- It satisfies the identity of Proposition 7.3, $\lambda(M, N, H, r) = \frac{1}{2}r(M + N - r) + \lambda(M - r, N - r, H - r, 0)$ with equal multiplicities, for all 2754 tuples with $M, N, H \leq 9$; since the proposition proves the identity for the true values, the formula's zero-truth part determines the rest.
- It gives the four values of Corollary 7.4, and $\lambda = 4$, $m = 1$ for $M = N = H = 3$, $r = 1$, which the proposition reduces to the zero-truth case $(2, 2, 2)$ with $\lambda_0 = \frac{3}{2}$.

- For $M = N = 6$, $r = 3$ and $H = 3,4,5,6$ it gives $\frac{27}{2}$, 15, 16, 17 with $m = 1,2,1,2$, the theoretical values of Watanabe (2013, Table 3).
- It reproduces the ten entries of Drton and Plummer (2017, Table 1), 5 responses and 3 covariates, and their statement that $m = 2$ only for model rank 3 at true rank 0.

DEEP LINEAR NETWORKS. For $y = W_L \cdots W_1 x + \varepsilon$ with widths H_0 (input), $H_1, \ldots, H_L$ (output), Aoyagi (2024) computed λ and m exactly. We state the result in the form of Aoyagi (2025, Definition 4 and Remark 1), which includes the multiplicity; Chapter 13 uses it to validate estimators.

Theorem 7.6 (Aoyagi 2024). *Let $W_j \in \mathbb{R}^{H_j \times H_{j-1}}$, the noise be Gaussian, the inputs have a density positive on a set of positive volume, the prior be smooth with compact support and positive at the true parameter, and the true product have rank r; put $\Delta_j = H_j - r$, $j = 0, \ldots, L$. If some $\Delta_j = 0$, then $\lambda = \frac{1}{2}(r(H_0 + H_L) - r^2)$ and $m = 1$. Otherwise let $\Gamma \subseteq \{0, \ldots, L\}$, $|\Gamma| = \ell + 1 \geq 2$, be such that $\Delta_j < \Delta_k$ for $j \in \Gamma$ and $k \notin \Gamma$, and such that $S = \sum_{j \in \Gamma} \Delta_j$ satisfies $S \geq \ell \Delta_j$ for $j \in \Gamma$ and $S < \ell \Delta_k$ for $k \notin \Gamma$. With $q = \lceil S/\ell \rceil$ and $p = S - (q-1)\ell$,*

$$\lambda = \frac{r(H_0 + H_L) - r^2}{2} + \frac{p(\ell - p)}{4\ell} + \frac{S^2}{4\ell} - \frac{1}{4} \sum_{j \in \Gamma} \Delta_j^2, \qquad m = p(\ell - p) + 1.$$

For $L = 2$ the formula agrees with Theorem 7.5 on the 2754 tuples above, and for widths $1, \ldots, 1$ it gives the chain. A consequence that we derive here and have not found stated is that depth beyond the width leaves λ fixed and raises m:

Corollary 7.7. *If all widths equal h, the truth is 0 and $L \geq h$, then $\lambda = \frac{1}{4}h(h+1)$ and $m = h(L-h) + 1$, against $\frac{d}{2} = \frac{1}{2}Lh^2$.*

PROOF. All $\Delta_j = h$, and two indices with equal Δ are both in Γ or both out, so Γ is the whole index set, $\ell = L$ and $S = (L+1)h$. Since $L \geq h$, $h < S/\ell = h + h/L \leq h + 1$, so $q = h + 1$ and $p = (L+1)h - hL = h$. Theorem 7.6 gives $m = h(L-h) + 1$ and

$$\lambda = \frac{h(L-h)}{4L} + \frac{(L+1)^2 h^2}{4L} - \frac{(L+1)h^2}{4} = \frac{hL - h^2 + (L+1)h^2}{4L} = \frac{h(h+1)}{4}. \qquad \blacksquare$$

For width 2, $\lambda = \frac{3}{2}$ at every depth $L \geq 2$, with $m = 2L - 3$: 3 at depth 3 and 5 at depth 4, against $\frac{d}{2} = 6$ and 8.[5] Aoyagi (2025) notes that the coefficients decrease as layers are added (for equal widths, by Corollary 7.7, only until the depth reaches the width) and reads this as one theoretical reason for the effectiveness of deep linear networks. The values are global: at a generic trained parameter the local value can be larger (Chapter 13).

[5] The script `07-catalogue-formulas.py` checks the corollary against Theorem 7.6 for all widths $h \leq 6$ and depths $h \leq L \leq 12$.

ONE HIDDEN LAYER OF TANH UNITS. For $f(x) = \sum_{h=1}^{H} a_h \tanh(b_h x)$ with one input in $[-1, 1]$, one output, the truth 0 and a positive prior, Aoyagi

obtained

$$\lambda = \frac{H + k^2 + k}{4k + 2}, \quad k = \lfloor \sqrt{H} \rfloor, \qquad m = 2 \text{ if } k^2 = H, \quad m = 1 \text{ otherwise},$$

in papers of 2005, with Watanabe and in Japanese, and of 2006, which she cites in Aoyagi (2019); the end of §4 there restates the formula for several outputs from a paper of 2009. It gives $\frac{1}{2}$ ($m = 2$), $\frac{2}{3}$, $\frac{5}{6}$ and 1 ($m = 2$) for $H = 1, 2, 3, 4$, and grows like $\frac{1}{2}\sqrt{H}$, against $\frac{d}{2} = H$. The mechanism is a singularity of Vandermonde type: near the origin K is governed by the moments $\sum_h a_h b_h^p$ for odd p (Aoyagi 2010a; Aoyagi and Nagata 2012). For several inputs, values for $H \le 4$ are listed in Aoyagi (2019, Theorem 6),[6] and upper bounds for local values in Kurumadani (2027).

[6] We could not reconcile one entry of that list with the one-input formula. For $N = 1$ input, $M = 1$ output, $H = 4$ and $Q = 2$ (the tanh case), the minimum of the list for $H = 4$ is $\frac{2}{3}$, attained by its term $\frac{4N+Q(2(N-2)+2M)}{2(Q+1)}$, while the paper's own formula for one input gives 1. The negative factor $N - 2$ suggests that the list assumes more inputs. We do not use it.

Mixtures. The book's mixture $(1-a)\mathcal{N}(0,1) + a\mathcal{N}(b,1)$ at the truth $\mathcal{N}(0,1)$ has $\lambda = \frac{1}{2}$ and $m = 2$ by comparison with a^2b^2 (Chapter 2), consistent with the free-energy computation of Kariya and Watanabe (2022); its K is not analytic at $a = 0$, so the theorems of Part III do not apply verbatim. With both means free, $\alpha\mathcal{N}(\mu_1, 1) + (1-\alpha)\mathcal{N}(\mu_2, 1)$ has $\lambda = \frac{3}{4}$, $m = 1$, at a truth $\mathcal{N}(\mu, 1)$, and the regular $\frac{3}{2}$ at a genuine two-component truth (Drton and Plummer (2017, Example 2.1), citing Aoyagi (2010a)). For normal mixtures in general there are upper bounds (Yamazaki and Watanabe 2003).

Boltzmann machines and Bayesian networks. For a restricted Boltzmann machine with M binary visible units, N binary hidden units, no biases and $M > N$, Aoyagi (2010b, Theorem 3) gives $\lambda = \frac{1}{4}MN$ at the zero weights, with $m = 1$ if $M > N + 1$ and $m = M$ if $M = N + 1$. For $M = 3$ and $N = 1$ this is $\frac{3}{4}$, which a direct computation confirms: the model is $p(x) = \frac{1}{8}(1 + \sum_{i<j} b_i b_j x_i x_j)$ with $b_i = \tanh a_i$, so near the origin K is comparable to $\sum_{i<j} a_i^2 a_j^2$, whose Newton polyhedron meets the diagonal at $t = \frac{4}{3}$ (Chapter 6).[7] For naive Bayesian networks with $k \ge 3$ binary features and two hidden classes, $d = 2k + 1$, Rusakov and Geiger (2005, Theorem 4) prove that the coefficient of $\log n$ is $\frac{1}{2}(2k + 1)$ at regular points, $\frac{1}{2}(2k - 1)$ at their singularities of type 1, and $\frac{1}{2}(k + 1)$ where all features are independent, each with $m = 1$; for $k = 2$ it is $\frac{3}{2}$ at the independence set, with an extra $2 \log\log n$ in the log marginal likelihood, so $m = 3$. Binary latent tree models (Zwiernik 2011) and Gaussian latent tree and forest models (Drton et al. 2017) have formulas in terms of the tree, for the cases those papers treat. For hidden Markov models we found upper bounds (Yamazaki and Watanabe 2005) and no exact values.

[7] One entry of Theorem 2 of Aoyagi (2010b) gives $\lambda = \frac{1}{2}$ for $M = 3$ and $N = 1$ at true weights $b^* \neq 0$ with a zero coordinate. At $b^* = (c, 0, 0)$ the distribution is uniform, as at the zero weights, so for small c this local value cannot lie below the value $\frac{3}{4}$ at the origin; the same direct computation gives 1 there, and also at $(c, d, 0)$ with $cd \neq 0$. We do not use the entry.

Factorizations, ReLU networks, and recent results. Non-negative matrix factorization has upper bounds that are exact in special cases (Hayashi and Watanabe 2017), and latent Dirichlet allocation an exact formula with the case structure of Theorem 7.5 (Hayashi 2021). ReLU networks are not analytic. For deep ReLU networks with more layers than the truth needs, Nagayasu and Watanabe (2023, a preprint) bound the free energy by a quantity that does not grow with the extra layers, and Aoyagi (2025, a preprint) claims exact values for one hidden ReLU layer, as sums of deep-linear values over the regions of input space where the network

model	truth	$d/2$	λ	m	status
product model $y = abx$	0	1	1/2	2	proved (Ch. 3)
chain, $L = 3$	0	3/2	1/2	3	proved
rank one, $M = N = 3$	0	3	3/2	2	proved
	rank 1	3	5/2	1	proved
reduced rank 3×3, $H = 2$	0	6	5/2	1	source
	rank 1	6	7/2	2	proved
reduced rank 3×3, $H = 3$	rank 1	9	4	1	source, reduced
reduced rank 6×6, $H = 4$	rank 3	24	15	2	source
deep linear 2-2-2-2	0	6	3/2	3	source
deep linear 3-3-3-3	0	27/2	3	1	source
tanh units, $H = 2$	0	2	2/3	1	source, checked
tanh units, $H = 4$	0	4	1	2	source
mixture, 2 parameters	$\mathcal{N}(0, 1)$	1	1/2	2	proved (Ch. 2)
mixture, 3 parameters	$\mathcal{N}(\mu, 1)$	3/2	3/4	1	source
RBM, 3 visible, 1 hidden	0	3/2	3/4	1	source, checked
RBM, 4 visible, 2 hidden	0	4	2	1	source
naive Bayes, 3 features	independent	7/2	2	1	source
naive Bayes, 2 features	independent	5/2	3/2	3	source

Table 7.1. Learning coefficients of small instances against half the number of parameters. *Proved*: in this chapter (Computation 7.2, Theorem 7.2, Corollary 7.4) or where indicated. *Source*: taken from the cited source, which proves it or cites a proof (Theorems 7.5 and 7.6; Aoyagi (2010b, 2019) and Rusakov and Geiger (2005); Drton and Plummer (2017), citing Aoyagi (2010a), for the three-parameter mixture). *Checked*: also verified here independently (the comparable polynomial of Exercise 7.7; a Newton polyhedron). *Reduced*: brought to a zero truth by Proposition 7.3. Inputs uniform on $[-1, 1]$ for the tanh units; priors positive at the most singular true parameters. Values from `07-catalogue-formulas.py`.

is linear. Since 2020 exact values have appeared for factor analysis (Drton et al. 2025, a preprint), and local values at several critical points of the toy model of superposition (Chen et al. 2023, a preprint). As of October 2026 we found no exact value or rigorous bound for a transformer or an attention layer, only estimates (Chapters 13 and 14).

THE SINGULAR BIC. Learning coefficients enter model selection through the singular BIC of Drton and Plummer (2017) (Definition 8.7), which gives each model $\mathcal{M}_i$ an approximate marginal likelihood that averages the free energy formulas $L'_{ij} = P(X^n \mid \hat{w}_i, \mathcal{M}_i)(\log n)^{m_{ij}-1} n^{-\lambda_{ij}}$ over the submodels $\mathcal{M}_j$ that could contain the truth; for reduced-rank regression Theorem 7.5 supplies every λ_{ij} and m_{ij}. As a worked example, take $M = N = 3$ and compare $\mathcal{M}_0 = \{C = 0\}$ with the rank-one model $\mathcal{M}_1$, using $\lambda_{00} = 0$, $\lambda_{10} = \frac{3}{2}$ with $m_{10} = 2$ (Theorem 7.2), and $\lambda_{11} = \frac{5}{2}$ (Corollary 7.4). Let T be twice the maximized log-likelihood ratio of $\mathcal{M}_1$ against $\mathcal{M}_0$, $A = e^{T/2}$, $b = 1 - An^{-5/2}$ and $c = An^{-3/2} \log n$. Dividing every L' by the maximized likelihood of $\mathcal{M}_0$, the recursion of Drton and Plummer gives

$$\mathrm{sBIC}(\mathcal{M}_1) - \mathrm{sBIC}(\mathcal{M}_0) = \log \tfrac{1}{2}\left(-b + \sqrt{b^2 + 4c}\right).$$

For small T this is close to $\frac{T}{2} - \frac{3}{2} \log n + \log \log n$, the free energy formula at a zero truth; for large T, to BIC's $\frac{T}{2} - \frac{5}{2} \log n$. At $n = 100$ the sBIC prefers

the rank-one model once $T > 12.14$, while BIC with the dimension 5 of the rank-one matrices needs $T > 23.03$, and with $d = 6$, $T > 27.63$. At $T = 0$ the difference is -5.39, against -5.38 from the zero-truth formula; at $T = 40$ it is 8.57, against BIC's 8.49. Chapter 8 runs the full rank-selection experiment.

A WARNING ABOUT PRE-ASYMPTOTIC RANGES. The values above are limits as $\varepsilon \to 0$, and the approach can be slow. For the two tanh units of Section 7.2, with inputs uniform on $[-1, 1]$ and the uniform prior on $[-1, 1]^4$, `07-catalogue-tanh.py` estimates $V(\varepsilon)$ from 2×10^7 prior draws and fits exponents 0.484 over $[10^{-5}, 3 \times 10^{-2}]$, 0.519 over $[10^{-5}, 10^{-3}]$ and 0.551 over $[10^{-5}, 10^{-4}]$, against $\lambda = \frac{2}{3}$: a Monte Carlo study would report about $\frac{1}{2}$, the value of a single neuron. The reason is visible to third order, $K \approx 0.167\,(S_1 - 0.2S_3)^2 + 0.00127\,S_3^2$ with $S_p = a_1 b_1^p + a_2 b_2^p$: the second coefficient is 0.0076 times the first, and the constraint that makes $\lambda = \frac{2}{3}$ binds only far below $\varepsilon = 10^{-5}$. For the comparable polynomial $S_1^2 + S_3^2$, 4×10^7 draws give 0.660 over $[10^{-8}, 10^{-4}]$.

7.5 Measuring it

Volumes are of no use in twelve dimensions, so we estimate λ as Watanabe (2013) does, from posterior samples at two temperatures (Chapter 11),

$$\hat{\lambda} = \frac{\mathbb{E}_w^{\beta_1}[nL_n(w)] - \mathbb{E}_w^{\beta_2}[nL_n(w)]}{1/\beta_1 - 1/\beta_2}, \qquad \beta_1 = \frac{1}{\log n}, \quad \beta_2 = \frac{1.5}{\log n},$$

which converges to λ in probability (Watanabe 2013, Corollary 3). As in his experiment on reduced-rank regression, we sample at β_1 only, by random-walk Metropolis (Section A.6), and obtain the mean at β_2 by reweighting with $e^{-(\beta_2-\beta_1)nL_n}$ (Watanabe 2013, §6.2 and eq. (20)). If the tempered means behave like their population versions, $\mathbb{E}_w^{\beta}[nK] = \frac{1}{\beta}\big(\lambda - \frac{m-1}{\log(n\beta)+\kappa}\big)$ up to smaller terms (Proposition 11.4), with κ a constant of the model and the prior, then

$$\hat{\lambda} \approx \lambda - (m-1)\Big(\frac{3}{\ell_n + \kappa} - \frac{2}{\ell_n + \log 1.5 + \kappa}\Big), \qquad \ell_n = \log \frac{n}{\log n}, \tag{7.3}$$

a shortfall of about $(m-1)/\log n$, as the appendix records (Section A.6). For the product model on $[-1, 1]^2$ and rank-one regression with $M = N = 3$ on $[-1, 1]^6$ the population expansion is exact up to terms of order $e^{-n\beta/2}$, because their volumes are known exactly for $\varepsilon \le \frac{1}{2}$; then $\kappa = \gamma + \log 2 = 1.270$ and $\kappa = \frac{8}{\pi} J_1 - \log 2 - \psi(\frac{3}{2}) = 0.040$ (Exercise 7.2).

VALIDATION. The script `07-catalogue-validate.py` checks the estimator where exact tempered means are available, at $n = 10^3$, 10^4 and 10^5, with uniform priors on $[-1, 1]^d$ and the truth 0. On the regular plane, 20 data sets per n with 16 chains each give 1.002, 1.000, 1.000 (Student-t half-widths at most 0.010), against 1.000 from exact quadrature on the same data; per data set the two differ by 0.014 to 0.017 on average. On the product model they give 0.323, 0.371, 0.393 (half-widths 0.005), against 0.323, 0.369, 0.396 by exact quadrature in $c = ab$ and 0.320, 0.368, 0.396 from (7.3): the shortfall

from $\frac{1}{2}$ is a property of the posterior, not of the sampler. On rank-one regression with $M = N = 3$ at the truth 0 without noise, the population posterior, 1024 chains as long as those of the reduced-rank runs below give 1.269, 1.347, 1.391 (half-widths at most 0.008) against the exact 1.271, 1.342, 1.381. The last is 0.010 high, a small bias of the random walk at the largest n: a Gibbs sampler that redraws b given a and a given b from their exact conditional laws, truncated Gaussians, gives 1.381 ± 0.003.

REDUCED-RANK REGRESSION. The script `07-catalogue-rrr.py` runs the estimator for $M = N = 3$, $H = 1, 2$, the truths 0 and $\frac{1}{2}e_1e_1^\top$, the uniform prior on $[-1, 1]^{6H}$ and $n = 10^3, 10^4, 10^5$, with 12 data sets per case and 16 chains per data set (Figure 7.1). Reweighting keeps an effective sample size of 67 to 86 percent of the draws,[8] and in all twelve cases the estimates from the two halves of the kept draws differ by less than twice their standard error. Every estimate lies far below $\frac{d}{2}$; the ones at a zero truth also lie below half the dimension of the set of distributions, $\frac{5}{2}$ and 4. With $m = 1$ the estimates reach λ by $n = 10^5$, 2.502 ± 0.029 and 2.501 ± 0.033 against $\frac{5}{2}$, but not at $n = 10^3$: 2.737 for the rank-one truth with $H = 1$ and 2.341 for the zero truth with $H = 2$, a shortfall that shrinks steadily with n. The excess belongs to the tempered posterior, not to the data or the sampler: without noise the same sampler gives 2.742, 2.524, 2.504,[9] and quadrature of the population posterior 2.747, 2.529, 2.504. At $n\beta_1 = 145$ the posterior still feels the prior near the truth, mostly through the box: at a fixed size s, the prior weighs a rank-one matrix $s\,uv^\top$ ($|u| = |v| = 1$) by the range of $\log t$ over which its factorizations $(a/t, tb)$ stay inside the box, which is $\log \frac{1}{s}$ along the coordinate axes, where the truth lies, and longer off them. With the uniform prior on a product of balls, where the range is $\log \frac{1}{s}$ in every direction, the population value at $n = 10^3$ falls to 2.601; the rest of the excess comes from the growth of $\log \frac{1}{s}$ toward the singular point $s = 0$ and from the curvature of the set of rank-one matrices (Exercise 7.12). With $m = 2$ the estimates stay below λ. For $\lambda = \frac{3}{2}$ they are 1.294, 1.367, 1.399, where (7.3) with the exact κ predicts 1.271, 1.342, 1.381; the excess of about one half-width at every n is absent from the noiseless runs, apart from the sampler's 0.010 at $n = 10^5$, so the data shrink the shortfall a little. For $\lambda = \frac{7}{2}$ they are 3.427, 3.356, 3.438, where κ is unknown and $\kappa = 0$ gives 3.269, 3.341, 3.380. Watanabe's estimates for $M = N = 6$, true rank 3 and $n = 500$ (Watanabe 2013, Table 3) also fall short, 14.80, 15.72 and 16.55 against 15, 16 and 17, the first and last with $m = 2$; he attributes the differences to terms of lower order than $\log n$, and for $m = 2$ also to $\log\log n$.

[8] If β_1 times the energy above its minimum had the Gamma law of shape λ, its limit law in the population version when $m = 1$, the weights $e^{-(\beta_2-\beta_1)nL_n}$ would keep the fraction $(8/9)^\lambda$: 0.84, 0.75 and 0.66 for $\lambda = \frac{3}{2}, \frac{5}{2}, \frac{7}{2}$.

[9] Without noise the energy is $nK(w)$ itself; `07-catalogue-validate.py` runs the same sampler on it and computes the population values by quadrature, integrating b out in closed form.

7.6 The ledger

For rank-one regression the zeta function is a product, and Theorem 7.2 proves $\lambda = \frac{1}{2}\min(M, N)$, with $m = 2$ exactly when $M = N$. For reduced-rank regression Proposition 7.3 removes the rank of the truth, which with the rank-one theorem proves the small cases of Corollary 7.4, and the Aoyagi–Watanabe formula, cited and checked, gives the rest. Deep linear networks, one-layer tanh networks, simple mixtures, Boltzmann machines

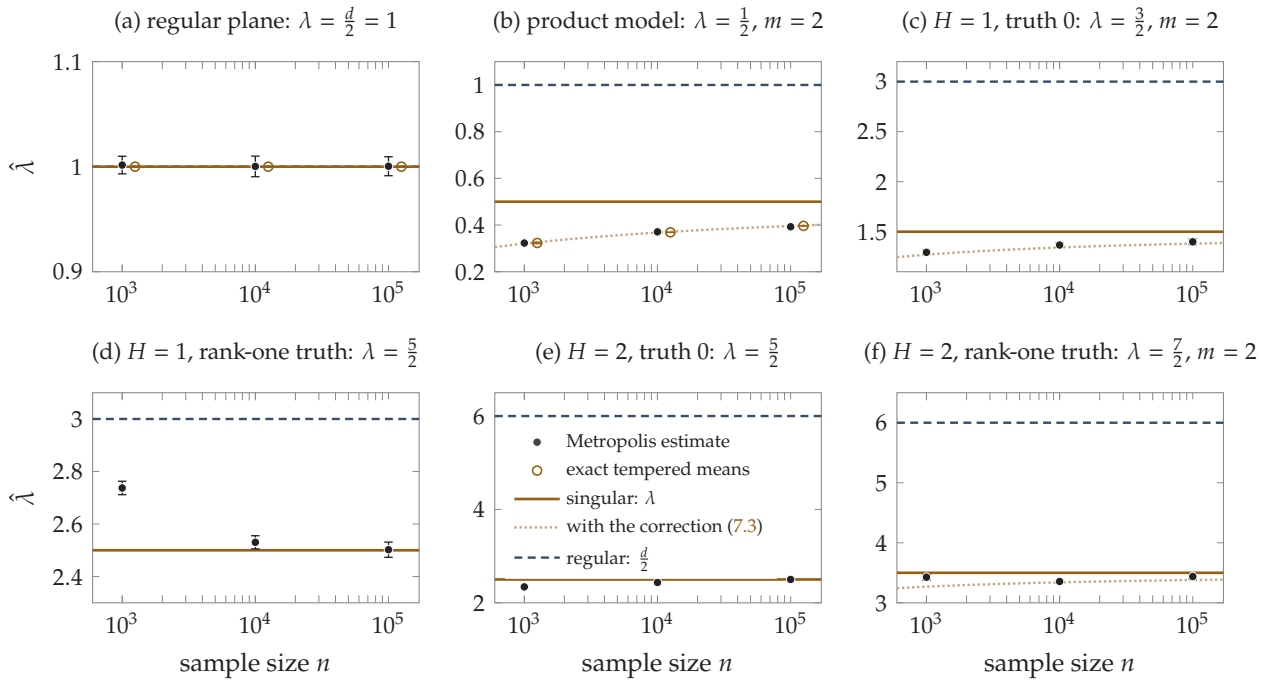


Figure 7.1. Two-temperature estimates of the learning coefficient, $\beta_1 = 1/\log n$ and $\beta_2 = 1.5/\log n$, with Student-t 95% intervals over data sets; uniform priors on $[-1, 1]^d$. (a, b) Validation, 20 data sets per n at the truth 0 (seeds $71000 + 1000k + j$ and $72000 + 1000k + j$ for the data set j at $n = 10^{k+3}$): dots, Metropolis with reweighting; circles, the same estimator from exact tempered means on the same data, drawn slightly to the right. (c)–(f) Reduced-rank regression with $M = N = 3$, $H = 1, 2$, the truths 0 and $\frac{1}{2}e_1e_1^\top$, 12 data sets per n (seeds $75000 + 10000c + 1000k + j$ for the case $c = 0, \ldots, 3$ in the order (c)–(f)), 16 chains per data set, 8000 adaptation steps, 16000 kept. Bronze: λ from Theorems 7.2 and 7.5 and Corollary 7.4; dotted: (7.3) with $\kappa = \gamma + \log 2$ (b), $\kappa = 0.040$ (c), and $\kappa = 0$ (f), where the true constant is unknown; slate dashed: $\frac{d}{2}$. Scripts `07-catalogue-validate.py` and `07-catalogue-rrr.py`.

and Bayesian networks have exact values in the literature, all far below $\frac{d}{2}$; most other architectures have bounds only. Estimates from posterior samples agree with the formulas at $n = 10^5$ once the multiplicity correction is counted; at $n = 10^3$ the prior near the truth can still move them by a quarter.

LEDGER · $y = BAx$	
QUANTITY	the learning coefficient λ and multiplicity m of models in use, at their most singular true parameters
REGULAR THEORY	$\frac{d}{2}$, or half the dimension of the set of distributions; $m = 1$
SINGULAR THEORY	rank-one regression: $\frac{1}{2}\min(M, N)$, $m = 2$ iff $M = N$ (Theorem 7.2); reduced-rank regression: the Aoyagi–Watanabe formula (Theorem 7.5), reduced to a zero truth by Proposition 7.3; other families in Table 7.1
RUNNING EXAMPLE	rank-one regression, $M = N = 3$: $\lambda = \frac{3}{2}$, $m = 2$ at the truth 0 and $\frac{5}{2}$ at a rank-one truth, against $\frac{d}{2} = 3$

MEASURED	two-temperature Metropolis estimates at $n = 10^5$: 1.40, 2.50, 2.50, 3.44 for $\lambda = \frac{3}{2}, \frac{5}{2}, \frac{5}{2}, \frac{7}{2}$; with $m = 2$ short of λ by about the correction (7.3)
RESTS ON	realizability; a prior positive at a most singular true parameter; Gaussian noise of known covariance; analytic (here polynomial) losses; $n \to \infty$ with the model fixed; global values, attained at the most singular points

7.7 Where the theory stops

WHAT WAS ASSUMED. Every value in this chapter belongs to a fixed model and a realizable truth as $n \to \infty$. Reduced-rank regression with $H < r$ cannot realize its truth; Watanabe (2013, §6.2) then uses the regular value $\frac{1}{2}H(M+N-H)$ of the best approximation, and Theorem 7.5 does not apply. The values need a prior positive at a most singular true parameter: a prior that vanishes there can raise the coefficient, and Jeffreys' prior raises it to at least $\frac{d}{2}$ (Watanabe 2024); a box that the true set meets only at corners raises it too, to 1 for the product model at the truth $ab = 1$ on $[-1, 1]^2$. They assume Gaussian noise with known covariance. Watanabe's statistical theorems need analytic losses: those of reduced-rank regression and deep linear networks are polynomials, tanh networks are analytic only with bounded inputs (Chapter 2), and the book's mixture and all ReLU networks are not analytic, so for them the values describe volumes and zeta functions, and Part III does not apply verbatim. Finally, the catalogue lists global values, minima over the true set, while a trained network sits at one true parameter, where the local value can be larger: $\frac{M}{2}$ against $\frac{N}{2}$ on the branch $\{b = 0\}$ of rank-one regression with $N < M$ (Chapter 13).

WHERE IT BREAKS. Exact values are known for few architectures: linear networks of every depth, one hidden tanh layer with a single input, a few mixtures, restricted Boltzmann machines with more visible than hidden units at the zero weights, naive Bayesian networks and latent trees. For networks with two or more nonlinear hidden layers, for hidden Markov models and for ReLU networks only bounds are published, and the exact ReLU values of Aoyagi (2025) are, as of October 2026, an unrefereed preprint. Even exact values can mislead at practical scales: for two tanh units, volumes measured down to $\varepsilon = 10^{-5}$ suggest $\frac{1}{2}$ instead of $\frac{2}{3}$, and the two-temperature estimator at $n = 10^3$ is off by up to a quarter in either direction, from the multiplicity and from lower-order terms that depend on the prior near the truth (Section 7.5). Published values also need care: in two theorems of the literature we found entries that we could not reconcile with independent computations (the side notes of Section 7.4).

OPEN PROBLEMS. Exact learning coefficients are open for networks with one tanh layer, several inputs and outputs and more than four hidden units, and for networks with two nonlinear layers beyond the smallest (Exercise 7.14); Aoyagi (2019) poses the general Vandermonde case. They are open for hidden Markov models, for non-negative matrix factorization outside its special cases (Hayashi and Watanabe 2017), and for attention layers, and

the exact ReLU values claimed by Aoyagi (2025) await independent checks. Resolution of singularities is constructive in principle, but no general method computes λ for a model with more than a handful of parameters; exact algorithms exist for two-dimensional models (Sergeant-Perthuis et al. 2026, a preprint). The sBIC needs λ_{ij} for every pair of nested models, which Theorem 7.6 supplies for deep linear networks; whether the consistency conditions of Drton and Plummer hold when the hidden width of a deep linear network is chosen is, as far as we know, unexamined (Exercise 7.13).

Exercises

7.1 CONSOLIDATION. (Unequal sides.) For rank-one regression with $(M, N) = (3, 2)$ and the uniform prior on the product of unit balls, show that $V(\varepsilon) = 3\delta^2 - 2\delta^3$ with $\delta = \sqrt{2\varepsilon}$, so that $\lambda = 1$ and $m = 1$. With the uniform prior on $[-1, 1]^5$, show from Theorem 7.2 that $V(\varepsilon) \sim C\varepsilon$ with $C = \frac{\pi}{2}\mathbb{E}|b|^{-2}$, b uniform on $[-1, 1]^3$; with $\mathbb{E}|b|^{-2} = 1.9185$, $C = 3.014$.

7.2 CONSOLIDATION. (The exact volume and the constant κ.) (a) For $M = N = 3$ and $A = B = [-1, 1]^3$, carry the proof of Theorem 7.2(ii) to the end for $\delta \le 1$: the part $|b| \ge \delta$ contributes $\frac{\pi}{6}\delta^3(\frac{\pi}{2}\log\frac{1}{\delta} + J_1)$ and the part $|b| < \delta$ contributes $\frac{\pi}{2}\delta^3 J_2$, with $J_2 = \int_1^\infty t^{-4}\mathbb{P}(|a| \le t)\,\mathrm{d}t$. Deduce (7.2) from $J_2 = \frac{1}{3}J_1 + \frac{\pi}{18}$, and prove this identity by writing $r(u) = 1/\max_i u_i$ for the distance to the boundary of the cube along a direction u in the positive octant of the sphere, so that $J_1 = \int \log r(u)\,\mathrm{d}\sigma(u)$. (b) From (7.2), show that $\int e^{-sK}\varphi = C's^{-3/2}(\log s + \kappa) + O(e^{-s/2})$ with $\kappa = \frac{8}{\pi}J_1 - \log 2 - \psi(\frac{3}{2}) = 0.040$, so that the population tempered mean is $\frac{1}{\beta}\big(\frac{3}{2} - \frac{1}{\log(n\beta)+\kappa}\big)$ up to exponentially small terms. Do the same for the product model and find $\kappa = \gamma + \log 2$.

7.3 CONSOLIDATION. (The identity by hand.) Check that Theorem 7.5 satisfies the identity of Proposition 7.3 for $(M, N, H, r) = (4, 4, 2, 1)$ and $(3, 5, 2, 1)$, finding $\lambda = 5$ with $m = 2$ and $\lambda = \frac{9}{2}$ with $m = 1$, and say which case of the theorem each side uses.

7.4 CONSOLIDATION. (Depth.) Use Corollary 7.7 for deep linear networks of width 3 and depths $3, 4, 5$ at the truth 0 (answers: $\lambda = 3$ with $m = 1, 4, 7$), and show from Theorem 7.6 that width 3 and depth 2, reduced-rank regression with $M = N = H = 3$, give $\lambda = \frac{7}{2}$ with $m = 2$, as Theorem 7.5 does.

7.5 CONSOLIDATION. (The singular BIC by hand.) In the worked example of Section 7.4, derive from Definition 8.7 the quadratic equation for $L'(\mathcal{M}_1)$ and the stated difference of the sBIC values, and show that at $n = 1000$ the sBIC prefers $\mathcal{M}_1$ once $T > 18.24$, while BIC with the dimension 5 needs $T > 34.54$.

7.6 BY HAND. (A second hidden unit.) Prove part (ii) of Corollary 7.4 in detail for $(M, N) = (2, 3)$, where $\lambda = \frac{5}{2}$ and $m = 1$, and check the

value against Theorem 7.5. Which true parameters are deepest when $C_0 = e_1 e_1^\top$?

7.7 BY HAND. ★ (Two tanh units.) Let $f(x) = a_1 \tanh(b_1 x) + a_2 \tanh(b_2 x)$, $x \sim \mathrm{Unif}[-1,1]$, the truth 0, and $S_p = a_1 b_1^p + a_2 b_2^p$. (a) Show that near the origin K is comparable to $S_1^2 + S_3^2$. (b) Show that $S_3 - b_1^2 S_1 = a_2 b_2 (b_2^2 - b_1^2)$, so that $S_1^2 + S_3^2$ is comparable to $S_1^2 + a_2^2 b_2^2 (b_2^2 - b_1^2)^2$. (c) In the chart $(b_1, b_2) = (t, ts)$ of the blow-up of the b-plane, with Jacobian $|t|$, and with $a_1' = a_1 + a_2 s$, show that the loss becomes comparable to $t^2 a_1'^2 + t^6 a_2^2$ away from $s \in \{0, \pm 1\}$, and compute the volume of $\{t^2 a_1'^2 + t^6 a_2^2 < \varepsilon\}$ with the weight $|t|$ to find $\lambda = \frac{2}{3}$ and $m = 1$ at the origin, after checking that the directions $s \in \{0, \pm 1\}$ and the other chart change nothing. *(Hint: For (a), $S_{p+4} = (b_1^2 + b_2^2) S_{p+2} - b_1^2 b_2^2 S_p$, and the Gram matrix of x and x^3 under* $\mathrm{Unif}[-1,1]$ *is positive definite. For (c), split the t-range at $\varepsilon^{1/2}$ and $\varepsilon^{1/6}$.)*

7.8 BY HAND. (Two Bernoullis.) The mixture $(1-a)\,\mathrm{Bern}(p_1) + a\,\mathrm{Bern}(p_2)$ with the uniform prior on $[0,1]^3$ is fitted to $\mathrm{Bern}(q_0)$ data, $0 < q_0 < 1$. Show that the model is $\mathrm{Bern}(q)$ with $q = (1-a)p_1 + a p_2$, that the true set is a smooth surface meeting the interior of the cube, and that $\lambda = \frac{1}{2}$ and $m = 1$, against $\frac{d}{2} = \frac{3}{2}$.

7.9 BY HAND. (Hessians.) For $K = \frac{1}{2}\|ba^\top - C_0\|_F^2$ with $C_0 = b_0 a_0^\top \neq 0$, compute the Hessian at (a_0, b_0) and show that its kernel is spanned by $(-a_0, b_0)$. At the zero truth compute the Hessians at the origin and at $(a^*, 0)$, and compare their ranks with the local learning coefficients.

7.10 VARIATION. (A Gaussian prior.) Let $a \sim \mathcal{N}(0, I_N)$ and $b \sim \mathcal{N}(0, I_M)$ be independent. Show that $\zeta(z) = 2^z \Gamma(z + \frac{N}{2}) \Gamma(z + \frac{M}{2}) / (\Gamma(\frac{N}{2}) \Gamma(\frac{M}{2}))$, so that λ and m are those of Theorem 7.2, and find all the poles. For which M, N do later poles collide?

7.11 VARIATION. (An upper bound from sub-models.) Write BA as the sum of the rank-one terms $b_h a_h^\top$, $h = 1, \dots, H$, show that the squared Frobenius norm of BA is at most $H \sum_h |a_h|^2 |b_h|^2$, and deduce from the sum and comparison rules that at the zero truth $\lambda \leq \frac{H}{2} \min(M, N)$. Compare with Theorem 7.5 for $M = N = 3$ and $H = 2, 3$.

7.12 MEASUREMENT. (How close is the singularity?) Replace the rank-one truth $\frac{1}{2} e_1 e_1^\top$ of the script `07-catalogue-rrr.py` by $\sigma e_1 e_1^\top$ with $\sigma = 0.25$ and $\sigma = 0.9$, and rerun the case $H = 1$. Explain the size and sign of the deviation from $\frac{5}{2}$ at $n = 10^3$ and 10^4 in terms of $n\beta_1\sigma^2$ and of the range $\log \frac{1}{\sigma}$ of $\log t$ over which the true factorizations $(a/t, tb)$ stay inside $[-1,1]^6$, and what happens at $\sigma = 0.9$, where that range is short.

7.13 RESEARCH. (The singular BIC for deep linear networks.) For deep linear networks of a fixed depth $L \geq 3$ from N inputs to M outputs with all hidden widths equal to $h \leq \min(M, N)$, the models $h = 0, 1, 2, \dots$ are nested: they are the sets of matrices of rank at most h. Using Theorem 7.6, write down λ_{ij} and m_{ij} for choosing h, and check

whether they satisfy the conditions under which Drton and Plummer (2017) prove the consistency of the singular BIC, in particular that the complexities increase with the model. Why does choosing the depth at a fixed width not fit their framework? As far as we know this has not been examined.

7.14 RESEARCH. (Two nonlinear layers.) For $y = a\tanh(b\tanh(cx))$ with $x \sim \mathrm{Unif}[-1,1]$, the truth 0 and a box prior, show by comparison that $\lambda = \frac{1}{2}$ and $m = 3$, as for the chain. Then take two units in each of two hidden layers and find λ; we know of no exact value for a network with two nonlinear hidden layers of more than one unit each.

Notes and References

Reduced-rank regression goes back to Anderson (1951), who estimated linear restrictions on the coefficients of a multivariate normal regression. Its learning coefficients were computed by Aoyagi and Watanabe (2005); Watanabe (2013) used them, with their multiplicities, to test WBIC and the two-temperature estimator, and Drton and Plummer (2017) in the first experiment with the singular BIC, the choice of a rank. Theorem 7.2, with its proof by factorizing the zeta function, and Proposition 7.3, with its proof by normal form and homogeneity, are the book's own presentations; the reduction is the method of Aoyagi's papers written out for this case, and Corollary 7.7 is our consequence of Aoyagi's theorem.

Watanabe's first papers already treated layered networks (Watanabe 2001), and his books (Watanabe 2009, 2018) collect the general method. Aoyagi then computed one family after another: three-layer networks and normal mixtures through singularities of Vandermonde type (Aoyagi 2010a, 2019; Aoyagi and Nagata 2012), restricted Boltzmann machines (Aoyagi 2010b), deep linear networks (Aoyagi 2024), and, in a preprint, ReLU networks (Aoyagi 2025). In algebraic statistics, Rusakov and Geiger (2005), Zwiernik (2011) and Drton et al. (2017) computed marginal likelihood asymptotics for Bayesian networks and latent trees, and Drton et al. (2025) for factor analysis.

The deep-learning literature uses the catalogue as ground truth. Lau et al. (2025) validated their estimator of the local learning coefficient on deep linear networks against Aoyagi's values, which Hitchcock and Hoogland (2025, a preprint) point out are only lower bounds for the local values at generic trained parameters; Chapter 13 returns to both, and Chapter 11 to the two-temperature estimator.

TRANSLATING THE LITERATURE. Watanabe (2013) writes $x \in \mathbb{R}^M$ for the input and $y \in \mathbb{R}^N$ for the output, with A of size $H \times M$ and B of size $N \times H$, so his M and N are our N and M, and his true rank H_0 is our r. Drton and Plummer (2017) call the responses N and the covariates M, also the reverse of ours, and write $\lambda_i(\pi_0)$, $m_i(\pi_0)$ for our λ and m of model i at the truth π_0 and λ_{ij} for its generic value at truths of rank j; the formula of Theorem 7.5 is symmetric in M and N, so the swaps do not change it. Aoyagi (2025) writes c_{w_0} and θ for our local λ and m, $H^{(1)}, \dots, H^{(L+1)}$ for our widths in

the order output to input, $\mathcal{M}$ for our Γ, and a for our p. Rusakov and Geiger (2005) write N for the sample size and n for the number of features, our n and k, and expand the log marginal likelihood, $-F_n$, so that their coefficient of $-\log N$ is our λ.

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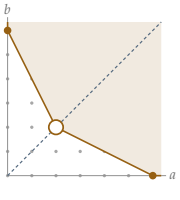
Part III

Watanabe's Theorems

With the learning coefficient in hand, the classical formulas can be rewritten for singular models: the free energy, the generalization error, the information criteria, and the response of the posterior to temperature. The posterior does not settle on one point; it moves between singularities as data arrive.

$K = a^6 + a^2b^2 + b^6$

$\lambda = \frac{1}{2}$



8 The Free Energy Formula

Model selection without the Laplace approximation

THE MODEL — $y = ax_1 + b^3x_2$

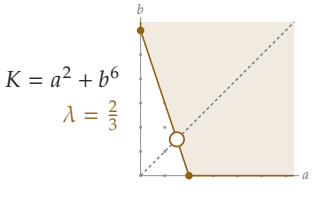
IN THIS CHAPTER. Bayesian model selection compares free energies, the negative logarithms of marginal likelihoods, and for regular models Schwarz's formula charges $\frac{1}{2}\log n$ per parameter. For the two-parameter model $y = ax_1 + b^3x_2$ we compute by hand that the volume of nearly true parameters is exactly $1.4457\,\varepsilon^{2/3}$, that the free energy without noise is $\frac{2}{3}\log n - 0.266$, and that the noise in the energy is at most its square root times one random factor, so that it shifts the free energy by a bounded amount. This sandwich proves $F_n = nS_n + \lambda \log n - (m-1)\log\log n + O_p(1)$ for every model linear in a coefficient matrix, given its volume function; Watanabe's theorem extends it to analytic models by resolution of singularities. Exact quadrature for five models gives slopes 0.998, 0.749, 0.667, 0.491 against $\lambda = 1, \frac{3}{4}, \frac{2}{3}, \frac{1}{2}$ for the four with $m = 1$, and 0.510 for the product model once its $\log\log n$ is added back; a singular BIC that uses learning coefficients finds the rank of a reduced-rank regression where BIC does not. The formula is for a fixed model and truth, and its bounded remainder holds all it leaves out.

8.1 The question: what a singular model pays

Bayesian model selection compares models by their marginal likelihoods: the probability that each model assigns to the data once its parameters have been integrated out against the prior. For neural networks this is the evidence with which MacKay (1992) chose weight decays and compared architectures of backpropagation networks, and which the Laplace approximation, revived for modern networks by Daxberger et al. (2021), still estimates: expand the log-likelihood to second order at a trained parameter and integrate the Gaussian. In this book the negative logarithm of the marginal likelihood is the *free energy*

$$F_n = -\log \int_W \varphi(w) \prod_{i=1}^{n} p(X_i \mid w)\, \mathrm{d}w = -\log \int_W \varphi(w) e^{-nL_n(w)}\, \mathrm{d}w.$$

The logarithm of the ratio of two marginal likelihoods, the log *Bayes factor*, is the difference of the two free energies (Kass and Raftery 1995). Since $L_n(w) = S_n + K_n(w)$ in the realizable case, the term nS_n is common to all models of the same data, and model comparison compares the *normalized*



free energies

$$F_n - nS_n = -\log \int_W \varphi(w) e^{-nK_n(w)}\, \mathrm{d}w.$$

THE REGULAR ANSWER · the free energy. If the truth is regular for a model with d parameters, then $F_n = nL_n(\hat{w}) + \frac{d}{2}\log n + O_p(1)$, with $\hat{w}$ the maximum-likelihood estimate (Haughton 1988; Schwarz 1978); Laplace's method, Theorem 1.4, identifies the bounded term. For a realizable truth $nS_n - nL_n(\hat{w}) = O_p(1)$, so the data charge the model $\frac{1}{2}\log n$ per parameter: $F_n - nS_n = \frac{d}{2}\log n + O_p(1)$.

The $\frac{d}{2}$ has a simple origin (Chapter 1):[1] near a nondegenerate minimum the posterior is a Gaussian of width $n^{-1/2}$ in each of the d directions, and each direction removes a factor $n^{-1/2}$ from the prior mass that the posterior keeps. The model of this chapter breaks that picture with two parameters. Fit $y = ax_1 + b^3x_2 + \varepsilon$, with inputs $x \sim \mathcal{N}(0, I_2)$, unit noise and the uniform prior on $[-1,1]^2$, to data whose truth is $(a,b) = (0,0)$, so that y is pure noise. The loss is

$$K(a,b) = \tfrac{1}{2}\mathbb{E}_x\left[(ax_1 + b^3x_2)^2\right] = \tfrac{1}{2}\left(a^2 + b^6\right),$$

whose Hessian at the truth is $\mathrm{diag}(1,0)$: the Fisher information is singular, Laplace's term $\frac{1}{2}\log\det J$ is $-\infty$, and the posterior is not Gaussian. In the direction of b the data see only b^3, so they locate b to within $n^{-1/6}$ rather than $n^{-1/2}$. Call this the *sextic model*. Its coefficient vector $c = (a, b^3)$ runs over the same square $[-1,1]^2$ as the coefficient vector (a,b) of the regular plane $y = ax_1 + bx_2$: the two models are the same set of distributions, with the same maximized likelihood and the same number of parameters, and Schwarz's formula cannot tell them apart. Their marginal likelihoods differ.

Chapter 3 computed the free energy of models whose loss depends on a single coefficient, and Chapter 4 the *population free energy*

$$\bar{F}(s) = -\log \int_W \varphi(w) e^{-sK(w)}\, \mathrm{d}w, \qquad s > 0,$$

in which the empirical K_n is replaced by its mean: if the volume function $V(\varepsilon) = \int_{K<\varepsilon} \varphi$ behaves like $C\varepsilon^{\lambda}(\log\frac{1}{\varepsilon})^{m-1}$, then $\bar{F}(n) = \lambda\log n - (m-1)\log\log n + O(1)$ by the Abelian lemma, Lemma 4.3. What is missing is the data. The empirical loss K_n fluctuates around K, and where the posterior lives, $K \approx n^{-1}$, the fluctuation is as large as K itself. Does the fluctuation change the coefficients of the free energy, that is, is $F_n - nS_n = \lambda\log n - (m-1)\log\log n + O_p(1)$ for singular models in general? And what does the answer do to model selection?

[1] Schwarz derived the criterion in 1978 for exponential families and linear submodels; Haughton (1988) extended the expansion to curved families and showed that a criterion built from its leading terms selects the true model with probability tending to one.

8.2 By hand: the sextic model with and without noise

We first remove the noise and compute the population free energy of the sextic model from its volume function, then add the noise back and bound its effect.

COMPUTATION 8.1 · THE POPULATION FREE ENERGY OF $y = ax_1 + b^3x_2$

1 The truth is $c_0 = 0$, the inputs are independent with unit variance, and so $K(a,b) = \frac{1}{2}(a^2+b^6)$. The set of true parameters is the single point $(0,0)$, at which the Hessian $\operatorname{diag}(1,0)$ is singular: the pair is singular in the sense of this book although W_0 is a point.

2 For $0 < \varepsilon \le \frac{1}{2}$ the set $\{K < \varepsilon\} = \{|b| < (2\varepsilon)^{1/6},\ |a| < \sqrt{2\varepsilon - b^6}\}$ lies inside the square, and the substitution $b = (2\varepsilon)^{1/6}v$ shows that its area $\int 2\sqrt{2\varepsilon - b^6}\,\mathrm{d}b$ is $2(2\varepsilon)^{2/3}I_6$, with

$$I_6 = \int_{-1}^{1} \sqrt{1-v^6}\,\mathrm{d}v = \frac{\Gamma(\frac{1}{6})\Gamma(\frac{3}{2})}{3\,\Gamma(\frac{5}{3})} = 1.8215$$

by the substitution $u = v^6$, which turns I_6 into $\frac{1}{3}B(\frac{1}{6},\frac{3}{2})$. With the prior density $\frac{1}{4}$,

$$V(\varepsilon) = 2^{-1/3}I_6\,\varepsilon^{2/3} = 1.4457\,\varepsilon^{2/3} \qquad (0 < \varepsilon \le \tfrac{1}{2}).$$

3 The volume is an exact power: $\lambda = \frac{2}{3}$ and $m = 1$. The coefficient a contributes $\frac{1}{2}$, as a regular parameter does, and b contributes $\frac{1}{6}$, because $\{b^6 < \varepsilon\}$ has length $2\varepsilon^{1/6}$. The Newton diagram at the foot of the chapter's first page gives the same number: its boundary through $(0,6)$ and $(2,0)$ meets the diagonal at $t = \frac{3}{2}$, and $1/t = \frac{2}{3}$ (Theorem 6.4).

4 The partition function factorizes, $\int e^{-nK}\varphi = Z_aZ_b$ with $Z_a = \frac{1}{2}\int_{-1}^{1} e^{-na^2/2}\,\mathrm{d}a$ and $Z_b = \frac{1}{2}\int_{-1}^{1} e^{-nb^6/2}\,\mathrm{d}b$. Extending both integrals to the whole line changes them by a relative error below $e^{-n/2}$ for $n \ge 2$, and the substitutions $a = (2/n)^{1/2}u$, $b = (2/n)^{1/6}v$ give $Z_a = \sqrt{\pi/(2n)}$ and $Z_b = \Gamma(\frac{7}{6})(2/n)^{1/6}$. Hence, up to an error below $3e^{-n/2}$,

$$\bar{F}(n) = \tfrac{2}{3}\log n - \log\Big(\sqrt{\tfrac{\pi}{2}}\,2^{1/6}\,\Gamma(\tfrac{7}{6})\Big) = \tfrac{2}{3}\log n - 0.2663,$$

in agreement with Lemma 4.3, which predicts the constant $-\log(C\Gamma(\lambda+1)) = -\log(1.4457 \times 0.9027) = -0.2663$. At $n = 10^4$ the sextic model pays 5.874; the regular plane, whose population free energy is $\log n - \log\frac{\pi}{2}$, pays 8.759.

Now the data. Write $\hat{\Sigma}_n = \frac{1}{n}\sum_i x_i x_i^{\top}$ for the empirical second moment of the inputs and $z_n = n^{-1/2}\sum_i x_i y_i$; at the truth $c_0 = 0$ the outputs are the noise, $y_i = \varepsilon_i$.

COMPUTATION 8.2 · THE NOISE IS AT MOST THE SQUARE ROOT OF THE ENERGY

1 Expanding the squares in $nK_n(w) = \sum_i \left[\frac{1}{2}(y_i - c^\top x_i)^2 - \frac{1}{2}y_i^2\right]$ with $c = (a, b^3)$ gives

$$nK_n(w) = t(w) - \sqrt{n}\, c^\top z_n, \qquad t(w) = \tfrac{n}{2}\, c^\top \hat{\Sigma}_n c.$$

The first term is the energy of w on this design, a deterministic function of w once the inputs are fixed; the second is the noise.

2 Given the inputs, $z_n = n^{-1/2} \sum_i x_i \varepsilon_i$ is Gaussian with covariance $\hat{\Sigma}_n$. So $\xi = \hat{\Sigma}_n^{-1/2} z_n$ is a standard Gaussian vector in $\mathbb{R}^2$, and $\rho^2 = |\xi|^2$ has the χ^2_2 law whatever n is.

3 Since $\sqrt{n}\, c^\top z_n = (\sqrt{n}\, \hat{\Sigma}_n^{1/2} c)^\top \xi$ and $|\sqrt{n}\, \hat{\Sigma}_n^{1/2} c|^2 = 2t(w)$, writing this inner product with the angle $\theta(w)$ between $\hat{\Sigma}_n^{1/2} c$ and ξ gives

$$nK_n(w) = t(w) - \sqrt{2t(w)}\, \rho \cos\theta(w).$$

Since $|\cos\theta| \le 1$, which is the Cauchy–Schwarz inequality, the noise is at most the square root of the energy, times one random number ρ that does not depend on w.

4 For $\eta \in (0, 1)$, the inequality $2uv \le \eta u^2 + v^2/\eta$ with $u = \sqrt{t}$ and $v = \rho/\sqrt{2}$ gives $\sqrt{2t}\rho \le \eta t + \rho^2/(2\eta)$, so that for every w at once

$$(1 - \eta)\, t(w) - \frac{\rho^2}{2\eta} \le nK_n(w) \le (1 + \eta)\, t(w) + \frac{\rho^2}{2\eta}.$$

5 Let $\Phi_n(\sigma) = -\log \int e^{-\sigma t(w)} \varphi(w)\, \mathrm{d}w$, the free energy of noiseless data with the same inputs. Exponentiating, integrating against φ and taking $-\log$ gives

$$\Phi_n(1 - \eta) - \frac{\rho^2}{2\eta} \le F_n - nS_n \le \Phi_n(1 + \eta) + \frac{\rho^2}{2\eta}.$$

For the typical design $\hat{\Sigma}_n = I_2$, Φ_n is the population free energy: $\Phi_n(\sigma) = \bar{F}(\sigma n) = \frac{2}{3}\log(\sigma n) - 0.266$.

6 With $n = 10^4$, $\eta = \frac{1}{2}$, $\hat{\Sigma}_n = I_2$ and $\rho^2 = 2$, its mean: $3.41 \le F_n - nS_n \le 8.14$. The band has width $\frac{2}{3}\log 3 + 4 = 4.73$ at every n, and 3.96 at the best $\eta = 0.775$; the mean of $F_n - nS_n$ over 200 simulated data sets at $n = 10^4$ is 5.17 (Section 8.5).

WHAT ONE HAD TO NOTICE. Two things. The volume of nearly true parameters is an exact power, because the loss is a sum of monomials in separate variables; this gives the population free energy, as in Chapter 4. And the noise enters the energy linearly in the coefficient vector, so that Cauchy–Schwarz bounds it, uniformly in w, by the square root of the quadratic part times one random factor whose law is fixed. Where the

energy is large its square root is negligible, and where it is small both are of order one: the noise moves the free energy by an additive amount but cannot change how fast the volume of $\{t < \varepsilon\}$ shrinks. For the product model of Chapter 3, whose coefficient is a scalar, the angle is 0 or π: $nK_n = t \mp \sqrt{2t}\,\rho$.

THE EXACT FREE ENERGY. The sandwich is crude, but it is the argument that generalizes. For linear models one can also complete the square. With $\hat{c} = \hat{\Sigma}_n^{-1} z_n/\sqrt{n}$, the least-squares coefficient,[2]

$$F_n - nS_n = -\frac{\rho^2}{2} - \log \int \exp\Big(-\frac{n}{2}(c-\hat{c})^{\top}\hat{\Sigma}_n(c-\hat{c})\Big)\varphi(w)\,\mathrm{d}w. \qquad (8.1)$$

[2] The square $\rho^2 = n\,\hat{c}^{\top}\hat{\Sigma}_n\hat{c}$ is twice the log-likelihood ratio of the unrestricted least-squares fit against the truth, the statistic of Wilks's likelihood-ratio test; see Lemma 8.3 below.

The integral is a Gaussian of width $n^{-1/2}$ centered at $\hat{c}$, averaged against the law of $c = (a, b^3)$ under the prior, whose density is $\frac{1}{12}|c_2|^{-2/3}$ on $[-1,1]^2$: the computation of Chapter 3 in two coordinates. Rescaling $u = \sqrt{n}\,c$ shows (Exercise 8.6) that $R_n = F_n - nS_n - \frac{2}{3}\log n$ converges in law to

$$\log 12 - \tfrac{1}{2}\log(2\pi) - \tfrac{1}{2}|\zeta|^2 - \log H(\zeta_2), \qquad H(z) = \int_{\mathbb{R}} e^{-(t-z)^2/2}|t|^{-2/3}\,\mathrm{d}t, \quad (8.2)$$

with $\zeta \sim \mathcal{N}(0, I_2)$; its mean is -0.991. For the regular plane the same rescaling gives Laplace's formula exactly, $R_n \to -\log\frac{\pi}{2} - \frac{1}{2}|\zeta|^2$, with mean -1.452.

8.3 The theorem: models linear in their coefficients

The computation used three properties of the sextic model: the log-likelihood ratio is a quadratic function of a finite coefficient vector; its random part is linear in the same vector, with Gaussian coefficients whose law does not depend on n; and the volume function has a known asymptotic form. Every model whose mean is linear in a coefficient matrix has the first two properties, and the third is what Part II computes.

Assumption 8.1 (Linear in the coefficients). The data are independent pairs (x_i, y_i), $i = 1, \dots, n$, with inputs $x_i \in \mathbb{R}^N$ distributed as x, where $\mathbb{E}\|x\|^2 < \infty$ and $\Sigma = \mathbb{E}[xx^{\top}]$ is positive definite, and outputs $y_i = C_0 x_i + \varepsilon_i \in \mathbb{R}^M$ with noise $\varepsilon_i \sim \mathcal{N}(0, I_M)$ independent of x_i. The model is $y = G(w)x + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, I_M)$, for a measurable map $G\colon W \to \mathbb{R}^{M\times N}$ on a measurable set $W \subseteq \mathbb{R}^d$, with a prior probability density φ on W. The truth is realizable, $C_0 = G(w_0)$ for some $w_0 \in W$. With $\Delta(w) = G(w) - C_0$, the volume function of

$$K(w) = \tfrac{1}{2}\operatorname{tr}\big(\Delta(w)\,\Sigma\,\Delta(w)^{\top}\big)$$

satisfies $V(\varepsilon) = \int_{\{K<\varepsilon\}} \varphi(w)\,\mathrm{d}w \sim C\varepsilon^{\lambda}(\log\frac{1}{\varepsilon})^{m-1}$ as $\varepsilon \to 0$, for some $C > 0$, $\lambda > 0$ and integer $m \geq 1$.

A scalar output is the case $M = 1$, $G(w) = g(w)^{\top}$, which gives the models $y = g(w)^{\top}x + \varepsilon$; the five running models of Section 8.5 are of this form, and rank-one and reduced-rank regression, with $G(w) = ba^{\top}$ or BA, are of the

matrix form. The law of the inputs is the same for every w, so it cancels from every likelihood ratio, and we write $L_n(w) = -\frac{1}{n}\sum_i \log p(y_i \mid x_i, w)$ and $S_n = L_n(w_0)$. The function K is the Kullback–Leibler divergence: $\mathbb{E}\left[\frac{1}{2}\|\varepsilon - \Delta x\|^2 - \frac{1}{2}\|\varepsilon\|^2\right] = \frac{1}{2}\operatorname{tr}(\Delta\Sigma\Delta^\top)$, because ε is independent of x with mean 0. The data enter through

$$\hat\Sigma_n = \frac{1}{n}\sum_{i=1}^n x_i x_i^\top, \qquad Z_n = \frac{1}{\sqrt n}\sum_{i=1}^n \varepsilon_i x_i^\top \in \mathbb{R}^{M\times N}, \qquad \rho_n^2 = \operatorname{tr}\big(Z_n \hat\Sigma_n^{-1} Z_n^\top\big),$$

the last when $\hat\Sigma_n$ is invertible. Expanding the squares as in Computation 8.2 gives, with $\Delta = \Delta(w)$,

$$nK_n(w) = \tfrac{n}{2}\operatorname{tr}\big(\Delta\hat\Sigma_n\Delta^\top\big) - \sqrt n \operatorname{tr}\big(\Delta Z_n^\top\big), \tag{8.3}$$

because $\sum_i \varepsilon_i^\top \Delta x_i = \operatorname{tr}(\Delta \sum_i x_i \varepsilon_i^\top)$.

Lemma 8.2 (The sandwich). *Let $P \in \mathbb{R}^{N\times N}$ be positive definite, $Z \in \mathbb{R}^{M\times N}$, $s > 0$ and $\eta \in (0,1)$. For $\Delta \in \mathbb{R}^{M\times N}$ put $t(\Delta) = \frac{s}{2}\operatorname{tr}(\Delta P \Delta^\top)$, and let $\rho^2 = \operatorname{tr}(ZP^{-1}Z^\top)$. Then for every Δ*

$$(1-\eta)\,t(\Delta) - \frac{\rho^2}{2\eta} \le t(\Delta) - \sqrt s \operatorname{tr}(\Delta Z^\top) \le (1+\eta)\,t(\Delta) + \frac{\rho^2}{2\eta}.$$

Proof. The bilinear form $\langle A, B\rangle = \operatorname{tr}(APB^\top)$ on $\mathbb{R}^{M\times N}$ is an inner product: $\langle A, A\rangle = \sum_j a_j^\top P a_j > 0$ for $A \neq 0$, with $a_j^\top$ the rows of A. In it, $\operatorname{tr}(\Delta Z^\top) = \operatorname{tr}(\Delta P P^{-1} Z^\top) = \langle \Delta, ZP^{-1}\rangle$, and $\langle ZP^{-1}, ZP^{-1}\rangle = \operatorname{tr}(ZP^{-1}Z^\top) = \rho^2$. The Cauchy–Schwarz inequality gives

$$\sqrt s\,|\operatorname{tr}(\Delta Z^\top)| \le \sqrt{s\langle\Delta,\Delta\rangle}\,\rho = \sqrt{2t(\Delta)}\,\rho,$$

and $2uv \le \eta u^2 + v^2/\eta$ with $u = \sqrt{t(\Delta)}$, $v = \rho/\sqrt 2$ bounds the right side by $\eta\,t(\Delta) + \rho^2/(2\eta)$. Subtract the bound from $t(\Delta)$, and add it, to obtain the two inequalities. ∎

Lemma 8.3 (The noise statistic). *Under Assumption 8.1, conditionally on inputs $x_1, \dots, x_n$ for which $\hat\Sigma_n$ is invertible, ρ_n^2 has the χ^2_{MN} law. Moreover $\rho_n^2 = 2\big(nL_n(w_0) - \min_C nL_n(C)\big)$, where the minimum is over all matrices $C \in \mathbb{R}^{M\times N}$ and $L_n(C)$ is the loss of the coefficient matrix C.*

Proof. Write ε_{ij} for the coordinates of ε_i. Given the inputs, the jth row of $\sqrt n Z_n$ is $u_j^\top = \sum_i \varepsilon_{ij} x_i^\top$, a Gaussian vector with mean 0 and covariance $\sum_i x_i x_i^\top = n\hat\Sigma_n$, and the M rows are independent because the coordinates ε_{ij} are. So the vectors $(n\hat\Sigma_n)^{-1/2}u_j$ are independent standard Gaussians in $\mathbb{R}^N$, and $\rho_n^2 = \sum_{j=1}^M u_j^\top (n\hat\Sigma_n)^{-1} u_j$ is a sum of MN independent squared standard Gaussians. For the second claim, (8.3) holds for every matrix C in place of $G(w)$, with $\Delta = C - C_0$; its right side is a convex quadratic function of Δ whose gradient $n\Delta\hat\Sigma_n - \sqrt n Z_n$ vanishes at $\hat\Delta = Z_n\hat\Sigma_n^{-1}/\sqrt n$, where it equals $\frac{1}{2}\rho_n^2 - \rho_n^2 = -\frac{1}{2}\rho_n^2$. ∎

Theorem 8.4 (The free energy formula for linear models). *Under Assumption 8.1,*

$$F_n - nS_n = \lambda \log n - (m-1)\log\log n + O_p(1).$$

More precisely, let $R_n = F_n - nS_n - \lambda\log n + (m-1)\log\log n$ *and* $\kappa = -\log\big(C\,\Gamma(\lambda+1)\big)$*, and for* $\eta \in (0,1)$ *let* $A_n(\eta)$ *be the event* $(1-\eta)\Sigma \le \hat\Sigma_n \le (1+\eta)\Sigma$*. Then* $\mathbb{P}(A_n(\eta)) \to 1$*, and there are numbers* $\epsilon_n(\eta) \to 0$ *such that on* $A_n(\eta)$

$$2\lambda\log(1-\eta) - \frac{\rho_n^2}{2\eta} - \epsilon_n(\eta) \le R_n - \kappa \le 2\lambda\log(1+\eta) + \frac{\rho_n^2}{2\eta} + \epsilon_n(\eta), \quad (8.4)$$

where ρ_n^2 *has the* χ^2_{MN} *law given the inputs (Lemma 8.3).*

Proof. The plan is to sandwich the random energy nK_n between two multiples of the population energy nK, up to the additive random constant $\rho_n^2/(2\eta)$, and to read the two population free energies off the Abelian lemma.

The event. The inputs have finite second moments, so $\hat\Sigma_n \to \Sigma$ almost surely by the strong law of large numbers. The event $A_n(\eta)$ says that the eigenvalues of $\Sigma^{-1/2}\hat\Sigma_n\Sigma^{-1/2}$ lie in $[1-\eta, 1+\eta]$, and since this matrix tends to I_N, the event holds for all large n almost surely; in particular $\mathbb{P}(A_n(\eta)) \to 1$. On $A_n(\eta)$ the matrix $\hat\Sigma_n$ is invertible.

The energies. For a positive semidefinite matrix Q and every Δ, $\operatorname{tr}(\Delta Q\Delta^\top) = \sum_j \delta_j^\top Q\delta_j \ge 0$, with $\delta_j^\top$ the rows of Δ. Applied to $Q = \hat\Sigma_n - (1-\eta)\Sigma$ and to $Q = (1+\eta)\Sigma - \hat\Sigma_n$, this shows that on $A_n(\eta)$ the energy $t(w) = \frac{n}{2}\operatorname{tr}(\Delta(w)\hat\Sigma_n\Delta(w)^\top)$ satisfies

$$(1-\eta)\,nK(w) \le t(w) \le (1+\eta)\,nK(w) \qquad \text{for every } w.$$

The sandwich. By (8.3), $nK_n(w) = t(w) - \sqrt{n}\operatorname{tr}(\Delta(w)Z_n^\top)$, and Lemma 8.2 with $P = \hat\Sigma_n$, $Z = Z_n$, $s = n$ bounds it between $(1-\eta)t(w) - \rho_n^2/(2\eta)$ and $(1+\eta)t(w) + \rho_n^2/(2\eta)$. Combined with the previous step, on $A_n(\eta)$

$$(1-\eta)^2 nK(w) - \frac{\rho_n^2}{2\eta} \le nK_n(w) \le (1+\eta)^2 nK(w) + \frac{\rho_n^2}{2\eta} \qquad \text{for every } w.$$

Integration. Exponentiating, integrating against φ and taking $-\log$ turns these bounds into bounds on $F_n - nS_n = -\log\int e^{-nK_n}\varphi$:

$$\bar F\big((1-\eta)^2 n\big) - \frac{\rho_n^2}{2\eta} \le F_n - nS_n \le \bar F\big((1+\eta)^2 n\big) + \frac{\rho_n^2}{2\eta}.$$

The Abelian lemma. The prior φ is a probability density and $K \ge 0$, so Lemma 4.3 applies to the hypothesis on V: $\bar F(s) = \lambda\log s - (m-1)\log\log s + \kappa + o(1)$ as $s \to \infty$ through the real numbers. For fixed $\sigma > 0$, $\log\log(\sigma n) - \log\log n = \log\big(1 + \log\sigma/\log n\big) \to 0$, and therefore

$$\bar F(\sigma n) = \lambda\log n - (m-1)\log\log n + \lambda\log\sigma + \kappa + o(1).$$

Apply this with $\sigma = (1 \mp \eta)^2$ and let $\epsilon_n(\eta)$ be the larger of the absolute values of the two $o(1)$ terms; subtracting $\lambda \log n - (m-1)\log\log n + \kappa$ from the bounds of the previous step gives (8.4).

Boundedness in probability. Take $\eta = \frac{1}{2}$, so that (8.4) gives $|R_n - \kappa| \le 2\lambda \log 2 + \rho_n^2 + \epsilon_n(\frac{1}{2})$ on $A_n(\frac{1}{2})$. Given $\delta > 0$, choose b_δ with $\mathbb{P}(\chi^2_{MN} > b_\delta) < \delta/2$, and n_0 such that $\mathbb{P}(A_n(\frac{1}{2})^c) < \delta/2$ and $\epsilon_n(\frac{1}{2}) \le 1$ for $n \ge n_0$. Since $A_n(\frac{1}{2})$ depends only on the inputs, Lemma 8.3 gives $\mathbb{P}(A_n(\frac{1}{2}) \cap \{\rho_n^2 > b_\delta\}) = \mathbb{E}\big[\mathbf{1}_{A_n(1/2)}\mathbb{P}(\rho_n^2 > b_\delta \mid x_1, \dots, x_n)\big] \le \mathbb{P}(\chi^2_{MN} > b_\delta)$. Hence $\mathbb{P}\big(|R_n - \kappa| > 2\lambda\log 2 + b_\delta + 1\big) < \delta$ for $n \ge n_0$, which is $R_n = O_p(1)$. ∎

The proof used little. Compactness of W, analyticity of G, an interior truth and a prior positive at the truth enter only through the volume hypothesis, where the geometry of Part II does its work; a Gaussian prior on $\mathbb{R}^d$ is allowed (Exercise 8.9). The Gaussian noise was used only in Lemma 8.3: for noise with mean zero and finite covariance, independent of x, (8.3) still holds, ρ_n^2 is bounded in probability by the central limit theorem, and the theorem holds with $nL_n(w_0)$ in place of nS_n. In one line, $F_n - nS_n = \bar{F}(n) + O_p(1)$: the data shift the free energy of a linear model by a bounded random amount. Watanabe states the same relation for analytic models, crediting it to his 2001 paper (Watanabe 2013, §7.3).[3]

[3] The proof is the book's own presentation of a standard idea; we have not found the sandwich in this form in the literature.

Corollary 8.5 (The running models). *With inputs $x \sim \mathcal{N}(0, I)$, unit noise, the truth 0 and the uniform prior on $[-1,1]^d$, $F_n - nS_n = \lambda \log n - (m-1)\log\log n + O_p(1)$ with $(\lambda, m) = (1,1)$ for the regular plane, $(\frac{3}{4}, 1)$ for the quartic model $y = ax_1 + b^2x_2$, $(\frac{2}{3}, 1)$ for the sextic model, $(\frac{1}{2}, 1)$ for the quartic cup $y = a^2x_1 + b^2x_2$, $(\frac{1}{2}, 2)$ for the product model $y = abx$, and $(\frac{1}{2}, L)$ for the chain $y = a_1 \cdots a_L x$.*

Proof. Each model is linear in its coefficients, so it suffices to check the volume hypothesis. For the sextic model it is Computation 8.1. For the regular plane $V(\varepsilon) = \frac{\pi}{2}\varepsilon$ for $\varepsilon \le \frac{1}{2}$; for the quartic model and the quartic cup V is an exact power for small ε, $1.4699\,\varepsilon^{3/4}$ and $1.3110\,\varepsilon^{1/2}$ (Exercise 8.2). For the chain, whose coefficient c has the density $(\log\frac{1}{|c|})^{L-1}/(2(L-1)!)$ on $[-1,1]$ (Chapter 3), $V(\varepsilon)$ is the integral of this density over $|c| < r = \sqrt{2\varepsilon}$; since $\int_0^r (\log\frac{1}{c})^{L-1}\,dc \sim r(\log\frac{1}{r})^{L-1}$ and $\log\frac{1}{r} \sim \frac{1}{2}\log\frac{1}{\varepsilon}$, $V(\varepsilon) \sim \frac{2^{1/2-(L-1)}}{(L-1)!}\,\varepsilon^{1/2}(\log\frac{1}{\varepsilon})^{L-1}$, which for $L = 2$ is the product model. ∎

8.4 The theorem: Watanabe's formula and model selection

For a model that is not linear in its coefficients, such as the tanh neuron or a mixture, the log-likelihood ratio is not a quadratic function of a finite vector, and its random part is not a single Gaussian vector. Watanabe's theorem says that after a change of coordinates both properties come back. Its hypotheses are stated with the log density ratio $f(x,w) = \log\big(p_0(x)/p(x \mid w)\big)$, so that $K(w) = \mathbb{E}_X[f(X,w)] = L(w) - L(w_0)$, and the condition that makes the random part small is the following: the pair has *relatively finite variance* if

there are $\varepsilon, c > 0$ with

$$\mathbb{E}_X\big[f(X,w)\big] \geq c\,\mathbb{E}_X\big[f(X,w)^2\big] \qquad \text{whenever } K(w) \leq \varepsilon.$$

The variance of $nK_n(w)$ is then at most $nK(w)/c$: the noise is at most the square root of the energy, as in Computation 8.2. Every realizable pair and every regular pair has relatively finite variance (Watanabe 2013, p. 872); for the sextic model $\mathbb{E}f = K$ and $\mathbb{E}f^2 = 2K + 3K^2$, so the constant $c = \frac{1}{5}$ works on the whole square (Exercise 8.4).

Theorem 8.6 (Watanabe's free energy formula). *Let the model $p(x \mid w)$, $w \in W \subseteq \mathbb{R}^d$, the prior φ and the truth q satisfy the following conditions.*

(i) Some w_0 in the interior of W minimizes L, and every minimizer of L gives the same density $p_0 = p(\cdot \mid w_0)$.

(ii) W is compact with nonempty interior, and $W = \{w : \pi_1(w) \geq 0, \ldots, \pi_k(w) \geq 0\}$ for analytic functions π_j.

(iii) $\varphi = \varphi_1\varphi_2$ with $\varphi_1 \geq 0$ analytic and $\varphi_2 > 0$ infinitely differentiable.

(iv) For some $s \geq 6$, the map $w \mapsto f(\cdot, w)$ extends to an analytic map from an open set $W' \supseteq W$ to $L^s(q)$.

(v) The pair has relatively finite variance.

Let λ and m be the learning coefficient and multiplicity of K with respect to φ. Then

$$F_n = nL_n(w_0) + \lambda \log n - (m-1)\log\log n + R_n,$$

with a remainder R_n that converges in law to a random variable as $n \to \infty$. If q is realizable, then $nL_n(w_0) = nS_n$.

This is Theorem 2 of Watanabe (2013), whose Fundamental Conditions (1)–(4) are our (ii)–(v) and whose standing assumptions are (i).[4] By Watanabe's account (Watanabe 2013, p. 876), the expectation $\mathbb{E}[F_n] = nS + \lambda \log n - (m-1)\log\log n + O(1)$ of the realizable case is in Watanabe (2001), the realizable case as a random variable in his 2009 book (§6.2), and the unrealizable case in Watanabe (2010a), which derives it from the limit of the rescaled posterior. The learning coefficient here is the real log canonical threshold of K with respect to φ; the two names denote the same number, which Watanabe defines through a resolution and Chapter 4 through the largest pole, $-\lambda$, of the zeta function (Theorem 5.3 connects the two).

[4] That paper is about WBIC (Chapter 11); it restates the free energy formula as the background result that WBIC is designed to estimate.

Proof sketch. The plan is the proof of Theorem 8.4, carried out on a parameter space on which K is a monomial. We sketch the statement with $O_p(1)$ in place of convergence in law, and say at the end what the convergence in law needs.

Localization. The process $\psi_n(w) = n^{-1/2}\sum_i (K(w) - f(X_i, w))$ (Watanabe's η_n) converges in law in the supremum norm (Theorems 5.9 and 5.10, as cited in Watanabe 2013), so $\sup_W |\psi_n| = O_p(1)$ and $nK_n \geq nK - \sqrt{n}\sup|\psi_n|$. On $\{K \geq \varepsilon\}$ the integrand e^{-nK_n} is therefore at most $e^{-n\varepsilon + O_p(\sqrt{n})}$, negligible against $n^{-\lambda}$ (Watanabe 2013, Lemma 1).

Resolution. On $\{K < \varepsilon\}$, Hironaka's theorem (Hironaka 1964), in the form of Theorem A.8, gives a manifold $\mathcal{M}$ and a proper analytic map $g\colon \mathcal{M} \to W$

such that in local coordinates u

$$K(g(u)) = u^{2k}, \qquad f(x, g(u)) = u^k a(x, u), \qquad \varphi(g(u))|g'(u)| = b(u)|u^h|,$$

with multi-indices k, h, an $L^s(q)$-valued analytic a, and a smooth $b > 0$ (Watanabe 2013, Theorem 1, the standard representation). The middle identity is where relatively finite variance enters: $\mathbb{E} f^2 \le \mathbb{E} f/c = u^{2k}/c$ forces f to be divisible by u^k.

The empirical process. With $\xi_n(u) = n^{-1/2} \sum_i \big(u^k - a(X_i, u)\big)$ and $t = nu^{2k}$,

$$nK_n(g(u)) = nu^{2k} - \sqrt{n}\, u^k \xi_n(u) = t - \sqrt{t}\, \xi_n(u) \cdot \operatorname{sgn}(u^k),$$

and ξ_n converges in law to a Gaussian process with $\mathbb{E} \sup \xi_n^2 \to \mathbb{E} \sup \xi^2$ (Theorems 6.1–6.3, as cited in Watanabe 2013); see Theorem A.11. The noise is again at most the square root of the energy, times $\sup_u |\xi_n(u)| = O_p(1)$; in the linear models $\xi_n(u)$ was $\sqrt{2}\, \rho_n \cos\theta$.

The sandwich. Since $\sqrt{t}|\xi_n(u)| \le \eta t + \sup \xi_n^2/(4\eta)$, the argument of Theorem 8.4 bounds $\int_{K<\varepsilon} e^{-nK_n}\varphi$ above and below by $e^{\pm \sup \xi_n^2/(4\eta)}$ times the population integrals at $(1 \mp \eta)n$, chart by chart. On each chart the volume function is that of a monomial with the weight $b(u)|u^h|$, so $V(\varepsilon) \sim C\varepsilon^\lambda (\log \frac{1}{\varepsilon})^{m-1}$ (Proposition 4.4 and Theorem 5.3), and the Abelian lemma gives $F_n - nL_n(w_0) = \lambda \log n - (m-1)\log\log n + O_p(1)$.

Convergence in law. The finer statement replaces the sandwich by an asymptotic expansion: $\delta(t - u^{2k})|u^h|\,\mathrm{d}u = t^{\lambda-1}(\log \frac{1}{t})^{m-1}\,\mathrm{d}u^* + O\big(t^{\lambda-1}(\log \frac{1}{t})^{m-2}\big)$ for a measure $\mathrm{d}u^*$ supported where the minimum λ is attained (Watanabe 2013, Lemma 4, citing Theorem 4.6 of the 2009 book). With the sign of u^k absorbed into ξ_n orthant by orthant, as Watanabe does by summing over the orthants, it gives

$$\int e^{-nK_n}\varphi = \frac{(\log n)^{m-1}}{n^\lambda}\Big(\int \mathrm{d}u^* \int_0^\infty t^{\lambda-1} e^{-t+\sqrt{t}\,\xi_n(u)}\,\mathrm{d}t + o_p(1)\Big),$$

and the convergence in law of ξ_n gives that of R_n, to minus the logarithm of the same double integral with ξ in place of ξ_n (Watanabe 2024, eq. (52) of the arXiv version).

What is omitted is the proof of the three black boxes: the standard representation (Hironaka's theorem and the division by u^k), the convergence of ξ_n on the resolved manifold, and the expansion of $\delta(t - u^{2k})|u^h|$. They occupy Chapters 4–6 of Watanabe (2009). ∎

SCHWARZ'S CRITERION OVERCHARGES. The Bayesian information criterion estimates F_n; in Watanabe's convention it is BIC $= nL_n(\hat{w}) + \frac{d}{2}\log n$. For the linear models of Theorem 8.4, $0 \le nS_n - \inf_W nL_n \le \frac{1}{2}\rho_n^2$ by Lemma 8.3, since a supremum of the likelihood over W is at most its maximum over all matrices (and $\inf_W$ replaces $nL_n(\hat{w})$ when W is not compact); for analytic models on a compact W the difference $nL_n(w_0) - nL_n(\hat{w})$ is bounded in probability as well (Watanabe 2013, citing Main Theorem 6.4 of the 2009 book). In both cases

$$F_n - \mathrm{BIC} = -\Big(\frac{d}{2} - \lambda\Big)\log n - (m-1)\log\log n + O_p(1). \tag{8.5}$$

When $\lambda < \frac{d}{2}$, as it is whenever the prior is continuous and positive at an interior true parameter where the Hessian of an analytic K is degenerate (then $\{K < \varepsilon\}$ contains a box with sides of order $\varepsilon^{1/2}$, and $\varepsilon^{1/3}$ along a null direction), Schwarz's criterion overestimates the free energy of a singular model by an amount that grows like $\log n$, and model selection by BIC is biased toward smaller models.

THE SAME DISTRIBUTIONS, TWO PRIORS. The regular plane and the sextic model are the same set of distributions, $\{\mathcal{N}(c^\top x, 1) : c \in [-1,1]^2\}$, with the same maximum likelihood and the same d, so their BIC values are equal. At the truth 0, Theorem 8.4 applied to both on the same data gives a log Bayes factor in favor of the sextic model of

$$F_n^{\text{reg}} - F_n^{\text{sext}} = \left(1 - \tfrac{2}{3}\right) \log n + O_p(1).$$

The sextic model wins because its prior, pushed forward to the coefficients, has the density $\frac{1}{12}|c_2|^{-2/3}$, which is infinite at the truth: it puts more prior mass near the truth, which is how the marginal likelihood implements Occam's razor (MacKay 1992). At a truth with $c_2 \neq 0$ both models are regular, and the log Bayes factor converges to $\log(|c_2|^{-2/3}/3)$, which favors the sextic model only when $|c_2| < 3^{-3/2} = 0.192$ (Exercise 8.10). The learning coefficient belongs to the model together with its prior, not to the set of distributions.[5]

[5] Skalse (2023) argues, in a blog post, that the inductive bias of a Bayesian learner is fixed by the prior and by the map from parameters to functions, and that singular learning theory abstracts away from both. The example shows that the learning coefficient depends on both: it records that bias without explaining it.

THE SINGULAR BIC. Since λ and m depend on the unknown truth, the free energy formula cannot be applied directly to choose a model: to know which formula holds one would need to know which model is true. Drton and Plummer (2017) call this the paradox of circular reasoning and resolve it by averaging. Let $\mathcal{M}_0 \subset \mathcal{M}_1 \subset \cdots \subset \mathcal{M}_I$ be nested models, and for $j \le i$ let λ_{ij} and m_{ij} be the learning coefficient and multiplicity of $\mathcal{M}_i$ at a generic truth in $\mathcal{M}_j$ (for $j = i$, typically half the dimension of $\mathcal{M}_i$, with $m_{ii} = 1$). Put

$$L'_{ij} = P(X^n \mid \hat{w}_i, \mathcal{M}_i) \, \frac{(\log n)^{m_{ij}-1}}{n^{\lambda_{ij}}},$$

the approximation of the marginal likelihood of $\mathcal{M}_i$ that Theorem 8.6 gives if the truth is a generic point of $\mathcal{M}_j$, with the likelihood at the truth replaced by the maximized likelihood $P(X^n \mid \hat{w}_i, \mathcal{M}_i)$.

Definition 8.7 (The singular BIC; Drton and Plummer 2017, Definition 3.1). The *singular Bayesian information criterion* of $\mathcal{M}_i$ is $\text{sBIC}(\mathcal{M}_i) = \log L'(\mathcal{M}_i)$, where $L'(\mathcal{M}_0), \ldots, L'(\mathcal{M}_I)$ is the solution with all components positive of

$$\sum_{j \le i} \left(L'(\mathcal{M}_i) - L'_{ij}\right) L'(\mathcal{M}_j) = 0, \qquad i = 0, \ldots, I.$$

Each $L'(\mathcal{M}_i)$ is the average of the L'_{ij}, $j \le i$, weighted by the approximate marginal likelihoods $L'(\mathcal{M}_j)$ themselves, as the posterior over submodels would weight them if the L' were exact and the models equally likely a

priori (their system also allows other priors on the models). The positive solution is unique and recursive: $L'(\mathcal{M}_0) = L'_{00}$, and

$$L'(\mathcal{M}_i) = \tfrac{1}{2}\Big(-b_i + \sqrt{b_i^2 + 4c_i}\Big), \qquad b_i = \sum_{j<i} L'(\mathcal{M}_j) - L'_{ii}, \quad c_i = \sum_{j<i} L'_{ij} L'(\mathcal{M}_j)$$

(Drton and Plummer 2017, Proposition 3.1). On the scale of this chapter −sBIC estimates F_n, and when every model is regular −sBIC is BIC. Under three conditions, that likelihood ratios of true models are bounded in probability, that false models lose likelihood exponentially fast, and that the complexities $n^{\lambda_{ij}}/(\log n)^{m_{ij}-1}$ increase with the model and decrease toward smaller truths, Drton and Plummer (2017, Theorems 4.1 and 4.2) prove that the sBIC selects the smallest true model with probability tending to 1, and that $\log L(\mathcal{M}_i) = \mathrm{sBIC}(\mathcal{M}_i) + O_p(1)$ for every true model at a generic truth: unlike BIC, the sBIC inherits the free energy formula. For reduced-rank regression with known noise covariance the coefficients λ_{ij} are those of Aoyagi and Watanabe (Theorem 7.5), and the model, $y = BAx + \varepsilon$, is linear in its coefficient matrix, so Theorem 8.4 covers it.[6]

[6] Drton and Plummer's Table 1 lists λ_{ij} for 5 responses and 3 covariates; our implementation of the Aoyagi–Watanabe formula reproduces all ten entries.

8.5 Measuring it

The script `08-freeenergy-models.py` computes $F_n - nS_n$ for five models with two parameters at the truth 0: the regular plane, the quartic, sextic and product models, and the quartic cup. Each is linear in its coefficients, so the data enter only through $\hat{\Sigma}_n$ and z_n, which the script draws exactly: $n\hat{\Sigma}_n$ from the Wishart law by the Bartlett decomposition,[7] and $z_n \sim \mathcal{N}(0, \hat{\Sigma}_n)$ given it. There are 200 data sets at each n in $10^2, 10^{2.5}, \ldots, 10^6$, shared by the five models (the product model uses the first input). The integral over a is done exactly with error functions where a enters linearly, and the rest by composite Gauss–Legendre quadrature on panels refined at the posterior scale $n^{-1/(2k)}$ of a coefficient b^k; for the product model it uses the kit's exact quadrature in $c = ab$. At $n = 100$ and 1000 the results agree to five decimals with the kit's grid quadrature on 1000^2 and 2000^2 points, and doubling the number of Gauss–Legendre nodes changes them by less than 10^{-14}. The run takes one to two minutes.

[7] It writes a 2×2 Wishart matrix with n degrees of freedom as $LL^\top$ with L lower triangular and independent entries, $L_{11}^2 \sim \chi^2_n$, $L_{22}^2 \sim \chi^2_{n-1}$ and $L_{21} \sim \mathcal{N}(0,1)$; then $z_n = L\zeta/\sqrt{n}$ with $\zeta \sim \mathcal{N}(0, I_2)$.

Figure 8.1 shows the result. The five models pay $\lambda \log n$, not $\log n$: by $n = 10^6$ the sextic model has paid 8.16 against the regular plane's 12.27. The bronze curves use the means of the limit laws, −1.452, −0.957, −0.991, −0.462 and −0.033 from top to bottom, computed from (8.2) and its analogues. For the four models with $m = 1$ the measured means lie within 0.15 of their curves at every n, and inside the 95% interval in 32 of the 36 cases; three of the four misses are at $n = 10^{4.5}$, where the data sets, shared by the models, happen to have a mean ρ^2 of 1.72 instead of 2, which by (8.1) raises $F_n - nS_n$ by about 0.14. The product model lies below its curve by 0.14 at $n = 100$ and 0.06 at $n = 10^6$: its remainder converges like $-\gamma/\log n$ (Exercise 8.5). Table 8.1 fits a line in $\log n$ to the five curves. For the four models with $m = 1$ the slope estimates λ to within 0.01, inside its interval. The product model's slope is 0.394, not $\frac{1}{2}$: over this range $\frac{1}{2}\log n - \log\log n$

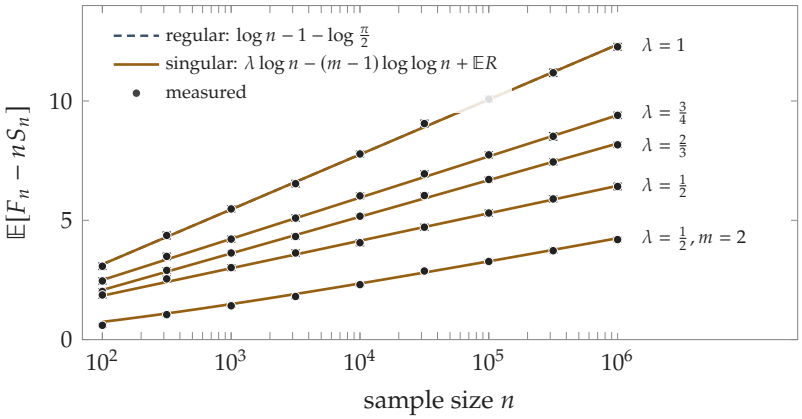


Figure 8.1. The normalized free energy of five models with two parameters at the truth 0, from top to bottom the regular plane, the quartic model, the sextic model, the quartic cup and the product model; inputs $\mathcal{N}(0, I_2)$, unit noise, uniform prior on $[-1, 1]^2$. Dots: means over 200 data sets at each n (seeds 80020, 80025, . . . , 80060), computed by exact one- and two-dimensional quadrature, with Student-t 95% intervals. Bronze: the singular prediction $\lambda \log n - (m-1)\log\log n + \mathbb{E}R$, with $\mathbb{E}R$ the mean of the limit law of the remainder. Slate dashed: the regular prediction for two parameters, Laplace's value for the regular plane, which coincides there with the singular one. Script: `08-freeenergy-models.py`.

has slope $\frac{1}{2} - 1/\log n \approx 0.40$, and adding $\log\log n$ back to the measured free energies gives 0.510. A measured slope of a free energy estimates λ only when m is known.

The same run checks the theorem's ingredients. The sandwich of Computation 8.2 with $\eta = \frac{1}{2}$, computed with the exact noiseless free energies, held in all 900 checks (20 data sets per n, five models), as a deterministic inequality must; the event $A_n(\frac{1}{2})$ held in 1799 of the 1800 data sets; and the mean of ρ^2 was 1.997. The log Bayes factor of the sextic model against the regular plane was 1.05 at $n = 100$ and 4.11 at $n = 10^6$ (intervals ±0.05), against the prediction $\frac{1}{3}\log n - 0.460$ of the two limit laws, 1.07 and 4.14, and it favored the sextic model in all 1800 data sets. The remainder R_n has standard deviation 0.97 for the regular plane, 0.77 for the sextic model and 0.58 for the product model, but on the same data much of it cancels: the log Bayes factor above has standard deviation about 0.33, and the one between the sextic and quartic models, whose learning coefficients differ by $\frac{1}{12}$, about 0.6; it favors the sextic model in 153 of 200 data sets at $n = 100$ and 191 at $n = 10^4$.

The second script, `08-freeenergy-sbic.py`, reruns the rank-selection experiment for reduced-rank regression of Drton and Plummer (2017, §5.1) with our own seeds: $M = 10$ outputs, $N = 15$ inputs, a coefficient matrix of rank 5 with singular values 1.2, 1.0, 0.8, 0.6, 0.4 and singular vectors drawn from the Haar measure for every data set, known noise covariance I_{10}, and candidate ranks $0, \ldots, 10$. The maximized likelihoods come from reduced-rank least squares; BIC charges $\frac{1}{2}\dim \mathcal{M}_i \cdot \log n$ with $\dim \mathcal{M}_i = i(25 - i)$, and the sBIC uses the Aoyagi–Watanabe coefficients, solved in logarithms

model	λ	m	fitted slope	with $(m-1)\log\log n$	slope of $\bar{F}$
regular plane	1	1	0.998 ± 0.015	0.998	1.000
quartic	$\frac{3}{4}$	1	0.749 ± 0.015	0.749	0.750
sextic	$\frac{2}{3}$	1	0.667 ± 0.012	0.667	0.667
quartic cup	$\frac{1}{2}$	1	0.491 ± 0.014	0.491	0.500
product	$\frac{1}{2}$	2	0.394 ± 0.009	0.510	0.400

Table 8.1. Least-squares slopes against $\log n$ of the means in Figure 8.1, $n = 10^2, \ldots, 10^6$, with 95% intervals from the standard errors of the means; the regular theory predicts 1 for all five. Fifth column: the slope after adding $(m-1)\log\log n$ to the means. Last column: the slope of the exact population free energy over the same range.

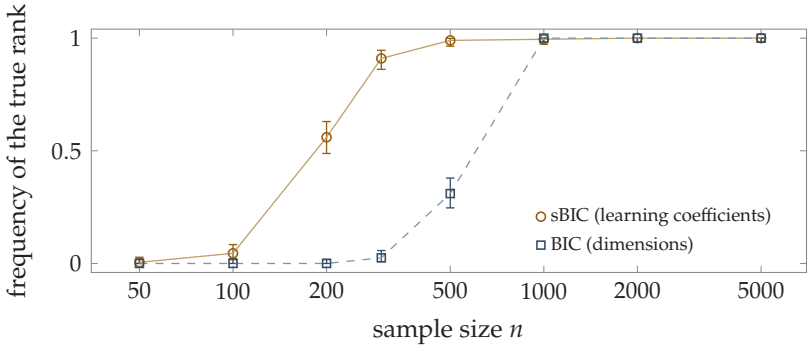


Figure 8.2. Choosing the rank of a reduced-rank regression (10 outputs, 15 inputs, true rank 5, singular values $1.2, 1.0, 0.8, 0.6, 0.4$): the fraction of 200 data sets at each n in which the criterion selects rank 5, with Clopper–Pearson 95% intervals; seeds $8{,}100{,}000 + n$. Circles: the singular BIC of Drton and Plummer (2017) with the Aoyagi–Watanabe learning coefficients. Squares: Schwarz's BIC with the dimension of the matrices of rank at most i. Script: `08-freeenergy-sbic.py`.

(checked against 60-digit arithmetic). In Figure 8.2, at $n = 300$ the sBIC selects rank 5 in 182 of 200 data sets and BIC in 5, choosing rank 4 in 161 and rank 3 in 34; both are correct in at least 199 data sets from $n = 1000$ on. Drton and Plummer report the same pattern (their Figure 5.1).

The reason is (8.5) at a smaller truth. While the fifth singular value is not yet resolved, the data look like a truth of rank 4, at which the rank-5 model has learning coefficient $\lambda_{54} = 45$, while the rank-4 model has $\lambda_{44} = 42$, half its dimension. The marginal likelihood charges the fifth rank $3\log n$; BIC charges $\frac{1}{2}(\dim \mathcal{M}_5 - \dim \mathcal{M}_4)\log n = 8\log n$, which at $n = 300$ is 46 nats, more than the fifth component of the signal is worth.

8.6 The ledger

The free energy of a singular model grows like $\lambda \log n - (m-1)\log\log n$, not like $\frac{d}{2}\log n$. For models linear in a coefficient matrix this is proved here for every prior whose volume function has the form $C\varepsilon^{\lambda}(\log\frac{1}{\varepsilon})^{m-1}$, with nothing but Cauchy–Schwarz and the Abelian lemma, because the noise

in the energy is at most its square root; Watanabe's theorem extends the formula to analytic models, where resolution of singularities restores that property. Schwarz's criterion overestimates the free energy of a singular model by $(\frac{d}{2} - \lambda)\log n$, and a criterion that uses learning coefficients selects better than one that counts dimensions.

LEDGER · $y = ax_1 + b^3x_2$

QUANTITY	the free energy $F_n = -\log$ (marginal likelihood), normalized as $F_n - nS_n$
REGULAR THEORY	$\frac{d}{2}\log n + O_p(1)$: Schwarz's BIC, Laplace's method (Theorem 1.4)
SINGULAR THEORY	$\lambda \log n - (m-1)\log\log n + O_p(1)$: proved for linear models (Theorem 8.4), Watanabe's theorem for analytic models (Theorem 8.6)
RUNNING EXAMPLE	the sextic model $y = ax_1 + b^3x_2$: $\lambda = \frac{2}{3}$, $m = 1$ against $\frac{d}{2} = 1$; $\bar{F}(n) = \frac{2}{3}\log n - 0.266$
MEASURED	slope 0.667 ± 0.012 (sextic); 0.998, 0.749, 0.491 for $\lambda = 1, \frac{3}{4}, \frac{1}{2}$; 0.394 for the product model; the sBIC finds the true rank where BIC does not (182 against 5 of 200 at $n = 300$)
RESTS ON	a fixed model and truth as $n \to \infty$; realizability or relatively finite variance; the volume asymptotics (analyticity and Hironaka's theorem in general); the exact posterior at $\beta = 1$

8.7 Where the theory stops

WHAT WAS ASSUMED. Both theorems take $n \to \infty$ with the model, the prior and the truth fixed. A truth at distance $n^{-1/2}$ from a singularity, which is what a slowly learned feature looks like, behaves like neither the singular nor the regular formula over a long range of n (Chapter 12). Watanabe's theorem asks for a compact parameter set, because infinity can itself act as a singularity: for the regression of x_2 on x_1 with mean $a\sin(bx_1)$, at a truth whose true parameters include the line $a = 0$, the maximum-likelihood estimator never exists (Watanabe 2010b, p. 5 of the arXiv version). Theorem 8.4 needs no compactness, but networks are not linear in their coefficients, and their priors are Gaussian on all of $\mathbb{R}^d$. The theorem asks for analyticity, which ReLU networks lack; Nagayasu and Watanabe (2023) bound the free energy of deep ReLU networks with more layers than needed (a preprint; we rely on its abstract). It asks for a prior that is analytic up to a smooth positive factor: if the only true parameter is $w = 0$ and the prior behaves like $e^{-1/\|w\|^2}$ there, vanishing to infinite order, the volume of nearly true parameters shrinks faster than every power of ε; in this example, also from the arXiv version of Watanabe (2010b), the generalization errors need not scale like $1/n$. It asks for relatively finite variance, which realizable and regular truths have and unrealizable singular ones can lack, and for essential uniqueness of the optimal distribution: when several distributions

are optimal, Nagayasu and Watanabe (2022) find asymptotic terms that differ from the formula's. And it describes the exact posterior at $\beta = 1$, while the posteriors of networks are approximated.

WHERE IT BREAKS. The cleanest failure is Watanabe's non-renormalizable example (Watanabe 2010a, §5):[8] the truth $q = \mathcal{N}(0, I_2)$ and the model $p(x, y \mid a) = \mathcal{N}\big((a, \sqrt{a^4 - a^2 + 1}), I_2\big)$, which cannot realize q. The energy is exactly $nK_n(a) = \frac{n}{2}a^4 - \sqrt{n}\,a\,\xi_1 - \sqrt{n}\,h(a)\,\xi_2$, with $h(a) = \sqrt{a^4 - a^2 + 1} - 1$ and independent standard Gaussians ξ_1, ξ_2: the noise $\sqrt{n}\,a\,\xi_1$ is not bounded by a multiple of the square root $\sqrt{n/2}\,a^2$ of the energy. Watanabe derives, by a sketch, learning curves of order $n^{-2/3}$ instead of $1/n$; the source does not treat the free energy, and we argue, without a full proof, as follows. Since the prior is a probability density, $F_n - nL_n(w_0) \geq \min_a nK_n(a)$, and a window of width $n^{-1/3}$ around the minimizer gives an upper bound larger by $O_p(\log n)$. The minimum of $\frac{n}{2}a^4 - \sqrt{n}\,a\,\xi_1$ is $-3 \cdot 2^{-7/3}|\xi_1|^{4/3}n^{1/3}$, and the term with $h(a) \approx -\frac{1}{2}a^2$ moves it by $O_p(n^{1/6})$. So $F_n - nL_n(w_0)$ decreases like $n^{1/3}$, with mean $-3c\,n^{1/3} + \frac{1}{3}\log n + O(1)$, $c = \Gamma(\frac{7}{6})/(2^{5/3}\sqrt{\pi}) = 0.165$. In `08-freeenergy-nonrenorm.py` (uniform prior on $[-1, 1]$, 400 data sets per n, means estimated with a control variate) the mean divided by $n^{1/3}$ is -0.447 ± 0.003 at $n = 10^6$ and -0.482 ± 0.001 at $n = 10^8$, within 0.001 of a Laplace approximation at the minimizer, on its slow way to $-3c = -0.495$. Since $\mathbb{E}[G_n] - L(w_0)$, for the Bayes generalization loss G_n, is the increment of $\mathbb{E}[F_n - nL_n(w_0)]$ from n to $n + 1$, this suggests the learning curve $-c\,n^{-2/3}$ that Chapter 10 derives and measures; the constants printed in the source are larger by the factor $2^{4/3}$.

Within the theory, the bounded remainder carries everything the formula leaves out, and at practical sample sizes it competes with the differences the formula is meant to detect. It depends on the scale of the prior: widening the uniform prior of the sextic model from $[-1, 1]^2$ to $[-10, 10]^2$ raises its free energy by $2\log 10 = 4.6$ at every n, up to exponentially small terms, because the posterior lives inside the smaller square; between two models whose learning coefficients differ by $\frac{1}{6}$, that is the whole difference $\frac{1}{6}\log n$ at $n = 10^{12}$. It converges slowly when $m > 1$: for the product model the mean remainder is $-0.033 - \gamma/\log n + O(1/\log^2 n)$, with Euler's γ, and still -0.035 at $n = 10^{100}$ (Exercise 8.5). Without compactness the remainder of (8.5) can diverge: the log-likelihood ratios of Gaussian mixtures with unbounded means can grow like $\log\log n$, as Drton and Plummer (2017) note, citing Hartigan (1985). Finally, the marginal likelihood is not a measure of predictive performance: in deep-learning model selection it can be negatively correlated with generalization (Lotfi et al. 2022).

OPEN PROBLEMS. The theorems fix the dimension d while n grows; neural networks are trained with d comparable to n or larger. For linear models the sandwich of Lemma 8.2 holds for every number $k = MN$ of coefficients, but its random term $\rho_n^2 \sim \chi_k^2$ is of order k, so the sandwich pins the free energy down only to $O_p(k)$, and the comparison of $\hat{\Sigma}_n$ with Σ needs $N \ll n$ (Exercise 8.13). How the free energy of a singular model behaves when d grows with n, and when $\lambda \log n$ still dominates its fluctuation, is open; non-

[8] Watanabe takes the word from physics, where a family is renormalizable if some rescaling reveals a universal law (Watanabe 2010a, §6.2); here the rescaling is $t = nK$, and the law is the limit of the rescaled posterior, from which the free energy formula and the learning curves follow.

asymptotic PAC-Bayes bounds for Gibbs posteriors whose prior integrals are analyzed with the tools of this book (Wang and Yang 2026, a 2026 preprint, we rely on its abstract) approach the question from another side. The limit law of the remainder is known (Theorem 8.6), but Drton and Plummer (2017, §2) knew of no higher-order approximations in statistically relevant singular settings, and estimating the remainder from data is open as far as we know. Finally, the sBIC needs λ_{ij} for every pair of nested models, and these are known for few families; Imai (2019) proposes estimating them from the variance of the energy in thermodynamic integration and inserting the estimates into the sBIC (a preprint), and computing them for networks of practical size is open (Chapter 7).

Exercises

8.1 CONSOLIDATION. (Wilks's statistic.) Complete the square to prove the second claim of Lemma 8.3, and deduce (8.1) from the identity, with $\hat{\Delta} = Z_n \hat{\Sigma}_n^{-1}/\sqrt{n}$,

$$nK_n(w) = \tfrac{n}{2} \operatorname{tr}\big((\Delta - \hat{\Delta})\hat{\Sigma}_n(\Delta - \hat{\Delta})^{\top}\big) - \tfrac{1}{2}\rho_n^2.$$

8.2 CONSOLIDATION. (The quartic model and the quartic cup.) For $K = \frac{1}{2}(a^2 + b^4)$ and $K = \frac{1}{2}(a^4 + b^4)$ with the uniform prior on $[-1,1]^2$, show that for small ε, $V(\varepsilon) = \frac{1}{2} I_4 (2\varepsilon)^{3/4} = 1.4699\,\varepsilon^{3/4}$ with $I_4 = \int_{-1}^{1} \sqrt{1 - v^4}\,\mathrm{d}v = 1.7480$, and $V(\varepsilon) = \frac{\sqrt{2}}{4} \frac{\Gamma(\frac{1}{4})\Gamma(\frac{5}{4})}{\Gamma(\frac{3}{2})}\,\varepsilon^{1/2} = 1.3110\,\varepsilon^{1/2}$. Check that $\bar{F}(n) = \frac{3}{4}\log n - 0.301$ and $\frac{1}{2}\log n - 0.150$, by the Abelian lemma and by factorizing the integral.

8.3 CONSOLIDATION. (The Gibbs bound.) For every model, prior and truth with a minimizer w_0 of L, show that $\mathbb{E}[F_n - nL_n(w_0)] \le \bar{F}(n)$, assuming $\mathbb{E}|f(X,w)|$ is bounded on W. *(Hint: for every probability density $\tilde{\varphi}$ on W, $-\log \int e^{-H}\varphi \le \int H\tilde{\varphi} + \mathrm{KL}(\tilde{\varphi}\,\|\,\varphi)$ by Jensen's inequality; take $H = nK_n$ and $\tilde{\varphi} \propto e^{-nK}\varphi$, and use $\mathbb{E}[K_n(w)] = K(w)$.)* From the limit laws, the gap $\mathbb{E}[F_n - nS_n] - \bar{F}(n)$ tends to -1 for the regular plane, -0.725 for the sextic model and $-\frac{1}{2}$ for the product model; derive the first and the last.

8.4 CONSOLIDATION. (Relatively finite variance.) For the sextic model at the truth 0, with $f = \frac{1}{2}(c^{\top}x)^2 - \varepsilon\, c^{\top}x$ and $c = (a, b^3)$, show that $\mathbb{E}f = \frac{1}{2}|c|^2 = K$ and $\mathbb{E}f^2 = |c|^2 + \frac{3}{4}|c|^4 = 2K + 3K^2$, so that $\mathbb{E}f \ge \frac{1}{5}\mathbb{E}f^2$ on $[-1,1]^2$, and that every constant below $\frac{1}{2}$ works in place of $\frac{1}{5}$ near the truth.

8.5 CONSOLIDATION. (The slow remainder of the product model.) For the product model, the exact computation of Chapter 3 (step 5 of its first computation box) gives, up to an exponentially small error, $F_n - nS_n = \frac{1}{2}\log S_{xx} - \log\big(\frac{1}{2}\log S_{xx} - h(Z_n)\big) - \frac{Z_n^2}{2} + \frac{1}{2}\log\frac{2}{\pi}$, with $S_{xx} = \sum_i x_i^2$, $Z_n = \sum_i x_i y_i/\sqrt{S_{xx}} \sim \mathcal{N}(0,1)$ and $h(z) = \mathbb{E}\log|z + U|$ for $U \sim \mathcal{N}(0,1)$. Show that $\mathbb{E}h(Z_n) = -\gamma/2$, with Euler's constant

γ, and, expanding the logarithm in powers of $1/\log S_{xx}$, that $\mathbb{E}R_n = \frac{1}{2}\log\frac{8}{\pi} - \frac{1}{2} - \gamma/\log n + O(1/\log^2 n)$: -0.095 at $n = 10^4$ and -0.035 at $n = 10^{100}$, against the limit -0.033. (Chapter 3 proves the corresponding expansion of R_n in probability.)

8.6 BY HAND. ★ (The limit law of the sextic model.) Prove (8.2). *(Hint: in (8.1) substitute $u = \sqrt{n}\,c$; the law of u under the prior has density $\frac{1}{12}n^{-2/3}|u_2|^{-2/3}$ on $[-\sqrt{n},\sqrt{n}]^2$; show that $\sqrt{n}\,\hat{c} - \xi \to 0$ in probability, with $\xi \sim \mathcal{N}(0, I_2)$ the vector of Computation 8.2, and use dominated convergence.)* Compute $H(0) = 2^{1/6}\Gamma(\frac{1}{6}) = 6.248$ and, numerically, $\mathbb{E}\log H(\zeta_2) = 1.557$ and the mean -0.991.

8.7 BY HAND. (Rank-one regression.) Let $y = b\,a^{\top}x + \varepsilon$ with $a, b \in \mathbb{R}^2$ uniform on $[-1,1]^2$ each, $x \sim \mathcal{N}(0, I_2)$, $\varepsilon \sim \mathcal{N}(0, I_2)$ and the truth 0. Show that $K = \frac{1}{2}|a|^2|b|^2$ and that $V(\varepsilon) \sim \frac{\pi^2}{8}\,\varepsilon\log\frac{1}{\varepsilon}$, and conclude from Theorem 8.4 that $F_n - nS_n = \log n - \log\log n + O_p(1)$. How much do BIC with $d = 4$ and BIC with the dimension 3 of the rank-one matrices overcharge? *(Hint: $\mathbb{P}(|a| < r) = \frac{\pi}{4}r^2$ for $r \le 1$.)*

8.8 BY HAND. (The singular BIC of two models.) Let $\mathcal{M}_0 = \{y = \varepsilon\}$ and $\mathcal{M}_1$ the product model, whose distributions form a one-dimensional family: $\lambda_{00} = 0$, $\lambda_{10} = \frac{1}{2}$ with $m_{10} = 2$, and $\lambda_{11} = \frac{1}{2}$ with $m_{11} = 1$. With T twice the maximized log-likelihood ratio and $A = e^{T/2}n^{-1/2}$, show that $\mathrm{sBIC}(\mathcal{M}_1) - \mathrm{sBIC}(\mathcal{M}_0) = \log\big(\frac{1}{2}(A - 1 + \sqrt{(1-A)^2 + 4A\log n})\big)$, and that for $n > e$ this lies between $\log A$, which is minus the difference of our BIC values when $\mathcal{M}_1$ is charged its dimension 1, and $\log A + \log\log n$. Evaluate it at $n = 100$ for $T = 0, 2, 10, 20$ (answers: -1.01, -0.21, 2.87, 7.70).

8.9 VARIATION. (A Gaussian prior.) Give the sextic model the prior $\mathcal{N}(0, I_2)$ on $\mathbb{R}^2$. Show that $V(\varepsilon) \sim \frac{2^{2/3}I_6}{\pi}\,\varepsilon^{2/3} = 0.920\,\varepsilon^{2/3}$, that Theorem 8.4 applies although W is not compact, and that the constant of $\bar{F}(n)$ increases by $\log\frac{\pi}{2}$.

8.10 VARIATION. (A generic truth.) Let the truth of the sextic model and the regular plane be $c_0 = (0, c_2)$ with $0 < |c_2| < 1$. Show that both pairs are regular, and that the log Bayes factor $F_n^{\mathrm{reg}} - F_n^{\mathrm{sext}}$ converges in probability to $\log(|c_2|^{-2/3}/3)$. For which c_2 does the marginal likelihood prefer the sextic model?

8.11 MEASUREMENT. (Fitting λ and m.) In `08-freeenergy-models.py`, fit $\alpha\log n - \beta\log\log n + \kappa$ to the product model's means with α, β, κ all free. How well are α and β determined over $n = 10^2, \ldots, 10^6$, and why? Repeat with n up to 10^8.

8.12 MEASUREMENT. (A weaker signal.) In `08-freeenergy-sbic.py`, halve the singular values. Since the eigenvalues of the explained sum of squares grow like n times the squared singular values, predict how the sample size at which each criterion is right in 90% of the data sets changes, and check.

8.13 RESEARCH. (Growing dimension.) Let $y = c^\top x + \varepsilon$ with $c \in [-1,1]^k$ uniform, $x \sim \mathcal{N}(0, I_k)$ and the truth 0, and let $k = k_n \to \infty$ with $k_n/n \to 0$. Using Lemma 8.2 and the fact that the extreme eigenvalues of $\hat\Sigma_n$ then tend to 1 in probability, show that $F_n - nS_n = \frac{k_n}{2}\log n + O_p(k_n)$. Formulate the analogous question for a singular family whose learning coefficient stays bounded while d grows, such as $y = \big(\sum_{j\le k} a_j b_j\big)x$, and say what would have to be controlled. As far as we know, the singular case is open.

Notes and References

The free energy formula is Watanabe's. It appears first in Watanabe (1999), the paper in which, by Watanabe's own account (Watanabe 2024), the real log canonical threshold was found to determine the accuracy of Bayesian estimation, and in full, for the mean, $\lambda\log n - (m-1)\log\log n + O(1)$, in Watanabe (2001), with an algorithm for λ and m by resolution. The random-variable form with a remainder converging in law is in Section 6.2 of the 2009 book (Watanabe 2009),[9] the unrealizable case under the renormalizable condition in Watanabe (2010a), and the statement we quote in Watanabe (2013). Watanabe (2024) reviews the whole theory, and the 2018 textbook (Watanabe 2018) is the place to read it with exercises; the algebraic-statistics side, Bayesian integrals of algebraic models, is in Drton et al. (2009, Chapter 5) and Lin (2011).

[9] Drton and Plummer (2017) cite it as Theorem 6.7 of the book; we have not checked the number against the book.

Theorem 8.4 and its sandwich are the book's own presentation of a standard argument; the idea that the noise of a log-likelihood ratio is at most the square root of its mean is Watanabe's relatively finite variance, a form of what Watanabe (2010a) calls the renormalizable condition. For regular models, Clarke and Barron (1990) showed that the expected normalized free energy, the relative entropy between the truth and the Bayes mixture of n observations, is $\frac{d}{2}\log n$ plus a constant determined by the prior density and the Fisher information at the truth; Kass and Raftery (1995) review Bayes factors and BIC.

The singular BIC is due to Drton and Plummer (2017), whose examples are reduced-rank regression, factor analysis and mixtures; it uses the reduced-rank values of Aoyagi and Watanabe (2005), and Watanabe (2013, §7.2) describes a related two-step method that he attributes to Drton. Watanabe's own answer to the paradox of circular reasoning is WBIC, which needs no learning coefficients (Chapter 11). Rao (2026), a 2026 preprint that we know from its abstract, studies linear-Gaussian rank models with exact marginal likelihoods and reports that the error of Laplace and BIC grows like $(\frac{d}{2} - \lambda)\log n$, as (8.5) says. Wei et al. (2023) argue for deep learning that dividing by the Hessian determinant, as the Laplace approximation does, is inappropriate for singular models.

TRANSLATING THE LITERATURE. Watanabe (2013) writes $\mathcal{F}$ for our F_n, L_n, K, K_n and BIC $= nL_n(\hat w) + \frac{d}{2}\log n$ as we do, and calls λ the real log canonical threshold; his Fundamental Conditions (1)–(4) are our conditions (ii)–(v) of Theorem 8.6, and his η_n is our ψ_n. Watanabe (2010a) writes H_n, L_0 and

B_g for our K_n, $L(w_0)$ and G_n. Section 6.2 of the 2009 book, "Evidence and stochastic complexity", uses the older names of the marginal likelihood and of its negative logarithm, and $f(x, w) = \log(q(x)/p(x \mid w))$. Watanabe (2024) writes θ, Θ, $\pi(\theta)$ for our w, W, $\varphi(w)$. Drton and Plummer (2017) work on the log-likelihood scale: their $L(\mathcal{M}_i)$ is our e^{-F_n}, their BIC is minus ours, and $\lambda_i(\pi_0)$, $m_i(\pi_0)$ are our learning coefficient and multiplicity at the truth π_0; in their reduced-rank regression N counts the responses and M the covariates, the reverse of our M outputs and N inputs, and π is our coefficient matrix.

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9 Generalization

The error λ/n and the equations of state

THE MODEL — $y = ab^3x$

IN THIS CHAPTER. The regular theory says that the Bayes predictive distribution of a model with d parameters has expected test loss $\frac{d}{2n}$ above the entropy of the truth. For the cubic cusp $y = ab^3x$ we compute by hand that the excess is half the squared posterior mean of the rescaled coefficient, a shrinkage estimator whose mean square is exactly $\frac{1}{3}$: the excess is $\frac{1}{6n}$, not $\frac{1}{n}$. Two integrations by parts, a virial identity in the parameter and Stein's identity in the noise, prove $n(\mathbb{E}[G_n] - S) \to \lambda$ for every one-coefficient regression with Gaussian inputs and noise, and three of the four limits in Watanabe's equations of state for these models; his general theorem rests on the same two integrations by parts. Exact quadrature over 1000 data sets per sample size confirms λ for three models and, for the product model with $m = 2$, shows how slowly the limit is reached. All of this concerns the exact posterior average at a fixed truth.

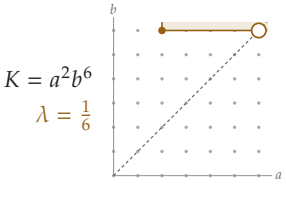
9.1 The question: what the posterior costs at test time

A model trained on n examples is judged on fresh ones, and when its output is a predictive distribution the natural score is the test cross-entropy, which no model can push below the entropy of the noise in the labels. How fast does the score approach that floor, and what sets the rate? For a Bayesian, who predicts with the posterior average of the models, the classical answer counts parameters. Whether that average is worth computing for neural networks is debated (Izmailov et al. 2021; Wenzel et al. 2020); this chapter asks what it buys in the simplest singular models, where it can be computed exactly.

Fix a model $p(x \mid w)$, $w \in W$, a prior φ, and data X^n from the truth q. The *predictive distribution* is $p^*(x) = \mathbb{E}_w\big[p(x \mid w)\big]$, and its *Bayes generalization loss* and Bayes training loss are

$$G_n = -\mathbb{E}_X\big[\log p^*(X)\big], \qquad T_n = -\frac{1}{n}\sum_{i=1}^{n} \log p^*(X_i),$$

with X a fresh draw from q. If the model realizes the truth, $q = p(\cdot \mid w_0)$, the *generalization error* $G_n - S = \mathrm{KL}(q \,\|\, p^*)$ is the Kullback–Leibler divergence from the truth to the predictive; in a regression $p(y \mid x, w)$ the input density cancels, and $G_n - S$ averages over a test input the divergence between the conditional laws of y. The function $n \mapsto \mathbb{E}[G_n] - S$ is the *learning curve*.



THE REGULAR ANSWER · the generalization error. If the truth is regular for a model with d parameters and the model realizes it, then the Bayes predictive distribution has $\mathbb{E}[G_n] - S = \frac{d}{2n} + o(\frac{1}{n})$ (Theorem 1.7) and $\mathbb{E}[T_n] - S = -\frac{d}{2n} + o(\frac{1}{n})$ (Watanabe 2010c); the maximum-likelihood plug-in has the same two values (Amari and Murata 1993). Their difference $\frac{d}{n}$ is the penalty of Akaike's criterion on the per-sample scale (Akaike 1974).

The model of this chapter is the *cubic cusp* $y = ab^3x + \varepsilon$, with inputs $x \sim \mathcal{N}(0,1)$, unit noise, the uniform prior on $[-1,1]^2$, and the truth $(a,b) = (0,0)$, so that y is pure noise. It has two parameters, and the box predicts an error of $\frac{1}{n}$.[1] Its distributions $\mathcal{N}(cx,1)$, $c = ab^3$, are those of the regular line $y = cx$, $|c| \le 1$, which has one parameter and the error $\frac{1}{2n}$. Neither answer is right: the error is $\frac{1}{6n}$. Four models with the same distributions have three different limits (Corollary 9.7): with the uniform prior on $[-1,1]^d$, the coefficient c has the density $\frac{1}{2}$ for the regular line $y = ax$, $\frac{1}{2}\log\frac{1}{|c|}$ for the product model $y = abx$, $\frac{1}{2}(|c|^{-1/2} - 1)$ for the cusp $y = ab^2x$, and $\frac{1}{4}(|c|^{-2/3} - 1)$ for the cubic cusp, and the errors are $\frac{1}{2n}$, $\frac{1}{2n}$, $\frac{1}{4n}$ and $\frac{1}{6n}$, against $\frac{d}{2n} = \frac{1}{2n}, \frac{1}{n}, \frac{1}{n}, \frac{1}{n}$. What tells them apart is the prior that each induces on c near the truth $c_0 = 0$, where the densities of the cusps blow up like a power of $|c|$, and the product model's only like a logarithm: the parametrization bets on the truth, and the posterior average collects the bet. How does the generalization error of a singular model depend on n, and through which property of the model and the prior? The answer is $\mathbb{E}[G_n] - S = \lambda/n + o(1/n)$ at the temperature $\beta = 1$ of Bayes' rule, with a companion number ν for the training error and for other temperatures.

[1] Akaike's criterion is usually written $-2\log L + 2d$, with L the maximized likelihood. That is $2n$ times $-\frac{1}{n}\log L + \frac{d}{n}$: the per-sample training loss plus the penalty $\frac{d}{n}$.

9.2 By hand: half the squared posterior mean

A *one-coefficient model* is a regression $y = g(w)x + \varepsilon$ whose mean depends on the parameters only through one number $c = g(w)$. Its posterior is the posterior of c under the prior density ρ that φ induces on c (Chapter 3), and its predictive at an input x is a mixture of Gaussians, $p^*(y \mid x) = \mathbb{E}_w\big[\phi(y - cx)\big]$, with ϕ the standard normal density.

COMPUTATION 9.1 · THE DIVERGENCE FROM THE TRUTH TO THE PREDICTIVE

1 Fix the data and a test input x; write $u = (c - c_0)x$ for the error of a coefficient drawn from the posterior and $\varepsilon = y - c_0x \sim \mathcal{N}(0,1)$ for the test noise. Since $\phi(y - cx)/\phi(y - c_0x) = e^{u\varepsilon - u^2/2}$,

$$D_x = \mathrm{KL}\big(q(\cdot \mid x) \,\|\, p^*(\cdot \mid x)\big) = -\mathbb{E}_\varepsilon \log \mathbb{E}_w\Big[e^{u\varepsilon - u^2/2}\Big].$$

2 The probabilists' Hermite polynomials $\mathrm{He}_0 = 1$, $\mathrm{He}_1(\varepsilon) = \varepsilon$, $\mathrm{He}_2(\varepsilon) = \varepsilon^2 - 1$, ... have the generating function $e^{u\varepsilon - u^2/2} =$

$\sum_{k\ge 0} \mathrm{He}_k(\varepsilon) u^k/k!$, so $\mathbb{E}_w\left[e^{u\varepsilon - u^2/2}\right] = 1 + A(\varepsilon)$ with

$$A(\varepsilon) = \mathbb{E}_w[u]\,\varepsilon + \tfrac{1}{2}\mathbb{E}_w\left[u^2\right](\varepsilon^2 - 1) + \tfrac{1}{6}\mathbb{E}_w\left[u^3\right]\mathrm{He}_3(\varepsilon) + \cdots .$$

3 The polynomials are orthogonal under $\mathcal{N}(0,1)$, with $\mathbb{E}[\mathrm{He}_j \mathrm{He}_k] = k!$ if $j = k$ and 0 otherwise. So $\mathbb{E}A = 0$, and $\mathbb{E}A^2 = \sum_{k\ge 1} \mathbb{E}_w\left[u^k\right]^2/k!$.

4 Expanding $\log(1+A) = A - \frac{1}{2}A^2 + \frac{1}{3}A^3 - \cdots$, $D_x = \frac{1}{2}\mathbb{E}A^2 - \frac{1}{3}\mathbb{E}A^3 + \cdots = \frac{1}{2}\mathbb{E}_w[u]^2 +$ (terms of degree 4 in u); terms of odd degree vanish because $\mathrm{He}_k(-\varepsilon) = (-1)^k \mathrm{He}_k(\varepsilon)$. The second moment $\mathbb{E}_w\left[u^2\right]$, which measures the spread of the posterior, multiplies He_2, whose mean is 0: the spread does not enter at second order.

5 The data locate c to within about $n^{-1/2}$, so the terms of degree 4 are of order n^{-2}. Averaging over the test input, with $\mathbb{E}x^2 = 1$,

$$G_n - S = \mathbb{E}_x D_x = \tfrac{1}{2}\left(\mathbb{E}_w[c] - c_0\right)^2 + O(n^{-2}).$$

The generalization error is half the squared error of the posterior mean; a Gibbs predictor, which draws one coefficient from the posterior, pays $\frac{1}{2}\mathbb{E}_w\left[(c - c_0)^2\right]$, the spread included. What made the computation possible was to read the likelihood ratio $e^{u\varepsilon - u^2/2}$ of a Gaussian shift as the generating function of an orthogonal family.

COMPUTATION 9.2 · THE REGULAR LINE, EXACTLY

1 Take $y = ax$ with the flat prior on $\mathbb{R}$. With $S_{xx} = \sum_i x_i^2$ and $\hat{c} = \sum_i x_i y_i / S_{xx}$, the posterior of $c = a$ is $\mathcal{N}(\hat{c}, S_{xx}^{-1})$, and the predictive at x is $\mathcal{N}(\hat{c}x, 1 + v)$ with $v = x^2/S_{xx}$, so

$$D_x = \tfrac{1}{2}\left(\log(1+v) + \frac{1 + (\hat{c} - c_0)^2 x^2}{1+v} - 1\right).$$

2 At the truth $Z_n = S_{xx}^{1/2}(\hat{c} - c_0)$ is standard normal and independent of S_{xx} (Lemma 9.2), and $(\hat{c} - c_0)^2 x^2 = Z_n^2 v$. Averaging over Z_n turns the fraction into exactly 1, and $\mathbb{E}[D_x \mid S_{xx}, x] = \frac{1}{2}\log(1+v)$.

3 The squares $x^2 \sim \chi_1^2$ and $S_{xx} \sim \chi_n^2$ are independent, so $B = x^2/(x^2 + S_{xx})$ has the Beta law with parameters $\frac{1}{2}$ and $\frac{n}{2}$, and $1 + v = 1/(1 - B)$. With $\mathbb{E}\log(1 - B) = \psi(\frac{n}{2}) - \psi(\frac{n+1}{2})$, ψ the digamma function,

$$\mathbb{E}[G_n] - S = \tfrac{1}{2}\left(\psi\left(\tfrac{n+1}{2}\right) - \psi\left(\tfrac{n}{2}\right)\right) = \frac{1}{2n} + \frac{1}{4n^2} + O(n^{-3}),$$

the regular answer with $d = 1$; at $n = 100$, $n(\mathbb{E}[G_n] - S) = 0.50250$. The rule of Computation 9.1 gives $\frac{1}{2}\mathbb{E}[(\hat{c} - c_0)^2] = \frac{1}{2(n-2)}$, which agrees to first order (Exercise 9.1).

COMPUTATION 9.3 · SHRINKAGE IN THE CUBIC CUSP

1 For independent a, b uniform on $[-1,1]$ and $0 < s \le 1$, $\mathbb{P}(|a||b|^3 \le s) = \int_0^1 \min\big(1, (s/r)^{1/3}\big)\,\mathrm{d}r = \frac{3}{2}s^{1/3} - \frac{1}{2}s$, so $c = ab^3$ has the density $\rho(c) = \frac{1}{4}(|c|^{-2/3} - 1)$ on $[-1,1]$. Near $c_0 = 0$, $\rho(c) = |c|^{-2/3}\ell(c)$ with $\ell(c) \to \frac{1}{4}$, and the exponent $-\frac{2}{3} = 2\lambda - 1$ gives the learning coefficient $\lambda = \frac{1}{6}$ of $K = \frac{1}{2}a^2b^6$.

2 Rescale the coefficient by the data, $t = S_{xx}^{1/2}c$. The energy is $nK_n = \frac{1}{2}t^2 - tZ_n$ (Lemma 9.2), so the posterior of t is proportional to $e^{-(t-Z_n)^2/2}(|t|^{-2/3} - S_{xx}^{-1/3})$ on $|t| \le S_{xx}^{1/2}$, and as $n \to \infty$ it tends to the density proportional to $e^{-(t-Z)^2/2}|t|^{-2/3}$: a Gaussian likelihood centered at the least-squares coordinate Z, times a prior that is infinite at the truth. By Computation 9.1, $n(G_n - S) \approx \frac{1}{2}\bar{t}^2$, with $\bar{t}$ the posterior mean of t.

3 For a Gaussian likelihood of unit variance and every prior weight w, *Tweedie's formula* (Efron 2011; Robbins 1956) gives the posterior mean $M(Z) = Z + h'(Z)/h(Z)$, with $h(Z) = \int e^{-(t-Z)^2/2}w(t)\,\mathrm{d}t$ (Theorem A.3). For $w(t) = |t|^{-2/3}$, expanding e^{tZ} in $h(Z) = e^{-Z^2/2}\int |t|^{-2/3}e^{-t^2/2+tZ}\,\mathrm{d}t$, integrating term by term and applying Kummer's transformation give $h(Z) = 2^{1/6}\Gamma(\frac{1}{6})\,{}_1F_1(\frac{1}{3};\frac{1}{2};-\frac{Z^2}{2})$, with ${}_1F_1$ Kummer's confluent hypergeometric function, and therefore

$$M(Z) = Z\left(1 - \frac{2}{3}\,\frac{{}_1F_1\left(\frac{4}{3};\frac{3}{2};-\frac{Z^2}{2}\right)}{{}_1F_1\left(\frac{1}{3};\frac{1}{2};-\frac{Z^2}{2}\right)}\right).$$

4 The posterior pulls the least-squares coordinate toward the truth: $M(Z) \approx \frac{1}{3}Z$ near 0, where both functions ${}_1F_1$ equal 1; $M(1) = 0.412$, $M(2.5) = 1.975$, $M(4) = 3.804$; and $M(Z) = Z - \frac{2}{3Z} + O(Z^{-3})$ for large $|Z|$. The posterior mean is a *shrinkage* estimator of Z, strong near the truth and weak far from it.

5 Quadrature over $Z \sim \mathcal{N}(0,1)$ gives $\mathbb{E}[M(Z)^2] = 0.333333\ldots$, which is $\frac{1}{3}$ to ten digits. Hence $n(\mathbb{E}[G_n] - S) \to \frac{1}{6}$: the cubic cusp pays $\frac{1}{6n}$, against $\frac{1}{n}$ for two regular parameters and $\frac{1}{2n}$ for the regular line.

Two things had to be noticed: at order $\frac{1}{n}$ only the posterior mean matters, and after the rescaling $t = S_{xx}^{1/2}(c - c_0)$ the problem forgets n, leaving a Gaussian location problem with one observation Z and the prior weight $|t|^{2\lambda-1}$ (Figure 9.1). Why should the mean square of this shrinkage estimator be exactly 2λ?[2]

[2] Shrinkage estimators of a normal mean, and the identity used below to compute their risk, are the subject of Stein (1981). Here the shrinkage is not designed: it is what the prior induced by the parametrization does, and it helps because the truth sits where that prior is singular.

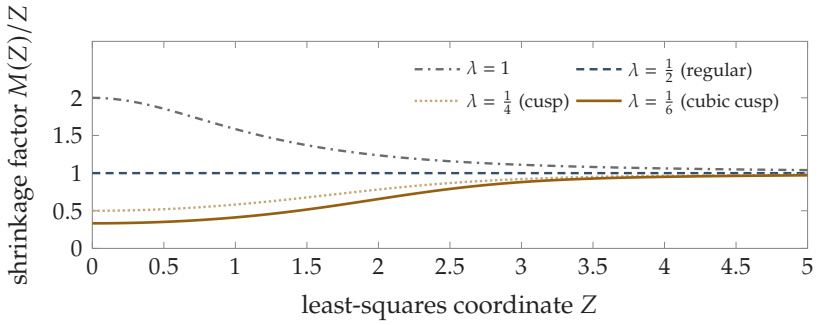


Figure 9.1. The shrinkage factor $M(Z)/Z$ of the mean of the limit posterior $\propto e^{-(t-Z)^2/2}|t|^{2\lambda-1}$, by quadrature at 101 values of $Z \geq 0$ (M is odd). It starts at $M'(0) = v(0) = 2\lambda$ and tends to 1: a prior weight that is infinite at the truth ($\lambda < \frac{1}{2}$) shrinks the least-squares coordinate toward 0, one that vanishes there ($\lambda = 1$) pushes it away. In every case $\mathbb{E}[M(Z)^2] = 2\lambda$ (Lemma 9.5). Script: `09-generalization-quartet.py`.

9.3 The theorem: λ/n for one-coefficient models

We prove the three facts the computation used, at every inverse temperature β, which Section 9.4 needs. At inverse temperature β the posterior is proportional to $\varphi(w)e^{-n\beta L_n(w)}$, with expectations $\mathbb{E}_w^\beta[\cdot]$, the predictive is $p_\beta^*(x) = \mathbb{E}_w^\beta\big[p(x \mid w)\big]$, and $G_n(\beta) = -\mathbb{E}_X \log p_\beta^*(X)$, so that $G_n = G_n(1)$.

Assumption 9.1 (One-coefficient regression). The data are independent pairs (x_i, y_i), $i = 1, \dots, n$, with $x_i \sim \mathcal{N}(0,1)$ and $y_i = c_0 x_i + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0,1)$ independent of x_i. The model is $y = g(w)x + \varepsilon$, $\varepsilon \sim \mathcal{N}(0,1)$, for a measurable g on $W \subseteq \mathbb{R}^d$ with $C_g = \sup_W |g - c_0| < \infty$ and a prior probability density φ on W; the truth is realizable, $c_0 = g(w_0)$ for some $w_0 \in W$. The law of $c = g(w)$ under φ has a density ρ, and there are $\eta_0 \in (0,1)$, $\lambda > 0$, an integer $m \geq 1$, and a function ℓ whose one-sided limits $\ell_\pm = \lim_{c \to c_0 \pm} \ell(c)$ are both positive, such that

$$\rho(c) = |c - c_0|^{2\lambda-1}\Big(\log \frac{1}{|c-c_0|}\Big)^{m-1}\ell(c) \qquad \text{for } 0 < |c - c_0| < \eta_0.$$

The exponent $\alpha = 2\lambda - 1 > -1$ is fixed from now on, and ρ vanishes outside $[c_0 - C_g, c_0 + C_g]$. The form of ρ is that of Theorem 3.4, except that the density may approach different constants on the two sides of the truth. The numbers λ and m are the learning coefficient and multiplicity of $K(w) = \frac{1}{2}(g(w) - c_0)^2$: the volume $\int_{|c-c_0|<\sqrt{2\varepsilon}} \rho(c)\,dc$ is a constant times $\varepsilon^\lambda(\log \frac{1}{\varepsilon})^{m-1}(1 + o(1))$ (Chapter 4).[3]

[3] The learning coefficient is the real log canonical threshold of K with respect to φ: two names for one number.

Lemma 9.2 (The data enter through two numbers). *Under Assumption 9.1 let $S_{xx} = \sum_i x_i^2$, $\kappa_n = S_{xx}^{1/2}$ and $Z_n = \kappa_n^{-1}\sum_i x_i\varepsilon_i$, and for $w \in W$ let $t = \kappa_n(g(w) - c_0)$.*

(i) The variables $\kappa_n^2 \sim \chi_n^2$ and $Z_n \sim \mathcal{N}(0,1)$ are independent.

(ii) $nK_n(w) = \frac{1}{2}t^2 - tZ_n$ *for every* w.
(iii) *At inverse temperature* β *the posterior law of* t *has the density proportional to* $e^{-\beta(t-Z_n)^2/2}\rho(c_0 + t/\kappa_n)$.
(iv) $G_n(\beta) - S = \mathbb{E}_x\big[D(\mu_x)\big]$, *where* $x \sim \mathcal{N}(0,1)$, μ_x *is the posterior law of* $u = (g(w) - c_0)x$, *and for a probability distribution* μ *on* $\mathbb{R}$ *and* $\varepsilon \sim \mathcal{N}(0,1)$

$$D(\mu) = -\mathbb{E}_\varepsilon \log \int e^{u\varepsilon - u^2/2}\mu(\mathrm{d}u).$$

Proof. The plan is to condition on the inputs and expand the squares. (i) Given the inputs, $\sum_i x_i\varepsilon_i$ is Gaussian with mean 0 and variance S_{xx}, so Z_n is standard normal given the inputs; its conditional law does not depend on them, which makes Z_n independent of the inputs, and of κ_n. (ii) With $c = g(w)$, $y_i - cx_i = \varepsilon_i - (c - c_0)x_i$; the normalizing constants cancel in $nK_n = nL_n(w) - nL_n(w_0)$, and

$$nK_n(w) = \tfrac{1}{2}\sum_i \big((\varepsilon_i - (c - c_0)x_i)^2 - \varepsilon_i^2\big) = \tfrac{1}{2}t^2 - tZ_n.$$

(iii) The posterior is proportional to $\varphi(w)e^{-n\beta K_n(w)}$, which depends on w only through c. For bounded F the definition of ρ gives, with $t = \kappa_n(c - c_0)$,

$$\int F(g(w))\, e^{-n\beta K_n(w)}\varphi(w)\,\mathrm{d}w = \int F(c)\, e^{-\beta(t^2/2 - tZ_n)}\rho(c)\,\mathrm{d}c,$$

and completing the square gives (iii). (iv) The input density cancels, and for a test pair with $\varepsilon = y - c_0x$, $\phi(y - cx)/\phi(y - c_0x) = e^{u\varepsilon - u^2/2}$, so $p^*_\beta(y \mid x)/q(y \mid x) = \int e^{u\varepsilon - u^2/2}\mu_x(\mathrm{d}u)$, with $\varepsilon \sim \mathcal{N}(0,1)$ under the truth. ∎

The function $D(\mu)$ is the Kullback–Leibler divergence from $\mathcal{N}(0,1)$ to the mixture $\int \mathcal{N}(u,1)\,\mu(\mathrm{d}u)$, and Computation 9.1 becomes the following inequality.

Lemma 9.3 (The divergence to a Gaussian mixture). *Let* μ *be a probability distribution on* $[-a,a]$, $a > 0$, *with mean* $\bar{u}$ *and second moment* $\gamma = \int u^2\mu(\mathrm{d}u)$. *Then*
(i) $0 \le D(\mu) \le \frac{1}{2}\gamma$, *and hence* $\big|D(\mu) - \frac{1}{2}\bar{u}^2\big| \le \frac{1}{2}\gamma$;
(ii) $\big|D(\mu) - \frac{1}{2}\bar{u}^2\big| \le \frac{1}{4}(1 + a^2)^2 e^{9a^2}\gamma^2$.

Proof. The plan is to split off the Gaussian at the mean, $\mathcal{N}(\bar{u},1)$, whose divergence from $\mathcal{N}(0,1)$ is exactly $\frac{1}{2}\bar{u}^2$, and to bound the rest with a quadratic bound for the logarithm, after Jensen's inequality has kept the mixture away from 0.

(i) The divergence is nonnegative. By Jensen's inequality for the logarithm, $-\log\int e^{u\varepsilon - u^2/2}\,\mathrm{d}\mu \le \int(\frac{1}{2}u^2 - u\varepsilon)\,\mathrm{d}\mu$, whose mean over ε is $\frac{1}{2}\gamma$; and $0 \le \frac{1}{2}\bar{u}^2 \le \frac{1}{2}\gamma$.

(ii) Write $r = u - \bar{u}$, so that $\int r\,\mathrm{d}\mu = 0$, $\gamma_0 = \int r^2\,\mathrm{d}\mu = \gamma - \bar{u}^2$, $|r| \le 2a$ and $|\bar{u}| \le a$. Since $u\varepsilon - \frac{1}{2}u^2 = (\bar{u}\varepsilon - \frac{1}{2}\bar{u}^2) + (r(\varepsilon - \bar{u}) - \frac{1}{2}r^2)$,

$$D(\mu) = \tfrac{1}{2}\bar{u}^2 + T, \qquad T = -\mathbb{E}_\varepsilon \log N(\varepsilon - \bar{u}), \qquad N(z) = \int e^{rz - r^2/2}\mu(\mathrm{d}u).$$

Jensen's inequality and $\int r\,\mathrm{d}\mu = 0$ give $N(z) \ge e^{-\gamma_0/2}$, and $N(z) \le e^{2a|z|}$, so T is finite. We use $|e^y - 1| \le |y|e^{|y|}$, $|e^y - 1 - y| \le \frac{1}{2}y^2 e^{|y|}$, $|r\bar{u}| \le 2a^2$, and $|rr'| \le 4a^2$ for an independent copy r' of r.

Lower bound. Since $-\log N \ge 1 - N$ and $\mathbb{E}_\varepsilon e^{r(\varepsilon-\bar{u}) - r^2/2} = e^{-r\bar{u}}$ (Tonelli's theorem exchanges the integrals),

$$T \ge -\int \left(e^{-r\bar{u}} - 1 + r\bar{u}\right)\mathrm{d}\mu \ge -\tfrac{1}{2}e^{2a^2}\bar{u}^2\gamma_0 \ge -\tfrac{1}{8}e^{2a^2}\gamma^2,$$

because $\bar{u}^2\gamma_0 \le \frac{1}{4}(\bar{u}^2 + \gamma_0)^2 = \frac{1}{4}\gamma^2$.

Upper bound. Taylor's theorem gives $-\log y = (1-y) + \frac{(1-y)^2}{2\xi^2}$ with ξ between $y > 0$ and 1; for $y = N(z)$, $\xi \ge e^{-\gamma_0/2}$, so $-\log N \le (1-N) + \frac{1}{2}e^{\gamma_0}(1-N)^2$. The mean of $1 - N(\varepsilon - \bar{u})$ is $1 - \int e^{-r\bar{u}}\,\mathrm{d}\mu \le 0$ by Jensen's inequality, so $T \le \frac{1}{2}e^{\gamma_0}\mathbb{E}_\varepsilon(N(\varepsilon - \bar{u}) - 1)^2$, and expanding the square with $\mathbb{E}_\varepsilon e^{(r+r')(\varepsilon-\bar{u}) - (r^2+r'^2)/2} = e^{rr' - (r+r')\bar{u}}$,

$$\mathbb{E}_\varepsilon(N-1)^2 = \iint \left(e^{rr'} - 1\right)e^{-(r+r')\bar{u}}\,\mathrm{d}\mu\,\mathrm{d}\mu + \left(\int e^{-r\bar{u}}\,\mathrm{d}\mu - 1\right)^2.$$

The part of the double integral with rr' in place of $e^{rr'} - 1$ is $\left(\int r(e^{-r\bar{u}} - 1)\,\mathrm{d}\mu\right)^2 \le e^{4a^2}\bar{u}^2\gamma_0^2$; the rest of it is at most $\frac{1}{2}\iint (rr')^2 e^{|rr'| + |r\bar{u}| + |r'\bar{u}|} \le \frac{1}{2}e^{8a^2}\gamma_0^2$; the last square is at most $\frac{1}{4}e^{4a^2}\bar{u}^4\gamma_0^2$. With $\bar{u}^2 \le a^2$ the sum is at most $\gamma_0^2 e^{8a^2}(\frac{1}{2} + a^2 + \frac{1}{4}a^4) \le \frac{1}{2}(1+a^2)^2 e^{8a^2}\gamma_0^2$, and since $\gamma_0 \le \gamma \le a^2$, $T \le \frac{1}{4}(1 + a^2)^2 e^{9a^2}\gamma^2$. ∎

The next lemma controls the rescaled posterior of Lemma 9.2(iii) uniformly in the data and identifies its limit.

Lemma 9.4 (The rescaled posterior). *Under Assumption 9.1 let $\beta > 0$. For $\kappa > 1$ and $z \in \mathbb{R}$ let $\pi_{\kappa,z}$ be the probability density proportional to $e^{-\beta(t-z)^2/2}\rho(c_0 + t/\kappa)$, and π_z the one proportional to $e^{-\beta(t-z)^2/2}w(t)$, with $w(t) = \ell_+ t^\alpha$ for $t > 0$ and $w(t) = \ell_-|t|^\alpha$ for $t < 0$. There are constants κ_1 and C such that:*

(i) $\int t^2\pi_z(t)\,\mathrm{d}t \le C(1+z^2)$ for every z, a bound that uses only $\alpha > -1$ and $\ell_\pm > 0$; and $\int t^2\pi_{\kappa,z}(t)\,\mathrm{d}t \le C(1+z^2)$ whenever $\kappa \ge \kappa_1$ and $|z| \le \sqrt{\kappa} - 1$;

(ii) for every z and $k \in \{1,2\}$, $\int t^k\pi_{\kappa,z}(t)\,\mathrm{d}t \to \int t^k\pi_z(t)\,\mathrm{d}t$ as $\kappa \to \infty$.

Proof. The plan is to rescale the prior density so that it converges to w, to bound it above and below on the scales that matter, and to compare the numerators and denominators of the posterior moments.

The rescaled density. Shrinking η_0, we may assume $\frac{1}{2}\ell_\pm \le \ell(c) \le 2\ell_\pm$ for $0 < \pm(c-c_0) < \eta_0$; put $\bar\ell = 2\max \ell_\pm$, $\underline{\ell} = \frac{1}{2}\min \ell_\pm$, and $\rho_\kappa(t) = \kappa^\alpha(\log\kappa)^{1-m}\rho(c_0 + t/\kappa)$, so that $\pi_{\kappa,z}$ is proportional to $e^{-\beta(t-z)^2/2}\rho_\kappa(t)$. For $0 < |t| < \eta_0\kappa$,

$$\rho_\kappa(t) = |t|^\alpha\Big(1 - \frac{\log|t|}{\log\kappa}\Big)^{m-1}\ell(c_0 + t/\kappa).$$

Let $\kappa \ge e$. The bracket lies between 1 and $1 + \log\frac{1}{|t|}$ if $|t| \le 1$, in $(0,1]$ if $1 \le |t| < \eta_0\kappa$, and is at least $\frac{1}{2}$ if $|t| \le \sqrt\kappa$. Hence, with $\log^+ y = \max(\log y, 0)$,

$$\rho_\kappa(t) \le \bar\ell\,|t|^\alpha\big(1 + \log^+\tfrac{1}{|t|}\big)^{m-1} \qquad (0 < |t| < \eta_0\kappa),$$

$$\rho_\kappa(t) \ge 2^{1-m}\underline{\ell}\,|t|^\alpha \qquad (0 < |t| \le \sqrt\kappa < \eta_0\kappa), \qquad \int_{|t|\ge\eta_0\kappa}\rho_\kappa(t)\,\mathrm{d}t \le \kappa^{1+\alpha}.$$

For each $t \ne 0$, $\rho_\kappa(t) \to w(t)$ as $\kappa \to \infty$; and w satisfies the two pointwise bounds for every $t \ne 0$.

Denominators. Let $|z| \le \sqrt\kappa - 1$, let ς be 1 if $z \ge 0$ and -1 otherwise, and let J be the interval between $z + \frac{1}{2}\varsigma$ and $z + \varsigma$. On J, $\frac{1}{2}(1+|z|) \le |t| \le 1 + |z| \le \sqrt\kappa$ and $e^{-\beta(t-z)^2/2} \ge e^{-\beta/2}$, so

$$\int e^{-\beta(t-z)^2/2}\rho_\kappa(t)\,\mathrm{d}t \ge c_1(1+|z|)^\alpha, \qquad c_1 = \tfrac{1}{2}e^{-\beta/2}2^{1-m-|\alpha|}\underline{\ell},$$

and the same holds with w in place of ρ_κ for every z.

Numerators. Let $\kappa_1 \ge e$ be so large that, for $\kappa \ge \kappa_1$, $\sqrt\kappa \le \frac{1}{2}\eta_0\kappa$ and $\kappa^{1+\alpha}s^2e^{-\beta s^2/8} \le 1$ for every $s \ge \eta_0\kappa$. On $|t| < \eta_0\kappa$ the upper bound reduces the integral of $t^2e^{-\beta(t-z)^2/2}\rho_\kappa(t)$ to $\bar\ell I(z)$, where, splitting at $|t| = 1$,

$$\begin{aligned} I(z) &= \int |t|^{2+\alpha}\big(1 + \log^+\tfrac{1}{|t|}\big)^{m-1}e^{-\beta(t-z)^2/2}\,\mathrm{d}t \\ &\le \int_{-1}^{1}|t|^\alpha\big(1 + \log\tfrac{1}{|t|}\big)^{m-1}\,\mathrm{d}t + \int |t|^{2+\alpha}e^{-\beta(t-z)^2/2}\,\mathrm{d}t \le C_2(1+|z|)^{2+\alpha} \end{aligned}$$

for a constant C_2: the first integral is finite because $\alpha > -1$, and in the second $|t|^{2+\alpha} \le 2^{1+\alpha}(|z|^{2+\alpha} + |t-z|^{2+\alpha})$ by convexity, as $2+\alpha \ge 1$. On $|t| \ge \eta_0\kappa$, $|t - z| \ge \frac{1}{2}|t|$ because $|z| \le \frac{1}{2}\eta_0\kappa$, and the integral is at most 1.

Conclusion of (i). Dividing, $\int t^2\pi_{\kappa,z} \le \big(\bar\ell C_2(1+|z|)^{2+\alpha} + 1\big)/\big(c_1(1+|z|)^\alpha\big) \le C(1+z^2)$, since $(1+|z|)^{-\alpha} \le (1+|z|)^2 \le 2(1+z^2)$. The same computation without the far part bounds the moments of π_z.

Convergence. Fix z; for large κ, $|z| \le \frac{1}{2}\eta_0\kappa$. For $k = 0, 1, 2$ the integrand $t^ke^{-\beta(t-z)^2/2}\rho_\kappa(t)\mathbf{1}\{|t| < \eta_0\kappa\}$ converges to $t^ke^{-\beta(t-z)^2/2}w(t)$ for $t \ne 0$ and is dominated by the integrable $\bar\ell|t|^{k+\alpha}(1 + \log^+\frac{1}{|t|})^{m-1}e^{-\beta(t-z)^2/2}$, while the far part tends to 0. By dominated convergence the three moment integrals converge, the zeroth to a positive limit, and so do the ratios. ∎

The weight of the limit posterior is homogeneous of degree α, $w(st) = s^\alpha w(t)$ for $s > 0$. Two integrations by parts compute the mean square of the posterior mean: one in the parameter, which uses the homogeneity and which we call the *virial identity*, after its counterpart in mechanics;[4] and one in the noise, *Stein's identity* $\mathbb{E}[Zf(Z)] = \mathbb{E}[f'(Z)]$ for $Z \sim \mathcal{N}(0,1)$ and absolutely continuous f with $\mathbb{E}|f'(Z)| < \infty$ (Theorem A.3; Stein (1981)).

[4] The virial theorem averages $x \cdot \nabla H$ against a Gibbs measure. Here $H = -\log\pi_z$, and the average of $t\,\partial_t H = \beta t(t-z) - \alpha$ is 1, by one integration by parts: that is (i) of Lemma 9.5.

Lemma 9.5 (Two integrations by parts). *Let $\alpha > -1$, $\ell_\pm > 0$, $\beta > 0$, and let π_z and w be as in Lemma 9.4, with mean $M(z)$ and variance $v(z)$. Then:*

(i) (virial identity) $\beta \int t(t-z)\pi_z(t)\,\mathrm{d}t = 1+\alpha$ for every $z \in \mathbb{R}$;
(ii) (Tweedie) M is continuously differentiable and $M'(z) = \beta v(z)$;
(iii) (Stein) for $Z \sim \mathcal{N}(0,1)$, $\mathbb{E}[ZM(Z)] = \beta\mathbb{E}[v(Z)]$, all terms being finite;
(iv) $\mathbb{E}[M(Z)^2] = \frac{1+\alpha}{\beta} - (1-\beta)\mathbb{E}[v(Z)]$; in particular $\mathbb{E}[M(Z)^2] = 1+\alpha = 2\lambda$ at $\beta = 1$.

Proof. The plan is to integrate by parts in t for (i), to differentiate under the integral sign for (ii), and to combine the two with Stein's identity. Write $\psi_z(t) = e^{-\beta(t-z)^2/2}$ and $h_k(z) = \int t^k \psi_z(t) w(t)\,\mathrm{d}t$, finite for $k \geq 0$ because $\alpha > -1$; then $M = h_1/h_0$ and $v = h_2/h_0 - M^2$.

(i) On $(0,\infty)$, $\beta(t-z)\psi_z = -\partial_t \psi_z$, and $tw(t) = \ell_+ t^{1+\alpha}$ tends to 0 as $t \downarrow 0$, because $1+\alpha > 0$, and has the derivative $(1+\alpha)w(t)$. Integrating by parts, the boundary terms vanish and

$$\beta \int_0^\infty t(t-z)\psi_z(t)w(t)\,\mathrm{d}t = (1+\alpha)\int_0^\infty \psi_z(t)w(t)\,\mathrm{d}t.$$

On $(-\infty,0)$, $tw(t) = -\ell_- |t|^{1+\alpha}$ also has the derivative $(1+\alpha)w(t)$, and the same computation applies. Add the two halves and divide by $h_0(z)$.

(ii) For $|z| \leq A$, $|t-z|\psi_z(t) \leq C_A e^{-\beta t^2/8}$, because $(t-z)^2 \geq \frac{1}{2}t^2 - A^2$; so the h_k may be differentiated under the integral sign, with $h_k' = \beta(h_{k+1} - z h_k)$, and $M' = (h_1' h_0 - h_1 h_0')/h_0^2 = \beta(h_2/h_0 - h_1^2/h_0^2) = \beta v$, which is continuous. At $\beta = 1$, $h_0'/h_0 = M - z$ is Tweedie's formula.

(iii) By Lemma 9.4(i), $M(z)^2 \leq \int t^2 \pi_z \leq C(1+z^2)$ and $0 \leq v(z) \leq C(1+z^2)$. So $\mathbb{E}|M'(Z)| = \beta\mathbb{E}v(Z) < \infty$, and Stein's identity gives $\mathbb{E}[ZM(Z)] = \mathbb{E}[M'(Z)] = \beta\mathbb{E}[v(Z)]$.

(iv) Since $\int t^2 \pi_z = M^2 + v$, (i) says $M(z)^2 + v(z) - zM(z) = (1+\alpha)/\beta$ for every z. Put $z = Z$ and take expectations, which are finite by (iii); then (iii) gives

$$\mathbb{E}[M(Z)^2] = \frac{1+\alpha}{\beta} - \mathbb{E}[v(Z)] + \mathbb{E}[ZM(Z)] = \frac{1+\alpha}{\beta} - (1-\beta)\mathbb{E}[v(Z)]. \qquad \blacksquare$$

For the symmetric weight $|t|^\alpha$ at $\beta = 1$, the virial identity is the heat-equation identity $h'' = \alpha h - z h'$ of Theorem A.3(iv) in other words: with Tweedie's $M = z + r$, $r = h'/h$, both say $r' = \alpha - zr - r^2$ (Exercise 9.2). The virial form needs neither the symmetry nor $\beta = 1$.

Theorem 9.6 (λ/n for one-coefficient models). *Under Assumption 9.1, at $\beta = 1$:*

(i) $n\big(\mathbb{E}[G_n] - S\big) \to \lambda$ as $n \to \infty$, the expectation being over the data;
(ii) $n(G_n - S)$ converges in law to $\frac{1}{2}M(Z)^2$, with $Z \sim \mathcal{N}(0,1)$ and M the mean of the limit posterior π_z of Lemma 9.4 at $\beta = 1$.

Proof. The plan is to replace the divergence by half the squared posterior mean (Lemma 9.3), to control the posterior on an event of overwhelming

probability (Lemma 9.4), and to pass to the limit, where Lemma 9.5 evaluates the mean square. The first three steps hold at every $\beta > 0$, and we prove them so, for Proposition 9.8. Write $\pi_n = \pi_{\kappa_n, Z_n}$ for the posterior of $t = \kappa_n(c - c_0)$ (Lemma 9.2(iii)), $\bar{t}_n$ and Θ_n for its mean and second moment, and $M = M_\beta$.

From the divergence to the posterior mean. For a test input x, the posterior law μ_x of $u = tx/\kappa_n$ has mean $x\bar{t}_n/\kappa_n$ and second moment $x^2\Theta_n/\kappa_n^2$, and lives on $|u| \le C_g|x|$. Put $\Delta_x = \left|D(\mu_x) - x^2\bar{t}_n^2/(2\kappa_n^2)\right|$. Lemma 9.3(i) gives $\Delta_x \le x^2\Theta_n/(2\kappa_n^2)$ for every x, and part (ii) with $a = C_gK$ gives, for $|x| \le K$,

$$\Delta_x \le C_K \frac{x^4\Theta_n^2}{\kappa_n^4}, \qquad C_K = \tfrac{1}{4}(1 + C_g^2K^2)^2 e^{9C_g^2K^2}.$$

Average over $x \sim \mathcal{N}(0,1)$, with $\mathbb{E}x^2 = 1$, $\mathbb{E}x^4 = 3$ and $e_K = \mathbb{E}[x^2\mathbf{1}\{|x| > K\}]$, and use Lemma 9.2(iv):

$$\left| n\big(G_n(\beta) - S\big) - \frac{n\bar{t}_n^2}{2\kappa_n^2} \right| \le 3C_K \frac{n\Theta_n^2}{\kappa_n^4} + \frac{e_K}{2}\frac{n\Theta_n}{\kappa_n^2}. \tag{9.1}$$

A good event. Let A_n be the event that $\kappa_n^2 \ge n/2$ and $|Z_n| \le (n/2)^{1/4} - 1$. Chernoff's bound gives $\mathbb{P}(\chi_n^2 < \frac{n}{2}) \le e^{n/4}\mathbb{E}e^{-\chi_n^2/2} = e^{n/4}2^{-n/2} \le e^{-n/11}$, and $\mathbb{P}(|Z_n| > s) \le 2e^{-s^2/2}$; so $n^2\mathbb{P}(A_n^c) \to 0$. Let n be so large that $\sqrt{n/2} \ge \kappa_1$. On A_n, $\kappa_n \ge \kappa_1$ and $|Z_n| \le \sqrt{\kappa_n} - 1$, so $\Theta_n \le C(1 + Z_n^2)$ by Lemma 9.4(i), while $n/\kappa_n^2 \le 2$; on every data set $|t| \le C_g\kappa_n$ under π_n, so $\Theta_n \le C_g^2\kappa_n^2$. Therefore

$$\mathbb{E}\left[\frac{n\Theta_n}{\kappa_n^2}\right] \le 2C\,\mathbb{E}[1 + Z^2] + nC_g^2\mathbb{P}(A_n^c) \to 4C,$$
$$\mathbb{E}\left[\frac{n\Theta_n^2}{\kappa_n^4}\right] \le \frac{4C^2}{n}\mathbb{E}\big[(1 + Z^2)^2\big] + nC_g^4\mathbb{P}(A_n^c) \to 0.$$

Taking expectations in (9.1), $\limsup_n \left|n\mathbb{E}[G_n(\beta) - S] - \frac{1}{2}\mathbb{E}[n\bar{t}_n^2/\kappa_n^2]\right| \le 2Ce_K$ for every K, and $e_K \to 0$ as $K \to \infty$.

The posterior mean converges. Let $\bar{t}(\kappa, z)$ be the mean of $\pi_{\kappa,z}$, a deterministic function, and $\Psi(\kappa) = \int_{|z| \le \sqrt{\kappa} - 1} \bar{t}(\kappa, z)^2\phi(z)\,\mathrm{d}z$. For $\kappa \ge \kappa_1$ the integrand is at most $C(1 + z^2)\phi(z)$ by Lemma 9.4(i), and it converges to $M(z)^2\phi(z)$ by part (ii); so $\Psi \le 2C$ and $\Psi(\kappa) \to \mathbb{E}[M(Z)^2]$ by dominated convergence. Let $B_n \supseteq A_n$ be the event that $\kappa_n^2 \ge n/2$ and $|Z_n| \le \sqrt{\kappa_n} - 1$. Since $Z_n \sim \mathcal{N}(0,1)$ independently of κ_n (Lemma 9.2(i)),

$$\mathbb{E}\left[\frac{n\bar{t}_n^2}{\kappa_n^2}\mathbf{1}_{B_n}\right] = \mathbb{E}\left[\frac{n}{\kappa_n^2}\mathbf{1}\{\kappa_n^2 \ge n/2\}\,\Psi(\kappa_n)\right].$$

The variable on the right is at most $4C$ and converges in probability to $\mathbb{E}[M(Z)^2]$, because $\kappa_n^2/n \to 1$ in probability by the law of large numbers; by bounded convergence the right side tends to $\mathbb{E}[M(Z)^2]$. Off B_n, $\bar{t}_n^2 \le \Theta_n \le C_g^2\kappa_n^2$, and $nC_g^2\mathbb{P}(B_n^c) \to 0$. With the previous step,

$$n\big(\mathbb{E}[G_n(\beta)] - S\big) \to \tfrac{1}{2}\mathbb{E}\big[M_\beta(Z)^2\big] \qquad \text{for every } \beta > 0. \tag{9.2}$$

Conclusion. At $\beta = 1$, Lemma 9.5(iv) gives $\frac{1}{2}\mathbb{E}[M(Z)^2] = \frac{1}{2}(1 + \alpha) = \lambda$, which is (i). For (ii), the expectation of the left side of (9.1) has limit superior

at most $2Ce_K$ for every K, so it tends to 0; with $n/\kappa_n^2 \to 1$ in probability and $\bar{t}_n^2 \le \Theta_n = O_p(1)$, $n(G_n - S) - \frac{1}{2}\bar{t}_n^2 \to 0$ in probability. The function $\Xi(\kappa) = \int \min(1, |\bar{t}(\kappa, z) - M(z)|)\phi(z)\,dz$ tends to 0 by Lemma 9.4(ii) and bounded convergence, and by independence $\mathbb{E}\min(1, |\bar{t}_n - M(Z_n)|) = \mathbb{E}\,\Xi(\kappa_n) \to 0$. So $n(G_n - S) - \frac{1}{2}M(Z_n)^2 \to 0$ in probability, and $\frac{1}{2}M(Z_n)^2$ has the law of $\frac{1}{2}M(Z)^2$ for every n; Slutsky's lemma (Definition A.13) gives (ii). ∎

Corollary 9.7 (The running models). *With inputs $\mathcal{N}(0,1)$, unit noise and the uniform prior on $[-1,1]^d$, at the truth $c_0 = 0$, $n(\mathbb{E}[G_n] - S)$ tends to $\frac{1}{2}$ for the regular line $y = ax$, the product model $y = abx$ and the chain $y = a_1 \cdots a_L x$, to $\frac{1}{4}$ for the cusp $y = ab^2x$, and to $\frac{1}{6}$ for the cubic cusp $y = ab^3x$. At a truth with $0 < |c_0| < 1$ it tends to $\frac{1}{2}$ for all five.*

Proof. The densities of c, given in Section 9.1 and Computation 9.3 and equal to $(\log \frac{1}{|c|})^{L-1}/(2(L-1)!)$ for the chain (Chapter 3), have the form of Assumption 9.1 at $c_0 = 0$ with $(\lambda, m) = (\frac{1}{2}, 1), (\frac{1}{2}, 2), (\frac{1}{2}, L), (\frac{1}{4}, 1), (\frac{1}{6}, 1)$ and $\ell(0\pm) > 0$; they are continuous and positive at every c_0 with $0 < |c_0| < 1$, the form with $\lambda = \frac{1}{2}$, $m = 1$; and $|c| \le 1$. ∎

The multiplicity entered only through the factor $(1 - \log|t|/\log\kappa)^{m-1}$, which tends to 1: the limit is λ whatever m is, but for $m \ge 2$ the approach is slow (Section 9.5). Gaussianity entered through the exact law of κ_n and the independence of Z_n and κ_n. The proof is the book's own, the one-coefficient case of the two integrations by parts in Watanabe's general proof (Section 9.4).[5]

[5] We have not found the argument in the literature. The chapter's script confirms $\mathbb{E}[M(Z)^2] = 1 + \alpha$ to ten digits for $\lambda = \frac{1}{6}, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1$ and for asymmetric weights.

9.4 The theorem: the equations of state

A Bayesian can score the posterior average (Bayes) or a single posterior draw (*Gibbs*), on test or training data: at inverse temperature β, besides $G_n(\beta)$, there are the Bayes training loss $T_n(\beta) = -\frac{1}{n}\sum_i \log p^*_\beta(X_i)$ and the Gibbs losses

$$G_{n,g}(\beta) = \mathbb{E}^\beta_w[L(w)]\,, \qquad T_{n,g}(\beta) = \mathbb{E}^\beta_w[L_n(w)]\,.$$

In the realizable case the four errors are $G_n(\beta) - S$, $T_n(\beta) - S_n$, $G_{n,g}(\beta) - S = \mathbb{E}^\beta_w[K(w)]$ and $T_{n,g}(\beta) - S_n = \mathbb{E}^\beta_w[K_n(w)]$; only the training losses can be computed from the data alone. The *functional variance* is $V_n = \sum_i \mathrm{Var}^\beta_w[\log p(X_i \mid w)]$.

Proposition 9.8 (The four errors of a one-coefficient model). *Under Assumption 9.1, let $\beta > 0$, let M and v be the mean and variance of the limit posterior π_z of Lemma 9.4, and let $\nu(\beta) = \frac{\beta}{2}\mathbb{E}[v(Z)]$ with $Z \sim \mathcal{N}(0,1)$. As $n \to \infty$,*

$$n\mathbb{E}\big[G_n(\beta) - S\big] \to \frac{\lambda - \nu}{\beta} + \nu,$$

$$n\mathbb{E}\big[G_{n,g}(\beta) - S\big] \to \frac{\lambda}{\beta} + \nu, \qquad n\mathbb{E}\big[T_{n,g}(\beta) - S_n\big] \to \frac{\lambda}{\beta} - \nu.$$

PROOF. The plan is to read the three limits off the proof of Theorem 9.6 and to evaluate them with Lemma 9.5. By (9.2) the first limit is $\frac{1}{2}\mathbb{E}[M(Z)^2]$, and Lemma 9.5(iv) with $\mathbb{E}v = 2\nu/\beta$ turns it into $\frac{1+\alpha}{2\beta} - \frac{(1-\beta)\nu}{\beta} = \frac{\lambda-\nu}{\beta} + \nu$. For the Gibbs errors, $K(w) = \frac{1}{2}(c-c_0)^2\mathbb{E}x^2 = t^2/(2\kappa_n^2)$ and $nK_n = \frac{1}{2}t^2 - tZ_n$ (Lemma 9.2), so

$$n\big(G_{n,g}(\beta) - S\big) = \frac{n\Theta_n}{2\kappa_n^2}, \qquad n\big(T_{n,g}(\beta) - S_n\big) = \tfrac{1}{2}\Theta_n - Z_n\bar{t}_n.$$

The step "the posterior mean converges" of that proof, with $\int t^2\pi_{\kappa,z}$ and $z\,\bar{t}(\kappa,z)$ in place of $\bar{t}(\kappa,z)^2$, both at most a constant times $1+z^2$ for $\kappa \ge \kappa_1$ and $|z| \le \sqrt{\kappa} - 1$, shows that the expectations on the good event converge to $\frac{1}{2}\mathbb{E}[M^2+v]$ and $\frac{1}{2}\mathbb{E}[M^2+v] - \mathbb{E}[ZM]$. Off it, $n\Theta_n/\kappa_n^2 \le nC_g^2$, $\Theta_n \le C_g^2\kappa_n^2$ and $|Z_n\bar{t}_n| \le C_g|Z_n|\kappa_n$, whose expectations on A_n^c are at most $nC_g^2\mathbb{P}(A_n^c)$, $C_g^2(\mathbb{E}\kappa_n^4)^{1/2}\mathbb{P}(A_n^c)^{1/2}$ and $C_g\sqrt{n}\,\mathbb{P}(A_n^c)^{1/2}$ by the Cauchy–Schwarz inequality; they tend to 0 because $\mathbb{E}\kappa_n^4 = n^2 + 2n$ and $n^2\mathbb{P}(A_n^c) \to 0$. Finally $\frac{1}{2}\mathbb{E}[M^2+v] = \frac{\lambda-\nu}{\beta} + \nu + \frac{\nu}{\beta} = \frac{\lambda}{\beta} + \nu$, and $\mathbb{E}[ZM] = \beta\mathbb{E}v = 2\nu$ by Lemma 9.5(iii). ■

THE BAYES TRAINING ERROR. Write $\Lambda_i = \log\big(p(X_i \mid w)/q(X_i)\big)$, which is $(c-c_0)x_i\varepsilon_i - \frac{1}{2}(c-c_0)^2x_i^2$. Then $n(T_n(\beta) - S_n) = -\sum_i \log \mathbb{E}_w^\beta\big[e^{\Lambda_i}\big]$. The cumulant expansion $\log \mathbb{E}_w^\beta\big[e^{\Lambda_i}\big] \approx \mathbb{E}_w^\beta[\Lambda_i] + \frac{1}{2}\operatorname{Var}_w^\beta(\Lambda_i)$, whose third-order terms total $O(n^{-1/2})$, gives $n(T_n(\beta) - S_n) \approx n(T_{n,g}(\beta) - S_n) - \frac{1}{2}V_n$, and $V_n \approx \operatorname{Var}_w^\beta(c)\sum_i x_i^2\varepsilon_i^2 \approx \operatorname{Var}_{\pi_n}(t)$, which tends in law to $v(Z)$. So $n\mathbb{E}[T_n(\beta) - S_n] \to \frac{\lambda}{\beta} - \nu - \frac{\nu}{\beta} = \frac{\lambda-\nu}{\beta} - \nu$, and $\frac{\beta}{2}\mathbb{E}V_n \to \nu$: our ν is Watanabe's singular fluctuation in both of his definitions below. ◇

The four limits satisfy two linear relations that do not involve λ: the generalization errors exceed the training errors by 2ν, for the Bayes and for the Gibbs pair, which is 2β times the gap ν/β between the Gibbs and the Bayes training errors. These are Watanabe's *equations of state*, named so because they hold for every true distribution, model, prior and singularity; here they are Stein's identity. The sum $2\lambda/\beta$ of the Gibbs limits is the virial identity, the only relation in which the singularity appears. The *singular fluctuation* ν is not determined by λ; Table 9.1 lists it for prior densities $\propto |c|^{2\lambda-1}$.

In the regular row $\nu = \lambda = \frac{1}{2}$ at every temperature, and at $\beta = 1$ the Gibbs error is twice the Bayes error. The product model has the same row, because the logarithm in its prior disappears in the limit: at the level of $\frac{1}{n}$ it is a regular one-parameter model, whose singularity shows only through $\lambda < \frac{d}{2}$ and the slow corrections of $m = 2$. In the cusps $\nu \ne \lambda$: the cubic cusp generalizes at $\frac{1}{6n}$, but its training error lies $\frac{2\nu}{n} = \frac{0.557}{n}$ below, the gap that WAIC estimates (Chapter 10). And the Bayes error depends on β through ν: for the cusp its limits are 0.192, 0.250 and 0.323 at $\beta = \frac{1}{2}, 1, 2$ (Exercise 9.8).

The two integrations by parts also work in $\mathbb{R}^k$, where a weight homogeneous of degree $\alpha > -k$ gives $\lambda = \frac{k+\alpha}{2}$. The last row of Table 9.1 is the variant $y = a\,(b^\top x) + \varepsilon$, $x \sim \mathcal{N}(0, I_2)$, with a uniform on $[-1,1]$ and b

prior density at the truth	λ	$\nu(\frac{1}{2})$	$\nu(1)$	$\nu(2)$	$\lambda - 2\nu(1)$	$\lambda + \nu(1)$
$\lvert c\rvert^{-2/3}$, cubic cusp	$\frac{1}{6}$	0.2263	0.2784	0.3458	−0.3901	0.4450
$\lvert c\rvert^{-1/2}$, cusp	$\frac{1}{4}$	0.3075	0.3487	0.3967	−0.4474	0.5987
constant, regular line	$\frac{1}{2}$	0.5	0.5	0.5	−0.5	1
$\lvert c\rvert^{1/2}$	$\frac{3}{4}$	0.6500	0.6099	0.5727	−0.4699	1.3599
$\lvert c\rvert^{-1}$ on $\mathbb{R}^2$, variant	$\frac{1}{2}$	0.6157	0.6997	0.7976	−0.8993	1.1997

Table 9.1. The singular fluctuation $\nu(\beta) = \frac{\beta}{2}\mathbb{E}[v(Z)]$ of the limit posterior $\propto e^{-\beta(t-Z)^2/2}|t|^{2\lambda-1}$, and the limits of n times the Bayes training and Gibbs generalization errors at $\beta = 1$ (the Bayes generalization limit is λ, the Gibbs training limit $\lambda - \nu$). The product model has the row of the regular line; the last row is the variant described in the text. Quadrature with 120 Gauss–Hermite nodes in Z, under which the virial and Stein identities hold to 10^{-12} (for the variant, 160 Gauss–Laguerre nodes in $\frac{1}{2}|Z|^2$). Script: `09-generalization-quartet.py`.

uniform on the unit disk: ab has the density $\frac{1}{\pi}(|c|^{-1} - 1)$ on the disk, so $k = 2$, $\alpha = -1$, $\lambda = \frac{1}{2}$, $m = 1$, and $\nu > \lambda$. We did not extend Lemma 9.4 to $\mathbb{R}^k$, so for this model the limits are argued, not proved.

Watanabe's theorem states the same structure for analytic models. We quote it with the hypotheses of the paper that proves it.

Theorem 9.9 (Watanabe's generalization theorem and the equations of state). *Let the model $p(x \mid w)$, the prior φ and the truth q satisfy the following conditions, with $f(x,w) = \log\big(q(x)/p(x \mid w)\big)$.*

(i) W is compact, the closure of an open set, and $W = \{w : \pi_1(w) \geq 0, \dots, \pi_k(w) \geq 0\}$ for analytic functions π_j; $\varphi = \varphi_0\varphi_1$ with $\varphi_0 > 0$ infinitely differentiable and $\varphi_1 \geq 0$ analytic.

(ii) For some $s \geq 6$, $w \mapsto f(\cdot, w)$ is an analytic map from an open set $W' \subset \mathbb{C}^d$ containing W to $L^s(q)$.

(iii) The truth is realizable, $W_0 = \{w : p(\cdot \mid w) = q\} \neq \emptyset$; on an open set $W^* \subset \mathbb{C}^d$ containing W the envelope $\bar f(x) = \sup_{W^*}|f(x,w)|$ has $\mathbb{E}_X[\bar f(X)^s] < \infty$, and $\int \bar f(x)^2 \sup_{K(w)\leq\epsilon} p(x \mid w)\,\mathrm{d}x < \infty$ for some $\epsilon > 0$.

Let $\beta > 0$, let λ be the learning coefficient, and write $B_{\mathrm{g}} = G_n(\beta) - S$, $B_{\mathrm{t}} = T_n(\beta) - S_n$, $G_{\mathrm{g}} = G_{n,\mathrm{g}}(\beta) - S$, $G_{\mathrm{t}} = T_{n,\mathrm{g}}(\beta) - S_n$. Then:

(a) nB_{g}, nB_{t}, nG_{g}, nG_{t} converge in law to random variables B^*_{g}, B^*_{t}, G^*_{g}, G^*_{t}, and their expectations converge to those of the limits;

(b) (the equations of state) $\mathbb{E}B^*_{\mathrm{g}} - \mathbb{E}B^*_{\mathrm{t}} = 2\beta(\mathbb{E}G^*_{\mathrm{t}} - \mathbb{E}B^*_{\mathrm{t}})$ and $\mathbb{E}G^*_{\mathrm{g}} - \mathbb{E}G^*_{\mathrm{t}} = 2\beta(\mathbb{E}G^*_{\mathrm{t}} - \mathbb{E}B^*_{\mathrm{t}})$;

(c) $nG_{\mathrm{g}} + nG_{\mathrm{t}} \to 2\lambda/\beta$ in probability, so $\mathbb{E}G^*_{\mathrm{g}} + \mathbb{E}G^*_{\mathrm{t}} = 2\lambda/\beta$;

(d) with the singular fluctuation $\nu = \nu(\beta) = \beta(\mathbb{E}G^*_{\mathrm{t}} - \mathbb{E}B^*_{\mathrm{t}})$,

$$\mathbb{E}B^*_{\mathrm{g}} = \frac{\lambda-\nu}{\beta}+\nu, \quad \mathbb{E}B^*_{\mathrm{t}} = \frac{\lambda-\nu}{\beta}-\nu, \quad \mathbb{E}G^*_{\mathrm{g}} = \frac{\lambda}{\beta}+\nu, \quad \mathbb{E}G^*_{\mathrm{t}} = \frac{\lambda}{\beta}-\nu.$$

In particular, at $\beta = 1$, $\mathbb{E}[G_n] - S = \frac{\lambda}{n} + o(\frac{1}{n})$.

This is Theorem 1, Theorem 2, Theorem 3 and Corollary 3 of Watanabe

(2010c), numbered as in its arXiv version, whose conditions (A.1)–(A.3) are our (i)–(iii); the value $\mathbb{E}B_g^* = \lambda$ at $\beta = 1$ ends his Corollary 2, which he says was first found in his papers of 1999 and 2001 (Watanabe 1999, 2001). Watanabe (2010a, Lemma 3) states the limits of (d) for the Bayes errors without realizability, with the proof for unrealizable singular truths in Watanabe (2010b), under relatively finite variance (Theorem 8.6), essential uniqueness and $s \geq 8$, with $L(w_0)$, $L_n(w_0)$ in place of S, S_n and $\nu = \lim_n \frac{\beta}{2}\mathbb{E}[V_n]$; at $\beta = 1$ this is $\mathbb{E}[G_n] = L(w_0) + \frac{\lambda}{n} + o(\frac{1}{n})$ and $\mathbb{E}[T_n] = L(w_0) + \frac{\lambda - 2\nu}{n} + o(\frac{1}{n})$ (Watanabe 2024).

Proof sketch. The plan is the proof of Theorem 9.6 on a parameter space on which K is a monomial, with the Gaussian observation Z replaced by a Gaussian process. As in the proof of Theorem 8.6, the posterior concentrates on $\{K < \epsilon\}$, Hironaka's theorem (Theorem A.8) gives local coordinates with $K = u^{2k}$ and $f = u^k a(x,u)$, and with $t = nu^{2k}$ the energy is $nK_n = t - \sqrt{t}\,\xi_n(u)$ up to the sign of u^k, where ξ_n converges in law to a Gaussian process ξ (Theorem A.11). To leading order the posterior becomes the measure $t^{\lambda-1}e^{-\beta t + \beta\sqrt{t}\,\xi(u)}\,dt\,du^*$, the analogue of π_Z, with averages $\langle\cdot\rangle$, and a cumulant expansion writes n times each error as such an average up to $o_p(1)$ (Watanabe 2024, §5). The integration by parts $\int_0^\infty \partial_t(t^\lambda e^{-\beta t+\beta\sqrt{t}\,\xi})\,dt = 0$ is the virial identity $\beta\langle t\rangle = \lambda + \frac{\beta}{2}\langle\sqrt{t}\,\xi\rangle$, which gives (c) since $nG_g \approx \langle t\rangle$ and $nG_t \approx \langle t - \sqrt{t}\,\xi\rangle$. Watanabe represents ξ as the limit in law of $n^{-1/2}\sum_i a(X_i,u)g_i$ with independent standard Gaussian g_i, and Stein's identity in each g_i gives (b); he writes that Theorems 2 and 3 are "essentially proved by partial integration" (Watanabe 2010c, §5). Then (d) is linear algebra, and moment bounds on $\sup_u|\xi_n|$ give (a). Omitted are the resolution, the empirical processes on the resolved space and the moment bounds: §4 and the appendix of Watanabe (2010c), and Chapters 5 and 6 of Watanabe (2009). ■

In a one-coefficient model the sketch's t is our $\frac{1}{2}t^2$ to leading order and its $\sqrt{t}\,\xi$ is our tZ. The free energy of Chapter 8 gives λ by a different road, and predicts the slow convergence measured in Section 9.5.

From the free energy to the generalization error. Since $F_{n+1} - F_n = -\log p^*(X_{n+1})$, the mean over X_{n+1} gives the exact identity $\mathbb{E}[G_n] = \mathbb{E}[F_{n+1}] - \mathbb{E}[F_n]$. Differencing the expansion $\mathbb{E}[F_n] - nS = \lambda\log n - (m-1)\log\log n + c + o(1)$ (Watanabe 2024) term by term, which needs an unchecked regularity condition, gives $\mathbb{E}[G_n] - S \approx \frac{\lambda}{n} - \frac{m-1}{n\log n}$. In one-coefficient models a second argument agrees: to first order in $1/\log\kappa_n$ the rescaled weight $|t|^\alpha(1 - \log|t|/\log\kappa_n)^{m-1}$ is $|t|^{\alpha'}$ with $\alpha' = \alpha - (m-1)/\log\kappa_n$, and Lemma 9.5(iv) gives $\frac{1}{2}(1+\alpha') \approx \lambda - \frac{m-1}{\log n}$. ◇

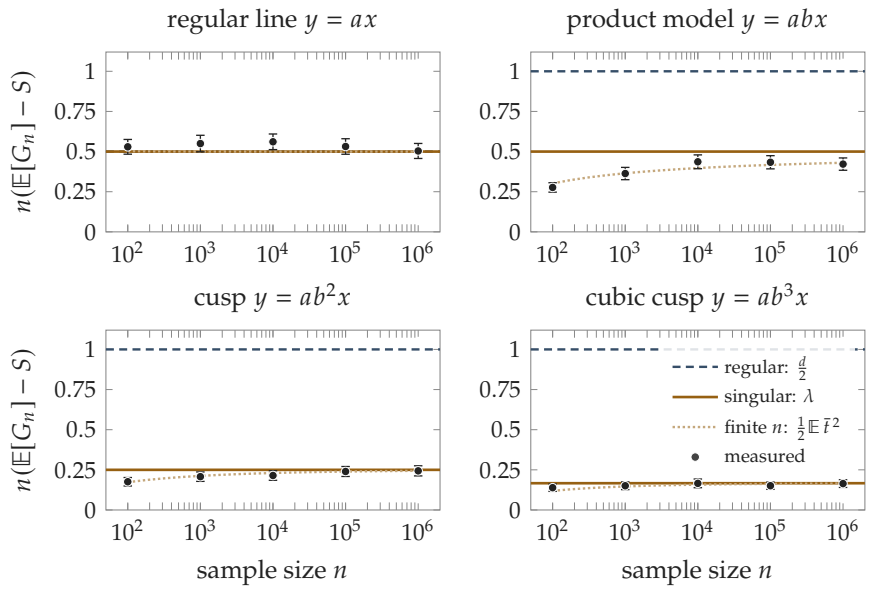


Figure 9.2. The Bayes generalization error, times n, of four one-coefficient models at the truth 0; inputs $\mathcal{N}(0,1)$, unit noise, uniform prior on $[-1,1]^d$. Dots: means over 1000 independent data sets at each n, each by exact quadrature, with Student-t 95% intervals (seeds $900 + 10j + \log_{10} n$). Bronze: the limit λ of Theorem 9.6. Dotted bronze: the finite-n prediction $\frac{1}{2}\mathbb{E}[\bar{t}(\sqrt{n}, Z)^2]$, half the mean square of the exact posterior mean of t at $\kappa_n = \sqrt{n}$ (for the regular line, the closed form of Computation 9.2). Slate dashed: the regular prediction $\frac{d}{2}$, equal to λ for the regular line. Script: `09-generalization-onecoef.py`.

9.5 Measuring it

The script `09-generalization-onecoef.py` computes the Bayes generalization error of the regular line, the product model, the cusp and the cubic cusp, data set by data set, without sampling. By Lemma 9.2 a data set enters only through the independent $\kappa_n^2 \sim \chi_n^2$ and $Z_n \sim \mathcal{N}(0,1)$, so the script draws 1000 independent data sets through these two statistics for each model and each $n \in \{10^2, 10^3, \ldots, 10^6\}$ (seed $900 + 10j + \log_{10} n$, $j = 0, \ldots, 3$ in the order just listed).[6] It computes $G_n - S = \mathbb{E}_x[D(\mu_x)]$ by 16-point Gauss–Hermite rules in x and ε, and composite Gauss–Legendre rules in t after the substitution $t = \pm u^p$, $p = 1, 4, 2, 3$, which removes the singularity of ρ at 0. Tripling the rules changes $n(G_n - S)$ by less than 3×10^{-9}; on the regular line, with the uniform prior on $[-1,1]$, the result matches the flat-prior closed form of Computation 9.2 to within 1.8×10^{-7} on each of its 5000 data sets. The run takes about a minute.

Figure 9.2 shows the result. None of the three two-parameter models comes near the regular prediction 1: at $n = 10^6$ the cubic cusp measures 0.164 ± 0.024 against $\lambda = \frac{1}{6}$ and the cusp 0.244 ± 0.031 against $\frac{1}{4}$ (the product model is discussed below), while the one-parameter regular line measures 0.504 ± 0.047 against $\frac{1}{2}$. Nineteen of the twenty intervals contain the dotted prediction; the twentieth, the regular line at $n = 10^4$ with 0.561 ± 0.049 against the exact 0.50003, is 2.5 standard errors high, as one interval in

[6] Drawing the two statistics instead of n pairs changes nothing in law, and at $n = 10^6$ saves 2×10^9 Gaussian draws per model.

twenty may be. Half the squared posterior mean accounts for nearly all of the error: its mean differs from the exact error by at most 0.009 at $n = 100$, 0.001 at $n = 1000$, and 10^{-4} from $n = 10^4$ on, as an $O(n^{-2})$ remainder should.

The singular models approach λ slowly. The product model has $m = 2$: its finite-n prediction is 0.302, 0.365, 0.397, 0.417 and 0.430 at $n = 10^2, \dots, 10^6$, close to the heuristic $\frac{1}{2} - \frac{1}{\log n}$ (0.283, 0.355, 0.391, 0.413, 0.428), and the measured values 0.277 ± 0.029, 0.363 ± 0.038, 0.436 ± 0.043, 0.433 ± 0.041 and 0.422 ± 0.038 follow it; the prediction is still 0.4785 at $n = 10^{20}$ and 0.4957 at $n = 10^{100}$.[7] The cusps have $m = 1$ but a second term in their prior densities, the -1 in $\frac{1}{2}(|c|^{-1/2} - 1)$, which multiplies the rescaled weight by $1 - |t|^{1/2}\kappa_n^{-1/2}$, and by $1 - |t|^{2/3}\kappa_n^{-2/3}$ for the cubic cusp. Their predictions approach λ like $n^{-1/4}$ and $n^{-1/3}$: $\frac{1}{4} - 0.2303 = 0.0197$ at $n = 10^4$ and 0.0059 at 10^6, and $\frac{1}{6} - 0.1648 = 0.0019$ at 10^6. The second term is the next pole of the zeta function (Chapter 4), at $-\frac{1}{2}$, and the rate is the distance between the poles; we know of no theorem that says this in general. A single data set's error does not concentrate: at $n = 10^6$ its standard deviation is 0.75, 0.51 and 0.38 for the regular line, the cusp and the cubic cusp, against 0.71, 0.51 and 0.42 for the limit law (Exercise 9.4).

[7] At $n = 10^{100}$ the product model is still 0.9% below its limit, as $\frac{1}{2} - \frac{1}{\log n}$ predicts.

9.6 The ledger

The expected generalization error of the Bayes predictive distribution is λ/n, not $\frac{d}{2n}$. For one-coefficient regressions with Gaussian inputs and noise this is proved here completely, for every learning coefficient and multiplicity: the divergence to the predictive is half the squared posterior mean, the rescaled posterior tends to a Gaussian location posterior with a homogeneous prior weight, and two integrations by parts show that the mean square of its mean is 2λ. At other temperatures the same proof gives $(\frac{\lambda-\nu}{\beta} + \nu)\frac{1}{n}$, and the Gibbs errors take the values of Watanabe's equations of state, which in these models are Stein's identity (the Bayes training error is argued); Watanabe's theorem proves all four for analytic models. The cubic cusp generalizes three times better than one regular parameter; the product model, at the level of $\frac{1}{n}$, exactly as well as one.

LEDGER · $y = ab^3x$	
QUANTITY	the Bayes generalization error $\mathbb{E}[G_n] - S$, the expected divergence from the truth to the predictive
REGULAR THEORY	$\frac{d}{2n} + o(\frac{1}{n})$ (Theorem 1.7); training error $-\frac{d}{2n}$; Gibbs error $\frac{d}{n}$
SINGULAR THEORY	$\frac{\lambda}{n} + o(\frac{1}{n})$ at $\beta = 1$: proved for one-coefficient models (Theorem 9.6), Watanabe's theorem in general (Theorem 9.9); $(\frac{\lambda-\nu}{\beta} + \nu)\frac{1}{n}$ at temperature β; training error $(\lambda - 2\nu)\frac{1}{n}$
RUNNING EXAMPLE	the cubic cusp $y = ab^3x$: $\frac{1}{6n}$ against $\frac{1}{n}$ for $d = 2$; the error is half the mean square of the posterior mean $M(Z)$, and $\mathbb{E}M^2 = \frac{1}{3}$; $\nu(1) = 0.278$

MEASURED	$n(\mathbb{E}[G_n] - S)$ at $n = 10^6$: 0.164 ± 0.024 (cubic cusp), 0.244 ± 0.031 (cusp), 0.504 ± 0.047 (regular line), 0.422 ± 0.038 (product model, $m = 2$, approaching $\frac{1}{2}$ like $\frac{1}{2} - \frac{1}{\log n}$)
RESTS ON	the exact posterior average at $\beta = 1$; a fixed realizable truth as $n \to \infty$; one coefficient, Gaussian inputs and noise (the proof); analyticity, compactness, and realizability or relatively finite variance (Watanabe)

9.7 Where the theory stops

WHAT WAS ASSUMED. The theorems describe the exact posterior average at $\beta = 1$. A point estimate does not inherit λ/n: for the cubic cusp the maximum-likelihood estimate, also the maximum a posteriori estimate under the uniform prior on (a, b), has error $\frac{1}{2n}$, three times the Bayes error (Exercise 9.6), and for analytic models the maximum-likelihood error is governed by the supremum of a Gaussian process over the true parameters (Watanabe 2024). The truth is fixed as n grows: at a truth $c_0 \neq 0$ of the cubic cusp, $\lambda = \frac{1}{2}$ and the gain disappears. For a truth at distance δ from the singularity, the limit posterior with its truth moved to $\sqrt{n}\,\delta$ gives an error of about $\frac{1}{6n}$ while $n\delta^2$ is small and $\frac{1}{2n}$ once it is large, but $0.77/n$ at $\sqrt{n}\,\delta = 2.8$: near the singularity the bet is lost, in a crossover like those of Chapter 12. Our proof used one coefficient and Gaussian inputs and noise; Watanabe's theorem needs analyticity, which ReLU networks lack, a compact parameter set, moments of order 6, and a prior of the form $\varphi_0\varphi_1$: with a smooth but non-analytic prior factor $e^{-1/\|w\|^2}$ at the only true parameter the errors need not be proportional to $\frac{1}{n}$ (Watanabe 2010c, §2). And the theorem is about the mean over data sets: under the limit law of Theorem 9.6(ii), a data set of the cubic cusp still has an error above $\frac{1}{2n}$ with probability 0.084.

WHERE IT BREAKS. The $o(\frac{1}{n})$ term can be large at every practical sample size: the product model's prediction is 0.430 at $n = 10^6$ against its limit $\frac{1}{2}$, and a second term in the prior slows the cusps to $n^{-1/4}$ and $n^{-1/3}$. In unrealizable pairs without relatively finite variance the theorem fails: for Watanabe's non-renormalizable example, whose free energy Chapter 8 measures, he argues that the learning curve is of order $n^{-2/3}$ (Watanabe 2010b). The theorem says nothing when d is comparable to n, and trained networks are not posterior averages (Chapter 15). In Bayesian deep learning, Wenzel et al. (2020) found carefully sampled posterior predictives worse than SGD point estimates and improved by cold posteriors, $\beta > 1$,[8] while Izmailov et al. (2021) found tempering unnecessary and attribute the effect largely to data augmentation. Even in a one-coefficient model the best temperature depends on where the prior puts its mass (Proposition 9.8); whether ν has anything to do with the cold-posterior effect has, as far as we know, not been tested.

[8] Cold means a temperature $T = 1/\beta$ below 1: the likelihood is raised to the power $\beta > 1$, as though each observation were counted β times.

OPEN PROBLEMS. Our computations suggest a remainder $-\frac{m-1}{n \log n}$ when $m \geq 2$, and of order $n^{-1-(\lambda_2-\lambda)}$ when the next pole of the zeta function is

at $-\lambda_2$; we know of no general theorem on the rate of the $o(\frac{1}{n})$ term. The singular fluctuation is not determined by λ: Watanabe conjectured that it is governed by a generalized Fisher information on the set of true parameters (Watanabe 2010c, §5), and we know of no closed form for ν even for the power-law class of Table 9.1. Non-asymptotic bounds on $\mathbb{E}[G_n] - S$ for singular models, and the learning curve when d grows with n, are open.

Exercises

9.1 CONSOLIDATION. (The fourth order.) Continue Computation 9.1 to degree 4: with $\bar{u}$ and γ_0 the posterior mean and variance of u, show that $D_x = \frac{1}{2}\bar{u}^2 + \frac{1}{4}\gamma_0^2 - \frac{1}{2}\bar{u}^2\gamma_0 + O(u^6)$. Check it against Computation 9.2 expanded in v, and show that its mean, $-\frac{3}{4n^2} + O(n^{-3})$, is the difference between the exact $\frac{1}{2n} + \frac{1}{4n^2}$ and $\frac{1}{2(n-2)}$. *(Hint: the terms of degree* 4 *come from* $\frac{1}{2}\mathbb{E}A^2$, $\frac{1}{3}\mathbb{E}A^3$ *and* $\frac{1}{4}\mathbb{E}A^4$; $\mathbb{E}[\varepsilon^2 \mathrm{He}_2(\varepsilon)] = 2$.*)*

9.2 CONSOLIDATION. (The Riccati equation.) For the weight $|t|^\alpha$ at $\beta = 1$ let $r = h'/h$, so that $M = z + r$. Show that $r' = \alpha - zr - r^2$ (Theorem A.3(iv)) is Lemma 9.5(i) rewritten with $M' = v$, and derive $\mathbb{E}[M(Z)^2] = 1 + \alpha$ by applying Stein's identity to r.

9.3 CONSOLIDATION. (Gibbs and Bayes on the regular line.) The limit posterior of the regular line at inverse temperature β is $\mathcal{N}(Z, 1/\beta)$. Compute the four limits of Theorem 9.9(d) directly, $\frac{1}{2}$, $-\frac{1}{2}$, $\frac{1}{2} + \frac{1}{2\beta}$ and $\frac{1}{2\beta} - \frac{1}{2}$, and check the equations of state with $\nu = \frac{1}{2}$.

9.4 CONSOLIDATION. (One data set.) From Theorem 9.6(ii), show that the limit law of $n(G_n - S)$ has standard deviation $\frac{1}{\sqrt{2}}$ for the regular line (0.423 for the cubic cusp, numerically), and that the limit probability of $n(G_n - S) > \frac{1}{2}$ is $2\mathbb{P}(Z > z_*)$ with $M(z_*) = 1$: 0.084 for the cubic cusp, where $z_* = 1.730$, against 0.317 for the regular line.

9.5 BY HAND. (A one-sided coefficient.) For $y = a^2 x$ with a uniform on $[-1, 1]$ and the truth 0, show that $c = a^2$ has the density $\frac{1}{2\sqrt{c}}$ on $(0, 1]$, that $\lambda = \frac{1}{4}$, and that the virial identity holds for the one-sided weight $t^{-1/2}\mathbf{1}\{t > 0\}$, so that $\mathbb{E}[M(Z)^2] = \frac{1}{2}$. ★ Adapt the denominator bound of Lemma 9.4, which must allow a factor $e^{-z^2/2}$ for $z < 0$, and prove $n(\mathbb{E}[G_n] - S) \to \frac{1}{4}$. *(Hint: for* $z < 0$, *integrate over* $0 < t < 1/(1 + |z|)$.*)*

9.6 BY HAND. (Maximum likelihood against Bayes.) For the cubic cusp at the truth 0, show that the maximum-likelihood estimate of $c = ab^3$, clipped to $[-1, 1]$, is also the maximum a posteriori estimate under the uniform prior on (a, b), and that $n\mathbb{E}[\mathrm{KL}(q \,\|\, p(\cdot \mid \hat{w}))] \to \frac{1}{2}$. For which λ in the power-law class is the Bayes predictive worse?

9.7 BY HAND. (The chain.) For $y = a_1 a_2 a_3 x$ with the uniform prior on $[-1, 1]^3$ and the truth 0, $\rho(c) = \frac{1}{4}(\log \frac{1}{|c|})^2$. Show that the effective-exponent argument gives $n(\mathbb{E}[G_n] - S) \approx \frac{1}{2} - \frac{2}{\log n}$, and compare with the finite-n predictions 0.303, 0.363, 0.396 at $n = 10^4, 10^6, 10^8$.

9.8 VARIATION. (Temperature.) For the weight $|t|^\alpha$, show that π_z at inverse temperature β is the law of $\beta^{-1/2}t'$, with t' from the limit posterior at $\beta = 1$ given the observation $\sqrt{\beta}z$. Deduce $\lim_n n\mathbb{E}[G_n(\beta) - S] = \frac{1}{2\beta}\mathbb{E}[M_1(\sqrt{\beta}Z)^2]$, which tends to $2\lambda^2$ as $\beta \to 0$ and to $\frac{1}{2}$ as $\beta \to \infty$. Which temperature is best for the cusp at its singular truth, and which for $\lambda = \frac{3}{4}$?

9.9 VARIATION. (An asymmetric prior.) Let $\ell_+ \neq \ell_-$ in Assumption 9.1. Explain why the limit of Theorem 9.6 does not depend on ℓ_+/ℓ_- while ν does; compute $\nu(1)$ for $\lambda = \frac{1}{4}$ and $\ell_-/\ell_+ = 0.2$ (answer: 0.3065, against 0.3487).

9.10 MEASUREMENT. (The training error.) Extend the script of Section 9.5 to the Bayes training error of the cubic cusp, on simulated data sets of size $n = 10^3$ and 10^4 (the training points are now needed). Compare $n\mathbb{E}[T_n - S_n]$ with $\lambda - 2\nu = -0.390$ and $\frac{1}{2}\mathbb{E}[V_n]$ with $\nu = 0.278$.

9.11 RESEARCH. ★ (Rates.) Show that for the cusp $n(\mathbb{E}[G_n] - S) - \frac{1}{4}$ is of exact order $n^{-1/4}$. Conjecture the rate in terms of the first two poles of the zeta function and their orders, test it on the cubic cusp, the chain, and the density $\frac{3}{16}|c|^{-1/2}(1 + |c|)$ on $[-1, 1]$, whose second pole is at $-\frac{3}{4}$, and find what the literature proves.

Notes and References

For regular models the learning curve $\frac{d}{2n}$ is classical: with the training error $-\frac{d}{2n}$ it makes up Akaike's penalty $\frac{d}{n}$ (Akaike 1974), and Amari and Murata (1993) proved the entropic errors $H_0 \pm \frac{m}{2t}$ of a faithful regular model with m parameters and t examples for maximum likelihood, with similar relations for Bayes and Gibbs learning. Summed over n, the learning curve is the redundancy of the Bayes mixture, $\frac{d}{2}\log n$ plus an explicit constant (Clarke and Barron 1990).

For singular models Watanabe obtained λ/n first as the increase of the free energy (Watanabe 1999, 2001). The equations of state, posted in 2007 and published as Watanabe (2010c), prove it directly; the 2009 book counts them among its four main formulas (Watanabe 2009). Watanabe (2010a) states the Bayes limits for unrealizable truths and proves WAIC asymptotically equivalent to cross-validation (Chapter 10); Watanabe (2024) reviews the theory at $\beta = 1$, and the 2018 textbook (Watanabe 2018) develops it with exercises.

Tweedie's formula was first published by Robbins (1956), who credits personal correspondence with Maurice Kenneth Tweedie (Efron 2011); Efron bases empirical Bayes corrections for selection bias on it and gives the second-order form, our $M' = \beta v$. The proofs of Theorem 9.6 and Proposition 9.8 are the book's own. The local learning coefficient of Lau et al. (2025) estimates λ near a trained network (Chapter 13); its link to that network's test loss runs through a posterior average that SGD does not compute (Chapter 15).

TRANSLATING THE LITERATURE. Watanabe (2010c) writes E_w for our $\mathbb{E}_w^\beta[\cdot]$; his Bayes and Gibbs errors B_g, B_t, G_g, G_t are our $G_n(\beta) - S$, $T_n(\beta) - S_n$, $G_{n,g}(\beta) - S$, $T_{n,g}(\beta) - S_n$, his B_g^* is the limit in law of nB_g, and his losses BL_g, BL_t, GL_g, GL_t are our $G_n(\beta)$, $T_n(\beta)$, $G_{n,g}(\beta)$, $T_{n,g}(\beta)$. Watanabe (2010a) writes $BgL(n)$ and $BtL(n)$ for our $G_n(\beta)$ and $T_n(\beta)$, $Bg(n) = BgL(n) - L(w_0)$, and $V(n)$ for our V_n. Watanabe (2024) works at $\beta = 1$ with θ and $\pi(\theta)$ for our w and $\varphi(w)$, and $\langle\cdot\rangle$ for the renormalized posterior. Efron (2011) writes $z \mid \mu \sim \mathcal{N}(\mu, \sigma^2)$ and $l = \log f$ for the log marginal density: his z, μ, σ^2 and f are our Z, t, $1/\beta$ and h up to a constant factor.

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10 WAIC

Estimating generalization without knowing λ

THE MODEL — $(1-a)\mathcal{N}(0,1) + a\mathcal{N}(b,1)$

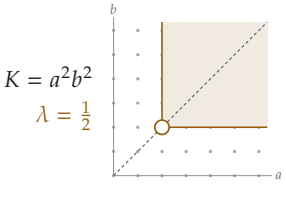
IN THIS CHAPTER. The expected generalization error of a singular model is λ/n, and λ is rarely known; this chapter asks whether the error can be estimated from the data and one posterior. By hand we compute the Bayes training loss, the functional variance V_n and leave-one-out cross-validation for the regular line and the product model, where V_n tends to 1 and not to the parameter count 2, and we find that all three are values of one cumulant generating function. We prove that leave-one-out cross-validation and WAIC $= T_n + V_n/n$ differ by $O(n^{-2})$ in mean, under a moment condition that we verify for the product model and that Watanabe states under general hypotheses; with the equations of state this makes WAIC unbiased for the Bayes generalization loss (Theorem 10.8). Over 500 data sets per sample size the bias of WAIC is zero within its interval for the mixture and the product model, the penalty d/n of AIC overstates the error by about $1/n$, and the errors of leave-one-out cross-validation and of the predictive have correlation below -0.98. The mixture itself violates the analyticity that the theorems assume, and without relatively finite variance the learning curve changes its order.

10.1 The question: the test loss of a posterior

A Bayesian neural network sampled by Markov chain Monte Carlo, or by stochastic-gradient Langevin dynamics (Welling and Teh 2011), ends as a cloud of parameter vectors drawn, approximately, from a posterior, and it predicts by averaging over the cloud. How well will it predict data it has not seen? A held-out set answers at the price of data, and leave-one-out cross-validation at the price of n posteriors. As of 2026, one computes instead, from the single cloud, the widely applicable information criterion of Watanabe (2010c),[1] or leave-one-out cross-validation estimated by importance sampling; the R package `loo` computes both (Vehtari et al. 2017), and both come with an "effective number of parameters" (Gelman et al. 2014). This chapter asks what these numbers estimate when the model is singular, and why they need no knowledge of the learning coefficient.

Throughout, the inverse temperature is $\beta = 1$: $\mathbb{E}_w[\cdot]$ is the ordinary posterior expectation, Var_w the posterior variance, and $p^*(x) = \mathbb{E}_w\big[p(x \mid w)\big]$ the predictive density. We write $\ell_i(w) = \log p(X_i \mid w)$ for the log-likelihood of one observation.

[1] Gelman et al. (2014) read the acronym as the *Watanabe–Akaike* information criterion; "widely applicable" is Watanabe's own name.



Definition 10.1 (Generalization, training, WAIC and leave-one-out). The *Bayes generalization loss* is $G_n = -\mathbb{E}_X[\log p^*(X)]$, the *Bayes training loss* is $T_n = -\frac{1}{n}\sum_{i=1}^n \log p^*(X_i)$, the *functional variance* is $V_n = \sum_{i=1}^n \mathrm{Var}_w[\ell_i(w)]$, and the *widely applicable information criterion* is $\mathrm{WAIC} = T_n + V_n/n$. The *Bayes leave-one-out cross-validation loss* is

$$\mathrm{LOO} = -\frac{1}{n}\sum_{i=1}^n \log p(X_i \mid X_{-i}^n), \qquad p(x \mid X_{-i}^n) = \mathbb{E}_w^{(-i)}[p(x \mid w)],$$

with $\mathbb{E}_w^{(-i)}$ the posterior given the $n-1$ observations other than X_i. Subtracting $L(w_0)$ from G_n, and $L_n(w_0)$ from T_n, WAIC and LOO, gives the corresponding *errors*. In the realizable case $L(w_0) = S$, $L_n(w_0) = S_n$, and $G_n - S = \mathrm{KL}(q \,\|\, p^*)$.

The four losses live on the per-sample scale of the log loss; the book never multiplies WAIC by $2n$ (Section 10.4.1 converts to the scale of the statistical software). Only G_n needs the unknown q. The others need the data and the posterior, and LOO seems to need n further posteriors as well.

THE REGULAR ANSWER · estimating the generalization loss. If the model is regular and realizable, the training loss of the maximum-likelihood fit $\hat{w}$ underestimates its generalization loss by $\frac{d}{n}$ on average, and $\mathrm{AIC} = L_n(\hat{w}) + \frac{d}{n}$ corrects it (Akaike 1974). The Bayes predictive obeys the same rule: $\mathbb{E}[G_n] - S = \frac{d}{2n} + o(\frac{1}{n})$ (Theorem 1.7) and $\mathbb{E}[T_n] - S = -\frac{d}{2n} + o(\frac{1}{n})$ (Watanabe 2010c), so $T_n + \frac{d}{n}$ is unbiased to $o(\frac{1}{n})$. Leave-one-out cross-validation with maximum-likelihood fits chooses models as AIC does, asymptotically (Stone 1977).

For a singular model both halves of this answer fail. The first half fails in its penalty: the Bayes training loss underestimates the Bayes generalization loss by $2\nu/n$ on average, with ν the singular fluctuation of Chapter 9, and not by d/n. For the product model $y = abx$ and for the mixture $(1-a)\mathcal{N}(0,1) + a\mathcal{N}(b,1)$ at the truth $\mathcal{N}(0,1)$, both with $d = 2$, we will find $2\nu = 1$: the optimism of the training loss is half of what AIC charges (Section 10.2). The second half fails in its target. Watanabe's theorem gives $\mathbb{E}[G_n] - S = \lambda/n + o(1/n)$ (Theorem 9.9), and λ is what we do not know. Even when we know it, the formula is poor at practical sample sizes: for the product model, with $\lambda = \frac{1}{2}$, the measured $n\,\mathbb{E}[G_n - S]$ is 0.40 at $n = 10^3$ and still 0.39 at $n = 10^5$ (Section 10.5), because the multiplicity $m = 2$ adds a correction of order $1/(n \log n)$.

So the question of this chapter is: *is there a statistic, computed from the data and one posterior, whose expectation is $\mathbb{E}[G_n]$ up to $o(1/n)$ whatever λ, ν and m are; and how is it related to leave-one-out cross-validation, which seems to need n posteriors?*

10.2 By hand: three readings of one function

The statistics T_n, V_n and LOO are posterior averages of functions of the $\ell_i(w)$. For the regular line the posterior is Gaussian and they come out in closed form; for the product model the posterior of the coefficient $c = ab$ is a Gaussian multiplied by a logarithm; and leave-one-out cross-validation turns out to need no further posterior at all.

COMPUTATION 10.1 · WAIC AND LEAVE-ONE-OUT FOR THE REGULAR LINE

1 The model is $y = ax + \varepsilon$ with $x, \varepsilon \sim \mathcal{N}(0,1)$ and the flat prior on $\mathbb{R}$; the uniform prior on $[-1,1]$ changes what follows by terms exponentially small in n when the truth is interior. With $S = \sum_i x_i^2$ and $\hat{a} = \sum_i x_i y_i / S$ the posterior is $\mathcal{N}(\hat{a}, 1/S)$. Write $r_i = y_i - \hat{a}x_i$ for the residuals and $h_i = x_i^2/S$ for the *leverages*; note $\sum_i h_i = 1$.

2 The predictive at x is $\mathcal{N}(\hat{a}x, 1 + x^2/S)$, so

$$T_n = \frac{1}{n}\sum_{i=1}^{n}\Big[\tfrac{1}{2}\log 2\pi + \tfrac{1}{2}\log(1+h_i) + \frac{r_i^2}{2(1+h_i)}\Big].$$

3 Put $a = \hat{a} + Z/\sqrt{S}$ with $Z \sim \mathcal{N}(0,1)$ under the posterior. Then $\ell_i(a) = -\tfrac{1}{2}\log 2\pi - \tfrac{1}{2}r_i^2 + r_i x_i Z/\sqrt{S} - \tfrac{1}{2}x_i^2 Z^2/S$, and since $\operatorname{Var} Z = 1$, $\operatorname{Cov}(Z, Z^2) = 0$ and $\operatorname{Var} Z^2 = 2$,

$$V_n = \sum_{i=1}^{n}\big(r_i^2 h_i + \tfrac{1}{2}h_i^2\big).$$

Here $\sum_i r_i^2 h_i = \sum_i r_i^2 x_i^2 / \sum_i x_i^2 \to \mathbb{E}[\varepsilon^2 x^2]/\mathbb{E}[x^2] = 1$ and $\sum_i h_i^2 \le \max_i h_i \to 0$ in probability: $V_n \to 1 = d$.

4 Without X_i the posterior is $\mathcal{N}(\hat{a}_{-i}, 1/(S - x_i^2))$. Its predictive at x_i has variance $1 + x_i^2/(S - x_i^2) = 1/(1-h_i)$, and $y_i - \hat{a}_{-i}x_i = r_i/(1-h_i)$, the classical leave-one-out residual. Hence

$$\mathrm{LOO} = \frac{1}{n}\sum_{i=1}^{n}\Big[\tfrac{1}{2}\log 2\pi - \tfrac{1}{2}\log(1-h_i) + \frac{r_i^2}{2(1-h_i)}\Big].$$

5 Subtracting the two previous steps from this one,

$$n(\mathrm{LOO} - \mathrm{WAIC}) = \sum_{i=1}^{n}\Big[-\tfrac{1}{2}\log(1-h_i^2) - \tfrac{1}{2}h_i^2 + \frac{r_i^2 h_i^3}{1-h_i^2}\Big],$$

in which every term is $O(h_i^3)$, and $h_i^3 \approx x_i^6/n^3$. The leading part is $\sum_i r_i^2 h_i^3 \approx \sum_i \varepsilon_i^2 x_i^6/n^3$, whose expectation is $15/n^2$ because $\mathbb{E}[x^6] = 15$: so $n^3\,\mathbb{E}[\mathrm{LOO} - \mathrm{WAIC}] \to 15$.

6 Expanding $\log(1+h_i)$ and $1/(1+h_i)$ to first order in h_i, the second step gives $nT_n = nL_n(\hat{a}) + \tfrac{1}{2}\big(1 - \sum_i r_i^2 h_i\big) + O_p(\tfrac{1}{n})$, and the bracket is $O_p(n^{-1/2})$. So $T_n = L_n(\hat{a}) + O_p(n^{-3/2})$ and $\mathrm{WAIC} = L_n(\hat{a}) + \frac{1}{n} + O_p(n^{-3/2})$: for a regular model WAIC is AIC.

Three facts are visible. The functional variance counts the parameter, $V_n \to d = 1$. Leave-one-out cross-validation needs no refitting: the leverages of the single fit give it exactly. And leave-one-out and WAIC are much closer to each other, $O(n^{-3})$, than either is to its target, whose fluctuations are of order $n^{-1/2}$. The measured $n^3\,\mathbb{E}[\mathrm{LOO} - \mathrm{WAIC}]$ is 14.99 ± 0.05 at $n = 10^5$ (Section 10.5).

The product model is the same computation with a twist. Its likelihood depends on (a, b) only through $c = ab$, and under the uniform prior on $[-1, 1]^2$ the coefficient c has the density $\rho(c) = \frac{1}{2}\log\frac{1}{|c|}$ on $[-1, 1]$ (Chapter 3). A Gaussian likelihood of width $n^{-1/2}$ against this density is a Gaussian with a slowly varying weight, and that is the whole difference from the regular line.

COMPUTATION 10.2 · THE FUNCTIONAL VARIANCE OF $y = abx$

1 The truth is $c = 0$. With $S = \sum_i x_i^2$ and $\hat{c} = \sum_i x_i y_i / S$, the posterior of c is proportional to $\rho(c)\exp(-\frac{S}{2}(c - \hat{c})^2)$ on $[-1, 1]$, and every posterior average of a function of the ℓ_i is an average over this one-dimensional posterior.

2 Rescale: $t = \sqrt{S}\,c$, $\mu = \sqrt{S}\,\hat{c}$, $L = \frac{1}{2}\log S$. Since $\log\frac{1}{|c|} = L - \log|t|$, the posterior density of t is proportional to

$$(L - \log|t|)\, e^{-(t-\mu)^2/2} \qquad (|t| \le \sqrt{S}).$$

Under the truth $\mu = \sum_i x_i y_i / \sqrt{S}$ is exactly $\mathcal{N}(0, 1)$ given the inputs. As $n \to \infty$ the weight $L - \log|t|$ varies little, relative to its size L, where the Gaussian factor lives: the posterior of t tends to $\mathcal{N}(\mu, 1)$, the posterior of a regular one-parameter model.

3 With $r_i = y_i - \hat{c}x_i$ we have $\ell_i(c) - \ell_i(\hat{c}) = r_i x_i (c - \hat{c}) - \frac{1}{2}x_i^2(c - \hat{c})^2$, and as in Computation 10.1,

$$V_n = \mathrm{Var}(t)\,\frac{\sum_i r_i^2 x_i^2}{S} + O_p(\tfrac{1}{n}),$$

the posterior moments of t being bounded in probability. Hence $V_n \to 1$: the product model has the functional variance of one parameter, half its parameter count.

4 To first order in $1/L$ the weight shifts the variance: $\mathrm{Var}(t) = 1 - \mathrm{Cov}\big((U-\mu)^2, \log|U|\big)/L + O(L^{-2})$ with $U \sim \mathcal{N}(\mu, 1)$. Differentiating $h_0(\mu) = \mathbb{E}\log|\mu + Z|$, $Z \sim \mathcal{N}(0, 1)$, twice under the Gaussian shows that this covariance is $h_0''(\mu)$, and $h_0'(\mu) = \sqrt{2}\,D(\mu/\sqrt{2})$ with Dawson's function $D(u) = e^{-u^2}\int_0^u e^{s^2}\,\mathrm{d}s$, so $h_0''(\mu) = 1 - \mu h_0'(\mu)$. At $\mu = 0$ it is 1; averaged over $\mu \sim \mathcal{N}(0, 1)$ it is $\frac{1}{2}$ (Exercise 10.6). Therefore

$$\mathbb{E}[V_n] = 1 - \frac{1}{\log n} + O\big((\log n)^{-2}\big),$$

which is 0.891 at $n = 10^4$; the script measures 0.896 ± 0.007.

5 With t approximately $\mathcal{N}(\mu, 1)$, the product model behaves at order $1/n$ like the regular line in the coordinate c: $n(G_n - S) \approx \frac{1}{2}\mu^2$, $n(T_n - S_n) \approx -\frac{1}{2}\mu^2$ and $n(\mathrm{LOO} - S_n) \approx 1 - \frac{1}{2}\mu^2$. The expectations are $\frac{1}{2}$, $-\frac{1}{2}$ and $\frac{1}{2}$, and $n(G_n - S) + n(\mathrm{LOO} - S_n) \approx 1 = 2\lambda$ for every data set.

The last two steps are expansions, not proofs; Theorem 10.8 below makes the limits exact, and the measurements of Section 10.5 check the first-order corrections. What the computation shows is the size of WAIC's penalty: $V_n/n \approx 1/n$, while AIC charges $d/n = 2/n$. The penalty is set by the posterior's fluctuation, not by the number of coordinates.

The chapter's model, the mixture $p(x \mid a, b) = (1-a)\phi(x) + a\phi(x-b)$ with $a \in [0,1]$, $b \in [-1,1]$, uniform prior and truth $\phi = \mathcal{N}(0,1)$, reduces to the same picture. Writing $g_{a,b}(x) = \log(p(x \mid a,b)/\phi(x)) = \log\bigl(1 + a(e^{bx-b^2/2} - 1)\bigr)$, for each fixed x we have $g_{a,b}(x) = abx + O(ab^2 + a^2b^2)$ near $(0,0)$: to first order the mixture is a regression on the single coefficient $c = ab$, and c again has the density $\frac{1}{2}\log\frac{1}{|c|}$ on $[-1,1]$ (this is why we take $b \in [-1,1]$ rather than the range $[-2,2]$ of Chapter 2, a change that affects neither λ nor m). Its Kullback–Leibler divergence $K(a,b) = \frac{1}{2}a^2(e^{b^2} - 1)(1 + O(a))$ is comparable to a^2b^2, so $\lambda = \frac{1}{2}$ and $m = 2$ (Chapter 2). There is one difference that will matter. For $b \neq 0$, K is infinitely differentiable but not analytic in a at $a = 0$: the coefficient of a^k in its Taylor series is $(-1)^k\mathbb{E}[(e^{bX-b^2/2} - 1)^k]/k$, which grows like $e^{k(k-1)b^2/2}$, so the series has radius of convergence zero. The likelihood ratio e^{bx} is unbounded in x, and the analyticity that Watanabe's theorems assume fails.[2]

The third computation needs no model at all.

[2] The same failure occurs for the tanh neuron with Gaussian inputs, where the poles of tanh at $\pm i\pi/2$ meet the unbounded support of the input; with inputs on $[-1,1]$ it disappears.

Lemma 10.2 (Leave-one-out from one posterior). *Let $n \geq 2$, let $p(x \mid w) > 0$ for all x and w, and suppose that $Z = \int \varphi(w) \prod_{j=1}^n p(X_j \mid w)\,\mathrm{d}w$ and $Z_{-i} = \int \varphi(w) \prod_{j\neq i} p(X_j \mid w)\,\mathrm{d}w$ are finite and positive for each i. Then $p(X_i \mid X_{-i}^n) = 1/\mathbb{E}_w\bigl[1/p(X_i \mid w)\bigr]$, so that*

$$\mathrm{LOO} = \frac{1}{n}\sum_{i=1}^n \log \mathbb{E}_w\left[\frac{1}{p(X_i \mid w)}\right].$$

If moreover $\mathbb{E}|\log p(X_i \mid X_{-i}^n)| < \infty$, then $\mathbb{E}[\mathrm{LOO}] = \mathbb{E}[G_{n-1}]$ exactly, where G_{n-1} is the generalization loss of the posterior from $n-1$ observations.

Proof. The posterior given X_{-i}^n has density $\varphi \prod_{j\neq i} p(X_j \mid w)/Z_{-i}$, so integrating $p(X_i \mid w)$ against it gives $p(X_i \mid X_{-i}^n) = Z/Z_{-i}$; the full posterior density times $1/p(X_i \mid w)$, finite because $p > 0$, is $\varphi \prod_{j\neq i} p(X_j \mid w)/Z$, whose integral is $\mathbb{E}_w\bigl[1/p(X_i \mid w)\bigr] = Z_{-i}/Z$. For the second statement, the pair (X_{-i}^n, X_i) has the law of (X^{n-1}, X) with $X \sim q$ independent of X^{n-1}, so $\mathbb{E}[-\log p(X_i \mid X_{-i}^n)] = \mathbb{E}[-\log p_{n-1}^*(X)] = \mathbb{E}[G_{n-1}]$; average over i. ∎

The identity $p(X_i \mid X_{-i}^n) = 1/\mathbb{E}_w\bigl[1/p(X_i \mid w)\bigr]$ is *importance-sampling leave-one-out*: the full posterior, reweighted by $1/p(X_i \mid w)$, is the posterior without

X_i.[3] With exact posterior averages it is exact, not an approximation. The second statement says that leave-one-out cross-validation is an unbiased estimate of the expected generalization loss with $n-1$ observations, for every model, prior and n; it needs neither regularity nor λ.

What one had to notice is that the three statistics are three readings of one function. Let $\kappa_i(\alpha) = \log \mathbb{E}_w\left[p(X_i \mid w)^\alpha\right]$. Then

$$T_n = -\frac{1}{n}\sum_{i=1}^{n} \kappa_i(1), \qquad V_n = \sum_{i=1}^{n} \kappa_i''(0), \qquad \mathrm{LOO} = \frac{1}{n}\sum_{i=1}^{n} \kappa_i(-1), \tag{10.1}$$

the first by definition, the second because $\kappa_i''(0)$ is the posterior variance of ℓ_i, the third by Lemma 10.2. In $\kappa_i(1) + \kappa_i(-1)$ the odd powers of α cancel and the first surviving term is $\kappa_i''(0)$. That is why the training loss plus the variance, which is WAIC, reproduces leave-one-out cross-validation; for the regular line all derivatives of order three and higher are tiny, which is Computation 10.1 in one line.

[3] Watanabe attributes the importance-sampling form to Gelfand et al. (1992); cross-validation itself goes back at least to Geisser (1975) and Stone (1977).

10.3 The theorem: leave-one-out is WAIC to order n^{-2}

The derivatives of κ_i at points other than 0 are cumulants of ℓ_i under reweighted posteriors.

Definition 10.3 (Tilted posteriors). For data X^n, an index i and $\alpha \in \mathbb{R}$, the *tilted posterior* is

$$\mathbb{E}_w^{(i,\alpha)}[g] = \frac{\mathbb{E}_w\left[g(w)\, p(X_i \mid w)^\alpha\right]}{\mathbb{E}_w\left[p(X_i \mid w)^\alpha\right]}.$$

It is the posterior in which X_i has weight $1+\alpha$: at $\alpha = -1$ the posterior without X_i, at $\alpha = 1$ the posterior with X_i counted twice. With p_0 the density of the optimal parameter, $f(x,w) = \log p_0(x) - \log p(x \mid w)$ is the log density ratio, so that $K(w) = \mathbb{E}_X[f(X,w)]$.

Lemma 10.4 (Tilted cumulants). *Suppose that $w \mapsto \ell_i(w)$ is bounded on W. Then κ_i is infinitely differentiable on $\mathbb{R}$, and $\kappa_i^{(k)}(\alpha)$ is the k-th cumulant of $\ell_i(w)$ under $\mathbb{E}_w^{(i,\alpha)}$. In particular $\kappa_i''(0) = \mathrm{Var}_w[\ell_i(w)]$, and for every α and every constant c,*

$$|\kappa_i^{(4)}(\alpha)| \le 32\, \mathbb{E}_w^{(i,\alpha)}\left[(\ell_i(w) - c)^4\right].$$

Proof. The plan is to recognize the increments of κ_i as a cumulant generating function, and then to compare the fourth cumulant with the fourth moment. Since ℓ_i is bounded, $\mathbb{E}_w\left[e^{\alpha \ell_i}\right]$ is finite and positive for every real α, and differentiation under the expectation is justified by dominated convergence; so κ_i is smooth. For fixed α and real s,

$$\kappa_i(\alpha + s) - \kappa_i(\alpha) = \log \frac{\mathbb{E}_w\left[e^{s\ell_i} e^{\alpha \ell_i}\right]}{\mathbb{E}_w\left[e^{\alpha \ell_i}\right]} = \log \mathbb{E}_w^{(i,\alpha)}\left[e^{s \ell_i(w)}\right],$$

the cumulant generating function of ℓ_i under the tilted posterior; its derivatives at $s = 0$ are the tilted cumulants. For the bound, let $Y = \ell_i(w) - c$ under $\mathbb{E}_w^{(i,\alpha)}$; cumulants of order four do not change when a constant is added, so $\kappa_i^{(4)}(\alpha)$ is the fourth cumulant of Y. In terms of the central moments $\mu_k = \mathbb{E}(Y - \mathbb{E}Y)^k$ it is $\mu_4 - 3\mu_2^2$, and $0 \le \mu_2^2 \le \mu_4$ by Jensen's inequality; hence $-2\mu_4 \le \mu_4 - 3\mu_2^2 \le \mu_4$. Finally Minkowski's inequality gives $\mu_4^{1/4} \le (\mathbb{E}Y^4)^{1/4} + |\mathbb{E}Y| \le 2(\mathbb{E}Y^4)^{1/4}$, so $2\mu_4 \le 32\,\mathbb{E}Y^4$. ∎

The hypothesis of the theorem is a bound on the fourth moments of the log density ratio under the tilted posteriors, averaged over the observations and over the data.

Assumption 10.5 (Tilted fourth moments). There are C and n_0 such that for every $n \ge n_0$

$$M_n := \sup_{|\alpha| \le 1} \mathbb{E}\Big[\frac{1}{n}\sum_{i=1}^{n} \mathbb{E}_w^{(i,\alpha)}\big[f(X_i, w)^4\big]\Big] \le \frac{C}{n^2}, \tag{10.2}$$

where $\mathbb{E}$ is the expectation over the data X^n.

The scaling is the natural one: over the bulk of the posterior $f(X_i, w)$ is of order $n^{-1/2}$, so its fourth moment is of order n^{-2}. The supremum over α is outside the expectation, which matters for verifying the assumption.

Theorem 10.6 (Leave-one-out equals WAIC to order n^{-2}; Watanabe 2010a, Theorem 1 and Corollary 1, at $\beta = 1$). *Let $\beta = 1$, and suppose that $w \mapsto \log p(x \mid w)$ is bounded on W for every x. Then for every data set*

$$\mathrm{LOO} - \mathrm{WAIC} = \frac{1}{n}\sum_{i=1}^{n} \int_0^1 \frac{(1-t)^3}{6}\big[\kappa_i^{(4)}(t) + \kappa_i^{(4)}(-t)\big]\,\mathrm{d}t, \tag{10.3}$$

and, with M_n as in (10.2),

$$\mathbb{E}|\mathrm{LOO} - \mathrm{WAIC}| \le \tfrac{8}{3} M_n.$$

Under Assumption 10.5, $\mathbb{E}|\mathrm{LOO} - \mathrm{WAIC}| \le \frac{8C}{3n^2}$ for $n \ge n_0$; in particular $\mathrm{LOO} = \mathrm{WAIC} + O_p(n^{-2})$.

Proof. The plan is to write both statistics through the functions κ_i, expand $\kappa_i(1) + \kappa_i(-1)$ by Taylor's formula with integral remainder, and bound the remainder by Lemma 10.4. By (10.1) and the definition $\mathrm{WAIC} = T_n + V_n/n$,

$$\mathrm{LOO} - \mathrm{WAIC} = \frac{1}{n}\sum_{i=1}^{n}\big[\kappa_i(1) + \kappa_i(-1) - \kappa_i''(0)\big].$$

The function κ_i is smooth by Lemma 10.4, and $\kappa_i(0) = 0$. Taylor's formula with integral remainder, applied to κ_i on $[0,1]$ and to $\alpha \mapsto \kappa_i(-\alpha)$ on $[0,1]$,

gives

$$\kappa_i(\pm 1) = \pm\kappa_i'(0) + \tfrac{1}{2}\kappa_i''(0) \pm \tfrac{1}{6}\kappa_i'''(0) + \int_0^1 \frac{(1-t)^3}{6}\,\kappa_i^{(4)}(\pm t)\,\mathrm{d}t.$$

Adding the two signs cancels the odd terms, and summing over i proves (10.3).

For the bound, take absolute values in (10.3), then expectations over the data. The integrand is nonnegative and continuous in t, so Tonelli's theorem exchanges $\mathbb{E}$ and $\int_0^1 \mathrm{d}t$; for each fixed t the expectation of $\frac{1}{n}\sum_i|\kappa_i^{(4)}(\pm t)|$ is at most the supremum over $|\alpha| \le 1$; and $\int_0^1 (1-t)^3/6\,\mathrm{d}t = 1/24$. Hence

$$\mathbb{E}|\mathrm{LOO} - \mathrm{WAIC}| \le \frac{2}{24}\sup_{|\alpha|\le 1}\mathbb{E}\Big[\frac{1}{n}\sum_{i=1}^{n}|\kappa_i^{(4)}(\alpha)|\Big].$$

Lemma 10.4 with $c = \ell_i(w_0) = \log p_0(X_i)$, for which $\ell_i(w) - c = -f(X_i, w)$, bounds each $|\kappa_i^{(4)}(\alpha)|$ by $32\,\mathbb{E}_w^{(i,\alpha)}\big[f(X_i,w)^4\big]$, and $32/12 = 8/3$. Under Assumption 10.5 the right side is at most $8C/(3n^2)$, and Markov's inequality turns the bound in mean into $O_p(n^{-2})$. ∎

Three remarks place the theorem.[4]

Where the assumption comes from. Assumption 10.5 is the case $k = 4$ of Watanabe (2010a, Lemma 1(2), eq. (14)), which bounds $\limsup_n n^{k/2}M_k(n)$ for $k \le 4$, where $M_k(n)$ is the supremum over $|\alpha| \le 1 + \beta$ of the expected average of $\mathbb{E}_w\big[|f|^k e^{\alpha f}\big]/\mathbb{E}_w\big[e^{\alpha f}\big]$. Since $e^{\alpha f} = (p_0/p)^\alpha$, these are our tilted moments with the tilt $-\alpha$, and the range $|\alpha| \le 2$ contains ours. Watanabe states the lemma under the hypotheses of Theorem 10.8 and refers its proof to eq. (32) of Watanabe (2010c) and eq. (6.59) of Watanabe (2009) for realizable singular models, where an analogous remainder is bounded, to Watanabe (2010d) for regular unrealizable models, and to Watanabe (2010b) for singular unrealizable ones. His proof of Theorem 1 writes the Taylor remainder at an intermediate point that depends on the data; the integral form used here keeps the supremum over α outside the expectation, where eq. (14) has it.

Other temperatures and the leading term. At inverse temperature β the third cumulants no longer cancel, and $\mathrm{LOO} - \mathrm{WAIC} = \frac{\beta-\beta^2}{2}Y_3 + O_p(n^{-2})$ with $Y_3 = \frac{1}{n}\sum_i \kappa_i'''(0) = O_p(n^{-3/2})$ (Watanabe 2010a, Corollary 1); see Exercise 10.1. At $\beta = 1$, with sixth moments bounded in the same way, the argument gives $\mathrm{LOO} - \mathrm{WAIC} = \frac{1}{12}Y_4 + O(n^{-3})$ in mean, $Y_4 = \frac{1}{n}\sum_i \kappa_i^{(4)}(0)$, the form of Watanabe's Corollary 1.

Gaussian posteriors do better. For the regular line Computation 10.1 gives n^{-3}: a Gaussian posterior makes the fourth cumulant of ℓ_i smaller than the bound by a further factor $1/n$.

The assumption can be checked by hand for the product model, so that Theorem 10.6 is completely proved there.

[4] The bound is in mean, which is stronger than the statement in probability of Watanabe's Corollary 1; it needs the assumption only in the averaged form (10.2).

Proposition 10.7 (The product model satisfies the assumption). *For $y = abx + \varepsilon$ with $x, \varepsilon \sim \mathcal{N}(0,1)$, the truth $ab = 0$ and the uniform prior on $[-1,1]^2$, Assumption 10.5 holds with $n_0 = 18$. Consequently $\mathbb{E}|\mathrm{LOO} - \mathrm{WAIC}| = O(n^{-2})$.*

Proof. The plan is to reduce to the coefficient $c = ab$, to show that the tilted posteriors of c are Gaussians weighted by ρ, to bound their moments by Gaussian moments, and to average over the data. Write $s_i = x_i y_i$, $t_i = x_i^2$, $S = \sum_j t_j$, $S_{xy} = \sum_j s_j$, and for each i let $S^{(i)} = S - t_i$ and $U_i = S_{xy} - s_i$ be the sums without the i-th observation.

Reduction. The density of x is common to model and truth, so $f(X_i, w) = \log\phi(y_i) - \log\phi(y_i - cx_i) = -cs_i + \frac{1}{2}c^2 t_i$, and $f^4 \le 8(c^4 s_i^4 + \frac{1}{16}c^8 t_i^4)$. The likelihood and f depend on w only through c, whose prior density is $\rho(c) = \frac{1}{2}\log\frac{1}{|c|}$ on $[-1,1]$. Multiplying $\exp(cS_{xy} - \frac{1}{2}c^2 S)$ by $\exp(\alpha(cs_i - \frac{1}{2}c^2 t_i))$ and completing the square shows that under $\mathbb{E}_w^{(i,\alpha)}$ the coefficient c has density proportional to $\rho(c)\exp(-\frac{1}{2}S_\alpha(c - m_\alpha)^2)$ on $[-1,1]$, with $S_\alpha = S + \alpha t_i$ and $m_\alpha = (S_{xy} + \alpha s_i)/S_\alpha$. For $|\alpha| \le 1$ we have $S_\alpha \ge S^{(i)}$ and $|m_\alpha| \le (2|s_i| + |U_i|)/S^{(i)}$.

Moments of a Gaussian weighted by ρ. Let $\sigma \le 1$, $m \in \mathbb{R}$, $k \in \{2,4\}$, and let π have density proportional to $\rho(c)e^{-(c-m)^2/(2\sigma^2)}$ on $[-1,1]$. We claim $\mathbb{E}_\pi[c^{2k}] \le 3\cdot 2^{2k-1}\big(m^{2k} + (2k-1)!!\,\sigma^{2k}\big)$. Let γ be $\mathcal{N}(m,\sigma^2)$ conditioned on $[-1,1]$. Under γ, $\rho(c)$ is a nonincreasing and c^{2k} a nondecreasing function of $|c|$, and for two such functions g, h of one random variable $(g(u) - g(v))(h(u) - h(v)) \le 0$, whose expectation over two independent copies gives $\mathbb{E}[gh] \le \mathbb{E}[g]\mathbb{E}[h]$, *Chebyshev's association inequality*. Hence $\mathbb{E}_\pi[c^{2k}] = \mathbb{E}_\gamma[\rho c^{2k}]/\mathbb{E}_\gamma[\rho] \le \mathbb{E}_\gamma[c^{2k}]$. If $|m| > 1$, then $\mathbb{E}_\gamma[c^{2k}] \le 1 < m^{2k}$. If $|m| \le 1$, then $[-1,1]$ contains $[m-1, m]$ or $[m, m+1]$, each of $\mathcal{N}(m,\sigma^2)$-probability at least $\mathbb{P}(0 \le Z \le 1) > \frac{1}{3}$; so $\mathbb{E}_\gamma[c^{2k}] \le 3\,\mathbb{E}[(m + \sigma Z)^{2k}] \le 3\cdot 2^{2k-1}(m^{2k} + \sigma^{2k}\mathbb{E}Z^{2k})$, and $\mathbb{E}Z^{2k} = (2k-1)!!$.

Averaging. On the event $\{S^{(i)} \ge 1\}$ the claim applies with $\sigma^2 = 1/S_\alpha \le 1/S^{(i)}$ and gives, uniformly in $|\alpha| \le 1$,

$$\mathbb{E}_w^{(i,\alpha)}\big[f^4\big] \le 192\, s_i^4\left[\frac{(2|s_i| + |U_i|)^4}{(S^{(i)})^4} + \frac{3}{(S^{(i)})^2}\right] + 192\, t_i^4\left[\frac{(2|s_i| + |U_i|)^8}{(S^{(i)})^8} + \frac{105}{(S^{(i)})^4}\right];$$

on its complement $c^2 \le 1$ gives $\mathbb{E}_w^{(i,\alpha)}\big[f^4\big] \le 8s_i^4 + \frac{1}{2}t_i^4$. Now take expectations. The observation (x_i, y_i) is independent of $(S^{(i)}, U_i)$; $S^{(i)}$ has the χ^2_{n-1} distribution; and given the inputs, $U_i \sim \mathcal{N}(0, S^{(i)})$, so $\mathbb{E}[U_i^{2k} \mid x] = (2k-1)!!\,(S^{(i)})^k$. After $(2|s| + |U|)^{2k} \le 2^{2k-1}((2s)^{2k} + U^{2k})$, every term is a finite moment of (x_i, y_i) times $\mathbb{E}[(\chi^2_{n-1})^{-r}] = 2^{-r}\Gamma(\frac{n-1}{2} - r)/\Gamma(\frac{n-1}{2})$ with $r \in \{2,4,8\}$, finite for $n \ge 18$ and of order n^{-r}. The terms with $r = 2$, namely $s_i^4 U_i^4/(S^{(i)})^4$ and $s_i^4/(S^{(i)})^2$, are of order n^{-2}; all others are of order n^{-4} or smaller. The complement has probability $\mathbb{P}(\chi^2_{n-1} < 1) \le (e/(n-1))^{(n-1)/2}$. The resulting bound does not depend on i or α, which proves (10.2); Theorem 10.6 applies because $|c| \le 1$ makes $\log p(x \mid w)$ bounded. ∎

10.4 The theorem: what WAIC estimates

Theorem 10.6 is deterministic algebra plus a moment bound; it says nothing yet about the generalization loss. The connection is Lemma 10.2, $\mathbb{E}[\mathrm{LOO}] = \mathbb{E}[G_{n-1}]$, together with Watanabe's equations of state (Theorem 9.9). The statement collects Watanabe's hypotheses.

Theorem 10.8 (WAIC and leave-one-out estimate the generalization loss; Watanabe 2010a, Lemma 1, Corollary 1, Lemma 3 and Theorem 2). *Let $W = \{\pi_1 \geq 0, \dots, \pi_k \geq 0\} \subset \mathbb{R}^d$ be compact with nonempty interior, the π_j analytic; let $\varphi = \varphi_1\varphi_2$ with $\varphi_1 \geq 0$ analytic and $\varphi_2 > 0$ of class C^∞; let L attain its minimum on W, all minimizers giving the same density p_0; let $w \mapsto f(\cdot, w)$ be an analytic map into $L^s(q)$ for some $s \geq 8$; and let $K(w) \geq c\,\mathbb{E}_X[f(X,w)^2]$ for some $c > 0$ whenever $K(w) \leq \varepsilon$* (relatively finite variance)*. Let $\beta = 1$, let λ be the learning coefficient, $L_0 = L(w_0)$, and ν the singular fluctuation of Chapter 9. Then:*

(i) $\mathbb{E}[\mathrm{LOO}] = \mathbb{E}[G_{n-1}]$ *for every* $n \geq 2$, *and* $\mathbb{E}[\mathrm{WAIC}] = \mathbb{E}[G_n] + o(1/n)$;

(ii) $\mathrm{LOO} = \mathrm{WAIC} + O_p(n^{-2})$;

(iii) $n(\mathbb{E}[G_n] - L_0) \to \lambda$, $n(\mathbb{E}[T_n] - L_0) \to \lambda - 2\nu$, *and* $\mathbb{E}[V_n] \to 2\nu$;

(iv) $(G_n - L_0) + (\mathrm{LOO} - L_n(w_0)) = \frac{2\lambda}{n} + o_p(\frac{1}{n})$, *and the same with* WAIC *in place of* LOO.

Realizability is not assumed.

Proof sketch. Part (i), first half, is Lemma 10.2. Part (ii) is Theorem 10.6, whose assumption is Watanabe's Lemma 1(2), and whose boundedness hypothesis holds almost surely: analyticity into $L^s(q)$ makes $\sup_{w\in W}|f(x,w)|$ finite for q-almost every x. Part (iii) is Lemma 3 of Watanabe (2010a): $n\mathbb{E}[G_n - L_0]$ and $n\mathbb{E}[T_n - L_n(w_0)]$ converge to λ and $\lambda - 2\nu$, where $2\nu = \lim \mathbb{E}[V_n]$ defines ν, and $\mathbb{E}[L_n(w_0)] = L_0$. Adding, $n(\mathbb{E}[\mathrm{WAIC}] - L_0) \to \lambda - 2\nu + 2\nu = \lambda$, which with the first limit is the second half of (i). For (iv), Lemma 3 also states $n(G_n - L_0) + n(T_n - L_n(w_0)) + V_n \to 2\lambda$ in probability, and by (ii) $\mathrm{LOO} - L_n(w_0) = T_n - L_n(w_0) + V_n/n + O_p(n^{-2})$. What this sketch omits is the content of Watanabe's Lemmas 1 and 3: that the rescaled errors converge in law with their expectations, and the equations of state. Their proofs, in Watanabe (2010c) for realizable models, Watanabe (2010d) for regular unrealizable ones and Watanabe (2010b) for singular unrealizable ones, use resolution of singularities and the empirical process on the resolved space (Theorem A.11); Chapter 9 sketches them. ∎

The product model satisfies every hypothesis: $W = [-1,1]^2$ is cut out by $1-a^2$ and $1-b^2$, the prior is constant, $f = -abxy + \frac{1}{2}a^2b^2x^2$ is a polynomial in (a,b) with coefficients in every $L^s(q)$, and a realizable truth has relatively finite variance and a unique optimal density. The mixture does not satisfy the analyticity hypothesis, as Section 10.2 showed; for it Theorem 10.8 is a prediction, which Section 10.5 tests.[5]

Three consequences deserve words.

[5] Watanabe treats mixtures separately in Watanabe (2021b); we have not used that paper.

WAIC estimates the learning curve at the sample size in hand. By part (i), the bias of leave-one-out cross-validation as an estimate of $\mathbb{E}[G_n]$ is $\mathbb{E}[G_{n-1}] - \mathbb{E}[G_n]$, the decrease of the learning curve in one step, and WAIC differs from it by $O(n^{-2})$. If $\mathbb{E}[G_n] - L_0 = \lambda/n - (m-1)/(n \log n) + \cdots$ is smooth in n, that decrease is about λ/n^2: WAIC tracks the finite-n learning curve, slowly converging corrections included, and never needs λ. This is the precise sense of the chapter's title. (The smoothness is an assumption here; the identity is exact.)

WAIC is anticorrelated with the error it estimates. Part (iv) is a statement about each data set. For the product model $n(G_n - S) \approx \frac{1}{2}\mu^2$ and $n(\mathrm{LOO} - S_n) \approx 1 - \frac{1}{2}\mu^2$ with the same $\mu \sim \mathcal{N}(0,1)$ (Computation 10.2): a data set on which the predictive generalizes badly is one on which WAIC reports a small error, and the correlation of the two errors tends to -1. WAIC estimates what the posterior does on average over data sets, $\mathbb{E}[G_n]$, not what it did on the data set at hand.[6] As an estimate of the realized G_n, WAIC is worse than the constant $\mathbb{E}[G_n]$ would be, if one knew it.

The penalty is 2ν, not 2λ. The functional variance estimates 2ν, and ν is a second invariant, distinct from λ in general. In the product model and the mixture the coefficient's limit posterior is Gaussian and $\nu = \lambda = \frac{1}{2}$. In the cusp model $y = ab^2x$ the limit posterior of $\sqrt{n}\,c$ is proportional to $|t|^{-1/2}e^{-(t-\mu)^2/2}$, and a one-dimensional quadrature gives $2\nu \approx 0.697$ against $2\lambda = \frac{1}{2}$ (Exercise 10.8). WAIC's penalty is therefore not a function of λ, and a criterion that replaced V_n by 2λ would be biased.

[6] Watanabe (2010a) reports, for a three-layer tanh network with $n = 200$ and 100 data sets sampled by Metropolis, correlations -0.854 between the generalization error and the importance-sampling leave-one-out error, -0.873 with WAIC, and 0.996 between leave-one-out and WAIC.

10.4.1 WAIC in practice

Three practical points. First, scales: Gelman et al. (2014) put WAIC on the deviance scale, $-2(\mathrm{lppd} - p_{\mathrm{WAIC}})$, with $\mathrm{lppd} = \sum_i \log p^*(X_i) = -nT_n$ and the recommended penalty $p_{\mathrm{WAIC2}} = \sum_i \mathrm{Var}_w[\ell_i] = V_n$; so their WAIC is $2n$ WAIC, and the "effective number of parameters" is the functional variance, which for our running models tends to 1, not to $d = 2$. Vehtari et al. (2017) report $\mathrm{elpd}_{\mathrm{loo}} = \sum_i \log p(X_i \mid X_{-i}^n) = -n\,\mathrm{LOO}$ and $\mathrm{elpd}_{\mathrm{waic}} = -n$ WAIC. Second, with posterior draws in place of exact averages, Lemma 10.2 becomes an importance-sampling estimate whose weights $1/p(X_i \mid w_s)$ can have heavy tails; *Pareto-smoothed importance sampling* fits a generalized Pareto distribution to the largest weights, and its estimated shape $\hat{k}$ flags observations whose estimate cannot be trusted (Vehtari et al. 2017, 2024). Third, the deviance information criterion (Spiegelhalter et al. 2002) plugs the posterior mean into the likelihood; in a singular model that mean need not lie near the true parameters, and its expectation then differs from the Bayes generalization loss (Watanabe 2010a, Section 5.3).

10.5 Measuring it

The scripts compute every quantity of Definition 10.1 exactly, up to quadrature error, for 500 independent data sets at each sample size, and report means with Student-t 95% intervals. Because $\mathbb{E}[S_n] = S$, the bias of an estimate $\widehat{G}$ of G_n equals $\mathbb{E}[(\widehat{G} - S_n) - (G_n - S)]$; working with these differences

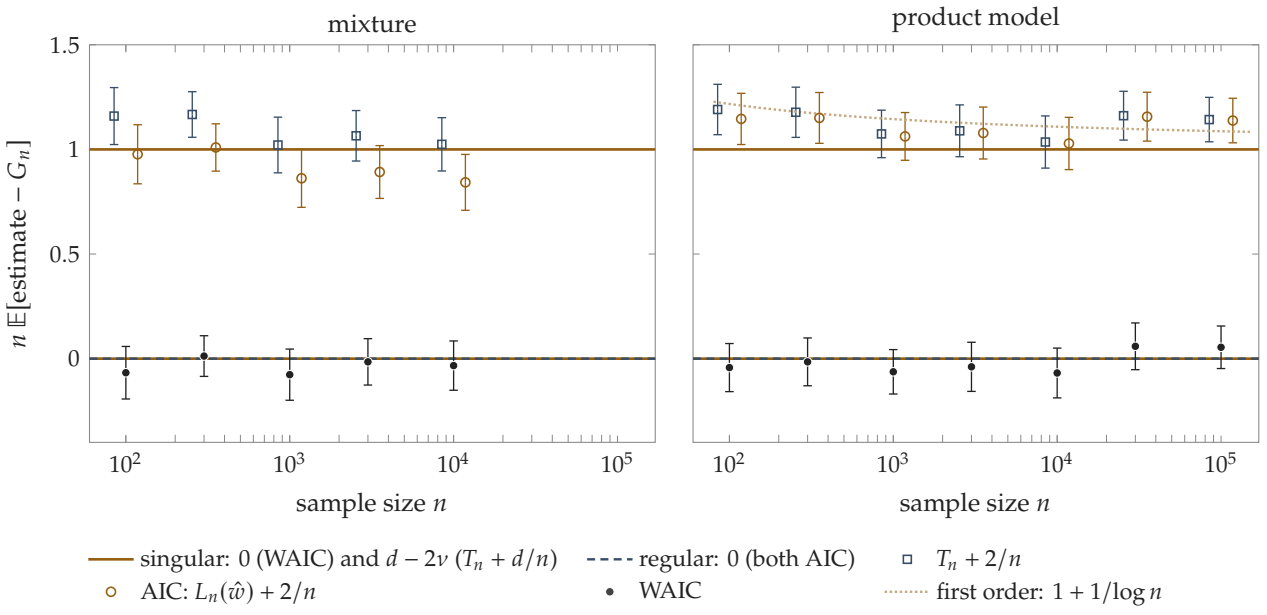


Figure 10.1. The bias of three estimates of the Bayes generalization loss, times n: WAIC (dots), the training loss with AIC's penalty, $T_n + d/n$ (squares, drawn slightly left), and Akaike's AIC $L_n(\hat{w}) + d/n$ (circles, drawn slightly right), with $d = 2$, for the mixture (left) and the product model (right) at their singular truths. Means over 500 data sets per sample size (seeds $1000 + n$ and $2000 + n$) with Student-t 95% intervals. Bronze: the singular prediction, 0 for WAIC (Theorem 10.8) and $d - 2\nu = 1$ for $T_n + d/n$ (Computation 10.2); dotted: the first-order correction $2 - \mathbb{E}[V_n] \approx 1 + 1/\log n$ for the product model. Slate dashed: the regular prediction, 0 for both AIC estimates. Scripts `10-waic-mixture.py` and `10-waic-product.py`.

of errors removes the fluctuation of $S_n - S$, of order $n^{-1/2}$, which is common to all estimates and to all models fitted to the same data.[7]

For the mixture (`figures/scripts/10-waic-mixture.py`; data seeds $1000 + n$) the posterior is computed by midpoint quadrature on a 96×192 grid in (τ, η) with $a = \tau^3$, $b = \eta^3$, which is dense near the set $\{a = 0\} \cup \{b = 0\}$ where the posterior lives. The posterior averages of $e^{\pm g}$, g and g^2, smooth functions of the one number x_i, are computed at 96 Chebyshev nodes on $[-7, 7]$ and interpolated to the data, which agrees with direct evaluation to 10^{-9}. Doubling the grid to 192×384 changes $n\mathbb{E}[G_n - S]$, $\mathbb{E}[V_n]$ and the other errors by at most 7×10^{-4}, and n^2(LOO − WAIC) by at most 4×10^{-3}. For the product model (`10-waic-product.py`; seeds $2000 + n$) the posterior of c is handled exactly, by Gauss–Legendre quadrature in c for $n \le 300$ and above that by closed forms through $h_0(\mu) = \mathbb{E}\log|\mu + Z|$ (the two agree to 4×10^{-8}); on one data set with $n = 10^3$, 20,000 draws from the kit's exact sampler give $V_n = 0.754$ against the exact 0.762, within Monte Carlo error. The regular line uses the formulas of Computation 10.1. The runs take under two minutes each.

Figure 10.1 is the chapter's answer in one picture. At every sample size, for both models, the bias of WAIC is zero within its interval: $n\,\mathbb{E}[\text{WAIC} - G_n]$

[7] With real data S_n is unknown, but it cancels when two models fitted to the same data are compared by their WAIC.

model	n	$n\,\mathbb{E}[G_n - S]$	$\mathbb{E}[V_n]$	$n\,\mathbb{E}[\mathrm{err}_G + \mathrm{err}_{\mathrm{LOO}}]$	corr
mixture	10^2	0.334	0.773	0.601	−0.9975
	10^3	0.402	0.902	0.728	−0.9982
	10^4	0.413	0.943	0.793	−0.9989
product	10^2	0.319	0.767	0.598	−0.9834
	10^3	0.398	0.863	0.733	−0.9970
	10^4	0.431	0.896	0.793	−0.9997
	10^5	0.390	0.912	0.834	−1.0000
limit (Theorem 10.8)		0.5	1	1	−1

Table 10.1. The singular quantities behind Figure 10.1: the generalization error, the functional variance, the sum of the generalization error $\mathrm{err}_G = G_n - S$ and the leave-one-out error $\mathrm{err}_{\mathrm{LOO}} = \mathrm{LOO} - S_n$, and the correlation of the two errors, over 500 data sets. The 95% half-widths are about 0.055 for $n\mathbb{E}[G_n - S]$, 0.005 to 0.023 for $\mathbb{E}[V_n]$, and 0.001 to 0.017 for the sum. The limits are λ, 2ν, 2λ and -1.

ranges from -0.08 ± 0.12 to $+0.01 \pm 0.10$ for the mixture and from -0.07 ± 0.12 to $+0.06 \pm 0.11$ for the product model. This holds at $n = 100$, where $n\mathbb{E}[G_n - S]$ is only 0.33 (mixture) and 0.32 (product) against its limit $\lambda = \frac{1}{2}$: WAIC follows the learning curve, not its limit. The regular recipe $T_n + 2/n$ overestimates G_n by 1.02 to 1.19 units of $1/n$, close to the singular prediction $(d - 2\nu)/n = 1/n$ plus the first-order excess $1/\log n$, and far from the regular prediction 0. Akaike's own criterion does no better: its bias is 0.84 to 1.01 units for the mixture and 1.03 to 1.16 for the product model. The bias of the regular recipe is exactly $(d - \mathbb{E}[V_n])/n$ plus that of WAIC, so the measured $\mathbb{E}[V_n]$ tells the same story more precisely: it rises from 0.77 to 0.94 for the mixture and from 0.77 to 0.91 for the product model as n goes from 10^2 to 10^4 and 10^5, and for the product model it agrees with $1 - 1/\log n$ to within 0.02 at every n (Table 10.1).

Table 10.1 shows how slowly the λ-dependent limits arrive. The sum of the two errors in part (iv) of Theorem 10.8 concentrates sharply, its standard deviation across data sets being 0.04 (mixture) and 0.02 (product) at $n = 10^4$, but around 0.79, not $2\lambda = 1$. Since leave-one-out is unbiased, the mean of the sum is about $2n\mathbb{E}[G_n - S]$; and since $p^*(X_{n+1})$ is the ratio of the marginal likelihoods of $n + 1$ and n observations, $\mathbb{E}[G_n] = \mathbb{E}[F_{n+1}] - \mathbb{E}[F_n]$ exactly. The product model's free energy $\frac{1}{2}\log n - \log\log n + O_p(1)$ (Theorem 3.4) then suggests $n\mathbb{E}[G_n - S] \approx \frac{1}{2} - 1/\log n$ and a sum of $1 - 2/\log n = 0.78$; this is an expansion, since the differences of the bounded remainder are not controlled. The correlation of the two errors is below -0.996 from $n = 10^3$ on. Figure 10.2 shows the 500 pairs at $n = 10^4$ for both models: they lie on a line of slope -1, parallel to the singular prediction and far from the regular one.

Figure 10.3 tests Theorem 10.6 itself. For the regular line $n^2\mathbb{E}[\mathrm{LOO} - \mathrm{WAIC}]$ falls like $15/n$, as Computation 10.1 predicts. For the product model, where Proposition 10.7 proves the bound, it stays below 0.2 and drifts down, from 0.18 ± 0.03 at $n = 10^2$ to 0.06 ± 0.01 at $n = 10^5$, consistent with the theorem's $O(n^{-2})$; the drift is the slow approach of the posterior of $\sqrt{n}\,c$ to

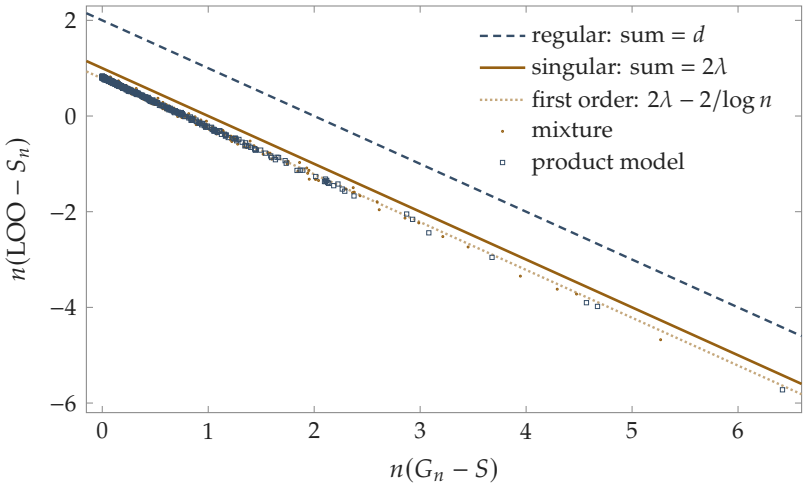


Figure 10.2. The generalization error and the leave-one-out error of the same 500 data sets at $n = 10^4$, for the mixture (bronze dots) and the product model (slate squares); the WAIC errors are indistinguishable at this scale. Lines are where $n(G_n - S) + n(\mathrm{LOO} - S_n)$ equals $d = 2$ (slate dashed, regular), $2\lambda = 1$ (bronze, Theorem 10.8) and its first-order value $1 - 2/\log n$ (dotted; see the text). Seeds $1000 + n$ and $2000 + n$.

a Gaussian. For the mixture it rises over the whole range, from 0.05 ± 0.01 at $n = 10^2$ to 0.76 ± 0.10 at $n = 10^4$. The script locates the cause. At $n = 10^4$ the contribution of one observation to $n^2(\mathrm{LOO} - \mathrm{WAIC})$, as a function of its value x, grows by a factor of 15 to 40 per unit of x between $x = 3$ and $x = 5$, and the one percent of observations with the largest $|x|$ supply 84% of the sum (median over data sets). This is the unbounded likelihood ratio e^{bx} again, the same feature that destroys analyticity: the average over the data is dominated by rare observations. The rise is not an artifact of sampling them: averaging the same function of x over $x \sim \mathcal{N}(0, 1)$ gives 0.08 at $n = 10^2$ and 0.62 at $n = 10^4$, and the share of $|x| > 4$ in it grows from 3% to 25%. Whether $n^2\mathbb{E}|\mathrm{LOO} - \mathrm{WAIC}|$ stays bounded for the mixture is not decided by these runs; at $n = 10^4$ the difference itself is 8×10^{-9}, far below everything else in the problem.

Two cautions. The bias of WAIC expected at finite n, about λ/n^2, is 0.005 units at $n = 100$, a twentieth of our interval; the runs cannot see it. And the mixture's agreement with Theorem 10.8 is an observation about a model that the theorem does not cover.

10.6 The ledger

The chapter established that the generalization loss of a Bayesian posterior can be estimated from the data and the posterior alone, without the learning coefficient, the singular fluctuation, or the multiplicity. Leave-one-out cross-validation estimates $\mathbb{E}[G_{n-1}]$ exactly; it can be computed from one posterior by importance sampling; and it equals WAIC $= T_n + V_n/n$ up to $O(n^{-2})$ in mean, which we proved completely for the product model

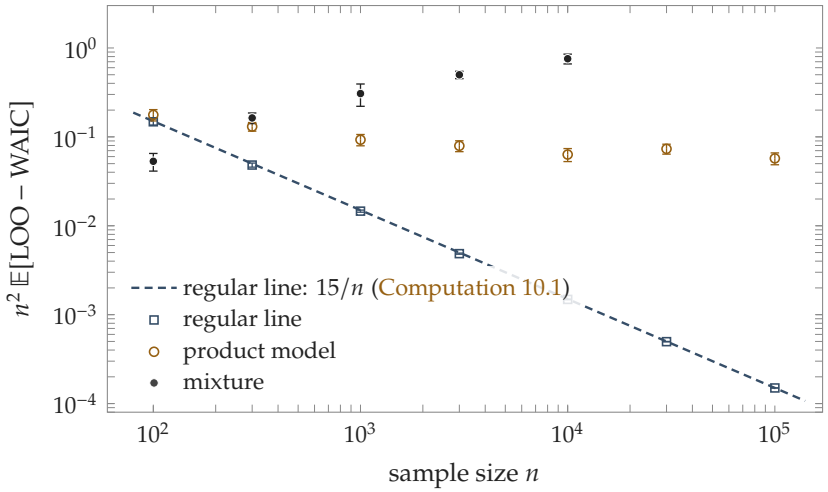


Figure 10.3. Leave-one-out minus WAIC, times n^2, at $\beta = 1$: means over 500 data sets with 95% intervals (all values are positive). Theorem 10.6 bounds $n^2\mathbb{E}|\mathrm{LOO} - \mathrm{WAIC}|$ under Assumption 10.5, which Proposition 10.7 proves for the product model; the mixture does not meet the analyticity hypothesis under which Watanabe states the assumption. Slate dashed: the exact leading term $15/n$ for the regular line, whose posterior is Gaussian.

and under Watanabe's moment bound in general. With the equations of state, WAIC is unbiased to $o(1/n)$ (Theorem 10.8). Its penalty is the functional variance, an estimate of $2v$; for our running models $2v = 1$ where AIC charges $d = 2$. The price is variance: the error of WAIC mirrors the generalization error, with correlation tending to -1, so WAIC measures the expected performance of the posterior, not its performance on the data set in hand.

LEDGER · $(1-a)\mathcal{N}(0,1) + a\mathcal{N}(b,1)$	
QUANTITY	estimating the Bayes generalization loss G_n from the training data and one posterior
REGULAR THEORY	AIC: $T_n + \frac{d}{n}$ or $L_n(\hat{w}) + \frac{d}{n}$, unbiased to $o(\frac{1}{n})$; leave-one-out agrees with AIC
SINGULAR THEORY	$\mathbb{E}[\mathrm{WAIC}] = \mathbb{E}[G_n] + o(\frac{1}{n})$, $\mathrm{LOO} = \mathrm{WAIC} + O_p(n^{-2})$, penalty V_n/n with $\mathbb{E}[V_n] \to 2v$, and $(G_n - L_0) + (\mathrm{LOO} - L_n(w_0)) = \frac{2\lambda}{n} + o_p(\frac{1}{n})$ (Theorems 10.6 and 10.8)
RUNNING EXAMPLE	mixture and $y = abx$ at the singular truth: $2v = 2\lambda = 1$, so AIC's penalty $2/n$ overstates the optimism by $1/n$
MEASURED	bias of WAIC 0 within $\pm 0.13/n$ for $10^2 \le n \le 10^5$; bias of $T_n + 2/n$ between $1.02/n$ and $1.19/n$; $\mathbb{E}[V_n] = 0.94$ (mixture) and 0.91 (product) at $n = 10^4, 10^5$; correlation of the errors below -0.996 from $n = 10^3$
RESTS ON	exact posterior averages at $\beta = 1$; independent data; compactness, analyticity (which the mixture violates), $s \ge 8$ moments, relatively finite variance

10.7 Where the theory stops

WHAT WAS ASSUMED. Exact posterior averages at $\beta = 1$: our scripts compute them by quadrature, while a network's posterior is sampled, by Markov chain Monte Carlo or stochastic-gradient Langevin dynamics, whose bias enters V_n and the importance weights directly; Watanabe (2010a, Section 5.2) warns that leave-one-out from separate posteriors, its importance-sampling form, and WAIC can then disagree. Independent data: the exact identity of Lemma 10.2 is exchangeability. A fixed model as n grows, compact W, and an analytic log-likelihood ratio with $s \geq 8$ moments; the chapter's own mixture violates analyticity, and so do the tanh neuron with Gaussian inputs and networks with ReLU units. Relatively finite variance, which holds for realizable and for regular truths but can fail for truths that are neither.

WHERE IT BREAKS. Watanabe's non-renormalizable example shows what happens without relatively finite variance (Watanabe 2010b, Section 5). Let $q = \mathcal{N}(0, I_2)$ on (x, y) and $p(x, y \mid a) = \mathcal{N}\big((a, \sqrt{a^4 - a^2 + 1}), I_2\big)$. Then $K(a) = L(a) - L_0 = a^4/2$, while the log density ratio $f = -ax - (\sqrt{a^4 - a^2 + 1} - 1)y + a^4/2$ has $\mathbb{E}_q[f^2] \approx a^2$: the fluctuation is not bounded by the mean. The posterior concentrates near $|a| \approx n^{-1/6}$ with width $n^{-1/3}$, the learning curve is of order $n^{-2/3}$ instead of $1/n$, and $\mathbb{E}[V_n]$ grows like $n^{1/3}$ instead of converging to $2v$. Watanabe derives these orders by a sketch, and states that the equations of state still hold at order $n^{-2/3}$. Carrying out the sketch, we find $\mathbb{E}[G_n] - L_0 \approx -c\, n^{-2/3}$ and $\mathbb{E}[V_n] \approx 4c\, n^{1/3}$ with $c = \Gamma(\frac{7}{6})/(2^{5/3}\sqrt{\pi}) \approx 0.165$ (Exercise 10.10).[8] The generalization loss is below L_0: the predictive, a mixture of members of the model, approximates q better than every member does. Table 10.2 confirms both rates and shows WAIC still unbiased at the new order; what breaks is the universal law λ/n, the convergence of V_n, and the rate of Theorem 10.6: the median of $n^2|\mathrm{LOO} - \mathrm{WAIC}|$ grows from 0.7 to 8.7 over $10^2 \leq n \leq 10^4$. Leverage points break importance sampling: Peruggia (1997) gives necessary and sufficient conditions, in terms of leverage and influence, for the case-deletion weights of the Bayesian linear model to have finite variance, with real data sets that violate them, and Epifani et al. (2008) give conditions for a central limit theorem; with a leverage point, leave-one-out and WAIC also move apart. When the inputs of a regression are fixed or dependent, leave-one-out does not estimate the conditional generalization loss and WAIC does (Watanabe 2018b, 2021a); Watanabe (2024) summarizes both points.

OPEN PROBLEMS. Is $\mathbb{E}|\mathrm{LOO} - \mathrm{WAIC}| = O(n^{-2})$ for models such as the mixture, whose log-likelihood ratio is smooth but not analytic, and whose likelihood ratio grows exponentially in the data? How do the errors of an approximate sampler propagate into V_n and into importance-sampling leave-one-out, so that WAIC computed from stochastic-gradient Langevin samples of a network comes with a guarantee? What replaces Theorem 10.8 without relatively finite variance? Watanabe (2010b) suggests that a more general universal law covers both the renormalizable and the non-renormalizable case. And since WAIC is anticorrelated with the error it estimates, which

[8] At $\beta = 1$ the source prints $\mathbb{E}[B_g] \cong L_0 - \frac{Q}{2} n^{-2/3}$ and $\mathbb{E}[Y_t] \cong 2Q\, n^{-2/3}$ (its Y_t is our V_n/n), with $Q = \mathbb{E}|\xi_1|^{4/3} = 2^{7/3}c$: both are $2^{4/3}$ times our values. With its own minimizer $a_* = (\xi_1/2\sqrt{n})^{1/3}$, the Gibbs error $\mathbb{E}[a_*^4/2]$ is $c\, n^{-2/3}$, where the source prints $\frac{Q}{2} n^{-2/3}$. The simulation of Table 10.2 agrees with c.

n	$n^{2/3}\mathbb{E}[G_n - L_0]$	$\mathbb{E}[V_n]/n^{1/3}$	$n^{2/3}\mathbb{E}[\mathrm{WAIC} - G_n]$
10^2	-0.102 ± 0.029	0.600 ± 0.032	0.024 ± 0.090
$3 \cdot 10^2$	-0.127 ± 0.032	0.634 ± 0.040	0.037 ± 0.094
10^3	-0.134 ± 0.035	0.635 ± 0.044	0.010 ± 0.098
$3 \cdot 10^3$	-0.175 ± 0.044	0.676 ± 0.063	0.094 ± 0.120
10^4	-0.168 ± 0.059	0.674 ± 0.087	0.047 ± 0.155
limit	-0.165	0.659	0

Table 10.2. The non-renormalizable example of Watanabe (2010b), with the uniform prior on $[-1.5, 1.5]$: means with 95% half-widths over 400 data sets (300 at $n = 3\cdot10^3$, 200 at $n = 10^4$; seeds $4000+n$), posterior by Gauss–Legendre quadrature. The limits are $-c$, $4c$ and 0 with $c = \Gamma(\frac{7}{6})/(2^{5/3}\sqrt{\pi})$. Script `10-waic-nonrenorm.py`.

estimators of G_n have smaller variance at the same bias? Watanabe (2023) proposes an adjusted cross-validation that combines leave-one-out with hold-out validation for this purpose, and compares its variance with that of leave-one-out only for regular models, leaving the singular case open.

Exercises

10.1 CONSOLIDATION. Let the posterior have inverse temperature β. Show that leave-one-out is $\frac{1}{n}\sum_i[\kappa_i(-\beta) - \kappa_i(1-\beta)]$ and WAIC is $-\frac{1}{n}\sum_i \kappa_i(1) + \beta V_n/n$, with κ_i now defined by the tempered posterior. Expand both to third order and derive $\mathrm{LOO} - \mathrm{WAIC} = \frac{\beta-\beta^2}{2}Y_3 + O_p(n^{-2})$, $Y_3 = \frac{1}{n}\sum_i \kappa_i'''(0)$, under a fourth-moment bound for tilts with $|\alpha| \le 1+\beta$. Why does the third cumulant cancel only at $\beta = 1$? *(Hint: at β, the posterior without X_i is the posterior tilted by $p(X_i \mid w)^{-\beta}$.)*

10.2 CONSOLIDATION. Show that $|\kappa_4| \le 2\mu_4$ is sharp: for $Y = \pm 1$ with probability $\frac{1}{2}$ each, $\kappa_4 = -2$ and $\mu_4 = 1$. For which distributions is $\kappa_4 = \mu_4$?

10.3 CONSOLIDATION. For the regular line, carry out the expansions behind the last step of Computation 10.1: show $T_n = L_n(\hat{a}) + O_p(n^{-3/2})$, $\mathrm{WAIC} = \mathrm{AIC} + O_p(n^{-3/2})$, and $n^3\,\mathbb{E}[\mathrm{LOO} - \mathrm{WAIC}] \to 15$.

10.4 CONSOLIDATION. Gelman, Hwang and Vehtari's mean-based penalty is $p_{\mathrm{WAIC1}} = 2\sum_i(\log \mathbb{E}_w\big[p(X_i \mid w)\big] - \mathbb{E}_w[\ell_i])$. Show that $p_{\mathrm{WAIC1}} = 2\sum_i[\kappa_i(1) - \kappa_i'(0)] = V_n + O_p(n^{-1/2})$ under Assumption 10.5, and that $p_{\mathrm{WAIC1}} = 2n(G_{\mathrm{t}} - T_n)$, with $G_{\mathrm{t}} = -\frac{1}{n}\sum_i \mathbb{E}_w[\ell_i]$ the Gibbs training loss.

10.5 BY HAND. For the Gaussian mean $x_i \sim \mathcal{N}(\theta, 1)$ with the prior $\theta \sim \mathcal{N}(0, \tau^2)$, compute T_n, V_n and LOO exactly. Show that they are the formulas of Computation 10.1 with every leverage equal to $h = 1/(n + \tau^{-2})$ and $r_i = x_i - \mathbb{E}_w[\theta]$, that $V_n \to 1$, and that $n^3\mathbb{E}[\mathrm{LOO} - \mathrm{WAIC}] \to 1$.

10.6 BY HAND. Let $h_0(\mu) = \mathbb{E}\log|\mu + Z|$ with $Z \sim \mathcal{N}(0,1)$. Show that $h_0(0) = -\frac{1}{2}(\gamma + \log 2)$, that $h_0'(\mu) = \sqrt{2}\,D(\mu/\sqrt{2})$ as a principal value

$\mathbb{E}[1/(\mu+Z)]$, that $h_0'' = 1-\mu h_0'$, and that $\mathbb{E}_\mu[h_0''(\mu)] = \frac{1}{2}$ for $\mu \sim \mathcal{N}(0,1)$. Deduce the first-order formula $\mathbb{E}[V_n] \approx 1-1/\log n$ of Computation 10.2. *(Hint: $h_0''(\mu) = \mathbb{E}[(Z^2-1)\log|\mu+Z|]$; for the average, condition on $\mu+Z$.)*

10.7 BY HAND. For the chain $y = a_1a_2a_3x$ at the truth 0, the coefficient density is $\frac{1}{4}(\log\frac{1}{|c|})^2$. Show that the posterior of $t = \sqrt{n}\,c$ still tends to $\mathcal{N}(\mu,1)$, so $V_n \to 1$, and that to first order $\mathbb{E}[V_n] \approx 1 - 2/\log n$: the multiplicity $m = 3$ doubles the correction.

10.8 BY HAND. For the cusp $y = ab^2x$ at the truth 0 the coefficient density is $\frac{1}{2}(|c|^{-1/2}-1)$ (Chapter 3). Show that the posterior of $t = \sqrt{n}\,c$ tends to the density proportional to $|t|^{-1/2}e^{-(t-\mu)^2/2}$, and compute numerically $2\nu = \lim \mathbb{E}[V_n] = \mathbb{E}_\mu[\mathrm{Var}(t)] \approx 0.697$. Conclude that WAIC's penalty is about $0.70/n$, while $2\lambda/n = 0.5/n$ and $d/n = 2/n$, and check with the same quadrature that $\mathbb{E}_\mu[(\mathbb{E}t)^2]/2 = \lambda = \frac{1}{4}$.

10.9 VARIATION. Take the product model at the generic truth $ab = \frac{1}{2}$. Predict the limits of $\mathbb{E}[V_n]$, $n\mathbb{E}[G_n - S]$ and $n\mathbb{E}[T_n + 2/n - G_n]$, and explain why the approach to them is faster than at the singular truth.

10.10 VARIATION. ★ In the non-renormalizable example, approximate $nK_n(a)$ near its minimum $a_* = (\xi_1/(2\sqrt{n}))^{1/3}$, where $\xi_1 = n^{-1/2}\sum_i x_i \sim \mathcal{N}(0,1)$, by a quadratic with curvature $6na_*^2$. Derive $\mathbb{E}[K(a_*)] \approx c\,n^{-2/3}$ and $\mathbb{E}[V_n] \approx 4c\,n^{1/3}$ with $c = \Gamma(\frac{7}{6})/(2^{5/3}\sqrt{\pi})$, and $\mathbb{E}[G_n] - L_0 \approx -c\,n^{-2/3}$. *(Hint: $\mathbb{E}|\xi|^{4/3} = 2^{2/3}\Gamma(\frac{7}{6})/\sqrt{\pi}$ and $\mathbb{E}|\xi|^{-2/3} = 2^{-1/3}\Gamma(\frac{1}{6})/\sqrt{\pi}$; at order $n^{-2/3}$, $G_n - L_0 \approx \mathbb{E}_w[K(a)] - \frac{1}{2}\mathbb{E}_X\,\mathrm{Var}_w[\log p(X \mid a)]$.)*

10.11 MEASUREMENT. Replace the exact averages for the product model at $n = 10^3$ by S posterior draws from `sk.sample_product_posterior`; compute WAIC with `sk.waic`, and importance-sampling leave-one-out from the same draws by Lemma 10.2. How large must S be before the Monte Carlo error of LOO − WAIC is smaller than the exact value, about 10^{-7}? What does this say about checking Theorem 10.6 with a sampler?

10.12 RESEARCH. Prove or refute $\mathbb{E}|\mathrm{LOO} - \mathrm{WAIC}| = O(n^{-2})$ for the mixture of this chapter, whose log-likelihood ratio is not analytic at $a = 0$. A first step is to bound the tilted moments $\mathbb{E}_w^{(i,\alpha)}\big[f^4\big]$ through the size of $a(e^{bx-b^2/2}-1)$, which for $|b| \le 1$ grows like $e^{|x|}$.

Notes and References

The widely applicable information criterion was introduced by Watanabe in the paper on the equations of states, posted in 2007 and published as Watanabe (2010c), where it appears in two forms built from the Bayes and Gibbs training errors; Section 8.3 of Watanabe (2009) is titled "Widely applicable information criteria". The form with the functional variance is already in Watanabe (2010c,d); its asymptotic equivalence with Bayes leave-one-out cross-validation as random variables is proved in Watanabe (2010a), whose Theorem 1, Corollary 1 and Theorem 2 this chapter follows;

Watanabe (2018a) carries the comparison to higher order, the second book has a chapter on information criteria (Watanabe 2018b, Chapter 8), and the review Watanabe (2024) restates the results at $\beta = 1$. The proof of Theorem 10.6 is Watanabe's, with the Taylor remainder written in integral form; Proposition 10.7 and the first-order computations of Computation 10.2 are the book's own.

Cross-validation grew up alongside the information criteria: the predictive sample reuse of Geisser (1975), and the theorem of Stone (1977) that leave-one-out with maximum-likelihood fits chooses models as AIC (Akaike 1974) does. The deviance information criterion (Spiegelhalter et al. 2002) is an older Bayesian criterion (Section 10.4.1).

Gelman et al. (2014) compared WAIC with AIC, DIC and cross-validation on the deviance scale, and Vehtari et al. (2017) argued that Pareto-smoothed importance-sampling leave-one-out is more robust than WAIC in finite samples with weak priors or influential observations, and implemented both in the R package `loo`; the smoothing method is analyzed in Vehtari et al. (2024).

The renormalizable condition and the non-renormalizable example are from Watanabe (2010b); the regular unrealizable case, where WAIC reduces to Takeuchi's criterion, is Watanabe (2010d). Watanabe (2021b) treats WAIC and WBIC for mixture models.

Translating the literature. Watanabe (2010a) writes $\mathrm{BgL}(n)$, $\mathrm{BtL}(n)$, $\mathrm{CvL}(n)$, $V(n)$ and $\mathrm{WAIC}(n) = \mathrm{BtL}(n) + \beta V(n)/n$ for our G_n, T_n, LOO, V_n and WAIC, and the errors $\mathrm{Bg}(n) = \mathrm{BgL}(n) - L_0$, $\mathrm{Bt}(n) = \mathrm{BtL}(n) - L_n$ and $\mathrm{Cv}(n) = \mathrm{CvL}(n) - L_n$, where his L_n is the empirical loss of the optimal density, our $L_n(w_0)$. His generating function $F(\alpha)$ is our $\frac{1}{n}\sum_i \kappa_i(\alpha)$, his functional cumulants are $Y_k(n) = F^{(k)}(0)$, his $E_w^{(i)}$ is our $\mathbb{E}_w^{(-i)}$, the tilted posterior with $\alpha = -1$, and his real log canonical threshold λ is our learning coefficient, the same number. Watanabe (2010c) uses the errors B_g, B_t, G_g, G_t, with q in place of p_0. Watanabe (2010b) uses the same letters for the losses (its B_g, B_t are our G_n, T_n), Y_t for our V_n/n; its Q is $\mathbb{E}|\xi_1|^{4/3} = 2^{7/3}c$. Watanabe (2024) and, as far as we know, Watanabe (2018b) write G_n, T_n, C_n, W_n for our G_n, T_n, LOO, WAIC and V_θ for Var_w. The notation of Gelman et al. (2014) and Vehtari et al. (2017), with $p_{\mathrm{WAIC2}} = p_{\mathrm{waic}} = V_n$, is translated in Section 10.4.1.

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11 Temperature

WBIC and the thermodynamics of the posterior

THE MODEL — $y = ax_1 + b^2x_2$

IN THIS CHAPTER. A sampler sees posterior averages, not the free energy; but the derivative of the free energy in the inverse temperature β is a posterior average, the mean energy $\mathbb{E}_w^\beta[nL_n]$. By hand, the quartic model $y = ax_1 + b^2x_2$ holds mean energy $\frac{3}{4\beta}$, not the $\frac{1}{\beta}$ of equipartition, and the product model $\frac{1}{2\beta} - \frac{1}{\beta(\log n\beta+\gamma+\log 2)}$. We prove that the population mean energy is $\frac{\lambda}{\beta} - \frac{m-1}{\beta\log(n\beta)}$ up to smaller terms, that one temperature reproduces the free energy exactly, and, completely for linear models, Watanabe's theorem that WBIC, the mean energy at $\beta = 1/\log n$, is the free energy up to $O_p(\sqrt{\log n})$, with the $\sqrt{\log n}$ term computed and shown to vanish under a sign symmetry. Two-temperature estimates of λ follow the predictions for four models up to $n = 10^8$, including a bias of $-1/\log n$ in the product model and a $\sqrt{\log n}$ response of WBIC to the noise in the quartic model. Everything rests on exact tempered posteriors and on corrections of relative size $1/\log n$, which no practical n makes small.

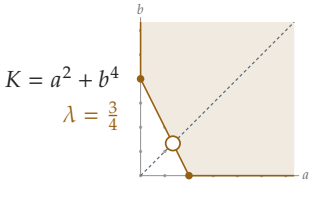
11.1 The question: what a sampler can see of the free energy

Suppose two networks are candidates for the same data, one with three hidden units and one with four, and we want to compare their marginal likelihoods, as Chapter 8 recommends. A marginal likelihood is a normalizing constant, and its negative logarithm is the free energy F_n. A Markov chain Monte Carlo sampler, the tool we have for the posteriors of models with more than a few parameters, produces draws $w_1, w_2, \ldots$ from the posterior and with them posterior averages $\frac{1}{T}\sum_t f(w_t) \approx \mathbb{E}_w\big[f\big]$. It does not produce the normalizing constant: multiplying the integrand by a constant changes neither the posterior nor the draws.

Temperature makes the constant visible. For $\beta \geq 0$ let

$$F_n(\beta) = -\log \int_W \varphi(w)\, e^{-\beta n L_n(w)}\, \mathrm{d}w,$$

the free energy at *inverse temperature* β, with $F_n(1) = F_n$; its normalized integrand is the *tempered posterior*, with expectations $\mathbb{E}_w^\beta[\cdot]$. At $\beta = 1$ it is the posterior; for $\beta < 1$ it is hotter and wider, for $\beta > 1$ colder. Since φ is a



probability density, $F_n(0) = 0$, and differentiating under the integral sign (Lemma 11.1) gives

$$F_n'(\beta) = \mathbb{E}_w^\beta[nL_n], \qquad F_n = \int_0^1 \mathbb{E}_w^\beta[nL_n]\,\mathrm{d}\beta. \tag{11.1}$$

The right side is an integral of posterior means, each of which a sampler run at its own temperature estimates. This is *thermodynamic integration*, and the vocabulary is that of statistical physics: $nL_n(w)$ is the energy of the configuration w, the tempered posterior is the Gibbs measure at inverse temperature β, $\mathbb{E}_w^\beta[nL_n]$ is the *mean energy*, and $\beta^2 \operatorname{Var}_w^\beta(nL_n) = -\beta^2 F_n''(\beta)$ is the heat capacity.[1]

What should the mean energy be? For a regular model the physicist's answer is the equipartition theorem.

> THE REGULAR ANSWER · equipartition. If the model is regular, the tempered posterior is asymptotically normal with covariance $(n\beta J)^{-1}$, each of the d coordinates holds mean energy $\frac{1}{2\beta}$, and for each fixed $\beta > 0$, $\mathbb{E}_w^\beta[nL_n] = nL_n(\hat{w}) + \frac{d}{2\beta} + o_p(1)$ by Laplace's method (as in Theorem 1.4). At $\beta = 1/\log n$ this is BIC, $nL_n(\hat{w}) + \frac{d}{2}\log n$ (Schwarz 1978), up to $o_p(1)$ (Watanabe 2013, Theorem 5).

Integrated over β by (11.1), equipartition returns the classical free energy: the energy $\frac{d}{2\beta}$ holds from $\beta = 1$ down to about $\beta = 1/n$, where the posterior meets the prior, and its integral is $\frac{d}{2}\log n$. For a singular model none of this survives. The posterior of the product model $y = abx$ at its singular truth is not normal but spreads over all scales along the cross $ab = 0$ (Theorem 2.5), and its mean energy with the noise removed, computed below, is $\frac{1}{2\beta} - \frac{1}{\beta(\log n\beta + \gamma + \log 2)}$: about half of what two parameters should hold, less a term that decays only like $1/\log(n\beta)$. The quartic model $y = ax_1 + b^2x_2$, the running model of this chapter, holds $\frac{3}{4\beta}$. So: what does the mean energy measure in a singular model? Can one temperature replace the integral in (11.1), so that one run of a sampler estimates the free energy? And can two temperatures give the learning coefficient without knowing the truth?

[1] In physics the heat capacity is the derivative of the mean energy with respect to the temperature $1/\beta$, which is β^2 times the variance of the energy when Boltzmann's constant is 1.

11.2 By hand: the mean energy of the running models

We first replace the empirical energy $nK_n = nL_n - nL_n(w_0)$ by its average nK, as Chapter 4 did for the free energy. For $s > 0$ let μ_s be the probability measure with density proportional to $e^{-sK(w)}\varphi(w)$, the *population* tempered posterior at $s = n\beta$. Its mean energy is

$$\mathbb{E}_{\mu_{n\beta}}[nK] = \frac{\mathcal{E}(n\beta)}{\beta}, \qquad \mathcal{E}(s) := s\,\mathbb{E}_{\mu_s}[K] = \frac{s\int_W K\,e^{-sK}\varphi\,\mathrm{d}w}{\int_W e^{-sK}\varphi\,\mathrm{d}w}. \tag{11.2}$$

We call $\mathcal{E}$ the dimensionless mean energy. It depends on n and β only through their product, so heating the posterior by a factor of ten has the same effect as discarding nine tenths of the data. Equipartition says $\mathcal{E}(s) \to \frac{d}{2}$.

COMPUTATION 11.1 · THE MEAN ENERGY OF THE QUARTIC MODEL

1 The loss is $K = \frac{1}{2}(a^2 + b^4)$ and the prior is uniform on $[-1,1]^2$, so μ_s is the product of a law for a, with density proportional to $e^{-sa^2/2}$ on $[-1,1]$, and a law for b, with density proportional to $e^{-sb^4/2}$. Hence $\mathcal{E}(s) = s\mathbb{E}[\frac{a^2}{2}] + s\mathbb{E}[\frac{b^4}{2}]$.

2 For a put $t = sa^2/2$; on $a \in [0,1]$, $\mathrm{d}a \propto t^{-1/2}\,\mathrm{d}t$, so t has density proportional to $t^{1/2-1}e^{-t}$ on $[0, s/2]$, a Gamma law with shape $\frac{1}{2}$, truncated at $s/2$. For b put $t = sb^4/2$; then $\mathrm{d}b \propto t^{-3/4}\,\mathrm{d}t$, and t is Gamma with shape $\frac{1}{4}$, truncated at $s/2$. A Gamma law with shape α has mean α, and conditioning on $[0,T]$ lowers it by $T^\alpha e^{-T}/\int_0^T t^{\alpha-1}e^{-t}\,\mathrm{d}t$ (integrate by parts): `11-temperature-energy.py` gives $\mathcal{E}(10) = 0.7387$, $\mathcal{E}(30) = 0.749999$, and $\mathcal{E}(100) = 0.75$ to ten digits.

3 Hence $\mathbb{E}_{\mu_{n\beta}}[nK] = \frac{3}{4\beta}$ once $n\beta \gtrsim 30$, not the $\frac{2}{2\beta}$ of equipartition: a holds $\frac{1}{2\beta}$, the quartic coordinate b only $\frac{1}{4\beta}$.

4 Thermodynamic integration of the population energy, (11.1) with K in place of K_n, gives the population free energy as an area: $\bar{F}(n) = \int_0^1 \mathbb{E}_{\mu_{n\beta}}[nK]\,\mathrm{d}\beta = \int_0^n \mathcal{E}(s)\frac{\mathrm{d}s}{s} = \int_{-\infty}^{\log n} \mathcal{E}(e^x)\,\mathrm{d}x$. The integrand is about $\frac{4s}{15}$ for $s \ll 1$ (the prior mean of K is $\frac{4}{15}$) and $\frac{3}{4}$ for $s \gtrsim 30$, so $\bar{F}(n) = \frac{3}{4}\log n + O(1)$. Directly, $\bar{F}(n) = -\log\big(\frac{1}{2}\int_{-1}^1 e^{-na^2/2}\,\mathrm{d}a \cdot \frac{1}{2}\int_{-1}^1 e^{-nb^4/2}\,\mathrm{d}b\big) = \frac{3}{4}\log n + \frac{1}{2}\log\frac{2}{\pi} - \frac{1}{4}\log 2 - \log\Gamma(\frac{5}{4})$ up to $O(e^{-n/2})$, that is, $\frac{3}{4}\log n - 0.3008$; the script evaluates both sides, which agree to ten digits at $n = 10^2, 10^4, 10^6$.

What one had to notice is that the population posterior factorizes and that each factor becomes a Gamma law in the variable $t = s \times$ (the coordinate's loss). A coordinate whose loss grows like $|w_j|^{2k}$ holds $\frac{1}{2k\beta}$ (Exercise 11.4): equipartition is a statement about quadratic losses. In Figure 11.1 each curve rises from 0, where the posterior is the prior, to a plateau at the learning coefficient, and the area under it up to $\log n$ is the population free energy: the plateau's height is λ and its length is $\log n$. Three plateaus are reached by $s \approx 30$; the product model's is not reached at $s = 10^{12}$.

COMPUTATION 11.2 · THE MEAN ENERGY OF THE PRODUCT MODEL

1 The loss of $y = abx$ is $K = \frac{1}{2}c^2$ with $c = ab$, and the prior density of c is $\rho(c) = \frac{1}{2}\log\frac{1}{|c|}$ on $[-1,1]$ (Chapter 3). So $\mathcal{E}(s) = s\int \frac{c^2}{2}e^{-sc^2/2}\rho(c)\,\mathrm{d}c \,/ \int e^{-sc^2/2}\rho(c)\,\mathrm{d}c$.

2 Substitute $u = \sqrt{s}\,c$ and put $\Lambda = \frac{1}{2}\log s$, so that $\rho(u/\sqrt{s}) = \frac{1}{2}(\Lambda - \log|u|)$. Extending the range $|u| \le \sqrt{s}$ to the whole line changes both integrals by amounts exponentially small in s, and $\mathcal{E}(s) = \frac{1}{2}\mathbb{E}[Z^2(\Lambda - \log|Z|)]/\mathbb{E}[\Lambda - \log|Z|]$ with $Z \sim \mathcal{N}(0,1)$.

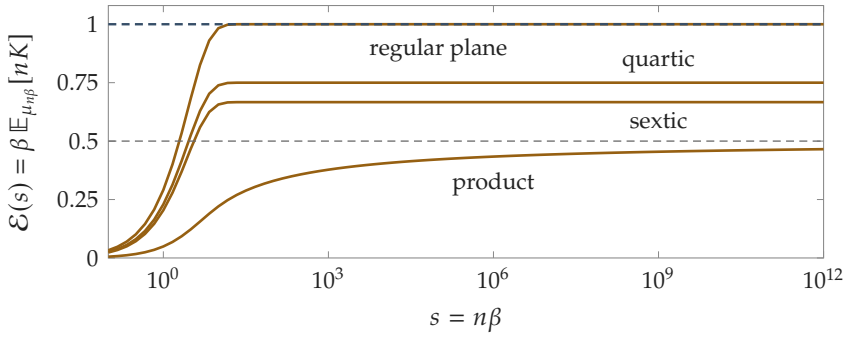


Figure 11.1. The dimensionless mean energy $\mathcal{E}(s) = s\,\mathbb{E}_{\mu_s}[K]$ of the population tempered posterior for four running models with $d = 2$ and the uniform prior on $[-1,1]^2$, computed exactly (incomplete Gamma functions; adaptive quadrature over $c = ab$ for the product model; script `11-temperature-energy.py`). Solid: the exact curves, whose plateaus are the learning coefficients $1, \frac{3}{4}, \frac{2}{3}$; the product model approaches $\lambda = \frac{1}{2}$ (gray) like $\frac{1}{2} - 1/(\log s + \gamma + \log 2)$. Dashed: equipartition, $\frac{d}{2} = 1$. Against the logarithmic axis, the area under a curve up to $s = n$ is the population free energy $\bar{F}(n)$.

3 Differentiating $\mathbb{E}|Z|^p = 2^{p/2}\Gamma(\frac{p+1}{2})/\sqrt{\pi}$ in p gives $\mathbb{E}[|Z|^p \log|Z|] = \mathbb{E}|Z|^p \cdot \frac{1}{2}\big(\log 2 + \psi(\frac{p+1}{2})\big)$, with $\psi = \Gamma'/\Gamma$. Since $\psi(\frac{1}{2}) = -\gamma - 2\log 2$ and $\psi(\frac{3}{2}) = \psi(\frac{1}{2}) + 2$, with Euler's constant $\gamma = 0.5772$, we get $\mathbb{E}\log|Z| = -\kappa$ and $\mathbb{E}[Z^2 \log|Z|] = 1 - \kappa$, where $\kappa = \frac{1}{2}(\gamma + \log 2) = 0.6352$.

4 Therefore $\mathcal{E}(s) = \frac{1}{2} \cdot \frac{\Lambda - 1 + \kappa}{\Lambda + \kappa} = \frac{1}{2} - \frac{1}{2\Lambda + 2\kappa} = \frac{1}{2} - \frac{1}{\log s + \gamma + \log 2}$, up to exponentially small terms.

5 Numbers: $\mathcal{E}(10^3) = 0.3777$, $\mathcal{E}(10^6) = 0.4337$, $\mathcal{E}(10^{12}) = 0.4654$, equal to the quadrature of the first step to ten digits. In terms of n and β, $\mathbb{E}_{\mu_{n\beta}}[nK] = \frac{1}{2\beta} - \frac{1}{\beta(\log n\beta + \gamma + \log 2)}$.

The leading coefficient $\frac{1}{2}$ is the learning coefficient of $y = abx$.[2] The deficit $1/\log s$ is its multiplicity at work: $m - 1 = 1$, and it comes from the factor $\log \frac{1}{|c|}$ in the prior density of c, the trace of the cross. Equipartition fails twice here, in the plateau and in the slowness of the approach: at $s = 10^6$ the product model still holds 13% less than $\frac{1}{2\beta}$.

[2] The digamma function $\psi = \Gamma'/\Gamma$ satisfies $\psi(x+1) = \psi(x) + \frac{1}{x}$, $\psi(1) = -\gamma$ and $\psi(\frac{1}{2}) = -\gamma - 2\log 2$; these give every constant of this chapter that involves γ.

With data the energy is nK_n, which differs from nK by a fluctuation of order $\sqrt{nK}$ (Chapter 8), that is, of order $\sqrt{\log n}$ at the temperature $1/\log n$ of WBIC (Section 11.4). Whether it survives the posterior average depends on a sign.

COMPUTATION 11.3 · WHERE WBIC FEELS THE NOISE: THE QUARTIC MODEL WITH DATA

1 Take data from the truth $(a, b) = (0, 0)$, so $y_i = \varepsilon_i$, with $x \sim \mathcal{N}(0, I_2)$; let $Z_n = (Z_{n,1}, Z_{n,2}) = n^{-1/2} \sum_i \varepsilon_i x_i$, approximately

$\mathcal{N}(0, I_2)$, and treat $\frac{1}{n}\sum_i x_i x_i^\top$ as the identity (Theorem 11.9 handles its fluctuation). Expanding the squares in $\sum_i[(y_i - ax_{i1} - b^2 x_{i2})^2 - y_i^2]/2$ gives $nK_n = \frac{n}{2}(a^2 + b^4) - \sqrt{n}(aZ_{n,1} + b^2 Z_{n,2})$.

2. So the tempered posterior is μ_s, $s = n\beta$, tilted by e^τ with $\tau = \beta\sqrt{n}(aZ_{n,1} + b^2 Z_{n,2})$: $\mathbb{E}_w^\beta[f] = \mathbb{E}_{\mu_s}[fe^\tau]/\mathbb{E}_{\mu_s}[e^\tau]$. Under μ_s the variables $\sqrt{s}\,a$ and $\sqrt{s}\,b^2$ are of order one, so $\tau = \sqrt{\beta}(\sqrt{s}\,aZ_{n,1} + \sqrt{s}\,b^2 Z_{n,2})$ is of order $\sqrt{\beta}$, small at $\beta = 1/\log n$. To first order in τ, $\mathbb{E}_w^\beta[f] = \mathbb{E}_{\mu_s}[f] + \mathrm{Cov}_{\mu_s}(f, \tau)$.

3. Apply this to $\beta nK_n = s(a^2 + b^4)/2 - \tau$ and divide by β. The terms in a vanish because μ_s is symmetric under $a \mapsto -a$, and WBIC $-nS_n = \mathbb{E}_w^\beta[nK_n] \approx \mathbb{E}_{\mu_s}[nK] + \sqrt{n}\big[\beta\,\mathrm{Cov}_{\mu_s}(nK, b^2) - \mathbb{E}_{\mu_s}[b^2]\big]Z_{n,2}$.

4. With $t = sb^4/2$, Gamma with shape $\frac{1}{4}$: $\sqrt{n}\,b^2 = \sqrt{2t/\beta}$, and $\beta\,\mathrm{Cov}_{\mu_s}(nK, b^2) = \mathrm{Cov}(t, b^2)$ because a and b are independent. Since $\mathbb{E}\sqrt{t} = \Gamma(\frac{3}{4})/\Gamma(\frac{1}{4})$ and $\mathrm{Cov}(t, \sqrt{t}) = \mathbb{E}t^{3/2} - \mathbb{E}t\,\mathbb{E}\sqrt{t} = (\frac{3}{4} - \frac{1}{4})\Gamma(\frac{3}{4})/\Gamma(\frac{1}{4})$, the bracket times $\sqrt{n}$ is $\sqrt{2/\beta}\,\big(\mathrm{Cov}(t, \sqrt{t}) - \mathbb{E}\sqrt{t}\big) = -\Gamma(\frac{3}{4})/(\Gamma(\frac{1}{4})\sqrt{2\beta})$.

5. With $\mathbb{E}_{\mu_s}[nK] = \frac{3}{4}\log n$ and $\kappa_Q = \Gamma(\frac{3}{4})/\Gamma(\frac{1}{4}) = 0.3380$, at $\beta = 1/\log n$: WBIC $-nS_n = \frac{3}{4}\log n - \kappa_Q\sqrt{\frac{1}{2}\log n}\,Z_{n,2} + O_p(1)$. The coefficient of the noise is -0.513 at $n = 10^2$, -0.725 at 10^4, -1.026 at 10^8.

6. The free energy has no such term: $F_n - nS_n = \frac{3}{4}\log n + O_p(1)$ (Theorem 8.4, with $m = 1$). So $F_n - \mathrm{WBIC} = \kappa_Q\sqrt{\frac{1}{2}\log n}\,Z_{n,2} + O_p(1)$, a gap whose standard deviation grows like $\sqrt{\log n}$.

What one had to notice is that $b^2 \geq 0$: the hot posterior gives b^2 a positive mean of order $\sqrt{\log n/n}$, so the noise $Z_{n,2}$ enters the average loss linearly, with a gain of order $\sqrt{\log n}$. In $c_2 = b^2$ the quartic model is the regular model $y = ax_1 + c_2 x_2$ restricted to $c_2 \geq 0$, with the truth on the boundary.[3] In the product model and in the sextic model $y = ax_1 + b^3 x_2$ the corresponding means vanish, because ab and b^3 change sign with a or b, and so does the $\sqrt{\log n}$ term.

[3] Watanabe (2013, §7.3) makes this observation about a model with a parameter restricted to $c \geq 0$ and the unrestricted model with $c = b^2$, and conjectures that his "even" models are related to restricted ones.

11.3 The theorem: the mean energy and the optimal temperature

Thermodynamic integration needs only a bounded energy.

Lemma 11.1 (Thermodynamic identities). *Let the data be fixed, let φ be a probability density on W, and suppose that nL_n is bounded on W (as it is for continuous losses on a compact W). Then $F_n(\beta)$ is infinitely differentiable on*

$\mathbb{R}$, $F_n(0) = 0$, *and for every* β,

$$F_n'(\beta) = \mathbb{E}_w^\beta[nL_n], \qquad F_n''(\beta) = -\operatorname{Var}_w^\beta(nL_n).$$

In particular $F_n = \int_0^1 \mathbb{E}_w^\beta[nL_n]\,\mathrm{d}\beta$.

PROOF. Write $H = nL_n$, $M = \sup_W |H|$, and $\mathcal{Z}(\beta) = \int_W \varphi e^{-\beta H}\,\mathrm{d}w$. For $|\beta| \le b$ the k-th derivative in β of the integrand, $(-H)^k \varphi e^{-\beta H}$, is bounded by $M^k e^{bM}\varphi$, which is integrable; by dominated convergence $\mathcal{Z}$ is infinitely differentiable with $\mathcal{Z}^{(k)}(\beta) = \int (-H)^k \varphi e^{-\beta H}\,\mathrm{d}w$. Since $\mathcal{Z} > 0$, the function $F_n = -\log \mathcal{Z}$ is infinitely differentiable too, $F_n' = -\mathcal{Z}'/\mathcal{Z} = \mathbb{E}_w^\beta[H]$, and $F_n'' = -\mathcal{Z}''/\mathcal{Z} + (\mathcal{Z}'/\mathcal{Z})^2 = -(\mathbb{E}_w^\beta[H^2] - \mathbb{E}_w^\beta[H]^2)$. Finally $\mathcal{Z}(0) = \int \varphi = 1$, so $F_n(0) = 0$, and the last identity is the fundamental theorem of calculus. ∎

Theorem 11.2 (The optimal temperature; Watanabe 2013, Theorem 3). *Under the hypotheses of Lemma 11.1, suppose that nL_n is not constant φ-almost everywhere. Then $\beta \mapsto \mathbb{E}_w^\beta[nL_n]$ is strictly decreasing, and there is exactly one $\beta^* \in (0,1)$ with*

$$F_n = \mathbb{E}_w^{\beta^*}[nL_n].$$

PROOF. The plan is to show that $F_n(\beta)$ is strictly concave and to apply the mean value theorem. The tempered posterior has density proportional to $\varphi e^{-\beta n L_n}$, which is positive exactly where φ is, so the two measures have the same null sets. Since nL_n is not φ-almost everywhere constant, it is not almost surely constant under the tempered posterior, and its variance is positive. By Lemma 11.1, $F_n'' < 0$ on $\mathbb{R}$, so $F_n' = \mathbb{E}_w^\beta[nL_n]$ is strictly decreasing. By the mean value theorem there is $\beta^* \in (0,1)$ with $F_n'(\beta^*) = F_n(1) - F_n(0) = F_n$, and it is unique because F_n' is injective. ∎

Watanabe states the hypothesis as "L_n is not a constant function"; the proof uses the almost-everywhere version. The theorem does not locate β^*; the rest of the chapter asks how far it is from $1/\log n$.

For the population mean energy we need one more term than the volume asymptotics of Lemma 4.3 provide.

Assumption 11.3 (Volume expansion). For the loss $K \ge 0$ and the prior φ, the volume $V(\varepsilon) = \varphi\{w : K(w) < \varepsilon\}$ satisfies, for some $\lambda > 0$, an integer $m \ge 1$, a polynomial $P(x) = \sum_{j=0}^{m-1} p_j x^j$ with $p_{m-1} > 0$, and constants $\eta, B, \varepsilon_0 > 0$,

$$\left| V(\varepsilon) - \varepsilon^\lambda P\!\left(\log \tfrac{1}{\varepsilon}\right) \right| \le B\,\varepsilon^{\lambda+\eta} \qquad (0 < \varepsilon \le \varepsilon_0). \tag{11.3}$$

The expansion implies $V(\varepsilon) \sim p_{m-1}\varepsilon^\lambda (\log \tfrac{1}{\varepsilon})^{m-1}$; and if K is bounded, the expansion inserted into the Mellin formula of Theorem 4.6 continues ζ meromorphically to $\Re z > -\lambda - \eta$, where its only pole is $-\lambda$, of order m. So λ and m are the learning coefficient and its multiplicity; as an

invariant of K and φ, λ is the real log canonical threshold, the same number. For $\varepsilon \le \frac{1}{2}$ the running models satisfy (11.3) with no remainder: $V(\varepsilon) = \frac{\pi}{2}\varepsilon$ for the regular plane, $1.4699\,\varepsilon^{3/4}$ for the quartic model,[4] and $\sqrt{2}\,\varepsilon^{1/2}(\frac{1}{2}\log\frac{1}{\varepsilon} + 1 - \frac{1}{2}\log 2)$ for the product model. When K is analytic and φ satisfies Watanabe's conditions, the state density V' has an asymptotic expansion in the functions $t^{\lambda_k - 1}(\log\frac{1}{t})^j$ read off from the poles of the zeta function (Watanabe 2009, Chapter 4), and integrating it gives (11.3).

[4] The region $\{a^2 + b^4 < 2\varepsilon\}$ has area $2(2\varepsilon)^{3/4}\int_{-1}^{1}(1-u^4)^{1/2}\,\mathrm{d}u$, and with the prior density $\frac{1}{4}$ this gives $V(\varepsilon) = 2^{-5/4}B(\frac{1}{4},\frac{3}{2})\,\varepsilon^{3/4}$, with Euler's Beta function B.

Proposition 11.4 (The tempered mean energy). *Under Assumption 11.3, let*

$$Q(x) = \sum_{j=0}^{m-1} p_j \sum_{i=0}^{j} \binom{j}{i} (-1)^i\, \Gamma^{(i)}(\lambda+1)\, x^{j-i},$$

a polynomial of degree $m-1$ with leading coefficient $p_{m-1}\Gamma(\lambda+1) > 0$, and write $x = \log s$. As $s \to \infty$, the dimensionless mean energy (11.2) and the heat capacity $C(s) := s^2 \operatorname{Var}_{\mu_s}(K)$ satisfy

$$\mathcal{E}(s) = \lambda - \frac{Q'(x)}{Q(x)} + O(s^{-\eta}), \qquad C(s) = \lambda - \frac{Q'(x)}{Q(x)} + \Big(\frac{Q'}{Q}\Big)'(x) + O(s^{-\eta}). \tag{11.4}$$

Consequently, as $n\beta \to \infty$,

$$\mathbb{E}_{\mu_{n\beta}}[nK] = \frac{\lambda}{\beta} - \frac{m-1}{\beta\log(n\beta)} + O\Big(\frac{1}{\beta\log^2(n\beta)}\Big), \tag{11.5}$$
$$\beta^2 \operatorname{Var}_{\mu_{n\beta}}(nK) = \lambda - \frac{m-1}{\log(n\beta)} + O\Big(\frac{1}{\log^2(n\beta)}\Big),$$

and when $m = 1$ the remainders are $O((n\beta)^{-\eta}/\beta)$ and $O((n\beta)^{-\eta})$.

Proof. The plan is to write $\mathcal{E}$ and C through Laplace transforms of the volume, to compute those transforms exactly for the model function $t^\lambda P(\log\frac{1}{t})$, and to show that the remainder in (11.3) changes them by a relative $O(s^{-\eta})$.

Laplace transforms. Let $\mathcal{Z}(s) = \int e^{-sK}\varphi$, $N(s) = \int Ke^{-sK}\varphi$, and $M(s) = \int K^2 e^{-sK}\varphi$, so that $\mathcal{E} = sN/\mathcal{Z}$ and $C = s^2 M/\mathcal{Z} - \mathcal{E}^2$. For every $k \ge 0$,

$$e^{-sk} = \int_k^\infty s e^{-st}\,\mathrm{d}t, \qquad ke^{-sk} = \int_k^\infty (st-1)e^{-st}\,\mathrm{d}t,$$
$$k^2 e^{-sk} = \int_k^\infty (st^2 - 2t)e^{-st}\,\mathrm{d}t,$$

as one checks by differentiating $-e^{-st}$, $-te^{-st}$ and $-t^2e^{-st}$. Insert $k = K(w)$, integrate against φ, and exchange the integrals, which Fubini's theorem allows because the absolute values of the integrands have finite integrals over $W \times (0,\infty)$ (for instance $\int_0^\infty (st+1)e^{-st}\,\mathrm{d}t = 2/s$). Since $\int_W \mathbf{1}\{K(w) < t\}\varphi(w)\,\mathrm{d}w = V(t)$,

$$\mathcal{Z}(s) = s\int_0^\infty e^{-st}V(t)\,\mathrm{d}t, \qquad N(s) = \int_0^\infty (st-1)e^{-st}V(t)\,\mathrm{d}t,$$
$$M(s) = \int_0^\infty (st^2-2t)e^{-st}V(t)\,\mathrm{d}t. \tag{11.6}$$

The model function. Let $V_P(t) = t^\lambda P(\log \frac{1}{t})$ for $t > 0$ and define $\mathcal{Z}_P, N_P, M_P$ by (11.6) with V_P in place of V. Substituting $t = u/s$, so that $\log \frac{1}{t} = \log s - \log u$, expanding $(\log s - \log u)^j$ by the binomial theorem, and using $\Gamma^{(i)}(\lambda+1) = \int_0^\infty u^\lambda (\log u)^i e^{-u}\,\mathrm{d}u$ (Definition A.4) gives $\mathcal{Z}_P(s) = s^{-\lambda} Q(\log s)$. Differentiating (11.6) under the integral sign gives $N_P = -\mathcal{Z}_P'$ and $M_P = \mathcal{Z}_P''$, and differentiating $s^{-\lambda}Q(\log s)$ gives

$$sN_P(s) = s^{-\lambda}\big(\lambda Q - Q'\big)(x),$$
$$s^2 M_P(s) = s^{-\lambda}\big(\lambda(\lambda+1)Q - (2\lambda+1)Q' + Q''\big)(x).$$

Dividing by $\mathcal{Z}_P$ and using $(Q'/Q)' = Q''/Q - (Q'/Q)^2$, the model function has dimensionless mean energy $\lambda - Q'/Q$ and heat capacity $\lambda(\lambda+1) - (2\lambda+1)Q'/Q + Q''/Q - (\lambda - Q'/Q)^2 = \lambda - Q'/Q + (Q'/Q)'$, both at x.

The remainder. The difference $R = V - V_P$ satisfies $|R(t)| \le Bt^{\lambda+\eta}$ on $(0, \varepsilon_0]$; on (ε_0, ∞), $|\log \frac{1}{t}| \le \log \frac{1}{\varepsilon_0} + t$, so $|R(t)| \le 1 + |V_P(t)| \le C(1+t)^{\lambda+m}$, and for $s \ge 1$ its contributions to the three integrals, multiplied by 1, s and s^2, are at most $C' e^{-s\varepsilon_0/4}$. On $(0, \varepsilon_0]$ we use $\int_0^\infty (st)^a e^{-st} t^{\lambda+\eta}\,\mathrm{d}t = \Gamma(\lambda+\eta+a+1)s^{-\lambda-\eta-1}$ for $a = 0, 1, 2$. Together,

$$|\mathcal{Z} - \mathcal{Z}_P| + s|N - N_P| + s^2|M - M_P| \le C'' s^{-\lambda-\eta} \qquad (s \ge 1).$$

The ratios. Since Q has a positive leading coefficient, there are $c, x_0 > 0$ with $Q(x) \ge c\,x^{m-1} \ge c$ for $x \ge x_0$, while $\lambda Q - Q'$ and $\lambda(\lambda+1)Q - (2\lambda+1)Q' + Q''$ are $O(x^{m-1})$. Dividing the last display by $s^{-\lambda}Q(x)$ shows $\mathcal{Z} = s^{-\lambda}Q(x)(1 + O(s^{-\eta}))$, $sN = s^{-\lambda}((\lambda Q - Q')(x) + O(s^{-\eta}))$, and the same for $s^2 M$, whence (11.4). If $m \ge 2$, write $Q(x) = q_{m-1}x^{m-1} + q_{m-2}x^{m-2} + \dots$; then $Q'/Q = (m-1)/x + O(x^{-2})$ and $(Q'/Q)' = O(x^{-2})$, and (11.5) follows from $\mathbb{E}_{\mu_{n\beta}}[nK] = \mathcal{E}(n\beta)/\beta$ and $\beta^2 \operatorname{Var}_{\mu_{n\beta}}(nK) = C(n\beta)$. If $m = 1$, Q is a positive constant and both corrections vanish. ∎

For the product model, $P(x) = \frac{\sqrt{2}}{2}x + \sqrt{2}(1 - \frac{1}{2}\log 2)$, so $Q(x) = \Gamma(\frac{3}{2})\big(p_1 x + p_0 - p_1\psi(\frac{3}{2})\big)$ and $Q'/Q = 1/(x + \gamma + \log 2)$, which is Computation 11.2 again; for a regular model, $\lambda = \frac{d}{2}$ and $m = 1$, which is equipartition. The proposition is the derivative in β of the population free energy $\lambda \log(n\beta) - (m-1)\log\log(n\beta) + O(1)$ of Lemma 4.3, but asymptotic formulas cannot be differentiated term by term, and this one cannot under the hypothesis of that lemma alone (Exercise 11.10).

11.4 The theorem: WBIC

Definition 11.5 (WBIC; Watanabe 2013). The *widely applicable Bayesian information criterion* is the mean energy of the tempered posterior at inverse temperature $1/\log n$:

$$\mathrm{WBIC} = \mathbb{E}_w^\beta[nL_n(w)], \qquad \beta = \frac{1}{\log n}.$$

It costs one run of a sampler at one temperature.[5] Theorem 11.2 says that some β^* reproduces F_n exactly; WBIC bets that $\beta^* \approx 1/\log n$, and Watanabe's theorem, under his "fundamental conditions", says how good the bet is.

[5] The paper was submitted to the *Journal of Machine Learning Research* in August 2012, revised in February 2013 and published in March 2013, according to its header.

Theorem 11.6 (WBIC; Watanabe 2013, Theorem 4 and Corollaries 1–3). *Assume that $W \subset \mathbb{R}^d$ is compact with nonempty interior and $W = \{\pi_1 \geq 0, \dots, \pi_k \geq 0\}$ for analytic π_j; that $\varphi = \varphi_1\varphi_2$ with $\varphi_1 \geq 0$ analytic and $\varphi_2 > 0$ smooth; that the map $w \mapsto f(\cdot, w) = \log(p_0/p(\cdot \mid w))$ extends to an analytic map, on an open set containing W, into the functions with finite sixth moment under q; that $\mathbb{E}_X[f(X,w)] \geq c\,\mathbb{E}_X[f(X,w)^2]$ whenever $K(w) \leq \varepsilon$, for some $c, \varepsilon > 0$ (relatively finite variance); and that L has a minimizer w_0 in the interior of W and all its minimizers give the same distribution p_0. Let $\beta = \beta_0/\log n$ with a constant $\beta_0 > 0$. Then:*

(i) there are random variables U_n with $\mathbb{E}[U_n] = 0$, converging in law to a Gaussian random variable, such that

$$\mathbb{E}_w^\beta[nL_n] = nL_n(w_0) + \frac{\lambda \log n}{\beta_0} + U_n\sqrt{\frac{\lambda \log n}{2\beta_0}} + O_p(1),$$

and $\mathbb{E}[U_n^2] < 1$ if the truth is realizable, $q = p_0$;

(ii) if the parity of the model is odd, then $U_n = 0$;

(iii) the optimal temperature of Theorem 11.2 satisfies $\beta^* \log n = 1 + U_n/\sqrt{2\lambda \log n} + o_p(1/\sqrt{\log n})$;

(iv) for $\beta_j = \beta_{0j}/\log n$ with constants $\beta_{01} \neq \beta_{02}$,

$$\hat\lambda := \frac{\mathbb{E}_w^{\beta_1}[nL_n] - \mathbb{E}_w^{\beta_2}[nL_n]}{1/\beta_1 - 1/\beta_2} \longrightarrow \lambda \quad \text{in probability.}$$

In particular $\mathrm{WBIC} = nL_n(w_0) + \lambda \log n + U_n\sqrt{\lambda \log n/2} + O_p(1)$, *and with Theorem 8.6,*

$$F_n - \mathrm{WBIC} = -(m-1)\log\log n - U_n\sqrt{\tfrac{1}{2}\lambda \log n} + O_p(1) = O_p\big(\sqrt{\log n}\big). \tag{11.7}$$

The *parity* is defined on a resolution of singularities $w = g(u)$ (Theorem A.8) that puts K and the prior in the normal-crossing form $K(g(u)) = u^{2k}$, $\varphi(g(u))|g'(u)| = b(u)|u^h|$: it is odd if some resolution has, in every chart attaining λ with multiplicity m, a coordinate u_j attaining λ with k_j odd and with the u_j-axis near 0 in the support of the prior, and even otherwise (Watanabe 2013, eq. (17)). Regular models with a prior positive at the truth are odd (Watanabe 2013, Lemma 3), and so is reduced-rank regression for every truth and prior (Watanabe 2013, §6.1). The product model is odd: $K = \frac{1}{2}a^2b^2$ is already in normal-crossing form with $k = (1,1)$. The quartic model is even: a coordinate attaining $\lambda = \frac{3}{4} = \frac{h_j+1}{2k_j}$ has $3k_j = 2(h_j+1)$, so k_j is even in every resolution (the weighted blow-up $(a,b) = (uv^2, v)$ of Chapter 5 gives $K = \frac{1}{2}v^4(1+u^2)$). When the truth is realizable the parity does not depend on the resolution (Watanabe 2013, Lemma 2).

For linear models the theorem can be proved with the tools of Chapter 8, and the proof shows where U_n comes from.

Assumption 11.7 (Linear models). (i) The pairs (x_i, y_i) are independent and identically distributed, $x_i \in \mathbb{R}^k$ with $\Sigma = \mathbb{E}[xx^\top]$ positive definite and $\mathbb{E}\|x\|^4 < \infty$, and $y_i = g(w_0)^\top x_i + \varepsilon_i$ with $\varepsilon_i \sim \mathcal{N}(0,1)$ independent of x_i.

(ii) The model is $y = g(w)^\top x + \mathcal{N}(0,1)$ noise, with w in a compact set $W \subset \mathbb{R}^d$ that contains w_0, g continuous, and a bounded prior density φ. Write $\delta(w) = g(w) - g(w_0)$ and $K(w) = \frac{1}{2}\delta(w)^\top \Sigma\, \delta(w)$.

(iii) K and φ satisfy Assumption 11.3.

These hypotheses contain those of Theorem 8.4. The losses are conditional on the inputs, $L_n(w) = -\frac{1}{n}\sum_i \log p(y_i \mid x_i, w)$, so $nL_n(w_0) = nS_n$, and the data enter through $\Sigma_n = \frac{1}{n}\sum_i x_i x_i^\top$ and $Z_n = n^{-1/2}\sum_i \varepsilon_i x_i$.

Lemma 11.8 (Moments of the population posterior). *Under Assumption 11.3, let $T = sK$ under μ_s. For every integer $j \geq 0$ and $\theta \in [0,1)$, $\mathbb{E}_{\mu_s}[T^j e^{\theta T}]$ is bounded for $s \geq 1$, and T converges in law, as $s \to \infty$, to the Gamma distribution with shape λ and scale 1.*

Proof. The proof of Proposition 11.4 shows $\mathcal{Z}(s) = s^{-\lambda} Q(\log s)(1 + o(1))$, so $\mathcal{Z}(as)/\mathcal{Z}(s) \to a^{-\lambda}$ for each fixed $a > 0$. Put $\theta' = \frac{1}{2}(1+\theta) < 1$. Since $T^j e^{\theta T} \leq C_{j,\theta} e^{\theta' T}$, we get $\mathbb{E}_{\mu_s}[T^j e^{\theta T}] \leq C_{j,\theta}\mathcal{Z}((1-\theta')s)/\mathcal{Z}(s)$, which is continuous in s and converges to $C_{j,\theta}(1-\theta')^{-\lambda}$, hence is bounded on $[1,\infty)$. For $u \geq 0$, $\mathbb{E}_{\mu_s}[e^{-uT}] = \mathcal{Z}((1+u)s)/\mathcal{Z}(s) \to (1+u)^{-\lambda}$, the Laplace transform of the Gamma law with shape λ, and convergence of the Laplace transforms of nonnegative random variables implies convergence in law. ∎

Theorem 11.9 (WBIC for linear models). *Under Assumption 11.7, let $\beta = \beta_0/\log n$ with a constant $\beta_0 > 0$, put $s = n\beta$, and define the deterministic vector*

$$c_n = \beta \operatorname{Cov}_{\mu_s}\big(nK, \sqrt{n}\,\delta\big) - \mathbb{E}_{\mu_s}\big[\sqrt{n}\,\delta\big] \in \mathbb{R}^k.$$

(i) $\mathbb{E}_w^\beta[nL_n] = nL_n(w_0) + \dfrac{\lambda \log n}{\beta_0} + c_n^\top Z_n + O_p(1)$.

(ii) $\mathbb{E}[c_n^\top Z_n] = 0$ *and* $\operatorname{Var}(c_n^\top Z_n) = c_n^\top \Sigma c_n = O(\log n)$.

(iii) If a measurable map $\iota : W \to W$ carries the prior distribution to itself and $\delta \circ \iota = -\delta$, then $c_n = 0$.

(iv) If, under μ_s, the pair (sK, ϑ) with $\vartheta = \Sigma^{1/2}\delta/\|\Sigma^{1/2}\delta\|$ converges in law to the product of the Gamma law with shape λ and a law ν on the unit sphere, then $U_n := c_n^\top Z_n / \sqrt{\lambda \log n/(2\beta_0)}$ converges in law to $\mathcal{N}(0,\sigma^2)$ and $\mathbb{E}[U_n^2] \to \sigma^2$, where

$$\sigma^2 = \left(\frac{\Gamma(\lambda + \frac{1}{2})}{\Gamma(\lambda)}\right)^2 \frac{\|\bar{\vartheta}\|^2}{\lambda} < 1, \qquad \bar{\vartheta} = \int \vartheta\, \nu(\mathrm{d}\vartheta).$$

In particular, for $\beta_0 = 1$ and by Theorem 8.4, $F_n - \mathrm{WBIC} = -(m-1)\log\log n - c_n^\top Z_n + O_p(1)$.

Proof. The plan is to write the tempered posterior as the population posterior μ_s tilted by a factor e^{τ} with τ small, to expand to first order in τ, and to read off the deterministic term from Proposition 11.4. Write $\mu = \mu_s$, $T = sK$, $v = \Sigma^{1/2}\delta$ (so $\|v\|^2 = 2K$), $\zeta_n = \Sigma^{-1/2}Z_n$, and $D_n = \Sigma^{-1/2}\Sigma_n\Sigma^{-1/2} - I$. By the central limit theorem and Chebyshev's inequality, $\zeta_n = O_p(1)$ and $\|D_n\| = O_p(n^{-1/2})$.

The tilt. Expanding the squares as in Computation 11.3, $nK_n = \frac{n}{2}\delta^{\top}\Sigma_n\delta - \sqrt{n}\,\delta^{\top}Z_n = nK - A_n$ with

$$A_n = \sqrt{n}\,v^{\top}\zeta_n - \tfrac{n}{2}v^{\top}D_n v, \qquad |A_n| \le \sqrt{2nK}\,\|\zeta_n\| + nK\|D_n\|.$$

Hence the tempered posterior has density proportional to $e^{-T+\tau}$ with respect to φ, where $\tau = \beta A_n$, and $\mathbb{E}_w^{\beta}\big[f\big] = \mathbb{E}_{\mu}[f e^{\tau}]/\mathbb{E}_{\mu}[e^{\tau}]$.

Bounds. Let $\rho_n = \sqrt{\beta}\,\|\zeta_n\| + \|D_n\|$, which is $O_p((\log n)^{-1/2})$, and work on the event $\{\|D_n\| \le \frac{1}{4}\}$, whose probability tends to one. There $|\tau| \le \sqrt{2\beta T}\|\zeta_n\| + T\|D_n\| \le \sqrt{2}(1+T)\rho_n$, and since $\sqrt{2\beta T}\|\zeta_n\| \le \frac{T}{4} + 2\beta\|\zeta_n\|^2$, also $|\tau| \le \frac{T}{2} + 2\beta\|\zeta_n\|^2$. With Lemma 11.8 for $\theta = \frac{1}{2}$ this gives, for all $j, k \ge 0$,

$$\mathbb{E}_{\mu}\big[T^j|\tau|^k e^{|\tau|}\big] \le C_{j,k}\, e^{2\beta\|\zeta_n\|^2}\rho_n^k = O_p(\rho_n^k).$$

First order. By Taylor's theorem $e^{\tau} = 1 + \tau + R$ with $|R| \le \frac{1}{2}\tau^2 e^{|\tau|}$. For f equal to T or τ put $\Delta_f = \mathrm{Cov}_{\mu}(f, R) - \mathrm{Cov}_{\mu}(f, \tau)\,\mathbb{E}_{\mu}[\tau + R]$; expanding numerator and denominator gives the identity

$$\mathbb{E}_w^{\beta}\big[f\big] - \mathbb{E}_{\mu}[f] - \mathrm{Cov}_{\mu}(f, \tau) = \frac{\Delta_f}{\mathbb{E}_{\mu}[e^{\tau}]}.$$

By the bounds, $\Delta_T = O_p(\rho_n^2)$ and $\Delta_\tau = O_p(\rho_n^3)$, while $\mathbb{E}_{\mu}[e^{\tau}] \ge e^{-\mathbb{E}_{\mu}|\tau|}$ by Jensen's inequality is bounded below in probability; and $\mathrm{Cov}_{\mu}(\tau, \tau) = O_p(\rho_n^2)$. Therefore

$$\mathbb{E}_w^{\beta}[T] = \mathbb{E}_{\mu}[T] + \mathrm{Cov}_{\mu}(T, \tau) + O_p(\rho_n^2), \qquad \mathbb{E}_w^{\beta}[\tau] = \mathbb{E}_{\mu}[\tau] + O_p(\rho_n^2).$$

Assembly. Since $\beta n K_n = T - \tau$, dividing by β gives $\mathbb{E}_w^{\beta}[nK_n] = \mathbb{E}_{\mu}[nK] + \mathrm{Cov}_{\mu}(T, A_n) - \mathbb{E}_{\mu}[A_n] + O_p(\rho_n^2/\beta)$, and $\rho_n^2/\beta \le 2\|\zeta_n\|^2 + 2\|D_n\|^2\log n/\beta_0 = O_p(1)$. In A_n, the term with D_n satisfies $|\frac{n}{2}v^{\top}D_n v| \le \|D_n\|T/\beta$, so its contribution to $\mathrm{Cov}_{\mu}(T, A_n) - \mathbb{E}_{\mu}[A_n]$ is $O_p(\|D_n\|\log n) = o_p(1)$. The term with ζ_n contributes $\big(\mathrm{Cov}_{\mu}(T, \sqrt{n}\,v) - \mathbb{E}_{\mu}[\sqrt{n}\,v]\big)^{\top}\zeta_n = (\Sigma^{1/2}c_n)^{\top}\Sigma^{-1/2}Z_n = c_n^{\top}Z_n$. Hence $\mathbb{E}_w^{\beta}[nK_n] = \mathbb{E}_{\mu}[nK] + c_n^{\top}Z_n + O_p(1)$.

The deterministic term. By Proposition 11.4, $\mathbb{E}_{\mu}[nK] = \frac{1}{\beta}\big(\lambda - \frac{m-1}{\log s} + O(\log^{-2} s)\big)$, and $\log s = \log n - \log\log n + \log\beta_0 \ge \frac{1}{2}\log n$ for large n, so $\mathbb{E}_{\mu}[nK] = \frac{\lambda\log n}{\beta_0} + O(1)$. Adding $nL_n(w_0)$ to $\mathbb{E}_w^{\beta}[nK_n]$ proves (i).

(ii)–(iv). The mean and the covariance of Z_n are 0 and Σ, which gives the first two claims of (ii). Next, $\sqrt{n}\,v = \sqrt{2T/\beta}\,\vartheta$, so

$$\sqrt{\beta}\,\Sigma^{1/2}c_n = \mathrm{Cov}_{\mu}\big(T, \sqrt{2T}\vartheta\big) - \mathbb{E}_{\mu}\big[\sqrt{2T}\vartheta\big] = \mathbb{E}_{\mu}\big[\sqrt{2T}\,\vartheta\,(T - \mathbb{E}_{\mu}T - 1)\big],$$

whose norm is bounded by Lemma 11.8; hence $c_n^{\top}\Sigma c_n = O(1/\beta) = O(\log n)$. For (iii), $K \circ \iota = K$, so μ is invariant under ι; then $\mathbb{E}_{\mu}[\delta] = \mathbb{E}_{\mu}[\delta \circ \iota] = -\mathbb{E}_{\mu}[\delta]$,

likewise $\mathbb{E}_\mu[K\delta] = -\mathbb{E}_\mu[K\delta]$, both vanish, and so does c_n. For (iv), the variables in the last display are uniformly integrable by Lemma 11.8, and $\mathbb{E}_\mu T \to \lambda$, so the display converges to $\sqrt{2}\,\mathbb{E}[\sqrt{t}(t-\lambda-1)]\,\bar{\vartheta}$ with t Gamma with shape λ. Since $\mathbb{E}t^{3/2} = (\lambda+\frac{1}{2})\mathbb{E}\sqrt{t}$ and $\mathbb{E}\sqrt{t} = \Gamma(\lambda+\frac{1}{2})/\Gamma(\lambda)$, the limit is $\gamma_\infty = -\Gamma(\lambda+\frac{1}{2})\bar{\vartheta}/(\sqrt{2}\,\Gamma(\lambda))$. Now $U_n = \sqrt{2/\lambda}\,(\sqrt{\beta}\,\Sigma^{1/2}c_n)^\top\zeta_n$, where ζ_n converges in law to $\mathcal{N}(0,I)$ and has covariance I; Slutsky's lemma gives the limit law $\mathcal{N}(0, 2\|\gamma_\infty\|^2/\lambda) = \mathcal{N}(0,\sigma^2)$, and $\mathbb{E}[U_n^2] = \frac{2}{\lambda}\|\sqrt{\beta}\,\Sigma^{1/2}c_n\|^2 \to \sigma^2$. Finally, the Cauchy–Schwarz inequality for $t^{(\lambda-1)/2}e^{-t/2}$ and $t^{\lambda/2}e^{-t/2}$, which are not proportional, gives $\Gamma(\lambda+\frac{1}{2})^2 < \Gamma(\lambda)\Gamma(\lambda+1) = \lambda\Gamma(\lambda)^2$, and since $\|\bar{\vartheta}\| \le 1$, it follows that $\sigma^2 \le \Gamma(\lambda+\frac{1}{2})^2/(\lambda\Gamma(\lambda)^2) < 1$. The last sentence of the theorem combines (i) with Theorem 8.4, whose hypotheses Assumption 11.7 contains. ∎

The symmetry of (iii) holds for the product model ($\iota(a,b) = (-a,b)$), the sextic model and the regular plane ($\iota(w) = -w$), and the chain; there $U_n = 0$, as odd parity predicts. In the quartic model ϑ is a function of the sign of a and of $X/(X+Y)$, with $X = sa^2/2$ and $Y = sb^4/2$ independent Gamma variables in the limit, so ϑ is independent of $T = X+Y$ and (iv) applies,[6] with $c_n = -\kappa_Q e_2/\sqrt{2\beta}$ up to exponentially small terms, as in Computation 11.3, and $\mathbb{E}[U_n^2] \to \frac{4}{3}\kappa_Q^2 = 0.152$ (Exercise 11.3).

[6] For independent Gamma variables X and Y with a common scale, $X+Y$ and $X/(X+Y)$ are independent: the change of variables $(X,Y) \mapsto (X+Y, X/(X+Y))$ factors their joint density.

Corollary 11.10 (Learning coefficients from temperatures). *Under Assumption 11.7, let $\beta_j = \beta_{0j}/\log n$ with constants $\beta_{01} \neq \beta_{02}$, and let $\hat{\lambda}$ be the two-temperature estimator of Theorem 11.6.*

(i) $\hat{\lambda} = \lambda + O_p((\log n)^{-1/2})$, and $\hat{\lambda} = \lambda + O_p(1/\log n)$ if the vectors c_n of both temperatures vanish.

(ii) The heat-capacity estimator $\hat{\lambda}_{\mathrm{C}} := \beta_1^2 \operatorname{Var}_w^{\beta_1}(nL_n)$ satisfies $\hat{\lambda}_{\mathrm{C}} = \lambda + O_p((\log n)^{-1/2})$.

(iii) Their population versions, with $\mathbb{E}_{\mu_{n\beta_j}}$ in place of $\mathbb{E}_w^{\beta_j}$ and K in place of K_n, equal $\lambda - \frac{m-1}{\log n} + O\big(\frac{\log\log n}{\log^2 n}\big)$.

Proof. The plan is to subtract or to square the expansions of the theorem. For (i), subtract the expansions of Theorem 11.9(i) at β_1 and β_2 and divide by $1/\beta_1 - 1/\beta_2 = (1/\beta_{01} - 1/\beta_{02})\log n$: the terms $nL_n(w_0)$ cancel, the leading terms give λ, the terms $c_n^\top Z_n$ give $O_p((\log n)^{1/2})/\log n$, and the remainders $O_p(1)/\log n$. For (ii), $\beta_1^2 \operatorname{Var}_w^{\beta_1}(nL_n) = \operatorname{Var}_w^{\beta_1}(T-\tau)$ in the notation of the last proof. For every f, $\mathbb{E}_w^\beta[f] - \mathbb{E}_\mu[f] = \operatorname{Cov}_\mu(f, e^\tau - 1)/\mathbb{E}_\mu[e^\tau]$ and $|e^\tau - 1| \le |\tau|e^{|\tau|}$, so the bounds of that proof give $\mathbb{E}_w^\beta[f] = \mathbb{E}_\mu[f] + O_p(\rho_n)$ for $f = (T-\tau)^2$ and $f = T-\tau$; hence $\operatorname{Var}_w^{\beta_1}(T-\tau) = \operatorname{Var}_\mu(T-\tau) + O_p(\rho_n) = \operatorname{Var}_\mu(T) + O_p(\rho_n)$, and $\operatorname{Var}_\mu(T) = C(s) = \lambda + O(1/\log s)$ by Proposition 11.4. For (iii), insert $\log(n\beta_j) = \log n - \log\log n + \log\beta_{0j}$ into (11.5): then $1/(\beta_j\log(n\beta_j)) = \frac{\log n}{\beta_{0j}}\big(\frac{1}{\log n} + O(\frac{\log\log n}{\log^2 n})\big)$, and both estimators become $\lambda - \frac{m-1}{\log n} + O(\frac{\log\log n}{\log^2 n})$. ∎

The bias $-(m-1)/\log n$ of (iii) belongs to the population: it remains when the sampler is perfect and the noise is averaged away. For the product

model at $n = 10^3$ it is -0.145 to leading order, and the exact population value of $\hat\lambda$ with $\beta_{01} = 1$ and $\beta_{02} = 1.5$, from Computation 11.2, is 0.320 instead of $\frac{1}{2}$. The heat-capacity estimator was proposed by Imai (2019a), in a preprint, with a consistency argument that differentiates Watanabe's expansion in β; part (ii) proves it for linear models.

Proof sketch of Theorem 11.6. The plan is that of Theorem 11.9, with a resolution of singularities in place of linearity; we follow Watanabe (2013, §5) and omit the empirical-process estimates. After localizing to $\{K < n^{-r}\}$, $r < \frac{1}{2}$ (his Lemma 1), a resolution gives $K(g(u)) = u^{2k}$, $f(x, g(u)) = u^k a(x, u)$ and $\varphi\,\mathrm{d}w = b(u)|u^h|\,\mathrm{d}u$ (his Theorem 1; relatively finite variance makes f divisible by u^k), so $nK_n(g(u)) = nu^{2k} - \sqrt{n}\,u^k\xi_n(u)$ with an empirical process ξ_n that converges in law to a Gaussian process. As $t \to 0$ the measure $\delta(t - u^{2k})|u^h|\,\mathrm{d}u$ behaves like $t^{\lambda-1}(\log\frac{1}{t})^{m-1}\,\mathrm{d}u^*$ for a measure $\mathrm{d}u^*$ on the set where the coordinates attaining λ vanish (his Lemma 4). With $t = n\beta u^{2k}$ the tilt is $e^{\pm\sqrt{\beta t}\,\xi_n(u)} = 1 \pm \sqrt{\beta t}\,\xi_n(u) + O_p(\beta)$, with the sign of u^k: our expansion with $\xi_n(u)$ in place of $\sqrt{2}\,\vartheta^\top\zeta_n$, and the Gamma moments give $\mathbb{E}^\beta_w[nK_n] = \frac{\lambda}{\beta} - \frac{\Theta\,\Gamma(\lambda+\frac{1}{2})}{2\sqrt{\beta}\,\Gamma(\lambda)} + O_p(1)$, with Θ the $\mathrm{d}u^*$-average of ξ_n times the signs of u^k, a sum that odd parity makes vanish. In the realizable case $\mathbb{E}[\xi_n(u)^2] = 2$ where $u^{2k} = 0$, so $\mathbb{E}[\Theta^2] \le 2$, and $\mathbb{E}[U_n^2] < 1$ by the Gamma inequality above. The general proof adds Hironaka's theorem, the empirical process on the resolved manifold, and non-Gaussian data; analyticity provides the volume expansion. ∎

11.5 Measuring it

The script `11-temperature-lambda.py` measures $\hat\lambda$ (with $\beta_{01} = 1$, $\beta_{02} = 1.5$, as in Watanabe's experiment), the heat-capacity estimate, and F_n − WBIC for four running models at the truth $w_0 = 0$: the regular plane ($\lambda = 1$), the quartic model ($\frac{3}{4}$), the sextic model ($\frac{2}{3}$) and the product model ($\frac{1}{2}$, $m = 2$), all with $\frac{d}{2} = 1$. For each model and each $n = 10^2, 10^3, \ldots, 10^8$ it draws 200 data sets, which enter only through $S = \sum_i x_i x_i^\top$ and $r = \sum_i x_i y_i$, sampled exactly,[7] and it integrates the tempered posteriors by quadrature (in closed form over a, by Gauss–Legendre nodes over b or $c = ab$), which agrees with the kit's 2000×2000 grid to six digits. It runs in about a minute.

In Figure 11.2 the regular plane gives $\hat\lambda = 1.000$ at every $n \ge 10^3$: its posterior is Gaussian, and equipartition holds exactly. The quartic model gives λ within its interval at every n (0.745 [0.738, 0.752] at $n = 10^2$, 0.750 [0.746, 0.754] at 10^8): $m = 1$ and its volume is an exact power. The sextic model lies just above $\frac{2}{3}$, by 0.0045 at $n = 10^2$ and 0.0003 at 10^8, outside its narrow intervals: the $O_p(1/\log n)$ error of Corollary 11.10(i), from the data, not the population. The product model sits far below $\frac{1}{2}$, at 0.324 for $n = 10^3$ and 0.438 for 10^8, and for $n \ge 10^3$ its means follow the population curve to within 0.004 (0.320 and 0.438; 0.013 at $n = 10^2$): the shortfall is the multiplicity's $-(m-1)/\log n$, not noise. The heat-capacity estimator gives 0.317 and 0.437, against 0.314 and 0.437. Single estimates scatter as Theorem 11.9 predicts: in the quartic model with standard deviation

[7] S has the Wishart distribution with n degrees of freedom, sampled by Bartlett's decomposition into chi-square and normal variables, and r given S is $\mathcal{N}(0, S)$; so $n = 10^8$ costs no more than $n = 10^2$.

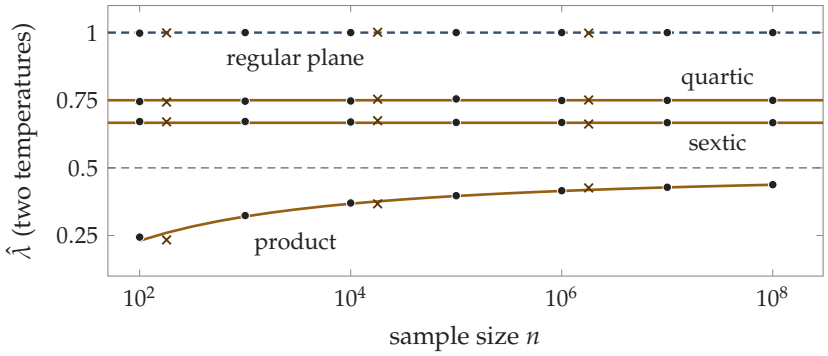


Figure 11.2. Two-temperature estimates $\hat\lambda$ ($\beta_1 = 1/\log n$, $\beta_2 = 1.5/\log n$) for four models with $d = 2$ at the truth 0. Dots: means over 200 data sets per n (seed 11000 + 100×model+ $\log_{10} n$), by quadrature, with Student-t 95% intervals (most are smaller than the dots). Crosses (shifted right): means of random-walk Metropolis estimates on (a, b) for 40 other data sets at $n = 10^2, 10^4, 10^6$ (seed 21000 + 100×model+ $\log_{10} n$). Solid: the learning coefficients $\frac{3}{4}$ and $\frac{2}{3}$ (for the regular plane $\lambda = 1$ is the dashed line), and for the product model the population value of $\hat\lambda$ (Computation 11.2), which tends to $\frac{1}{2}$ (gray) like $\frac{1}{2} - 1/\log n$. Dashed: the regular prediction $\frac{d}{2} = 1$. Script: `11-temperature-lambda.py`.

$0.132/\sqrt{\log n}$ (Exercise 11.8), 0.043 at $n = 10^4$ and 0.031 at 10^8, against measured 0.039 and 0.029; in the odd product model only 0.0051 at $n = 10^4$.

The crosses are the same estimates computed as for a model without sufficient statistics: random-walk Metropolis on (a, b) at β_1, with 16 chains per data set started from the prior, 2000 adaptation steps toward acceptance rate 0.3, then 8000 steps thinned by 4, and $\mathbb{E}_w^{\beta_2}$ obtained by reweighting the draws (Watanabe 2013, eq. (20)). On the same data sets the paired difference from quadrature is at most 0.006 in size in eleven of the twelve cases (95% half-widths 0.002 to 0.008); one of them, -0.004 $[-0.006, -0.001]$ for the product model at $n = 10^4$, excludes zero. In the twelfth, the product model at $n = 10^6$, Metropolis overestimates $\hat\lambda$ by 0.010 $[0.007, 0.013]$: its chains move along the arms of the cross in steps of the arms' width and spend too little time near the center, where the factor $\log \frac{1}{|c|}$ behind the multiplicity correction lives (7.1% of their draws have $\max(|a|, |b|) < 0.05$, against 8.2% of exact draws). Chains four times as long reach 8.0% and an overestimate of 0.003 $[0.001, 0.006]$. A sampler that does not explore a singular set biases $\hat\lambda$, here toward the value of a regular model with one parameter.

Figure 11.3(a) shows the mean of $F_n - \mathrm{WBIC}$. For the regular plane it converges to $\log \frac{2}{\pi} = -0.452$: $\mathrm{WBIC} = \mathrm{BIC} + o_p(1)$ (Watanabe 2013, Theorem 5), and F_n differs from BIC by $-\frac{d}{2}\log 2\pi + \frac{1}{2}\log\det J - \log\varphi(w_0)$ (Theorem 1.4), here $-\log 2\pi + \log 4$. For the quartic and sextic models it stays near -0.31 ($m = 1$, $\mathbb{E}[U_n] = 0$). For the product model it falls like $-\log\log n$: -0.55 at $n = 10^3$ and -1.43 at 10^8, against -0.47 and -1.45 from $-\log\log n + \frac{1}{2}\log\frac{8}{\pi} + 1$. The constant combines the limit $\frac{1}{2}\log\frac{8}{\pi} - \frac{1}{2}Z^2$ of $F_n - nS_n - \frac{1}{2}\log n + \log\log n$ (Chapter 3) with the limit $-1 - \frac{1}{2}Z^2$ of $\mathrm{WBIC} - nS_n - \frac{1}{2}\log n$, whose -1 comes from Computation 11.2 and whose $-\frac{1}{2}Z^2$ is the second-order term of a Gaussian posterior in $c = ab$ (argued,

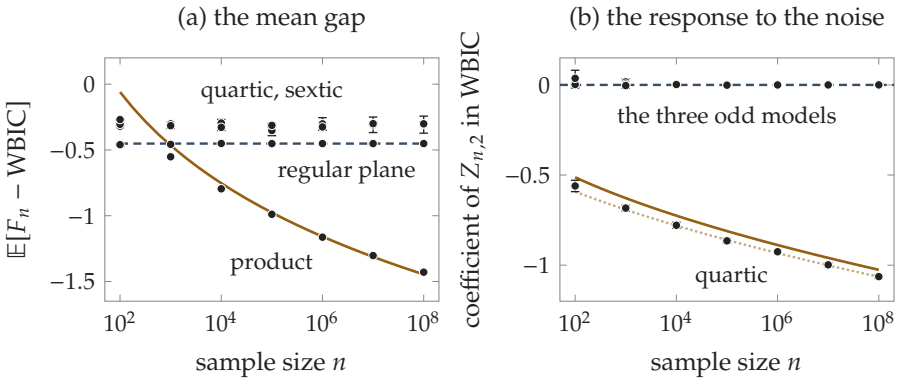


Figure 11.3. WBIC against the free energy, over the data sets of Figure 11.2, with 95% intervals. (a) The mean of F_n − WBIC. Dashed: the regular limit $\log \frac{2}{\pi}$ of the regular plane. Solid: $-\log\log n + \frac{1}{2}\log\frac{8}{\pi} + 1$ for the product model. (b) The coefficient of $Z_{n,2}$ in a least-squares fit of WBIC $-nS_n$ on 1, $Z_{n,2}$, $Z_{n,2}^2$ and $Z_{n,1}^2$ (for the product model, whose noise is one variable, on 1, Z_n and Z_n^2), with intervals from heteroscedasticity-consistent (HC3) standard errors. Solid: the leading term $-\kappa_Q\sqrt{\frac{1}{2}\log n}$ of Computation 11.3; dotted: the average slope computed by quadrature with $\Sigma_n = I$ and Gauss–Hermite averaging over the noise; dashed: 0, the value for every odd model.

not proved here).

Figure 11.3(b) isolates the $\sqrt{\log n}$ term; the squares in the fit absorb the $O_p(1)$ terms of Theorem 11.9, which are even in the noise to leading order. For the three odd models the coefficient of $Z_{n,2}$ is below 0.04 in size at every n (below 0.02 for $n \geq 10^3$), as part (iii) of the theorem says. For the quartic model it grows from −0.56 at $n = 10^2$ to −1.06 at 10^8, against the leading term −0.51 to −1.03. The dotted curve keeps all orders; it differs from the leading term by 0.08 at $n = 10^2$ and 0.04 at 10^8, shrinking like $(\log n)^{-1/2}$, and for $n \geq 10^3$ every measured coefficient lies on it within its interval. The $\sqrt{\log n}$ term of Watanabe's theorem is real and belongs to single data sets: it has mean zero, so it does not show in panel (a).

11.6 The ledger

The free energy is the integral over the inverse temperature of the posterior mean energy, which for the population posterior is $\frac{\lambda}{\beta} - \frac{m-1}{\beta\log n\beta}$: λ replaces equipartition's $\frac{d}{2}$, and the multiplicity enters as a slowly vanishing correction (Proposition 11.4). One temperature, $1/\log n$, recovers the free energy's leading term, with an error of order $\sqrt{\log n}$ unless a sign symmetry kills it (Theorem 11.9); two temperatures, or the heat capacity at one, estimate λ with a bias $-(m-1)/\log n$ (Corollary 11.10).

LEDGER · $y = ax_1 + b^2x_2$	
QUANTITY	the mean energy $\mathbb{E}_w^\beta[nL_n]$ of the tempered posterior; WBIC, its value at $\beta = 1/\log n$; $\hat{\lambda}$, its slope in $1/\beta$
REGULAR THEORY	equipartition, $nL_n(\hat{w}) + \frac{d}{2\beta}$; WBIC = BIC $+ o_p(1)$; $\hat{\lambda} \to \frac{d}{2}$
SINGULAR THEORY	population mean energy $\frac{\lambda}{\beta} - \frac{m-1}{\beta \log n\beta} + \dots$ (Proposition 11.4); $F_n - \text{WBIC} = -(m-1)\log\log n + O_p(\sqrt{\log n})$, the $\sqrt{\log n}$ term zero under odd parity (Theorems 11.6 and 11.9)
RUNNING EXAMPLE	quartic model: $\frac{3}{4\beta}$, and WBIC's noise term $-0.338\sqrt{\frac{1}{2}\log n}\, Z_{n,2}$ (even parity); product model: $\frac{1}{2\beta} - \frac{1}{\beta(\log n\beta + \gamma + \log 2)}$, none
MEASURED	$\hat{\lambda}$ at $n = 10^8$: quartic 0.750, sextic 0.667, regular plane 1.000, product 0.438 (population 0.438; at $n = 10^3$, 0.324 against 0.320); WBIC's coefficient on $Z_{n,2}$: -0.56 to -1.06 (leading term -0.51 to -1.03)
RESTS ON	exact tempered posteriors; the volume expansion; realizable linear models with Gaussian noise (complete proof); Watanabe's conditions (general theorem)

11.7 Where the theory stops

WHAT WAS ASSUMED. Every result of this chapter is about integrals against $\varphi e^{-\beta n L_n}$, which we computed by quadrature; in practice they are Monte Carlo averages, and our Metropolis check showed the first way in which that fails. The population results need the volume expansion, which analytic models satisfy. The complete proof of the WBIC theorem is for realizable linear models with Gaussian noise and inputs with four moments, a compact parameter set, a fixed model and truth as n grows, and fixed β_0; Watanabe's general theorem needs analyticity with six moments, relatively finite variance, an interior minimizer, and a unique optimal distribution. Networks with ReLU activations are not analytic, their posteriors are never computed exactly, and they grow with the data.

WHERE IT BREAKS. The bias $-(m-1)/\log n$ does not disappear at practical sizes: in the product model $\hat{\lambda}$ is 0.32 at $n = 10^3$ and 0.44 at $n = 10^8$ for $\lambda = 0.5$, and in Watanabe's experiment with reduced-rank regression ($n = 500$) the three largest models gave means 14.80, 15.72 and 16.55 for $\lambda = 15$, 16 and 17, with $m = 2$, 1 and 2, which he attributes to lower-order terms and, for $m = 2$, to $\log\log n$ (Watanabe 2013, Table 3 and §6.2). With even parity the error of WBIC is in general of order $\sqrt{\log n}$ in a single data set, and its expectation differs from that of F_n by $O(\log\log n)$ when $m > 1$ (Watanabe 2013, eq. (32)). WBIC also drops the prior's contribution of order one: in a regular model it equals BIC up to $o_p(1)$, while F_n contains $-\log\varphi(w_0)$. Friel et al. (2017) found that WBIC performs adequately with informative priors but can systematically overestimate the evidence, particularly for small samples; Imai (2019b), in a preprint, identified the overestimating

term. Unrealizable singular pairs can violate relatively finite variance (Watanabe 2010), and then the theorem does not apply. When the tempered posterior moves between regions of different loss and learning coefficient as β changes (Chapter 12), the mean energy changes abruptly and a sampler at one temperature mixes poorly; exchange Monte Carlo (Hukushima and Nemoto 1996) runs all temperatures at once. Finally, the local learning coefficient of Lau et al. (2025) is this chapter's dimensionless mean energy at $\beta = 1/\log n$, measured from a trained network w^* as $\beta\,\mathbb{E}^{\beta}_{w}[nL_n - nL_n(w^*)]$, localized near it and sampled by stochastic-gradient Langevin dynamics; it inherits the multiplicity bias of Proposition 11.4 and adds those of its sampler and localization (Chapter 13).

Open problems. Watanabe (2013, p. 875) conjectures that the parity is a birational invariant for every truth; he proves it when the truth is realizable and reduces the general case to showing that every nonnegative analytic function is the Kullback–Leibler divergence of some truth and model. We found no resolution of the conjecture. Every error term in this chapter is a power of $1/\log n$ or $1/\sqrt{\log n}$ with unspecified constants; non-asymptotic bounds for WBIC and $\hat{\lambda}$, even for linear models, are open. As of 2026 we found no estimator of λ with a proved error of smaller order than $1/\log n$ when $m \geq 2$ is unknown; the estimators of Imai (2019a) and Takio and Suzuki (2025), both preprints, reduce bias or variance in experiments, and their analyses give consistency at the rate $(\log n)^{-1/2}$ but do not treat the bias $-(m-1)/\log n$ (Exercise 11.12).

Exercises

11.1 consolidation. (Bounds from concavity.) Under the hypotheses of Theorem 11.2, show that $\mathbb{E}^{1}_{w}[nL_n] < F_n < \mathbb{E}_{\varphi}[nL_n]$, where $\mathbb{E}_{\varphi}$ is the prior expectation. Show that the right inequality is Jensen's inequality for $-\log \int \varphi e^{-nL_n}$, and that the gap on the left, $F_n - \mathbb{E}^{1}_{w}[nL_n]$, is the Kullback–Leibler divergence of the posterior from the prior.

11.2 consolidation. (Heat capacity.) Show that $C(s) = \mathcal{E}(s) - s\,\mathcal{E}'(s)$. Deduce from Computation 11.1 that $C(s) = \frac{3}{4}$ up to exponentially small terms for the quartic model (a Gamma law with shape α has variance α), and from Computation 11.2 that $C(s) = \frac{1}{2} - \frac{1}{x'} - \frac{1}{x'^2}$ with $x' = \log s + \gamma + \log 2$ for the product model.

11.3 consolidation. (Signs.) Verify the symmetry of Theorem 11.9(iii) for the product, sextic and regular models and for the chain $y = a_1 a_2 a_3 x$. For the quartic model, show that c_n differs from $-\kappa_{\mathrm{Q}} e_2/\sqrt{2\beta}$ by an amount exponentially small in s, that $\mathbb{E}\sqrt{Y/(X+Y)} = 4\Gamma(\frac{3}{4})^2/\Gamma(\frac{1}{4})^2$ for independent Gamma variables X, Y with shapes $\frac{1}{2}, \frac{1}{4}$ *(Hint: a Beta law)*, and that $\sigma^2 = \frac{4}{3}\kappa_{\mathrm{Q}}^2 = 0.152$. Check that this U_n agrees with Watanabe's $U_n = -\Theta\Gamma(\lambda + \frac{1}{2})/(\sqrt{2\lambda}\,\Gamma(\lambda))$ in the chart $(a, b) = (uv^2, v)$, where $\mathrm{d}u^*$ is proportional to $(1+u^2)^{-3/4}\,\mathrm{d}u$.

11.4 BY HAND. (Generalized equipartition.) Let $K = \frac{1}{2}\sum_{j=1}^{d} w_j^{2k_j}$ on $[-1,1]^d$ with prior density proportional to $\prod_j |w_j|^{h_j}$. Show that $\mathcal{E}(s) = \sum_j \frac{h_j+1}{2k_j}$ up to exponentially small terms. Which learning coefficient does this give, and how does it compare with the rule for sums of functions of separate variables in Proposition 5.4?

11.5 BY HAND. (Thermodynamic integration for the regular line.) For $y = ax$ with $K = \frac{1}{2}a^2$ and the uniform prior on $[-1,1]$, show that $\mathcal{E}(s) = \frac{1}{2} - \sqrt{s/2}\, e^{-s/2}/(\sqrt{\pi}\,\operatorname{erf}\sqrt{s/2})$, and verify by differentiation that $\int_{-\infty}^{\log n} \mathcal{E}(e^x)\,dx = \frac{1}{2}\log\frac{2n}{\pi} - \log\operatorname{erf}\sqrt{n/2} = \bar{F}(n)$. How large must s be for $\mathcal{E}(s)$ to be within 10^{-4} of $\frac{1}{2}$?

11.6 BY HAND. (The cusp and the chain.) For $y = ab^2x$ the prior density of $c = ab^2$ is $\frac{1}{2}(|c|^{-1/2} - 1)$ on $[-1,1]$. Show that $\mathcal{E}(s) = \frac{1}{4} - \frac{r/4}{s^{1/4}-r}$ up to exponentially small terms, with $r = 2^{1/4}\sqrt{\pi}/\Gamma(\frac{1}{4}) = 0.5814$, so that $\mathcal{E}(s)$ is within 1% of $\lambda = \frac{1}{4}$ only for $s \gtrsim 1.2\times 10^7$, and relate the speed of approach to the second term of the volume expansion. For the chain $y = a_1a_2a_3x$, whose density of c is $\frac{1}{4}(\log\frac{1}{|c|})^2$, show that $\mathcal{E}(s) = \frac{1}{2} - \frac{2x}{x^2+\pi^2/2}$ with $x = \log s + \gamma + \log 2$ *(Hint:* $\mathbb{E}[\log^2|Z|] = \kappa^2 + \frac{\pi^2}{8}$ *and* $\mathbb{E}[Z^2\log^2|Z|] = (1-\kappa)^2 + \frac{\pi^2}{8} - 1$*).*

11.7 VARIATION. (Jeffreys' prior.) Give b the density $|b|$ on $[-1,1]$, Jeffreys' prior of $y = b^2x_2$. Show that the quartic model then has $\mathcal{E}(s) \to 1 = \frac{d}{2}$, equipartition restored, but $c_n = -e_2/\sqrt{2\pi\beta}$: WBIC keeps its noise term.

11.8 VARIATION. (Temperature pairs.) In the quartic model, show that the leading fluctuation of $\hat{\lambda}$ is $-\kappa_Q Z_{n,2}/\big(\sqrt{2\log n}\,(\beta_{01}^{-1/2} + \beta_{02}^{-1/2})\big)$, with standard deviation $0.132/\sqrt{\log n}$ for $(\beta_{01},\beta_{02}) = (1, 1.5)$. What limits the choice of small β_{0j}? For the product model, expand the population bias of $\hat{\lambda}$ to the order $\frac{\log\log n}{\log^2 n}$ and compare it with Figure 11.2.

11.9 MEASUREMENT. (The optimal temperature.) Modify the script of Section 11.5 to compute β^* for each data set by solving $\mathbb{E}_w^{\beta}[nK_n] = F_n - nS_n$. For the quartic model at $n = 10^3, 10^6, 10^9$ you should find means of $\beta^*\log n$ near 1.06, 1.03, 1.02 and regression coefficients on $Z_{n,2}$ near -0.02, -0.03, -0.03, smaller than the -0.12, -0.09, -0.07 of Theorem 11.6(iii). Explain why: F_n depends on $Z_{n,2}$ at order one, with coefficient about -0.59.

11.10 MEASUREMENT. (An oscillation.) For $\varepsilon < e^{-4}$ let

$$V(\varepsilon) = \varepsilon^{1/2}\Big(1 + \frac{\sin(\log\frac{1}{\varepsilon})}{2\log\log\frac{1}{\varepsilon}}\Big).$$

Show that V is increasing there and $V(\varepsilon) \sim \varepsilon^{1/2}$. Compute $\mathcal{E}(s) - \frac{1}{2}$ for $\log s$ between 20 and 200 by quadrature (substitute $t = u/s$ in (11.6)), compare it with $1/\log s$, and explain with the Mellin transform why

the oscillation survives. Then prove that $\mathcal{E}(s) \to \lambda$ under $V(\varepsilon) \sim C\varepsilon^{\lambda}(\log \frac{1}{\varepsilon})^{m-1}$ alone, by the argument of Lemma 4.3 applied to $\mathrm{d}V$ and to $t\ \mathrm{d}V(t)$.

11.11 RESEARCH. (Parity.) State Watanabe's conjecture that the parity is a birational invariant for every truth, and his reduction of it to the realization of every nonnegative analytic function as a Kullback–Leibler divergence (Watanabe 2013, §3). Show that a function $K = \sum_j f_j^2$ with analytic f_j and a zero in W is the Kullback–Leibler divergence of a realizable Gaussian regression model, so that the conjecture holds for it. We found no proof of the conjecture beyond such cases.

11.12 RESEARCH. (Removing the multiplicity bias.) Under Assumption 11.7, the mean energies at several temperatures $\beta_{0j}/\log n$ determine $\mathcal{E}$ at several values of s up to the noise terms of Theorem 11.9. Propose an estimator of (λ, m) from them, using (11.4), and analyze its bias and variance for the product model and the chain. Whether an estimator of λ with error $o(1/\log n)$ exists when $m \geq 2$ is, as far as we found, open, and so is its behavior with samples from stochastic-gradient Langevin dynamics.

Notes and References

Thermodynamic integration came to statistics from physics: Gelman and Meng (1998) presented it as path sampling, noting that numerical analysts call it Ogata's method (Ogata 1989), and Friel and Pettitt (2008) computed marginal likelihoods with tempered posteriors, their power posteriors.

WBIC is due to Watanabe (2013), who proves Theorem 11.2 (his Theorem 3), Theorem 11.6 (Theorem 4 with Corollaries 1–3) and the regular case (Theorem 5), introduces the parity, and illustrates WBIC on reduced-rank regression. Watanabe's textbook (Watanabe 2018) has a chapter on information criteria, and Suzuki (2023) works through WAIC and WBIC with exercises in R and Stan. Friel et al. (2017) studied the accuracy of WBIC; Imai (2019b) and Takio and Suzuki (2025) (preprints) propose a correction and another estimator of λ. When the learning coefficients of submodels are known, the singular BIC of Drton and Plummer (2017) is an alternative for model selection (Chapter 8).

Proposition 11.4, with its correction term and the polynomial Q, and Theorem 11.9 with its proof are the book's own. The proof follows Watanabe's, with the population posterior and the linear structure of the model in place of a resolution; for the quartic model its U_n agrees with Watanabe's formula (Exercise 11.3). The heat-capacity estimator is due to Imai (2019a); Corollary 11.10(ii) proves its consistency for linear models. The local learning coefficient of Lau et al. (2025) uses the temperature $1/\log n$ of WBIC (Chapter 13).

TRANSLATING THE LITERATURE. Watanabe (2013) uses our notation ($\mathbb{E}_w^{\beta}$, L_n, K_n, λ, m, U_n, β^*), with F for our F_n and $Q(K, \varphi)$ for the parity (1 for odd). Imai (2019a) writes t for β, E_θ^t and V_θ^t for the tempered mean and variance,

and $\log L(M)$ for our $-F_n$; his WBIC(M) is our $-$WBIC, and his $\hat{\lambda}^1_V$ our $\hat{\lambda}_C$. Friel et al. (2017) write t for β and speak of powered posteriors. Lau et al. (2025) write $\beta^* = 1/\log n$ for the WBIC temperature, not for our optimal β^*.

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12 Phase Transitions

How the posterior chooses between singularities

THE MODEL — $y = ax_1 + b^2x_2,\ y = abx_1 + (2b^4 - b^2)x_2$

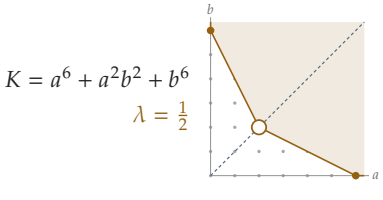
IN THIS CHAPTER. When several regions of parameter space nearly fit the data, the posterior puts its mass where the local free energy $nL_n(w_\alpha) + \lambda_\alpha \log n$ is smallest, trading fit against learning coefficient. By hand, two models fitted to the same truth $y = \delta x_2$ show two ways the trade plays out: in the quartic model $y = ax_1 + b^2x_2$ the effective learning coefficient crosses over from $\frac{3}{4}$ to 1 along a universal curve of $n\delta^2$ that dips and overshoots, and in a double-well model the posterior jumps from a segment of misfitting parameters with $\lambda = \frac{1}{2}$ to two regular true points once $n\delta^2$ reaches about $\log n$. We prove the local free energy formula for models linear in their coefficients, that the posterior chooses the region whose local free energy is smallest by a growing margin and moves toward better fit and larger learning coefficient, and the limiting crossover curve and posterior odds of the two examples. Quadrature over thousands of data sets confirms the curves, including a switch near $n = 3000$, where the regular theory, which charges every region the same, gives the better-fitting region most of the posterior already at $n = 100$. The theory does not say where the regions come from, the switch point of a single data set is random, and whether training passes through the same regions is open.

12.1 The question: where does the posterior go?

Train a small network on more and more data and look at where its weights end up. In the toy model of superposition of Elhage et al. (2022),[1] a ReLU network asked to store six sparse features in two hidden dimensions, the critical points of the loss include regular polygons: four features stored as a square, five as a pentagon, six as a hexagon; and training curves rest near such configurations before dropping to the next (Chen et al. 2023). Chen et al. (2023) asked the Bayesian version of the question: as the number n of examples grows, does the posterior move between the same configurations, and at which n? A configuration that stores fewer features fits worse, but the weights of the unused feature are partly free, so it is more degenerate. The posterior has to trade fit against degeneracy.

The tool is local. For a measurable set $U \subseteq W$ with positive prior mass,

[1] Elhage et al. (2022) introduced the model to study how a network can represent more features than it has dimensions, which they call superposition.



the *local free energy* is

$$F_n(U) = -\log \int_U \varphi(w)\, e^{-nL_n(w)}\, \mathrm{d}w, \tag{12.1}$$

the free energy of the model restricted to U, with the prior not renormalized. The posterior density is $\varphi\, e^{-nL_n} e^{F_n}$, so the posterior mass of U is exactly

$$\mathbb{P}(U \mid X^n) = e^{F_n - F_n(U)}. \tag{12.2}$$

The posterior prefers the region with the smallest local free energy. Following physics and Chen et al. (2023), we call a region whose local free energy we follow a *phase*, and a change of the region that carries most of the posterior mass a *phase transition*.[2]

> THE REGULAR ANSWER · where the posterior goes. If the model is regular, then the posterior concentrates in a neighborhood of the maximum-likelihood estimate $\hat{w}$ of radius of order $n^{-1/2}$ and is asymptotically normal there (van der Vaart 1998, Theorem 10.1). Applied region by region, Laplace's approximation charges every region the same $\frac{d}{2}\log n$ (Theorem 1.4), so the posterior odds of two regions are their maximized likelihood ratio up to a bounded factor: the region that fits better wins as soon as n times the difference of their losses is large.

Singular models break the second sentence. Watanabe's formula, applied to the model restricted to a region (Theorem 8.6), charges it $\lambda_\alpha \log n - (m_\alpha - 1)\log\log n$ with its own learning coefficient. A region that fits worse by Δ per data point but has a smaller learning coefficient keeps the posterior until $n\Delta$ catches up with the difference of the $\lambda_\alpha \log n$. In the double-well model of this chapter at $n = 3162$, equal penalties predict that the better-fitting region carries 93% of the posterior; it carries 53%. The regular answer also fails more quietly: near a singular truth a region does not have one learning coefficient over the range of n that matters. This chapter asks which region carries the posterior at sample size n, when and how sharply it switches, and what replaces the free energy formula when the truth sits close to a singularity.

[2] In statistical physics a phase transition is a point at which a free energy, in the limit of a large system, fails to be analytic in a parameter such as the temperature. Here the parameter is n: the minimum $\min_\alpha(n\Delta_\alpha + \lambda_\alpha \log n)$ of the leading terms of the local free energies has a kink where the minimizing region changes, as a free energy does at a first-order transition. In the statistical mechanics of learning, where the numbers of parameters and of examples grow together, learning curves can jump (Engel and Van den Broeck 2001, ch. 7, "Discontinuous learning"); that is not the limit of this chapter.

12.2 By hand: one truth, two models

Both models of this section have inputs $x = (x_1, x_2) \sim \mathcal{N}(0, I_2)$, unit noise, the uniform prior on $[-1, 1]^2$, and the truth

$$y = \delta x_2 + \varepsilon, \qquad 0 < \delta < 1.$$

Both are linear in a coefficient vector, $y = g(w)^\top x + \varepsilon$, so $K(w) = \frac{1}{2}|g(w) - c_0|^2$ with $c_0 = (0, \delta)$, as in Chapter 8. They differ only in how they reach c_0.

A CROSSOVER. In the quartic model $y = ax_1 + b^2 x_2$ the truth is reached at the two points $(0, \pm\sqrt{\delta})$; at $\delta = 0$ the same model is singular, with $\lambda = \frac{3}{4}$ (Chapter 5). The two answers cannot both hold at every n.

COMPUTATION 12.1 · THE QUARTIC MODEL NEAR ITS SINGULAR TRUTH

1 The loss $K = \frac{1}{2}a^2 + \frac{1}{2}(b^2 - \delta)^2$ vanishes at $(0, \pm\sqrt{\delta})$, where its Hessian is $\mathrm{diag}(1, 4\delta)$: both true parameters are regular, and $\lambda = 1 = \frac{d}{2}$.

2 The population free energy $\bar{F}(n) = -\log \int \varphi e^{-nK}$ factorizes: $\int \varphi e^{-nK} = \frac{1}{4} I_a I_b$, with $I_a = \int_{-1}^{1} e^{-na^2/2}\,\mathrm{d}a = \sqrt{2\pi/n}\,(1 - 2\bar{\Phi}(\sqrt{n}))$, $\bar{\Phi}$ the standard normal tail, and $I_b = \int_{-1}^{1} e^{-n(b^2-\delta)^2/2}\,\mathrm{d}b$.

3 Substitute $b = n^{-1/4}u$, the scale of the singular model's posterior. Then $n(b^2 - \delta)^2 = (u^2 - \tau)^2$ with $\tau = \sqrt{n}\,\delta$, and $I_b = n^{-1/4} \int_{|u| \le n^{1/4}} e^{-(u^2-\tau)^2/2}\,\mathrm{d}u$: the truth enters only through τ.

4 Let $h(s) = \int_{\mathbb{R}} e^{-(u^2-s)^2/2}\,\mathrm{d}u$. Up to the tails beyond $|u| = n^{1/4}$ and $|a| = 1$, which are exponentially small (Proposition 12.8),

$$\bar{F}(n) \approx \tfrac{3}{4}\log n + \log 4 - \tfrac{1}{2}\log(2\pi) - \log h(\sqrt{n}\,\delta).$$

5 At $\tau = 0$, $h(0) = 2^{5/4}\Gamma(\frac{5}{4}) = 2.156$ and $\bar{F}(n) = \frac{3}{4}\log n + O(1)$, the singular formula. For large τ the integrand of h has two Gaussian peaks at $\pm\sqrt{\tau}$ of width $1/(2\sqrt{\tau})$, so $h(\tau) \sim \sqrt{2\pi/\tau}$ and $\bar{F}(n) = \log n + \frac{1}{2}\log\delta + \log\frac{2}{\pi} + o(1)$, Laplace's formula at the two regular points.

6 The *effective learning coefficient* $\lambda_{\mathrm{eff}}(n) = \mathrm{d}\bar{F}/\,\mathrm{d}\log n$ is therefore, up to exponentially small terms, a function of $n\delta^2$ alone:

$$\lambda_{\mathrm{eff}}(n) = \Lambda_{\mathrm{pop}}(\sqrt{n}\,\delta), \qquad \Lambda_{\mathrm{pop}}(\tau) = \frac{3}{4} - \frac{\tau}{2}\,\frac{h'(\tau)}{h(\tau)}.$$

The script `12-phases-crossover.py` gives $\Lambda_{\mathrm{pop}}(0.1) = 0.730$, $\Lambda_{\mathrm{pop}}(1) = 0.803$, $\Lambda_{\mathrm{pop}}(10) = 1.004$, and $\Lambda_{\mathrm{pop}} = \frac{7}{8}$ at $n\delta^2 = 1.60$.

What one had to notice is the scale: the singular posterior of b lives at $n^{-1/4}$, and on that scale the true value $\sqrt{\delta}$ is invisible until $\sqrt{n}\,\delta$ is of order one. A coefficient within $O(n^{-1/2})$ of the singular one behaves like the singular point, one much farther away like a regular point, and in between the free energy follows neither formula.[3] The passage takes about four decades of $n\delta^2$, and it is not monotone: Λ_{pop} dips to 0.708 at $n\delta^2 = 0.13$, returns to $\frac{3}{4}$ where h is largest ($\tau = 0.76$), and overshoots to 1.084 at $n\delta^2 = 6.7$ (Figure 12.2).[4] Heuristically, the hump between the wells is responsible: along $a = 0$ the loss has a local maximum $\frac{1}{2}\delta^2$ at $b = 0$, so the state density of Chapter 4 is infinite at $\varepsilon = \frac{1}{2}\delta^2$, and a posterior, which lives where K is of order $1/n$, meets the hump when $\frac{1}{2}n\delta^2$ is of order one. First the flattened valley lets it spread, then the wells separate and it narrows at the regular rate. This is a *crossover*: one region whose effective learning coefficient changes. The point $(0, 0)$ is a saddle of K, and nothing separates it from the true points.

[3] The function h is a parabolic cylinder function, $h(s) = \sqrt{\pi}\,e^{-s^2/4}D_{-1/2}(-s)$, which we checked numerically; nothing below uses the identity.

[4] Watanabe and Amari (2003) studied layered models under this scaling, a truth at Kullback–Leibler distance of order $1/n$ from the singular ones, and found that the Bayes generalization error can then exceed the regular one, an effect of the same kind as the overshoot.

A TRANSITION. The *double-well model*

$$y = ab\,x_1 + q(b)\,x_2 + \varepsilon, \qquad q(b) = 2b^4 - b^2,$$

reaches c_0 differently. Read b as the input weight of a unit, a as its output weight on the first feature, and let the same input weight drive the coefficient $q(b)$ of the second feature: q is 0 at $b = 0$, negative for $0 < |b| < 1/\sqrt{2}$ with minimum $q(\pm\frac{1}{2}) = -\frac{1}{8}$, and $q(\pm 1) = 1$. To fit δx_2 the unit must pass through weights at which it fits worse than not at all.

COMPUTATION 12.2 · TWO REGIONS OF THE DOUBLE-WELL MODEL

1 The loss is $K = \frac{1}{2}a^2b^2 + \frac{1}{2}(q(b) - \delta)^2$. It vanishes only where $ab = 0$ and $q(b) = \delta$; since $q \le 0$ for $|b| \le 1/\sqrt{2}$, this forces $a = 0$ and $b = \pm b_\delta$ with $b_\delta^2 = \frac{1}{4}(1 + s)$, $s = \sqrt{1 + 8\delta}$. There the Jacobian of $g(a, b) = (ab, q(b))$ is $\mathrm{diag}(\pm b_\delta, \pm q'(b_\delta))$, of determinant $D = b_\delta q'(b_\delta) = \frac{1}{2}s(1 + s)$ in absolute value: both true parameters are regular, $\lambda = 1$. For $\delta = 0.05$, $b_\delta = 0.7388$ and $D = 1.2916$.

2 Region $A = \{|b| \le \frac{1}{2}\}$. There $q(b) \le 0$, so $K \ge \frac{1}{2}\delta^2 + \frac{1}{2}a^2b^2 \ge \frac{1}{2}\delta^2$, with equality exactly on the segment $\{b = 0\}$, for every a: the unit is off, the misfit is $\Delta = \frac{1}{2}\delta^2$ per data point, and a is not identified. Across the segment $K - \Delta = (\delta + \frac{1}{2}a^2)b^2 + O(b^4)$, so the segment is a set of minima with nondegenerate curvature across it, and $\lambda = \frac{1}{2}$, $m = 1$ (Proposition 5.5).

3 Region $B = \{|b| \ge \frac{1}{2}\}$ contains both true parameters. On the ridge $|b| = \frac{1}{2}$, $K \ge \frac{1}{2}(\delta + \frac{1}{8})^2$; for $\delta = 0.05$ that is 0.0153 against $\Delta = 0.00125$, a barrier of height $\eta = 0.0141$.

4 The local population free energies. In A, for fixed a the integral over b is $\sqrt{\pi/(n(\delta + a^2/2))}$ to leading order, and $\int_{-1}^{1}(\delta + \frac{1}{2}a^2)^{-1/2}\,\mathrm{d}a = 2\sqrt{2}\,\mathrm{arsinh}(1/\sqrt{2\delta})$, so

$$\bar{F}_A(n) = n\Delta + \tfrac{1}{2}\log n - \log\!\left(\sqrt{\pi/2}\,\mathrm{arsinh}(1/\sqrt{2\delta})\right) + o(1).$$

In B, Laplace's method gives each true point $\frac{1}{4}\cdot 2\pi/(nD)$, so $\bar{F}_B(n) = \log n + \log\frac{D}{\pi} + o(1)$. For $\delta = 0.05$ the constants are -0.851 and -0.889.

5 The switch: $\bar{F}_A = \bar{F}_B$ when $n\Delta = \frac{1}{2}\log n - 0.038$, at $\bar{n}_c = 3198$ (3232 from the leading terms alone). With the population loss in place of the data, the posterior mass of B is 0.11, 0.49 and 0.94 at $n = 1000$, 3162 and 5623.

What one had to notice is that the misfit turns the segment into a trap: from each of its points, the first move in b makes the coefficient of x_2 negative, away from the truth. The segment is the network without the unit, embedded in the larger network as a line of local minima, as Fukumizu and Amari (2000) describe for multilayer perceptrons.[5] The truth that the quartic model approached through a crossover is now reached by a jump,

[5] They show that a critical point of a perceptron with $H - 1$ hidden units gives lines of critical points of the perceptron with H units, which can be local minima or saddles, using only the hierarchical structure of the model.

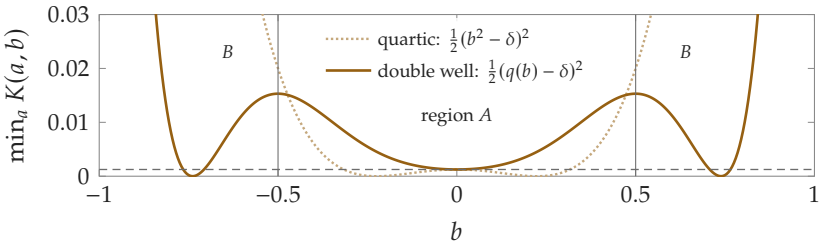


Figure 12.1. The loss along the line $a = 0$, which is the minimum of $K(a, b)$ over a, for the two models at the truth $y = 0.05\,x_2$. Dotted: the quartic model, whose singular point $b = 0$ is a local maximum at height $\frac{1}{2}\delta^2$ between the true parameters ± 0.224. Solid: the double-well model, whose segment $b = 0$ is a valley floor at the same height Δ (gray), separated from the true parameters ± 0.739 by ridges at $b = \pm\frac{1}{2}$ (vertical lines).

when the misfit $n\Delta = \frac{1}{2}\tau^2$ has caught up with the extra $\frac{1}{2}\log n$; the quartic model needed only τ of order one. At the same height $\frac{1}{2}\delta^2$, the singular point of the quartic model is a hilltop and the segment of the double-well model a valley floor (Figure 12.1). The regular theory charges both regions $\log n$, so it prefers the better-fitting region B at every n; with data it gives B most of the posterior already at $n = 100$ (Section 12.5).

12.3 The theorem: local free energies and the posterior's choice

Both computations used the population loss. With data we must show that each local free energy has the expansion the computations suggest, that the posterior then sits in the region with the smallest one, and, in Section 12.4, the limiting curves of the two examples.

Lemma 12.1 (Local free energies). *Let $U_1, \dots, U_k$ be measurable subsets of W with positive prior mass that cover W and overlap only in sets of prior measure zero. Then*

$$F_n = -\log \sum_{\alpha=1}^{k} e^{-F_n(U_\alpha)}, \qquad \min_\alpha F_n(U_\alpha) - \log k \le F_n \le \min_\alpha F_n(U_\alpha),$$

and $\mathbb{P}(U_\alpha \mid X^n) = e^{-F_n(U_\alpha)} / \sum_\beta e^{-F_n(U_\beta)}$ for each α.

Proof. The integral of φe^{-nL_n} over W is the sum of its integrals over the U_α, because the overlaps carry no prior mass; this is the identity. The sum lies between its largest term $e^{-\min_\alpha F_n(U_\alpha)}$ and k times that term, and the last formula is (12.2). ∎

The free energy is a soft minimum of the local ones. We prove the expansion of the local free energies for the models of Theorem 8.4, in a region that need not contain the truth.

Assumption 12.2 (A region of a linear model).

(i) The data are independent pairs (x_i, y_i). The inputs $x_i \in \mathbb{R}^k$ have the law of x, where $\Sigma = \mathbb{E}[xx^\top]$ is positive definite and $\mathbb{E}|x|^4 < \infty$, and the outputs are $y_i = c_0^\top x_i + \varepsilon_i$, with standard normal noise independent of x_i.

(ii) The model is $y = g(w)^\top x + \varepsilon$ with standard normal noise, for a measurable $g\colon W \to \mathbb{R}^k$ on $W \subseteq \mathbb{R}^d$, with a prior density φ. So $L(w)$ is $\frac{1}{2}|g(w) - c_0|_\Sigma^2$ plus a constant, where $|v|_\Sigma^2 = v^\top \Sigma v$.

(iii) The region $U \subseteq W$ is measurable, and L attains its infimum over U at a point $w_U \in U$. Write $K_U = L - L(w_U) \geq 0$ on U.

(iv) (Relatively finite variance on U.) For some $c_U > 0$, $K_U(w) \geq c_U |g(w) - g(w_U)|_\Sigma^2$ for every $w \in U$.

(v) (Local volume.) $V_U(\varepsilon) = \int_{U \cap \{K_U < \varepsilon\}} \varphi \, dw \sim C_U \varepsilon^{\lambda_U} (\log \frac{1}{\varepsilon})^{m_U - 1}$ as $\varepsilon \to 0$, for some $C_U > 0$, $\lambda_U > 0$ and integer $m_U \geq 1$.

Nothing requires the truth to lie in U: the best point w_U may have a misfit $L(w_U) - L(w_0) > 0$, as the segment of the double-well model does. Condition 4 is Watanabe's relatively finite variance for the model restricted to U, with w_U in the role of the truth: for $f_U = \log p(y \mid x, w_U) - \log p(y \mid x, w)$ one computes $\mathbb{E} f_U = K_U(w)$ and $\mathbb{E} f_U^2 \geq |g(w) - g(w_U)|_\Sigma^2$, with a matching upper bound when g is bounded. When U contains a true parameter, $K_U = \frac{1}{2}|g - g(w_U)|_\Sigma^2$ and $c_U = \frac{1}{2}$. The condition fails when U contains a whole curve of distinct, equally good distributions (Exercise 12.7).

Theorem 12.3 (Local free energies and the posterior's choice). *(i) Under Assumption 12.2,*

$$F_n(U) = n L_n(w_U) + \lambda_U \log n - (m_U - 1) \log \log n + O_p(1).$$

More precisely, let $R_n = F_n(U) - n L_n(w_U) - \lambda_U \log n + (m_U - 1) \log \log n$ *and* $\kappa_U = -\log(C_U \Gamma(\lambda_U + 1))$*; let* $\Sigma_n = \frac{1}{n}\sum_i x_i x_i^\top$, $Z_n = n^{-1/2} \sum_i x_i \varepsilon_i$, $\tilde{Z}_n = Z_n - \sqrt{n}(\Sigma_n - \Sigma)(g(w_U) - c_0)$, $\tilde{\rho}_n^2 = \tilde{Z}_n^\top \Sigma^{-1} \tilde{Z}_n$*; and let* $A_n(\eta)$ *be the event* $(1 - \eta)\Sigma \preceq \Sigma_n \preceq (1 + \eta)\Sigma$*. Then* $\mathbb{P}(A_n(\eta)) \to 1$*,* $\tilde{\rho}_n^2$ *is bounded in probability, and for every* $\eta \in (0, \frac{1}{2} c_U]$ *there are numbers* $\epsilon_n(\eta) \to 0$ *such that on* $A_n(\eta)$

$$|R_n - \kappa_U| \leq \frac{2 \lambda_U \eta}{c_U} + \frac{\tilde{\rho}_n^2}{2\eta} + \epsilon_n(\eta). \tag{12.3}$$

(ii) Let $U_1, \dots, U_k$ *be as in Lemma 12.1, each satisfying Assumption 12.2 with its own* w_α*,* λ_α*,* m_α*, and put* $\Phi_\alpha(n) = n L_n(w_\alpha) + \lambda_\alpha \log n - (m_\alpha - 1) \log \log n$*. If* $\Phi_\beta(n) - \Phi_\alpha(n) \to \infty$ *in probability for every* $\beta \neq \alpha$*, then* $\mathbb{P}(U_\alpha \mid X^n) \to 1$ *in probability.*

Proof. The plan is the proof of Theorem 8.4 with the region's best point w_U in the place of the truth: sandwich the random energy on U between

two multiples of nK_U and read the two population free energies off the Abelian lemma. Write $\delta_U = g(w_U) - c_0$ and $e(w) = g(w) - g(w_U)$.

The energy. Since $y_i - g(w)^\top x_i = \varepsilon_i - (\delta_U + e)^\top x_i$, expanding the squares in $nL_n(w) = \frac{1}{2}\sum_i (y_i - g(w)^\top x_i)^2 + \frac{n}{2}\log(2\pi)$ gives $nL_n(w) - nL_n(w_U) = n\,e^\top \Sigma_n \delta_U + \frac{n}{2} e^\top \Sigma_n e - \sqrt{n}\, e^\top Z_n$, while $nK_U(w) = n\,e^\top \Sigma \delta_U + \frac{n}{2} e^\top \Sigma e$. Subtracting,

$$nL_n(w) - nL_n(w_U) = nK_U(w) + \tfrac{n}{2}\, e^\top (\Sigma_n - \Sigma) e - \sqrt{n}\, e^\top \tilde{Z}_n \qquad (w \in U). \tag{12.4}$$

The noise. $\tilde{Z}_n = n^{-1/2}\sum_i \xi_i$ with $\xi_i = x_i \varepsilon_i - (x_i x_i^\top - \Sigma)\delta_U$, independent and identically distributed with mean 0, because ε_i is independent of x_i with mean 0, and finite covariance, because $\mathbb{E}|x|^4 < \infty$. So $\mathbb{E}\tilde{\rho}_n^2 = \mathbb{E}[\xi_1^\top \Sigma^{-1} \xi_1]$ does not depend on n, and $\tilde{\rho}_n^2$ is bounded in probability by Markov's inequality. The law of large numbers gives $\Sigma_n \to \Sigma$ almost surely, hence $\mathbb{P}(A_n(\eta)) \to 1$.

The sandwich. On $A_n(\eta)$, $|e^\top(\Sigma_n - \Sigma)e| \le \eta |e|_\Sigma^2$, and by the Cauchy–Schwarz inequality $\sqrt{n}\,|e^\top \tilde{Z}_n| \le \sqrt{n}|e|_\Sigma \tilde{\rho}_n \le \frac{\eta}{2} n |e|_\Sigma^2 + \tilde{\rho}_n^2/(2\eta)$. So the last two terms of (12.4) are at most $\eta n |e|_\Sigma^2 + \tilde{\rho}_n^2/(2\eta)$ in absolute value, and condition 4 bounds $\eta n |e|_\Sigma^2$ by $\theta n K_U$, with $\theta = \eta / c_U \le \frac{1}{2}$. On $A_n(\eta)$, for every $w \in U$,

$$(1-\theta)\, nK_U(w) - \frac{\tilde{\rho}_n^2}{2\eta} \le nL_n(w) - nL_n(w_U) \le (1+\theta)\, nK_U(w) + \frac{\tilde{\rho}_n^2}{2\eta}.$$

Integration. Exponentiating, integrating against φ over U and taking $-\log$ gives, with $\bar{F}_U(s) = -\log \int_U \varphi\, e^{-sK_U}$,

$$\bar{F}_U\big((1-\theta)n\big) - \frac{\tilde{\rho}_n^2}{2\eta} \le F_n(U) - nL_n(w_U) \le \bar{F}_U\big((1+\theta)n\big) + \frac{\tilde{\rho}_n^2}{2\eta}.$$

The Abelian lemma. Lemma 4.3, applied to the prior restricted to U, whose mass is at most 1, gives $\int_U \varphi e^{-sK_U} \sim C_U \Gamma(\lambda_U + 1) s^{-\lambda_U} (\log s)^{m_U - 1}$ as $s \to \infty$ through the real numbers. Since $\log\log(\sigma n) - \log\log n \to 0$, $\bar{F}_U(\sigma n) = \lambda_U \log n - (m_U - 1)\log\log n + \lambda_U \log \sigma + \kappa_U + o(1)$ for fixed $\sigma > 0$. Take $\sigma = 1 \pm \theta$, use $|\log(1 \pm \theta)| \le 2\theta$, and let $\epsilon_n(\eta)$ be the larger $o(1)$ term: this is (12.3). With $\eta = \frac{1}{2} c_U$ its right side is bounded in probability, and $\mathbb{P}(A_n(\frac{1}{2}c_U)) \to 1$, so $R_n = O_p(1)$.

The choice. By Lemma 12.1, $\mathbb{P}(U_\alpha \mid X^n) = \big(1 + \sum_{\beta \ne \alpha} e^{-(F_n(U_\beta) - F_n(U_\alpha))}\big)^{-1}$, and by part (i) $F_n(U_\beta) - F_n(U_\alpha) = \Phi_\beta(n) - \Phi_\alpha(n) + O_p(1) \to \infty$ in probability. ∎

When U contains a true parameter, part (i) is Theorem 8.4 for the restricted model. When it does not, the leading term is random at the scale $\sqrt{n}$: by (12.4) with the truth in the place of w_U, $nL_n(w) - nS_n = nK(w) + O_p(\sqrt{n})$ for each fixed w. A misfitting region competes with a true one through fluctuations of that size, which is why the switch point of a single data set is random (Section 12.5). For a fixed truth the fluctuations lose in the end.

Corollary 12.4 (Large samples). *In the setting of Theorem 12.3(ii), suppose that some region contains a true parameter, a point w_0 with $g(w_0) = c_0$, and put $\Delta_\alpha = L(w_\alpha) - L(w_0)$. Then $\mathbb{P}(U_\alpha \mid X^n) \to 0$ in probability whenever*

$\Delta_\alpha > 0$; and if, among the regions with $\Delta_\beta = 0$, the region α alone has the smallest learning coefficient, or alone the largest multiplicity among those with the smallest learning coefficient, then $\mathbb{P}(U_\alpha \mid X^n) \to 1$ in probability.

PROOF. If $\Delta_\beta = 0 < \Delta_\alpha$, then $\Phi_\alpha(n) - \Phi_\beta(n) = n\Delta_\alpha + O_p(\sqrt{n}) + O(\log n) \to \infty$, by the remark above, and $\mathbb{P}(U_\alpha \mid X^n) \le e^{-(F_n(U_\alpha) - F_n(U_\beta))} \to 0$. If $\Delta_\alpha = \Delta_\beta = 0$, then $g(w_\alpha) = g(w_\beta) = c_0$, so $L_n(w_\alpha) = L_n(w_\beta)$ and $\Phi_\beta - \Phi_\alpha = (\lambda_\beta - \lambda_\alpha)\log n - (m_\beta - m_\alpha)\log\log n$, which tends to ∞ under the condition on α. Part (ii) finishes the proof. ∎

Before that limit the posterior moves, and at leading order the order of its moves is forced. Ignoring the fluctuations and the $\log\log n$ terms, each region is a point $(\lambda_\alpha, \Delta_\alpha)$ that pays $n\Delta_\alpha + \lambda_\alpha \log n$.

Proposition 12.5 (Fit against complexity). *Let $\bar\Phi_\alpha(n) = n\Delta_\alpha + \lambda_\alpha \log n$ for $\alpha = 1, \dots, k$, with $\Delta_\alpha \ge 0$ and $\lambda_\alpha > 0$, and let $e < n_1 < n_2$. If $\bar\Phi_\alpha(n_1) < \bar\Phi_\beta(n_1)$ and $\bar\Phi_\beta(n_2) < \bar\Phi_\alpha(n_2)$, then $\Delta_\beta < \Delta_\alpha$ and $\lambda_\beta > \lambda_\alpha$, and on (e, ∞) the two functions cross exactly once, where*

$$\frac{n}{\log n} = \frac{\lambda_\beta - \lambda_\alpha}{\Delta_\alpha - \Delta_\beta}.$$

In particular, as n increases through (e, ∞), every change of the unique minimizer lowers Δ and raises λ.

PROOF. Put $D(n) = (\bar\Phi_\beta(n) - \bar\Phi_\alpha(n))/\log n = (\Delta_\beta - \Delta_\alpha)\, n/\log n + (\lambda_\beta - \lambda_\alpha)$. On (e, ∞) the function $n/\log n$ has derivative $(\log n - 1)/\log^2 n > 0$, so D is strictly monotone unless $\Delta_\beta = \Delta_\alpha$, when it is constant. Since $D(n_1) > 0 > D(n_2)$, D decreases, so $\Delta_\beta < \Delta_\alpha$, and $\lambda_\beta - \lambda_\alpha = D(n_1) + (\Delta_\alpha - \Delta_\beta)\, n_1/\log n_1 > 0$; a strictly decreasing function has at most one zero. The last claim is the first applied to consecutive unique minimizers. ∎

Geometrically, $\bar\Phi_\alpha(n)/n = \Delta_\alpha + t\lambda_\alpha$ with $t = \log n/n$, which decreases from $1/e$ to 0: the preferred region minimizes a linear function of $(\lambda_\alpha, \Delta_\alpha)$ whose slope shrinks as data arrive. The regions ever preferred lie on the lower boundary of the convex hull of the points and are visited from small λ and large misfit toward large λ and small misfit; a point above that boundary is never preferred (Exercise 12.3).[6] Chen et al. (2023, App. C) derive the switch equation and show, for transitions that leave the constant terms unchanged, that slopes increase along a sequence of transitions. This is the precise sense in which a growing data set makes the posterior jump to more complex singularities: a region of smaller learning coefficient wins only while its worse fit has not yet been multiplied by enough data. The $\log\log n$ terms, the constants and the fluctuations of $nL_n(w_\alpha)$ shift the switch points, and Section 12.4 computes them in the examples.

[6] For $c > e$ the equation $n/\log n = c$ has the solution $n = -c\, W_{-1}(-1/c)$ on (e, ∞), with W_{-1} the lower branch of Lambert's function; numerically n lies between $1.27\, c \log c$ and $1.41\, c \log c$ for c between 10^2 and 10^4.

THE GENERAL FORMULA. For a model that is not linear in its coefficients, the local formula is Theorem 8.6, in its form for truths that the model does not realize (Watanabe 2010), applied to the model restricted to U: if the restricted

model satisfies its conditions with $p(\cdot \mid w_U)$ as the optimal distribution (among them, U compact and defined by analytic inequalities, w_U interior with the only optimal distribution in U, and relatively finite variance with respect to it), then $F_n(U) = nL_n(w_U) + \lambda_U \log n - (m_U - 1) \log\log n + O_p(1)$, with λ_U and m_U those of K_U on U. This is the local free energy formula of Chen et al. (2023, eq. (8)), who glue the regions with a partition of unity instead of indicator functions and call the comparison of local free energies *internal model selection*.

> Observation 12.6 (Chen et al. 2023). In the toy model of superposition with 2 hidden dimensions and 6 features, $f(x, w) = \mathrm{relu}(V^\top V x + c)$ with $V \in \mathbb{R}^{2\times 6}$ and $c \in \mathbb{R}^6$ ($d = 18$), the losses $\mathbb{E}\|x - f(x, w)\|^2$ and the learning coefficients of the pentagon (0.06874, $\lambda = 7$) and the hexagon (0.04819, $\lambda = 8.5$), with constants from the prior density at the two polygons, predict that the posterior moves from the pentagon to the hexagon at $n \approx 600$ (their switch equation gives 601). Posterior samples from NUTS, with the prior $\mathcal{N}(0, I)$, six chains of 5000 iterations after 500 of burn-in, and 10 data sets for each n, show the switch between $n = 600$ and $n = 700$. Below $n \approx 400$ the classification of samples becomes unreliable. The authors flag a tension in comparing asymptotic formulas at finite n. The same transition was not observed in their stochastic gradient descent runs.

The study is a preprint, and the agreement is less sharp than it looks. The switch equation charges each polygon n times the loss $\mathbb{E}\|x - f\|^2$, but the likelihood that their Appendix F says the sampler used, $p(x \mid w) \propto \exp(-\frac{1}{2}\|x - f(x, w)\|^2)$ (their eq. (3)), has half that energy, an apparent inconsistency by a factor of two. With the halved energy and the same constants the switch is at $n \approx 1317$; without the constants it is at 445, or 1010 with the halved energy (our computations from their tables); Chapter 14, which also tries the asymptotic constants, finds predicted switches from about 110 to about 1300. Their pentagon with an extra positive bias, which has the hexagon's learning coefficient 8.5 but the higher loss 0.06180, is never preferred at leading order.

12.4 The theorem: the crossover and the transition

With data, the computations of Section 12.2 become limit theorems in which the fluctuations of the data are kept rather than bounded. The tool is a form of dominated convergence for integrals that depend on a few statistics of the data.

> **Lemma 12.7** (Continuous convergence). *Let $\xi_n \to \xi^*$ in probability for a constant ξ^*, and let ζ_n be bounded in probability. Let f_n and f be measurable, f continuous, with $f_{n_j}(x_j, z_j) \to f(z)$ whenever $n_j \to \infty$, $x_j \to \xi^*$ and $z_j \to z$. Then $f_n(\xi_n, \zeta_n) - f(\zeta_n) \to 0$ in probability.*

Proof. The plan is to make the convergence uniform on compact sets and then use tightness. Since $\xi_n \to \xi^*$ in probability, there are $r_n \to 0$ with $\mathbb{P}(|\xi_n - \xi^*| > r_n) \to 0$: take N_j increasing with $\mathbb{P}(|\xi_n - \xi^*| > 1/j) < 1/j$

for $n \ge N_j$, and $r_n = 1/j$ for $N_j \le n < N_{j+1}$. Fix $M > 0$ and let s_n be the supremum of $|f_n(x,z) - f(z)|$ over $|x - \xi^*| \le r_n$, $|z| \le M$. If $s_n \not\to 0$, then there are $\epsilon > 0$, $n_j \to \infty$ and such (x_j, z_j) with $|f_{n_j}(x_j, z_j) - f(z_j)| \ge \epsilon$; along a subsequence $z_j \to z$, and then $f_{n_j}(x_j, z_j) \to f(z)$ and $f(z_j) \to f(z)$, a contradiction. Hence

$$\mathbb{P}\big(|f_n(\xi_n, \zeta_n) - f(\zeta_n)| > \epsilon\big) \le \mathbb{P}(|\xi_n - \xi^*| > r_n) + \mathbb{P}(|\zeta_n| > M) + \mathbf{1}\{s_n > \epsilon\},$$

whose upper limit is at most $\sup_n \mathbb{P}(|\zeta_n| > M)$, small for large M. ∎

Proposition 12.8 (The crossover). *Consider the quartic model $y = ax_1 + b^2 x_2$ with inputs $\mathcal{N}(0, I_2)$, unit noise, the uniform prior on $[-1,1]^2$ and the truth $y = \delta x_2 + \varepsilon$, $0 \le \delta < 1$. Let $h(s) = \int_{\mathbb{R}} e^{-(u^2-s)^2/2}\,\mathrm{d}u$, $\tau = \sqrt{n}\,\delta$, $Z \sim \mathcal{N}(0,1)$, $\Psi(\tau) = -\mathbb{E}\log h(\tau + Z)$, and*

$$\Lambda_{\mathrm{pop}}(\tau) = \frac{3}{4} - \frac{\tau}{2}\frac{h'(\tau)}{h(\tau)}, \qquad \Lambda_{\mathrm{data}}(\tau) = \frac{3}{4} + \frac{\tau}{2}\Psi'(\tau).$$

(i) (Population.) For every $n \ge 1$, $\bar{F}(n) = \frac{3}{4}\log n + \log 4 - \frac{1}{2}\log(2\pi) - \log h(\tau) + r_n$ with $r_n \ge 0$; for each $\bar{\delta} < 1$, $\sup_{\delta \le \bar{\delta}} r_n$ and $\sup_{\delta \le \bar{\delta}} |\lambda_{\mathrm{eff}}(n) - \Lambda_{\mathrm{pop}}(\tau)|$ tend to 0 faster than every power of n, where $\lambda_{\mathrm{eff}} = \mathrm{d}\bar{F}/\,\mathrm{d}\log n$.

(ii) (Data, a truth scaled with n.) For fixed $\tau \ge 0$ and $\delta = \tau/\sqrt{n}$, $F_n - nS_n - \frac{3}{4}\log n$ converges in law to $\log 4 - \frac{1}{2}\log(2\pi) - \frac{1}{2}Z_1^2 - \frac{1}{2}Z_2^2 - \log h(\tau + Z_2)$, with Z_1, Z_2 independent standard normal, and

$$\mathbb{E}[F_n - nS_n] = \tfrac{3}{4}\log n + \log 4 - \tfrac{1}{2}\log(2\pi) - 1 + \Psi(\tau) + o(1).$$

(iii) For $r > 1$ and fixed $\tau > 0$, as $n \to \infty$ with rn an integer and $\delta = \tau/\sqrt{n}$, the slope $\big(\mathbb{E}[F_{rn} - rnS_{rn}] - \mathbb{E}[F_n - nS_n]\big)/\log r$ tends to $\frac{3}{4} + (\Psi(\sqrt{r}\,\tau) - \Psi(\tau))/\log r$, and this tends to $\Lambda_{\mathrm{data}}(\tau)$ as $r \downarrow 1$.

Proof. The plan is to compute (i) exactly, bounding the tails cut off by the box, and to obtain (ii) from the rescaled integral of Computation 12.1 with data, by Lemma 12.7; uniform integrability gives the means, and (iii) follows from (ii).

(i). By Computation 12.1, $r_n = -\log(1 - 2\bar{\Phi}(\sqrt{n})) - \log(1 - t_n/h(\tau)) \ge 0$ with $t_n = \int_{|u|>n^{1/4}} e^{-(u^2-\tau)^2/2}\,\mathrm{d}u$. First, $2\bar{\Phi}(\sqrt{n}) \le e^{-n/2}$. For $|u| \ge n^{1/4}$, $u^2 - \tau \ge \sqrt{n}(1-\delta) > 0$, and convexity of the square with $u^2 - \sqrt{n} \ge 2n^{1/4}(|u| - n^{1/4})$ gives $(u^2-\tau)^2 \ge n(1-\delta)^2 + 4n^{3/4}(1-\delta)(|u| - n^{1/4})$, so $t_n \le e^{-n(1-\delta)^2/2}/(n^{3/4}(1-\delta))$. And $h(\tau) \ge e^{-1/2}/\sqrt{1+\tau}$, since the integrand of h is at least $e^{-1/2}$ on $\sqrt{\tau} \le |u| \le \sqrt{\tau+1}$, a set of length at least $1/\sqrt{1+\tau}$; with $\tau \le \sqrt{n}$ this bounds r_n. For the slope, $\lambda_{\mathrm{eff}}(n) = n\mathbb{E}_{\mu_n}[K]$ with $\mu_n \propto e^{-nK}\varphi$, a product of laws of a and b. The law of a contributes $\frac{1}{2}\mathbb{E}[v^2 \mid |v| \le \sqrt{n}]$ for standard normal v, within $\sqrt{n}\,e^{-n/2}$ of $\frac{1}{2}$ for $n \ge 2$. With $u = n^{1/4}b$ and $f(u) = \frac{1}{2}(u^2-\tau)^2$, the law of b contributes the mean of f under the density proportional to e^{-f}, conditioned on $|u| \le n^{1/4}$. Without the conditioning, integration by parts gives $\mathbb{E}[u\,f'(u)] = 1$, and $u\,f'(u) = 4f(u) + 2\tau(u^2 - \tau)$

with $\mathbb{E}[u^2-\tau]=h'(\tau)/h(\tau)$, so $\frac{1}{2}+\mathbb{E}f=\Lambda_{\mathrm{pop}}(\tau)$. The conditioning moves $\mathbb{E}f$ by at most $(\mathbb{E}[f;|u|>n^{1/4}]+\mathbb{E}[f]\mathbb{P}(|u|>n^{1/4}))/\mathbb{P}(|u|\le n^{1/4})$, which the same tail bound, applied to $f/2$ through $fe^{-f}\le e^{-f/2}$, makes smaller than every power of n.

(ii), the limit law. By (12.4) with the truth in the place of w_U, $nK_n(w)=\frac{n}{2}e^\top\Sigma_n e-\sqrt{n}\,e^\top Z_n$ with $e=(a,b^2-\delta)$. Substituting $a=v/\sqrt{n}$ and $b=n^{-1/4}u$ makes $\sqrt{n}\,e=\zeta=(v,u^2-\tau)$. With $D_n=[-\sqrt{n},\sqrt{n}]\times[-n^{1/4},n^{1/4}]$,

$$F_n-nS_n-\tfrac{3}{4}\log n=\log 4-\log\int_{D_n}e^{-\frac{1}{2}\zeta^\top\Sigma_n\zeta+\zeta^\top Z_n}\,\mathrm{d}v\,\mathrm{d}u=:f_n(\Sigma_n,Z_n).$$

Let $f(z)$ be the same expression with $\Sigma_n=I$ and $D_n=\mathbb{R}^2$; completing the squares in v and in $u^2-\tau$ gives $f(z)=\log 4-\frac{1}{2}\log(2\pi)-\frac{1}{2}z_1^2-\frac{1}{2}z_2^2-\log h(\tau+z_2)$, which is continuous. If $S_j\to I$, $z_j\to z$ and $n_j\to\infty$, then for large j the integrand is at most $\exp(-\frac{1}{4}|\zeta|^2+(|z|+1)|\zeta|)\le\exp(-\frac{1}{8}|\zeta|^2+2(|z|+1)^2)$, integrable over $\mathbb{R}^2$, and dominated convergence gives $f_{n_j}(S_j,z_j)\to f(z)$. Since $\Sigma_n\to I$ in probability and Z_n has mean 0 and covariance I, Lemma 12.7 gives $F_n-nS_n-\frac{3}{4}\log n-f(Z_n)\to 0$ in probability; the central limit theorem, continuous mapping and Slutsky's lemma (Definition A.13) give the limit law.

(ii), the means. With $\rho_n^2=Z_n^\top\Sigma_n^{-1}Z_n$, Cauchy–Schwarz bounds the exponent above by $-\frac{1}{4}\lambda_{\min}(\Sigma_n)|\zeta|^2+\rho_n^2$ and below by $-\lambda_{\max}(\Sigma_n)|\zeta|^2-\frac{1}{2}\rho_n^2$. Integrating the upper bound over $\mathbb{R}^2$, using $\int e^{-s|\zeta|^2}=\sqrt{\pi/s}\,(2s)^{-1/4}h(\sqrt{2s}\,\tau)$ and that h is bounded, and the lower bound over $[-1,1]^2\subseteq D_n$,

$$\left|F_n-nS_n-\tfrac{3}{4}\log n\right|\le C\big(1+\rho_n^2+\lambda_{\max}(\Sigma_n)+|\log\lambda_{\min}(\Sigma_n)|\big).$$

Each term has a second moment bounded in n: ρ_n^2 is χ^2_2 given the inputs; $\lambda_{\max}(\Sigma_n)\le\operatorname{tr}\Sigma_n$; and $\lambda_{\min}(\Sigma_n)\ge\det\Sigma_n/\operatorname{tr}\Sigma_n$, where $n^2\det\Sigma_n$ is a product of independent χ^2_n and χ^2_{n-1} variables by Bartlett's decomposition. So the sequence is uniformly integrable, and its mean converges to $\mathbb{E}f(Z)$.

(iii). At sample size rn the same truth is $\delta=\sqrt{r}\,\tau/\sqrt{rn}$, so (ii) applies with $\sqrt{r}\,\tau$; subtract. Since h'/h grows at most linearly, $\Psi'(\tau)=-\mathbb{E}[(h'/h)(\tau+Z)]$ by dominated convergence, and the difference quotient tends to $\frac{3}{4}+\frac{\tau}{2}\Psi'(\tau)$. ∎

With data the free energy depends on $\tau+Z_2$: the noise in the coefficient of x_2 has the same order $n^{-1/2}$ as the truth and blurs the hump, so Λ_{data} dips lower and overshoots less than Λ_{pop}, with a minimum 0.675 at $n\delta^2=0.29$, a maximum 1.067 at $n\delta^2=12.8$, and the value $\frac{7}{8}$ at $n\delta^2=3.1$.

Proposition 12.9 (The transition). *Consider the double-well model $y=abx_1+q(b)x_2$, $q(b)=2b^4-b^2$, with inputs $\mathcal{N}(0,I_2)$, unit noise, the uniform prior on $[-1,1]^2$ and the truth $y=\delta x_2+\varepsilon$, $0<\delta<1$. Let $U_A=\{|b|\le\frac{1}{2}\}$, $U_B=\{|b|\ge\frac{1}{2}\}$, $S_{xx}=\sum_i x_ix_i^\top$, $S_{xy}=\sum_i x_iy_i$, $\kappa_B=\log(D/\pi)$ with D as in Computation 12.2, and*

$$\kappa_A(z)=-\log\Big(\frac{1}{4}\int_{-1}^{1}\sqrt{\frac{\pi}{\delta+a^2/2}}\,\exp\Big(\frac{a^2z^2}{4\delta+2a^2}\Big)\,\mathrm{d}a\Big).$$

(i) $F_n(U_A) = nL_n(0) + \frac{1}{2}\log n + \kappa_A\big((S_{xy})_1/\sqrt{n}\big) + o_p(1)$, *with* $L_n(0)$ *the loss on the segment* $b = 0$.

(ii) $F_n(U_B) = nL_n(0) - \frac{1}{2}S_{xy}^\top S_{xx}^{-1} S_{xy} + \log n + \kappa_B + o_p(1)$.

(iii) *With* $\omega_n = \frac{1}{2}S_{xy}^\top S_{xx}^{-1} S_{xy} - \frac{1}{2}\log n + \kappa_A\big((S_{xy})_1/\sqrt{n}\big) - \kappa_B$,

$$\log \frac{\mathbb{P}(U_B \mid X^n)}{\mathbb{P}(U_A \mid X^n)} = \omega_n + o_p(1),$$

so $\mathbb{P}(U_B \mid X^n) - \operatorname{sig}(\omega_n) \to 0$ *in probability, where* $\operatorname{sig}(t) = 1/(1 + e^{-t})$.

Proof. The plan is to obtain (ii) from Laplace's approximation for regular models, (i) from Lemma 12.7 with an explicit dominating bound, and (iii) as their difference.

Region B. The map $(a, b) \mapsto (-a, -b)$ preserves the prior and g, so $F_n(U_B) = F_n(U_B^+) - \log 2$ with $U_B^+ = [-1, 1] \times [\frac{1}{2}, 1]$. On U_B^+, conditions (R1)–(R6) of Theorem 1.4 hold for the prior renormalized to U_B^+, which is uniform with density 1: L has the unique minimizer $(0, b_\delta)$, interior because $\frac{1}{2} < b_\delta < 1$, with Hessian $Dg^\top Dg$ of determinant D^2, and the log-likelihood is a polynomial in w with coefficients of finite moments. Since U_B^+ has prior mass $\frac{1}{4}$, $F_n(U_B^+) = nL_n(\hat{w}) + \log\frac{n}{2\pi} + \log D + \log 4 + o_p(1)$, with $\hat{w}$ a minimizer of L_n over U_B^+. As g maps a neighborhood of $(0, b_\delta)$ diffeomorphically onto a neighborhood of c_0 and $S_{xx}^{-1}S_{xy} \to c_0$ in probability, with probability tending to one $nL_n(\hat{w})$ is the minimum over all $c \in \mathbb{R}^2$ of $nL_n(0) + \frac{1}{2}c^\top S_{xx} c - c^\top S_{xy}$, which is $nL_n(0) - \frac{1}{2}S_{xy}^\top S_{xx}^{-1} S_{xy}$. This gives (ii).

Region A. On U_A, $g = (ab, q(b))$ vanishes on the segment, so $nL_n(w) - nL_n(0) = \frac{1}{2}g^\top S_{xx} g - g^\top S_{xy}$. Substitute $b = u/\sqrt{n}$ and write $q_n(u) = q(u/\sqrt{n}) = -u^2/n + 2u^4/n^2$; then

$$F_n(U_A) - nL_n(0) - \tfrac{1}{2}\log n = \log 4 - \log \int_{-1}^{1} \int_{|u| \le \sqrt{n}/2} e^{-H_n(a,u)}\, \mathrm{d}u\, \mathrm{d}a,$$

$$H_n = \frac{(S_{xx})_{11}}{2n} a^2u^2 + \frac{(S_{xx})_{12}}{\sqrt{n}} au\, q_n(u) + \frac{(S_{xx})_{22}}{2n} n\, q_n(u)^2 - \frac{(S_{xy})_1}{\sqrt{n}} au - \frac{(S_{xy})_2}{n} n\, q_n(u).$$

Let $\xi_n = (\Sigma_n, (S_{xy})_2/n)$, which tends to (I, δ) in probability, and $\zeta_n = (\sqrt{n}(\Sigma_n)_{12}, \tilde{Z}_n)$ with $\tilde{Z}_n = (S_{xy} - nc_0)/\sqrt{n}$, which is bounded in probability by the central limit theorem; since $c_0 = (0, \delta)$, the first coordinate of $\tilde{Z}_n$ is $(S_{xy})_1/\sqrt{n}$. Because $q_n(u) \to 0$ and $nq_n(u) \to -u^2$, $H_n(a, u) \to (\delta + \frac{1}{2}a^2)u^2 - zau$ along statistics with $\xi \to (I, \delta)$ and $(S_{xy})_1/\sqrt{n} \to z$. For the domination, let the coefficients $\sqrt{n}(\Sigma_n)_{12}$ and $(S_{xy})_1/\sqrt{n}$ of H_n be at most T in absolute value, $(S_{xy})_2/n \ge \frac{1}{2}\delta$, and $(\Sigma_n)_{11}, (\Sigma_n)_{22} \ge 0$, as near the limits. On $|u| \le \frac{1}{2}\sqrt{n}$, $-nq_n(u) = u^2(1 - 2u^2/n) \ge \frac{1}{2}u^2$ and $|u\, q_n(u)| \le |u|^3/n \le u^2/(2\sqrt{n})$; the first and third terms of H_n are nonnegative, so $H_n \ge \frac{1}{4}\delta u^2 - Tu^2/(2\sqrt{n}) - T|u| \ge$

$\frac{1}{8}\delta u^2 - T|u|$ once $\sqrt{n} \geq 4T/\delta$, a bound integrable over $[-1,1]\times\mathbb{R}$. Dominated convergence verifies the hypothesis of Lemma 12.7, with the continuous limit $\log 4 - \log \int_{-1}^{1}\int_{\mathbb{R}} e^{-(\delta+a^2/2)u^2+zau}\,du\,da = \kappa_A(z)$, and the lemma gives (i).

The odds. By Lemma 12.1 the log odds are $F_n(U_A) - F_n(U_B)$; subtract (ii) from (i), and use that sig is Lipschitz. ∎

Here $\frac{1}{2}S_{xy}^{\mathsf{T}}S_{xx}^{-1}S_{xy} = nL_n(0) - \min_c nL_n(c)$ is the log-likelihood ratio of the least-squares fit against the zero function, and ω_n is that ratio minus the singular penalty $\frac{1}{2}\log n$, plus constants; the regular theory charges no penalty. With the population statistics $S_{xx} = nI$, $S_{xy} = nc_0$, $\omega_n = n\Delta - \frac{1}{2}\log n + \kappa_A(0) - \kappa_B$, whose zero is $\bar{n}_c$. With data the ratio fluctuates around $n\Delta$ with standard deviation about $\sqrt{2n\Delta}$, which is 2.7 at $n = 3000$ and $\delta = 0.05$, comparable to the penalty $\frac{1}{2}\log n = 4$.

12.5 Measuring it

THE CROSSOVER. The script `12-phases-crossover.py` measures the effective learning coefficient of the quartic model with data, for $\delta = 10^{-2}, 10^{-3}, 10^{-4}$. For each δ it draws 3000 data sets through their sufficient statistics, S_{xx} from the Wishart law by Bartlett's decomposition[7] and $\sum_i x_i\varepsilon_i \sim \mathcal{N}(0, S_{xx})$ given it, and grows each data set by doubling n, from the larger of 100 and e^{-6}/δ^2 to $n\delta^2 \approx e^{6.5}$ (17 to 19 sizes, up to $n = 6.5\times 10^{10}$). Each free energy is a one-dimensional integral: for fixed b the energy is quadratic in a, so the integral over $a \in [-1,1]$ is a Gaussian integral with error functions, and the integral over b a trapezoid rule on 1200 points of the scale $u = n^{1/4}b$, which doubling changes by less than 2×10^{-13}. The measured coefficient between consecutive sizes is the difference of the means of $F_n - nS_n + \frac{1}{2}\rho_n^2 - 1$ divided by $\log 2$; since ρ_n^2 is χ^2_2 given the inputs, this control variate leaves the mean unchanged and cuts the standard deviation per data set from 1.33 to about 0.45. The intervals are Student-t 95% intervals over the paired differences (Section A.8). The run takes 50 to 75 seconds.

[7] A 2×2 Wishart matrix with n degrees of freedom is LL^{T} with L lower triangular and independent entries, $L_{11}^2 \sim \chi^2_n$, $L_{22}^2 \sim \chi^2_{n-1}$ and $L_{21} \sim \mathcal{N}(0,1)$; nested data sets add independent Wishart increments.

In Figure 12.2 the three series fall on one curve: over eight orders of magnitude of n, the effective learning coefficient depends on n and δ only through $n\delta^2$, and it follows the curve with data, not the population curve. Of the 52 measured slopes, 51 lie within their intervals of the prediction of Proposition 12.8(iii) for doubling, which differs from Λ_{data} by at most 0.004; the exception, 0.656 ± 0.023 against 0.680 at $n\delta^2 = 0.45$, is about what 52 intervals at 95% lead one to expect. The dip and the overshoot are measured: the smallest slopes of the three series are 0.656, 0.666, 0.672 and the largest 1.064, 1.065, 1.067, against the predicted extremes 0.675 and 1.067. For $n\delta^2 < 0.01$ the coefficient is within 0.03 of $\frac{3}{4}$, for $n\delta^2 > 100$ within 0.01 of 1; in between, which for $\delta = 10^{-3}$ means $10^4 \lesssim n \lesssim 10^8$, neither formula describes the free energy.

THE TRANSITION. The script `12-phases-transition.py` measures the posterior mass of region B of the double-well model with $\delta = 0.05$, for 500 nested data sets (seed 1210) at 25 sample sizes from 10^2 to 10^5, equally spaced in $\log n$. It integrates over a in closed form and over b by the trapezoid

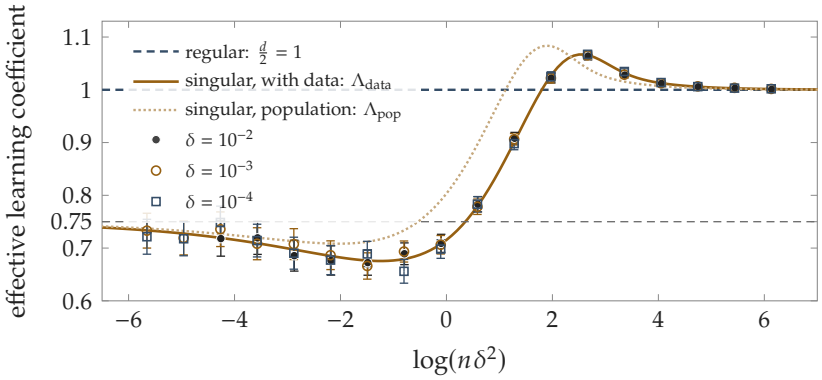


Figure 12.2. The effective learning coefficient of the quartic model at the truth $y = \delta x_2$ against $\log(n\delta^2)$; inputs $\mathcal{N}(0, I_2)$, unit noise, uniform prior on $[-1, 1]^2$. Points: slopes of the mean normalized free energy between n and $2n$, over 3000 nested data sets for each δ (seed 1200), by exact quadrature, with Student-t 95% intervals. Solid bronze: Λ_{data} of Proposition 12.8. Dotted bronze: Λ_{pop}, which the population free energies of the three values of δ reproduce to 5×10^{-8}. Dashed slate: the regular value 1. Gray: $\frac{3}{4}$, the learning coefficient at $\delta = 0$. Script: `12-phases-crossover.py`.

rule on 8001 points of $[-1, 1]$, the boundaries $\pm\frac{1}{2}$ among them (doubling changes the logarithms of the region integrals by less than 6×10^{-8}). The prediction of Proposition 12.9 is the mean of $\text{sig}(\omega_n)$ over 10^5 independent draws of (S_{xx}, S_{xy}) for each n; the regular prediction is the mean of sig of the maximized log-likelihood ratio of the two regions, the odds that Laplace's approximation gives when both regions pay $\log n$. The run takes 55 to 95 seconds.

In Figure 12.3, from $n \approx 400$ on, the measured masses and the prediction agree within the intervals at all 20 sample sizes; at $n = 3162$ the measured mass is 0.534 ± 0.031 and the prediction 0.542. Below $n \approx 400$ the prediction is low by 0.03 to 0.045, the $o_p(1)$ of Proposition 12.9 made visible: the log odds of the quadrature exceed ω_n by 0.25 on average at $n = 100$, 0.044 at $n = 1000$ and 0.005 at $n = 10^4$ (standard deviations 0.33, 0.084, 0.026). The measured curve first falls, from 0.26 at $n = 100$ to 0.20 at $n = 422$, because at $n = 0$ the posterior is the prior, which gives each region half its mass: the first move of the posterior is out of the prior and into the singular region. It then rises through $\frac{1}{2}$ near $n = 3000$ and reaches 0.97 at $n = 10^4$. The regular prediction is 0.62 already at $n = 100$ and 0.93 at $n = 3162$.

Each data set switches sharply, but at its own n. The fraction of data sets that give B more than half the posterior mass is 0.17, 0.51 and 0.83 at $n = 1000$, 3162 and 5623, against 0.18, 0.53 and 0.81 for the fraction of draws with $\omega_n > 0$, whose median is at $n = 2972$; the averaged transition is therefore much wider than the population one, which goes from 0.19 to 0.94 between $n = 1778$ and 5623. The ridge separates the regions once $n\eta$ is large: the posterior mass of the band $0.4 \le |b| \le 0.6$, averaged over the data sets, is 8.8% at $n = 100$, 1.3×10^{-4} at $n = 1000$ and 3×10^{-12} at $n = 3162$.

Figure 12.4 shows one data set before, at and after its switch. At $n = 300$

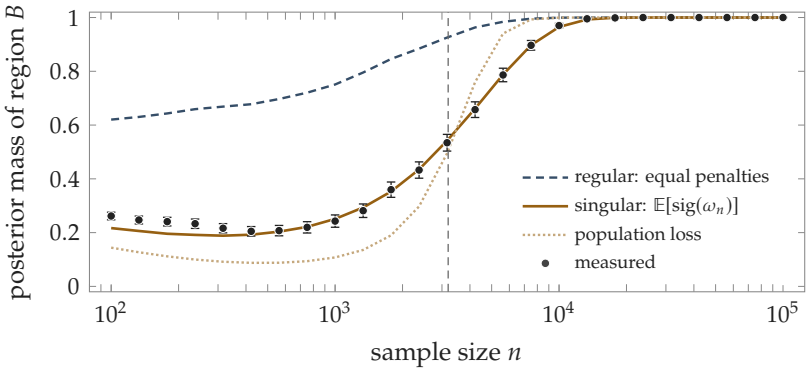


Figure 12.3. The posterior mass of the regular region $B = \{|b| \geq \frac{1}{2}\}$ of the double-well model at the truth $y = 0.05\,x_2$, against n; inputs $\mathcal{N}(0, I_2)$, unit noise, uniform prior on $[-1, 1]^2$. Dots: means over 500 nested data sets (seed 1210) of the exact posterior mass, by quadrature, with Student-t 95% intervals. Solid bronze: the prediction $\mathbb{E}[\mathrm{sig}(\omega_n)]$ of Proposition 12.9. Dotted bronze: the posterior mass with the population loss, which switches at $\bar{n}_c = 3198$ (gray line). Dashed slate: the regular prediction, the mean of sig of the maximized log-likelihood ratio of the regions. Script: `12-phases-transition.py`.

most draws lie along the segment $b = 0$, over all values of a, thicker where a is small because the valley across the segment has curvature $2\delta + a^2$. At $n = 3000$ half the mass has moved to two small clusters at $(0, \pm b_\delta)$, with no draws in between; at $n = 30000$ the segment is empty. The draws in B all have a near 0: the true parameters lie on the other arm, $a = 0$, of the product model's cross.

12.6 The ledger

Where the posterior goes is decided by local free energies, which for models linear in their coefficients are $nL_n(w_\alpha) + \lambda_\alpha \log n - (m_\alpha - 1) \log\log n$ up to a bounded term, whether or not the region contains the truth. At leading order a growing sample moves the posterior along the lower convex hull of the points $(\lambda_\alpha, \Delta_\alpha)$, toward better fit and larger learning coefficient. Near a singular truth the effective learning coefficient is a universal, non-monotone function of $n\delta^2$; between separated regions the posterior odds are a likelihood ratio minus the difference of the $\lambda \log n$ terms, with a fluctuation of order $\sqrt{n}$ that makes each data set switch at its own n.

LEDGER · $y = ax_1 + b^2x_2,\ y = abx_1 + (2b^4 - b^2)x_2$	
QUANTITY	the posterior mass of a region U, through its local free energy $F_n(U)$
REGULAR THEORY	every region pays $\frac{d}{2} \log n$; the posterior sits in the region of the maximum-likelihood estimate once n times the loss gap is large

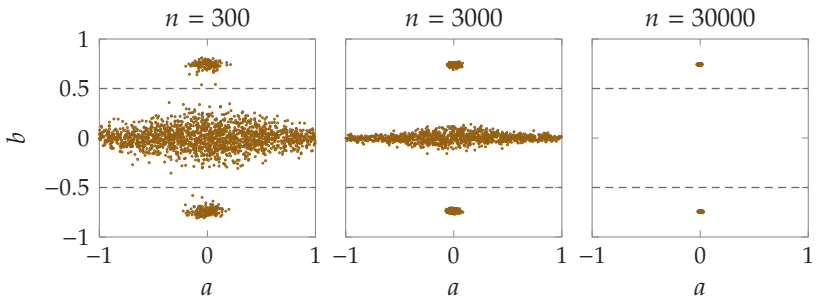


Figure 12.4. Draws from the posterior of the double-well model for one data set at the truth $y = 0.05\,x_2$, at $n = 300$, 3000 and 30000 (nested; seed 1221), 2000 draws each: b from its marginal by the inverse distribution function on 40001 points, then a from its conditional law, a truncated normal. Gray: the ridges $b = \pm\frac{1}{2}$. The posterior mass of region B is 0.15, 0.49 and 1.00. Script: `12-phases-transition.py`.

SINGULAR THEORY	$F_n(U_\alpha) = nL_n(w_\alpha) + \lambda_\alpha \log n - (m_\alpha - 1)\log\log n + O_p(1)$, and the posterior takes the smallest (Theorem 12.3); transitions in n lower the loss and raise λ (Proposition 12.5)
RUNNING EXAMPLE	the quartic model at $y = \delta x_2$: λ_{eff} from $\frac{3}{4}$ to 1 along a universal curve of $n\delta^2$, dipping to 0.68 and overshooting to 1.07; the double well at $\delta = 0.05$: from a segment with $\lambda = \frac{1}{2}$ to regular points with $\lambda = 1$ at $n \approx 3000$
MEASURED	51 of 52 slopes within their intervals; posterior mass of B 0.24, 0.53, 0.97 at $n = 10^3$, 3162, 10^4, against 0.25, 0.54, 0.96 predicted and 0.75, 0.93, 1.00 from the regular theory
RESTS ON	a fixed model and truth, or a truth scaled with n; regions with relatively finite variance and known local volumes; linear models in the proofs, Watanabe's conditions in general; exact posteriors; transitions in n, not in training time

12.7 Where the theory stops

WHAT WAS ASSUMED. The local formula is asymptotic for fixed regions and a fixed truth, and not uniform in the truth: in the quartic model it holds with $\lambda = 1$ for every $\delta > 0$, yet for $n\delta^2 \lesssim 100$ the free energy grows at a rate between 0.68 and 1.07 instead, a mispricing over a range of n that grows like δ^{-2}. The regions are an input: the theorem does not say how to find them in a network, and Chen et al. (2023) define theirs by classifying posterior samples. The proofs cover models linear in their coefficients; in general each region needs Watanabe's conditions, including analyticity, which the ReLU of the toy model of superposition violates. Everything concerns the exact posterior at inverse temperature 1, as a function of n; nothing here is about training time.

WHERE IT BREAKS. The terms below $n\Delta + \lambda \log n$ are not small where transitions happen. In the double-well model the constants nearly cancel (3198 against 3232), but in the toy model of superposition the constants and the convention for the energy move the predicted switch from about 110 to about 1300 (Chapter 14); and in Chapter 3 a $\log\log n$ term keeps a mixed prior's posterior in the region of larger learning coefficient until $n \approx 10^{13}$. The fluctuation of order $\sqrt{n}$ makes the switch of a single data set random: in Figure 12.3 the fraction of data sets that have switched rises from 0.17 to 0.83 while n grows by a factor of 5.6. In a preprint on transformers trained for in-context linear regression, Carroll et al. (2025) read the move from a ridge-regression solution to one fitted to the finite set of training tasks as a trade-off of this kind; above a threshold of task diversity the move does not happen, although the Bayesian account predicts it, and they leave the reason open, from capacity and under-training to a real difference between Bayesian and dynamical selection. Finally, the posterior mass of a region is easy to define and hard to sample: a ridge of height η holds a local sampler for a time that grows like $e^{n\eta}$, and at the switch of the double-well model $n\eta \approx 45$ (Exercise 12.9); our figures use quadrature.[8]

OPEN PROBLEMS. Chen et al. (2023) conjecture that the transitions met during training have Bayesian antecedents, transitions of the posterior in n between the same regions; what links the two, and when training follows the order of Proposition 12.5, is open as of 2026, and Chapter 14 returns to it. The dynamics of gradient learning near singularities, which produce plateaus (Amari et al. 2006; Cousseau et al. 2008; Fukumizu and Amari 2000; Wei et al. 2008), are understood for small networks; their relation to the posterior's phases is not. Uniform asymptotics near singular truths are open beyond examples and the layered models of Watanabe and Amari (2003): which scaling variable replaces $n\delta^2$ for a family of truths approaching a singular one, and is the crossover function fixed by the geometry of the singularity? The lower-order terms of local free energies, which decide transitions at practical n, are unknown in general, and Carroll et al. (2025) list them among the open questions of the loss–complexity account.

[8] Exchange Monte Carlo, a method from statistical physics that swaps chains run at several temperatures, was brought to singular learning machines by Nagata and Watanabe (2008) because Metropolis sampling of their posteriors is costly.

Exercises

12.1 CONSOLIDATION. (The crossover curve.) Prove that $\mathbb{E}_\tau[\frac{1}{2}(u^2-\tau)^2] = \frac{1}{4} - \frac{\tau}{2}h'(\tau)/h(\tau)$ for the density proportional to $e^{-(u^2-\tau)^2/2}$; show that $\Lambda_{\text{pop}}(0) = \frac{3}{4}$ by identifying the law of $\frac{1}{2}u^4$ at $\tau = 0$ as a Gamma law with shape $\frac{1}{4}$, and that for $\tau > 0$, $\Lambda_{\text{pop}}(\tau) < \frac{3}{4}$ exactly where $h'(\tau) > 0$. Check numerically that h is largest at $\tau = 0.765$.

12.2 CONSOLIDATION. (The other side of the singularity.) Fit the quartic model to $y = -\delta x_2 + \varepsilon$, which it cannot realize. Show that the best parameter is $(0, 0)$, with misfit $\frac{1}{2}\delta^2$, and that $\bar{F}(n) - \frac{1}{2}n\delta^2$ has effective coefficient $\frac{3}{4} + \frac{\tau}{2}\mathbb{E}[u^2]$ under the density proportional to $e^{-\tau u^2 - u^4/2}$, which tends to 1; check numerically that, unlike Λ_{pop}, it increases monotonically (0.773, 0.895, 0.996 at $\tau = 0.1, 1, 10$).

12.3 CONSOLIDATION. (Lower convex hull.) Show that a region that is the unique preferred one at leading order for some $n > e$ has its point $(\lambda_\alpha, \Delta_\alpha)$ at a vertex of the lower boundary of the convex hull of all the points, and deduce the statement of Chen et al. (2023, App. C.4) that the slopes $(\Delta_\beta - \Delta_\alpha)/(\lambda_\beta - \lambda_\alpha)$ increase along a sequence of transitions.

12.4 CONSOLIDATION. (The tempered posterior.) For the posterior at a fixed inverse temperature β (Chapter 11), show under Assumption 12.2 that the local free energy $-\log \int_U \varphi e^{-\beta n L_n}$ is $\beta n L_n(w_U) + \lambda_U \log(n\beta) - (m_U - 1)\log\log(n\beta)$ up to $O_p(1)$. Deduce that at fixed n heating the posterior moves it toward simpler regions; for the double-well model at $n = 10^4$ and $\delta = 0.05$ the population switch is at $\beta = 0.32$.

12.5 BY HAND. (Three regions.) Regions have $(\lambda, \Delta) = (\frac{1}{2}, 3\epsilon)$, $(\frac{3}{4}, \epsilon)$ and $(1, 0)$, with $\epsilon = 10^{-3}$. Using the leading terms, find the order in which they are preferred and the switch points (answers: $n = 842$ and $n = 1885$). Replace the middle region by $(\frac{3}{4}, 2\epsilon)$, and show that it is never preferred and that the posterior jumps from the first region to the last at $n = 1179$.

12.6 BY HAND. (The emblem.) Show from its Newton polygon (Theorem 6.4) that the chapter's emblem $K = a^6 + a^2b^2 + b^6$ has $\lambda = \frac{1}{2}$ and $m = 2$, and that region A of the double-well model at $\delta = 0$, where $K = \frac{1}{2}a^2b^2 + \frac{1}{2}b^4 + O(b^6)$, has the same values (compare K with $a^2b^2 + b^4$). Show that the constant $-\log(\sqrt{\pi/2}\,\operatorname{arsinh}(1/\sqrt{2\delta}))$ of $\bar{F}_A$ behaves like $-\log\log\frac{1}{\delta}$ as $\delta \to 0$.

12.7 BY HAND. ★ (A circle of equally good distributions.) Let $y = (\cos\theta)x_1 + (\sin\theta)x_2 + \varepsilon$ with θ uniform on $[0, 2\pi)$, the truth $y = \varepsilon$, and a design with $\frac{1}{n}\sum_i x_i x_i^\top = I_2$ exactly. Show that every θ has misfit $\frac{1}{2}$, that relatively finite variance fails, and, with I_0 the modified Bessel function and $Z_n = n^{-1/2}\sum_i x_i\varepsilon_i$, that

$$F_n - nS_n = \tfrac{n}{2} - \log I_0\big(\sqrt{n}\,|Z_n|\big) = \tfrac{n}{2} - \sqrt{n}\,|Z_n| + \tfrac{1}{4}\log n + O_p(1).$$

Which conclusions of Theorem 12.3 fail? *(Hint:* $nL_n(\theta) - nS_n = \frac{n}{2} - \sqrt{n}\,|Z_n|\cos(\theta - \theta_n)$ *for an angle* θ_n*.)*

12.8 VARIATION. (Why not the mixture.) For the mixture $(1-a)\mathcal{N}(0,1) + a\mathcal{N}(b,1)$ with a truth (a_0, b_0), $0 < a_0 < 1$, $b_0 \neq 0$, show that $\partial L/\partial a = -a_0(e^{bb_0} - 1)$ at $(0, b)$, so that the arm $a = 0$ of the cross is not a set of local minima where $bb_0 > 0$, and argue that no ridge separates the cross from the true point. What makes the segment of the double-well model different?

12.9 MEASUREMENT. (Sampling across the ridge.) Run the kit's random-walk Metropolis (`sk.metropolis`) on the posterior of the double-well model at $\delta = 0.05$ and $n = 10^4$: 64 chains started on the segment $b = 0$, steps tuned toward acceptance 0.3 for 10^4 discarded steps, then 10^5 steps. How many chains reach $|b| \geq \frac{1}{2}$, although region B has posterior

mass near 0.97? Relate the answer to $n\eta \approx 140$. Then fix the step at 0.3: why do the chains now cross, and at what cost? Try tempering.

12.10 RESEARCH. (Scaling near singular truths.) For a family of truths $c_0(\delta)$ approaching a singular truth, find conditions under which the population free energy is $\lambda_0 \log n + \Psi(n^\gamma \delta) + o(1)$, and relate them to the Newton polygon of K at the singular truth. A test case: the product model with the truth $c_0 = \delta$, where only the multiplicity changes.

12.11 RESEARCH. (Antecedents.) Gradient descent started near the segment of the double-well model stays there, while the posterior leaves at $n \approx 3000$: a Bayesian transition without a dynamical one. Design a two-parameter model with a dynamical transition without a Bayesian antecedent, or argue that none exists under stated conditions.

Notes and References

Chen et al. (2023), for whom jumps of the posterior between regions are universal in singular learning theory, build their definitions on Watanabe (2009, §7.6), "Singular learning process", Watanabe (2018, §9.4) and a 2020 talk of Watanabe's; they point to the second and to Carroll's master's thesis (Carroll 2021) for transitions in a hyperparameter of the prior or the truth instead of n, and state the local free energy formula, the switch equation, internal model selection and the Bayesian antecedent hypothesis. The position paper of Pepin Lehalleur et al. (2025) uses the same comparison of local free energies, and Hoogland et al. (2025), followed in Chapter 14, divide the training of transformers into stages at critical points of the estimated local learning coefficient (Lau et al. 2025). The blog sequence of Carroll (2023) is an informal introduction to phases and transitions. On the dynamical side, the segment of the double-well model is one of the lines of critical points that a smaller perceptron leaves in a larger one (Fukumizu and Amari 2000), and Chapter 15 compares gradient descent with the posterior.

Theorem 12.3, a localization of the sandwich of Theorem 8.4 to regions that need not contain the truth, the crossover curves of Proposition 12.8, and the double-well model with its statistic ω_n are the book's own; we have not found them in the literature, although Watanabe and Amari (2003) studied the same scaling for layered models.

TRANSLATING THE LITERATURE. Chen et al. (2023) write W_α for our regions U_α and call them phases, w^*_α for our w_α, and $F_n(W_\alpha)$ for the local free energy, with the prior multiplied by a partition of unity and renormalized, so that our $F_n(U_\alpha)$ is their $F_n(W_\alpha) + v_\alpha$; λ_α and m_α are ours. In their switch equation $n\Delta L + \Delta\lambda \log n + \Delta c = 0$, ΔL is a difference of their losses $\mathbb{E}\|x - f\|^2$, twice our $\Delta_\beta - \Delta_\alpha$ for the likelihood of their eq. (3), and c_α collects v_α and the lower-order terms; n_{cr} is the critical sample size and $\alpha \to \beta$ a local phase transition. Their toy model writes W and b for our V and c. Watanabe's 2009 book calls the free energy the stochastic complexity.

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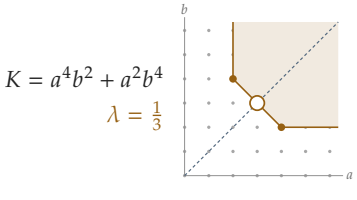
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Part IV

Deep Learning

A trained network is not a posterior, and a large one is not in the asymptotic regime of the theorems. Still, the learning coefficient can be estimated at a trained network, and it moves during training. This part says what such measurements show, and what singular learning theory does not explain.



The Local Learning Coefficient

13

Measuring a trained network

THE MODEL — $y = a\,\mathrm{relu}(bx)$, deep linear networks

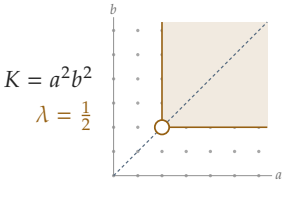
IN THIS CHAPTER. A trained network is one point w^*; how many of its parameters do the data pay for there? The local learning coefficient $\lambda(w^*)$, the learning coefficient of the loss near w^*, answers: by hand it is $\frac{1}{2}$ at every point of the ReLU neuron's cross and $\frac{3}{2}$ at two very different points of a six-parameter rank-one regression. We prove that the rank r of the Hessian sandwiches it, $\frac{r}{2} \le \lambda(w^*) \le \frac{r}{2} + \frac{d-r}{4}$, with equality on the left exactly when the true set is a manifold of dimension $d - r$, and that it is invariant under reparametrization, unlike the Hessian's eigenvalues. The estimator in use, an average over a localized posterior sampled by stochastic-gradient Langevin dynamics, is consistent with a compact localizer; we compute its finite-n bias exactly, prove that a fixed Gaussian localizer can make it converge to another point's value, and measure it on fourteen points with known values. No theorem covers the estimator as it is used on large networks.

13.1 The question: how complex is one trained network?

Training ends at a point. Stochastic gradient descent has moved the weights of a network to some w^* near a minimum of the training loss, and every later question, about generalization, compression, or what the network has learned, is asked about that one point, not about a posterior. The oldest measure of how complex the point is counts parameters, each charged $\frac{1}{2}\log n$ by the Bayesian information criterion. Finer measures look at the basin around w^*: flat minima were proposed as simple ones by Hochreiter and Schmidhuber (1997), and MacKay's effective number of parameters counts the directions that the data determine, through the eigenvalues of the Hessian of the loss (MacKay 1992).[1] All of these descend from the Laplace approximation.

THE REGULAR ANSWER · the complexity of one minimum. If w^* is a nondegenerate minimum of L in the interior of W, with $\varphi(w^*) > 0$, then the local form of Theorem 1.4 gives, for the free energy $F_n(\mathcal{B}) = -\log \int_{\mathcal{B}} \varphi(w) e^{-nL_n(w)}\,dw$ of a small ball $\mathcal{B}$ around w^* and the minimizer

[1] MacKay's count is $\sum_i h_i/(h_i + \alpha)$, for a Gaussian prior of precision α and the eigenvalues h_i of the Hessian of the data term: a direction with $h_i \gg \alpha$ counts one, a flat one zero.



$\hat{w}$ of L_n in $\mathcal{B}$,

$$F_n(\mathcal{B}) = nL_n(\hat{w}) + \frac{d}{2}\log\frac{n}{2\pi} + \frac{1}{2}\log\det\nabla^2 L(w^*) - \log\varphi(w^*) + o_p(1)\,:$$

each direction costs $\frac{1}{2}\log n$ and its curvature h_i adds $\frac{1}{2}\log h_i$. When the Hessian is degenerate the theory says nothing, and a common practice is to keep its r nonzero eigenvalues: the minimum is said to have r effective parameters and to cost $\frac{r}{2}\log n$.

Degenerate Hessians are the rule: the Hessians of trained networks have a bulk of eigenvalues near zero (Sagun et al. 2016), and for ReLU networks with more than one layer the Hessian of the population loss is degenerate at every true parameter (Wei et al. 2023). The ReLU neuron $y = a\,\mathrm{relu}(bx)$ shows what goes wrong. At $w^* = (0,0)$, the unit switched off, the Hessian is zero: the rank charges nothing, the parameter count charges $\log n$, and the truth (Chapter 3) is $\frac{1}{2}\log n - \log\log n$. At $w^* = (1,0)$ the rank, 1, happens to be right, but the curvature that Laplace also needs is not a property of the function: the weights $(a/s, sb)$ give the same function for every $s > 0$ and move the nonzero eigenvalue from $\frac{1}{2}$ to $1/(2s^2)$.[2]

Singular learning theory replaces the count by a volume. The *local learning coefficient* $\lambda(w^*)$ is the learning coefficient of the loss restricted to a small neighborhood of w^*, the exponent with which the volume of nearly optimal parameters near w^* shrinks; its multiplicity $m(w^*)$ is the power of the accompanying logarithm. It replaces $\frac{d}{2}$ in the free energy of the neighborhood, $F_n(\mathcal{B}) = nL_n(w^*) + \lambda(w^*)\log n - (m(w^*)-1)\log\log n + O_p(1)$, as Lau et al. (2025) sketch by following Watanabe's proof (their Appendix B). They introduced the quantity for deep learning, with an estimator that they validated on deep linear networks with up to 10^8 parameters. This chapter asks two questions. What is $\lambda(w^*)$ at the points of our models, and what does the Hessian say about it? And what does the estimator estimate, and how well?

[2] Dinh et al. (2017) used this rescaling to make the minima of rectifier networks arbitrarily sharp without changing the function they compute.

13.2 By hand: the cast at a point

We compute local learning coefficients where the Hessian sees everything and where it sees nothing. The volume is the tool, as in Chapter 4: if the volume $V(\varepsilon)$ of the parameters near w^* with loss below ε behaves like $\varepsilon^\lambda(\log\frac{1}{\varepsilon})^{m-1}$, then $\lambda(w^*) = \lambda$ and $m(w^*) = m$ (Definition 13.1 makes this precise).

COMPUTATION 13.1 · THE RELU NEURON AT THE ORIGIN AND ON ITS CROSS

1 For $b \geq 0$, $\mathrm{relu}(bx) = b\,\mathrm{relu}(x)$, and for $b < 0$, $\mathrm{relu}(bx) = |b|\,\mathrm{relu}(-x)$. Since $\mathbb{E}[\mathrm{relu}(x)^2] = \mathbb{E}[x^2\mathbf{1}\{x > 0\}] = \frac{1}{2}$ for $x \sim \mathcal{N}(0,1)$, $\mathbb{E}[\mathrm{relu}(bx)^2] = b^2/2$, and the truth $y =$ noise gives $K(a,b) = \frac{1}{2}\mathbb{E}[(a\,\mathrm{relu}(bx))^2] = \frac{1}{4}a^2b^2$, the product model's loss

up to a factor $\frac{1}{2}$. The true set is the cross $\{ab = 0\}$.

2 The Hessian is $\nabla^2 K(a,b) = \begin{pmatrix} b^2/2 & ab \\ ab & a^2/2 \end{pmatrix}$. At the origin it is 0, so $r = 0$. At $(a^*, 0)$ with $a^* \neq 0$ it is $\mathrm{diag}(0, a^{*2}/2)$, so $r = 1$; likewise $r = 1$ at $(0, b^*)$.

3 At the origin, with the uniform prior on $[-1,1]^2$, $V(\varepsilon) = \mathbb{P}(|ab| < 2\sqrt{\varepsilon}) = s(1+\log\frac{1}{s})$ with $s = 2\sqrt{\varepsilon}$, because $\mathbb{P}(|a||b| \le s) = s(1-\log s)$ for independent uniforms on $[-1,1]$ (Chapter 3). So $V(\varepsilon) \asymp \varepsilon^{1/2}\log\frac{1}{\varepsilon}$: $\lambda(0,0) = \frac{1}{2}$ and $m(0,0) = 2$. The same exponents hold on every ball around the origin, because K is homogeneous (Lemma 13.6).

4 Near $(a^*, 0)$, on the box $\{|a - a^*| \le |a^*|/2,\ |b| \le \delta\}$, the factor $a^2/4$ lies between $a^{*2}/16$ and $9a^{*2}/16$, so $\{K < \varepsilon\}$ is the set $|b| < 2\sqrt{\varepsilon}/|a|$, of area $\int 4\sqrt{\varepsilon}/|a|\,\mathrm{d}a \asymp \varepsilon^{1/2}$: $\lambda(a^*, 0) = \frac{1}{2}$ with $m = 1$, and likewise at $(0, b^*)$. The bounds of Theorem 13.3 below, $[0, \frac{1}{2}]$ at the origin and $[\frac{1}{2}, \frac{3}{4}]$ on the arms, are attained at the top and the bottom; the parameter count $\frac{d}{2} = 1$ is wrong at both.

COMPUTATION 13.2 · RANK-ONE REGRESSION WITH $M = N = 3$ AT TWO POINTS

1 The model is $y = b\,(a^\top x)$ with $a \in \mathbb{R}^3$, $b \in \mathbb{R}^3$, $x \sim \mathcal{N}(0, I_3)$, and the truth is pure noise. Then $K(a,b) = \frac{1}{2}\mathbb{E}|b\,a^\top x|^2 = \frac{1}{2}|b|^2\mathbb{E}[(a^\top x)^2] = \frac{1}{2}|a|^2|b|^2$, with $d = 6$. The true set $\{a = 0\} \cup \{b = 0\}$ is two 3-planes in $\mathbb{R}^6$ meeting at the origin.

2 The Hessian has blocks $\partial^2_{aa} K = |b|^2 I_3$, $\partial^2_{bb} K = |a|^2 I_3$, $\partial^2_{ab} K = 2ab^\top$. At the origin it vanishes: $r = 0$. At $(a^*, 0)$ with $a^* \neq 0$ it is $\mathrm{diag}(0, |a^*|^2 I_3)$: $r = 3$.

3 At the origin, take Lebesgue measure on the product of unit balls and polar coordinates $\rho = |a|$, $\sigma = |b|$, so that $\{K < \varepsilon\} = \{\rho\sigma < s\}$ with $s = \sqrt{2\varepsilon}$:

$$V(\varepsilon) = \int_0^1 4\pi\rho^2 \,\frac{4\pi}{3}\min\Big(1, \frac{s}{\rho}\Big)^3 \mathrm{d}\rho = \frac{16\pi^2}{3}\Big(\frac{s^3}{3} + s^3\log\frac{1}{s}\Big).$$

So $V(\varepsilon) \asymp \varepsilon^{3/2}\log\frac{1}{\varepsilon}$: $\lambda(0) = \frac{3}{2}$ and $m(0) = 2$, the value $\min(M,N)/2$ of Theorem 7.2.

4 Near $(a^*, 0)$ the factor $|a|^2$ is pinched between positive constants, so $K \asymp |b|^2$, nondegenerate in the three coordinates of b with those of a free: $V(\varepsilon) \asymp \varepsilon^{3/2}$, $\lambda = \frac{3}{2}$, $m = 1$. The Hessian bounds, $[0, \frac{3}{2}]$ at the origin and $[\frac{3}{2}, \frac{9}{4}]$ at $(a^*, 0)$, are attained at the top and the bottom; the parameter count $\frac{d}{2} = 3$ is twice the truth at both.

5 For general M, N the same computations give $\lambda(0) = \min(M,N)/2$, $\lambda(a^*, 0) = M/2$ and $\lambda(0, b^*) = N/2$. When $N < M$ the origin is strictly more singular than the points $(a^*, 0)$ of the

same true set; when $N \geq M$ the two have the same λ, although their Hessian ranks are still 0 and M.

What one had to notice is that the Hessian sees only the quadratic part of K, while the volume sees everything. At $(a^*, 0)$ the quadratic part controls the directions it constrains and the others are exactly flat, so the rank is the answer; at the origin the quadratic part vanishes, the quartic a^2b^2 decides, and the rank says nothing beyond $\lambda \geq 0$. The global coefficient of Chapter 7 is the minimum of these local ones over the true set (Proposition 5.4), and a trained network sits at one point of that set, not necessarily the minimizing one.

13.3 The theorem: the Hessian sandwich

Throughout this section w^* is a local minimum of the population loss L, and $K(w) = L(w) - L(w^*)$, which is the book's K when w^* is a true parameter and in general is nonnegative on a neighborhood U of w^*. Balls are closed and centered at w^*; for a ball $\mathcal{B} \subset U$ and a prior density φ,

$$V_{\mathcal{B}}(\varepsilon) = \int_{\mathcal{B} \cap \{K < \varepsilon\}} \varphi(w)\, \mathrm{d}w.$$

We write $f \asymp g$ for two positive functions of ε when $c\, g \leq f \leq C\, g$ for some constants $0 < c \leq C$ and all small $\varepsilon > 0$.

Definition 13.1 (Local learning coefficient). Let φ be continuous on U with $\varphi(w^*) > 0$. If there are a number $\lambda \geq 0$, an integer $m \geq 1$ and a radius $\rho_0 > 0$ such that

$$V_{\mathcal{B}}(\varepsilon) \asymp \varepsilon^{\lambda} \Big(\log \frac{1}{\varepsilon}\Big)^{m-1} \qquad (\varepsilon \to 0)$$

for every ball $\mathcal{B}$ of radius at most ρ_0, then $\lambda(w^*) = \lambda$ is the *local learning coefficient* of K at w^* and $m(w^*) = m$ its *local multiplicity*.

The pair is unique when it exists, because $\varepsilon^{\lambda}(\log \frac{1}{\varepsilon})^{m-1} \asymp \varepsilon^{\lambda'}(\log \frac{1}{\varepsilon})^{m'-1}$ forces $\lambda = \lambda'$ and then $m = m'$. It does not depend on φ, which is pinched between two positive constants on small balls, so Lebesgue measure gives the same pair; Lau et al. (2025) use Lebesgue measure on one ball on which w^* minimizes L (their Definition 1).

Remark 13.2 (Analytic losses). If K is real analytic near w^*, the pair exists, $\lambda(w^*)$ is rational, $V_{\mathcal{B}}(\varepsilon) = c_{\mathcal{B}}\, \varepsilon^{\lambda}(\log \frac{1}{\varepsilon})^{m-1}(1 + o(1))$, and $-\lambda(w^*)$ is the largest pole of the local zeta function $\int_{\mathcal{B}} K^z \varphi$, of order $m(w^*)$ (Theorem 4.6). This is resolution of singularities (Theorem A.8) near w^*: on one resolution over a fixed ball the exponent of $V_{\mathcal{B}}$ takes finitely many values $(h_j + 1)/(2k_j)$ and can only grow as $\mathcal{B}$ shrinks, so it is constant for small balls. The global learning coefficient is the minimum of the local ones over the true set (Lin 2017, Proposition 2.5); so every point of $\{K = 0\}$ in a small ball has local coefficient at least $\lambda(w^*)$, the extra condition that Lau et al. impose for the

local free energy (their Appendix B). The real log canonical threshold of K at w^* is the same number as $\lambda(w^*)$.

The main result of the chapter compares $\lambda(w^*)$ with the rank of the Hessian.

Theorem 13.3 (The Hessian sandwich). *Let K be four times continuously differentiable on a neighborhood of $w^* \in \mathbb{R}^d$, with $K \geq 0$ there and $K(w^*) = 0$, let φ be continuous with $\varphi(w^*) > 0$, and let r be the rank of $\nabla^2 K(w^*)$. There are a radius $\rho_0 > 0$ and constants $0 < c \leq C$ such that for every ball $\mathcal{B}$ of radius at most ρ_0 and all sufficiently small $\varepsilon > 0$*

$$c\,\varepsilon^{\frac{r}{2}+\frac{d-r}{4}} \;\leq\; V_{\mathcal{B}}(\varepsilon) \;\leq\; C\,\varepsilon^{\frac{r}{2}}. \tag{13.1}$$

Consequently, if $\lambda(w^*)$ exists, in particular if K is analytic, then

$$\frac{r}{2} \;\leq\; \lambda(w^*) \;\leq\; \frac{r}{2} + \frac{d-r}{4},$$

and if $\lambda(w^*) = \frac{r}{2}$, then $m(w^*) = 1$.

Proof. The plan is to bound the sublevel set $\{K < \varepsilon\}$ from outside by a slab of width $\sqrt{\varepsilon}$ in the directions the Hessian sees, and from inside by a box of widths $\sqrt{\varepsilon}$ and $\varepsilon^{1/4}$. Translate w^* to the origin and rotate so that $H = \nabla^2 K(0) = \operatorname{diag}(h_1, \dots, h_r, 0, \dots, 0)$ with $h_1, \dots, h_r > 0$; the Hessian is positive semidefinite because 0 is a local minimum of K, and rotations about the center preserve balls and volumes. Write $w = (u, z)$ with $u \in \mathbb{R}^r$ and $z \in \mathbb{R}^{d-r}$, let $h = \min_i h_i$ and $h^* = \max_i h_i$, and let ω_k be the volume of the unit ball of $\mathbb{R}^k$. If $r = 0$ there is nothing to prove on the right of (13.1), since $V_{\mathcal{B}}(\varepsilon) \leq \int_{\mathcal{B}} \varphi$; if $r = d$ there are no z-coordinates, and the argument below applies with them omitted.

The upper bound on $V_{\mathcal{B}}$. The block $\partial^2_{uu} K$ is continuous and equals $\operatorname{diag}(h_1, \dots, h_r)$ at 0, so there is $\delta > 0$ such that on the cube $Q = [-\delta, \delta]^d$ the prior φ is continuous and positive, $K \geq 0$, and $\partial^2_{uu} K(u, z) \geq \frac{h}{2} I_r$. Fix $z \in [-\delta, \delta]^{d-r}$ and let $k_z(u) = K(u, z)$ on the convex set $Q_u = [-\delta, \delta]^r$. Taylor's formula with integral remainder and the bound on $\partial^2_{uu} K$ give $k_z(u') \geq k_z(u) + \nabla k_z(u) \cdot (u' - u) + \frac{h}{4}|u' - u|^2$ for $u, u' \in Q_u$. Let u_z minimize k_z on the compact set Q_u. Along the segment from u_z to $u' \in Q_u$ the function k_z is smallest at u_z, so $\nabla k_z(u_z) \cdot (u' - u_z) \geq 0$, and therefore, since $k_z \geq 0$,

$$k_z(u') \;\geq\; k_z(u_z) + \tfrac{h}{4}|u' - u_z|^2 \;\geq\; \tfrac{h}{4}|u' - u_z|^2.$$

Hence the slice $\{u \in Q_u : K(u, z) < \varepsilon\}$ lies in the ball of radius $2\sqrt{\varepsilon/h}$ around u_z, of volume $\omega_r (4\varepsilon/h)^{r/2}$. Integrating over z (Fubini) and bounding φ by its maximum $\varphi_{\max}$ on Q, every ball $\mathcal{B} \subset Q$ satisfies

$$V_{\mathcal{B}}(\varepsilon) \;\leq\; \varphi_{\max} (2\delta)^{d-r} \omega_r (4/h)^{r/2}\, \varepsilon^{r/2}.$$

The cubic term vanishes on the kernel. By Taylor's theorem there are $\delta' \in (0, 1]$ and C_4 with $K(w) = \frac{1}{2} w^\top H w + T(w) + R(w)$ and $|R(w)| \leq C_4 |w|^4$ for $|w| \leq \delta'$, where T is the cubic form of third derivatives at 0. Fix z and let $s \to 0$: the

quadratic term vanishes on $(0,z)$, so $0 \le K(0,sz) = s^3T(0,z) + R(0,sz)$, and dividing by $|s|^3$ and letting $s \to 0$ from both sides gives $T(0,z) \ge 0$ and $-T(0,z) \ge 0$. So $T(0,z) = 0$ for every z.

The lower bound on $V_{\mathcal{B}}$. By the last step every monomial of T contains a coordinate of u; the monomials with exactly one such coordinate are bounded by $C_3|u||z|^2$, and those with two or three by $C_3|u|^2|w|$, for a constant C_3. For $|w| \le \delta' \le 1$ we use $|u||z|^2 \le \frac{1}{2}(|u|^2 + |z|^4)$, $|u|^2|w| \le |u|^2$ and $|w|^4 \le 2|u|^4 + 2|z|^4 \le 2|u|^2 + 2|z|^4$; with $A = \frac{h^*}{2} + \frac{3}{2}C_3 + 2C_4$, which we may take positive, this gives

$$K(u,z) \;\le\; \tfrac{h^*}{2}|u|^2 + C_3|u||z|^2 + C_3|u|^2|w| + C_4|w|^4 \;\le\; A\big(|u|^2 + |z|^4\big).$$

On the set $P_\varepsilon = \{|u| < (\varepsilon/2A)^{1/2},\ |z| < (\varepsilon/2A)^{1/4}\}$ this gives $K < \varepsilon$, and P_ε lies in $\mathcal{B}$ and in $\{|w| \le \delta'\}$ once ε is small. With $\varphi_{\min} > 0$ the minimum of φ on Q,

$$V_{\mathcal{B}}(\varepsilon) \;\ge\; \varphi_{\min}\,\omega_r\,\omega_{d-r}\,(2A)^{-\frac{r}{2}-\frac{d-r}{4}}\;\varepsilon^{\frac{r}{2}+\frac{d-r}{4}}.$$

Taking $\rho_0 = \delta$ proves (13.1).

The consequences. Suppose $V_{\mathcal{B}}(\varepsilon) \asymp \varepsilon^\lambda(\log\frac{1}{\varepsilon})^{m-1}$. If $\lambda < \frac{r}{2}$, then $V_{\mathcal{B}}(\varepsilon)/\varepsilon^{r/2} \ge c'\varepsilon^{\lambda-r/2} \to \infty$, against the upper bound; if $\lambda = \frac{r}{2}$ and $m \ge 2$, then $V_{\mathcal{B}}(\varepsilon)/\varepsilon^{r/2} \ge c'\log\frac{1}{\varepsilon} \to \infty$, against it again; and if $\lambda > \frac{r}{2} + \frac{d-r}{4}$, then $V_{\mathcal{B}}(\varepsilon)/\varepsilon^{r/2+(d-r)/4} \le C'\varepsilon^{\lambda-r/2-(d-r)/4}(\log\frac{1}{\varepsilon})^{m-1} \to 0$, against the lower bound. ∎

Both bounds are attained, as Computations 13.1 and 13.2 showed: the lower one on the arms of the cross and at $(a^*,0)$ in rank-one regression, the upper one at the origin of both models, with multiplicity 2, and at the origin of the quartic model $y = ax_1 + b^2x_2$ of Chapter 5 ($r = 1$, $d = 2$, $\lambda = \frac{3}{4}$, $m = 1$). The lower bound on λ needs only two derivatives (Exercise 13.1). The upper bound needs four: killing the cubic term on the kernel is what gives a quarter per kernel direction; with three derivatives the argument gives only $\frac{r}{2} + \frac{d-r}{3}$, and a kernel direction can then contribute more than a quarter (Exercise 13.2).[3] Our proof works with volumes, needs no splitting lemma and no analyticity for (13.1), and so applies to a C^4 loss, such as the ReLU neuron's polynomial K, whenever its exponents exist.

When is the lower bound attained? The splitting lemma (Theorem A.9) answers this. In its real-analytic form it says: if f is analytic near $0 \in \mathbb{R}^d$ with $f(0) = 0$, $\nabla f(0) = 0$ and $\operatorname{rank}\nabla^2 f(0) = r$, then there is an analytic diffeomorphism ψ of a neighborhood of 0 with $\psi(0) = 0$ and $f(\psi(u,z)) = \sum_{i\le r} \pm u_i^2 + g(z)$, where g is analytic near $0 \in \mathbb{R}^{d-r}$ and vanishes at 0 with its first and second derivatives.

Corollary 13.4 (Minimally singular points). *Let K be analytic near w^*, with $K \ge 0$ and $K(w^*) = 0$, and let $r = \operatorname{rank}\nabla^2 K(w^*)$. Then $\lambda(w^*) = \frac{r}{2}$ if and only if, near w^*, the set $\{K = 0\}$ is an analytic submanifold of dimension $d - r$. In that case $m(w^*) = 1$.*

Proof. The plan is to move to the coordinates of the splitting lemma, where the question becomes whether g vanishes. Apply the lemma to $f = K$ at w^*. The Hessian of $K \circ \psi$ at 0, $D\psi(0)^\top \nabla^2 K(w^*) D\psi(0)$, is positive semidefinite

[3] The upper bound, in the language of ideals and with the rank of a Jacobian in place of the Hessian, is Corollary 4.15 of Lin (2017), a paper first posted to arXiv in 2010; see also Lin (2011). Bushnaq (2024) rediscovered it in a blog post of 8 August 2024, first with 3 in the denominator; an edit dated 16 August credits Dmitry Vaintrob with the improvement to 4. The post calls the lower bound known; a footnote of Wang et al. (2025, Appendix D.3) sketches it through the splitting lemma.

and equals $\mathrm{diag}(\pm 2, \dots, \pm 2, 0, \dots, 0)$, so every sign is $+$: $K \circ \psi = |u|^2 + g(z)$ on a box $Q_u \times Q_z$ around 0, with $g(z) = K(\psi(0, z)) \geq 0$. By Proposition 13.5 below we may compute λ and m for $K \circ \psi$ with Lebesgue measure on such boxes; $\{K = 0\}$ corresponds to $\{u = 0,\ g(z) = 0\}$.

If $g \equiv 0$ on a neighborhood of 0, then $\{K \circ \psi = 0\}$ is the plane $\{u = 0\}$ near 0, whose image under ψ is an analytic submanifold of dimension $d - r$, and $V(\varepsilon) = \omega_r \varepsilon^{r/2} \mathrm{vol}(Q_z)$ for small ε, so $\lambda = \frac{r}{2}$ and $m = 1$. Otherwise the analytic function g is not identically zero on the connected box Q_z, so its zero set has measure zero (Definition A.7) and contains no open piece of the plane, and $V_g(\varepsilon) = \mathrm{vol}\{z \in Q_z : g(z) < \varepsilon\}$ decreases to 0. Since $|u|^2 < \varepsilon - g(z)$ forces $g(z) < \varepsilon$, we get $V(\varepsilon) \leq \omega_r \varepsilon^{r/2} V_g(\varepsilon)$, so $V(\varepsilon)/\varepsilon^{r/2} \to 0$ and $\lambda(w^*) \neq \frac{r}{2}$; and $\{K \circ \psi = 0\} = \{0\} \times \{g = 0\}$ is not a $(d - r)$-dimensional submanifold near 0. ∎

The lower bound is attained exactly at the *minimally singular* points of Wei et al. (2023), where the loss is a nondegenerate quadratic transversally to a smooth true set (Proposition 5.5); Chapter 14 uses them for the critical points of a toy model of superposition. The next proposition says why the local learning coefficient, unlike the Hessian's eigenvalues, is a property of the function the network computes.

Proposition 13.5 (Reparametrization invariance). *Let ψ be a C^1 diffeomorphism from a neighborhood of w' onto a neighborhood of $w^* = \psi(w')$, and let $K' = K \circ \psi$. Then $\lambda(K', w')$ and $m(K', w')$ exist if and only if $\lambda(w^*)$ and $m(w^*)$ do, and they are equal. If ψ is C^2, the Hessians satisfy $\nabla^2 K'(w') = D\psi(w')^\top \nabla^2 K(w^*) D\psi(w')$: they have the same rank, but not in general the same eigenvalues.*

Proof. For a small ball $\mathcal{B}'$ around w', the set $\psi(\mathcal{B}')$ contains a ball $\mathcal{B}_1$ and lies in a ball $\mathcal{B}_2$ around w^*, both as small as we like. By the change of variables formula, $V'_{\mathcal{B}'}(\varepsilon) = \int_{\psi(\mathcal{B}') \cap \{K < \varepsilon\}} \varphi'(\psi^{-1}(w)) |\det D\psi^{-1}(w)| \, \mathrm{d}w$, and on the compact closure the integrand lies between two positive multiples of $\varphi(w)$, for priors φ', φ continuous and positive at the two points. So $V'_{\mathcal{B}'}$ lies between positive multiples of $V_{\mathcal{B}_1}$ and $V_{\mathcal{B}_2}$, both $\asymp \varepsilon^{\lambda(w^*)} (\log \frac{1}{\varepsilon})^{m(w^*)-1}$ if these exist; the converse follows with ψ^{-1}. Differentiating $K \circ \psi$ twice at a critical point gives the Hessian formula, and congruent matrices have the same rank. ∎

For the ReLU neuron the rescaling $\psi_s(a, b) = (a/s, sb)$ of Section 13.1 is such a diffeomorphism, with $K \circ \psi_s = K$: at $(a^*/s, 0)$ the rank, 1, and $\lambda = \frac{1}{2}$ stay, while the eigenvalue $a^{*2}/(2s^2)$ takes every positive value. The invariance is Appendix C of Lau et al. (2025).[4]

The coefficients of Chapter 7 are global, minima over the true set, and an estimator measures a local one. At the center of a homogeneous loss the two coincide.

Lemma 13.6 (Homogeneous losses). *If $K(w^* + s(w - w^*)) = s^k K(w)$ for all $s > 0$ and some $k > 0$, then the exponents of $V_{\mathcal{B}}$ are the same for all balls*

[4] They also checked their estimator on small ReLU networks rescaled over eight orders of magnitude, with a preconditioner set by hand (their Appendix O).

around w^*, and $\lambda(w^*)$, $m(w^*)$ are the learning coefficient and multiplicity of K for every bounded, compactly supported prior that is positive and continuous near w^*.

PROOF. Take $w^* = 0$ and Lebesgue measure. The substitution $w = \rho v$ gives $\mathrm{vol}(\mathcal{B}_\rho \cap \{K < \varepsilon\}) = \rho^d \,\mathrm{vol}(\mathcal{B}_1 \cap \{K < \varepsilon\rho^{-k}\})$ for the balls $\mathcal{B}_\rho$ of radius ρ, which therefore share their exponents; and such a prior lies between positive multiples of the indicators of a small and a large ball. ■

For a deep linear network $y = W_L \cdots W_1 x$ with zero truth and $x \sim \mathcal{N}(0, I)$, the loss $K = \frac{1}{2}\|W_L \cdots W_1\|_F^2$ is homogeneous of degree $2L$, so its local coefficient at 0 is the global one. Aoyagi (2024) computed the global coefficient for every truth; Theorem 7.6 of Chapter 7 states her theorem in the form of Aoyagi (2025, Definition 4), with the multiplicity $m = p(\ell - p) + 1$, and we take λ and m from it. The restatement by Lau et al. (their Theorem 1) leaves this integer (their a) undefined and omits the multiplicity. Widths $(N, 1, M)$ give rank-one regression, $\lambda = \min(M, N)/2$ with $m = 2$ exactly when $M = N$ (Theorem 7.2), and widths $(1, \ldots, 1)$ give the chain, $\lambda = \frac{1}{2}$ with $m = L$; our implementation of the theorem reproduces both.[5] At points of the true set other than the most degenerate ones the local coefficient can exceed the global one, so the experiments of Section 13.5 sit at the origin, or at a point that Hitchcock and Hoogland (2025), following Pepin Lehalleur and Rimányi (2024), show to be most degenerate: for equal widths, every layer the partial identity with r ones.

[5] The theorem uses a set Γ of layers with prescribed properties. In 3000 random architectures with up to six layers of widths up to nine, such a set existed and was unique whenever one was needed.

13.4 The theorem: what the estimator estimates

Volumes in 10^8 dimensions cannot be computed. The estimator of Lau et al. (2025) replaces the volume by an average energy, the localized form of WBIC (Theorem 11.6).

Definition 13.7 (The LLC estimator; Lau et al. 2025). Let w^* be a local minimum of L, $\beta = 1/\log n$ and $\gamma > 0$. The *localized tempered posterior* is the probability density on $\mathbb{R}^d$

$$p(w \mid w^*, \beta, \gamma) \;\propto\; \exp\Big\{-n\beta L_n(w) - \frac{\gamma}{2}|w - w^*|^2\Big\},$$

and the *LLC estimator* is $\hat{\lambda}(w^*) = n\beta\big(\mathbb{E}_{w|w^*,\beta,\gamma}[L_n(w)] - L_n(w^*)\big)$. To compute the expectation, stochastic-gradient Langevin dynamics (SGLD, Section A.7) starts at $w^{(0)} = w^*$ and iterates, for $k = 0, 1, 2, \ldots$ and independent $\xi_k \sim \mathcal{N}(0, I_d)$,

$$w^{(k+1)} = w^{(k)} - \frac{\epsilon}{2}\Big(n\beta\,\nabla L_{B_k}(w^{(k)}) + \gamma(w^{(k)} - w^*)\Big) + \sqrt{\epsilon}\,\xi_k,$$

with the minibatch loss $L_B(w) = -\frac{1}{|B|}\sum_{i\in B}\log p(X_i \mid w)$ and step size ϵ; the data are shuffled once and cut into consecutive minibatches B_k. After a burn-in the average of $L_n(w^{(k)})$ replaces the expectation; their Algorithm 1 averages the minibatch losses $L_{B_k}(w^{(k)})$ already computed for the gradients instead.

We keep Lau et al.'s conventions throughout: $n\beta = n/\log n$ and a localizing drift $-\frac{\epsilon}{2}\gamma(w - w^*)$.[6] Later work changed both, so numbers from different papers are not directly comparable: version 1.3 of the `devinterp` library (van Wingerden et al. 2024) sets $n\beta = |B|/\log|B|$ by default, with $|B|$ the minibatch size, Wang et al. (2025) use $n\beta = 30$, and Hoogland et al. (2025) write the localizing drift of their SGLD step with $\gamma/2$ in place of γ.

[6] The step size ϵ is not the tolerance ε of a volume.

What does $\hat{\lambda}(w^*)$ estimate? Replace the empirical loss by the population loss, write $t = n\beta$, and replace the Gaussian by a weight $\Psi \geq 0$, for instance the indicator of a small ball times a prior. The number $t\,\mathbb{E}_t[K]$, with $\mathbb{E}_t[\cdot]$ the expectation under the density proportional to $e^{-tK}\Psi$, is the *population skeleton* of the estimator, the subject of Proposition 11.4, whose proof uses only the volume function; we record the leading term.

Lemma 13.8 (The skeleton with a compact localizer). *Let $\Psi \geq 0$ be bounded, measurable and compactly supported, $K \geq 0$ continuous on its support, and suppose $V_\Psi(\varepsilon) = \int_{\{K<\varepsilon\}} \Psi\,\mathrm{d}w = c\,\varepsilon^\lambda(\log\frac{1}{\varepsilon})^{m-1}(1 + o(1))$ as $\varepsilon \to 0$, with $c, \lambda > 0$. Then $t\,\mathbb{E}_t[K] \to \lambda$ as $t \to \infty$.*

Proof. The plan is to write both expectations through V_Ψ and pass to the limit under the integral. Let μ be the image of $\Psi(w)\,\mathrm{d}w$ under K, so that $V_\Psi(\varepsilon) = \mu([0, \varepsilon))$. Write e^{-tx} and xe^{-tx} as the integrals of te^{-ts} and $(ts - 1)e^{-ts}$ over $s > x$, exchange the order of integration and substitute $s = \sigma/t$:

$$t\,\mathbb{E}_t[K] = \frac{t\int xe^{-tx}\,\mathrm{d}\mu(x)}{\int e^{-tx}\,\mathrm{d}\mu(x)} = \frac{\int_0^\infty(\sigma - 1)e^{-\sigma}V_\Psi(\sigma/t)\,\mathrm{d}\sigma}{\int_0^\infty e^{-\sigma}V_\Psi(\sigma/t)\,\mathrm{d}\sigma}.$$

Divide both integrals by $V_\Psi(1/t)$; for fixed σ the ratio $V_\Psi(\sigma/t)/V_\Psi(1/t)$ tends to σ^λ. Choose $\varepsilon_0 < 1$ such that $V_\Psi(\varepsilon)$ is within a factor 2 of $c\,\varepsilon^\lambda(\log\frac{1}{\varepsilon})^{m-1}$ for $\varepsilon \leq \varepsilon_0$. Then the ratio is at most 1 for $\sigma \leq 1$, because V_Ψ is nondecreasing, at most $4\sigma^\lambda$ for $1 < \sigma \leq \varepsilon_0 t$, because $\log(t/\sigma) \leq \log t$, and at most $\mu(\mathbb{R})/V_\Psi(1/t) \leq Ct^\lambda$ beyond, where its integral against $(1 + \sigma)e^{-\sigma}$ tends to 0. By dominated convergence the two integrals, divided by $V_\Psi(1/t)$, tend to $\Gamma(\lambda + 2) - \Gamma(\lambda + 1) = \lambda\Gamma(\lambda + 1)$ and $\Gamma(\lambda + 1)$. ∎

With Ψ the indicator of a small ball and K analytic, the limit is $\lambda(w^*)$, because the volume then has the required form (Remark 13.2). The convergence is slow when $m \geq 2$: for the expansion of the volume in powers of $\log\frac{1}{\varepsilon}$ that analytic losses have, Proposition 11.4 gives $t\,\mathbb{E}_t[K] = \lambda - \frac{m-1}{\log t} + o(\frac{1}{\log t})$, a bias of order $1/\log n$ that no better sampler removes. On the ReLU neuron the skeleton can be computed exactly.

COMPUTATION 13.3 · THE SKELETON AT THE RELU NEURON'S ORIGIN

1 Localize with the book's prior box $[-1, 1]^2$. By symmetry $Z(t) = \frac{1}{4}\iint_{[-1,1]^2} e^{-ta^2b^2/4}\,\mathrm{d}a\,\mathrm{d}b = \int_0^1\int_0^1 e^{-ta^2b^2/4}\,\mathrm{d}b\,\mathrm{d}a$, and $t\,\mathbb{E}_t[K] = -t\frac{\mathrm{d}}{\mathrm{d}t}\log Z(t)$.

2 The inner integral is $\int_0^1 e^{-ta^2b^2/4}\,\mathrm{d}b = \frac{\sqrt{\pi}}{a\sqrt{t}}\operatorname{erf}\!\left(\frac{a\sqrt{t}}{2}\right)$, and the substitution $v = a\sqrt{t}/2$ gives $Z(t) = \sqrt{\pi/t}\;G(\sqrt{t}/2)$ with $G(X) = \int_0^X \operatorname{erf}(v)\frac{\mathrm{d}v}{v}$.

3 For large X, $G(X) = \log X + \int_0^\infty \big(\operatorname{erf}(v) - \mathbf{1}\{v > 1\}\big)\frac{\mathrm{d}v}{v} + \int_X^\infty \operatorname{erfc}(v)\frac{\mathrm{d}v}{v}$. Integrating by parts, the constant is $-\frac{2}{\sqrt{\pi}}\int_0^\infty \log(v)e^{-v^2}\,\mathrm{d}v = -\frac{1}{2\sqrt{\pi}}\Gamma'(\tfrac12) = \frac{\gamma_E}{2} + \log 2$, by $\Gamma'(\tfrac12)/\Gamma(\tfrac12) = -\gamma_E - 2\log 2$ (γ_E is Euler's constant), and the last integral is $O(e^{-X^2})$.

4 So $Z(t) = \frac{\sqrt{\pi}}{2\sqrt{t}}(\log t + \gamma_E) + O(e^{-t/4})$, with a derivative of the same exponentially small order, and

$$t\,\mathbb{E}_t[K] = \frac{1}{2} - \frac{1}{\log t + \gamma_E} + O(e^{-t/4}).$$

5 At $n = 10^4$, $t = n/\log n = 1086$ and the skeleton is 0.368, 26% below $\lambda = \frac12$; at $n = 10^6$ it is 0.415, and at $n = 10^{12}$ still 0.460. The leading correction is $-\frac{m-1}{\log t}$ with $m = 2$, as Proposition 11.4 says.

6 At $(1, 0)$, with the box $[\frac12, \frac32] \times [-\frac12, \frac12]$, which avoids the origin, the inner integral is $\frac{2\sqrt{\pi}}{a\sqrt{t}}\operatorname{erf}\!\left(\frac{a\sqrt{t}}{4}\right)$, so $Z(t) \propto t^{-1/2}(1 + O(e^{-t/64}))$ and the skeleton is $\frac12$ up to exponentially small terms. Same λ, multiplicity 1: no logarithmic bias.

What one had to notice is that the posterior over the cross is a mixture over scales, as in Chapter 2: the factor $\log t + \gamma_E$ counts the scales of $|a|$ the posterior can occupy, and the estimator pays for them with a bias that decays like $1/\log n$, even with an exact sampler. With noisy data there is a second bias: at the true parameter each identifiable direction contributes $-\beta Z^2/2$ with $Z \sim \mathcal{N}(0,1)$, on average $-1/(2\log n)$ (Exercise 13.4). Both biases show in the normal-crossing test of the first arXiv version of Lau et al. (2025): for $y = w_1w_2x$ with $n = 1000$ ($\lambda = \frac12$, $m = 2$; noise variance $\frac14$, a Gaussian localizer with $\gamma = 1$ at the origin) it reports means of 0.29 to 0.33 from MCMC and SGLD (its Table 2), and the exact estimator with the same localizer, a one-dimensional integral over $c = w_1w_2$, has mean 0.32 over 2000 data sets (`13-llc-ideal.py`): the shortfall is the estimator's, not the sampler's.

With data and a compact localizer, Watanabe's theorem on WBIC applies to the problem restricted to the ball, as Hitchcock and Hoogland (2025) note (their Theorem 3.2).

Theorem 13.9 (Watanabe 2013, Theorem 4, applied to a ball). *Let $\mathcal{B}$ be a small closed ball around w^* on which the model, the truth and the restricted prior satisfy Watanabe's fundamental conditions (an analytic log-likelihood ratio with six moments, relatively finite variance, an analytic*

times a smooth positive prior, w^* an interior minimizer of L on $\mathcal{B}$), and let $\hat{\lambda}_{\mathcal{B}} = n\beta(\mathbb{E}^{\beta}_{\mathcal{B}}[L_n] - L_n(w^*))$ with $\beta = 1/\log n$ and the tempered posterior restricted to $\mathcal{B}$. Then there are random variables U_n with $\mathbb{E}[U_n] = 0$, converging in law to a Gaussian random variable and equal to 0 when the parity of the problem is odd, such that

$$\hat{\lambda}_{\mathcal{B}} = \lambda(w^*) + U_n\sqrt{\frac{\lambda(w^*)}{2\log n}} + O_p\Big(\frac{1}{\log n}\Big).$$

In particular $\hat{\lambda}_{\mathcal{B}} \to \lambda(w^*)$ in probability.

This is Theorem 11.6 divided by $\log n$, for the restricted problem, whose learning coefficient is $\lambda(w^*)$ once $\mathcal{B}$ is small (Remark 13.2). Practice leaves it in three ways: a Gaussian localizer, a trained w^* found on the same data, and a short constant-step SGLD chain. For the Gaussian localizer Lau et al. (2025) state, as a direct extension of Watanabe's proofs, that the localized WBIC matches the first two terms of the localized free energy, with some coefficient $\tilde{\lambda}(w^*)$, and conjecture that $\tilde{\lambda}(w^*) = \lambda(w^*)$ under conditions on γ to be found (their Appendix D). The next proposition shows what such conditions must accomplish.

Proposition 13.10 (What a fixed Gaussian localizer measures). *In rank-one regression, $K(a,b) = \frac{1}{2}|a|^2|b|^2$ with $a \in \mathbb{R}^N$, $b \in \mathbb{R}^M$, let $w^* = (a^*, 0)$ with $a^* \neq 0$, so that $\lambda(w^*) = M/2$, and let $\mathbb{E}_t[\cdot]$ be the expectation under the density proportional to $\exp\{-tK(w) - \frac{\gamma}{2}|w - w^*|^2\}$ on $\mathbb{R}^{N+M}$.*

(i) $t\,\mathbb{E}_t[K] = \frac{M}{2}\,\mathbb{E}\Big[\frac{t|a|^2}{t|a|^2+\gamma}\Big]$, where a has the density proportional to $(t|a|^2 + \gamma)^{-M/2}e^{-\gamma|a-a^*|^2/2}$ on $\mathbb{R}^N$.

(ii) For every fixed $\gamma > 0$, $t\,\mathbb{E}_t[K] \to \min(M,N)/2$ as $t \to \infty$.

In particular, if $N < M$ the population estimate converges to $N/2$, the local learning coefficient of the origin, and not to $\lambda(w^*) = M/2$.

Proof. The plan is to integrate out b exactly and then let $t \to \infty$ in the remaining integral over a. For fixed a the exponent is $-\frac{1}{2}(t|a|^2 + \gamma)|b|^2 - \frac{\gamma}{2}|a - a^*|^2$, a centered Gaussian in b with precision $\kappa = t|a|^2 + \gamma$; so $\int e^{-\kappa|b|^2/2}\,\mathrm{d}b = (2\pi/\kappa)^{M/2}$, and $\int \frac{t}{2}|a|^2|b|^2 e^{-\kappa|b|^2/2}\,\mathrm{d}b = \frac{M}{2}\frac{t|a|^2}{\kappa}(2\pi/\kappa)^{M/2}$. Dividing gives (i).

For (ii) write $h_t(a) = (t|a|^2 + \gamma)^{-M/2}e^{-\gamma|a-a^*|^2/2}$, $I_t = \int h_t$ and $I_t - J_t = \int \frac{\gamma}{t|a|^2+\gamma}h_t$, so that $t\,\mathbb{E}_t[K] = \frac{M}{2}J_t/I_t$; there are three cases.

If $N < M$, substitute $a = s/\sqrt{t}$. The factor $e^{-\gamma|s/\sqrt{t}-a^*|^2/2}$ is at most 1 and tends to $e^{-\gamma|a^*|^2/2}$ for each s, and $(|s|^2 + \gamma)^{-M/2}$ is integrable on $\mathbb{R}^N$, so by dominated convergence $t^{N/2}I_t \to e^{-\gamma|a^*|^2/2}\int_{\mathbb{R}^N}(|s|^2+\gamma)^{-M/2}\,\mathrm{d}s$, and likewise for $t^{N/2}J_t$ with the extra factor $|s|^2/(|s|^2+\gamma)$. In polar coordinates both limits are Beta integrals, $\int_0^\infty \rho^{p-1}(\rho^2 + \gamma)^{-q}\,\mathrm{d}\rho = \frac{1}{2}\gamma^{p/2-q}B(\frac{p}{2}, q - \frac{p}{2})$, and their ratio is $B(\frac{N}{2}+1, \frac{M-N}{2})/B(\frac{N}{2}, \frac{M-N}{2}) = N/M$. So $t\,\mathbb{E}_t[K] \to \frac{M}{2}\cdot\frac{N}{M} = \frac{N}{2}$.

If $N > M$, then $t^{M/2}h_t(a) = (|a|^2 + \gamma/t)^{-M/2}e^{-\gamma|a-a^*|^2/2}$ increases to $|a|^{-M}e^{-\gamma|a-a^*|^2/2}$, which is integrable on $\mathbb{R}^N$, so $t^{M/2}I_t$ has a finite positive limit by monotone convergence. Since $\frac{\gamma}{t|a|^2+\gamma} \le \min\bigl(1, \frac{\gamma}{t|a|^2}\bigr)$, dominated convergence with the same integrable bound gives $t^{M/2}(I_t - J_t) \to 0$. So $J_t/I_t \to 1$ and the limit is $\frac{M}{2}$.

If $N = M$, fix $\delta \in (0, |a^*|)$. On $|a| > \delta$ the integrand of $t^{N/2}I_t$ is at most $\delta^{-N}e^{-\gamma|a-a^*|^2/2}$, so that part is bounded. On $|a| \le \delta$ the Gaussian factor lies between $e^{-\gamma(|a^*|+\delta)^2/2}$ and $e^{-\gamma(|a^*|-\delta)^2/2}$, and the substitution $a = s/\sqrt{t}$ turns the rest into $S_{N-1}\int_0^{\delta\sqrt{t}} \rho^{N-1}(\rho^2+\gamma)^{-N/2}\,d\rho = \frac{1}{2}S_{N-1}\log t + O(1)$, with S_{N-1} the area of the unit sphere. So $t^{N/2}I_t \to \infty$, while $t^{N/2}(I_t - J_t) \le \int_{\mathbb{R}^N} \gamma(|s|^2 + \gamma)^{-N/2-1}\,ds < \infty$. Again $J_t/I_t \to 1$, and the limit is $\frac{M}{2} = \frac{\min(M,N)}{2}$. ■

The mechanism is the competition of Chapter 12: the Gaussian gives the origin a weight $e^{-\gamma|a^*|^2/2}$, a constant, while the volume near the origin wins a factor $t^{(M-N)/2}$ that grows without bound. Comparing the two local free energies gives a rule of thumb.

HEURISTIC DERIVATION. Let w' be a point of the true set at distance δ from w^* with $\lambda(w') < \lambda(w^*)$. In the localized posterior its neighborhood has free energy about $\frac{\gamma}{2}\delta^2 + \lambda(w')\log t$, against $\lambda(w^*)\log t$ near w^* (Theorem 12.3, with the localizer in the prior), so the posterior stays near w^* only while $\gamma\delta^2 \gtrsim 2(\lambda(w^*) - \lambda(w'))\log t$. And γ must stay small against the curvature $t|a^*|^2$ of the identifiable directions, or part (i) shows a shrinkage factor $t|a^*|^2/(t|a^*|^2 + \gamma)$. For $M = 3$, $N = 2$, $|a^*| = 1$ and $n = 10^4$ this leaves the window $7 \lesssim \gamma \ll 10^3$. ◇

So whatever conditions make the conjecture of Lau et al. true, in this example they cannot be conditions on a fixed γ. The same holds for their coefficient $\tilde{\lambda}(w^*)$ itself when $N < M$: for fixed γ the Gaussian weight is bounded below on a box around the origin, the free energy of that box with a uniform prior is $nS_n + \frac{N}{2}\log n + O_p(1)$ (Theorems 7.2 and 8.4), and restricting the integral to the box gives $\tilde{\lambda}(w^*) \le \frac{N}{2} < \lambda(w^*)$ whenever their expansion holds with a remainder of smaller order than $\log n$. The localizer must tighten as n grows, by the heuristic at least like $\log n$; Lau et al. leave the conditions open, so this constrains their conjecture rather than refuting it. With a fixed γ every more singular point of the true set is eventually within reach; there is none at the origin of a deep linear network with zero truth (Lemma 13.6).

> OBSERVATION 13.11 (Lau et al. 2025). On random deep linear networks with 10^3 to 10^8 parameters (noiseless labels, $n = 10^5$ or 10^6, $\gamma = 1$, minibatches of 500, 90% burn-in), SGLD estimates at true parameters, and at minima found by SGD, tracked Aoyagi's learning coefficient across five orders of magnitude. The step size had to be retuned for each size class, from 5×10^{-7} to 2×10^{-8}, and some chains diverged.

The comparison was with the global coefficient, a lower bound for the local one at random true parameters, as Lau et al. note (their Appendix I). In the benchmark of Hitchcock and Hoogland (2025), a preprint with exactly

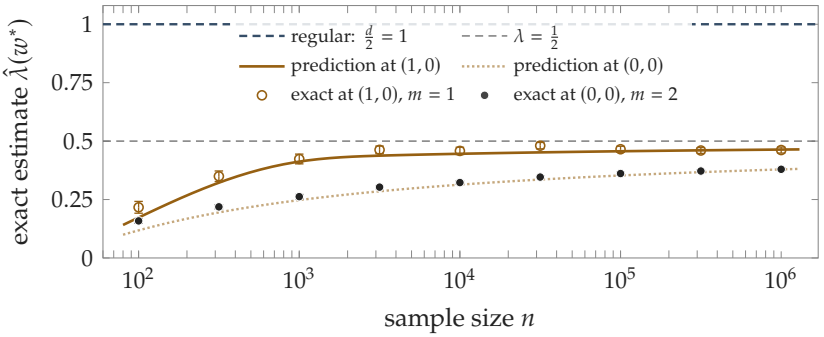


Figure 13.1. The LLC estimator of the ReLU neuron with an exact posterior, at the origin (dots; prior box $[-1, 1]^2$, $m = 2$) and at $(1, 0)$ (circles; box $[\frac{1}{2}, \frac{3}{2}] \times [-\frac{1}{2}, \frac{1}{2}]$, $m = 1$), with $\beta = 1/\log n$ and noisy labels: means over 200 data sets per n with Student-t 95% intervals (seed 1301; sufficient statistics drawn exactly). Solid and dotted bronze: the skeletons of Computation 13.3 minus $1/(2 \log n)$. Script `13-llc-ideal.py`.

known local values, plain SGLD underestimates at the most degenerate ("special") points and is accurate at generic ones, and SGLD preconditioned by RMSProp or Adam is less sensitive to the step size than plain SGLD; the bias of Computation 13.3, largest at high multiplicity, may explain part of the underestimate.

13.5 Measuring it

Three scripts measure the estimator where the values are known: without a sampler, with SGLD at fourteen points of nine models, and with a varying localizer.

THE ESTIMATOR WITHOUT A SAMPLER. The script `13-llc-ideal.py` computes $\hat{\lambda}(w^*)$ exactly for the ReLU neuron with noisy labels: on each half-plane $b > 0$, $b < 0$ the energy depends on (a, b) only through $c = ab$, so every posterior average is a one-dimensional integral (Chapter 3). The localizers are the boxes of Computation 13.3. At the origin, where $m = 2$, the exact estimate is 0.32 at $n = 10^4$ and 0.38 at $n = 10^6$ for a true value of $\frac{1}{2}$; at $(1, 0)$, where $m = 1$, it is 0.46 at both (Figure 13.1). The bronze curves, the skeletons of Computation 13.3 minus the noise term $1/(2 \log n)$ of Exercise 13.4, agree with the measured means to within 0.03 for $n \geq 10^4$ and to within 0.005 for $n \geq 3 \cdot 10^5$; at small n they are too low, because the noise term is only a leading-order description. The regular count, $\frac{d}{2} = 1$, is off by a factor of two to six.

SGLD AT FOURTEEN POINTS. The script `13-llc-sgld.py` follows Lau et al.'s deep-linear-network protocol at laptop scale (noiseless labels, $x \sim \mathcal{N}(0, I)$, minibatches of 500, $\gamma = 1$; the rest in Table 13.1) at five step sizes.[7] The target is the skeleton, exact where we can compute it (Proposition 13.10; the ReLU neuron is the case $N = M = 1$ with $K = \frac{1}{4}a^2b^2$), otherwise $\lambda - (m - 1)/\log(n\beta)$. At $(1, 0)$ with $\gamma = 1$ the Gaussian reaches the origin, whose multiplicity pulls the target down to 0.389 (the case $M = N$ of the

[7] The models are linear in a feature vector, so the losses come from minibatch sufficient statistics, and the 840 chains run in about a minute.

model	w^*	d	r	$[\frac{r}{2}, \frac{r}{2}+\frac{d-r}{4}]$	λ (m)	$\frac{d}{2}$	target	$\hat{\lambda}$, $\epsilon = 10^{-4}$	$\hat{\lambda}$, diagnostic (ϵ)
regular plane	0	2	2	[1, 1]	1 (1)	1	0.999	1.01 ± 0.03	1.10 ± 0.03 ($3 \cdot 10^{-4}$)
ReLU neuron	(0, 0)	2	0	[0, 0.5]	$\frac{1}{2}$ (2)	1	0.372	0.36 ± 0.12	0.45 ± 0.14 (10^{-3})
	(1, 0)	2	1	[0.5, 0.75]	$\frac{1}{2}$ (1)	1	0.389	0.40 ± 0.14	0.50 ± 0.18 (10^{-3})
	(1, 0)†	2	1	[0.5, 0.75]	$\frac{1}{2}$ (1)	1	0.474	0.47 ± 0.06	0.54 ± 0.04 (10^{-3})
rank-one 3×3	0	6	0	[0, 1.5]	$\frac{3}{2}$ (2)	3	1.347	1.38 ± 0.19	1.51 ± 0.11 (10^{-3})
	(e_1, 0)	6	3	[1.5, 2.25]	$\frac{3}{2}$ (1)	3	1.355	1.38 ± 0.07	1.57 ± 0.26 (10^{-3})
	(e_1, 0)†	6	3	[1.5, 2.25]	$\frac{3}{2}$ (1)	3	1.412	1.43 ± 0.14	1.63 ± 0.13 (10^{-3})
DLN 2-2-2	0	8	0	[0, 2]	$\frac{3}{2}$ (1)	4	1.5	1.49 ± 0.09	1.62 ± 0.24 ($3 \cdot 10^{-4}$)
DLN 3-3-3	0	18	0	[0, 4.5]	$\frac{7}{2}$ (2)	9	3.36	3.36 ± 0.12	the same (10^{-4})
DLN 3-3-3-3	0	27	0	[0, 6.75]	3 (1)	13.5	3	2.94 ± 0.23	the same (10^{-4})
DLN 4-3-2-4	0	26	0	[0, 6.5]	3 (3)	13	2.71	2.68 ± 0.19	the same (10^{-4})
DLN 3-4-5-3	0	47	0	[0, 11.75]	4 (1)	23.5	4	4.15 ± 0.18	3.29 ± 0.18 ($3 \cdot 10^{-5}$)
DLN 2-2-2-2-2	0	16	0	[0, 4]	$\frac{3}{2}$ (5)	8	0.93	1.00 ± 0.28	1.27 ± 0.09 (10^{-3})
DLN 3-3-3‡	special	18	5	[2.5, 5.75]	4 (1)	9	4	4.12 ± 0.19	the same (10^{-4})

Table 13.1. SGLD estimates at $n = 10^4$ ($n\beta = 1086$, noiseless labels, $\gamma = 1$, or 10 where marked †; 4 data sets × 3 chains of 12,000 steps, half discarded; seed 1302) with Student-t 95% half-widths over data sets. DLN H_0-$\cdots$-H_L: deep linear network, zero truth except ‡ (rank-one truth, special point); λ, m from Theorem 7.6; r computed numerically. Target: the exact skeleton (three decimals) or $\lambda - (m-1)/\log(n\beta)$. Script `13-llc-sgld.py`.

proposition, Exercise 13.8); $\gamma = 10$ gives 0.474. The script also records the diagnostic that Lau et al. (2025) recommend for tuning ϵ, the Metropolis acceptance probability of the Langevin step with full-batch gradients, to be kept between 0.9 and 0.95; the diagnostic rule takes the largest step with mean at least 0.9.

At $\epsilon = 10^{-4}$, the middle of the grid, all fourteen estimates in Table 13.1 lie within their intervals of the target, order the points as the targets do apart from one near-tie, and are far below the parameter counts; at the points of multiplicity 3 and 5 they are 11% and 33% below λ, as $\lambda - (m-1)/\log(n\beta)$ predicts. Across the grid, though, the errors run from −34% to +31% for DLN 3-3-3-3 and from −82% to −15% for DLN 2-2-2-2-2 (Figure 13.2). Small steps leave the chain near w^*, where the loss is lowest, and underestimate; large steps inflate the posterior's width and overestimate. For the regular plane the second effect is exact: on a quadratic loss the recursion has stationary variance $1/(\kappa(1-\epsilon\kappa/4))$ instead of $1/\kappa$, $\kappa = n\beta+\gamma$ (Exercise 13.3), which predicts 1.027, 1.088 and 1.372 at the three largest steps; we measured 1.014, 1.103 and 1.375. The diagnostic rule picks 10^{-3} for the ReLU neuron and rank-one regression, 12 to 28% above the target, and $3 \cdot 10^{-5}$ for the largest network, 18% below. On these fourteen points it never beats the fixed step against the target, and beats it against λ only where the upward bias of large steps cancels the skeleton's downward bias, which nobody without the true value can rely on. This is no verdict on the rule for large networks: for the ReLU neuron the acceptance stayed above 0.95 even at 10^{-3}, and for DLN 3-4-5-3 Lau et al.'s band lies between two steps of our grid (Exercise 13.9).

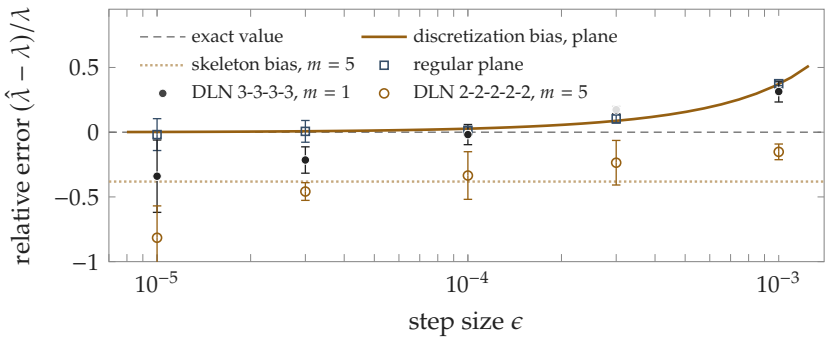


Figure 13.2. Relative error of the SGLD estimate against the step size at three of the points of Table 13.1 (same runs and intervals). Solid bronze: the exact discretization bias for the regular plane, $(1-\epsilon(n\beta+\gamma)/4)^{-1}n\beta/(n\beta+\gamma)-1$. Dotted bronze: the predicted bias $-(m-1)/(\lambda\log(n\beta))$ of the skeleton at the multiplicity-5 point. Script `13-llc-sgld.py`.

THE LOCALIZER. The script `13-llc-localize.py` computes the skeleton of Proposition 13.10 for rank-one regression at $w^* = (e_1, 0)$ by quadrature and runs SGLD there ($\epsilon = 10^{-4}$, 30,000 steps; Figure 13.3). With $(M, N) = (3, 2)$ and $n = 10^4$ the skeleton is 0.99 at $\gamma = 1$, the value of the origin, not of w^*; it rises to 1.44 at $\gamma = 30$ and falls as the localizer takes over, to 1.36 at $\gamma = 100$ and 1.17 at $\gamma = 300$. The SGLD estimates follow the skeleton within their intervals except at $\gamma = 30$; there and at $\gamma = 100$ they lie 2 to 3% above it, the discretization bias $(1-\epsilon\kappa/4)^{-1}$ with $\kappa = n\beta+\gamma$. At fixed γ every curve ends at $N/2 = 1$ as t grows; $\gamma = 30$ holds out until $t \approx 10^{10}$. For $M = N = 3$, where both points have $\lambda = \frac{3}{2}$, the skeleton moves by less than 10% over $0.3 \le \gamma \le 100$.

These measurements are small ($d \le 47$, $n = 10^4$, noiseless labels): they show that the estimator can be right on such models, and how it goes wrong, not that it is right with 10^8 parameters.

13.6 The ledger

The local learning coefficient is the exponent of the volume of nearly optimal parameters near one point. It lies between half the rank of the Hessian and that plus a quarter of the nullity, equals the lower end exactly at minimally singular points, and survives the reparametrizations that move Hessian eigenvalues at will. Its estimator is consistent in the setting of Watanabe's theorem, but biased by about $(m-1)/\log n$ even with an exact sampler; with SGLD it moves with the step size by tens of percent, and with a fixed Gaussian localizer it can measure another point. At scale there are no known values to check it against.

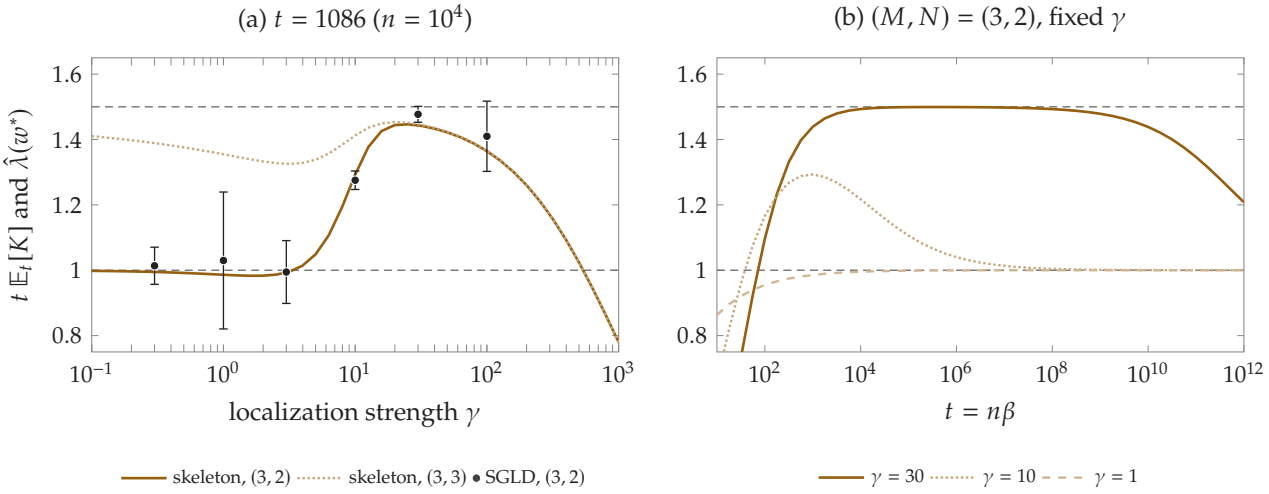


Figure 13.3. What a Gaussian localizer measures in rank-one regression at $w^* = (e_1, 0)$, where $\lambda(w^*) = \frac{3}{2}$; the origin has $\lambda = \min(M, N)/2$ (gray lines at $\frac{3}{2}$ and 1). (a) The skeleton of Proposition 13.10 against γ, and SGLD estimates for $(M, N) = (3, 2)$ ($\epsilon = 10^{-4}$, 30,000 steps, half discarded, noiseless labels, 3 data sets $\times$ 4 chains, Student-t 95% intervals, seed 1303). (b) The skeleton against t at fixed γ. Script `13-llc-localize.py`.

LEDGER · $y = a\,\mathrm{relu}(bx)$, deep linear networks

QUANTITY	the local learning coefficient $\lambda(w^*)$ and multiplicity $m(w^*)$ at a minimum w^*, and the estimate $\hat{\lambda}(w^*)$
REGULAR THEORY	Laplace at w^*: $\frac{d}{2}$, or $\frac{r}{2}$ with r the Hessian rank when the Hessian is degenerate
SINGULAR THEORY	$\frac{r}{2} \le \lambda(w^*) \le \frac{r}{2} + \frac{d-r}{4}$ (Theorem 13.3); $= \frac{r}{2}$ iff the true set is a $(d-r)$-manifold near w^* (Corollary 13.4); invariant under reparametrization (Proposition 13.5)
RUNNING EXAMPLE	rank-one regression, $M = N = 3$: $\lambda = \frac{3}{2}$ at 0 ($r = 0$, $m = 2$) and at $(e_1, 0)$ ($r = 3$, $m = 1$); $\frac{d}{2} = 3$
MEASURED	exact estimator for the ReLU neuron at $n = 10^4$: 0.32 ($m = 2$) and 0.46 ($m = 1$) for $\lambda = \frac{1}{2}$; SGLD at $\epsilon = 10^{-4}$: all 14 points within their intervals of the target, but errors from −82% to +118% across step sizes; fixed Gaussian localizer: 0.99 instead of $\frac{3}{2}$
RESTS ON	w^* a local minimum of a C^4 (for existence, analytic) loss; for the estimator: a compact localizer, exact posterior averages, $n \to \infty$; no guarantee for SGLD

13.7 Where the theory stops

WHAT WAS ASSUMED. The definition and the Hessian sandwich need little: a local minimum of the population loss and four derivatives there. The estimator needs much more: Theorem 13.9 assumes Watanabe's conditions on a ball (an analytic model; relatively finite variance, which Lau et al. (2025) call unclear for real data), a compact localizer, exact posterior averages, and $n \to \infty$ with the model fixed. Practice departs from each. The point w^* is a trained parameter, near a minimum of L_n rather than of L (Lau et al. saw no adverse effect on deep linear networks). ReLU networks are not analytic; our ReLU neuron escapes because its K is a polynomial and each half-plane is a one-coefficient model. The localizer is Gaussian. The averages come from a few thousand steps of constant-step minibatch SGLD, whose stationary law is biased by the step size and the gradient noise (Brosse et al. 2018); the consistency results for SGLD assume decreasing steps and growth conditions (Teh et al. 2016) that degenerate losses violate (Hitchcock and Hoogland 2025).

WHERE IT BREAKS. With a Gaussian localizer of fixed width the estimator can converge to the learning coefficient of a more singular point, not of w^* (Proposition 13.10): at $n = 10^4$ and Lau et al.'s $\gamma = 1$, rank-one regression at $(e_1, 0)$ gives 0.99 computed and 1.03 ± 0.21 measured, for $\lambda(w^*) = 1.5$ (Figure 13.3). Even with an exact sampler and a compact localizer, the estimate at the ReLU neuron's origin, a point of multiplicity 2, is 26% low at $n = 10^4$ and 17% low at $n = 10^6$ without noise (Computation 13.3), and 35% and 24% low with noisy labels (Figure 13.1); the bias grows with m. The step size moves the answer by tens of percent either way (Figure 13.2), and chains diverge on large models (Lau et al. 2025).[8] At points that are not minima the estimate can be negative (Chen et al. 2023; Lau et al. 2025), yet it is routinely computed along training, away from minima (Hoogland et al. 2025). At scale there is no ground truth: Wang et al. (2025) write that they cannot guarantee the accuracy of their estimates and rely on their ordering. The Hessian sandwich is hard to use at scale for another reason: the numerical rank of a large Hessian depends on a threshold, and Wang et al. (their Appendix D) found Hessian-rank estimates sensitive to it and inconclusive.

[8] The unadjusted Langevin algorithm diverges when the gradient of the log-density grows faster than linearly, as it does for deep linear networks, by a theorem of Hutzenthaler, Jentzen and Kloeden that Hitchcock and Hoogland (2025) invoke.

OPEN PROBLEMS. (1) The conjecture of Lau et al. (2025, Appendix D): find conditions on γ, necessarily depending on n by Proposition 13.10, under which the Gaussian-localized estimator converges to $\lambda(w^*)$. (2) Explain why stochastic-gradient MCMC chains recover local volume exponents at all, without the global convergence guarantees that degenerate losses rule out; Hitchcock and Hoogland (2025) pose this, and ask what determines the effective support of the chains. (3) Error bars: our intervals cover the variation over data sets and chains, not the bias, and we know of no method that gives valid confidence intervals for $\lambda(w^*)$ in a large network. (4) Ground truth beyond deep linear networks, which are only mildly singular (their coefficients are half the codimension of the true set, Pepin Lehalleur and Rimányi (2024) as restated by Hitchcock and Hoogland (2025)). (5)

Estimators corrected for the $(m-1)/\log n$ bias.

Exercises

13.1 CONSOLIDATION. (Two derivatives suffice below.) Check that the first part of the proof of Theorem 13.3 uses only that K is twice continuously differentiable, and conclude that $\lambda(w^*) \geq \frac{r}{2}$ whenever $K \in C^2$ and $\lambda(w^*)$ exists.

13.2 CONSOLIDATION. (The original bound.) (a) Show that if K is only three times continuously differentiable, the argument of the proof gives $\lambda(w^*) \leq \frac{r}{2} + \frac{d-r}{3}$. (b) Show that $K(a,b) = a^2 + |b|^{7/2}$ is C^3 but not C^4, has $r = 1$ at the origin, and has $\lambda = \frac{1}{2} + \frac{2}{7} > \frac{3}{4}$; so the fourth derivative is needed for the quarter.

13.3 CONSOLIDATION. (Discretization bias.) For $K = \frac{h}{2}|w|^2$ on $\mathbb{R}^d$ and the full-gradient recursion of Definition 13.7, show that the stationary law is $\mathcal{N}(0, vI_d)$ with $v = 1/(\kappa(1-\epsilon\kappa/4))$, $\kappa = th + \gamma$, if $\epsilon\kappa < 4$, so that the estimate is $\frac{d}{2}\frac{th}{\kappa}(1-\epsilon\kappa/4)^{-1}$. Evaluate it for the regular plane at $t = 1086$, $\gamma = 1$, $\epsilon = 10^{-4}, 3\cdot 10^{-4}, 10^{-3}$.

13.4 CONSOLIDATION. (Noise at the true parameter.) For the regular line $y = ax + \varepsilon$ with truth $a = 0$ and the prior uniform on $[-1,1]$, show that the exact estimator at $w^* = 0$ is $\hat{\lambda} = \frac{1}{2} - \beta Z^2/2$ up to terms that are exponentially small in βS_{xx} on the event $|S_{xy}/S_{xx}| < \frac{1}{2}$, where $Z = S_{xy}/\sqrt{S_{xx}} \sim \mathcal{N}(0,1)$. Deduce that its mean is close to $\frac{1}{2} - 1/(2\log n)$, and that evaluating at the empirical minimizer instead removes this term.

13.5 BY HAND. (The chain.) For $y = a_1a_2a_3x$ with truth 0, find λ, m, the Hessian rank and the bounds of Theorem 13.3 at $(1,0,0)$, at $(1,1,0)$ and at the origin, and say which bounds are attained.

13.6 BY HAND. (The cusp.) For $y = ab^2x$, so $K = \frac{1}{2}a^2b^4$, do the same at the origin, at $(1,0)$ and at $(0,1)$. Show that the Hessian vanishes at every point of the axis $\{b = 0\}$ while $\lambda = \frac{1}{4}$ there, strictly inside the bounds.

13.7 BY HAND. (Depth.) For a deep linear network of depth L with zero truth, show that $K(w) \leq \frac{1}{2L^L}|w|^{2L}$ and deduce $\lambda(0) \leq \frac{d}{2L}$. Compare with Theorem 7.6 for widths 3-3-3-3 and for the chain, where the bound is attained.

13.8 VARIATION. (Equal sizes.) In Proposition 13.10 with $M = N$, show that $(\frac{M}{2} - t\,\mathbb{E}_t[K])\log t \to 1$, the multiplicity correction $(m-1)/\log t$ with $m = 2$. *(Hint: $t^{N/2}(I_t - J_t) \to e^{-\gamma|a^*|^2/2} S_{N-1}\int_0^\infty \gamma\rho^{N-1}(\rho^2+\gamma)^{-N/2-1}\,\mathrm{d}\rho$, and the integral is $1/N$.)*

13.9 MEASUREMENT. (Choosing the step.) In `13-llc-sgld.py`, refine the step-size grid between $3\cdot10^{-5}$ and 10^{-3} for the deep linear networks 3-3-3-3 and 3-4-5-3, and plot the relative error against the mean acceptance diagnostic. Is there a diagnostic level at which the error crosses zero

for both? What does the answer say about tuning the step size without ground truth?

13.10 RESEARCH. (Conditions on the localizer.) For rank-one regression with $N < M$, find conditions on a sequence γ_n under which the population estimate, and then the estimator with data, converges to $M/2$, and formulate the analogous condition for an analytic K in terms of the distance from w^* to the points of $\{K = 0\}$ with smaller local coefficient. Lau et al. (2025) conjecture that some such conditions exist; the general question is open.

Notes and References

Watanabe's learning coefficient appeared in Watanabe (1999) and, with the free energy formula, in Watanabe (2001); his monograph (Watanabe 2009) develops it. Wei et al. (2023) argued that it, not the parameter count, measures the effective dimension of a neural network. The local learning coefficient of a trained network and the SGLD estimator are due to Lau et al. (2025), whose first arXiv version (2023) was titled *Quantifying degeneracy in singular models via the learning coefficient* and whose second absorbed Furman and Lau (2024). Stochastic-gradient Langevin dynamics is due to Welling and Teh (2011); Teh et al. (2016) proved its consistency for decreasing steps, and Brosse et al. (2018) showed that with a constant step its stationary law can stay far from the posterior.

The upper half of the Hessian sandwich is Corollary 4.15 of Lin (2017), also in his thesis (Lin 2011), stated for the ideal of functions $f_1, \dots, f_s$ with the rank of their Jacobian, the rank of the Hessian of $K = \sum f_i^2$; Bushnaq (2024) found it again. Our volume proof of Theorem 13.3, and Corollary 13.4, Lemma 13.6, Computation 13.3, and Proposition 13.10 with the bound on $\tilde{\lambda}(w^*)$ that follows it, are the book's own; we did not find them in the literature.

Aoyagi's formula (Aoyagi 2024) extends the reduced-rank regression formula of Aoyagi and Watanabe (2005); Pepin Lehalleur and Rimányi (2024) study the fibers of the multiplication map, from which Hitchcock and Hoogland (2025) compute local coefficients at generic points.

TRANSLATING THE LITERATURE. Lau et al. (2025) write ϵ for both the step size and the tolerance of the volume, β^* for our $\beta = 1/\log n$, m for the minibatch size (our $|B|$), t for the step index (our k), and $B(w^*)$ for a ball on which w^* minimizes L. Lin (2017) writes RLCT for the pair (threshold, multiplicity); the threshold of an ideal $\langle f_1, \dots, f_s \rangle$ is twice our λ for $K \asymp \sum f_i^2$, so his bound $\frac{1}{2}(r + d)$ is our $\frac{r}{2} + \frac{d-r}{4}$. Chapter 7 translates the notation of Aoyagi (2025). In `devinterp` (van Wingerden et al. 2024), `nbeta` is our $n\beta$ and `localization` our γ.

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14 Stages

Learning coefficients along training

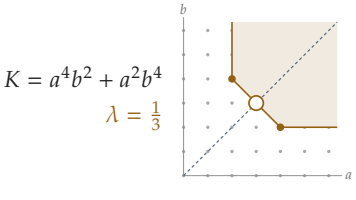
THE MODEL — the toy model of superposition

IN THIS CHAPTER. Training curves of small networks have long plateaus and sudden drops. In the toy model of superposition, an autoencoder with eighteen parameters, the plateaus sit at critical points shaped like regular polygons, and we compute their learning coefficients by hand: 7 at the 5-gon, whose set of equally good parameters is four-dimensional, and $\frac{17}{2}$ at the 5^+-gon and the 6-gon, where it is a circle of rotations. The theorem behind the count: a manifold of critical points across which the Hessian is positive definite has learning coefficient half its codimension. Along runs of stochastic gradient descent the estimated coefficient steps up at each drop of the loss, from 4.6 to 7.6 and from 5.2 to 7.8 in our two runs; but at the sample size of the runs the estimator cannot tell the 5-gon from the 5^+-gon, and it reaches the hand-computed values only near $n = 10^6$. That training visits critical points in the order a Bayesian posterior would is a conjecture, and the results on transformers are observations.

14.1 The question: do the plateaus of training have learning coefficients?

Train a small network and plot its loss against the number of steps: the curve is often a staircase, with long plateaus that end in sudden drops. Plateaus of this kind were analyzed for multilayer perceptrons, where they arise near singularities of the parameter space, at parameters where two hidden units coincide or one is switched off (Amari et al. 2006; Fukumizu and Amari 2000); in transformers, induction heads form at a visible bump in the training loss (Olsson et al. 2022). In the toy model of superposition of Elhage et al. (2022), an autoencoder that stores sparse features in fewer dimensions, Chen et al. (2023) report, for six features in two dimensions, plateaus separated by sudden drops, during each of which the weights sit near a critical point whose columns form a regular polygon.

A plateau is a long stay near a critical point w^*, and Chapter 13 gave a local minimum a number, the local learning coefficient $\lambda(w^*)$ of Definition 13.1, the exponent with which the volume of nearly optimal parameters near w^* shrinks. Does it change at the drops? If so, its changes could mark stages of development that the loss curve hides; this is the program of Hoogland et al. (2025), who estimate it along the training of transformers.



THE REGULAR ANSWER · complexity along training. If every critical point that training visits is a nondegenerate minimum, the Laplace approximation charges each of them $\frac{d}{2}\log n$ (Theorem 1.4): a complexity read off the Hessian's rank is $\frac{d}{2}$ on every plateau and cannot mark stages. About the training dynamics the regular theory says nothing.

In the toy model with $d = 18$ parameters no critical point with $W \neq 0$ is nondegenerate: rotating the plane of the columns does not change the loss, so such a point lies on a circle of critical points, and the Hessian, where it exists, has a zero eigenvalue. Moreover the critical points on the staircase differ in their flat directions. At the 5-gon a switched-off sixth column moves freely in three directions, and we will find $\lambda = 7$; at the 6-gon only the rotation is free, and $\lambda = \frac{17}{2}$; both are below $\frac{d}{2} = 9$. This chapter asks two questions. What are the learning coefficients of the critical points that a training run visits, and how are they computed? And what is known, proved, argued or observed, about the order in which training moves between them?

14.2 By hand: the polygons of the toy model of superposition

The *toy model of superposition* maps $x \in \mathbb{R}^6$ to

$$f(x, w) = \mathrm{relu}\big(W^\top W x + b\big), \qquad W \in \mathbb{R}^{2\times 6}, \quad b \in \mathbb{R}^6,$$

so that $w = (W, b)$ has $d = 18$ coordinates; $W_1, \ldots, W_6 \in \mathbb{R}^2$ are the columns of W and $b_1, \ldots, b_6$ the biases. The inputs are in the *high-sparsity limit* of Chen et al. (2023): one feature is active, $x = \mu e_i$ with i uniform on $\{1, \ldots, 6\}$ and μ uniform on $[0, 1]$. Their loss is the mean squared reconstruction error, which we call $\ell(w) = \mathbb{E}\|x - f(x, w)\|^2$. Following them we treat the model as $p(x \mid w) \propto \exp(-\frac{1}{2}\|x - f(x, w)\|^2)$, ignoring that the normalizing constant depends on w; equivalently, as a regression of the target $y = x$ with unit Gaussian noise. The book's population loss is then $L = \frac{1}{2}\ell$ up to a constant, $K(w) = \frac{1}{2}(\ell(w) - \ell(w^*))$ near a local minimum w^*, and the energy of the posterior is nL_n.[1] Learning coefficients do not change when K is multiplied by a positive constant, so the factor $\frac{1}{2}$ matters only where n appears.

[1] The truth is not in the model: no parameter reconstructs every input, the smallest value of ℓ is positive, and K is measured from a local minimum w^*, as in Chapter 13.

COMPUTATION 14.1 · THE LOSS IN THE HIGH-SPARSITY LIMIT

1. For $x = \mu e_j$ the hidden vector is $Wx = \mu W_j$, and output o is $\mathrm{relu}(\mu\, W_o \cdot W_j + b_o)$. So $\|x - f\|^2 = (\mu - \mathrm{relu}(\mu\|W_j\|^2 + b_j))^2 + \sum_{o\neq j} \mathrm{relu}(\mu\, W_o \cdot W_j + b_o)^2$.
2. Averaging over j and μ, $\ell = \frac{1}{6}\sum_{o,j} I_{oj}$, where I_{oj} integrates over $\mu \in [0, 1]$ the term of output o for input j, a piecewise quadratic in μ with at most one kink. Write $\alpha = W_o \cdot W_j$ and $\beta = b_o$.
3. Interference, $o \neq j$. If $\beta \leq 0$, the ReLU is active on $[-\beta/\alpha, 1]$ when $\alpha + \beta > 0$, giving $3I_{oj} = (\alpha + \beta)^3/\alpha$, and nowhere otherwise,

giving 0. If $\beta > 0$, it is active on all of $[0,1]$ when $\alpha \geq -\beta$, giving $\alpha^2 + 3\alpha\beta + 3\beta^2$, and on $[0, -\beta/\alpha)$ when $\alpha < -\beta$, giving $-\beta^3/\alpha$.

4 Reconstruction, $o = j$, with $\alpha = \|W_o\|^2$, $\beta = b_o$. If $\beta \leq 0 \leq \alpha + \beta$, the error is μ for $\mu < \mu_0 = -\beta/\alpha$ and $(1-\alpha)\mu - \beta$ beyond, so

$$\begin{aligned} 3I_{oo} &= \mu_0^3 + \frac{(1-\alpha-\beta)^3 - \mu_0^3}{1-\alpha} \\ &= (1-\alpha)^2 - 3(1-\alpha)\beta + 3\beta^2 + \beta^3\Big(\frac{1}{\alpha^2} + \frac{1}{\alpha}\Big), \end{aligned}$$

the second line holding also at $\alpha = 1$. If $\beta \leq 0$ and $\alpha + \beta < 0$ the output never fires and $3I_{oo} = 1$; if $\beta > 0$ it always fires and $3I_{oo} = (1-\alpha)^2 - 3(1-\alpha)\beta + 3\beta^2$.

5 So $\ell = \frac{1}{18}H$, where $H = \sum_{o,j} 3I_{oj}$, the *TMS potential* of Chen et al. (2023), is a sum of 36 closed forms (their Lemma 3.1). Where strict inequalities fix the case of every term, it is a rational function of (W, b) with nonvanishing denominators: analytic. At three polygons and three random points it agrees with quadrature of the 36 integrals to 10^{-15}, and with Monte Carlo averages of 10^6 draws within 2.6 standard errors (script `14-stages-kgons.py`).

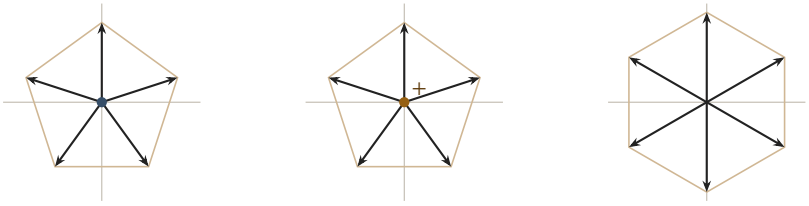
What one had to notice is that a single active feature turns the reconstruction into thirty-six one-dimensional ReLU integrals. The formula divides parameter space into *chambers*, according to which case of each integral applies; inside a chamber H is analytic, on the walls between chambers it need not be.

The polygons

In a *k-gon* the columns $W_1, \ldots, W_k$ have a common length l and point at the vertices of a regular k-gon, the biases $b_1, \ldots, b_k$ have a common value $b \leq 0$ (negative except at the 4-gons, where it is 0), and the other $6 - k$ columns are *vestigial*: zero, with negative biases, so that their outputs never fire. In a k^+-gon one vestigial bias is positive instead, in a k^{++}-gon two are (Figure 14.1). Two tools make the polygons computable by hand.

Lemma 14.1 (Symmetric points). *Let G be a group of orthogonal linear maps of $\mathbb{R}^d$, let $H\colon \mathbb{R}^d \to \mathbb{R}$ satisfy $H(gw) = H(w)$ for all $g \in G$ and $w \in \mathbb{R}^d$, and let $F = \{w : gw = w \text{ for all } g \in G\}$. If H is differentiable at a point $w \in F$, then $\nabla H(w) \in F$; so w is a critical point of H if and only if it is a critical point of the restriction of H to the subspace F.*

Proof. For $g \in G$ and $v \in \mathbb{R}^d$ we have $H(w + gv) = H(g(w+v)) = H(w+v)$, because $gw = w$. Differentiating in v at $v = 0$ gives $g^\top \nabla H(w) = \nabla H(w)$, and multiplying by $g = (g^\top)^{-1}$ gives $\nabla H(w) = g\nabla H(w)$ for every g, so the gradient lies in F. The gradient of the restriction at w is the orthogonal projection of $\nabla H(w)$ onto F, which is $\nabla H(w)$ itself. ∎



5-gon, $b_6 < 0$
$\ell = 0.0687$, $s = 4$, $\lambda = 7$

5^+-gon, $b_6 = \frac{1}{12}$
$\ell = 0.0618$, $s = 1$, $\lambda = \frac{17}{2}$

6-gon
$\ell = 0.0482$, $s = 1$, $\lambda = \frac{17}{2}$

Figure 14.1. The three polygons whose learning coefficients this chapter computes, to scale, in the plane of the columns W_i (arrows). The dot is a vestigial column $W_6 = 0$: with a negative bias its output never fires (left); with the bias $\frac{1}{12}$ it outputs a constant (center). Under each: the loss $\ell = \mathbb{E}\|x - f\|^2$, the dimension s of the set of equally good parameters nearby, and $\lambda = \frac{18-s}{2}$ (Proposition 14.4).

Lemma 14.2 (Circulant Hessians). *Let $\mathcal{H}$ be a real symmetric $kp \times kp$ matrix of $p \times p$ blocks $\mathcal{H}_{ij}$, indices modulo k, with $\mathcal{H}_{i+1,j+1} = \mathcal{H}_{ij}$; put $C_q = \mathcal{H}_{0q}$ and $\omega = e^{2\pi i/k}$. Then the eigenvalues of $\mathcal{H}$, with multiplicity, are those of the k Hermitian matrices $\hat{C}(r) = \sum_{q=0}^{k-1} \omega^{qr} C_q$, $r = 0, \dots, k-1$.*

Proof. The plan is to write down eigenvectors. The hypothesis gives $\mathcal{H}_{ij} = C_{j-i}$, and symmetry gives $C_q^\top = \mathcal{H}_{q0} = C_{-q}$, so $\hat{C}(r)^* = \sum_q \omega^{-qr} C_{-q} = \hat{C}(r)$. If $\hat{C}(r)u = \kappa u$, the vector v with blocks $v_j = \omega^{jr}u$ satisfies $(\mathcal{H}v)_i = \sum_j C_{j-i}\omega^{jr}u = \omega^{ir}\hat{C}(r)u = \kappa v_i$. Orthonormal eigenbases of the k matrices give kp such vectors; two with different r are orthogonal because $\sum_j \omega^{j(r-r')} = 0$, and two with the same r have k times the inner product of their u's. So they form an orthogonal basis of $\mathbb{C}^{kp}$. ∎

COMPUTATION 14.2 · THE 6-GON

1 The map g that rotates every column by 60° and shifts the indices, $W_i \mapsto RW_{i-1}$, $b_i \mapsto b_{i-1}$, is orthogonal and leaves H unchanged: rotations preserve the inner products $W_o \cdot W_j$, and the shift permutes the 36 terms. Its fixed points are the regular hexagons $W_i = R^{i-1}W_1$ with a common bias b, and by Lemma 14.1 a hexagon at which H is differentiable is critical if and only if it is critical for H restricted to them, a function of $a = \|W_1\|^2$ and b.

2 In the chamber $b < 0 < \frac{a}{2} + b$, neighbors interfere ($W_o \cdot W_j = \frac{a}{2}$) and the others do not ($-\frac{a}{2}$ and $-a$), and Computation 14.1 gives $H = 6\phi(a, b)$ with

$$\phi(a,b) = \frac{4(\frac{a}{2}+b)^3}{a} + (1-a)^2 - 3(1-a)b + 3b^2 + b^3\left(\frac{1}{a^2} + \frac{1}{a}\right). \quad (14.1)$$

3 The equations $\partial_a\phi = \partial_b\phi = 0$ have the solution $a^* = 1.74379$, $b^* = -0.61814$, so the columns have length $l^* = 1.32053$, as in Table A.1 of Chen et al. (2023). It lies inside the chamber,

$\frac{a^*}{2} + b^* = 0.2538$, and $\ell = H/18 = \phi(a^*, b^*)/3 = 0.048192$.

4 Describe a nearby parameter by the radial and tangential displacements of each column and the change of its bias. In these coordinates g shifts the triples cyclically, so the Hessian of H satisfies Lemma 14.2 with $k = 6$, $p = 3$; only neighbors interact, so its eigenvalues are those of $\hat{C}(r) = C_0 + \omega^r C_1 + \omega^{-r} C_1^{\top}$. The frames are orthonormal at the hexagon, so these are the eigenvalues in the coordinates (W, b) as well.

5 The script computes

$$C_0 = \begin{pmatrix} 14.105 & 0 & 7.756 \\ 0 & 6.334 & 0 \\ 7.756 & 0 & 6.146 \end{pmatrix}, \qquad C_1 = \begin{pmatrix} 1.322 & -2.291 & 0.985 \\ 2.291 & -3.167 & 1.706 \\ 0.985 & -1.706 & 0 \end{pmatrix}.$$

In mode $r = 0$ the tangential direction decouples with eigenvalue $6.334 + 2(-3.167) = 0$: the rotation. The radial–bias block $\left(\begin{smallmatrix} 16.750 & 9.727 \\ 9.727 & 6.146 \end{smallmatrix}\right)$ has trace 22.896 and determinant 8.337, so its eigenvalues 0.370 and 22.53 are positive. The other modes give 0.700, 2.068, 21.97 ($r = 1, 5$), 1.896, 7.084, 19.45 ($r = 2, 4$) and 2.436, 12.67, 15.17 ($r = 3$).

6 The Hessian is positive semidefinite with one zero eigenvalue, along the circle of rotations: rank 17, $s = 1$, and Theorem 14.3 below gives $\lambda = \frac{18-1}{2} = \frac{17}{2}$, $m = 1$.

COMPUTATION 14.3 · THE 5-GON AND THE 5^+-GON

1 For a regular pentagon of five columns only neighbors interfere again, with $W_o \cdot W_j = a \cos 72°$; the restricted potential is $5\phi_5$, where ϕ_5 is (14.1) with its first term replaced by $2(a \cos 72° + b)^3/(a \cos 72°)$. Its critical point is $a^* = 1.36998$, $b^* = -0.28230$, $l^* = 1.17046$, inside the chamber $(a^* \cos 72° + b^* = 0.1410)$; as a model with five features the pentagon has $\ell^{(5)} = \phi_5(a^*, b^*)/3 = 0.015826$.

2 Add a sixth column $W_6 = 0$ with bias $b_6 < 0$. Its output never fires, contributing $3I_{66} = 1$, and no output fires for input 6, since $b_j = b^* < 0$. These strict inequalities survive small changes, so with $H^{(5)}$ the potential of the five-feature model, on a neighborhood of the 5-gon

$$H = H^{(5)}(W_1, \dots, W_5, b_1, \dots, b_5) + 1,$$

and $\ell = (15\,\ell^{(5)} + 1)/18 = 0.068744$, independent of (W_6, b_6).

3 The Hessian of H is that of $H^{(5)}$, with one zero eigenvalue (the rotation) and fourteen from 1.035 to 19.02, bordered by a 3×3 block of zeros for (W_6, b_6): rank 14. The equally good parameters nearby form a circle of rotations times the three free coordinates

of the dead column: $s = 4$, $\lambda = \frac{18-4}{2} = 7$.

4 With $W_6 = 0$ and $b_6 > 0$ the sixth output fires for every input, adding $3b_6^2$ for each of the five other inputs and $1 - 3b_6 + 3b_6^2$ for its own: $18b_6^2 - 3b_6 + 1$, against the 1 of the 5-gon. The minimum is at $b_6 = \frac{1}{12}$, the 5^+-gon, where ℓ drops by $\frac{1}{144}$ to 0.061799; and $\partial^2 H/\partial b_6^2 = 36$ is an eigenvalue of the Hessian.

5 The gradient in W_6 is $3b_6 \sum_{j\le 5} W_j = 0$, since the pentagon's columns sum to zero. The script finds one zero eigenvalue and seventeen from 1.035 to 36: $s = 1$, $\lambda = \frac{17}{2}$.

What one had to notice is that λ is here a dimension count: the hexagon's columns are locked together and only a rotation moves them, while the 5-gon carries a dead column whose three parameters are free. The count is legitimate because the Hessian is positive definite in every other direction; that is the theorem. It says nothing at the 4-gons, whose biases sit at 0, on the walls between chambers, where H is not twice differentiable (Exercise 14.5).

14.3 The theorem: minimally singular critical sets

The computations used one principle: at a critical point whose Hessian is positive definite across a manifold of critical points, the learning coefficient is half the codimension of the manifold. Wei et al. (2023) call such points *minimally singular*; the function is Morse–Bott along the manifold, after Bott (1954). Proposition 5.5 proves the rule when the zero set of K is known to be the manifold. For the polygons we know less, a manifold of critical points from rotations and dead columns and a Hessian at one point, and the version we need concludes the rest. As in Chapter 13, $V_B(\varepsilon) = \int_{B\cap\{K<\varepsilon\}} \varphi \, dw$ for a ball B and a prior density φ.

Theorem 14.3 (Minimally singular critical sets). *Let L be twice continuously differentiable on an open set $U \subseteq \mathbb{R}^d$, and let $S \subseteq U$ be a C^2 submanifold of dimension $s < d$ on which ∇L vanishes. Let $w^* \in S$, suppose that $\nabla^2 L(w^*)$ is positive semidefinite of rank $d - s$, let φ be continuous near w^* with $\varphi(w^*) > 0$, and put $K = L - L(w^*)$. Then there is $\rho_0 > 0$ such that every closed ball B centered at w^* with radius at most ρ_0 satisfies:*

(i) $K \ge 0$ on B, and $\{w \in B : K(w) = 0\} = S \cap B$;

(ii) there are constants $0 < c_B \le C_B$ with $c_B\, \varepsilon^{(d-s)/2} \le V_B(\varepsilon) \le C_B\, \varepsilon^{(d-s)/2}$ for all sufficiently small $\varepsilon > 0$.

In particular w^* is a local minimum of L, $\lambda(w^*) = \frac{d-s}{2}$ and $m(w^*) = 1$.

The tangent space of S always lies in the kernel of the Hessian (differentiate $\nabla L \equiv 0$ along a curve in S), so the rank is at most $d - s$; the hypothesis is that it is as large as it can be and that the Hessian is positive semidefinite. That w^* is a minimum and that S is the whole zero set are not assumed; they follow.

PROOF. The plan is to write S as a graph over its tangent space, bound K from below on each normal slice by strong convexity and from above near S by Taylor's formula, and so trap the sublevel sets between two tubes of radius of order $\sqrt{\varepsilon}$ around S. Put $r = d - s$.

Coordinates. Translate and rotate so that $w^* = 0$ and the tangent space of S at 0 is $\{0\} \times \mathbb{R}^s$; balls, volumes, and the rank and definiteness of the Hessian do not change. Write $w = (u, z)$ with $u \in \mathbb{R}^r$, $z \in \mathbb{R}^s$. A C^2 submanifold is locally the graph of a C^2 function over its tangent space: there are $\delta_1 > 0$, a neighborhood N of 0, and a C^2 map g on $\{|z|_\infty < \delta_1\}$ with $g(0) = 0$, $Dg(0) = 0$, such that $S \cap N$ consists of the points $(g(z), z)$ that lie in N.

The normal block. The kernel of $A = \nabla^2 L(0)$ contains $\{0\} \times \mathbb{R}^s$, of dimension s, and A has rank r, so the kernel is $\{0\} \times \mathbb{R}^s$. For positive semidefinite A, $v^\top A v = 0$ forces $Av = 0$; hence $u^\top A_{uu} u = (u, 0)^\top A (u, 0) > 0$ for $u \neq 0$, and $A_{uu} = \nabla^2_{uu} L(0)$ is positive definite. Let $h > 0$ be its smallest eigenvalue.

A box. By continuity there is $\delta \in (0, \delta_1)$ such that the cube $Q = [-\delta, \delta]^d$ lies in $U \cap N$ and, on Q, $\nabla^2_{uu} L \geq \frac{h}{2} I$, the operator norm of $\nabla^2 L$ is at most some $M \geq h$, and $0 < \varphi_{\min} \leq \varphi \leq \varphi_{\max}$. Choose $\delta' \in (0, \delta]$ with $|g(z)|_\infty \leq \frac{\delta}{2}$ for $|z|_\infty \leq \delta'$, put $Q' = [-\delta, \delta]^r \times [-\delta', \delta']^s$, and let $\rho_0 = \delta'$, so that every ball of radius at most ρ_0 about 0 lies in Q'.

The lower bound on K; proof of (i). Fix z with $|z|_\infty \leq \delta'$, let $k(u) = L(u, z)$ on the convex cube $[-\delta, \delta]^r$, and $u_z = g(z)$. Since $(u_z, z) \in S$, $\nabla k(u_z) = 0$, and Taylor's formula with integral remainder and $\nabla^2 k \geq \frac{h}{2} I$ give $k(u) \geq k(u_z) + \frac{h}{4} |u - u_z|^2$. The function $z \mapsto L(g(z), z)$ has derivative $\nabla_u L \, Dg + \nabla_z L = 0$ on the connected cube, so it equals $L(0)$. Hence

$$K(u, z) \;\geq\; \frac{h}{4} |u - g(z)|^2 \qquad \text{on } Q'. \tag{14.2}$$

So $K \geq 0$ on Q', and $K(u, z) = 0$ forces $u = g(z)$, a point of S; conversely $K = 0$ at the points $(g(z), z)$, which make up $S \cap Q'$. This is (i).

The upper bound on V_B. By (14.2), for $|z|_\infty \leq \delta'$ the section $\{u : K(u, z) < \varepsilon\}$ lies in the ball of radius $2\sqrt{\varepsilon/h}$ around $g(z)$, of volume $\omega_r (4\varepsilon/h)^{r/2}$, where ω_r is the volume of the unit ball of $\mathbb{R}^r$. By Fubini every ball $B \subseteq Q'$ has $V_B(\varepsilon) \leq \varphi_{\max} (2\delta')^s \omega_r (4/h)^{r/2} \varepsilon^{r/2}$.

The lower bound on V_B. On Q', Taylor's formula from $(g(z), z)$, where K and ∇K vanish, gives $K(u, z) \leq \frac{M}{2} |u - g(z)|^2$. Let B have radius $\rho \leq \rho_0$, and choose $\eta \in (0, \delta']$ with $|g(z)| + |z| \leq \frac{\rho}{2}$ when $|z|_\infty \leq \eta$. For $\varepsilon \leq M\rho^2/8$ the tube $T_\varepsilon = \{(u, z) : |z|_\infty \leq \eta, \; |u - g(z)| < \sqrt{2\varepsilon/M}\}$ lies in B, since $|(u, z)| \leq |u - g(z)| + |g(z)| + |z| < \rho$, and $K < \varepsilon$ on it. By Fubini its volume is $(2\eta)^s \omega_r (2\varepsilon/M)^{r/2}$, so $V_B(\varepsilon) \geq \varphi_{\min} (2\eta)^s \omega_r (2/M)^{r/2} \varepsilon^{r/2}$. This is (ii); by (i) w^* is a local minimum, and by (ii) and Definition 13.1, $\lambda(w^*) = \frac{r}{2}$ and $m(w^*) = 1$. ∎

The proof needs two derivatives and no resolution of singularities. Its upper bound on the volume is that of the Hessian sandwich, Theorem 13.3, which gives $\lambda \geq \frac{r}{2}$ at every local minimum; the manifold of critical points raises the lower bound on the volume to the same order, which is why minimally singular points attain the lower end of the sandwich.

Proposition 14.4 (The polygons of the toy model). *Let ℓ be the loss of the toy model with six features in the high-sparsity limit, $L = \frac{1}{2}\ell$, and let φ be continuous and positive at the points below.*

(a) If (a^*, b^*) is a critical point of ϕ in (14.1) with $b^* < 0 < \frac{a^*}{2} + b^*$, the hexagon with $\|W_i\|^2 = a^*$ and $b_i = b^*$ is a critical point with $\ell = \phi(a^*, b^*)/3$. If its Hessian is positive semidefinite of rank 17, it is a local minimum with $\lambda = \frac{17}{2}$ and $m = 1$.

(b) If the pentagon (a^*, b^*) of Computation 14.3 is critical with $b^* < 0 < a^* \cos 72° + b^*$, then for every $b_6 < 0$ the 5-gon is a critical point; if its Hessian is positive semidefinite of rank 14, it is a local minimum with $\lambda = 7$ and $m = 1$.

(c) Under the hypotheses of (b) the 5^+-gon is a critical point; if its Hessian is positive semidefinite of rank 17, it is a local minimum with $\lambda = \frac{17}{2}$ and $m = 1$.

The script `14-stages-kgons.py` finds the critical points with gradients below $4 \cdot 10^{-15}$, and Hessians with exactly 1, 4 and 1 eigenvalues below 10^{-10} and all others at least 0.37, 1.03 and 1.03 (in units of $H = 18\ell$).

Proof. The plan is to exhibit at each polygon a manifold of critical points of the right dimension and apply Theorem 14.3. In each case strict inequalities fix which case of Computation 14.1 applies to each of the 36 terms, with no kink at an end of $[0, 1]$; they persist on a neighborhood, where H is a finite sum of rational functions with nonvanishing denominators, hence twice continuously differentiable.

(a) By Lemma 14.1 and Computation 14.2 the hexagon is critical. Rotations R_θ of the plane preserve H, so $\nabla H(R_\theta w) = R_\theta \nabla H(w) = 0$, and the orbit $\{R_\theta w\}$ is an embedded circle of critical points, since $W \neq 0$: $s = 1$.

(b) On the neighborhood of Computation 14.3, H does not depend on (W_6, b_6), and in the other coordinates it is the five-feature potential, critical at the pentagon by Lemma 14.1 with the rotation by 72°. The points $(R_\theta W_1, \dots, R_\theta W_5, b_1, \dots, b_5, W_6, b_6)$ with θ and (W_6, b_6) near their values form the image of an injective immersion of an open set of $\mathbb{R} \times \mathbb{R}^2 \times \mathbb{R}$, an embedded four-dimensional manifold of critical points: $s = 4$.

(c) Near the 5^+-gon the sixth output always fires, and its terms are $\sum_{j\le5}\big((W_6 \cdot W_j)^2 + 3(W_6 \cdot W_j)b_6 + 3b_6^2\big) + (1 - \|W_6\|^2)^2 - 3(1 - \|W_6\|^2)b_6 + 3b_6^2$. At $W_6 = 0$ their gradient is $(2W_6 \cdot W_j + 3b_6)W_6 = 0$ in each W_j, $3b_6 \sum_{j\le5} W_j = 0$ in W_6, and $36b_6 - 3 = 0$ in b_6 at $b_6 = \frac{1}{12}$; the outputs $j \le 5$ still never fire for input 6, and the other terms are those of the critical pentagon. So the 5^+-gon is critical, and its rotation orbit gives $s = 1$. ∎

Two steps are numerical: the solution (a^*, b^*) of two equations, and the eigenvalues of three 18×18 matrices. The residual of the computed solution is below 10^{-14} and the Hessian of the reduced function is positive definite there (eigenvalues 0.142 and 8.41 for the hexagon), so by the Newton–Kantorovich theorem, with a numerically estimated bound on the third derivatives of ϕ, an exact solution lies within 10^{-12}; and the gaps between

critical point	ℓ	s	Hessian rank	λ	$\frac{d}{2}$
6-gon	0.048192	1	17	$\frac{17}{2}$	9
5^+-gon	0.061799	1	17	$\frac{17}{2}$	9
5-gon	0.068744	4	14	7	9
4^{++}, 4^+, 4-gon	0.0972, 0.1042, $\frac{1}{9}$	–	–	unknown	9

Table 14.1. The critical points of the toy model with $d = 18$ that training visits: the loss $\ell = \mathbb{E}\|x - f\|^2$ from the closed form (Chen et al. print 0.04819, 0.06180, 0.06874), the dimension s of the set of equally good parameters nearby, and $\lambda = \frac{18-s}{2}$ (Proposition 14.4), computer-assisted here and in the source. The 4-gons lie on walls between chambers, where the potential is not analytic; only estimates of their coefficients exist. Script `14-stages-kgons.py`.

the zero eigenvalues, forced by symmetry and by the dead coordinates, and the smallest positive ones exceed the numerical errors by eight orders of magnitude. The proposition is in this sense computer-assisted, as Chen et al.'s results are: its implications are proved, but their hypotheses are checked in floating point, not certified with interval arithmetic, so the values $\frac{17}{2}$, 7, $\frac{17}{2}$ rest on numerical evidence.

WHAT IS PROVED IN THE SOURCE. Chen et al. (2023) prove the closed form of H (their Lemma 3.1) and, given a solution of two polynomial equations that they compute with Mathematica, that the standard k-gons are critical points for $k \in \{4, \dots, 8\}$ (their Appendix H, with Theorem H.1), and so is the 5^+-gon (Appendix I). The values $\lambda = 7, \frac{17}{2}, \frac{17}{2}$ of their Table 1 rest, like ours, on numerical Hessians: a computed non-degeneracy modulo rotations for the 6-gon (Appendix H.1), a reduction to the five-feature model for the 5-gon (H.3), and computed eigenvalues for the 5^+-gon (Lemma I.1). That c-gons are critical points for $c > 8$ exactly when c is a multiple of 4 is their conjecture. At the 4-gons the potential is not analytic (Appendix J); for four features they report the list $\{4, 4.5, 5, 5.5\}$ "when approached from these different subspaces", and for six features they state that the values are unknown and give SGLD estimates (Remark J.1, Table K.1). Table 14.1 summarizes.

14.4 The theorem: from learning coefficients to stages

What do the learning coefficients say about the order in which a posterior, or a training run, visits the polygons? For the posterior there is a principle with a proof in simple cases; for training, observations and a conjecture.

Phases of the posterior

Split the parameter space into regions U_α, the *phases*, each around one critical point w_α. The posterior mass of a phase is governed by its local free energy $F_n(U_\alpha) = -\log \int_{U_\alpha} \varphi e^{-nL_n}$, and Theorem 12.3 proves, for models linear in their coefficients, that $F_n(U_\alpha) = nL_n(w_\alpha) + \lambda_\alpha \log n - (m_\alpha - 1) \log\log n + O_p(1)$ and that the posterior concentrates on the phase with the smallest leading terms. The toy model is not of that form, and its loss is not analytic

everywhere, so that theorem does not cover it; Chen et al. (2023) assume Watanabe's conditions for each phase and use the same expansion.[2] With the lower-order terms absorbed into constants, $F_n(U_\alpha) \approx nL(w_\alpha) + \lambda_\alpha \log n + c_\alpha$, the posterior switches from phase α to phase β where

$$n\,\Delta L + \Delta\lambda\,\log n + \Delta c = 0, \tag{14.3}$$

with $\Delta L = L(w_\beta) - L(w_\alpha)$ and so on. If $\Delta c = 0$, a phase of lower loss loses to one of higher loss only at small n and only if its learning coefficient is larger: as n grows the posterior moves from simple, inaccurate phases to complex, accurate ones, which Chen et al. call *internal model selection*.

[2] They build on Watanabe's discussions of phase transitions, citing Section 7.6 of Watanabe (2009) and Section 9.4 of Watanabe (2018).

> OBSERVATION 14.5 (Chen et al. 2023, Section 4.2 and Appendices C and F). With a standard Gaussian prior on $\mathbb{R}^{18}$ and the constant $c_\alpha = \frac{1}{2}\|w_\alpha\|^2$ from the prior density at the critical point, the crossing (14.3) of the 5-gon and the 6-gon is at $n_{\text{cr}} \approx 600$ (601 in their Appendix C.2). With NUTS (six chains of 5000 draws after 500 of burn-in, thinned by 10; ten data sets per n), samples classified by the vertices of the convex hull of the columns and the number of positive biases show the dominant phase switch from the 5-gon to the 6-gon between $n = 600$ and $n = 700$; below $n \approx 400$ the classification is unreliable. The authors call it a "fundamental tension" that an asymptotic formula is applied at finite n.

Where the crossing lands depends on a factor of two and on the constants; `14-stages-kgons.py` recomputes it each way. As we read the preprint, its text is inconsistent about the factor: Table 1 and Appendix C.2 use our ℓ, for which Section 3 drops the $\frac{1}{2}$ of the likelihood $p(x \mid w) \propto e^{-\|x-f\|^2/2}$, while Section 4.1 and Appendix F say that the posterior and the sampler use that likelihood, whose energy is nL_n with $L = \frac{\ell}{2}$. With $\frac{\ell}{2}$ the same constants put the crossing at $n \approx 1317$, not 601. We have not seen the authors' code, which may scale the likelihood differently. The constant is a further choice, which Laplace's method makes exact as $n \to \infty$.

LAPLACE'S METHOD ON A CRITICAL SET. Near a compact piece of the set S of Theorem 14.3, $K(u, z)$ is $\frac{1}{2}(u - g(z))^{\mathsf{T}} A(z)(u - g(z))$ up to higher-order terms, with $A(z)$ the Hessian on the normal space. Integrating e^{-nK} over the normal directions gives $(2\pi/n)^{(d-s)/2} \det A(z)^{-1/2}$, and integrating over S against the prior, with $\mathrm{d}\sigma$ the volume on S, gives for a small neighborhood U of S

$$-\log \int_U \varphi e^{-nL} = nL(w^*) + \frac{d-s}{2}\log\frac{n}{2\pi} - \log \int_S \frac{\varphi}{\sqrt{\det A}}\,\mathrm{d}\sigma + o(1).$$

The integral over S is what c_α replaces by the prior density at one point. ◇

For the prior of Chen et al. the integral is computable. The 6-gon's set is 120 circles, one for each cyclic order of the labeled columns, each of length $2\pi\sqrt{6}\,l^*$; the 5-gon's is 144 pieces (a dead column and a cyclic order of the others), each a circle of length $2\pi\sqrt{5}\,l^*$ (the pentagon's l^*) times the region of the dead column's (W_6, b_6) where none of its ReLUs fires, of prior mass

0.0141. With the Hessian determinants of Computations 14.2 and 14.3, the constant becomes -5.81 (energy nL) or -4.77 (energy $n\ell$) instead of $+2.75$, and the 5-gon is preferred only for $88 < n < 225$, or $44 < n < 113$. The predicted crossing thus ranges from about 110 to about 1300, against an observed 600 to 700. The asymptotic constant is not a better finite-n approximation: beyond the walls of its region the loss rises only at third order in the distance, and beyond its tip it falls (Exercise 14.6), so at n of a few hundred the posterior spreads far beyond it. The agreement of 601 with the sampler is an observation about this model at these sample sizes, not a test of the free energy formula.

Stages of stochastic gradient descent

Training does not sample a posterior, and the order of the phases in n need not be the order of plateaus in time. Chen et al. (2023) measured both.

> OBSERVATION 14.6 (Chen et al. 2023, Section 5 and Appendices B and K). Minibatch SGD on $n = 1000$ samples (batch 20, learning rate 0.005, 4500 epochs) from a 4-gon plus Gaussian noise of standard deviation 0.01, with the local learning coefficient estimated along training by full-batch SGLD ($\epsilon = 0.001$, $\gamma = 1$, 500 steps; the inverse temperature is not stated). Plateaus sit at classified critical points, and the transitions seen are $4^{++-} \to 4^{+} \to 5 \to 5^{+}$ and three other sequences of 4-gon variants; each drop of the loss comes with a jump of the estimate, "opposing staircases". In 400 runs started from posterior samples at $n = 100$, lower-loss plateaus have higher estimates; at highly singular points the estimate is negative unless the sampler's hyperparameters are retuned, though its order is still right.

> **Conjecture 14.7** (Bayesian antecedent hypothesis; Chen et al. 2023). *"The dynamical transitions $\alpha \to \beta$ encountered in neural network training have Bayesian antecedents": for each such transition there is a Bayesian phase transition $\alpha \to \beta$ as n grows.*

The authors call it a hypothesis and offer evidence, not a proof: the observed transitions go from higher to lower loss and from lower to higher estimated coefficient, the direction in which (14.3) can have a solution, and an analysis with the constants c_α "suggests" antecedents for all but two, where it is inconclusive (their Appendix D.1). They do not claim the converse, and it fails here: the Bayesian $5 \to 6$ transition has not been seen in SGD (they note that it may occur with low probability per step), and in our runs no trajectory reached the 6-gon (Section 14.5). No theorem connects the plateaus of SGD with the phases of the posterior. For Langevin diffusion at a large inverse temperature on a nonnegative real-analytic potential, Larsson (2026), in a preprint, derives a candidate limiting evolution on the zero set and shows that the Dirichlet forms converge, in a certain sense, to a hierarchy on strata indexed by the local learning coefficient and multiplicity, strongly biased toward the more singular strata. The program's position paper states the bridge it uses, that the preference of SGD at step t evolves as

that of the posterior at some sample size n increasing with t, as a hypothesis (Pepin Lehalleur et al. 2025).

Stages in transformers

In transformers no learning coefficient is known, and the estimate is the only handle.

> OBSERVATION 14.8 (Hoogland et al. 2025). A two-layer attention-only language model (3,355,016 parameters, a subset of the Pile) and a transformer of about 50,000 parameters trained on in-context linear regression (Garg et al. 2022), with the coefficient estimated at checkpoints (for the language model 20 chains of 200 SGLD steps, $\epsilon = 0.001$, $\gamma = 100$, $n\beta = 21.7$). Critical points of the estimated curve divide training into five stages in each model, ending at steps 900, 6.5k, 8.5k, 17k, 50k and 1k, 40k, 126k, 320k, 500k, with changes of the estimate +26.4, +22.5, −1.57, +8.62, +1.77 and +21.4, +149, −12.3, −44.1, +3.56. Most stages coincide with changes in internal structure or behavior, such as the formation of induction heads (Olsson et al. 2022); similar divisions appear for other seeds; the trace and the largest eigenvalue of the Hessian detected only one of the four boundaries of the regression model. The authors call this "suggestive evidence", and note that the checkpoints are typically not local minima, violating the conditions under which the coefficient is defined.

> OBSERVATION 14.9 (Wang et al. 2025). On the same language model, estimates refined to the weights of one attention head, or computed on a different data distribution, separate the heads by the shapes of their curves over training; the clusters match functional types and revealed a previously unknown multigram circuit. The authors write that they cannot guarantee the accuracy of the estimates, whose definition presumes a local minimum, but have reason to believe that their ordering is correct.

Later work keeps this status. Carroll et al. (2025) (a preprint) find transformers on in-context regression first approximating ridge regression and then specializing, and read the estimates as a trade-off between loss and complexity. Elliott et al. (2026) (a preprint) prove that the coefficient of the regret governs a generalized posterior over policies in reinforcement learning, and observe opposing staircases in a gridworld. Baker et al. (2026) turn to susceptibilities, responses of local posterior averages to changes of the data distribution. None of these estimates can be checked against a known coefficient; as of 2026 exact benchmarks at scale exist only for deep linear networks (Hitchcock and Hoogland 2025).

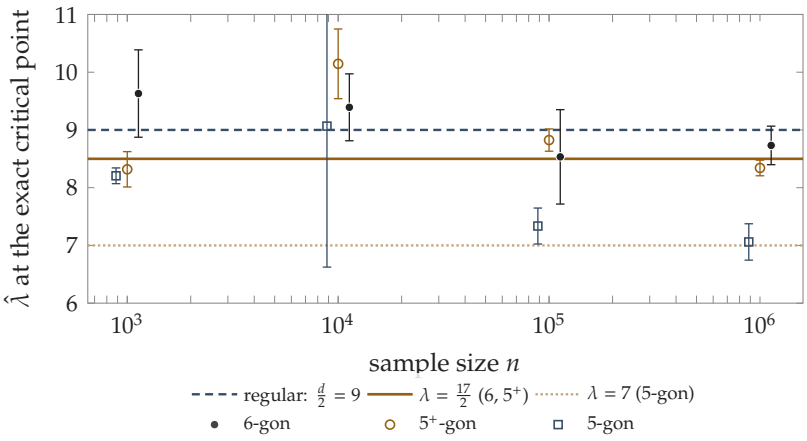


Figure 14.2. The estimator at the three polygons as the sample size grows: for each n and each of 4 data sets (seed $1440 + 10j + r$ for the rth data set at the jth sample size), gradient descent on L_n from the exact polygon to the nearby critical point of L_n, then 4 SGLD chains of 9000 steps (2250 discarded; sampler seed 1450) with $n\beta = n/\log n$, $\gamma = 1$, $\epsilon = 0.1/(n\beta)$; the 5-gon's dead column has bias -3. Means with Student-t 95% intervals over the data sets; points shifted sideways for legibility. Bronze: $\frac{17}{2}$ and 7 (Proposition 14.4); slate: $\frac{d}{2}$. Script `14-stages-calib.py`.

14.5 Measuring it

Three scripts check the chapter on the toy model, where the answers are known. The two that sample use the estimator of Lau et al. (2025) with the book's conventions (Chapter 13): $\hat{\lambda}(w^*) = n\beta\,(\overline{L_n} - L_n(w^*))$, with $\overline{L_n}$ the average of L_n along full-batch SGLD chains for the localized posterior $\propto \exp\{-n\beta L_n(w) - \frac{\gamma}{2}|w - w^*|^2\}$, $n\beta = n/\log n$, $\gamma = 1$ and step $\epsilon = 0.1/(n\beta)$. The empirical loss is exact and fast: for output o and input feature j the sum of $\mathrm{relu}(\mu\, W_o \cdot W_j + b_o)^2$ over the training points of that feature is a quadratic in $W_o \cdot W_j$ and b_o whose coefficients are sums of 1, μ and μ^2 over a tail of the sorted magnitudes, so a step costs $O(\log n)$.

`14-stages-kgons.py` (under 20 seconds) checks every number of Section 14.2; the Fourier blocks reproduce the full spectrum to $7 \cdot 10^{-10}$. Local minimization (L-BFGS) from 300 random starts ends at the loss of the 6-gon 31 times, of the 5^+-gon 60 times and of the 5-gon 75 times, and otherwise at losses of 4-gon variants or of configurations that ignore three or more features; no start found a loss below the 6-gon's.

THE ESTIMATOR AT THE POLYGONS. `14-stages-calib.py` (one to two minutes) runs the estimator at the exact polygons for n from 10^3 to 10^6 (Figure 14.2). At $n = 10^6$ it gives 7.06, 8.34 and 8.73 for the 5-gon, the 5^+-gon and the 6-gon, against 7, $\frac{17}{2}$ and $\frac{17}{2}$: the intervals of the 5-gon and the 6-gon contain the predictions, and that of the 5^+-gon, $[8.21, 8.48]$, ends just below. At smaller n the estimator is far off. At $n = 10^3$ the three read 8.21, 8.32 and 9.63, so the 5-gon and the 5^+-gon are not told apart; with three further sampler seeds the four means range over $[8.21, 8.56]$, $[8.14, 8.41]$ and $[9.11, 9.63]$, and the two pentagons swap places, so the intervals, which

cover the data sets for one seed, understate the uncertainty; at $n = 10^4$ the 5^+-gon and the 6-gon read 10.14 and 9.39, above $\frac{d}{2} = 9$, and the 5-gon 8.0 to 9.0 on three data sets and 11.3 on the fourth; the order is right only from $n = 10^5$ on. The same chains on the closed-form population loss, without data noise, give 8.68, 8.57 and 9.62 at $n\beta = 144.8$ and 7.07, 8.64 and 8.51 at $n = 10^6$: the error at small n comes from the geometry of the loss at the scale of the tempered posterior, not from the data. The Hessians show it. The softest normal direction of the 6-gon has eigenvalue 0.0103 in units of K, so at $n\beta = 145$ the posterior has width $1/\sqrt{1.49} \approx 0.8$ along it, a sizeable fraction of the columns' length 1.32, and the quadratic approximation fails there: with the localizer it predicts 7.69 at $\gamma = 1$ and 5.24 at $\gamma = 10$, where the chains on the population loss read 9.62 at $\gamma = 1$. The dead column of the 5-gon adds the walls of its region, which no quadratic approximation sees.

STAGES OF SGD. `14-stages-tms.py` (under two minutes) repeats the protocol of Chen et al. (2023) on 32 runs: $n = 1000$, minibatches of 20, step 0.005 on ℓ, 4500 epochs (225,000 steps), from the standard 4-gon plus $\mathcal{N}(0, 0.01^2)$ noise (data seeds 1000 to 1031, initialization seeds 2000 to 2031). After 5000 steps every run is on the plateau of a 4-gon variant: 9 at the 4-gon ($\ell = \frac{1}{9}$), 13 at the 4^+-gon (0.1042), 10 at the 4^{++}-gon (0.0972). By the end, 7 of the 13 4^+-runs have dropped to the 5-gon and 2 of the 10 4^{++}-runs to the 5^+-gon, after 32,500 to 135,000 steps; no other run dropped, and none reached the 6-gon. Figure 14.3 shows the two that dropped first. In both, the column that becomes the fifth vertex is a vestigial column with a positive bias near $\frac{1}{12}$ that grows slowly through the plateau, from norm 0.04 to 0.27 over 25,000 steps in run 17, before the loss falls within about 7500 steps: a slow escape along a nearly flat direction, as in the plateaus of Fukumizu and Amari (2000).

Along both runs the estimate steps up when the loss steps down: from a mean of 4.58 over the 4^+ plateau of run 17 (interval [4.32, 4.84] over 10 checkpoints) to 7.56 ([7.41, 7.72], 20 checkpoints) at the 5-gon, and from 5.20 ([5.05, 5.35]) at the 4^{++}-gon of run 13 to 7.83 ([7.62, 8.05]) at the 5^+-gon. These are the opposing staircases of Chen et al., reproduced. The levels must be read against the calibration, not against λ: on the same two data sets the estimator gives 8.21 and 8.00 at the exact 5-gon, 7.64 and 7.94 at the 5^+-gon, and 9.20 and 8.98 at the 6-gon, with chain-to-chain intervals of half-width up to 1.3; the plateau values agree with these. That the 5-gon plateau of run 17 reads 7.56, close to $\lambda = 7$, is no evidence that the estimator measures λ there, since on the same data it reads 7.64 at the 5^+-gon, whose coefficient is $\frac{17}{2}$. What the staircase shows reliably is a step of about 3 between the 4-gon plateaus and the 5-gon plateaus; it does not resolve 7 against $\frac{17}{2}$, and on the 4-gon plateaus, where the loss is not twice differentiable and from which these two runs escape, there is no λ to compare with. The regular count $\frac{d}{2} = 9$ is constant along both runs and wrong on every plateau.

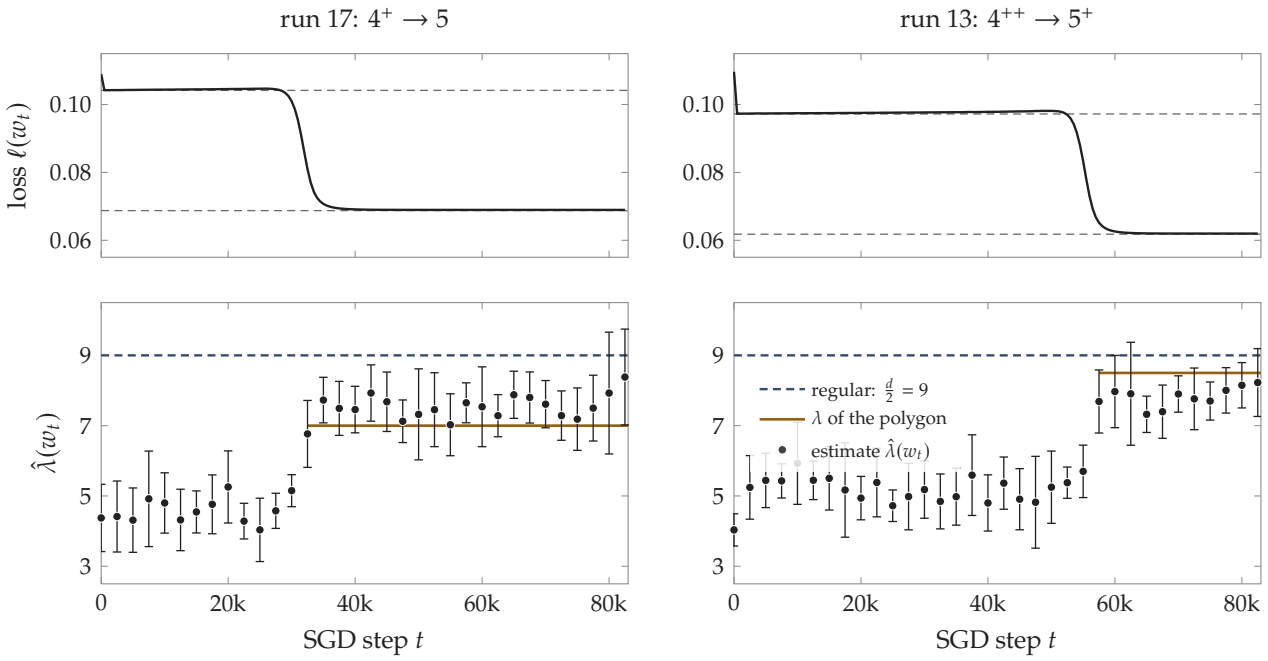


Figure 14.3. Two SGD runs of the toy model ($n = 1000$, batch 20, step 0.005 on ℓ, from the standard 4-gon plus noise), the two of 32 whose loss dropped first. Top: the population loss $\ell(w_t)$ every 500 steps, with the losses of the 4^+-gon and the 5-gon (left) and of the 4^{++}-gon and the 5^+-gon (right) in gray. Bottom: the estimate every 2500 steps from 6 SGLD chains of 6000 steps (1500 discarded; seed 1414) on the training data, $n\beta = 144.8$, $\gamma = 1$, $\epsilon = 6.9 \cdot 10^{-4}$, with Student-$t$ 95% intervals over the chains; bronze: the learning coefficient of the polygon reached; slate: $\frac{d}{2}$. Script `14-stages-tms.py`.

14.6 The ledger

The plateaus of the toy model of superposition are critical points with computable learning coefficients: the count of flat directions gives 7 at the 5-gon and $\frac{17}{2}$ at the 5^+-gon and the 6-gon, and Theorem 14.3 justifies the count whenever the Hessian is positive definite across the manifold of critical points. The posterior orders phases by $nL + \lambda \log n$, with a proof in simple cases, but where transitions are observed the switch point depends on constants and conventions. Along training the estimated coefficient steps up at each drop of the loss in the toy model, and in most stages of the transformers studied; at those sample sizes the estimator misses λ by a unit or more, and that training follows the posterior's order is a conjecture.

LEDGER · the toy model of superposition	
QUANTITY	the local learning coefficient $\lambda(w^*)$ at the critical points that training visits, and its estimate $\hat{\lambda}(w_t)$ along training
REGULAR THEORY	$\frac{d}{2} = 9$ at every nondegenerate minimum; nothing about the order of plateaus

SINGULAR THEORY	$\lambda = \frac{d-s}{2}$, $m = 1$ at a minimally singular critical set of dimension s (Theorem 14.3); phases ordered by $nL + \lambda \log n$ (Theorem 12.3, linear models); Bayesian antecedents conjectured (Conjecture 14.7)
RUNNING EXAMPLE	toy model of superposition, $d = 18$: 5-gon $\lambda = 7$ ($s = 4$); 5^+-gon and 6-gon $\lambda = \frac{17}{2}$ ($s = 1$) (Proposition 14.4)
MEASURED	at the exact polygons $\hat{\lambda} = 7.06, 8.34, 8.73$ at $n = 10^6$, but 8.21, 8.32, 9.63 at $n = 10^3$; along SGD at $n = 1000$, $4.58 \to 7.56$ ($4^+ \to 5$) and $5.20 \to 7.83$ ($4^{++} \to 5^+$); 9 of 32 runs dropped, none to the 6-gon
RESTS ON	a C^2 loss near a manifold of critical points; a Hessian rank computed in floating point; for estimates along training, SGLD at points that are not minima ($n\beta = n/\log n$, $\gamma = 1$); no theorem for SGD

14.7 Where the theory stops

WHAT WAS ASSUMED. The learning coefficients of the polygons need a twice differentiable loss near a manifold of critical points; the toy model provides it at the 5-, 5^+- and 6-gons, inside the chambers of its potential, but not at the 4-gons, which sit on the walls (Exercise 14.5). They also need the rank of an 18×18 Hessian, computed numerically. The Bayesian side needs more: a likelihood, which for this autoencoder is a convention (its normalizing constant depends on w and is dropped); a model that is not realizable, so that K is measured from a local minimum; Watanabe's conditions on each phase, which as far as we know nobody has checked for the toy model; and an asymptotic formula applied at n in the hundreds. The training side assumes that an estimate computed at a point that is not a minimum, from a few thousand SGLD steps with a Gaussian localizer, measures something stable; Hoogland et al. (2025) state that their checkpoints violate the conditions under which the coefficient is defined, and Cai (2026) (a preprint) points out that at transient checkpoints the baseline $L_n(w_t)$ of the estimator is itself ambiguous.

WHERE IT BREAKS. At the sample size of the training runs the estimator does not measure the coefficients computed here: at $n = 1000$ it puts the 5-gon at 8.21 and the 5^+-gon at 8.32, whose values are 7 and $\frac{17}{2}$, and it orders the polygons correctly only from $n \approx 10^5$ (Figure 14.2); the staircase of Figure 14.3 is real, but its heights are set by the geometry of the loss at the scale of the tempered posterior, by the localizer and by the step size. The Bayesian $5 \to 6$ prediction is not robust: the same losses and coefficients give a crossing at $n \approx 601$, 1317, 225 or 113, depending on the energy convention and the constant. The $5 \to 6$ transition, the clearest Bayesian transition of the model, is not seen in SGD (Chen et al. 2023); in our 32 runs no trajectory reached the 6-gon in 225,000 steps, though it is the lowest critical point found. In transformers, stages with decreasing estimates exist, -12.3 and -44.1 in two of the five stages of the regression model (Hoogland et al. 2025), so "loss down, complexity up" is not a rule

of training; and above a threshold of task diversity, transformers trained on in-context regression stay at ridge regression, against what Bayesian internal model selection suggests (Carroll et al. 2025).

OPEN PROBLEMS. (1) A theory of stages for SGD: conditions under which the plateaus of training visit critical points in the order of the posterior's phases, as the Bayesian antecedent hypothesis of Chen et al. (2023) conjectures (Conjecture 14.7), or a proof of the hypothesis that SGD at step t resembles the posterior at some $n(t)$ (Pepin Lehalleur et al. 2025). (2) The local learning coefficients of the 4-gons, where several chambers meet; Chen et al. state that they are unknown for more than four features. (3) Their conjecture that regular c-gons are critical points for $c > 8$ exactly when c is a multiple of 4. (4) Finite-n theory of phase transitions: the constants of local free energies at the sample sizes where transitions are observed, including regions, such as the dead column's, bounded by walls of higher order. (5) Calibrated estimates along training, with intervals that cover bias as well as variance.

Exercises

14.1 CONSOLIDATION. (The integrals.) Derive the cases of the interference and reconstruction integrals in Computation 14.1. Check that the standard 4-gon, with columns $\pm e_1, \pm e_2$, zero biases and two dead columns, has $\ell = \frac{1}{9}$, and that a positive dead bias $\frac{1}{12}$ lowers it by $\frac{1}{144}$.

14.2 CONSOLIDATION. (Modes of the hexagon.) From C_0 and C_1 in Computation 14.2, compute $\hat{C}(0)$ and $\hat{C}(3)$, check that the tangential entry of $\hat{C}(0)$ vanishes, and find the eigenvalues of both. Show that the reflection of the plane combined with the reversal of the indices also preserves H, and deduce a relation between C_1 and $C_1^\top$.

14.3 CONSOLIDATION. (The 5^+-gon.) Show that with $W_6 = 0$ the sixth output of the 5^+-gon contributes $18b_6^2 - 3b_6 + 1$ to H, find its minimum, and show that b_6 decouples in the Hessian with eigenvalue 36. Show that the block of W_6 alone is $(5a^* - 4 + 6b_6)I_2$, and evaluate it.

14.4 CONSOLIDATION. (Two derivatives.) Where does the proof of Theorem 14.3 use that S consists of critical points, and where that the Hessian is positive semidefinite? Give an example with $d = 2$, $s = 1$ in which the Hessian has rank 1 but is not positive semidefinite, and describe the sublevel sets.

14.5 BY HAND. (The standard 4-gon.) For four features, show that the standard 4-gon reconstructs every input exactly, that nearby the parameters with $\ell = 0$ form its circle of rotations, and that along the bias b_1 alone $H = 9b_1^2 + b_1^3$ for $b_1 > 0$ and $3b_1^2 + 2b_1^3$ for $b_1 < 0$. Conclude that H is not twice differentiable there and Theorem 14.3 does not apply.

14.6 BY HAND. (The dead column's region.) At the 5-gon, show that no ReLU involving the sixth feature fires for $\mu \in [0, 1]$ if and only if $b_6 \le 0$,

$\|W_6\|^2 \le |b_6|$ and $W_j \cdot W_6 \le \min(|b_6|, |b^*|)$ for $j \le 5$, and that the condition on $\|W_6\|^2$ follows from the others ($\max_j W_j \cdot W_6 \ge l^* \cos 36° \|W_6\|$). Show that beyond the last walls the loss rises at third order in the distance, and that beyond the tip $W_6 = 0$, $b_6 = 0$ a positive b_6 lowers it at first order. Estimate by Monte Carlo the mass of the region under the standard Gaussian prior on (W_6, b_6) (it is 0.0141).

14.7 BY HAND. (A circle of minima.) For $K(a, b) = (a^2 + b^2 - 1)^2$ with the uniform prior on $[-2, 2]^2$, show that $V(\varepsilon) = \frac{\pi}{8}\sqrt{\varepsilon}$ for $\varepsilon < 1$, and check $\lambda = \frac{1}{2}$, $m = 1$ against Theorem 14.3 with $d = 2$, $s = 1$.

14.8 BY HAND. (The 7-gon.) For seven features, write the reduced function of the regular 7-gon (neighbors interfere; second neighbors have $a \cos \frac{4\pi}{7} < 0$), solve its two equations numerically, and check $l^* = 1.44839$, $b^* = -0.96691$ and $\ell = 0.08327$. What is λ if the Hessian has rank 20?

14.9 VARIATION. (The prior and the crossing.) Replace the Gaussian prior by the uniform prior on a large box containing both polygons, so that Chen et al.'s c_α is the same for all phases, and solve (14.3) for the $5 \to 6$ crossing with $\Delta c = 0$, with the energy $n\ell$ and with $nL = n\ell/2$ (the script prints 445 and 1010).

14.10 MEASUREMENT. (The localizer.) From the eigenvalues of Computation 14.2, divided by 36 for the Hessian of K, compute the localized quadratic prediction $\sum_i \frac{1}{2} \frac{n\beta h_i}{n\beta h_i + \gamma}$ at $n\beta = 144.8$ for $\gamma = 1$ and $\gamma = 10$, and compare with `14-stages-calib.py` at $n = 10^3$ run with each γ. How much of the difference does the localizer explain?

14.11 MEASUREMENT. (Escape.) In `14-stages-calib.py`, set the dead bias of the 5-gon to -0.1 instead of -3, and look at $n = 10^5$. Record for each chain the fraction of steps with L_n below $L_n(w^*)$, and explain the change of the estimate with Computation 14.3 and Exercise 14.6.

14.12 RESEARCH. (The 4-gons.) Compute the local learning coefficient of the standard 4-gon for four features from the volume definition, chamber by chamber. Show that if each chamber's sublevel volume behaves like ε^{λ_C}, then λ is the smallest λ_C, and decide whether the list $\{4, 4.5, 5, 5.5\}$ of Chen et al. consists of such exponents. The case of six features is open.

Notes and References

The toy model of superposition is due to Elhage et al. (2022), who used it to study how networks store more sparse features than they have dimensions, and found the features arranged in polygons and polytopes. Chen et al. (2023) computed its loss in closed form in the high-sparsity limit, classified the polygon critical points with two hidden dimensions, compared Bayesian phase transitions in n with transitions during SGD, and proposed the Bayesian antecedent hypothesis; the preprint was submitted to ICLR 2024

and rejected (OpenReview), and as of October 2026 we found no refereed version. Plateaus near the singularities of multilayer perceptrons are classical (Amari et al. 2006; Fukumizu and Amari 2000).

The local learning coefficient and its SGLD estimator are those of Lau et al. (2025), built on Watanabe's widely applicable Bayesian information criterion (Watanabe 2013) and on stochastic-gradient Langevin dynamics (Welling and Teh 2011). Estimating it along training to find stages is the program of Hoogland et al. (2025), refined to components and data by Wang et al. (2025); the `devinterp` library implements the estimator (van Wingerden et al. 2024), and Hitchcock and Hoogland (2025) benchmark its samplers on deep linear networks.

Critical manifolds along which the Hessian is nondegenerate in the normal directions are the nondegenerate critical manifolds of Bott (1954); that half their codimension is the learning coefficient at a minimum is used by Wei et al. (2023), Section 4, and by Chen et al., and is Proposition 5.5 of this book. Finding symmetric critical points by restricting to symmetric configurations is the principle of symmetric criticality (Palais 1979), which for a group of orthogonal maps reduces to Lemma 14.1; block circulant matrices are diagonalized by the discrete Fourier transform. Theorem 14.3 in this form and its proof, the Fourier computation of the hexagon's Hessian, the Laplace constants of the phases, and the calibration of the estimator at the polygons are the book's own; we did not find them in the literature.

TRANSLATING THE LITERATURE. Chen et al. (2023) write L for our $\ell = \mathbb{E}\|x - f\|^2$, twice our L up to a constant; $H = 3cL$ for the TMS potential (our H, with $c = 6$); r and c for the numbers of hidden dimensions and features; (l, θ, b) for lengths, angles between consecutive columns and biases; and $k^{\sigma+,\phi-}$ for a k-gon with σ positive and ϕ large negative biases (4^{+-} has one of each). Their "non-degenerate" and "minimally singular" are meant modulo rotations and permutations; our s counts the rotations. Lau et al. (2025) write $\beta^* = 1/\log n$ for our β and ϵ for the step size; Hoogland et al. (2025) write ℓ for the population loss and report $n\beta$ directly, and the localizing drift of their SGLD update carries $\gamma/2$ where ours carries γ, although their localized posterior is ours (Chapter 13). In `devinterp`, `nbeta` is our $n\beta$ and `localization` our γ.

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15 SGD Is Not Bayes

What the theory does not explain

THE MODEL — $y = abx$ under gradient descent

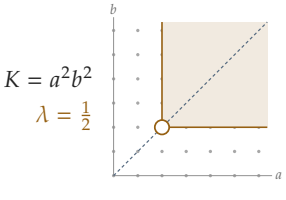
IN THIS CHAPTER. Singular learning theory describes the Bayesian posterior; deep learning trains one network by stochastic gradient descent. On the product model $y = abx$ both can be followed by hand: gradient flow conserves $a^2 - b^2$ and stops at the point of the curve of minimizers that this quantity selects, while the posterior spreads uniformly over scales along the curve. We prove that from the prior's starting points gradient flow ends at the two extreme scales 1 and $n^{-1/2}$, half the time each, from a small start at $n^{-1/4}$, and that finite steps break the conservation law toward balance. Scripts confirm this and show that minibatch noise can collapse the weights onto the singular point and that Langevin dynamics recovers the posterior while minibatch SGLD does not. The chapter ends with what the theory explains about trained networks and what it does not.

15.1 The question: one trained network, or a posterior?

Every theorem of the previous fourteen chapters is a statement about the posterior: an average over all parameters, weighted by the prior and the likelihood, for a fixed model, as the number n of examples grows. Deep learning does something else. It draws one initialization, runs stochastic gradient descent (SGD) for a budget of steps, and keeps the last iterate, often with more parameters than examples.[1] Chapters 13 and 14 used the theory anyway, through the local learning coefficient of a trained point, and said what that use assumes. This chapter asks the prior question: what does gradient descent find, and how is it related to what the posterior averages?

[1] Averaging the iterates is a common exception; Mandt et al. (2017) analyze Polyak averaging from the same stochastic-process view as their approximate inference.

THE REGULAR ANSWER · a trained point and a posterior draw. If the truth is regular, with a unique true parameter w_0 in the interior of W and an invertible Fisher information I, then with probability tending to one the maximum-likelihood estimate $\hat{w}$ is the only minimizer of L_n near w_0, and gradient descent started near w_0 converges to it. By the Bernstein–von Mises theorem (van der Vaart 1998, Theorem 10.1), the posterior law of $\sqrt{n}(w - \hat{w})$ converges in total variation to $\mathcal{N}(0, I^{-1})$, in probability (Chapter 1). The trained point is the center of the posterior, and a posterior draw differs from it by $O_p(n^{-1/2})$: to first order, training and Bayesian inference agree.



The product model breaks this answer twice. Its loss depends on (a, b) only through $c = ab$, so the minimizers of L_n form a curve, the hyperbola $ab = \hat{c}$, and the point that training finds depends on where training starts. And the posterior does not concentrate at a point: at the singular truth, $\log|a|/\log n$ is asymptotically uniform on $[-\frac{1}{2}, 0]$ under the posterior (Theorem 2.5). Training and inference cannot agree to first order; the question is how they differ. Where on the hyperbola does gradient descent stop, and how does the law of its end point compare with the posterior? And beyond this toy: what does singular learning theory say about networks trained by stochastic gradient descent, and what does it not say?

15.2 By hand: gradient flow on $y = abx$

The data enter the product model's loss through $S_{xx} = \sum_i x_i^2$ and $S_{xy} = \sum_i x_i y_i$. Expanding the square as in Chapter 2,

$$L_n(a, b) = \frac{\kappa}{2}(ab - \hat{c})^2 + C_n, \qquad \kappa = \frac{S_{xx}}{n}, \quad \hat{c} = \frac{S_{xy}}{S_{xx}}, \tag{15.1}$$

with C_n independent of (a, b). Gradient descent with step η replaces $w = (a, b)$ by $w - \eta\nabla L_n(w)$; as $\eta \to 0$, the first t/η steps follow the solution of the *gradient flow* $\dot{w} = -\nabla L_n(w)$ up to time t. The flow removes the step size from the problem, and Proposition 15.4 puts it back.

COMPUTATION 15.1 · GRADIENT FLOW ON THE PRODUCT MODEL

1. *The flow.* Differentiating (15.1), $\nabla L_n(a, b) = r\,(b, a)$ with the scaled residual $r = \kappa(ab - \hat{c})$; so $\dot{a} = -rb$ and $\dot{b} = -ra$.
2. *A conserved quantity.* $\frac{\mathrm{d}}{\mathrm{d}t}(a^2 - b^2) = 2a\dot{a} - 2b\dot{b} = -2rab + 2rab = 0$. So $D = a^2 - b^2$ keeps its initial value: each trajectory runs along one hyperbola $a^2 - b^2 = D$, or along one of the lines $a = \pm b$ when $D = 0$.
3. *Coordinates that decouple.* In $u = a + b$ and $v = a - b$ the flow reads $\dot{u} = -ru$ and $\dot{v} = rv$, so $u(t) = u(0)e^{-R(t)}$ and $v(t) = v(0)e^{R(t)}$ with $R(t) = \int_0^t r$. Neither u nor v changes sign, and $uv = D$.
4. *The product.* $c = ab$ obeys $\dot{c} = \dot{a}b + a\dot{b} = -\kappa(c - \hat{c})(a^2 + b^2)$, and $a^2 + b^2 = \sqrt{D^2 + 4c^2} \geq |D|$. If $D \neq 0$, then $|c(t) - \hat{c}| \leq |c(0) - \hat{c}|\, e^{-\kappa|D|t}$: the product reaches $\hat{c}$ exponentially fast.
5. *The end point.* The limit (a_∞, b_∞) has $a_\infty b_\infty = \hat{c}$ and $a_\infty^2 - b_\infty^2 = D$. Solving, $a_\infty^2 = \frac{1}{2}(D + \sqrt{D^2 + 4\hat{c}^2})$ and $b_\infty^2 = \frac{1}{2}(-D + \sqrt{D^2 + 4\hat{c}^2})$, with the signs of u and v kept from the start.
6. *Numbers.* Take $\kappa = 1$ and $\hat{c} = 0.01$, the size of $\hat{c}$ for 10^4 examples of pure noise. From $(0.6, 0.3)$, where $D = 0.27$, the flow stops at $(0.5200, 0.0192)$; from $(0.3, 0.6)$ at $(0.0192, 0.5200)$; from $(0.6, 0.6)$, on the diagonal, at the balanced point $(0.1, 0.1)$; and from $(0.6, -0.6)$, on the line $u = 0$, it runs into the origin, a saddle point. From the small start $(0.001, 0.002)$, where $D = -3 \times 10^{-6}$, it

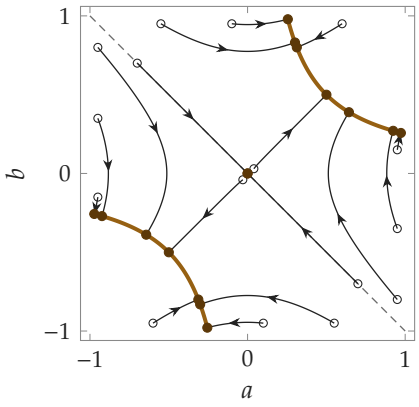


Figure 15.1. Gradient flow on $L_n = \frac{\kappa}{2}(ab - \hat{c})^2$ with $\kappa = 1$ and $\hat{c} = 0.25$, a value chosen large for visibility. Thin lines: the flow from sixteen starts (circles) to its end points (dots), each along a hyperbola $a^2 - b^2 = \text{const}$, from the closed form of Theorem 15.2. Bronze: the minimizers $ab = \hat{c}$. Dashed: the line $a = -b$, along which the flow runs into the saddle point at the origin; starts with $a + b > 0$ end on the upper branch, starts with $a + b < 0$ on the lower one. Script: `15-sgd-product.py`.

stops at $(0.09999, 0.10001)$, after a plateau: c grows like $c(0)e^{2\kappa\hat{c}t}$ from 2×10^{-6} and is halfway to $\hat{c}$ near $t = \log(\hat{c}/c(0))/(2\kappa\hat{c}) = 426$ (numerically, at $t = 420$).

What one had to notice is the symmetry. The rescalings $(a, b) \mapsto (sa, b/s)$ leave the loss unchanged; the vector $(a, -b)$ that generates them is the gradient of $\frac{1}{2}(a^2 - b^2)$ and is orthogonal to $\nabla L_n = r\,(b, a)$ at every point, so the flow can never change $\frac{1}{2}(a^2 - b^2)$. This is Noether's theorem in its simplest form, and it is how Saxe et al. (2014) found the conserved quantity for deep linear networks, whose dynamics in each mode is exactly this flow (Figure 15.1).[2]

The end point depends on the start only through D and the signs of $a + b$ and $a - b$. To compare it with the posterior we need a coordinate along the fiber $\{ab = c\}$, the set of parameters that compute the function $y = cx$.[3]

COMPUTATION 15.2 · TWO BIASES AT THE SINGULAR TRUTH

1. *A coordinate along the fiber.* On the fiber $\{ab = c\}$, $c \neq 0$, write $|a| = |c|^{1/2}e^{\theta}$ and $|b| = |c|^{1/2}e^{-\theta}$. The hyperbolic angle $\theta = \frac{1}{2}\log|a/b|$ says where on the fiber a point lies, and $a^2 - b^2 = 2|c|\sinh 2\theta$.
2. *Where the posterior puts θ.* Given c, the uniform prior restricted to the fiber makes $\log|a|$ uniform on $[\log|c|, 0]$ (Chapter 2), so θ is uniform on $[-\frac{1}{2}\log\frac{1}{|c|}, \frac{1}{2}\log\frac{1}{|c|}]$: the posterior spreads evenly along the whole fiber inside the prior box.
3. *Where gradient flow puts θ.* By step 5 of Computation 15.1, $2|\hat{c}|\sinh 2\theta_\infty = D$, so $\theta_\infty = \frac{1}{2}\operatorname{arcsinh}\frac{D}{2|\hat{c}|}$: one point, fixed by the start and the data.
4. *At the singular truth.* There $|\hat{c}|$ is of order $n^{-1/2}$, and the posterior's interval of θ has length about $\frac{1}{2}\log n$. A start drawn from the prior has D of order 1; since $\operatorname{arcsinh} x = \pm\log(2|x|) + o(1)$ as $x \to \pm\infty$, θ_∞ lies within $\frac{1}{2}|\log|D||$ of an end of that interval: $|a_\infty| \approx |D|^{1/2}$ if

[2] Saxe, McClelland and Ganguli write the error as $\frac{1}{2\tau}(s - ab)^2$, with s the strength of an input–output mode and τ a time constant, note that the dynamics "follows hyperbolas of constant $a^2 - b^2$", and solve it in hyperbolic coordinates in their Appendix A.

[3] For $c > 0$ the branch $a, b > 0$ of the fiber is $(a, b) = \sqrt{c}\,(e^{\theta}, e^{-\theta})$, as $(\cosh\theta, \sinh\theta)$ traces $x^2 - y^2 = 1$; hence the name hyperbolic angle for θ.

$D > 0$ and $|a_\infty| \approx |\hat{c}|/|D|^{1/2}$ if $D < 0$. A start so small that $|D| \ll |\hat{c}|$ has $\theta_\infty \approx 0$, the middle, where $|a_\infty| \approx |b_\infty| \approx |\hat{c}|^{1/2}$, of order $n^{-1/4}$.

5 *Numbers.* For $n = 10^6$ and a data set with $\kappa = 1$ and $\hat{c} = 10^{-3}$, the posterior gives the event $|a| > 0.1$ probability 0.324; gradient flow from a start uniform on $[-1, 1]^2$ gives it probability 0.483, the probability that $D > 0.0099$; gradient flow from a small start never reaches it, since then $|a_\infty| = 0.0316$. By symmetry, the event $\max(|a|, |b|) > 0.1$ has twice these probabilities: 0.647 and 0.965.

Both procedures end near the fiber $ab = \hat{c}$, so they agree about the function, $y = \hat{c}x$ up to $O_p(n^{-1/2})$; they disagree about the parameters. The posterior weights each point of the fiber by the prior volume of the nearly true parameters around it and forgets every start; gradient flow ignores volume and remembers its start through D. The balanced point is the posterior's center in logarithmic coordinates, since given c the posterior mean of $\log|a|$ is $\frac{1}{2}\log|c|$, but the posterior puts no more mass near it than at other scales. The two biases also differ in sharpness: at a point of the fiber the Hessian of L_n has the eigenvalues 0 and $\kappa(a^2 + b^2) = \kappa\sqrt{D^2 + 4\hat{c}^2}$ (Exercise 15.2), so the end point reached from $(0.6, 0.3)$ in Computation 15.1 is 13.5 times sharper than the balanced point, though both compute $y = 0.01x$; with this rescaling Dinh et al. (2017) showed that sharpness is not a property of the function a network computes.

15.3 The theorem: gradient flow conserves $a^2 - b^2$

The computation becomes a theorem once we know that the flow exists for all time and converges, and which starts are exceptional. One fact about scalar equations does the work.

Lemma 15.1 (A monotone scalar equation). *Let $g\colon \mathbb{R} \to \mathbb{R}$ be continuously differentiable and strictly decreasing, and let R solve $\dot{R} = g(R)$, $R(0) = 0$, on its maximal interval of existence. If g has a zero R^*, then R exists for all $t \geq 0$, is monotone, and converges to R^*. If $g > 0$ on $\mathbb{R}$, then R exists for all $t \geq 0$ and increases to $+\infty$; if $g < 0$ on $\mathbb{R}$, then it exists for all $t \geq 0$ and decreases to $-\infty$.*

Proof. The plan is to show that R moves monotonically toward the zero of g and cannot stop short of it. By the Picard–Lindelöf theorem the equation has a unique solution on a maximal interval $[0, T)$, and if $T < \infty$, then $|R(t)| \to \infty$ as $t \to T$. Suppose $g(R^*) = 0$ with $R^* > 0$. The constant R^* is a solution, so by uniqueness $R(t) < R^*$ for all t; then $\dot{R} = g(R) > 0$, so R increases and stays in $[0, R^*)$, which forces $T = \infty$. Its limit $L \leq R^*$ equals R^*: otherwise $\dot{R} \geq g(L) > 0$ for all t and $R(t) \geq g(L)\,t \to \infty$. The case $R^* < 0$ is symmetric, and $R \equiv 0$ if $R^* = 0$. If $g > 0$ on $\mathbb{R}$, then R increases, so $R \geq 0$ and $\dot{R} \leq g(0)$; thus $R(t) \leq g(0)\,t$ cannot blow up, $T = \infty$, and a bounded R would converge to some L with $\dot{R} \geq g(L) > 0$ for all t, which is absurd. The case $g < 0$ is symmetric. ∎

Theorem 15.2 (Gradient flow on the product model). *Let (x_i, y_i), $i = 1, \dots, n$, be data with $S_{xx} > 0$ and $\hat{c} \neq 0$, with κ and $\hat{c}$ as in (15.1), and let $w(t) = (a(t), b(t))$ solve the gradient flow $\dot{w} = -\nabla L_n(w)$ from $w(0) = (a_0, b_0) \in \mathbb{R}^2$. Put $D = a_0^2 - b_0^2$ and let $E = \{(a, b) : a = -\operatorname{sgn}(\hat{c})\, b\}$.*

(i) Conservation. *The solution exists for all $t \geq 0$, and $a(t)^2 - b(t)^2 = D$ for every t.*

(ii) Convergence. *As $t \to \infty$, $w(t)$ converges to a critical point $w_\infty = (a_\infty, b_\infty)$ of L_n. For every start outside the line E, a set of measure zero, $a_\infty b_\infty = \hat{c}$, so that w_∞ minimizes L_n; from a start on E the flow converges to the origin, a saddle point of L_n.*

(iii) The limit point. *If $w(0) \notin E$, then w_∞ is the unique point with $a_\infty b_\infty = \hat{c}$, $a_\infty^2 - b_\infty^2 = D$, $\operatorname{sgn}(a_\infty + b_\infty) = \operatorname{sgn}(a_0 + b_0)$ and $\operatorname{sgn}(a_\infty - b_\infty) = \operatorname{sgn}(a_0 - b_0)$, and*

$$a_\infty^2 = \frac{D + \sqrt{D^2 + 4\hat{c}^2}}{2}, \qquad b_\infty^2 = \frac{-D + \sqrt{D^2 + 4\hat{c}^2}}{2}; \tag{15.2}$$

equivalently, $\log|a_\infty / b_\infty| = \operatorname{arcsinh}\big(D/(2|\hat{c}|)\big)$.

(iv) The scale at the singular truth. *Let the x_i and y_i be independent standard normal variables (the product model at its singular truth), and let the start $w(0) = w_0^{(n)}$ be random and independent of the data, with $\mathbb{P}(D = 0) = 0$ and $\sqrt{n}\, D \to 0$ in probability. Then $\log|a_\infty|/\log n \to -\frac{1}{4}$ and $\log|b_\infty|/\log n \to -\frac{1}{4}$ in probability.*

Proof. The plan is to pass to the coordinates $u = a + b$, $v = a - b$ of Computation 15.1, in which the flow is multiplicative, and to reduce it to one scalar equation, to which Lemma 15.1 applies. Write $u_0 = a_0 + b_0$ and $v_0 = a_0 - b_0$, so that $u_0 v_0 = D$.

Reduction. Since $\partial_a L_n = -\frac{1}{n}\sum_i (y_i - abx_i)x_i b = \kappa(ab - \hat{c})\, b$, and likewise for b, the flow is $\dot{a} = -r(a,b)\, b$, $\dot{b} = -r(a,b)\, a$ with $r(a,b) = \kappa(ab - \hat{c})$. For $R \in \mathbb{R}$ let

$$g(R) = \kappa\Big(\tfrac{1}{4}\big(u_0^2 e^{-2R} - v_0^2 e^{2R}\big) - \hat{c}\Big),$$

let R solve $\dot{R} = g(R)$, $R(0) = 0$, on its maximal interval $[0, T)$, and put $u = u_0 e^{-R}$, $v = v_0 e^{R}$, $a = \frac{1}{2}(u + v)$ and $b = \frac{1}{2}(u - v)$. Then $ab = \frac{1}{4}(u^2 - v^2)$, so $\dot{R} = r(a,b)$, and

$$\dot{a} = \tfrac{1}{2}(-\dot{R}u + \dot{R}v) = -r(a,b)\, b, \qquad \dot{b} = \tfrac{1}{2}(-\dot{R}u - \dot{R}v) = -r(a,b)\, a.$$

So (a, b) solves the flow from (a_0, b_0) on $[0, T)$. The vector field $-\nabla L_n$ is a polynomial, hence locally Lipschitz, so this is the solution, and along it $a^2 - b^2 = uv = u_0 v_0 = D$.

Cases. If $u_0 = v_0 = 0$, the start is the origin, a critical point, and the flow stays there. Otherwise $g'(R) = -\frac{\kappa}{2}(u_0^2 e^{-2R} + v_0^2 e^{2R}) < 0$, so g is strictly decreasing. As $R \to -\infty$, $g(R) \to +\infty$ if $u_0 \neq 0$ and $g(R) \to -\kappa\hat{c}$ if $u_0 = 0$; as $R \to +\infty$, $g(R) \to -\infty$ if $v_0 \neq 0$ and $g(R) \to -\kappa\hat{c}$ if $v_0 = 0$. (a) If $u_0 \neq 0$ and $v_0 \neq 0$, then g takes every real value and has a zero R^*. By Lemma 15.1,

$T = \infty$ and $R \to R^*$, so $u \to u_\infty = u_0 e^{-R^*}$ and $v \to v_\infty = v_0 e^{R^*}$, both nonzero, with $u_\infty v_\infty = D$ and, since $g(R^*) = 0$, $\frac{1}{4}(u_\infty^2 - v_\infty^2) = \hat{c}$. (b) If $v_0 = 0 \neq u_0$, the start lies on the diagonal $a = b$, and g decreases from $+\infty$ to $-\kappa\hat{c}$. If $\hat{c} > 0$, then g has a zero, and as in (a) $u \to u_\infty$ with $u_\infty^2 = 4\hat{c}$ and $\operatorname{sgn} u_\infty = \operatorname{sgn} u_0$, while $v \equiv 0$. If $\hat{c} < 0$, then $g > 0$, $R \to +\infty$ by the lemma, and $u \to 0$: the flow converges to the origin. (c) If $u_0 = 0 \neq v_0$, the start lies on the line $a = -b$, and symmetrically: if $\hat{c} < 0$, then $v \to v_\infty$ with $v_\infty^2 = -4\hat{c}$ and $\operatorname{sgn} v_\infty = \operatorname{sgn} v_0$, while $u \equiv 0$; if $\hat{c} > 0$, then $R \to -\infty$ and the flow converges to the origin. In every case $T = \infty$, which completes (i), and $w(t)$ converges.

Identification. In (a), in (b) with $\hat{c} > 0$, and in (c) with $\hat{c} < 0$, the limit satisfies $a_\infty b_\infty = \frac{1}{4}(u_\infty^2 - v_\infty^2) = \hat{c}$ and $a_\infty^2 - b_\infty^2 = u_\infty v_\infty = D$, with $\operatorname{sgn} u_\infty = \operatorname{sgn} u_0$ and $\operatorname{sgn} v_\infty = \operatorname{sgn} v_0$. The remaining starts, those of (b) with $\hat{c} < 0$, those of (c) with $\hat{c} > 0$, and the origin, make up the line E, and from them the flow converges to the origin. There $\nabla L_n = 0$, and $\nabla^2 L_n(0)$ has the entries 0 on the diagonal and $-\kappa\hat{c}$ off it, with eigenvalues $\pm\kappa\hat{c}$: a saddle point. This proves (ii). The conditions in (iii) determine one point: $uv = D$ and $u^2 - v^2 = 4\hat{c}$ make u^2 and $-v^2$ the two roots of $X^2 - 4\hat{c}X - D^2 = 0$, so $u^2 = 2\hat{c} + \sqrt{D^2 + 4\hat{c}^2}$ and $v^2 = -2\hat{c} + \sqrt{D^2 + 4\hat{c}^2}$, and the signs of u and v are prescribed. Since $a^2 = \frac{1}{4}(u^2 + v^2 + 2uv)$ and $b^2 = \frac{1}{4}(u^2 + v^2 - 2uv)$, this is (15.2). With $x = D/(2|\hat{c}|)$ it gives $a_\infty^2/b_\infty^2 = (x + \sqrt{1+x^2})/(-x + \sqrt{1+x^2}) = (x + \sqrt{1+x^2})^2$, and $\log(x + \sqrt{1+x^2}) = \operatorname{arcsinh} x$.

The scale. Under the singular truth, given $x_1, \dots, x_n$, the statistic S_{xy} is normal with mean 0 and variance S_{xx}; so $Z_n = S_{xy}/\sqrt{S_{xx}} \sim \mathcal{N}(0,1)$ exactly (Chapter 2), $\hat{c} = Z_n/\sqrt{S_{xx}} \neq 0$ almost surely, and $S_{xx}/n \to 1$ almost surely by the strong law of large numbers. Since $E \subset \{D = 0\}$, the start avoids E almost surely, and (iii) gives

$$\log|a_\infty| = \tfrac{1}{2}\log|\hat{c}| + \tfrac{1}{2}\operatorname{arcsinh}\frac{D}{2|\hat{c}|}, \qquad \log|b_\infty| = \tfrac{1}{2}\log|\hat{c}| - \tfrac{1}{2}\operatorname{arcsinh}\frac{D}{2|\hat{c}|}.$$

The argument of arcsinh is $\sqrt{n}D \cdot \sqrt{S_{xx}/n}\,/(2|Z_n|)$: a factor that tends to 0 in probability, one that tends to 1, and one that is bounded in probability, because Z_n has the same law for every n and no atom at 0. So it tends to 0 in probability, and so does its arcsinh. And $\frac{1}{2}\log|\hat{c}| = \frac{1}{2}\log|Z_n| - \frac{1}{4}\log S_{xx}$, where $\log|Z_n|$ is bounded in probability and $\log S_{xx}/\log n \to 1$. Dividing by $\log n$ proves (iv). ∎

The same case analysis handles $\hat{c} = 0$, an event of probability zero (Exercise 15.3). Near w_∞ the residual decays like $e^{-\kappa\sqrt{D^2+4\hat{c}^2}\,t}$, at the rate of the nonzero eigenvalue of the Hessian there, which is smallest at the balanced point, where it is $2\kappa|\hat{c}|$, of order $n^{-1/2}$ at the singular truth; from a small start the flow first waits on a plateau of length about $\log(|\hat{c}|/|c(0)|)/(2\kappa|\hat{c}|)$, the plateau that the Amari school found near the singularities of networks.[4] Numerical integration of the flow from 24 starts on four data sets of size 10^4 agrees with (15.2) to 6×10^{-13} (`15-sgd-product.py`). Part (iv) needs a start much smaller than $n^{-1/4}$; starts of order 1 end at the two ends of the posterior's range of scales.

[4] Fukumizu and Amari (2000) showed that critical points of a network with fewer hidden units give lines of critical points of a larger one, which can cause plateaus; Chapter 2 has the history.

Corollary 15.3 (Two biases). *In the setting of Theorem 15.2(iv), let instead the start have a law μ that does not depend on n, with $\mu\{D=0\}=0$. Then $\log|a_\infty|/\log n + \frac{1}{2}\mathbf{1}\{D<0\} \to 0$ in probability. In particular, if μ is the prior, uniform on $[-1,1]^2$, then $\log|a_\infty|/\log n$ converges in law to $\frac{1}{2}\delta_{-1/2}+\frac{1}{2}\delta_0$, whereas under the posterior $\log|a|/\log n$ converges in law to $\mathrm{Unif}(-\frac{1}{2},0)$ (Theorem 2.5), and from a small start it converges to $-\frac{1}{4}$ (Theorem 15.2(iv)).*

Proof. The plan is to expand arcsinh for large arguments in the formula for $\log|a_\infty|$ of the last proof. For $x>0$, $\operatorname{arcsinh} x = \log(2x) + \log\frac{1+\sqrt{1+x^{-2}}}{2}$, and the last term lies in $[0,\frac{1}{4x^2}]$; and arcsinh is odd. Put $x_n = |D|/(2|\hat{c}|) = \sqrt{n}|D|\sqrt{S_{xx}/n}\,/(2|Z_n|)$. Since D has a fixed law without an atom at 0 and $|Z_n|$ is bounded in probability, $x_n \to \infty$ in probability. On $\{D>0\}$ this gives $\log|a_\infty| = \frac{1}{2}\log|\hat{c}| + \frac{1}{2}\log(D/|\hat{c}|) + o_p(1) = \frac{1}{2}\log D + o_p(1)$, which is bounded in probability; on $\{D<0\}$ it gives $\log|a_\infty| = \log|\hat{c}| - \frac{1}{2}\log|D| + o_p(1)$, and $\log|\hat{c}|/\log n \to -\frac{1}{2}$ in probability. Dividing by $\log n$ proves the first claim. For the prior, $\mu\{D<0\} = \mu\{D>0\} = \frac{1}{2}$ by the symmetry $(a,b)\mapsto(b,a)$, and $-\frac{1}{2}\mathbf{1}\{D<0\}$ has the law $\frac{1}{2}\delta_{-1/2}+\frac{1}{2}\delta_0$. ∎

So the middle half of the scales, $-\frac{3}{8} < \log|a|/\log n \le -\frac{1}{8}$, gets probability tending to 0 from gradient flow with the prior's starts, $\frac{1}{2}$ from the posterior, and 1 from gradient flow with a small start; starts of size $n^{-\gamma}$ interpolate (Exercise 15.7).

Finite steps. Gradient descent takes steps of a fixed size, often on mini-batches. The conservation law does not survive, but it fails in one direction, and the failure is exact.

Proposition 15.4 (Finite steps break the law toward balance). *For a nonempty list B of indices, repetitions allowed, let $L_B(a,b) = \frac{1}{2|B|}\sum_{i\in B}(y_i - abx_i)^2 + \frac{1}{2}\log(2\pi)$, and let $w_{t+1} = w_t - \eta\nabla L_{B_t}(w_t)$ with a step $\eta>0$ and minibatches B_t chosen in an arbitrary way ($B_t = \{1,\dots,n\}$ for gradient descent). With $r_t = \frac{1}{|B_t|}\sum_{i\in B_t}(x_i^2 a_t b_t - x_i y_i)$, $u_t = a_t + b_t$, $v_t = a_t - b_t$ and $D_t = a_t^2 - b_t^2$,*

$$u_{t+1} = (1-\eta r_t)\,u_t, \qquad v_{t+1} = (1+\eta r_t)\,v_t, \qquad D_{t+1} = (1-\eta^2 r_t^2)\,D_t. \tag{15.3}$$

Consequently: (i) the lines $a=b$ and $a=-b$, and the origin, are invariant; (ii) if $D_t\neq 0$ and $0<\eta^2 r_t^2<2$, then $|D_{t+1}|<|D_t|$; (iii) if $\eta|r_t|\le 1$ for all t, then $|D_t| \le |D_0|\exp(-\eta^2\sum_{s<t} r_s^2)$, so that $D_t\to 0$ whenever $\sum_t r_t^2 = \infty$.

Proof. As for L_n, $\nabla L_B(a,b) = r\,(b,a)$ with $r = \frac{1}{|B|}\sum_{i\in B}(x_i^2 ab - x_i y_i)$. Hence $a_{t+1}+b_{t+1} = a_t + b_t - \eta r_t(b_t+a_t)$ and $a_{t+1} - b_{t+1} = a_t - b_t + \eta r_t(a_t - b_t)$, which are the first two identities, and $D_{t+1} = u_{t+1}v_{t+1}$ gives the third. Part (i) follows from the first two, (ii) from $|1-x|<1$ for $0<x<2$, and (iii) from $0\le 1-x\le e^{-x}$ for $0\le x\le 1$. ∎

For gradient descent that converges to a point of the fiber, $r_t\to 0$ geometrically and the end point keeps the fraction $\prod_t(1-\eta^2 r_t^2)$ of D,

close to 1 for small η: from $(0.6, 0.3)$ in Computation 15.1, with $\eta = 0.1$, it keeps 99.6%.[5] Minibatches change this. At a minimizer of L_n, r_t is the disagreement of the minibatch with the full batch, of mean square about $\sigma_{xy}^2/|B|$ for minibatches of $|B|$ examples drawn with replacement, with σ_{xy}^2 the empirical variance of the products $x_i y_i$ (about 1 at the singular truth). So $|D_t|$ decays roughly like $e^{-\eta^2\sigma_{xy}^2 t/|B|}$: SGD forgets its start on the time scale $|B|/(\eta^2\sigma_{xy}^2)$ and drifts along the fiber to the balanced point. Kunin et al. (2021) showed that finite learning rates break the conservation laws of gradient flow, and Wang et al. (2022) that gradient descent with a large learning rate on matrix factorization ends nearly balanced. For full batches, (15.3) is the one-unit case of an identity in the proofs of Du et al. (2018, Appendix B.1); it gives the exact form of both effects for one unit. Once balanced, SGD meets a second effect.

[5] With momentum, $u_{t+1} = (1 - \eta r_t + \mu)u_t - \mu u_{t-1}$, so the lines $a = \pm b$ stay invariant when the start has no velocity, but D_t no longer changes by a single factor.

Heuristic derivation of stochastic collapse. On the balanced lines (15.3) gives $c_{t+1} = c_t(1 - \operatorname{sgn}(c_t)\,\eta r_t)^2$: the product evolves multiplicatively and keeps its sign. Near $c = 0$, $r_t \approx -\frac{1}{|B|}\sum_{i\in B} x_i y_i$ has mean $-\kappa\hat{c}$ and variance $\sigma_{xy}^2/|B|$ over minibatches, so to second order in η the expected change of $\log|c_t|$ per step is

$$\mathbb{E}\left[2\log|1 - \operatorname{sgn}(c_t)\,\eta r_t|\right] \approx 2\eta\kappa\,\operatorname{sgn}(c_t)\,\hat{c} - \frac{\eta^2\sigma_{xy}^2}{|B|}.$$

If $\operatorname{sgn}(c_t) = \operatorname{sgn}(\hat{c})$ and $|\hat{c}| > \eta\sigma_{xy}^2/(2\kappa|B|)$, then $\log|c_t|$ drifts up while c_t is small, and SGD fluctuates along the balanced line. With the full mean $\kappa(c_t - \hat{c})$ of r_t in place of $-\kappa\hat{c}$, the drift averages to zero where the mean of $|c_t|$ is about $|\hat{c}| - \eta\sigma_{xy}^2/(2\kappa|B|)$ (compare Exercise 15.8). If $|\hat{c}| < \eta\sigma_{xy}^2/(2\kappa|B|)$, then it drifts down: $c_t \to 0$, and the weights collapse onto the origin, the most singular point of the cross. The balanced line opposite to $\hat{c}$ is the invariant line E of Theorem 15.2, on which $c_t \to 0$ whatever $\hat{c}$; near it, above the threshold, SGD passes near the origin and leaves along the balanced line on the side of $\hat{c}$, as gradient flow does near E. ◇

At the singular truth $|\hat{c}|$ is of order $n^{-1/2}$, so for a fixed step and minibatch size the collapse condition holds with probability tending to one as n grows. This is the simplest instance of the *stochastic collapse* of Chen et al. (2023a), who give a sufficient condition, a competition between curvature and gradient noise, under which SGD is attracted to invariant sets of simpler subnetworks, such as vanishing units; here the invariant set is the origin, a saddle point. The posterior gives the scales below $n^{-1/2}$ vanishing mass (Theorem 2.5), and the origin none.

15.4 The theorem: what transfers from posteriors to training

On the product model the comparison is exact. For networks the literature asks three questions: whether some training algorithm samples the posterior, whether SGD as practiced does, and whether SGD nevertheless prefers what the posterior prefers. We state each result as its authors state it.

LANGEVIN DYNAMICS. The Langevin diffusion of Section A.7 adds noise to gradient flow so that the posterior becomes its equilibrium.

Proposition 15.5 (The tempered posterior is invariant under Langevin dynamics). *Let $U\colon \mathbb{R}^d \to \mathbb{R}$ be twice continuously differentiable with $\int e^{-U} < \infty$, and let $p_\infty = e^{-U}/\int e^{-U}$. Then p_∞ is a stationary solution of the Fokker–Planck equation $\partial_t p = \frac{1}{2}\nabla\cdot(p\nabla U) + \frac{1}{2}\Delta p$ of the diffusion $\mathrm{d}w = -\frac{1}{2}\nabla U(w)\,\mathrm{d}t + \mathrm{d}B_t$. For $U = n\beta L_n - \log\varphi$ with a smooth positive prior φ, p_∞ is the tempered posterior at inverse temperature β.*

PROOF. Since $\nabla p_\infty = -p_\infty \nabla U$, the field $p_\infty\nabla U + \nabla p_\infty$ vanishes, and so does its divergence, $\nabla\cdot(p_\infty\nabla U) + \Delta p_\infty$. ∎

Stationarity is the easy half: convergence to p_∞ needs growth conditions on U, the Euler scheme with a fixed step has an invariant law of its own (Chapter 13), and a uniform prior on a box makes the diffusion reflect at the boundary.[6] On the product model one Euler step $w \mapsto w - \frac{\epsilon}{2}\nabla U(w) + \sqrt{\epsilon}\,\xi$ is a step of gradient descent with $\eta = \epsilon n/2$ plus Gaussian noise of variance $\epsilon = 2\eta/n$ in each coordinate, and the noise is what moves the iterate along the fiber, where the gradient vanishes. Welling and Teh (2011) replaced ∇U by a minibatch estimate, which gives *stochastic-gradient Langevin dynamics* (SGLD), and argued that with step sizes annealed to zero the iterates converge to posterior samples; consistency for decreasing steps was proved later (Section A.7). With the usual constant step, proportional to $1/n$, Brosse et al. (2018) proved, for strongly log-concave posteriors, that the invariant law of SGLD comes close to that of SGD as n grows, and, in Bayesian linear regression, that it stays at a distance of order one from the posterior, whose width shrinks like $n^{-1/2}$.

[6] Reflection is the dynamical form of the uniform prior: without it the chains would leave $[-1,1]^2$, where the posterior vanishes.

CONSTANT-STEP SGD. Mandt et al. (2017) assume Gaussian minibatch gradient noise with a constant covariance $C/|B|$ for minibatches of $|B|$ examples, a stochastic differential equation in place of SGD, and a stationary law confined to where the loss is a quadratic with positive definite Hessian A (their Assumptions 1–4). Then the stationary law q of SGD with step η is Gaussian with covariance Σ solving $A\Sigma + \Sigma A = (\eta/|B|)\,C$, and the step $\eta^* = 2|B|d/(n\,\mathrm{tr}\,C)$, for n examples and d parameters, minimizes $\mathrm{KL}(q\,\|\,p)$ with p the posterior, which the quadratic loss makes Gaussian (their Theorem 1). The positive definite Hessian is what fails at a singular minimum: on the product model the Hessian at a point of the fiber and the covariance of the gradient noise both have rank 1, in the direction (b, a), so to first order the noise never moves SGD along the fiber; it does so only through the second-order effect of Proposition 15.4. In a blog post on small degenerate landscapes, Corlouer and Macé (2024) found that SGD converges more slowly to more degenerate critical points and that its end-of-training distribution can differ from the posterior.

OBSERVATION 15.6 (Izmailov et al. 2021). With full-batch Hamiltonian Monte Carlo on modern architectures, Bayesian neural networks could beat standard training and deep ensembles, posterior tempering was

not needed for near-optimal performance, and SGMCMC and deep ensembles gave predictive distributions distinct from those of Hamiltonian Monte Carlo.

About functions the agreement can be good: Mingard et al. (2021) found that the probability that SGD finds a function consistent with the training set correlates well with its Bayesian posterior probability, computed with Gaussian processes, for a range of architectures and data sets. On the product model, too, training and the posterior agree about the function and disagree about the parameters, and the quantities of singular learning theory are volumes in parameter space.

IMPLICIT BIASES. The product model has shown three, memory of the start, balance from finite steps, and collapse from gradient noise, and each has a general form. Gradient flow conserves the difference of the squared norms of the incoming and outgoing weights of every unit with a linear, ReLU or leaky ReLU activation (Du et al. 2018, Theorem 2.1), and every differentiable symmetry of the loss that satisfies a further condition, which translations, scalings and rescalings of weights do, gives a conservation law (Kunin et al. 2021, Theorem 1); gradient noise attracts SGD to invariant sets of simpler subnetworks (Chen et al. 2023a). For Langevin diffusion at a large inverse temperature on a nonnegative real-analytic potential, started on its zero set, Larsson (2026), in a preprint, derives a candidate limiting evolution on the zero set: the Dirichlet forms converge, in a certain sense, to a hierarchy on strata indexed by the local learning coefficient and its multiplicity, strongly biased toward the higher-dimensional, "more singular" strata. He offers this as a mechanism for the preference of stochastic gradient methods for singular solutions.

THE PROGRAM'S BRIDGE. The deep-learning applications of Chapters 13 and 14 rest on two hypotheses. A *dynamical transition* is a sudden move of SGD from near one critical point to near another; a *Bayesian transition* is a change, as n grows, of the region that dominates the posterior (Chapter 12). The first hypothesis is Conjecture 14.7 of Chapter 14, restated here without its phase labels.

Conjecture 15.7 (Bayesian antecedent hypothesis; Chen et al. 2023b). *The dynamical transitions encountered in neural network training have Bayesian antecedents: for each such transition between two phases there is a Bayesian transition between the same two phases as n grows.*

Conjecture 15.8 (Pepin Lehalleur et al. 2025, Figure 2). *Stochastic gradient descent finds parameters that the posterior prefers, and at step t its preference is that of the posterior at a sample size $n(t)$ that increases with t.*

The evidence points both ways. In the toy model of superposition, Chen et al. (2023b) found that the critical points they classify for the posterior's phases also explain SGD's plateaus, each drop of the loss accompanied by a rise of the estimated local learning coefficient, and their analysis suggests

that the observed dynamical transitions have Bayesian antecedents, with two inconclusive cases; but the posterior's transition from the 5-gon to the 6-gon was not observed in SGD. Carroll et al. (2025), in a preprint, found transformers trained on in-context linear regression moving first toward a simpler and then toward a more complex solution, as the posterior does when n grows; above a threshold of task diversity they never left the simpler one, whereas the Bayesian account predicts that it always eventually gives way. The authors list possible reasons, from limited capacity or training to a real difference between Bayesian and dynamical selection, and leave the question open. Read as a statement about the law of the parameters, Conjecture 15.8 fails on the product model, as we argue in Section 15.7.

15.5 Measuring it

Two scripts test the chapter's claims at the singular truth of the product model, against the exact posterior.

GRADIENT FLOW AGAINST THE POSTERIOR. The script `15-sgd-product.py` (a few seconds) simulates data sets of size $n = 10^{12}$ exactly through (S_{xx}, S_{xy}), draws exact posterior samples (Chapter 2), and computes gradient flow's end points by (15.2) from starts uniform on $[-1,1]^2$ and from small starts $\mathcal{N}(0, 10^{-12}I)$. In logarithmic coordinates (Figure 15.2a) the posterior draws spread along the segment $\log_{10}|a| + \log_{10}|b| \approx \log_{10}|\hat{c}| = -5.88$, while 48% of the flows from the prior's starts end with $\log_{10}|a_\infty| > -1$, 49% within a decade of the other end, and 3% between, and the flows from small starts all end at the middle, $\log_{10}|a_\infty| = -2.94$. Over 200 data sets (Figure 15.2b), the middle half of the scales, $-\frac{3}{8} < \log|a|/\log n \le -\frac{1}{8}$, has posterior mass 0.490, against 0.005 for the flow from the prior and 1.000 from small starts. The posterior points agree with the exact formula of Theorem 2.5 to 0.0012 and lie slightly above the limit law on the left, the slow $1/\log n$ convergence of Chapter 2. The flow from the prior has its lower atom spread over a window of width $O(1/\log n)$ above $-\frac{1}{2}$: its distribution function is 0.199 at $\tau = -\frac{1}{2}$ and 0.497 at $\tau = -\frac{3}{8}$. At the atom itself its limit is not $\frac{1}{2}$ but $\mathbb{P}(D < 0,\ Z_n^2 \le |D|) = 0.195$, with $Z_n \sim \mathcal{N}(0,1)$ as in the proof of Corollary 15.3.

FINITE STEPS, NOISE, AND LANGEVIN. The script `15-sgd-steps.py` (under a minute) generates 400 data sets of size $n = 10^4$ and runs on each, from one start uniform on $[-1,1]^2$: gradient descent with step $\eta = 0.1$ for 2×10^4 steps; SGD with the same step and minibatches of $|B| = 10$; SGLD at $\beta = 1$ with step $\epsilon = 2\eta/n$, the same minibatches and drift, and injected noise, reflected into the prior box; and Langevin dynamics, the same with full batches (Figure 15.3). Langevin dynamics reproduces the posterior: its distribution function of $\log|a|/\log n$ is within 0.025 of the exact one on $[-0.75, 0]$. SGLD with minibatches of 10 does not: its gradient noise, about $\eta(\sigma_{xy}^2/|B|)^{1/2}|w| \approx 0.03|w|$ per step, reaches seven times the injected noise $\sqrt{\epsilon} = 0.0045$, it drifts toward balance like SGD, and its law is too narrow, with a largest error of 0.234, the effect that Brosse et al. (2018) proved for strongly log-concave posteriors at large n. SGD forgets $a^2 - b^2$

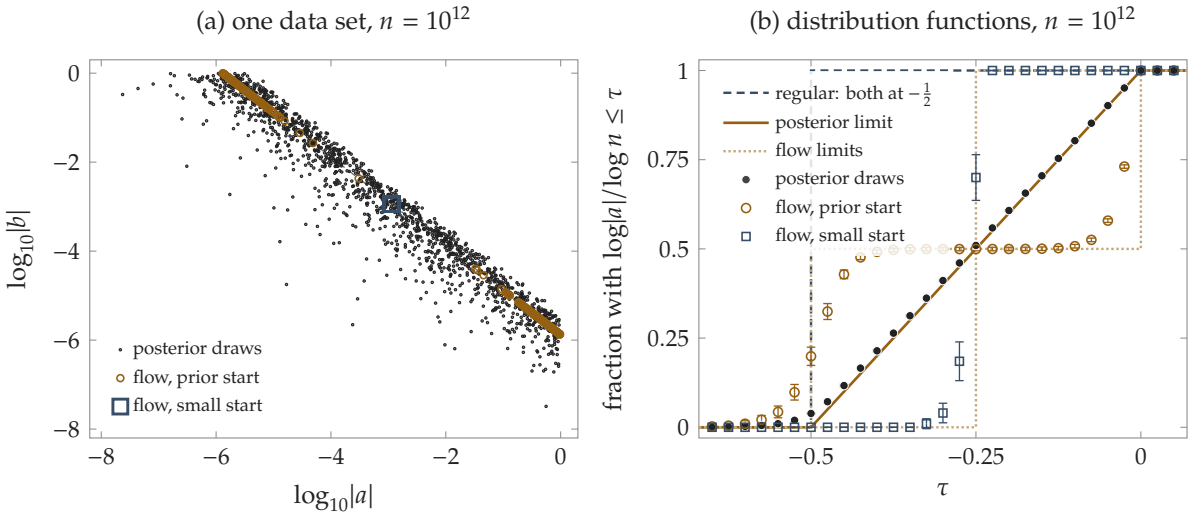


Figure 15.2. Where gradient flow stops and where the posterior goes, for the product model at its singular truth with $n = 10^{12}$. (a) One data set (seed 1503): 1500 exact posterior draws, and the end points of gradient flow from 300 starts uniform on $[-1, 1]^2$ and from 300 starts $\mathcal{N}(0, 10^{-12}I)$; dashed, the segment $\log_{10}|a|+\log_{10}|b| = \log_{10}|\hat{c}|$. A few draws with $|ab| \ll |\hat{c}|$ fall outside the frame. (b) The distribution function of $\log|a|/\log n$, averaged over 200 data sets (2000 draws and 2×200 flows each), with Student-t 95% intervals over data sets. Solid bronze: the posterior's limit Unif$(-\frac{1}{2}, 0)$ (Theorem 2.5); dotted bronze: the flow's limits, $\frac{1}{2}\delta_{-1/2} + \frac{1}{2}\delta_0$ from the prior and $\delta_{-1/4}$ from small starts (Corollary 15.3, Theorem 15.2); dashed slate: the regular plane, whose posterior and minimizer both sit at the scale $n^{-1/2}$. Script: `15-sgd-product.py`.

as Proposition 15.4 predicts: the median of $|D_t/D_0|$ is 0.359 after 1000 steps against $e^{-\eta^2\sigma_{xy}^2 t/|B|} = 0.368$, and 4.3×10^{-5} after 10^4 steps against 4.6×10^{-5}, while gradient descent keeps 0.9965 of it. And SGD collapses: 144 of the 400 runs end with $|ab| < 10^{-8}$. Of the 158 data sets with $|\hat{c}| < \eta\sigma_{xy}^2/(2\kappa|B|)$, the threshold of the heuristic, 83.5% collapsed within the 5×10^4 steps, of the others 5%, all of them within a factor 1.5 of the threshold. Every surviving run sits on a balanced line on the side of $\hat{c}$, with $|ab|$ over its last 10^4 steps averaging 0.0062 over the runs, against 0.0059 from the heuristic. These measurements on a model with two parameters show that the gap between training and the posterior is real where it can be computed; they say nothing about its size in a network with 10^8 parameters.

15.6 The ledger

On the product model, gradient flow conserves $a^2 - b^2$ and stops at the point of the hyperbola $ab = \hat{c}$ that this quantity and the signs of $a \pm b$ select (Theorem 15.2): from the prior's starts at the two extreme scales, from small starts at the middle one, $n^{-1/4}$ (Corollary 15.3), while the posterior spreads uniformly over all of them (Theorem 2.5). Finite steps shrink $a^2 - b^2$ by the exact factor $1 - \eta^2 r_t^2$ (Proposition 15.4), so SGD forgets its start and drifts

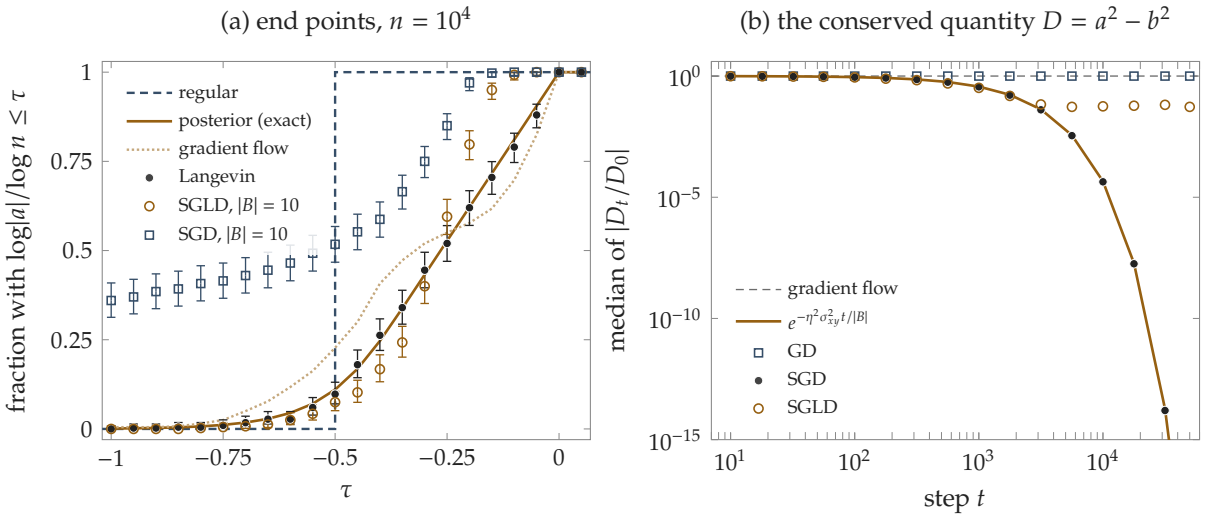


Figure 15.3. Finite steps and gradient noise on the product model at its singular truth, $n = 10^4$: 400 data sets, one run of each algorithm per data set from a common start uniform on $[-1, 1]^2$ (seeds 1511, 1512); step $\eta = 0.1$, minibatches of $|B| = 10$ drawn with replacement, SGLD and Langevin with $\epsilon = 2\eta/n$ at $\beta = 1$, reflected into the prior box. (a) Distribution functions of $\log|a|/\log n$ after 5×10^4 steps, with Clopper–Pearson 95% intervals; solid bronze, the exact posterior averaged over the data sets; dotted, the end points of gradient flow from the same starts; dashed slate, the regular prediction. SGD runs that collapsed onto the origin lie to the left of the frame. (b) Medians over the 400 runs of $|D_t/D_0|$ (gradient descent stops after 2×10^4 steps), with the prediction $e^{-\eta^2\sigma_{xy}^2 t/|B|}$ from Proposition 15.4 at the median $\sigma_{xy}^2 = 0.9997$ over the data sets; gradient flow keeps D exactly. Script: `15-sgd-steps.py`.

to balance, where gradient noise can collapse it onto the singular point. Langevin dynamics leaves the posterior invariant (Proposition 15.5) and reproduces it in our measurement; SGLD with small minibatches behaves like SGD. Training and the posterior find the same function and different parameters.

LEDGER · $y = abx$ under gradient descent

QUANTITY	what gradient descent finds against what the posterior averages: the law of $\log\|a\|/\log n$ at the end of training and under the posterior
REGULAR THEORY	they agree: the minimizer is the center of the asymptotically normal posterior, and both live at the scale $n^{-1/2}$ around $\hat{w}$
SINGULAR THEORY	they differ: gradient flow conserves $a^2 - b^2$ and stops where it selects (Theorem 15.2); finite steps multiply $a^2 - b^2$ by $1 - \eta^2 r_t^2$ (Proposition 15.4); the posterior spreads uniformly over scales (Theorem 2.5)

RUNNING EXAMPLE	$y = abx$ at its singular truth: flow from the prior $\to \frac{1}{2}\delta_{-1/2} + \frac{1}{2}\delta_0$, from a small start $\to -\frac{1}{4}$; posterior $\to$ Unif$(-\frac{1}{2}, 0)$
MEASURED	$n = 10^{12}$: mass of the middle half of the scales 0.490 (posterior), 0.005 (flow, prior start), 1.000 (flow, small start); $n = 10^4$: Langevin within 0.025 of the posterior, SGLD with $\lvert B\rvert = 10$ off by 0.234; SGD's $a^2 - b^2$ follows $e^{-\eta^2\sigma_{xy}^2 t/\lvert B\rvert}$, and 36% of SGD runs collapse onto the origin
RESTS ON	gradient flow or explicit steps on the product model and its one symmetry; the exact posterior at $\beta = 1$ with the uniform prior; Gaussian data at the singular truth; collapse argued, not proved

15.7 Where the theory stops

This chapter is where the book takes stock. We state the main critique of the theory's relevance to deep learning and the replies to it, list what the theory explains and what it does not (Table 15.1), and close with the three paragraphs that close every chapter.

THE CRITIQUE. In November 2023 Joar Skalse, who had attended a workshop on the theory that June, set out a critique in a blog post (Skalse 2023). The puzzle of generalization, he argues, is about small data relative to expressivity, and its answer must be an inductive bias toward simple functions, fixed by the prior and the parameter-function map; the theory abstracts from both, so its generalization theorem holds only relative to the machine's own implicit prior. His example fits polynomials of degree 4 with 15 parameters, $f(x) = \theta_1 x^4 + \theta_2\theta_3 x^3 + \cdots + \theta_{11}\cdots\theta_{15}$, to the one point $(1, 1)$; he argues that the fit of lowest learning coefficient is x^4, not the constant 1, so that a low learning coefficient can mean a complex function. He adds that the limit of infinite data is already covered by classical learning theory, and that the dominance of singular points needs both analyticity and infinite data. He does not object that networks are trained by gradient descent rather than by Bayesian updating, since he regards gradient descent as "empirically quite similar" to Bayesian sampling, citing his earlier summary of work he coauthored (Mingard et al. 2021).

THE REPLIES. Daniel Murfet (Murfet 2023a) separates (i) how it is possible for overparameterized models to generalize despite classical forms of Occam's razor such as BIC, from (ii) why neural networks in particular generalize on natural data. The theory, in his view, comes close to settling (i), modulo the validity of the asymptotic limit, the relation between Bayesian generalization error and the test error of a trained network (largely the posterior against SGD), and hypotheses such as relatively finite variance. It does not settle (ii), which it reduces to the question why natural data and networks give a low loss L_0, learning coefficient λ and singular fluctuation ν, through the Gibbs generalization error $L_0 + (\lambda + \nu)/n$.[7] In a second comment he grants that the expository posts may lead readers to think (ii) settled, and

[7] In Murfet's formula L_0 is roughly the loss of the best parameter and ν the singular fluctuation of Chapter 9.

claim	status	where
What singular learning theory explains		
Parameter counts are the wrong complexity for Bayesian learning in singular models: the free energy charges $\lambda \log n - (m-1)\log\log n$	proved	Theorems 3.4 and 8.6
The Bayes generalization error is λ/n, not $\frac{d}{2n}$; WAIC and WBIC estimate generalization and free energy	proved	Theorems 9.9, 10.8 and 11.6
The posterior of a singular model is not asymptotically normal; it spreads over scales	proved (product model)	Theorem 2.5; Watanabe (2018)
How overparameterized singular models *can* generalize under Bayesian learning (Murfet's (i))	proved as $n \to \infty$	Murfet (2023a) and Watanabe (2009)
The local learning coefficient as the complexity of a trained point	defined and bounded; estimates observed	Chapter 13; Lau et al. (2025)
Jumps of the estimated local learning coefficient at stage boundaries	observed in small models	Chapter 14; Hoogland et al. (2025)
What it does not explain		
Which minimizer SGD finds; whether SGD samples the posterior	not on $y = abx$ (flow proved, SGD observed); open for networks	Corollary 15.3 and Proposition 15.4; Mandt et al. (2017)
Why networks trained on natural data generalize (Murfet's (ii))	open	Murfet (2023a) and Skalse (2023)
That a low learning coefficient means a simple function	false in general; open for networks	Skalse (2023); Lau et al. (2025, App. F)
More parameters than data; finite n	open	Chapter 8; Nagayasu and Watanabe (2023)
Non-analytic architectures such as ReLU networks	open beyond special cases	Chapters 2 and 13
The role of training time: stages, $n(t)$	conjectured; evidence both ways	Conjectures 15.7 and 15.8; Carroll et al. (2025)

Table 15.1. What singular learning theory explains about deep learning and what it does not, as of October 2026, with the status of each claim and where it is proved or discussed.

places the theory's present value in phase transitions and emergent structure (Murfet 2023b). Dmitry Vaintrob replied that good generalization relative to the machine's own implicit prior is exactly what the theory claims, and is nontrivial, and that the distinction between SGD and sampling matters more than many people realize (Vaintrob 2023). In an appendix to the arXiv version of their paper, Lau et al. (2025, Appendix F) agree that the local learning coefficient is a model complexity, not a model-independent one: $f(x,w) = w^8 f_1(x) + (1-w^8) f_2(x)$ prefers f_2, of learning coefficient $\frac{1}{16}$, to f_1, of learning coefficient $\frac{1}{2}$, whatever their Kolmogorov complexities, and explaining simplicity bias by a bias toward low learning coefficient alone would be, as they say Skalse noted, tautological; a footnote adds that its role in inductive biases is "only rigorously established for Bayesian learning", though they suspect that it applies to SGD as well. Ege Erdil's discrete restatement of the theory took the stationary distribution of SGD to be proportional to $e^{-\beta L}$ (Erdil 2023); on the product model the long-run law of SGD concentrates near the balanced lines (Proposition 15.4) and is no

such density. A commenter, interstice, put the dynamical objection directly: what is specific to the theory is that volume concentrates at singular points as $n \to \infty$, and the commenter saw no mechanism by which SGD would be attracted to such points (interstice 2023); stochastic collapse (Chen et al. 2023a), of which our collapse onto the origin is an instance, and Larsson's result (Larsson 2026) can be read as partial answers.

FLATNESS AND DESCRIPTION LENGTH. Rescalings that keep the function change the sharpness (Dinh et al. 2017), as in Computation 15.2, but not the local learning coefficient (Chapter 13). Urdshals et al. (2026) prove that, under technical conditions, a realizable singular model has a two-part code with redundancy $\lambda \log n - (m-1) \log\log n + O_p(1)$, and observe that estimated local learning coefficients of language models track how coarsely their weights can be quantized, an asymptotic theorem and a correlational experiment.

WHAT WAS ASSUMED. Theorem 15.2 assumes gradient flow: infinitesimal steps, full batches, no momentum and no weight decay. Each of these changes the answer: finite steps and minibatches break the conservation law (Proposition 15.4), weight decay makes $a^2 - b^2$ decay to zero even under the flow (Exercise 15.8), and adaptive optimizers such as Adam were not treated. The theorem uses the product model's one continuous symmetry; a network has one rescaling symmetry for each positively homogeneous unit, each with its conserved quantity (Du et al. 2018), and permutations, which give no conservation law. The scale statements assume the singular truth, Gaussian data and a start independent of the data, and the comparison assumes the exact posterior at $\beta = 1$ with the uniform prior. Behind the whole book stand a fixed model as $n \to \infty$, realizability or relatively finite variance, analyticity, and here training run to convergence, which from a small start takes a time of order $n^{1/2} \log n$.

WHERE IT BREAKS. On the simplest singular model, training and the posterior disagree about the parameters in every respect we measured (Section 15.5): gradient flow from the prior's starts misses the middle scales that carry half the posterior mass, SGD collapses onto the origin in 36% of our runs, which the posterior never does, and only full-batch Langevin dynamics reproduces the posterior. For networks, the positive theorem on SGD as approximate inference (Mandt et al. 2017) assumes a positive definite Hessian, which a singular minimum lacks; constant-step SGLD departs from the posterior as n grows, even when the posterior is strongly log-concave (Brosse et al. 2018); SGMCMC and deep ensembles give predictive distributions distinct from Hamiltonian Monte Carlo (Izmailov et al. 2021); the Bayesian $5 \to 6$ transition of the toy model of superposition was not seen in SGD (Chen et al. 2023b); and the transformers of Carroll et al. (2025) stopped short of a transition that Bayesian model selection predicts. On the other side, SGD and the posterior agreed well about functions in the experiments of Mingard et al. (2021), as they do on our toy. Read as a statement about the law of the parameters, Conjecture 15.8 fails on the product model, we argue: under the posterior at every sample size the

position $\theta = \frac{1}{2}\log|a/b|$ along the fiber is spread out, with a density of order $1/\log(1/|c|)$ (Computation 15.2), while under SGD $D_t \to 0$ (Proposition 15.4 and Figure 15.3), so the law of θ_t concentrates at 0 and no choice of $n(t)$ keeps the two laws close in total variation.

OPEN PROBLEMS. (1) The long-run law of constant-step SGD on a singular model: on the product model, find the stationary law of (c_t, D_t) as $\eta \to 0$ with $\eta/|B|$ fixed, prove the collapse threshold, and compare the law with the posterior's local free energies; Corlouer and Macé (2024) pose the comparison for degenerate toy landscapes. (2) Conditions under which a dynamical transition has a Bayesian antecedent (Conjecture 15.7), and whether Conjecture 15.8 holds for functions, where the product model does not refute it. (3) Murfet's question (ii): why natural data and architectures give low L_0, λ and ν (Murfet 2023a). (4) Whether the local learning coefficient of neural networks is model-independent in some universality class, as Lau et al. (2025, Appendix F) suggest. (5) A theory that keeps the learning coefficient when the number of parameters is comparable to n or larger; bounds for redundant networks of fixed size, such as that of Nagayasu and Watanabe (2023) for deep ReLU networks with more layers than necessary, are a start.

Exercises

15.1 CONSOLIDATION. (Balancedness in deep linear networks.) For $y = W_L \cdots W_1 x$ with a square loss, show that gradient flow keeps every $W_{j+1}^\top W_{j+1} - W_j W_j^\top$ constant, and deduce that for the chain $y = a_1 \cdots a_L x$ every $a_i^2 - a_j^2$ is conserved. *(Hint: Write $\nabla_{W_j} L = P_j^\top G Q_j^\top$ with $P_j = W_L \cdots W_{j+1}$ and $Q_j = W_{j-1} \cdots W_1$, and use $P_j = P_{j+1} W_{j+1}$, $Q_{j+1} = W_j Q_j$.)*

15.2 CONSOLIDATION. (Sharpness along the fiber.) Show that at a point of the fiber $ab = \hat{c}$ the Hessian of L_n is $\kappa(b, a)^\top(b, a)$, with eigenvalues 0 and $\kappa\sqrt{D^2 + 4\hat{c}^2}$, so that the balanced points are the flattest minimizers, and that per unit length of the fiber the posterior density is proportional to $(a^2 + b^2)^{-1/2}$.

15.3 CONSOLIDATION. (The truth on the cross.) Show that if $\hat{c} = 0$, gradient flow converges from (a_0, b_0) to $(\operatorname{sgn}(a_0)\sqrt{D_+}, \operatorname{sgn}(b_0)\sqrt{D_-})$ with $D_\pm = \max(\pm D, 0)$, and that from a start on the lines $a = \pm b$ it approaches the origin only like $t^{-1/2}$.

15.4 CONSOLIDATION. (Langevin dynamics and Mandt's step.) (a) For the regular line, $U(c) = \frac{n\kappa}{2}(c - \hat{c})^2$, check Proposition 15.5 directly. (b) Model one SGD step on the same model as $c' - \hat{c} = (1 - \eta\kappa)(c - \hat{c}) + \eta\xi$ with independent ξ of variance $\sigma_{xy}^2/|B|$. Show that the stationary variance is $\eta\sigma_{xy}^2/(|B|\kappa(2 - \eta\kappa))$, and that it equals the posterior variance $1/(n\kappa)$ when $\eta = 2|B|/(n\sigma_{xy}^2 + |B|\kappa)$, the optimal step of Mandt et al. (2017) up to a relative error of order $|B|\kappa/(n\sigma_{xy}^2)$.

15.5 BY HAND. (Two arms with different learning coefficients.) For the cusp model $y = ab^2x$, show: (a) gradient flow conserves $a^2 - \frac{1}{2}b^2$; (b) the arm $\{b = 0\}$ of the true set has local learning coefficient $\frac{1}{4}$, the arm $\{a = 0\}$ has $\frac{1}{2}$; (c) assuming the flow converges, it ends near the arm $\{a = 0\}$ if $a_0^2 < \frac{1}{2}b_0^2$ and near $\{b = 0\}$ otherwise, so from a start uniform on $[-1, 1]^2$ it ends near $\{a = 0\}$ with probability $1/(2\sqrt{2}) \approx 0.354$ as $\hat{c} \to 0$; if $a_0\hat{c} < 0$ and $a_0^2 > \frac{1}{2}b_0^2$, then it ends at a critical point that is not a minimizer; (d) under the posterior at the singular truth, $\log|b|/\log n \to -\frac{1}{4}$ and $\log|a|/\log n \to 0$ in probability.*(Hint: Given $c = ab^2$, the uniform prior gives $|b|$ the density proportional to b^{-2} on $[|c|^{1/2}, 1]$.)*

15.6 BY HAND. (The chain.) For $y = a_1a_2a_3x$ at its singular truth, show that if gradient flow from a small, nearly balanced start converges to a minimizer, then it ends with $|a_i| \approx |\hat{c}|^{1/3}$, so that $\log|a_i|/\log n \to -\frac{1}{6}$ for starts small enough; and that under the posterior $\log|a_1|/\log n$ converges in law to the density $8(\tau + \frac{1}{2})$ on $[-\frac{1}{2}, 0]$, whose mean is $-\frac{1}{6}$.

15.7 BY HAND. (The size of the start.) Let $a_0 = n^{-\gamma}\xi_1$ and $b_0 = n^{-\gamma}\xi_2$ with independent standard normal ξ_1, ξ_2 and $\gamma \geq 0$. Show from (15.2) that at the singular truth $\log|a_\infty|/\log n$ tends in probability to $-\gamma$ on $\{|\xi_1| > |\xi_2|\}$ and to $-\frac{1}{2} + \gamma$ on its complement when $\gamma < \frac{1}{4}$, and to $-\frac{1}{4}$ when $\gamma \geq \frac{1}{4}$.

15.8 VARIATION. (Weight decay.) Add $\frac{\mu}{2}(a^2 + b^2)$ to L_n. Show that gradient flow now satisfies $\frac{\mathrm{d}}{\mathrm{d}t}(a^2 - b^2) = -2\mu(a^2 - b^2)$, and that the minimizers are the balanced points with $|ab| = |\hat{c}| - \mu/\kappa$ and $\mathrm{sgn}(ab) = \mathrm{sgn}(\hat{c})$ when $|\hat{c}| > \mu/\kappa$, and the origin otherwise. Compare the threshold μ/κ and the value of $|ab|$ at the minimizers with the collapse threshold $\eta\sigma_{xy}^2/(2\kappa|B|)$ of SGD and the mean of $|c_t|$ in the heuristic of stochastic collapse.

15.9 MEASUREMENT. (The collapse threshold.) In `15-sgd-steps.py`, vary $\eta \in \{0.05, 0.1, 0.2\}$ and $|B| \in \{5, 10, 20\}$, and compare the fraction of collapsed runs with the fraction of data sets with $|\hat{c}| < \eta\sigma_{xy}^2/(2\kappa|B|)$; which runs have not collapsed yet, and why? Then give SGLD minibatches of 100 and 1000, with fewer data sets to keep the run short, and watch its law approach the posterior's.

15.10 RESEARCH. (SGD and the learning coefficient.) On the cusp model of Exercise 15.5 the posterior concentrates on the arm of learning coefficient $\frac{1}{4}$. Where does constant-step SGD go in the long run, and does the answer depend on $\eta/|B|$? Formulate what would count as evidence for or against Conjecture 15.7 in a model with two arms, and test it; the question is open (Section 15.7).

Notes and References

The Amari school studied gradient learning near the singularities of networks long before singular learning theory reached deep learning (Chapter 2); Fukumizu and Amari (2000) showed how smaller networks create plateaus in larger ones. Saxe et al. (2014) solved the gradient dynamics of deep linear networks exactly and derived the conserved quantity $a^2 - b^2$ from the rescaling symmetry through Noether's theorem; they credit the balanced initial condition, under which the dynamics becomes a matrix Riccati equation, to a paper of Fukumizu at the 1998 International Conference on Neural Information Processing, which we have not seen. Arora et al. (2018) assumed balanced layers, $W_{j+1}^\top W_{j+1} = W_j W_j^\top$, at initialization and derived the dynamics of the end-to-end matrix (their Theorem 1); Du et al. (2018) proved the conservation laws for units with linear, ReLU or leaky ReLU activations (their Theorems 2.1 and 2.2), and their proofs for matrix factorization contain the full-batch form of (15.3); Kunin et al. (2021) derived conservation laws from symmetries, and Wang et al. (2022) proved the balancing effect of large learning rates. Theorem 15.2 is the case of one unit, with the convergence, the exceptional line and the scale $n^{-1/4}$ added; its proof through Lemma 15.1, Corollary 15.3, the minibatch form of (15.3) and the collapse heuristic are the book's own, and we did not find them in this form in the literature.

The debate on the theory's relevance took place on LessWrong in 2023, in Erdil (2023) and Skalse (2023) and their comment threads. As of October 2026 we know of no refereed paper whose main purpose is to criticize the theory's relevance to deep learning; the concessions are in the program's own papers, such as Appendix F of Lau et al. (2025), and in their limitations sections.

To read next: Watanabe (2018), the rigorous sequel, and Watanabe (2009); Suzuki (2023), exercises with Stan code; Carroll (2023), an informal introduction; and, in Japanese, Amari and Watanabe (2026), which joins singularities to statistical neurodynamics.

TRANSLATING THE LITERATURE. Saxe et al. (2014) write the error $\frac{1}{2\tau}(s - ab)^2$, with s the strength of an input–output mode and τ a time constant, for our $L_n - C_n$ with $\kappa = 1/\tau$ and $\hat{c} = s$; their hyperbolic coordinates parametrize the conserved hyperbolas, ours the fibers $ab = c$. Mandt et al. (2017) write ϵ, S, N and D for our η, $|B|$, n and d, and factor the noise covariance C as $BB^\top$; Welling and Teh (2011) write N and n for our n and $|B|$, and Section A.7 writes b for our $|B|$. Chen et al. (2023b) call a transition during training dynamical and one of the posterior as n grows Bayesian.

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Epilogue

The ledger

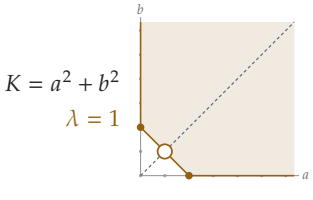
The prologue fitted the line $y = abx$ to pure noise and found that its two parameters learn like one: the free energy grows like $\frac{1}{2}\log n - \log\log n$, not like $\log n$. Fifteen chapters later the same replacement has been made in the formulas of Bayesian statistics that charge a model $\frac{d}{2}$ for its d parameters. Each chapter recorded its result on a card with the same six fields: the quantity, its value in the regular theory, its value in the singular theory, a running example, what was measured, and what the result rests on. Read one at a time, the cards summarize theorems. Read side by side, they show what the theory is: a geometric number and its multiplicity, which govern most of what a Bayesian asks of a model, under hypotheses that deep learning does not always meet. This epilogue sets the cards side by side, sets the hypotheses against practice, and collects the open problems.

The ledger

Tables E.1 and E.2 collect the cards of the fifteen chapters, with each field compressed to a phrase, the running example folded into the singular column, and what the results rest on gathered in Table E.3; the precise statements, with their hypotheses, are in the chapters. Read down the regular column, and one number recurs: $\frac{d}{2}$, which is half a unit of $\log n$ per parameter in the free energy, $\frac{1}{2n}$ per parameter in the generalization error, and $\frac{1}{2\beta}$ per parameter in the energy of a tempered posterior. Read down the singular column, and the learning coefficient λ takes its place in each of these roles, with its multiplicity m beside it.

What the ledger shows

One number in four roles. The regular theory does not distinguish the four places where $\frac{d}{2}$ appears, because there they are equal. The singular theory separates them and then proves that one number fills all four: the exponent of the volume of nearly true parameters (Chapter 4), the coefficient of $\log n$ in the free energy (Chapters 3 and 8), n times the expected Bayes generalization error (Chapter 9), and the slope of the tempered posterior's mean energy in $1/\beta$ (Chapter 11). The chapters measured it in all four roles, by procedures that share no computation: Monte Carlo volumes of the prior, quadrature of the marginal likelihood over many data sets, the divergence from the truth to the predictive distribution, and posterior energies at two temperatures. On the running models with $m = 1$ the four measurements agree with each other and with the value that Part II computes by changes of coordinates, and with $m \geq 2$ they agree once the correction described next is made. That agreement is the theory's main empirical content, and it is why one number can serve as a singular model's complexity. The



CHAPTER: QUANTITY	REGULAR THEORY	SINGULAR THEORY	MEASURED
PART I. WHERE REGULAR THEORY ENDS			
1 regular models: $F_n - nS_n$ and $\mathbb{E}[G_n] - S$	$\frac{d}{2}\log n + O_p(1)$ and $\frac{d}{2n} + o(\frac{1}{n})$: Laplace's method, BIC and AIC	the special case $\lambda = \frac{d}{2}$, $m = 1$	slopes 0.499, 0.998, 0.992 against $\log n$ for $d = 1, 2, 2$; n times Laplace's error on the mixture at a regular truth 0.596 at $n = 10^4$, against 0.594 predicted
2 singular models: the true set and the shape of the posterior	W_0 a point, the Fisher information invertible, the posterior close to a Gaussian of width $n^{-1/2}$	W_0 contains curves on which the Fisher information is singular; for $y = abx$, $\log\|a\|/\log n \to \text{Unif}(-\frac{1}{2}, 0)$ under the posterior; no Gaussian limit in C^1 coordinates	posterior quartiles of $\log\|a\|/\log n$ at $n = 10^{16}$: -0.3800, -0.2530, -0.1265, against -0.3809, -0.2539, -0.1270 predicted
3 one coefficient: $F_n - nS_n$ when the likelihood depends on w through $c = g(w)$	$\frac{d}{2}\log n$: $\log n$ for $y = abx$, $\frac{3}{2}\log n$ for $y = a_1a_2a_3x$	$\lambda \log n - (m-1)\log\log n + O_p(1)$, with λ and m read off the prior density of c at the truth, and the limit law of the remainder; $y = abx$: $\lambda = \frac{1}{2}$, $m = 2$	slopes 0.41, 0.32, 0.23, 0.16 over $10^2 \le n \le 10^8$ for $\lambda = \frac{1}{2}$ ($m = 2$), $\frac{1}{2}$ ($m = 3$), $\frac{1}{4}$, $\frac{1}{6}$
PART II. THE LEARNING COEFFICIENT			
4 volume: the prior mass $V(\varepsilon)$ of $\{K < \varepsilon\}$	$C\varepsilon^{d/2}$, the volume of an ellipsoid	$C\varepsilon^{\lambda}(\log\frac{1}{\varepsilon})^{m-1}$, with $-\lambda$ the largest pole of the zeta function and m its order; the quartic cup: $V(\varepsilon) = 1.311\,\varepsilon^{1/2}$ exactly for $\varepsilon \le \frac{1}{2}$, against ε^1	Monte Carlo exponents 1.0006, 0.5001, 0.7494, 0.6665 for $\lambda = 1, \frac{1}{2}, \frac{3}{4}, \frac{2}{3}$; for the product model 0.5001 with $\varepsilon^{1/2}(c_0 + c_1\log\frac{1}{\varepsilon})$ fitted, 0.39 for a straight line
5 blow-ups: λ and m on a resolution	$\frac{d}{2}$ and $m = 1$: by the Morse lemma, one blow-up resolves a nondegenerate minimum	$\min_j(h_j+1)/(2k_j)$ over the charts, m the most ties at one point; $\frac{1}{2}(a^2+b^4)$: two blow-ups, $\lambda = \frac{3}{4}$; $(a^2-b^3)^2$: three, $\frac{5}{12}$	chart sums equal the original volumes to 10^{-6}; local slopes 0.7500, 0.4134, 0.7500 at $\varepsilon \approx 10^{-14}$ for $\lambda = \frac{3}{4}, \frac{5}{12}, \frac{3}{4}$
6 Newton polygons: λ from the exponents of K	every exponent that matters is 2: $\frac{d}{2}$, $m = 1$	$\lambda = 1/t$, where the diagonal meets the Newton polygon at (t, t), and $m = 2$ if it meets a vertex (proved for sums of even monomials with positive coefficients in two variables, stated for nondegenerate losses); $\lambda \le 1/t$ in every coordinate system; $\frac{1}{2}(a^4+b^6)$: $\frac{5}{12}$	for thirty random sums, quadrature exponents within 3×10^{-7} of $1/t$; Monte Carlo slopes at $\varepsilon \ge 10^{-4}$ low by 0.09 when $m = 2$
7 the catalogue: λ and m of models in use	$\frac{d}{2}$, or half the dimension of the set of distributions	rank-one regression at the zero truth: $\frac{1}{2}\min(M, N)$, with $m = 2$ if and only if $M = N$; reduced-rank regression: the formula of Aoyagi and Watanabe, and deep linear networks: Aoyagi's, both stated and checked; $M = N = 3$: $\frac{3}{2}$ at the truth 0, against 3	two-temperature estimates at $n = 10^5$: 1.40, 2.50, 2.50, 3.44 for $\lambda = \frac{3}{2}, \frac{5}{2}, \frac{5}{2}, \frac{7}{2}$, below λ where $m = 2$ by somewhat less than the predicted correction

Table E.1. The ledger, Parts I and II. Each row compresses the card of its chapter; the chapters state the hypotheses. Here d is the number of parameters, $F_n - nS_n$ the normalized free energy, $\mathbb{E}[G_n] - S$ the Bayes generalization error, $K = L - L(w_0)$ the excess population loss, which vanishes on the true set W_0, $V(\varepsilon)$ the prior volume of $\{K < \varepsilon\}$, and M and N the numbers of outputs and inputs of rank-one regression; measured slopes are fitted against $\log n$ or $\log\frac{1}{\varepsilon}$.

CHAPTER: QUANTITY	REGULAR THEORY	SINGULAR THEORY	MEASURED
PART III. WATANABE'S THEOREMS			
8 free energy: $F_n - nS_n$	$\frac{d}{2}\log n + O_p(1)$: Schwarz's BIC	$\lambda \log n - (m-1)\log\log n + O_p(1)$: proved for regressions $y = G(w)x + \varepsilon$, Watanabe's theorem for analytic models; $y = ax_1 + b^3x_2$: $\frac{2}{3}$	slopes 0.998, 0.749, 0.667, 0.491 for $\lambda = 1, \frac{3}{4}, \frac{2}{3}, \frac{1}{2}$, and 0.394 for the product model; the singular BIC finds the true rank in 182 of 200 data sets at $n = 300$, BIC in 5
9 generalization: $\mathbb{E}[G_n] - S$	$\frac{d}{2n} + o(\frac{1}{n})$; training error $-\frac{d}{2n}$	$\frac{\lambda}{n} + o(\frac{1}{n})$: proved for one-coefficient regressions with Gaussian inputs and noise, Watanabe's theorem in general; by his theorem, training error $(\lambda - 2\nu)/n$; $y = ab^3x$: $\frac{1}{6n}$	$n(\mathbb{E}[G_n] - S)$ at $n = 10^6$: 0.164 ± 0.024, 0.244 ± 0.031, 0.504 ± 0.047 for $\lambda = \frac{1}{6}, \frac{1}{4}, \frac{1}{2}$; 0.422 ± 0.038 for the product model, which approaches $\frac{1}{2}$ like $\frac{1}{2} - 1/\log n$
10 WAIC: estimating G_n from one posterior	AIC, $L_n(\hat{w}) + \frac{d}{n}$ or $T_n + \frac{d}{n}$, unbiased to $o(\frac{1}{n})$; leave-one-out agrees	$\mathbb{E}[\text{WAIC}] = \mathbb{E}[G_n] + o(\frac{1}{n})$ (Watanabe); leave-one-out = WAIC $+O_p(n^{-2})$, proved for $y = abx$; penalty V_n/n with $\mathbb{E}[V_n] \to 2\nu$; at the singular truth of $y = abx$, $2\nu = 1$, half of AIC's 2	bias of WAIC zero within $\pm 0.13/n$ for $10^2 \le n \le 10^5$; bias of $T_n + 2/n$ between $1.02/n$ and $1.19/n$
11 temperature: the tempered mean energy, WBIC, $\hat{\lambda}$	equipartition, $\frac{d}{2\beta}$; WBIC = BIC $+o_p(1)$	population mean energy $\frac{\lambda}{\beta} - \frac{m-1}{\beta \log n\beta} + \cdots$; F_n − WBIC = $-(m-1)\log\log n + O_p(\sqrt{\log n})$: proved for realizable linear models, Watanabe's theorem in general; the quartic model: $\frac{3}{4\beta}$	$\hat{\lambda}$ at $n = 10^8$: 0.750, 0.667, 1.000 for $\lambda = \frac{3}{4}, \frac{2}{3}, 1$; 0.438 for the product model, its population value
12 phases: the posterior mass of a region	every region pays $\frac{d}{2}\log n$; the posterior goes to the region of the best fit	local free energies $nL_n(w_\alpha) + \lambda_\alpha \log n - (m_\alpha - 1)\log\log n$, proved for linear models, and the smallest wins; transitions in n lower the loss and raise λ; the double well: from $\lambda = \frac{1}{2}$ to 1 near $n = 3000$	posterior mass of the regular region 0.24, 0.53, 0.97 at $n = 10^3$, 3162, 10^4, against 0.25, 0.54, 0.96 predicted and 0.75, 0.93, 1.00 from the regular theory
PART IV. DEEP LEARNING			
13 local learning coefficient: $\lambda(w^*)$ at a minimum, and its estimate	Laplace at w^*: $\frac{d}{2}$, or $\frac{r}{2}$ for a Hessian of rank r	$\frac{r}{2} \le \lambda(w^*) \le \frac{r}{2} + \frac{d-r}{4}$, equal to $\frac{r}{2}$ exactly when the true set is a manifold of dimension $d - r$ near w^*; rank-one regression with $M = N = 3$: $\frac{3}{2}$ at 0 and at $(e_1, 0)$	the exact estimator for the ReLU neuron at $n = 10^4$: 0.32 ($m = 2$) and 0.46 ($m = 1$) for $\lambda = \frac{1}{2}$; SGLD errors from −82% to +118% across step sizes; a fixed Gaussian localizer gives 0.99 for $\frac{3}{2}$ when $N = 2$
14 stages: λ at the critical points that training visits	$\frac{d}{2} = 9$ at every nondegenerate minimum of the toy model of superposition	$\frac{d-s}{2}$, $m = 1$, at a minimally singular critical set of dimension s: 7 for the 5-gon, $\frac{17}{2}$ for the 5^+-gon and the 6-gon, computer-assisted; the order of stages conjectured from the posterior's	at the exact polygons 7.06, 8.34, 8.73 at $n = 10^6$, but 8.21, 8.32, 9.63 at $n = 10^3$; along SGD, 4.58 → 7.56 and 5.20 → 7.83; 9 of 32 runs dropped, none to the 6-gon
15 SGD is not Bayes: where training stops and where the posterior goes	the same place: the minimizer is the center of an asymptotically normal posterior	on $y = abx$, gradient flow conserves $a^2 - b^2$: $\log\lvert a\rvert/\log n \to \frac{1}{2}\delta_{-1/2} + \frac{1}{2}\delta_0$ from the prior, against Unif$(-\frac{1}{2}, 0)$ under the posterior	at $n = 10^{12}$, mass of the middle half of the scales 0.490 under the posterior, 0.005 under the flow from the prior; at $n = 10^4$, 36% of SGD runs collapse onto the origin

Table E.2. The ledger, Parts III and IV. Here ν is the singular fluctuation, T_n the Bayes training loss, V_n the functional variance, β the inverse temperature, $\hat{\lambda}$ the two-temperature estimate of λ, $L_n(w_\alpha)$ the empirical loss at the best point of a region, w^* a local minimum, and s the dimension of a set of critical points.

second number of Part III is not λ: the singular fluctuation ν, which λ does not determine, enters the training error and the Gibbs errors and sets the penalty of WAIC (Chapters 9 and 10). For the product model at its singular truth $2\nu = 1 = 2\lambda$, but for the cusp $y = ab^2x$, $2\nu \approx 0.70$ while $2\lambda = \frac{1}{2}$, and the chapters know of no closed form for ν, even for the power-law class of one-coefficient models (Chapter 9). WAIC needs neither number, which is what makes it useful.

THE MULTIPLICITY IS SLOW. Wherever $m \geq 2$, the measurements fall short of λ: the free-energy slope 0.39 of the product model over $10^2 \leq n \leq 10^6$ (the prologue and Chapter 8), its generalization error $0.422/n$ at $n = 10^6$ (Chapter 9), its two-temperature estimate 0.438 at $n = 10^8$ (Chapter 11), the rank-one estimates of Chapter 7, and the local estimate 0.32 at the origin of the ReLU neuron against 0.46 on one of its arms (Chapter 13). The shortfall is of the order of $(m-1)/\log n$, and where the chapters computed the correction, the corrected prediction fits the measurement, or nearly; none of these is a failure of the theory. But a correction of order $1/\log n$ does not go away at the sample sizes met in practice: at $n = 10^{12}$ it is still 0.036 for $m = 2$. An experimenter who reads λ off a slope without knowing m reads it low (Chapters 11 and 13), and the chapters know of no estimator of m with a guarantee (Chapters 3 and 4). Of what the headline formula leaves out, the multiplicity is the part that biases every measurement of λ in the same direction; the bounded terms are the part that moves the transitions of the posterior at practical sample sizes (Chapters 12 and 14).

WHERE THE POSTERIOR GOES. Three rows replace the regular picture of the posterior, a Gaussian at the maximum-likelihood estimate. The posterior of $y = abx$ spreads over scales along its true set instead of concentrating at the rate $n^{-1/2}$ (Chapter 2); when several regions nearly fit the data, it moves between them as n grows, trading loss against learning coefficient (Chapter 12); and gradient flow, which does stop at a point, stops at a point chosen by its dynamics, through a conserved quantity, not by volume (Chapter 15). In a regular model the optimizer and the posterior end up in the same place, and the theory of the one is a theory of the other. In a singular model they find the same function and part ways about the parameters. This is why Part IV measures the learning coefficient at a trained network instead of predicting it: the network is not a draw from the posterior, and Part III does not say which point training reaches.

WHAT IS PROVED AND WHAT IS MEASURED. The certainty of the ledger decreases down the page. Parts I and II prove their central results completely, or reduce them to Hironaka's theorem, stated in Chapter A; what they take from the literature, such as the generalization error of a general regular model (Chapter 1), the theorems on Newton polyhedra beyond positive sums in two variables (Chapter 6), and the formulas of Aoyagi and Watanabe (Chapter 7), they state with its source and check numerically where they can. Part III proves its theorems completely for classes of models in which the computation can be carried through, one coefficient (Chapter 9), the product model itself (Chapter 10), or regressions linear in the inputs (Chapters 8, 11

and 12), with one limit, the Bayes training error, only argued (Chapter 9), and states Watanabe's general theorems with their sources and an account of what the general proofs add. In Part IV the theorems concern the geometry of a minimum, the Hessian bounds (Chapter 13) and the coefficient of a minimally singular critical set (Chapter 14), and the dynamics of gradient flow on $y = abx$ (Chapter 15). Under Watanabe's conditions the estimator of the local learning coefficient is consistent for exact posteriors with a compact localizer, by his theorem on WBIC applied to a ball; what SGLD computes on a network has no guarantee, and the stages of training are measurements with a conjectured Bayesian explanation.

The hypotheses against practice

Every card ends with what its result rests on. Table E.3 collects the hypotheses that recur, sets each against how networks are trained, and says what the book shows without it. Two of them fail for every large network in use: the number of parameters is comparable to n or larger, not fixed while n grows, and the parameters come from an optimizer, not from the posterior. The first is the deeper gap, because every asymptotic statement of the book is a statement about $n \to \infty$ with d fixed; the second is the one that Chapter 15 makes concrete.

Open problems

The chapters end with open problems, and they fall into five groups.

Computing λ. Exact learning coefficients are known for rank-one and reduced-rank regression, for deep linear networks, for one tanh layer with a single input, and for a few other families, among them some mixtures, naive Bayesian networks, and latent trees; they are not known for one tanh layer with several inputs and outputs and more than four hidden units, for two nonlinear layers beyond the smallest, or for hidden Markov models or attention layers, and the exact ReLU values claimed by Aoyagi (2025) await independent checks (Chapter 7). No practical algorithm is known for the real log canonical threshold of a loss in many variables (Chapter 5). In three or more variables Greenblatt (2010, Section 1) asks when coordinates exist in which the Newton polyhedron gives sharp estimates; for learning, the question is for which architectures and truths a short, explicit sequence of blow-ups makes the loss nondegenerate in every chart (Chapter 6). For losses that are not analytic, such as those of ReLU networks, a theory of learning coefficients would need a replacement for Hironaka's theorem (Chapter 5).

Estimating λ and m. No estimator of the volume function of an analytic loss in high dimension is known to have controlled relative error at a cost polynomial in d and $\log \frac{1}{\varepsilon}$ (Chapter 4), and the chapters found no estimator of λ with a proved error of smaller order than $1/\log n$ when $m \geq 2$ is unknown: the analyses of Imai (2019) and Takio and Suzuki (2025) give consistency at the rate $(\log n)^{-1/2}$ and do not treat the bias $-(m-1)/\log n$

HYPOTHESIS	USED IN	IN DEEP LEARNING	WITHOUT IT
the model is fixed as $n \to \infty$	every asymptotic theorem	d is comparable to n or larger, and models grow with the data	for linear models with many fewer inputs than samples the free energy is pinned down only to $O_p(k)$, for k coefficients (Chapter 8); bounds for redundant networks of fixed size, such as that of Nagayasu and Watanabe (2023) for deep ReLU networks, are a start (Chapter 15); otherwise open
realizability, or relatively finite variance	Chapters 3 and 8 to 10	no network is the true distribution	Watanabe's theorems need only relatively finite variance and a unique optimal distribution; without the first, in his non-renormalizable example, the generalization error is of order $n^{-2/3}$ by his sketch, the free energy has a term of order $n^{1/3}$ by the book's argument, and WAIC stays unbiased in simulation (Chapters 8 and 10)
the loss is analytic near the true set	Chapters 5 and 8 to 10	ReLU networks are only piecewise analytic	λ and m keep their meaning as exponents of the volume where it has power-logarithm bounds (Chapter 5), which nothing in the book guarantees for a ReLU network (Chapter 4); the Hessian bounds need only a C^4 loss (Chapter 13); there is no resolution theorem to compute λ
a compact parameter set; a prior positive at the true set	Chapters 1, 8 and 13	Gaussian initialization, weight decay, unbounded weights	in rank-one regression with $N < M$, a fixed Gaussian localizer makes the population estimate converge to $\frac{N}{2}$, the value at a more singular point, not to $\lambda(w^*) = \frac{M}{2}$ (Proposition 13.10): the localization must depend on n
exact posterior averages	Chapters 2 to 12	parameters from SGD; averages over short SGLD chains	gradient flow selects its end point by a conservation law and SGD by drifting to balance, not by volume (Chapter 15); SGLD estimates have no guarantee and move with the step size (Chapter 13)
independent data from a fixed distribution	every chapter with data	augmentation, curricula, data that shift during training	nothing in this book, beyond a cited remark that with fixed or dependent inputs WAIC, unlike leave-one-out, still estimates the conditional generalization loss (Chapter 10)
transitions in n	Chapter 12	stages are seen in training time	that stages of training have Bayesian antecedents is a conjecture (Conjectures 14.7 and 15.7)
asymptotics in n, with corrections in $1/\log n$	Chapters 3, 9 and 11	n is finite, and $\log\log n$ is not small next to $\log n$	the multiplicity biases estimates of λ by about $(m-1)/\log n$ (Chapters 7, 11 and 13); constants of local free energies decide transitions at practical n (Chapters 12 and 14)

Table E.3. The hypotheses of the book against the practice of deep learning.

(Chapter 11). For the local learning coefficient, the conjecture of Lau et al. (2025, Appendix D) asks for conditions on the localizer, which must depend on n, under which the estimator converges to $\lambda(w^*)$ (Chapter 13); why stochastic-gradient MCMC chains recover local volume exponents at all is posed by Hitchcock and Hoogland (2025); and no method is known that gives confidence intervals covering the bias of an estimate, in a large network or along training (Chapters 13 and 14).

Finite n and growing d. Almost every error term in Part III is asymptotic. Non-asymptotic bounds for the generalization error (Chapter 9) and for WBIC and $\hat{\lambda}$ (Chapter 11) are open, and so is the behavior of the free energy when d grows with n (Chapter 8). The lower-order terms of local free energies, which decide transitions at practical sample sizes, are unknown in general, and Carroll et al. (2025) list them among the open questions of their account (Chapters 3, 12 and 14); so is the coefficient of $1/\log n$ in the mean free energy of a model with several coefficients and $m \geq 2$ (Chapter 3).

The posterior and training. Beyond one coefficient, for the tanh neuron, the mixture, and networks at reducible truths, the law of the posterior over scales is unknown (Chapter 2). That the transitions met in training have Bayesian antecedents, transitions of the posterior in n between the same regions, is the hypothesis of Chen et al. (2023), and what links the two is open (Chapters 12, 14 and 15); even for $y = abx$ the long-run law of stochastic gradient descent with a constant step, and its relation to the posterior's local free energies, are open (Chapter 15). And why natural data and architectures give low loss, learning coefficient, and singular fluctuation is the question of Murfet (2023) (Chapter 15).

Watanabe's conjectures. Watanabe conjectured that the singular fluctuation is governed by a generalized Fisher information on the set of true parameters (Watanabe 2010, Section 5) (Chapter 9), and that the parity on which the error of WBIC depends is a birational invariant for every truth (Watanabe 2013, p. 875) (Chapter 11). Whether WAIC agrees with leave-one-out cross-validation to order n^{-2} for models whose loss is smooth but not analytic, and what replaces its theorem without relatively finite variance, are open as well (Chapter 10).

What to read next

Watanabe's books are the sources of the general theory. Watanabe (2009) proves the free energy formula and the equations of state with resolution of singularities and empirical processes, and this book's appendix states what it uses from there; the unrealizable case, WBIC, and the equivalence of WAIC with cross-validation came later, in papers that the chapters cite. Watanabe (2018) is a graduate text with exercises, on WAIC, WBIC, and cross-validation among much else; the exercise books of Suzuki (2023a,b) work through the same theory with code; and Watanabe (2024) reviews the theory and its recent advances. For computing learning coefficients, the thesis of Lin (2011) develops the algebra behind Part II, and Drton and Plummer (2017), with its published discussion, is the statistician's

way into model selection with the singular BIC. For deep learning, Wei et al. (2023) argue for deep-learning readers that singularity is a feature of Bayesian learning, not a defect; Lau et al. (2025) introduce the local learning coefficient; Hoogland et al. (2025) use it to locate stages of development in transformers; and the sequence of Carroll (2023) is an informal introduction for readers from machine learning.

Two parameters, again

The prologue's model is the book in miniature. Its parameters fold onto the line of slopes, so its true set is a cross (Chapter 2). Its free energy pays $\frac{1}{2}\log n - \log\log n$ because the region of nearly true parameters is a thin cross whose arms thicken near the center (Chapters 3 and 4). Its loss is already a monomial, so the computations of Part II take one line, and the two exponents tie, which is where its multiplicity $m = 2$ comes from (Chapters 5 and 6). Its generalization error approaches $\frac{1}{2n}$ only like $(\frac{1}{2} - \frac{1}{\log n})\frac{1}{n}$ (Chapter 9); the penalty of WAIC is half of AIC's (Chapter 10); and its posterior spreads over scales while gradient descent picks one (Chapters 2 and 15). A network has many such folds, and they meet.

Singular learning theory does not explain why networks trained by stochastic gradient descent generalize, and Chapter 15 shows on the smallest example how training and the posterior part ways. What it does establish is narrower and firmer. In Bayesian learning the complexity of a model at a truth is a geometric number, not a count of parameters; the number can be computed in small models and estimated in large ones, though there the estimates have no guarantee and their biases are known only from small models; and the classical formulas that count parameters are the special case in which the geometry is trivial. Whether the networks we train live in the regime where that geometry decides what they learn is a question about networks, not about the theory. The theory makes it possible to ask it precisely.

Notes and References

The cards summarize the chapters, and the references are there. The tables quote the chapters' measured numbers as the scripts printed them; the chapters give their intervals and the number of data sets behind each.

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A Toolkit

A

Integrals, singularities, and samplers

This appendix collects the tools that several chapters share: Gaussian integrals and Laplace's method; the Gamma function, Mellin transforms and regular variation; analytic functions and the resolution of their singularities; the random process on which Watanabe's theorems rest; divergences and modes of convergence; and the samplers, estimators and intervals behind the book's figures. Short proofs are given in full. Long ones are replaced by a precise statement and a source, and the statement says what the book takes on trust; the two results of this kind that carry the most weight are Hironaka's theorem (Theorem A.8) and Watanabe's standard form of the log-likelihood ratio (Theorem A.11). Each item ends by saying which chapters use it. The appendix is a reference, not a course: each section can be read when a chapter points to it.

A.1 Gaussian integrals and Laplace's method

When f has a nondegenerate minimum, the integral $\int g\, e^{-nf}$ is a Gaussian integral to leading order; this is *Laplace's method*. The section ends with Stein's identity and Gaussian smoothing.

Theorem A.1 (Laplace's method). *Let $W \subset \mathbb{R}^d$ be a Borel set, let $f, g : W \to \mathbb{R}$ be measurable, and let w_0 be an interior point of W. Suppose that*

(i) f has four and g two continuous derivatives on an open set containing a closed ball $\bar{B}(w_0, r_0) \subset W$;

(ii) $\nabla f(w_0) = 0$, and the Hessian $H = \nabla^2 f(w_0)$ is positive definite;

(iii) the minimum is separated: $\eta(\delta) = \inf\{f(w) - f(w_0) : w \in W,\ |w - w_0| \geq \delta\} > 0$ for every $\delta > 0$;

(iv) $\int_W |g| e^{-n_0 f} < \infty$ for some n_0.

Then, as $n \to \infty$ through real values,

$$e^{n f(w_0)} \int_W g(w) e^{-n f(w)}\, \mathrm{d}w = \left(\frac{2\pi}{n}\right)^{d/2} (\det H)^{-1/2} \big(g(w_0) + O(n^{-1})\big). \tag{A.1}$$

In particular, if $g = \varphi$ with $\varphi(w_0) > 0$, then

$$-\log \int_W \varphi e^{-nf} = n f(w_0) + \frac{d}{2} \log \frac{n}{2\pi} + \frac{1}{2} \log \det H - \log \varphi(w_0) + O(n^{-1}).$$

Hypothesis (iii) fails when f comes back down to $f(w_0)$ at the boundary

of W or at infinity. The proof uses the following facts.

Lemma A.2 (Gaussian integrals, moments, and tails). *Let $Z \sim \mathcal{N}(0,1)$ and $Y \sim \mathcal{N}(0, I_d)$.*

(i) For a symmetric positive definite $d \times d$ matrix A and $b \in \mathbb{R}^d$,

$$\int_{\mathbb{R}^d} e^{-\frac{1}{2}x^\top A x + b^\top x}\,\mathrm{d}x = (2\pi)^{d/2}(\det A)^{-1/2} e^{\frac{1}{2}b^\top A^{-1} b}.$$

(ii) For real $s > -1$, $\mathbb{E}|Z|^s = 2^{s/2}\Gamma(\frac{s+1}{2})/\sqrt{\pi}$, so that $\mathbb{E}Z^{2k} = (2k-1)(2k-3)\cdots 1$. Every moment of $|Y|$ is finite, and every odd polynomial in Y has mean zero.

(iii) For $t > 0$, $\frac{t}{1+t^2}\frac{e^{-t^2/2}}{\sqrt{2\pi}} \le \mathbb{P}(Z > t) \le \min\bigl(\frac{e^{-t^2/2}}{t\sqrt{2\pi}}, \frac{1}{2}e^{-t^2/2}\bigr)$.

(iv) For $r \ge 0$, $\mathbb{P}(|Y| \ge r) \le 2^{d/2} e^{-r^2/4}$.

Proof. The plan is to reduce everything to the one-dimensional integral of $e^{-t^2/2}$. (i) By Fubini's theorem and polar coordinates, the square of $\int_{\mathbb{R}} e^{-t^2/2}\,\mathrm{d}t$ is $\int_0^{2\pi}\int_0^\infty e^{-r^2/2} r\,\mathrm{d}r\,\mathrm{d}\theta = 2\pi$. In general, completing the square gives $-\frac{1}{2}x^\top A x + b^\top x = -\frac{1}{2}(x-c)^\top A (x-c) + \frac{1}{2}b^\top A^{-1} b$ with $c = A^{-1}b$; translating by c and substituting $x = A^{-1/2}y$, whose Jacobian is $(\det A)^{-1/2}$, leaves $\int e^{-|y|^2/2}\,\mathrm{d}y = (2\pi)^{d/2}$. (ii) By symmetry and the substitution $u = t^2/2$, $\mathbb{E}|Z|^s = \frac{2}{\sqrt{2\pi}}\int_0^\infty t^s e^{-t^2/2}\,\mathrm{d}t = \frac{2}{\sqrt{2\pi}}\int_0^\infty (2u)^{(s-1)/2} e^{-u}\,\mathrm{d}u$, which is the claim; for $s = 2k$ use $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ and $\Gamma(x+1) = x\Gamma(x)$ (Definition A.4). The moments of $|Y|$ are finite because $|Y|^k \le d^{k/2}\sum_i |Y_i|^k$, and an odd polynomial P has $\mathbb{E}P(Y) = \mathbb{E}P(-Y) = -\mathbb{E}P(Y)$ because Y and $-Y$ have the same law. (iii) Let $\phi(x) = (2\pi)^{-1/2}e^{-x^2/2}$, so that $\phi'(x) = -x\phi(x)$. Since $x/t \ge 1$ on $x \ge t$, $\mathbb{P}(Z > t) \le \int_t^\infty \frac{x}{t}\phi(x)\,\mathrm{d}x = \phi(t)/t$. Since $\phi(t+u) = \phi(u)e^{-tu-t^2/2} \le \phi(u)e^{-t^2/2}$ for $u \ge 0$, $\mathbb{P}(Z > t) \le \frac{1}{2}e^{-t^2/2}$. Since the derivative of $-\phi(x)/x$ is $(1+x^{-2})\phi(x)$, $\phi(t)/t = \int_t^\infty (1+x^{-2})\phi(x)\,\mathrm{d}x \le (1+t^{-2})\mathbb{P}(Z > t)$. (iv) By (i) with $A = \frac{1}{2}I_d$, $\mathbb{E}e^{|Y|^2/4} = 2^{d/2}$, and Markov's inequality gives $\mathbb{P}(|Y| \ge r) \le e^{-r^2/4}\mathbb{E}e^{|Y|^2/4}$. ∎

Proof of Theorem A.1. The plan is to cut W into a small ball around w_0 and its complement. Outside the ball, hypothesis (iii) makes the integral exponentially small. Inside, the substitution $w = w_0 + u/\sqrt{n}$ and Taylor's theorem turn the integrand into a Gaussian times $g(w_0)$, plus a term of order $n^{-1/2}$ that is odd in u and integrates to zero, plus a remainder of order n^{-1}. We may assume that $f(w_0) = 0$. The degenerate case $g(w_0) = 0$ is included; the theorem then says that the integral is $O(n^{-d/2-1})$.

Let $\mu > 0$ be the smallest eigenvalue of H. On $\bar{B}(w_0, r_0)$ let G bound $|g|$, let C_1 and C_2 bound the norms of the first and second derivatives of g, and let C_3 and C_4 bound those of the third and fourth derivatives of f. Taylor's theorem gives $|f(w_0+v) - \frac{1}{2}v^\top H v| \le \frac{C_3}{6}|v|^3$ for $|v| \le r_0$; with $\delta = \min(r_0, 3\mu/(2C_3))$, or $\delta = r_0$ if $C_3 = 0$,

$$\bigl|f(w_0+v) - \tfrac{1}{2}v^\top H v\bigr| \le \tfrac{\mu}{4}|v|^2 \qquad (|v| \le \delta). \tag{A.2}$$

Outside the ball. For $n \ge n_0$ and $|w - w_0| \ge \delta$, (iii) gives $e^{-nf(w)} \le e^{-(n-n_0)\eta(\delta)} e^{-n_0 f(w)}$, so by (iv) the integral over $W \setminus B(w_0, \delta)$ is $O(e^{-n\eta(\delta)})$.

Inside the ball. Substitute $w = w_0 + u/\sqrt{n}$, and write $\rho_n(u) = nf(w_0 + u/\sqrt{n}) - \frac{1}{2}u^\top H u$ and $g_n(u) = g(w_0 + u/\sqrt{n})$. The integral over the ball becomes $n^{-d/2} \int_{|u|<\delta\sqrt{n}} g_n(u) e^{-\frac{1}{2}u^\top H u - \rho_n(u)}\, \mathrm{d}u$. Taylor's theorem gives $\rho_n = n^{-1/2}T + r_n$, with the cubic form $T(u) = \frac{1}{6}\sum_{ijk} \partial_{ijk} f(w_0) u_i u_j u_k$ and $|r_n(u)| \le C_4|u|^4/(24n)$, and $g_n = g(w_0) + n^{-1/2}\nabla g(w_0)\cdot u + s_n$ with $|s_n(u)| \le C_2|u|^2/(2n)$; also $|g_n(u) - g(w_0)| \le C_1|u|/\sqrt{n}$, and $|\rho_n(u)| \le \frac{\mu}{4}|u|^2$ by (A.2). Writing $e^{-\rho} = 1 - \rho + E(\rho)$ with $|E(\rho)| \le \frac{1}{2}\rho^2 e^{|\rho|}$ and collecting terms,

$$g_n e^{-\rho_n} = g(w_0) + n^{-1/2}P + R_n, \qquad P(u) = \nabla g(w_0)\cdot u - g(w_0)T(u),$$

with $R_n = s_n - g(w_0)r_n - (g_n - g(w_0))\rho_n + g_n E(\rho_n)$. For $n \ge 1$ and $|u| < \delta\sqrt{n}$ these bounds give $|R_n(u)| \le \frac{c}{n}(1 + |u|^8)e^{\mu|u|^2/4}$, with c independent of n and u; the largest term is $g_n E(\rho_n)$, in which $\rho_n^2 \le 2(C_3^2|u|^6/36 + C_4^2|u|^8/576)/n$. Since $e^{-\frac{1}{2}u^\top H u} \le e^{-\mu|u|^2/2}$, the remainder contributes $O(n^{-1})$, by part (ii) of Lemma A.2. The odd polynomial P integrates to zero against the even weight over the ball. By part (i), $\int_{|u|<\delta\sqrt{n}} e^{-\frac{1}{2}u^\top H u}\, \mathrm{d}u$ is $(2\pi)^{d/2}(\det H)^{-1/2}$ minus at most $\int_{|u|\ge\delta\sqrt{n}} e^{-\mu|u|^2/2}\, \mathrm{d}u$, which part (iv) bounds by $(4\pi/\mu)^{d/2} e^{-n\mu\delta^2/4}$ after the substitution $u = y/\sqrt{\mu}$.

Collecting. The integral over W is $n^{-d/2}\big((2\pi)^{d/2}(\det H)^{-1/2} g(w_0) + O(n^{-1})\big)$ plus exponentially small terms. This is (A.1), and taking logarithms gives the second display, because $\log(1 + O(n^{-1})) = O(n^{-1})$. ∎

Chapter 1 proves a random version of Theorem A.1, with $f = L_n$, for the free energy of a regular model (Theorem 1.4), and Chapter 3 uses the theorem itself in an exercise to integrate out a noise variance; Lemma A.2 serves wherever a Gaussian is integrated, as in Chapters 1, 3 and 8. At a singular point the proof breaks at its first step, because no ball satisfies (A.2); the rest of this appendix collects what replaces it.

Theorem A.3 (Stein's identity and Gaussian smoothing). *Let $Z \sim \mathcal{N}(0,1)$.*

(i) (Stein's identity.) If $f\colon \mathbb{R} \to \mathbb{R}$ is absolutely continuous on every bounded interval and $\mathbb{E}|f'(Z)| < \infty$, then $\mathbb{E}|Zf(Z)| < \infty$ and $\mathbb{E}[Zf(Z)] = \mathbb{E}[f'(Z)]$. If $Y \sim \mathcal{N}(0, I_d)$ and $f\colon \mathbb{R}^d \to \mathbb{R}$ is continuously differentiable with $\mathbb{E}|Y_i f(Y)|$ and $\mathbb{E}|\partial_i f(Y)|$ finite, then $\mathbb{E}[Y_i f(Y)] = \mathbb{E}[\partial_i f(Y)]$.

(ii) (The heat equation.) Let $\psi\colon \mathbb{R} \to \mathbb{R}$ be measurable with $\int |\psi(t)| e^{-\kappa t^2}\, \mathrm{d}t < \infty$ for some $\kappa \in (0, \frac{1}{2})$, and let $\Psi(s,z) = \mathbb{E}\,\psi(z + \sqrt{s}Z)$ for $0 < s < 1/(2\kappa)$ and $z \in \mathbb{R}$. Then Ψ is infinitely differentiable, its derivatives may be taken under the integral sign, and $\partial_s \Psi = \frac{1}{2}\partial_z^2 \Psi$.

(iii) (Tweedie's formula.) If moreover $\psi \ge 0$ and $\psi > 0$ on a set of positive measure, then $h(z) = \int e^{-(t-z)^2/2}\psi(t)\, \mathrm{d}t$ is positive, and the probability density proportional to $e^{-(t-z)^2/2}\psi(t)$ has mean $z +$

$h'(z)/h(z)$ *and variance* $1 + (\log h)''(z)$.

(iv) *(Homogeneity.) For* $\alpha > -1$, *the function* $h(z) = \int e^{-(t-z)^2/2}|t|^\alpha \, \mathrm{d}t$ *satisfies*

$$h''(z) = \alpha h(z) - z h'(z) \quad (z \in \mathbb{R}), \qquad h(0) = 2^{(\alpha+1)/2}\,\Gamma\big(\tfrac{\alpha+1}{2}\big), \tag{A.3}$$

and $r = h'/h$ *satisfies* $r' = \alpha - zr - r^2$.

Proof. The plan is to prove (i) by exchanging two integrals and (ii) by differentiating the heat kernel, and to deduce (iii) and (iv) from (ii). (i) Let ϕ be the standard normal density; since $\phi'(x) = -x\phi(x)$, $\phi(z) = \int_z^\infty x\phi(x)\,\mathrm{d}x = -\int_{-\infty}^z x\phi(x)\,\mathrm{d}x$. Hence

$$\mathbb{E}f'(Z) = \int_0^\infty f'(z) \int_z^\infty x\phi(x)\,\mathrm{d}x\,\mathrm{d}z - \int_{-\infty}^0 f'(z) \int_{-\infty}^z x\phi(x)\,\mathrm{d}x\,\mathrm{d}z.$$

With $|f'|$ in place of f' the right side is $\mathbb{E}|f'(Z)| < \infty$, so Fubini's theorem exchanges the integrals, and $\int_0^x f' = f(x) - f(0)$ turns the right side into $\mathbb{E}[Z(f(Z) - f(0))] = \mathbb{E}[Zf(Z)]$; the same computation bounds $\mathbb{E}|Z(f(Z) - f(0))|$ by $\mathbb{E}|f'(Z)|$. In the vector case, for almost every value of the coordinates other than Y_i the one-variable identity holds in Y_i (Fubini), and integrating over those coordinates gives the claim. (ii) Write $\Psi(s,z) = \int p_s(z-t)\psi(t)\,\mathrm{d}t$ with $p_s(x) = (2\pi s)^{-1/2}e^{-x^2/(2s)}$, for which $\partial_s p_s = (\frac{x^2}{2s^2} - \frac{1}{2s})p_s = \frac{1}{2}\partial_x^2 p_s$. Every derivative of $p_s(z-t)$ in (s,z) is a polynomial in $z-t$, $s^{-1/2}$ and s^{-1} times $p_s(z-t)$. For $s_1 \le s \le s_2 < 1/(2\kappa)$ and $|z| \le A$, the inequality $(t-z)^2 \ge (1-\theta)t^2 - (\theta^{-1}-1)A^2$, with $\theta > 0$ so small that $(1-\theta)/(2s_2) > \kappa$, bounds each such derivative by a constant times $e^{-\kappa t^2}$; so dominated convergence allows differentiation under the integral sign, and the heat equation passes from p_s to Ψ. (iii) Here $h(z) = \sqrt{2\pi}\,\Psi(1,z)$, which is positive. By (ii), $h'(z) = \int (t-z)e^{-(t-z)^2/2}\psi(t)\,\mathrm{d}t$ and $h''(z) = \int ((t-z)^2 - 1)e^{-(t-z)^2/2}\psi(t)\,\mathrm{d}t$, so under the density of (iii) the mean of $t-z$ is h'/h and the mean of $(t-z)^2$ is $1 + h''/h$; the variance is $1 + h''/h - (h'/h)^2 = 1 + (\log h)''$. (iv) The function $|t|^\alpha$ is locally integrable because $\alpha > -1$, so (ii) applies, and the substitution $t = \sqrt{s}\,\tau$ gives the homogeneity $\Psi(s,z) = (2\pi)^{-1/2}s^{\alpha/2}h(z/\sqrt{s})$. At $s = 1$, $\partial_s\Psi = (2\pi)^{-1/2}(\frac{\alpha}{2}h - \frac{z}{2}h')$ and $\partial_z^2\Psi = (2\pi)^{-1/2}h''$, so the heat equation reads $h'' = \alpha h - zh'$. Finally $h(0) = \sqrt{2\pi}\,\mathbb{E}|Z|^\alpha$, which part (ii) of Lemma A.2 evaluates, and $r' = h''/h - r^2 = \alpha - zr - r^2$. ∎

Identity (A.3) can also be checked directly, since $\alpha h - zh' - h''$ is the integral of the derivative of $t|t|^\alpha e^{-(t-z)^2/2}$. That integration by parts in t is the virial identity of Chapter 9, which proves the analogue of (A.3) for every inverse temperature and for unequal weights on the two sides of 0 (Lemma 9.5). There, for a one-coefficient model whose coefficient density behaves like $|c|^\alpha$ at the truth, the mean of the rescaled limit posterior is $M(Z) = Z + r(Z)$ with $r = h'/h$, as in (iii), and the virial identity together with Stein's identity (i), applied to M, gives $\mathbb{E}[M(Z)^2] = 1 + \alpha$, which is 2λ when $\alpha = 2\lambda - 1$. With (iv) the same value follows from (i) applied to r:

$\mathbb{E}[Zr] = \mathbb{E}[r'] = \alpha - \mathbb{E}[Zr] - \mathbb{E}[r^2]$, so $\mathbb{E}[(Z+r)^2] = 1 + 2\mathbb{E}[Zr] + \mathbb{E}[r^2] = 1 + \alpha$ (Exercise 9.2). We confirmed (A.3) and this value by quadrature for seven values of α between $-\frac{3}{4}$ and 2.[1]

[1] The identity underlies Stein (1981)'s unbiased estimate of risk; Efron (2011) discusses Tweedie's formula and its history.

A.2 The Gamma function, Mellin transforms, and regular variation

The learning coefficient enters the theory through two transforms of the law of $K(w)$ under the prior: the zeta function $\zeta(z) = \int K^z \varphi$, a Mellin transform, whose largest pole is $-\lambda$; and the partition function $\int e^{-nK}\varphi$, a Laplace transform, which behaves like $n^{-\lambda}(\log n)^{m-1}$.

Definition A.4 (The Gamma function). For complex s with $\Re s > 0$, the *Gamma function* is $\Gamma(s) = \int_0^\infty t^{s-1}e^{-t}\,\mathrm{d}t$. It has the following properties.

(i) $\Gamma(s+1) = s\Gamma(s)$ and $\Gamma(1) = 1$, so that $\Gamma(k+1) = k!$ for integers $k \geq 0$; and $\Gamma(\frac{1}{2}) = \sqrt{\pi}$, $\Gamma(\frac{3}{2}) = \frac{\sqrt{\pi}}{2}$.

(ii) For $n > 0$ and $\Re s > 0$, $\int_0^\infty t^{s-1}e^{-nt}\,\mathrm{d}t = \Gamma(s)\,n^{-s}$.

(iii) For $\Re s > -1$ and integers $k \geq 0$, $\int_0^1 t^s(\log\frac{1}{t})^k\,\mathrm{d}t = k!/(s+1)^{k+1}$.

Proofs. (i) Integrating by parts, $\Gamma(s+1) = [-t^s e^{-t}]_0^\infty + s\int_0^\infty t^{s-1}e^{-t}\,\mathrm{d}t = s\Gamma(s)$, and $\Gamma(1) = \int_0^\infty e^{-t}\,\mathrm{d}t = 1$; the substitution $t = u^2$ gives $\Gamma(\frac{1}{2}) = 2\int_0^\infty e^{-u^2}\,\mathrm{d}u = \sqrt{\pi}$ by part (i) of Lemma A.2. (ii) Substitute $t' = nt$. (iii) The substitution $t = e^{-u}$ turns the integral into $\int_0^\infty u^k e^{-(s+1)u}\,\mathrm{d}u$, which is $k!/(s+1)^{k+1}$ by (ii) for real $s > -1$; both sides are holomorphic in s on $\Re s > -1$, so they agree there. ∎

Integrals of this kind recur throughout the book: part (ii) gives the constant $\Gamma(\lambda+1)$ of Theorem A.6, and part (iii) the Mellin transforms below.

Definition A.5 (Mellin transform). The *Mellin transform* of a measurable function v on $(0,\infty)$ is $\mathcal{M}v(z) = \int_0^\infty t^z v(t)\,\mathrm{d}t$, at the complex z for which the integral converges absolutely; for a finite measure ν on $(0,\infty)$ it is $\int t^z \nu(\mathrm{d}t)$. It has the following properties.

(i) For real $s > -1$ and integers $k \geq 0$, the transform of $t^s(\log\frac{1}{t})^k \mathbf{1}_{(0,1]}(t)$ is $k!/(z+s+1)^{k+1}$ on $\Re z > -s-1$; it continues to a meromorphic function on $\mathbb{C}$ with one pole, of order $k+1$, at $z = -s-1$.

(ii) (Continuation by integrating by parts.) Let $N \geq 1$ and $k \geq 0$ be integers and let b have N continuous derivatives on $[0,1]$. Then $I_k(z) = \int_0^1 t^z(\log\frac{1}{t})^k b(t)\,\mathrm{d}t$, which is holomorphic on $\Re z > -1$, continues to a meromorphic function on $\Re z > -N-1$ whose poles lie among $-1,\dots,-N$; near $z = -(j+1)$ it equals $k!\,b^{(j)}(0)/\big(j!\,(z+j+1)^{k+1}\big)$ plus a holomorphic function. If b is infinitely differentiable, then I_k is meromorphic on $\mathbb{C}$.

(iii) (Uniqueness.) If two functions integrable on $(0,1]$ and zero on $(1,\infty)$ have transforms that agree on a vertical line $\Re z = \sigma \geq 0$, then the functions agree almost everywhere.

The convention is shifted by one from the usual $\int t^{s-1}v(t)\,\mathrm{d}t$, so that the zeta function, $\int K^z\varphi = \int t^z \nu(\mathrm{d}t)$ with ν the law of $K(w)$ under φ, is the Mellin transform of ν, and of the state density $v = V'$ when it exists. Part (ii) is the mechanism behind every pole in the book: a function that behaves like $t^s(\log\frac{1}{t})^k$ near 0 has its rightmost pole at $-(s+1)$, of order $k+1$, and nothing away from 0 contributes a pole; for a state density $v(t) \approx Ct^{\lambda-1}(\log\frac{1}{t})^{m-1}$ this is the pole of order m at $-\lambda$. Transforms also turn products into products: for independent positive random variables, $\mathbb{E}[(XX')^z] = \mathbb{E}[X^z]\,\mathbb{E}[X'^z]$. For two independent uniforms on $(0,1]$ the product therefore has transform $(z+1)^{-2}$, which by (i) and (iii) is the transform of the density $\log\frac{1}{t}$; in the same way the product of L of them has the density $(\log\frac{1}{t})^{L-1}/(L-1)!$, which Chapter 3 finds by the equivalent route that $-\log|a_1\cdots a_L|$ is a sum of L independent exponential variables.

Proof of (ii) and (iii). The plan is to integrate by parts once, which moves the half-plane of convergence one unit to the left, to iterate by induction on N, and to obtain the powers of the logarithm by differentiating in z. Write $I_0(z;b) = \int_0^1 t^z b(t)\,\mathrm{d}t$. For $\Re z > -1$, integrating $t^z = \frac{\mathrm{d}}{\mathrm{d}t}\frac{t^{z+1}}{z+1}$ by parts against b gives

$$I_0(z;b) = \frac{b(1)}{z+1} - \frac{I_0(z+1;b')}{z+1}; \tag{A.4}$$

the boundary term at 0 vanishes because $\Re(z+1) > 0$. Since b' is bounded, $I_0(z+1;b')$ is holomorphic on $\Re z > -2$, so (A.4) continues $I_0(\cdot;b)$ to $\Re z > -2$ with at most a simple pole at -1, of residue $b(1) - I_0(0;b') = b(0)$. This is the case $N = 1$, $k = 0$. Suppose the case $k = 0$ holds for $N-1$, and let b have N continuous derivatives. By the induction hypothesis applied to b', the function $z \mapsto I_0(z+1;b')$ is meromorphic on $\Re z > -N-1$, with poles among $-2,\dots,-N$ and residue $b^{(j)}(0)/(j-1)!$ at $-(j+1)$. By (A.4), $I_0(\cdot;b)$ is meromorphic there too, with residue $-\frac{1}{-j}\cdot\frac{b^{(j)}(0)}{(j-1)!} = \frac{b^{(j)}(0)}{j!}$ at $-(j+1)$. For $k \geq 1$, since $(\log\frac{1}{t})^k t^z = (-\partial_z)^k t^z$ and differentiation under the integral sign is allowed on $\Re z > -1$, $I_k = (-1)^k I_0^{(k)}$; and $(-\partial_z)^k$ maps $c/(z+j+1)$ to $k!\,c/(z+j+1)^{k+1}$, and holomorphic functions to holomorphic functions. This proves (ii). For (iii), the substitution $t = e^{-u}$ turns the transform on the line $\Re z = \sigma$ into the Fourier transform of the integrable function $u \mapsto e^{-(\sigma+1)u}v(e^{-u})$ on $(0,\infty)$, which determines it almost everywhere. ∎

Applied in each coordinate in turn, with the other coordinates as parameters, part (ii) shows that $\int_{[0,1]^d} u_1^{2k_1z+h_1}\cdots u_d^{2k_dz+h_d}\,b(u)\,\mathrm{d}u$, for b infinitely differentiable, is meromorphic on $\mathbb{C}$ with poles among the numbers $-(h_j+1+\ell)/(2k_j)$, $\ell = 0,1,2,\dots$; with Theorem A.8 this is why the poles of the zeta function are negative rational numbers. Chapter 3 meets the Mellin transform as the zeta function of a one-coefficient model (Proposition 3.7), Chapter 4 uses it for the relation between the zeta function and the volume

(Theorem 4.6), and Chapters 5 and 6 for the zeta function on the charts of a resolution.

A positive function ℓ on $(0,\infty)$ is *slowly varying* at infinity if $\ell(cx)/\ell(x) \to 1$ as $x \to \infty$ for every $c > 0$, as powers of $\log x$ are. A function V on $(0,\infty)$ is *regularly varying* at 0 with index λ if $V(\varepsilon) = \varepsilon^\lambda \ell(1/\varepsilon)$ with ℓ slowly varying.

Theorem A.6 (Karamata's theorem for Laplace transforms). *Let ν be a finite measure on $[0,\infty)$ with $\nu(\{0\}) = 0$, let $V(\varepsilon) = \nu([0,\varepsilon))$ for $\varepsilon > 0$, and let $Z(n) = \int e^{-nt}\nu(\mathrm{d}t)$ for $n > 0$.*

(i) (Abelian direction, the case of this book.) If $V(\varepsilon) \sim C\varepsilon^\lambda(\log\frac{1}{\varepsilon})^{m-1}$ as $\varepsilon \to 0$, with constants $C > 0$, $\lambda > 0$ and $m \geq 1$, then as $n \to \infty$

$$Z(n) \sim C\,\Gamma(\lambda+1)\,n^{-\lambda}(\log n)^{m-1},$$
$$\int te^{-nt}\nu(\mathrm{d}t) \sim C\lambda\,\Gamma(\lambda+1)\,n^{-\lambda-1}(\log n)^{m-1}.$$

Consequently $-\log Z(n) = \lambda\log n - (m-1)\log\log n - \log\bigl(C\Gamma(\lambda+1)\bigr) + o(1)$, and the mean of nt under the probability measure proportional to $e^{-nt}\nu(\mathrm{d}t)$ tends to λ.

(ii) (Abelian direction, general.) If V is regularly varying at 0 with index $\lambda \geq 0$, then $Z(n) \sim \Gamma(\lambda+1)\,V(1/n)$.

(iii) (Tauberian direction.) Conversely, if $Z(n) \sim \Gamma(\lambda+1)\,n^{-\lambda}\ell(n)$ as $n \to \infty$, with $\lambda \geq 0$ and ℓ slowly varying at infinity, then $V(\varepsilon) \sim \varepsilon^\lambda\ell(1/\varepsilon)$ as $\varepsilon \to 0$. In particular $Z(n) \sim C'n^{-\lambda}(\log n)^{m-1}$ implies $V(\varepsilon) \sim \frac{C'}{\Gamma(\lambda+1)}\varepsilon^\lambda(\log\frac{1}{\varepsilon})^{m-1}$.

When ν is the law of $K(w)$ under the prior, V is the book's volume function and $Z(n) = \int e^{-nK}\varphi$, so that $-\log Z(n)$ is the population free energy $\bar{F}(n)$.

Proof of (i). The plan is to write both transforms as integrals of V against explicit kernels, to rescale so that V is evaluated at v/n, and to pass to the limit by dominated convergence. For $t \geq 0$, $e^{-nt} = \int_t^\infty ne^{-ns}\,\mathrm{d}s$, so Tonelli's theorem gives $Z(n) = n\int_0^\infty e^{-ns}\nu([0,s))\,\mathrm{d}s = n\int_0^\infty e^{-ns}V(s)\,\mathrm{d}s$. In the same way $te^{-nt} = \int_t^\infty (ns-1)e^{-ns}\,\mathrm{d}s$, and Fubini's theorem applies because ν is finite; hence $\int te^{-nt}\nu(\mathrm{d}t) = \int_0^\infty (ns-1)e^{-ns}V(s)\,\mathrm{d}s$. With $J_k(n) = \int_0^\infty v^k e^{-v}V(v/n)\,\mathrm{d}v$, the substitution $s = v/n$ gives

$$Z(n) = J_0(n), \qquad \int te^{-nt}\nu(\mathrm{d}t) = \frac{J_1(n) - J_0(n)}{n}.$$

It therefore suffices to show that $J_k(n) \sim C\Gamma(\lambda+k+1)\,a_n$ for $k = 0, 1$, with $a_n = n^{-\lambda}(\log n)^{m-1}$; then $J_1 - J_0 \sim C(\Gamma(\lambda+2) - \Gamma(\lambda+1))a_n = C\lambda\Gamma(\lambda+1)a_n$, and the ratio of the two transforms gives the mean of nt.

Choose $\varepsilon_0 \in (0,1)$ with $V(\varepsilon) \leq 2C\varepsilon^\lambda(\log\frac{1}{\varepsilon})^{m-1}$ for $0 < \varepsilon < \varepsilon_0$, and let $n \geq e$. On $v \geq n\varepsilon_0$ we use only $V \leq M = \nu([0,\infty))$: that part of $J_k(n)$ is at most $M\int_{n\varepsilon_0}^\infty v^k e^{-v}\,\mathrm{d}v$, which decays exponentially and is $o(a_n)$. On $0 < v < n\varepsilon_0$, the integrand of $J_k(n)/a_n$ converges to $Cv^{k+\lambda}e^{-v}$ for each

v, because $V(v/n) \sim C(v/n)^{\lambda}(\log \frac{n}{v})^{m-1}$ and $\log \frac{n}{v}/\log n \to 1$. It is also dominated: since $\log \frac{n}{v}/\log n = 1 + \log \frac{1}{v}/\log n$ is at most 1 for $v \geq 1$ and at most $1 + \log \frac{1}{v}$ for $v < 1$, it is at most

$$2C\, v^{k+\lambda} e^{-v} \big(1 + \log^{+} \tfrac{1}{v}\big)^{m-1},$$

which is integrable on $(0, \infty)$ because $\lambda > 0$. Dominated convergence gives $J_k(n)/a_n \to C \int_0^{\infty} v^{k+\lambda} e^{-v}\, \mathrm{d}v = C\Gamma(\lambda + k + 1)$. ∎

Parts (ii) and (iii) are Karamata's theorem, which we state without proof. Bingham et al. (1987, Theorem 1.7.1) state it for a nondecreasing function at infinity and its Laplace–Stieltjes transform near 0, as does Feller (1971, Chapter XIII). The version at the other end, used here, has the same proof: the measures $\nu_\tau(A) = \nu(\tau^{-1}A)/Z(\tau)$ have the transforms $Z(s\tau)/Z(\tau)$, which tend to $s^{-\lambda}$ as $\tau \to \infty$ under the hypothesis of (iii), and the continuity theorem for Laplace transforms turns this into $\nu_\tau([0,t]) \to t^{\lambda}/\Gamma(\lambda+1)$, which at $t = 1$ is (iii). Part (ii) is proved like part (i), with Potter's bounds for slowly varying functions (Bingham et al. 1987) in place of the explicit form of V in the domination. For the zeta function, Chapter 4 proves a Tauberian statement of another kind, from the largest pole of ζ to the asymptotics of V, with a theorem of Delange (1954) (Theorem 4.6). The limit in (i) is approached slowly when $m \geq 2$: for the product model, $K = a^2b^2/2$ under the uniform prior on $[-1,1]^2$, $C = 1/\sqrt{2}$, $\lambda = \frac{1}{2}$, $m = 2$, and $Z(n) = \frac{1}{2}\sqrt{\pi/(2n)}\,(\log n + \gamma_{\mathrm{E}} + \log 2)$ up to an exponentially small error, with γ_{E} Euler's constant, so the ratio of $Z(n)$ to its limit form is still 1.07 at $n = 10^8$. Part (i) is Lemma 4.3 of Chapter 4, and Chapter 8 applies it to multiples of K; its second asymptotic, which says that the tempered posterior of the population loss holds the energy $\mathbb{E}_w^{\beta}[nK] \approx \lambda/\beta$, is refined in Proposition 11.4 and underlies the estimators of Chapters 11 and 13.

A.3 Analytic functions and the resolution of singularities

Watanabe's theory needs the loss to be analytic, because analytic functions are the functions whose singularities can be resolved. The splitting lemma at the end of the section separates the regular directions of a critical point from the singular ones.

Definition A.7 (Real analytic functions). Let $U \subset \mathbb{R}^d$ be open. A function $f : U \to \mathbb{R}$ is *real analytic* if every point $w^* \in U$ has a neighborhood on which f is the sum of a convergent power series $\sum_{\alpha} c_\alpha (w - w^*)^{\alpha}$, where α runs over the multi-indices in $\mathbb{N}^d$ and $(w-w^*)^{\alpha} = \prod_j (w_j - w_j^*)^{\alpha_j}$. A function on a set $W \subset \mathbb{R}^d$ that is not open, such as a compact parameter set, is analytic if it is the restriction of an analytic function on an open set containing W; a map into $\mathbb{R}^k$ is analytic if its coordinates are. An *analytic set* is a set $\{w \in U : f_1(w) = \cdots = f_k(w) = 0\}$ with analytic f_j; when K is analytic, the set of true parameters $W_0 = \{K = 0\}$ is one.

A d-dimensional real analytic manifold is a space covered by charts onto open subsets of $\mathbb{R}^d$ with analytic transition maps; an analytic isomorphism is an analytic bijection with an analytic inverse; and a continuous map is proper if the preimage of every compact set is compact. The book uses three facts.

(i) Analytic functions are infinitely differentiable, and their power series are their Taylor series. Sums, products, compositions, and quotients with nonvanishing denominators of analytic functions are analytic; polynomials, exp, tanh, and log on $(0,\infty)$ are analytic.

(ii) (Identity theorem.) If U is connected and f is analytic on U and vanishes on a nonempty open subset, then $f = 0$ on U.

(iii) If U is connected and f is analytic and not identically zero on U, then $\{f = 0\}$ has Lebesgue measure zero.

Proof of (ii) and (iii). The plan is to prove (ii) by a connectedness argument and to reduce (iii), through (ii), to the zeros of functions of one variable. (ii) The set of points at which every derivative of f vanishes is closed, because the derivatives are continuous; it is open, because at such a point the Taylor series is zero, so f and its derivatives vanish on a neighborhood; and it is nonempty, because it contains the open set on which f vanishes. Since U is connected, it is all of U. (iii) Because every open cover of U has a countable subcover, it suffices to show that each point w^* has a neighborhood in which $\{f = 0\}$ is null. By (ii), some derivative of f is nonzero at w^*; let k be the smallest order of such a derivative. Then $P(v) = \sum_{|\alpha|=k} \partial^\alpha f(w^*) v^\alpha/\alpha!$ is a nonzero polynomial, so some unit vector v has $P(v) \neq 0$, and the k-th derivative of f at w^* in the direction v, which is $k!\,P(v)$, is not zero. After a rotation, v is the first coordinate vector, and $\partial_1^k f \neq 0$ on a small cube Q around w^*. For fixed $(w_2, \dots, w_d)$, the function $w_1 \mapsto f(w)$ on the side of Q has a nonvanishing k-th derivative, so by Rolle's theorem it has at most k zeros; by Fubini's theorem, $\{f = 0\} \cap Q$ is null. ∎

By (iii), the set of true parameters of an analytic K that is not identically zero is null, so $V(\varepsilon) \to 0$ as $\varepsilon \to 0$ for every prior with a density. By (ii), an analytic function on $\mathbb{R}$ that vanishes on $(-1, 0)$ vanishes everywhere, so the ReLU is not analytic; the ReLU neuron of the running models escapes the problem because its $K = a^2b^2/4$ is a polynomial. Analyticity can also fail more quietly. For the tanh neuron with $x \sim \mathcal{N}(0,1)$, $F(b) = \mathbb{E}[\tanh^2(bx)]$ is infinitely differentiable, but its Taylor coefficients at 0 are those of $\tanh^2$, of order $k(2/\pi)^{2k}$, times $\mathbb{E}x^{2k} = (2k-1)(2k-3)\cdots 1$, so its Taylor series diverges for every $b \neq 0$ and $K = \frac{a^2}{2}F(b)$ is not analytic at the singular point (with inputs bounded by R it is analytic for $|b| < \pi/(2R)$). Its learning coefficient is still that of a^2b^2, by comparison, but theorems that assume analyticity do not apply to it directly. Chapters 2, 4, 5 and 13 use these facts.

Theorem A.8 (Hironaka's resolution of singularities, in the form used by Watanabe). *Let f be a real analytic function on a neighborhood of a point $w^* \in \mathbb{R}^d$, with $f(w^*) = 0$ and f not identically zero near w^*. Then there are an open neighborhood U of w^*, a d-dimensional real analytic manifold M, and a proper real analytic map $g\colon M \to U$ with the following properties.*

(i) Let $U_0 = \{w \in U : f(w) = 0\}$ and $M_0 = g^{-1}(U_0)$. Then g restricted to $M \setminus M_0$ is an analytic isomorphism onto $U \setminus U_0$.

(ii) Every point $P \in M_0$ has analytic local coordinates $u = (u_1, \dots, u_d)$ centered at P in which

$$f(g(u)) = S\, u_1^{k_1} \cdots u_d^{k_d}, \qquad \det g'(u) = b(u)\, u_1^{h_1} \cdots u_d^{h_d},$$

with a sign $S \in \{-1, 1\}$, nonnegative integers k_j and h_j, not all k_j zero, and an analytic function b with $b(0) \neq 0$.

The theorem is due to Hironaka (1964a,b), who proved resolution of singularities for algebraic varieties over fields of characteristic zero. The real analytic local form above, with the Jacobian, is the version that Lin (2011, Theorem 3.3) attributes to Atiyah (1970) and that, by Lin's account, Watanabe (2009) uses as his Theorem 2.3; we have not checked either attribution in the original. Watanabe (2024, Section 3) states the theorem directly for a loss and a prior, in the form (A.5) below. The book uses it as a black box. A function is *normal crossing* at a point P if, in some analytic coordinates centered at P, it equals a monomial $u_1^{k_1} \cdots u_d^{k_d}$ times an analytic function that does not vanish at P; near P its zero set is then the union of the coordinate hyperplanes $\{u_j = 0\}$ with $k_j > 0$, which cross each other transversally; the theorem makes every analytic function normal crossing, together with the Jacobian, by a change of variables that changes nothing away from the zero set. The product model's $K = a^2 b^2/2$ is normal crossing at the origin already; $a^2 + b^4$ is not, because its zero set near the origin is a single point, while the zero set of a normal-crossing function is a union of hypersurfaces; Chapter 5 resolves it by blow-ups.[2]

[2] Hironaka received the Fields Medal for this work in 1970 (International Mathematical Union n.d.).

For a loss $K \geq 0$ and a prior $\varphi = \varphi_1 \varphi_2$, with $\varphi_1 \geq 0$ analytic and $\varphi_2 > 0$ infinitely differentiable, the theorem takes the form in which the book uses it. Applied to the product $K\varphi_1$, it makes K and φ_1 normal crossing in the same coordinates, because a factor of a monomial times a unit is again a monomial times a unit. Since $K \circ g \geq 0$, each exponent of $K \circ g$ is even and its sign is +1, for otherwise $K \circ g$ would change sign with one coordinate. With compactness this gives what Watanabe (2013, Theorem 1) calls the *standard representation*: in each of finitely many coordinate cubes,

$$K(g(u)) = u^{2k} = u_1^{2k_1} \cdots u_d^{2k_d}, \qquad \varphi(g(u))\,|\det g'(u)| = b(u)\,|u^h|, \tag{A.5}$$

with $b > 0$; that paper states that its proof is given in Theorem 6.1 of the 2009 book. The learning coefficient λ is the minimum of $(h_j + 1)/(2k_j)$ over all cubes and all j with $k_j > 0$, and the multiplicity m is the largest number of indices j attaining it in one cube (Theorem 5.3). The zeta function becomes a finite sum of integrals of the type treated after Definition A.5,

so its poles are negative rational numbers; Bernshtein and Gel'fand (1969) and Atiyah (1970) used Hironaka's theorem in this way to continue $\int |f|^z \varphi$ meromorphically, answering a question that I. M. Gel'fand had asked in 1954 (Chapter 4 tells the story).

Chapters 5 and 6 carry out resolutions by hand and by toric changes of variables, Chapters 8 to 11 use the theorem through Theorem A.11, and Chapter 13 uses it to define the local learning coefficient.

Theorem A.9 (The splitting lemma). *Let f be real analytic near a point $w^* \in \mathbb{R}^d$ with $\nabla f(w^*) = 0$, and let the Hessian $\nabla^2 f(w^*)$ have rank r, with p positive and $r - p$ negative eigenvalues. Then there are an analytic function g near $0 \in \mathbb{R}^{d-r}$ with $g(0) = 0$, $\nabla g(0) = 0$ and $\nabla^2 g(0) = 0$, and an analytic isomorphism $w = \Phi(x, y)$, $x \in \mathbb{R}^r$, $y \in \mathbb{R}^{d-r}$, from a neighborhood of 0 onto a neighborhood of w^* with $\Phi(0) = w^*$, such that*

$$f(\Phi(x, y)) = f(w^*) + \sum_{i \le p} x_i^2 - \sum_{p < i \le r} x_i^2 + g(y).$$

If w^* is a local minimum of f, then $p = r$ and $g \ge 0$ near 0. For $r = d$ this is the Morse lemma: near a nondegenerate critical point, f is a sum of signed squares in suitable coordinates.

Proof sketch. After a linear change of coordinates, $\nabla^2 f(w^*)$ is diagonal with entries ± 2 in the first r places and 0 in the others; write $w - w^* = (x, y)$. Because $\partial_x^2 f(w^*)$ is invertible, the analytic implicit function theorem solves $\partial_x f(x, y) = 0$ by an analytic $x = \xi(y)$ with $\xi(0) = 0$, and $\xi'(0) = 0$ because the off-diagonal block of the Hessian vanishes. Taylor's formula with integral remainder in x around $\xi(y)$ writes $f(\xi(y) + x', y) - f(\xi(y), y) = x'^{\top} Q(x', y) x'$ with Q analytic and symmetric and $Q(0, 0) = \operatorname{diag}(\pm 1)$; completing the squares one variable at a time, which needs only square roots of analytic functions that do not vanish at 0, turns this into $\sum_{i \le r} \pm x_i''^2$ by an analytic change $x'' = T(x', y) x'$. The function $g(y) = f(\xi(y), y) - f(w^*)$ has vanishing first and second derivatives at 0 because $\partial_x f(\xi(y), y) = 0$, $\xi'(0) = 0$ and the y-block of the Hessian vanishes. At a local minimum the Hessian has no negative eigenvalue, so $p = r$, and $g(y) = f(\Phi(0, y)) - f(w^*) \ge 0$. These are the steps of the usual proofs of the splitting lemma for smooth functions, which Gromoll and Meyer (1969) prove on Hilbert spaces, and of the Morse lemma (Arnold et al. 1985); what we add is only that every step preserves analyticity. We omit two routine verifications: the induction that completes the squares with y as a parameter, and the check that the composite of the three changes of variables has an invertible derivative at 0, so that it is an analytic isomorphism near 0 by the analytic inverse function theorem. The analytic implicit and inverse function theorems are in Krantz and Parks (2002, Chapter 2). ∎

Chapters 5 and 6 use the case $r = d$, the Morse lemma, for their regular answers. Chapter 13 proves its bounds $\frac{r}{2} \le \lambda(w^*) \le \frac{r}{2} + \frac{d-r}{4}$ in terms of the rank r of the Hessian (Theorem 13.3) by comparing volumes, for every loss $K \ge 0$ with four continuous derivatives, without the splitting lemma and

without analyticity; it uses the lemma only in Corollary 13.4, which says that for analytic K the lower bound is attained exactly when the set of true parameters is, near w^*, an analytic submanifold of dimension $d - r$.

A.4 The log-likelihood ratio on a resolved space

Hironaka's theorem controls the deterministic part of the free energy. Watanabe's theorems also need the random part, the deviation of nK_n from nK, and on the parameter space itself its natural normalization fails exactly where it matters. Write $f(x, w) = \log\{p_0(x)/p(x \mid w)\}$ with $p_0 = p(\cdot \mid w_0)$, so that $K(w) = \mathbb{E}_X[f(X, w)]$ and $K_n(w) = \frac{1}{n}\sum_i f(X_i, w)$ are the book's K and K_n. Then

$$nK_n(w) = nK(w) - \sqrt{nK(w)}\,\eta_n(w), \qquad \eta_n(w) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \frac{K(w) - f(X_i, w)}{\sqrt{K(w)}},$$

and η_n is $0/0$ on the set of true parameters, where the posterior concentrates. After resolution, under the relatively finite variance of condition (iv) of Assumption A.10 below, the numerator becomes divisible by the square root of the denominator, and the quotient is a well-defined random process with a Gaussian limit (Watanabe 2024, Section 4.2).[3] The conditions are the following.

[3] Watanabe presents the theory as the solution of these two problems: the singular geometry of $\{K = 0\}$ and the $0/0$ in η_n (Watanabe 2024, Section 1).

Assumption A.10 (Watanabe's fundamental conditions).

(i) The parameter set $W \subset \mathbb{R}^d$ is compact with nonempty interior, and $W = \{w : \pi_1(w) \geq 0, \ldots, \pi_k(w) \geq 0\}$ for analytic functions π_j.

(ii) The prior is $\varphi = \varphi_1\varphi_2$ with $\varphi_1 \geq 0$ analytic and $\varphi_2 > 0$ infinitely differentiable.

(iii) Some w_0 in the interior of W minimizes L, and all minimizers give the same density p_0. For some $s \geq 6$, the map $w \mapsto f(\cdot, w)$ is analytic on an open set containing W as a map into the space $L^s(q)$ of functions with $\int |f(x)|^s q(x)\,\mathrm{d}x < \infty$: near each point it is a power series in w with coefficients in $L^s(q)$.

(iv) (Relatively finite variance.) There are constants $\varepsilon_0, c > 0$ such that $\mathbb{E}_X[f(X, w)] \geq c\,\mathbb{E}_X[f(X, w)^2]$ for every w with $K(w) \leq \varepsilon_0$.

These are the Fundamental Conditions (1)–(4) of Watanabe (2013, p. 872), with the interior minimizer and the uniqueness of p_0 that the paper assumes on p. 871. Condition (iv) holds when q is realizable by the model or regular for it (Watanabe 2013, p. 872); for an unrealizable and singular pair it can fail, and Watanabe (2010b) describes an example in which it fails and the learning curve decays like $n^{-2/3}$ instead of $1/n$.

Theorem A.11 (The standard form of the log-likelihood ratio). *Assume Assumption A.10, let $\varepsilon > 0$ be small enough, and let W_ε° be the interior of $\{w \in W : K(w) \leq \varepsilon\}$. Then there are a d-dimensional real analytic manifold*

M; a proper analytic map $g\colon M \to W_\varepsilon^\circ$ that maps $\{u : K(g(u)) \neq 0\}$ one-to-one onto $\{w \in W_\varepsilon^\circ : K(w) \neq 0\}$; a function $a(x,u)$, analytic in u with values in $L^s(q)$; and an infinitely differentiable function $b > 0$, such that in each local coordinate of M

$$K(g(u)) = u^{2k}, \qquad f(x, g(u)) = u^k a(x,u),$$
$$\varphi(g(u))\,|\det g'(u)| = b(u)\,|u^h|,$$

with multi-indices k and h of nonnegative integers, not all k_j zero. Finitely many coordinate neighborhoods, each identified with a cube, cover the preimage of a compact set. Let $\xi_n(u) = n^{-1/2} \sum_{i=1}^n \big(u^k - a(X_i,u)\big)$.

(i) For every u, $\mathbb{E}_X[a(X,u)] = u^k$, so that $\mathbb{E}\,\xi_n(u) = 0$, and

$$nK_n(g(u)) = nu^{2k} - \sqrt{n}\,u^k\,\xi_n(u). \tag{A.6}$$

(ii) As $n \to \infty$, ξ_n converges in law, as a random continuous function on each coordinate cube with the uniform norm, to a centered Gaussian process ξ with covariance $\mathbb{E}[\xi(u)\xi(v)] = \mathbb{E}_X[a(X,u)a(X,v)] - u^k v^k$, and $\mathbb{E}[\sup_u \xi_n(u)^2] \to \mathbb{E}[\sup_u \xi(u)^2]$.

(iii) If q is realizable by the model, then $\mathbb{E}[\xi_n(u)^2] = \mathbb{E}_X[a(X,u)^2] = 2$ at every u with $u^{2k} = 0$.

The quartet (M, g, a, b) and its formulas are Theorem 1 of Watanabe (2013), which states that its proof is given in Theorem 6.1 of the 2009 book (Watanabe 2009). The argument, as Watanabe (2024, Section 4.2) sketches it, combines Theorem A.8 with condition (iv): relatively finite variance is what makes $f(x, g(u))$ divisible by u^k. For parts (ii) and (iii), Watanabe (2013, p. 881) cites Theorems 6.1–6.3 of the 2009 book, which we know only through that citation; Watanabe (2024, Section 4.2, eqs. (42)–(43)) states the convergence in law, the covariance, and the convergence of $\mathbb{E}\sup_u \xi_n(u)^2$. Restricting to $\{K \le \varepsilon\}$ loses nothing: for $\beta = \beta_0/\log n$ and $0 \le r < \frac{1}{2}$, the integrals of $e^{-n\beta K_n}\varphi$ and of $nK_n e^{-n\beta K_n}\varphi$ over $\{K \ge n^{-r}\}$ are $o_p(e^{-\sqrt{n}})$ (Watanabe 2013, Lemma 1). The book takes (ii) and (iii) on trust; part (i) is a computation.

Proof of (i). Integrating $f(x, g(u)) = u^k a(x,u)$ against q gives $u^{2k} = K(g(u)) = u^k \mathbb{E}_X[a(X,u)]$, so $\mathbb{E}_X[a(X,u)] = u^k$ wherever $u^k \neq 0$, and by continuity everywhere. Then $nK_n(g(u)) = \sum_i f(X_i, g(u)) = u^k \sum_i a(X_i,u) = u^k(nu^k - \sqrt{n}\,\xi_n(u))$. ∎

In the variable $t = nu^{2k}$, identity (A.6) reads $nK_n = t \mp \sqrt{t}\,\xi_n(u)$, with the sign of u^k: the random part of nK_n is of the order of the square root of its mean, uniformly up to the true set. (The number 2 in (iii) comes from $\mathbb{E}_X[e^{-f}] = 1$ in the realizable case, which makes $K = \mathbb{E}_X[e^{-f} - 1 + f]$ equal to $\frac{1}{2}\mathbb{E}_X[f^2]$ to second order in f.) Substituting (A.5) and (A.6) into $\int e^{-nK_n}\varphi$ gives the renormalized posterior, with density in t proportional to $t^{\lambda-1}e^{-t+\sqrt{t}\,\xi}$ on the coordinates attaining λ (Watanabe 2024, eq. (48)), from

which the free energy, generalization, WAIC and WBIC theorems of Part III follow. Chapter 8 uses Theorem A.11 in its proof sketch of Theorem 8.6, Chapters 9 to 11 rely on it for Theorems 9.9, 10.8 and 11.6, and Chapter 13 uses its local version.

A.5 Divergences and modes of convergence

Definition A.12 (Entropy, cross-entropy, and Kullback–Leibler divergence). Let q and p be probability densities with respect to a measure on a space $\mathcal{X}$ (Lebesgue measure for continuous data, counting measure for discrete data), with distributions Q and P. The *entropy* of q is $S(q) = -\int q \log q$, the *cross-entropy* of p relative to q is $-\int q \log p$, and the *Kullback–Leibler divergence* is $\mathrm{KL}(q \,\|\, p) = \int q \log(q/p) \in [0, \infty]$, with $0 \log 0 = 0$ and $\mathrm{KL}(q \,\|\, p) = \infty$ when $Q(p = 0) > 0$. When the entropy is finite, the divergence is the cross-entropy minus the entropy. In the book's notation $L(w)$ is the cross-entropy of $p(\cdot \mid w)$ relative to the truth q, S is the entropy of q, and in the realizable case $K(w) = L(w) - S = \mathrm{KL}(q \,\|\, p(\cdot \mid w))$. The total variation distance is $\mathrm{TV}(q, p) = \sup_A |Q(A) - P(A)| = \frac{1}{2} \int |q - p|$.

(i) (Gibbs' inequality.) $\mathrm{KL}(q \,\|\, p) \geq 0$, with equality only if $p = q$ almost everywhere.

(ii) (Pinsker's inequality.) $\mathrm{TV}(q, p) \leq \sqrt{\mathrm{KL}(q \,\|\, p)/2}$.

(iii) $\mathrm{KL}(\mathcal{N}(\mu_1, \sigma^2) \,\|\, \mathcal{N}(\mu_2, \sigma^2)) = (\mu_1 - \mu_2)^2/(2\sigma^2)$. Consequently, for a regression $y = f(w, x) + \mathcal{N}(0, \sigma^2)$ with inputs $x \sim q(x)$ and the truth $p(\cdot \mid w_0)$, $K(w) = \mathbb{E}_x[(f(w, x) - f(w_0, x))^2]/(2\sigma^2)$.

Proofs. (i) If the divergence is infinite, then there is nothing to prove. Otherwise $\log x \leq x - 1$ gives $-\mathrm{KL}(q \,\|\, p) = \int_{q>0} q \log \frac{p}{q} \leq \int_{q>0} (p - q) = P(q > 0) - 1 \leq 0$, with equality only if $p = q$ almost everywhere on $\{q > 0\}$ and $P(q > 0) = 1$. (ii) The plan is to reduce to distributions on two points and then to compare two functions of one variable. Assume the divergence is finite, so that $q = 0$ almost everywhere on $\{p = 0\}$. For a set B with $P(B) > 0$, Jensen's inequality for the convex function $x \log x$ under the density $p\mathbf{1}_B/P(B)$ gives the log-sum inequality $\int_B q \log \frac{q}{p} \geq Q(B) \log \frac{Q(B)}{P(B)}$. Let $A = \{q > p\}$, $\alpha = Q(A)$ and $\beta = P(A)$, so that $\mathrm{TV}(q, p) = \alpha - \beta \geq 0$; if $P(A)$ or $P(A^c)$ is 0, then so is $\mathrm{TV}(q, p)$. Otherwise the log-sum inequality on A and on A^c gives $\mathrm{KL}(q \,\|\, p) \geq k(\beta)$, where

$$k(\beta) = \alpha \log \frac{\alpha}{\beta} + (1 - \alpha) \log \frac{1 - \alpha}{1 - \beta}.$$

The function $j(\beta) = k(\beta) - 2(\alpha - \beta)^2$ vanishes at $\beta = \alpha$, and for $0 < \beta \leq \alpha$ its derivative is $(\alpha - \beta)\big(4 - \frac{1}{\beta(1-\beta)}\big) \leq 0$, because $\beta(1 - \beta) \leq \frac{1}{4}$. So $j \geq 0$, and $\mathrm{KL}(q \,\|\, p) \geq 2(\alpha - \beta)^2 = 2\,\mathrm{TV}(q, p)^2$. (iii) Under $\mathcal{N}(\mu_1, \sigma^2)$ the log-ratio $\frac{(x-\mu_2)^2 - (x-\mu_1)^2}{2\sigma^2}$ has mean $\frac{(\mu_1-\mu_2)^2}{2\sigma^2}$. For the regression, the factor $q(x)$ cancels in the ratio of joint densities, and the divergence is the mean over x of the divergence between the two conditional normal densities. ∎

When the truth is realizable and the model is smooth, the Hessian of K at a true parameter is the Fisher information (Watanabe 2013, p. 872). Chapters 1 and 2 identify K with a divergence and with half the mean squared difference of the regression functions, and Chapters 9 and 10 measure generalization by the divergence from the truth to the predictive distribution.[4]

Definition A.13 (Modes of convergence). Let Y_n and Y be random vectors in $\mathbb{R}^k$. The sequence Y_n converges *in probability* to Y (on a common probability space) if $\mathbb{P}(|Y_n - Y| > \eta) \to 0$ for every $\eta > 0$, and *in law* if $\mathbb{E}h(Y_n) \to \mathbb{E}h(Y)$ for every bounded continuous h; for real random variables, convergence in law is convergence of the distribution functions at each continuity point of the limit. Random continuous functions on a compact set converge in law if the same holds for every bounded h that is continuous for the uniform norm. The sequence is $O_p(1)$ if for every $\eta > 0$ some C has $\mathbb{P}(|Y_n| > C) \le \eta$ for all n; $Y_n = O_p(a_n)$ means $Y_n/a_n = O_p(1)$, and $Y_n = o_p(a_n)$ means $Y_n/a_n \to 0$ in probability.

(i) Convergence in probability implies convergence in law; convergence in law to a constant implies convergence in probability; and a sequence that converges in law is $O_p(1)$.

(ii) (Continuous mapping.) If $Y_n \to Y$ in law and Φ is continuous, then $\Phi(Y_n) \to \Phi(Y)$ in law.

(iii) (Slutsky's lemma.) If $Y_n \to Y$ in law and $Z_n \to c$ in probability for a constant c, then $(Y_n, Z_n) \to (Y, c)$ in law. Hence $Y_n + Z_n \to Y + c$ and $Z_n Y_n \to cY$ in law, and, for real Z_n and $c \neq 0$, $Y_n/Z_n \to Y/c$.

(iv) $O_p(1)\, o_p(1) = o_p(1)$ and $O_p(1) + o_p(1) = O_p(1)$; and if $Y_n \to Y$ in law and $R_n = o_p(1)$, then $Y_n + R_n \to Y$ in law.

The book uses the *portmanteau lemma*: convergence in law holds as soon as $\mathbb{E}h(Y_n) \to \mathbb{E}h(Y)$ for every bounded Lipschitz h (van der Vaart 1998, Lemma 2.2). With it, the first claim of (i) follows from $|\mathbb{E}h(Y_n) - \mathbb{E}h(Y)| \le L\eta + 2B\,\mathbb{P}(|Y_n - Y| > \eta)$ for h bounded by B with Lipschitz constant L, and the second from $\mathbb{P}(|Y_n - c| > \eta) \le \mathbb{E}\min(1, |Y_n - c|/\eta)$; the third claim and (iv) are proved in van der Vaart (1998, Chapter 2); and (ii) holds because $h \circ \Phi$ is bounded and continuous whenever h is.

Proof of Slutsky's lemma. The plan is to check the portmanteau criterion for the pairs and then to apply continuous maps. Let h be bounded by B and Lipschitz with constant L. For every $\eta > 0$,

$$|\mathbb{E}h(Y_n, Z_n) - \mathbb{E}h(Y_n, c)| \le \mathbb{E}\min\big(L|Z_n - c|, 2B\big) \le L\eta + 2B\,\mathbb{P}(|Z_n - c| > \eta),$$

whose upper limit is at most $L\eta$; and $\mathbb{E}h(Y_n, c) \to \mathbb{E}h(Y, c)$, because $y \mapsto h(y, c)$ is bounded and continuous. Letting $\eta \to 0$ gives $\mathbb{E}h(Y_n, Z_n) \to \mathbb{E}h(Y, c)$, so $(Y_n, Z_n) \to (Y, c)$ in law, and the sum and product follow from (ii). For the quotient with $c > 0$, the function $\Phi(y, z) = y/\max(z, c/2)$ is

[4] Pinsker's inequality is named after Pinsker (1964); the form with the constant $\frac{1}{2}$ is in Kullback (1967).

continuous, so $\Phi(Y_n, Z_n) \to Y/c$ in law; and $Y_n/Z_n = \Phi(Y_n, Z_n)$ on the event $\{Z_n > c/2\}$, whose probability tends to 1, so the difference tends to 0 in probability and the sum rule gives $Y_n/Z_n \to Y/c$. The case $c < 0$ is the same with $\min(z, c/2)$. ■

The book's asymptotic statements about random quantities hold in law, in probability, or in expectation over the data; the proofs in Chapters 1, 3, 9 and 12 use Slutsky's lemma to discard $o_p(1)$ terms.

A.6 Samplers and estimators

The book's measurements are made on models with few parameters, so most of them use deterministic quadrature; Markov chain Monte Carlo appears where quadrature is impossible, and as a check of the samplers on models where it is possible. This section describes what the kit `figures/scripts/sltkit.py` implements and what can go wrong. The energy is $E(w) = nL_n(w)$, the prior is uniform on a box unless stated, and the tempered posterior at inverse temperature β has density proportional to $\varphi e^{-\beta E}$.

Quadrature on a grid. For $d \le 2$, `free_energy_grid` evaluates E at the midpoints w_j of a regular grid on the box, with cell volume $|\Delta|$, and computes

$$F_n(\beta) \approx -\log \sum_j \frac{|\Delta|}{\mathrm{vol}(W)} e^{-\beta E(w_j)}$$

as a log-sum of exponentials from which the largest exponent is subtracted first; e^{-E} itself underflows once n is in the hundreds. The function `tempered_mean_grid` averages with the weights $p_j \propto e^{-\beta E(w_j)}$, and `sample_grid` draws a cell with probability p_j and a uniform point inside it, an exact sample from the piecewise-constant approximation of the posterior. The midpoint rule is accurate when the integrand varies smoothly on the scale of the cells, and it degrades once the posterior is much narrower than a cell. For the product model the posterior hugs the cross in a band of width about $n^{-1/2}/|a|$; on one data set at each n, 1600 points per side of $[-1, 1]$ reproduce the exact one-dimensional free energy below to 10^{-7} up to $n = 10^6$, but are off by 2×10^{-3} at $n = 10^7$ and by 0.25 at $n = 10^8$, where the band is a fraction of a cell wide. A script should check its grid by doubling it.

One coefficient. When the energy depends on w only through a scalar $c = g(w)$ and the noise is Gaussian, it is quadratic in c: $E = \hat{E} + \frac{S_{xx}}{2\sigma^2}(c - \hat{c})^2$, with $S_{xx} = \sum_i x_i^2$, $\hat{c} = \sum_i x_i y_i / S_{xx}$, and $\hat{E}$ the minimum over all real c. Then $F_n(\beta) = \beta\hat{E} - \log \int e^{-\beta S_{xx}(c-\hat{c})^2/(2\sigma^2)} \rho(c)\, \mathrm{d}c$, with ρ the density of $g(w)$ under the prior: $(\log \frac{1}{|c|})^{L-1}/(2(L-1)!)$ for $c = a_1 \cdots a_L$ with independent uniform factors on $[-1, 1]$, and $(|c|^{-1/2} - 1)/2$ for $c = ab^2$. The functions `coefficient_free_energy` and `free_energy_chain` integrate over $\hat{c} \pm 40$ posterior standard deviations, split at the singular points of ρ, by adaptive quadrature to relative accuracy 10^{-10}.

EXACT DRAWS FOR $y = abx$. Under the uniform prior on $[-1,1]^2$, the change of variables $(a,b) \mapsto (a,c)$ with $c = ab$ has Jacobian $1/|a|$, so given c the density of a is proportional to $1/|a|$ on $|c| \le |a| \le 1$: $\log|a|$ is uniform on $[\log|c|, 0]$. The function `sample_product_posterior` draws c from its one-dimensional posterior by inverting a distribution function tabulated on about 8×10^4 points, dense near 0, then $\log|a|$, a random sign, and $b = c/a$; the draws are exact up to the interpolation of that table.

RANDOM-WALK METROPOLIS. The function `metropolis` runs many chains at once.[5] From the current point w it proposes $w' = w + h\,\xi$ with $\xi \sim \mathcal{N}(0, I_d)$ and accepts with probability $\min\{1, \pi(w')/\pi(w)\}$, where $\pi \propto \varphi e^{-\beta E}$; the scripts' log-densities are $-\infty$ outside the box, so a proposal outside it is rejected. Since $\pi(w)\min\{1, \pi(w')/\pi(w)\} = \min\{\pi(w), \pi(w')\}$ is symmetric in (w, w'), the chain satisfies detailed balance and leaves π invariant. During the first steps, which are discarded, each chain multiplies its h by $e^{0.05(1-0.3)}$ after an acceptance and by $e^{-0.05\cdot 0.3}$ after a rejection (for the default target 0.3), a recursion whose mean drift vanishes at acceptance rate 0.3; then h is frozen, so the kept draws come from an ordinary Metropolis chain. The rate 0.3 lies above 0.234, the optimal acceptance rate for densities with many independent coordinates (Roberts et al. 1997), because the book's posteriors have few. For a positive continuous density on a box the chain is irreducible and aperiodic, and its averages converge to posterior means (Tierney 1994), but slowly on a singular posterior: across an arm of the cross its width is about $n^{-1/2}/|a|$, along the arm it reaches the edge of the box (Theorem 2.5), and a step size accepted at the narrow end diffuses only slowly along $\log|a|$. Separated regions, such as the phases of Chapter 12, may not communicate at all. The defence is to run chains from dispersed starting points and compare them, for instance with the rank-normalized $\hat{R}$ of Vehtari et al. (2021), and to validate every Metropolis measurement on a running model whose answer is known.

[5] The algorithm is that of Metropolis et al. (1953).

HAMILTONIAN MONTE CARLO AND TEMPERING. The kit implements neither. Hamiltonian Monte Carlo (Neal 2011) adds a Gaussian momentum, follows Hamilton's equations for the energy $-\log\pi$ with the leapfrog scheme, and accepts by a Metropolis test; it scales better with dimension, but its step size is also set by the narrowest direction of the posterior. Tempering runs chains at a ladder of inverse temperatures, which helps when regions that do not communicate at $\beta = 1$ do at smaller β.

THERMODYNAMIC INTEGRATION. With a normalized prior, the tempered free energy $F_n(\beta) = -\log \int \varphi e^{-\beta E}$ vanishes at $\beta = 0$, and differentiation under the integral sign gives

$$F_n'(\beta) = \mathbb{E}_w^\beta[nL_n(w)]\,, \qquad F_n''(\beta) = -\operatorname{Var}_w^\beta\big(nL_n(w)\big) \le 0,$$

so that $F_n = \int_0^1 \mathbb{E}_w^\beta[nL_n(w)]\,\mathrm{d}\beta$. The function `thermodynamic_integration` applies the trapezoid rule on the ladder $\beta_k = (k/K)^5$, $k = 0, \dots, K$, of `power_ladder`, which is dense near $\beta = 0$, where the integrand changes fastest: its excess over $nL_n(w_0)$ is of order n at $\beta = 0$, where the tempered posterior is the prior, and about λ/β once $n\beta$ is large (Chapter 11), so the range

$1/n \ll \beta \le 1$ contributes the term $\lambda \log n$. This is the power posterior method of Friel and Pettitt (2008). Two errors add up: the discretization of the integral, largest near $\beta = 0$, and the sampling error of each rung, largest where the tempered posterior is wide or changes phase.

WAIC AND WBIC. From draws $w_1, \dots, w_S$ of the posterior at $\beta = 1$, `waic` computes

$$T_n = -\frac{1}{n}\sum_{i=1}^{n} \log\Big(\frac{1}{S}\sum_{s=1}^{S} p(X_i \mid w_s)\Big), \qquad V_n = \sum_{i=1}^{n} \widehat{\mathrm{Var}}_s\big[\log p(X_i \mid w_s)\big],$$
$$\mathrm{WAIC} = T_n + V_n/n,$$

with the inner mean taken as a log-sum of exponentials and $\widehat{\mathrm{Var}}_s$ the sample variance over the draws, with divisor $S - 1$. The formula assumes exact draws at $\beta = 1$, and correlated or few draws add Monte Carlo error. The book's WAIC is a loss per sample; the deviance scale of Gelman et al. (2014), used by common software, is $2n$ times it. The function `wbic` averages the energy over draws at $\beta = 1/\log n$ (Watanabe 2013); Theorem 11.6 says how close the result is to F_n.

TWO TEMPERATURES. Since $\mathbb{E}_w^{\beta}[nL_n]$ is about $nL_n(w_0) + \lambda/\beta$, the difference of two tempered means cancels the unknown $nL_n(w_0)$:

$$\hat{\lambda} = \frac{\mathbb{E}_w^{\beta_1}[nL_n(w)] - \mathbb{E}_w^{\beta_2}[nL_n(w)]}{1/\beta_1 - 1/\beta_2}, \qquad \beta_i = \frac{\beta_{0i}}{\log n},$$

which converges to λ in probability (Watanabe 2013, Corollary 3) and is computed by `lambda_two_temperatures`; the second mean can be obtained from draws at the first temperature by reweighting with $e^{-(\beta_2-\beta_1)nL_n}$ (Watanabe 2013, eq. (20)). For $m > 1$ the next term of the tempered mean, $-(m-1)/(\beta \log(n\beta))$ in the population version of Proposition 11.4, biases the estimate by about $-(m-1)/\log n$; it is the β-derivative of the term $-(m-1)\log\log(n\beta)$ of the free energy, and Watanabe (2013) notes that this term affected his estimates for reduced-rank regression with $m = 2$.

VOLUMES. The function `volume_fraction` estimates $V(\varepsilon)$ by the fraction of N uniform draws with $K < \varepsilon$, with Clopper–Pearson intervals. The relative error is about $(NV(\varepsilon))^{-1/2}$, so N must be many times $1/V(\varepsilon) \approx \varepsilon^{-\lambda}$. The least-squares slope of $\log V$ against $\log \varepsilon$, from `loglog_slope`, estimates λ only when $m = 1$: for $V \sim C\varepsilon^{\lambda}(\log \frac{1}{\varepsilon})^{m-1}$ the local slope is $\lambda - (m-1)/\log \frac{1}{\varepsilon}$.

Most chapters' scripts use the deterministic methods of this section. Metropolis with the two-temperature estimator, the second mean obtained by reweighting, serves Chapters 7 and 11; thermodynamic integration and WBIC serve Chapter 11, whose scripts compute them by quadrature in their own code; and `waic` serves Chapter 10.

A.7 Stochastic-gradient Langevin dynamics

The posterior of a trained network cannot be computed on a grid, and Metropolis chains on millions of parameters are too slow. Part IV therefore estimates the local learning coefficient from a localized tempered posterior sampled by stochastic-gradient Langevin dynamics, following Lau et al. (2025). For a parameter w^*, an inverse temperature β, and a localization strength $\gamma > 0$, the *localized tempered posterior* is the probability density on $\mathbb{R}^d$

$$p(w \mid w^*, \beta, \gamma) \propto \exp\bigl(-n\beta L_n(w) - \tfrac{\gamma}{2}|w - w^*|^2\bigr),$$

in which a Gaussian factor centered at w^* replaces the prior, and the estimator of the local learning coefficient is

$$\hat{\lambda}(w^*) = n\beta^*\bigl(\mathbb{E}_{w|w^*,\beta^*,\gamma}[L_n(w)] - L_n(w^*)\bigr), \qquad \beta^* = 1/\log n,$$

the WBIC idea applied near one point (Lau et al. 2025, eqs. (10) and (12)). Consider its population version, at a true parameter w^* and with K in place of K_n. If the volume of $\{K < \varepsilon\}$ under the Gaussian weight behaves like $C\varepsilon^{\tilde{\lambda}}(\log \frac{1}{\varepsilon})^{\tilde{m}-1}$, then part (i) of Theorem A.6, with that weight in place of the prior, shows that the estimate tends to $\tilde{\lambda}$ as $n\beta \to \infty$ with γ fixed. Because the weight is positive on all of $\mathbb{R}^d$, $\tilde{\lambda}$ is set by the most singular true parameters anywhere, not only near w^*; Lau et al. (2025) conjecture that their localized coefficient is the local learning coefficient under conditions on γ, and Chapter 13 discusses what is known.

The Langevin diffusion $\mathrm{d}w = -\frac{1}{2}\nabla U(w)\,\mathrm{d}t + \mathrm{d}B_t$, with B a standard Brownian motion, has stationary density proportional to e^{-U}, and its Euler discretization with step ϵ is $w_{t+1} = w_t - \frac{\epsilon}{2}\nabla U(w_t) + \eta_t$ with independent $\eta_t \sim \mathcal{N}(0, \epsilon I_d)$. *Stochastic-gradient Langevin dynamics* replaces ∇U by an unbiased estimate computed on a minibatch.[6] For $U(w) = n\beta L_n(w) + \frac{\gamma}{2}|w - w^*|^2$ and a minibatch B_t of b indices drawn at random at each step, the update is

$$w_{t+1} = w_t + \frac{\epsilon}{2}\Bigl(\frac{n\beta}{b}\sum_{i\in B_t}\nabla \log p(X_i \mid w_t) + \gamma(w^* - w_t)\Bigr) + \eta_t, \tag{A.7}$$

[6] Welling and Teh (2011) introduced the method, with step sizes that decrease to zero.

which is the update of Lau et al. (2025, Section 4.4) in the notation of this book (their minibatch size m is our b). The estimate replaces the posterior mean by the average of $L_n(w_t)$ over the steps kept after a burn-in; the authors recycle the minibatch losses computed for the gradients as unbiased estimates of $L_n(w_t)$.

The method has five known weaknesses. First, with a constant step the chain does not sample the target: decreasing steps give consistency and a central limit theorem (Teh et al. 2016), but practice uses constant steps, and with the usual step proportional to $1/n$ the invariant law of SGLD departs from the posterior as the data grow and comes to resemble that of stochastic gradient descent, as Brosse et al. (2018) prove for strongly log-concave posteriors. Second, too small a step does not equilibrate and underestimates, and too large a step is unstable and overestimates; Lau et al. (2025, Appendix H) tune ϵ with a Metropolis-adjusted acceptance diagnostic aimed at 0.9–0.95. Third, too large a γ overwhelms the likelihood, and

the Gaussian localizer, which is not compactly supported, lies outside the setting of the theory, as Hitchcock and Hoogland (2025) note in a preprint. Fourth, if w^* is not near a local minimum, then the chain finds lower loss and the estimate can be negative. Fifth, the global convergence guarantees for stochastic-gradient samplers assume conditions that degenerate landscapes do not satisfy (Hitchcock and Hoogland 2025): as of 2026 we know of no theorem that covers the chains used to estimate the local learning coefficient, and their support is empirical, from known values for deep linear networks and a comparison with the Metropolis-adjusted Langevin algorithm on a small network (Lau et al. 2025), and from the same preprint's benchmark of local samplers on deep linear networks with known local values. Conventions for $n\beta$ and γ differ between papers, so estimates are comparable only within one convention.

Chapter 13 defines the local learning coefficient and estimates it with (A.7), Chapter 14 follows the estimate along training, and Chapter 15 compares Langevin dynamics with the stochastic gradient descent it modifies.

A.8 Intervals

Most error bars in the book are 95% intervals of one of the three kinds below; a few are normal-approximation intervals, an estimate plus or minus 1.96 standard errors, or intervals for a fitted slope or a correlation. Every caption says which, and over how many independent replicates.

Student-t intervals. For R independent replicates $v_1, \dots, v_R$ of a quantity, such as a free energy on R data sets, with mean $\bar{v}$ and sample variance $s^2 = \sum_r (v_r - \bar{v})^2/(R-1)$, `mean_ci` reports $\bar{v} \pm t_{R-1} s/\sqrt{R}$, with t_{R-1} the 0.975 quantile of Student's t distribution with $R-1$ degrees of freedom. If the replicates are normal, then the coverage is exactly 95%, because $\sqrt{R}(\bar{v} - \mathbb{E}v)/s$ has that distribution; if they only have a finite variance, then the coverage tends to 95% as R grows, by the central limit theorem and Slutsky's lemma (Definition A.13). For skewed or heavy-tailed replicates and small R the coverage can miss 95%, and the interval is wrong for the dependent draws of one Markov chain, which are not replicates.

Clopper–Pearson intervals. For J successes in N independent trials with success probability θ, such as J uniform draws in $\{K(w) < \varepsilon\}$, `clopper_pearson` reports $[\theta_-(J), \theta_+(J)]$ (Clopper and Pearson 1934). Here $\theta_-(k)$ is the 0.025 quantile of the Beta distribution with parameters $(k, N-k+1)$, or 0 if $k = 0$, and $\theta_+(k)$ is the 0.975 quantile of the Beta distribution with parameters $(k+1, N-k)$, or 1 if $k = N$.

Proposition A.14. *For every $\theta \in [0,1]$, $\mathbb{P}_\theta\big(\theta_-(J) \le \theta \le \theta_+(J)\big) \ge 0.95$.*

Proof. The plan is to express binomial tails as Beta probabilities and then to use monotonicity. For $1 \le k \le N$, $\mathbb{P}_\theta(J \ge k) = \mathbb{P}(B_k \le \theta)$, where B_k has the Beta distribution with parameters $(k, N-k+1)$. Indeed, if $U_1, \dots, U_N$ are independent and uniform on $[0,1]$, then the number of $U_i \le \theta$ has the law of J, and it is at least k exactly when the k-th smallest U_i is at most

θ; that order statistic has density $\frac{N!}{(k-1)!(N-k)!}u^{k-1}(1-u)^{N-k}$, which is the Beta density with these parameters. The quantiles $\theta_+(k)$ increase with k. If some k has $\theta_+(k) < \theta$, then let k^* be the largest; it satisfies $k^* < N$, the event $\{\theta_+(J) < \theta\}$ is $\{J \le k^*\}$, and

$$\mathbb{P}_\theta(J \le k^*) = \mathbb{P}(B_{k^*+1} > \theta) \le \mathbb{P}\big(B_{k^*+1} > \theta_+(k^*)\big) = 0.025,$$

because $\theta_+(k^*)$ is the 0.975 quantile of B_{k^*+1} and $\theta > \theta_+(k^*)$. In the same way the event $\{\theta_-(J) > \theta\}$, if it is not empty, is $\{J \ge k_*\}$ for the smallest $k_* \ge 1$ with $\theta_-(k_*) > \theta$, and $\mathbb{P}_\theta(J \ge k_*) = \mathbb{P}(B_{k_*} \le \theta) \le \mathbb{P}(B_{k_*} \le \theta_-(k_*)) = 0.025$. The two failures together have probability at most 0.05. ■

The interval is conservative: the infimum of its coverage over θ, approached just outside an end point of one of the intervals, is 96.10% for $N = 10$ and 95.002% for $N = 200$. Chapters 4 and 6 use it for Monte Carlo volume fractions, and other chapters for the frequencies of events.

Bootstrap intervals. For a statistic that is not a mean, such as an exponent fitted to Monte Carlo volume fractions at several levels ε in Chapter 4, a script that reports a bootstrap interval resamples the R independent replicates (there, the uniform draws, whose resampling amounts to multinomial counts) with replacement B times, recomputes the statistic on each resample, and reports the 2.5% and 97.5% quantiles of the B values: the percentile interval of Efron (1979). Its coverage tends to 95% for smooth functions of means of independent replicates. It can fail for nonsmooth statistics and for parameters on the boundary of the parameter space, where the bootstrap is inconsistent (Andrews 2000); estimators at a singular point, whose limit laws change with the truth, are of the same nonregular kind, so a bootstrap interval for such an estimator needs a check by simulation. Resampling the correlated draws of one chain understates the error.

Notes and References

Wong (2001) treats Laplace's method and the Mellin transform technique for integrals. Bingham et al. (1987) is the standard reference on regular variation; the proof of part (i) of Theorem A.6 was written for this book, as the usual dominated-convergence argument made elementary by the explicit form of V. Krantz and Parks (2002) develop real analytic functions, and van der Vaart (1998) the modes of convergence. Hironaka's theorem, Watanabe's standard representation and the theorem on the empirical process are taken from the sources named where they are stated, and where we know a source only through another's citation we say so; WAIC is from Watanabe (2010a).

Translating the literature. Watanabe's 2009 book writes $f(x,w) = \log\{q(x)/p(x \mid w)\}$ in the realizable case, and his later papers $f(x,w) = \log\{p_0(x)/p(x \mid w)\}$; their K and K_n are ours, and Watanabe (2024) writes θ and $\pi(\theta)$ for our w and $\varphi(w)$. Bingham et al. (1987) write U for our V and $\hat{U}(s)$ for our $Z(n)$, and state Karamata's theorem at infinity rather than at 0. Lau et al. (2025) write m for the minibatch size that we call b (our m is the

multiplicity) and β^* for $1/\log n$. Gelman et al. (2014) report WAIC on the deviance scale, $2n$ times ours.

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How This Book Was Made

Colophon

Roles

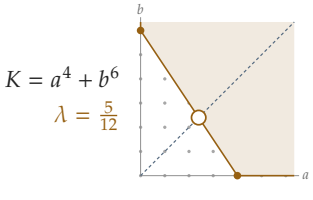
Singular Learning Theory for the Rest of Us was written by Claude Opus 5.5, an AI model made by Anthropic, in October 2026. Xiaoyu Li conceived the series *The Missing Textbooks* and proposed this volume: its subject and its audience, researchers in deep learning; its gap (a theory now used to measure how neural networks change during training, without a textbook that such readers find easy to learn from); its approach (one family of toy models, starting from the two-parameter model $y = abx$, on which every concept is computed by hand, with Hironaka's theorem used as a black box and the last chapters meeting the empirical results of deep learning); and its position relative to existing books (Watanabe's two books, the exercise books on WAIC and WBIC, and a series of blog posts). Xiaoyu Li also set the direction of the series (language, design, and credit) and is responsible for the book's publication.

Process

The book was written in one long session at maximum reasoning effort. A coordinating instance of the model first surveyed the literature with four research assistants, separate instances that verified, with web search and the papers themselves, Watanabe's core theory, the computation of learning coefficients, the work that applies the theory to deep learning, and the classical theory together with the books that already exist; they found no book, lecture notes, or course that teaches the theory to deep-learning researchers by computing every quantity on a family of small models. Their digests corrected or sharpened the plan in several places before a chapter was written: with Gaussian inputs the loss of the tanh neuron is smooth but not analytic, so the book draws its inputs from a uniform law; the mixture's loss is not analytic in its weight; the upper half of the Hessian bounds on the learning coefficient had been published in 2011 and 2017, before it was rediscovered in work on deep learning; WBIC differs from the free energy by a term of order $\sqrt{\log n}$, not by a bounded one; and that the estimator of the local learning coefficient with a Gaussian localizer recovers it is a conjecture of the estimator's authors, not a theorem.

The coordinator then wrote the plan (the thesis, the running models, the notation, fixed labels for results that chapters cite across the book, and a specification for each chapter), a style guide (the anatomy of a chapter, the voice, a proof style that follows Halmos, and the conventions of the scripts), the volume's typography, a shared kit for the scripts, and the prologue.

Separate instances of the same model drafted the fifteen chapters and the appendix, up to thirteen at a time. Each verified its references and the



claims it attributes to papers, wrote and ran the scripts behind its figures, checked its theorems and constants numerically, and built its chapter with the volume's checks for errors, references, layout, and style. Several authors found that their specifications were wrong, and said so. The correction term in the tempered mean energy of Chapter 11 does not follow from the asymptotics of the volume alone, as a slowly oscillating counterexample shows, so the chapter adds an assumption that analytic losses satisfy. The two regions of the mixture specified for Chapter 12 do not separate, and the chapter replaced the mixture with a double well. The digest's account of Watanabe's non-renormalizable example was wrong: the source states that the equations of state still hold, at order $n^{-2/3}$ (Chapter 10), and the author of Chapter 10 and the referees of Chapters 8 and 10 found that two constants printed in the source are too large by the factor $2^{4/3}$. Several results are, as far as the authors could find, new; the plan proposed some of them, which the authors proved and in several cases strengthened, and the authors found the others. Among them are the law of the posterior of $y = abx$ over scales (Chapter 2); the exact generalization error $1/(n-1)$ of the regular plane under a flat prior (Chapter 1); a proof that n times the expected generalization error tends to λ for every one-coefficient regression with Gaussian inputs and noise, through Tweedie's formula, a virial identity, and Stein's identity (Chapter 9); a proof that leave-one-out cross-validation and WAIC agree to order n^{-2} for $y = abx$ (Chapter 10); the tempered mean energy with its correction term and a complete proof of the WBIC formula for realizable linear models (Chapter 11); a proof of the local free energy formula for models linear in their coefficients, in regions that need not contain the truth (Chapter 12); a proof that with a fixed Gaussian localizer the population version of the estimator of the local learning coefficient can converge to the coefficient of another point (Chapter 13); a form of the rule that a minimally singular set of critical points has learning coefficient half its codimension, which derives that the set is a minimum instead of assuming it, with a Fourier computation of the Hessian at the hexagon of the toy model of superposition and a calibration of the estimator at its polygons (Chapter 14); and an analysis of where gradient flow on $y = abx$ stops, which adds convergence and a scale to the known conservation law (Chapter 15). The chapters mark these as the book's own.

Each chapter, and the prologue and appendix together, then received one independent referee pass from a separate instance, which re-derived the results line by line and checked them numerically, reran the scripts and compared every printed number with the text, worked the exercises, and checked the claims about papers against the papers; it corrected what it found directly. No referee found a main theorem that was false as stated, but the referees found real errors. Some hypotheses were wrong: a lemma on importance sampling lacked two conditions without which its identity can fail (Chapter 10), a theorem on Newton polyhedra did not require the prior to sit near the point (Chapter 6), a rule for products of losses omitted a convergence hypothesis (Chapter 5), and a lemma on symmetric criticality assumed a differentiability that the toy model of superposition, to

which it was applied, does not have (Chapter 14). A proof about a double well had a gap (Chapter 12). The transition map between two charts of a blow-up was wrong, and the second term in the volume of the cusp had the wrong sign and size (Chapter 5). A worked example claimed that Hessian ranks cannot show which of two points is more singular, which is false (Chapter 13). Some claims outran their evidence: one chapter said that it proves Watanabe's equations of state for one-coefficient models, where it proves three of the four limits they relate, and that it confirms the learning coefficient of a model whose measured interval excludes it (Chapter 9); three said that sets of estimates lay inside their intervals when some or all lie outside (Chapters 11, 13 and 14); and an error estimate presented as a bound was not a bound, and missed the typical error by five orders of magnitude (Chapter 1). In Chapter 7 a validation run contradicted a value proved in the same chapter, through the bias of a random-walk sampler, and a measured excess was attributed to the wrong cause; in Chapter 14 the calibration intervals ignored the variation between sampler seeds; and a table caption in Chapter 4 claimed checks that no script performed, until the referee wrote them. A heuristic for the collapse of stochastic gradient descent onto a saddle claimed more than holds off an invariant line (Chapter 15). Several exercises could not be solved as posed, and two were false. The referees of Chapters 12 and 14 confirmed that a crossing point reported for the toy model of superposition charges the loss twice what the paper's own likelihood does, which moves it from about 600 samples to about 1300; both chapters report it as an apparent inconsistency in a preprint. Two claims of novelty were too broad: the referee of Chapter 12 found work from 2003 on the scaling that the chapter had presented as new, and the referee of Chapter 15 found one of the chapter's identities in a 2018 paper; both claims are now qualified. Each error is corrected in the text. No referee found an arithmetic bug in a script, but many quoted numbers were misrounded or out of date with their scripts.

A usage limit stopped the running agents once, authors and referees alike, and each was resumed with its context. Once the chapters were drafted, the coordinator wrote the preface, the notation, and the epilogue. It then integrated the book: it reconciled cross-references between chapters, gave every work cited in several chapters a single bibliographic entry, rebuilt the epilogue's ledger from the chapters' final cards, and checked the whole book for layout, undefined references, and overflowing lines. A last referee pass checked the epilogue, the preface, the notation, this colophon, and the volume's README against the chapters.

Limits

No human has yet checked every proof or every attribution. Web search and the papers themselves were available while the book was written, but some full texts could not be opened: Watanabe's 2009 book is cited by chapter and section, and by theorem number only as later works, his own papers among them, cite it, and some claims about what a paper proves were checked against abstracts, the paper's own summary, or secondary

accounts; the chapters say so where it matters. One referee pass per chapter finds many errors but not all of them, and a referee that is an instance of the same model may share the author's blind spots. Results that the book presents as its own have been checked by the same model only. Much of the work that connects the theory to deep learning consists of preprints from 2023 to 2026, and most of the experiments behind it are small. The book's scripts are small too: most compute exact posteriors of models with a handful of parameters, and the rest run samplers or gradient descent on models with at most a few dozen, which confirms the arithmetic of the theorems, not their relevance to large networks. Corrections are welcome and will be recorded in the repository of *The Missing Textbooks*.

Typeset with pdfLaTeX in `missingbook`, the series class built on KOMA-Script, with the volume style `singular`; Palatino (newpx) for text and mathematics, Source Sans for labels, biblatex and biber for references, and Ti*k*Z and pgfplots for figures, whose data were computed with Python, NumPy, and SciPy, and checked with mpmath and SymPy. The title page carries 2400 draws from the posterior of $y = abx$ at its singular truth, computed on a grid of 1200×1200 cells, for 400 observations of pure noise (among the data sets with seeds 0 to 99, the one whose least-squares slope is closest to 0): the cross of Figure P.1. The foot of each chapter's opening page, and of each part page, carries the Newton polygon of a loss, the diagonal, and the point (t, t) at which the diagonal meets the polygon, which gives the learning coefficient $1/t$ (Chapter 6); this page carries that of the loss $a^4 + b^6$, whose coefficient is $\frac{5}{12}$. Both are computed or drawn by `figures/scripts/00-emblem.py` and `singular.sty`.